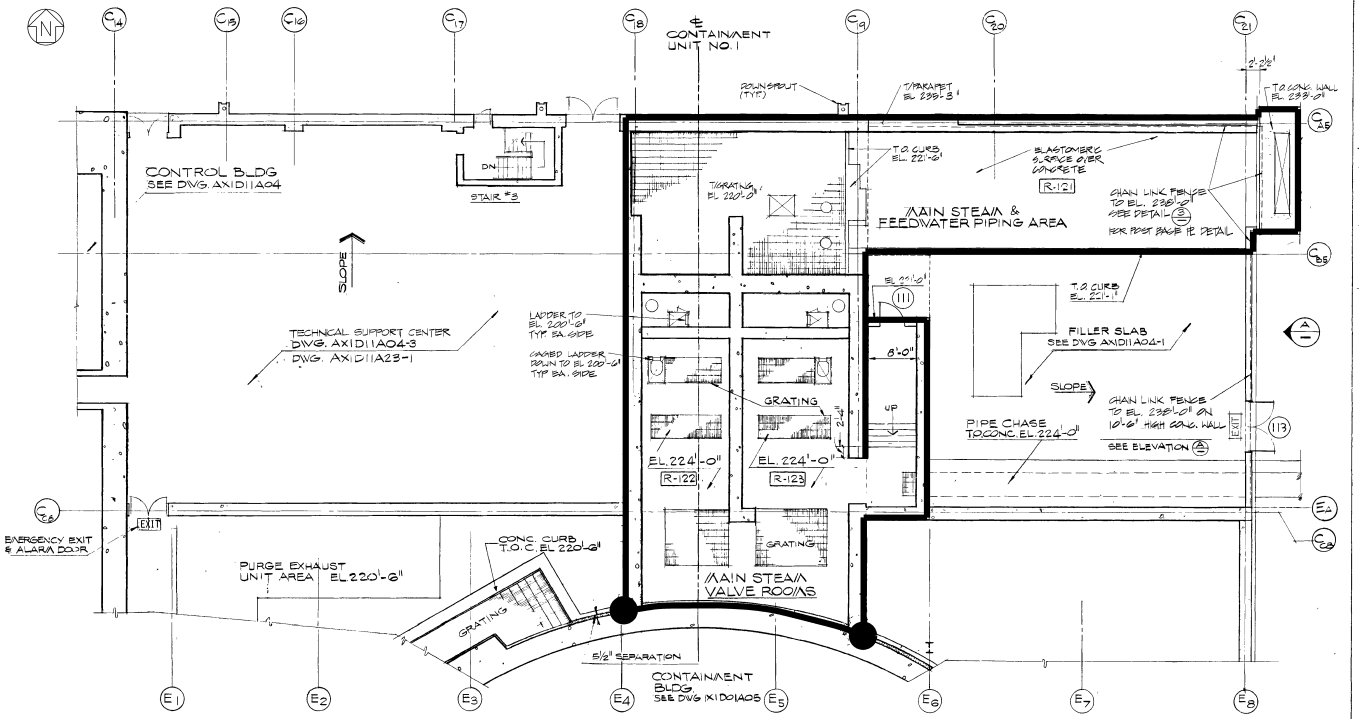
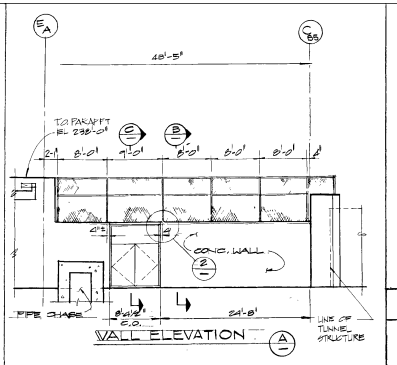
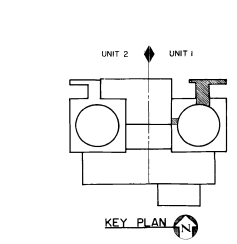


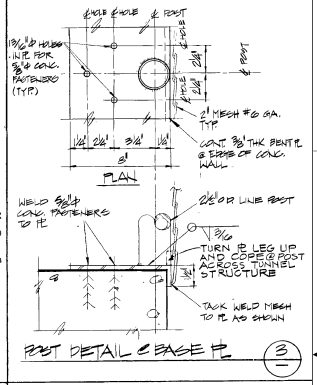
MAIN STEAM VALVE AREA
PARTIAL FLOOR PLAN EL. 220'-0"

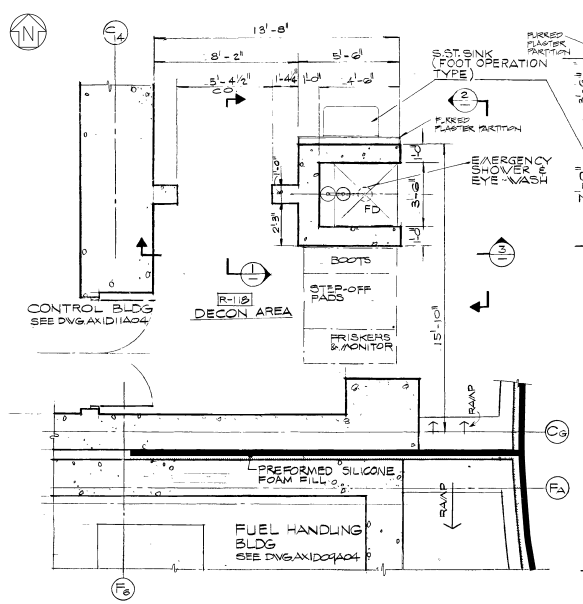


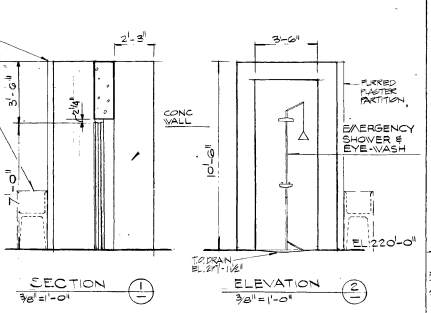


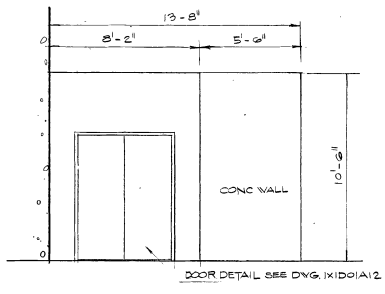
NOTES

FOR GENERAL NOTES SEE DWG. 1X1D01A01-1

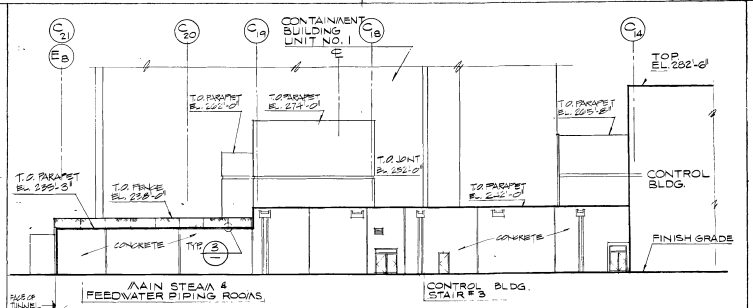




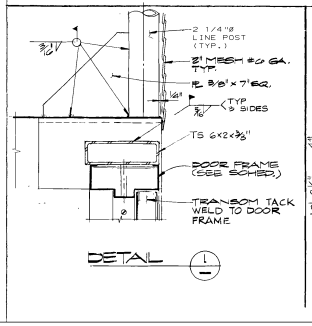


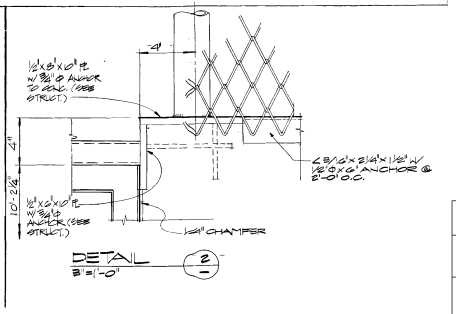


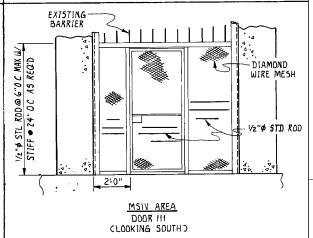
DECON. AREA
PARTIAL FLOOR PLAN EL. 220'-0"

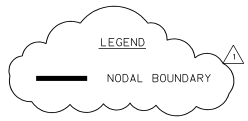


PARTIAL ELEVATION (NORTH)









SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

NODAL BOUNDARY
MSIV / MFIV AREA
CONTROL BUILDING

SCALE: NONE | DRAWING NO. 1X6DD302 | REV. 1

JOB NO. 10604

NO.	REVISIONS	DATE	DR	CHK	APPV	DTL
2						
1	REVISED PER REA 96-VAA054	10-1-98	GS	RBH	CBH	
0	ISSUED PER REA 96-VAA054	SEE MICROFILM FOR SIGNATURES				

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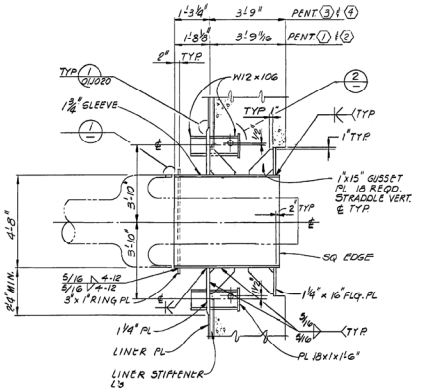
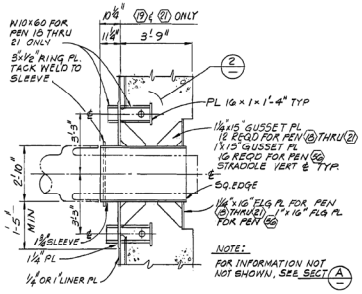
SIZE E 34x44 | CAD NAME 1X6DD302



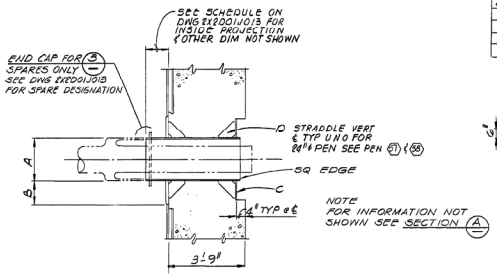
PENETRATIONS 1 THRU 4

SECTION A 1/2"=1'-0" — 01J010, 01J012, 01J013



PENETRATIONS 18 THRU 21 & 50

SECTION B 1/2"=1'-0" — 01J010, 01J012, 01J013

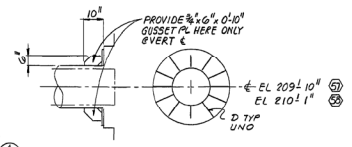


TYPICAL MECHANICAL PENETRATION

SECTION C 1/2"=1'-0" — 01J010, 01J011, 01J013

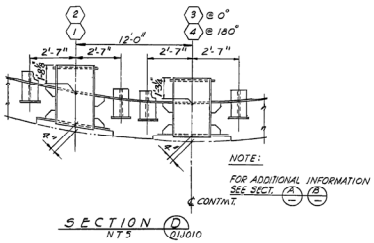
TYP MECHANICAL PEN SCHEDULE

A	B	C	D	NO. OF GUSSETS
20&24	13	1 x 11	¾ x 10	10
20	11	¾ x 9	½ x 8	8
18	11	¾ x 9	¾ x 8	8
18 & 10	9	½ x 7	½ x 6	6
16	11	¾ x 8	½ x 6	6



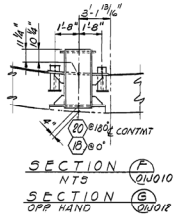
PENETRATION 57 & 58

SECTION H 1/2"=1'-0" — 01J010, 01J011



SECTION D NTS 01J010

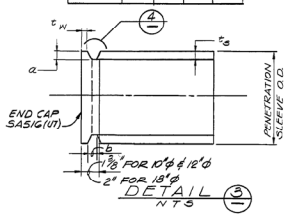
SECTION E OPP HAND 01J012



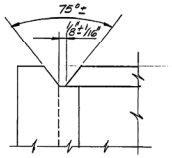
SECTION F NTS 01J010

SECTION G OPP HAND 01J012

SLEEVE SIZE	t s	a (+1/32)	b (+1/16)	t w (+1/16)
10" ø	.365	.865	.375	.5
12" ø	.375	.875	.375	.6
18" ø	.750	.750	.750	.875



DETAIL 3 NTS

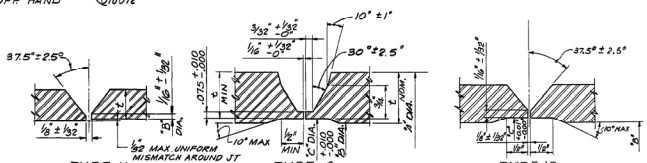


DETAIL 4 NTS

TABLE OF COUNTERBORE "C" DIMENSIONS FOR PENETRATION PIPE SLEEVES TYPES 1B AND 2 WELD END PREPARATIONS

PENETRATION NUMBER	PENETRATION PIPE SLEEVE OUTSIDE DIA	WALL THICKNESS SCHEDULE	"C" BORE DIMENSION	PIPE SLEEVE MATERIAL
1 THRU 4	26"	1.75" NOM	23.834"	SA-516, GR 70
18 THRU 21 & 50	24"	1.75" NOM	21.834"	
59 AND 60	26"	1" NOM	24.208"	
58	24"	1" NOM	22.208"	
63 AND 64	24"	0.5"/XS	23.084"	SA-333, GR 6
34 AND 35	20"	0.5"/XS	19.084"	
7 THRU 12, 69 THRU 74, 78, 80 AND 87	18"	0.5"/XS	17.084"	
91 THRU 98, 101 THRU 104	18"	0.75"/SCH 60	16.646"	
30, 31 AND 33	14"	0.5"/XS	13.192"	
57	24"	1" NOM	22.180	SA-516, GR 70

NOTES:
1. FOR GENERAL NOTES AND REFERENCE DWGS. SEE DWG. 1X2001J001.
2. MATERIAL SHOWN IN PHANTOM BY OTHERS.
3. COMPLETED ASSEMBLIES SHALL BE STRESS RELIEVED PER ASME SECTION III.
4. PLATES IDENTIFIED BY (U.T.) REQUIRE ULTRASONIC TESTING IN ACCORDANCE WITH SPECIFICATION X2AG06. REV 4
5. THE TORQUE ON THE 2 FLANGE CONNECTIONS SHOWN ON SHEET 2 FOR PENETRATION 5, 55 AND 90 SHOULD INITIALLY BE 320 FT. LBS. AND MAY BE INCREASED IF REQUIRED NOT TO EXCEED A MAXIMUM OF 448 FT. LBS.



TYPE 1A NOMINAL WALL (t) 3/16" THRU 3/8"

TYPE 2 NOMINAL WALL (t) OVER 1"

TYPE 1B NOMINAL WALL (t) OVER 3/8" THRU 1"

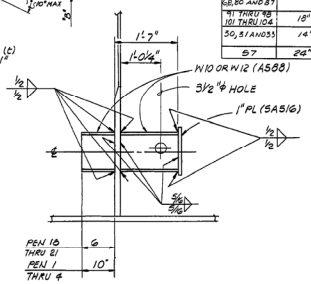
NOTE:
WELD-END PREPARATION TYPE 1B & TYPE 2 REQUIRE MACHINING TO COUNTERBORE DIMENSION "C" (SEE TABLE). "C" BORE TOLERANCE APPLICABLE WHILE PIECE IS IN THE SHOP JIG.

WHEN THE I.D. OF ACTUAL PIPE TO BE FABRICATED IS GREATER THAN REQUIRED FOR MINIMUM RING SEATING WHEN MACHINING TO DIMENSION "C" THE METAL REQUIRED FOR MACHINING MAY BE PROVIDED BY DEPOSITING WELD METAL ON THE I.D. OF THE PIPE IN THE AREA TO BE MACHINED.

ABBREVIATION
A. NOMINAL PIPE O.D., INCHES
B. NOMINAL PIPE I.D. INCHES
C. MACHINED PIPE I.D., INCHES.
t. NOMINAL WALL THK., INCHES

WELD PREPARATIONS

DETAIL 1 M.T.S.



DETAIL 2 1"=1'-0"

THIS DRAWING WAS RENUMBERED FROM 2X2001J019 (SHEET 1) 7.0
WORK THIS DRAWING WITH: 2X2001J019-2

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

CONTAINMENT
WALL LINER PLATE
PENETRATIONS DETAILS SHT. 1

SCALE: AS NOTED
JOB NO. 10604
DRAWING NO. 2X2001J019-1 7.0
SIZE E 34x44

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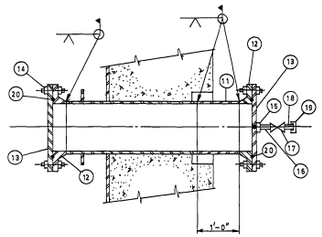


SECTION J 01J013
N.T.S.

PENETRATIONS (5) (55) (90)

NOTE:
FOR INFORMATION NOT SHOWN SEE SECTION C

SEE MODE 5 AND MODE 6 OPTION (OUTAGE "FIXTURE")

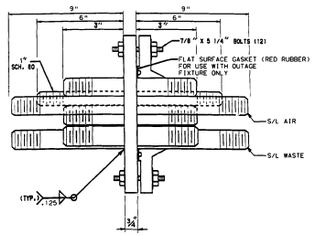
(11) EXTENSION SLEEVE FOR PENETRATION NRS 5, 55 & 90 – 10"ø PIPE, STD. WT, ASME SA-333 GR.6, CLASS 2.
(12) WELDING NECK FLANGE (10" RING JOINTS) – 10" 150 LBS, ASME SA-105 TO MATCH STD. WT. PIPE CLASS 2.
(13) BLIND FLANGE – 10" 150 LBS, WITH "R" RING JOINTS, ASME SA-105 GR.6, CLASS 2.
(14) NUTS & BOLTS – 7/8"ø ASME SA-194 GR.2H AND 7/8"ø X 5 1/4" LG. ASME SA-193 GR. B7.
(15) HALF COUPLING – 3/4" S.W. 3000 LBS., ASME SA-105, CLASS 2.
(16) 3/4"ø X 3" LONG NIPPLE (PBE.) – SCH. 80, ASME SA-106 GR.B, CLASS 2.
(17) PENS. 5 & 90 HAVE 3/4" GLOBE VALVE - MARK# 378 ASME SA-105, S.W. CLASS 2 PEN. 55 HAVE 3/4" GATE VALVE - MARK# 144 ASME SA-105, S.W. CLASS 2 USE VALVE TAG FOR VALVE TAG # 7.0
(18) 3/4"ø X 3" LG. NIPPLE (T.O.E.) – SCH 80 ASME SA-106 GR.B, CLASS 2.
(19) 3/4" THREATED CAP – 3000 LBS, ASME SA-105, CLASS 2.
(20) RING JOINT GASKET (R52) – STAINLESS STEEL.

~VALVE TABLE~

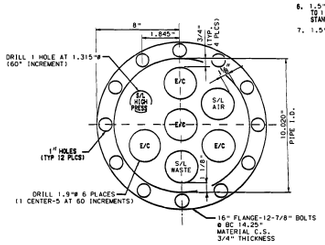
PENETRATION NR	VALVE TAG NR
5	2-1222-X4-017
55	2-1222-X4-018
90	2-1222-X4-019

(OUTAGE FIXTURE) NOTES:

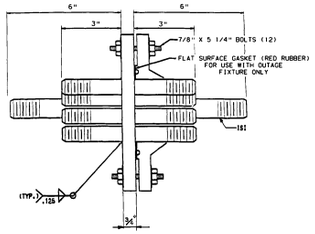
1. WELD ALL PIPE 1/8" FILLET BOTH SIDES OF 10"-150"ø FLANGE MACHINE TO 3/4" FLAT FACE (A-105).
2. TRUE FLANGE SEATING SURFACE AFTER WELDING (IF REQUIRED) MATE TO EXISTING FLANGE.
3. PROCESS IS GTAW ER70S-X OR SMAW E7018.
4. ALL PIPE IS TO BE THREADED ON BOTH ENDS. 1 1/2 TPI FOR 1.0"ø, 1.5"ø & 2.0"ø & 8 TPI - FOR 3.0"ø.
5. PIPE SLEEVES USED FOR CABLING TO BE SEALED WITH P-80 TO WITHIN 1/2" OF EACH END.
6. 1.5"ø PIPE OPENINGS NOT USED TO BE CAPPED WITH 1.5 COUPLING WELDED TO 1.5"ø SQ. DRIVE CAP. 3"ø & 2" SLEEVE PORTS TO BE CAPPED WITH STANDARD PIPE CAP WHEN NOT IN USE.
7. 1.5"ø, 2"ø & 3"ø PIPE SCH. 40, 1"ø SCH. 80 A-106.



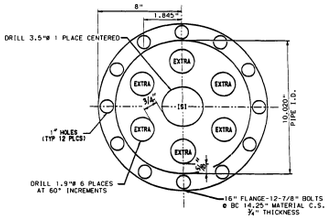
OUTAGE "FIXTURE" PROFILE
(1.5"ø & 1"ø PIPE)
(5) (55) OR (90)



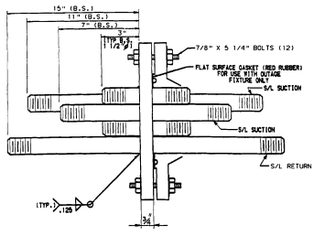
OUTAGE "FIXTURE" ELEVATION
(1.5"ø & 1"ø PIPE)
(5) (55) OR (90)



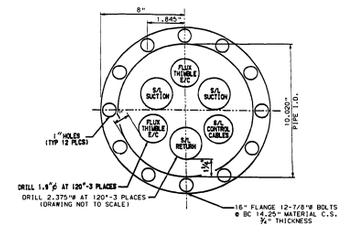
OUTAGE "FIXTURE" PROFILE
(3"ø & 1.5"ø PIPE)
(5) (55) OR (90)



OUTAGE "FIXTURE" ELEVATION
(3"ø & 1.5"ø PIPE)
(5) (55) OR (90)



OUTAGE "FIXTURE" PROFILE
(2"ø & 1.5"ø PIPE)
(5) (55) OR (90)



OUTAGE "FIXTURE" ELEVATION
(2"ø & 1.5"ø PIPE)
(5) (55) OR (90)

THIS DRAWING WAS RENUMBERED FROM 2X2D01J019 (SHEET 2) 7.0
WORK THIS DRAWING WITH: 2X2D01J019-1

0

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

CONTAINMENT
WALL LINER PLATE
PENETRATIONS DETAILS SHT. 1

SCALE: AS NOTED
JOB NO. 10604
DRAWING NO. 2X2D01J019-2 7.0
VER. 7.0
SIZE E 34X44

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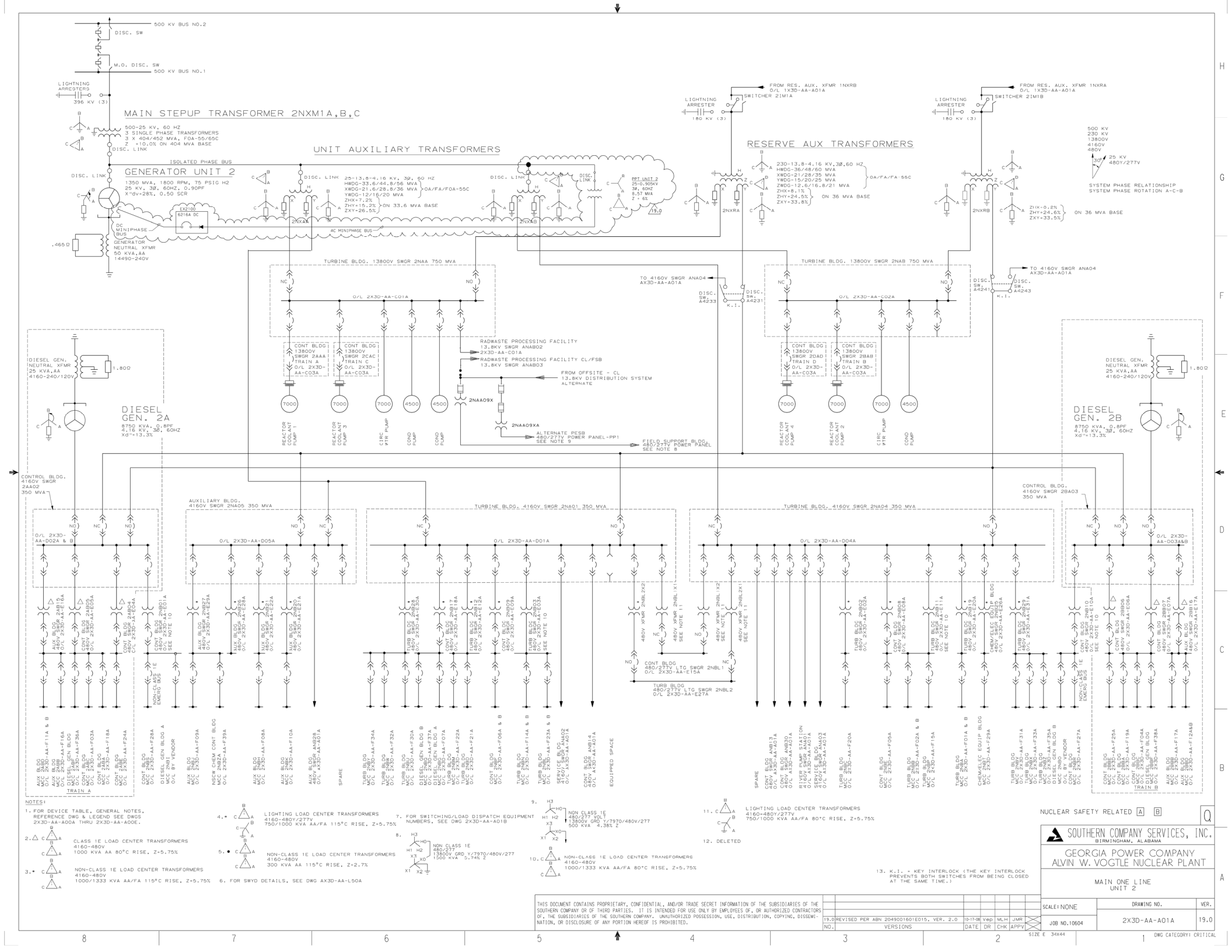
NO.	VERSIONS	DATE	DR	CHK	APPV
7.0	REVISED PER ABN 2100540301M003. VER 1.0	05/25/10	ELC	MWD	JMR


500 KV BUS NO.2
DISC. SW
M.O. DISC. SW
500 KV BUS NO.1
LIGHTNING ARRESTERS
396 KV (3)
MAIN STEPUP TRANSFORMER 2NXM1A,B,C
500-25 KV, 60 HZ
3 SINGLE PHASE TRANSFORMERS
3 X 404/452 MVA, FOA-55/65C
X"dv=28%, 0.50 SCR
Z =10.0% ON 404 MVA BASE
DISC. LINK
ISOLATED PHASE BUS
GENERATOR UNIT 2
1350 MVA, 1800 RPM, 75 PSIG H2
25 KV, 3Ø, 60HZ, 0.90PF
X"dv=28%, 0.50 SCR
EX2100
6216A DC
DC MINIPHASE BUS
.465Ω
GENERATOR NEUTRAL XFMR
50 KVA,AA
14490-240V
UNIT AUXILIARY TRANSFORMERS
25-13.8-4.16 KV, 3Ø, 60 HZ
HWDG-33.6/44.8/56 MVA
XWDG-21.6/28.8/36 MVA
YWDG-12/16/20 MVA
ZHX=7.2%
ZHY=15.2%
ZXY=26.5%
OA/FA/FOA-55C
ON 33.6 MVA BASE
2NXAA
2NXAB
AC MINIPHASE BUS
PPT UNIT 2
25-0.905KV
3Ø, 60HZ
8.57 MVA
Z = 6%
19.0
RESERVE AUX TRANSFORMERS
230-13.8-4.16 KV,3Ø,60 HZ
HWDG-36/48/60 MVA
XWDG-21/28/35 MVA
YWDG-15/20/25 MVA
ZWDG-12.6/16.8/21 MVA
ZHX=8.1%
ZHY=24.5%
ZXY=33.8%
OA/FA/FA-55C
ON 36 MVA BASE
2NXRA
2NXRB
FROM RES. AUX. XFMR 1NXRB
O/L 1X3D-AA-A01A
SWITCHER 21M1A
FROM RES. AUX. XFMR 1NXRA
O/L 1X3D-AA-A01A
SWITCHER 21M1B
ZIX=0.2%
ZHY=24.6%
ZXY=33.5%
ON 36 MVA BASE
500 KV
230 KV
13800V
4160V
480V
25 KV
480Y/277V
SYSTEM PHASE RELATIONSHIP
SYSTEM PHASE ROTATION A-C-B
TURBINE BLDG. 13800V SWGR 2NAA 750 MVA
TURBINE BLDG. 13800V SWGR 2NAB 750 MVA
O/L 2X3D-AA-C01A
O/L 2X3D-AA-C02A
TO 4160V SWGR ANA04
AX3D-AA-A01A
DISC. SW. A4233
DISC. SW. A4231
DISC. SW. A4241
DISC. SW. A4243
K.I.
CONT BLDG 13800V SWGR 2AAA TRAIN A
CONT BLDG 13800V SWGR 2CAC TRAIN C
CONT BLDG 13800V SWGR 2DAD TRAIN D
CONT BLDG 13800V SWGR 2BAB TRAIN B
O/L 2X3D-AA-C03A
RADWASTE PROCESSING FACILITY
13.8KV SWGR ANAB02
2X3D-AA-C01A
RADWASTE PROCESSING FACILITY CL/FSB
13.8KV SWGR ANAB03
FROM OFFSITE - CL
13.8KV DISTRIBUTION SYSTEM
ALTERNATE
2NAA09X
2NAA09XA
ALTERNATE PESB
480/277V POWER PANEL-PP1
SEE NOTE 9
FIELD SUPPORT BLDG.
480/277V POWER PANEL
SEE NOTE 8
REACTOR COOLANT PUMP 1
REACTOR COOLANT PUMP 3
REACTOR COOLANT PUMP 4
REACTOR COOLANT PUMP 2
CIRC WTR PUMP
COND PUMP
7000
4500
DIESEL GEN. NEUTRAL XFMR 25 KVA,AA 4160-240/120V
1.80Ω
DIESEL GEN. 2A
8750 KVA, 0.8PF
4.16 KV, 3Ø, 60HZ
Xd"=13.3%
DIESEL GEN. 2B
CONTROL BLDG. 4160V SWGR 2AA02 350 MVA
AUXILIARY BLDG. 4160V SWGR 2NA05 350 MVA
TURBINE BLDG. 4160V SWGR 2NA01 350 MVA
TURBINE BLDG. 4160V SWGR 2NA04 350 MVA
CONTROL BLDG. 4160V SWGR 2BA03 350 MVA
O/L 2X3D-AA-D02A & B
O/L 2X3D-AA-D05A
O/L 2X3D-AA-D01A
O/L 2X3D-AA-D04A
O/L 2X3D-AA-D03A&B
480V XFMR 2NBL1X1
480V XFMR 2NBL1X2
480V XFMR 2NBL2X1
480V XFMR 2NBL2X2
SEE NOTE 11
CONT BLDG 480/277V LTG SWGR 2NBL1
O/L 2X3D-AA-E15A
TURB BLDG 480/277V LTG SWGR 2NBL2
O/L 2X3D-AA-E27A
TRAIN A
TRAIN B
NON-CLASS 1E EMERG BUS
NON-CLASS 1E
SPARE
EQUIPPED SPACE
NOTES:
1. FOR DEVICE TABLE, GENERAL NOTES, REFERENCE DWG & LEGEND SEE DWGS 2X3D-AA-A00A THRU 2X3D-AA-A00E.
2. CLASS 1E LOAD CENTER TRANSFORMERS 4160-480V 1000 KVA AA 80°C RISE, Z=5.75%
3. NON-CLASS 1E LOAD CENTER TRANSFORMERS 4160-480V 1000/1333 KVA AA/FA 115°C RISE, Z=5.75%
4. LIGHTING LOAD CENTER TRANSFORMERS 4160-480Y/277V 750/1000 KVA AA/FA 115°C RISE, Z=5.75%
5. NON-CLASS 1E LOAD CENTER TRANSFORMERS 4160-480V 300 KVA AA 115°C RISE, Z=2.7%
6. FOR SWYD DETAILS, SEE DWG AX3D-AA-L50A
7. FOR SWITCHING/LOAD DISPATCH EQUIPMENT NUMBERS, SEE DWG 2X3D-AA-A01B
8. NON CLASS 1E 480/277 13800V GRD Y/7970/480V/277 1500 KVA 5.74% Z
9. NON-CLASS 1E 480/277 SOLT 13800V GRD Y/7970/480V/277 500 KVA 4.38% Z
10. NON-CLASS 1E LOAD CENTER TRANSFORMERS 4160-480V 1000/1333 KVA AA/FA 80°C RISE, Z=5.75%
11. LIGHTING LOAD CENTER TRANSFORMERS 4160-480Y/277V 750/1000 KVA AA/FA 80°C RISE, Z=5.75%
12. DELETED
13. K.I. - KEY INTERLOCK (THE KEY INTERLOCK PREVENTS BOTH SWITCHES FROM BEING CLOSED AT THE SAME TIME.)
NUCLEAR SAFETY RELATED A B
SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA
GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT
MAIN ONE LINE
UNIT 2
SCALE: NONE
DRAWING NO. 2X3D-AA-A01A
VER. 19.0
JOB NO. 10604
SIZE E 34x44
DWG CATEGORY: CRITICAL
19.0 REVISED PER ABN 204900160 1E015, VER. 2.0 10-17-08 Vep MLH JMR
NO. VERSIONS DATE DR CHK APPV
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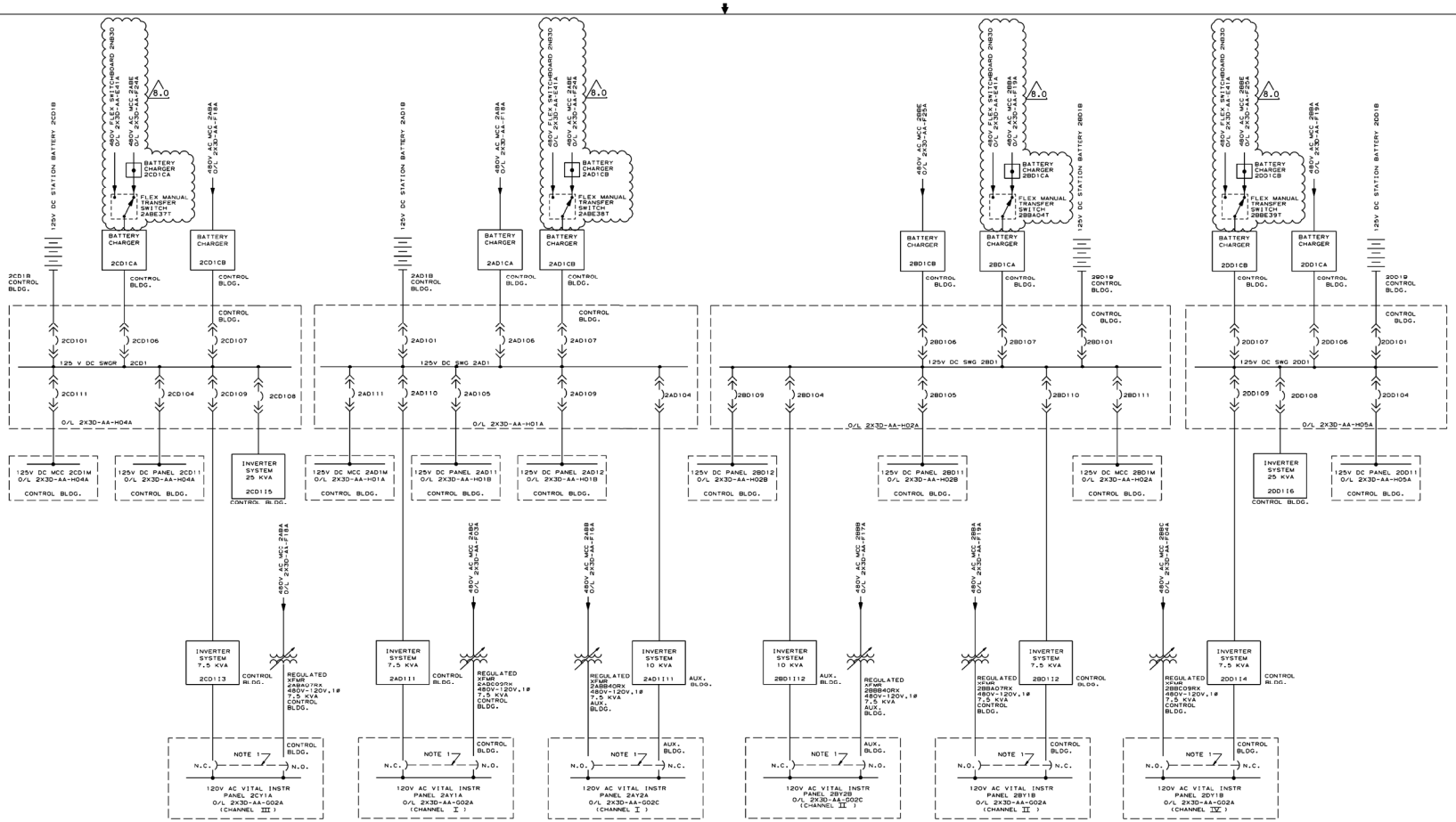

NOTE:
1. MECHANICAL INTERLOCKS SHALL BE PROVIDED SUCH THAT ONLY ONE BREAKER OF THE TWO INTERLOCKED IS CLOSED.

NUCLEAR SAFETY RELATED A B C D

0

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

MAIN ONE LINE
CLASS 1E 125V DC AND
120V VITAL AC SYSTEMS

SCALE: NONE

DRAWING NO. 2X3D-AA-G01A

VER. 8.0

JOB NO. 10604

SIZE E 34X44

DWG CATEGORY: CRITICAL

THIS DOCUMENT CONTAINS PROPRIETARY, CONFIDENTIAL, AND/OR TRADE SECRET INFORMATION OF THE SUBSIDIARIES OF THE SOUTHERN COMPANY OR OF THIRD PARTIES. IT IS INTENDED FOR USE ONLY BY EMPLOYEES OF, OR AUTHORIZED CONTRACTORS OF, THE SUBSIDIARIES OF THE SOUTHERN COMPANY. UNAUTHORIZED POSSESSION, USE, DISTRIBUTION, COPYING, DISSEMINATION, OR DISCLOSURE OF ANY PORTION HEREOF IS PROHIBITED.

NO.	VERSIONS	DATE	DR	CHK	APPV
8.0	REVISED PER ABN SNC475965E010, VER. 1.0	3/31/16	LEH	SO	WLN

EQUIPMENT TAG NUMBER	EQUIPMENT DESCRIPTION	PROJECT CLASSIFICATION (ELECTRICAL SYSTEM)	PROCUREMENT SPECIFICATION	BUS NUMBER	BUS VOLTAGE	NAMEPLATE RATING HP (OR KW)	REFERENCE DWG. 2X3D-	LOSS OF COOLANT ACCIDENT,SI PREFERRED OFFSITE SOURCE AVAILABLE: MANUAL PROCESS,ACCIDENT OR SEQUENCER START SEE TABLE 1	TIME (SECONDS) TO START (NOTE 7)	RUNNING TIME AFTER START (NOTE 4)	LOSS OF OFFSITE POWER & SUBSEQUENT SI: MANUAL PROCESS,ACCIDENT OR SEQUENCER START SEE TABLE 1	TIME (SEC) TO START AFTER D.G. BRKR IS CLOSED (NOTE 3.b)	RUNNING TIME AFTER START (NOTE 4)	LOSS OF OFFSITE POWER: MANUAL PROCESS,ACCIDENT OR SEQUENCER START SEE TABLE 1	TIME (SEC) TO START AFTER D.G. BRKR IS CLOSED (NOTE 3.c)	RUNNING TIME AFTER START (NOTE 4)
21202P4001M01	NUC SERV CLG WTR PP	11E	2AA02	4160V	700	BD-K04A		S	25.5	C	S	25.5	C	S	25.5	C
21202P4003M01	NUC SERV CLG WTR PP	11E	2AA02	4160V	700	BD-K04C		S	25.5	C	S	25.5	C	S	25.5	C
21202P4005M01	NUC SERV CLG WTR PP (SPARE)	11E	2AA02	4160V	700	BD-K04E		NOTE. 5	30.5	—	NOTE 5	30.5	—	NOTE 5	30.5	—
21203P4001M01	COMP CLG WTR PP	11E	2AA02	4160V	300	BD-L01A		S	20.5	C	S	20.5	C	S	20.5	C
21203P4003M01	COMP CLG WTR PP	11E	2AA02	4160V	300	BD-L01C		S	20.5	C	S	20.5	C	S	20.5	C
21203P4005M01	COMP CLG WTR PP (SPARE)	11E	2AA02	4160V	300	BD-L01E		NOTE 5	25.5	—	NOTE 5	25.5	—	NOTE 5	25.5	—
21204P6003M01	SAFETY INJ PP	11E	X6	2AA02	4160V	450	BD-D01C		S	5.5	C	S	5.5	C	M	>30.5	C
21205P6001M01	RESID HT REMVL PP	11E	X6	2AA02	4160V	400	BD-E01A		S	10.5	C	S	10.5	C	M	>30.5	C
21206P6001M01	CONTAINMENT SPRAY PP	11E	X6	2AA02	4160V	400	BD-J01A		NOTE 6	15.5	C	NOTE 6	15.5	C	M	>30.5	C
21208P6002M01	CENTRIFUGAL CHARG PP	11E	X6	2AA02	4160V	600	BD-C01A		S	0.5	C	S	0.5	C	S	0.5	C
21217P4001M01	AUX COMP CLG WTR PP	11E	X4	2AA02	4160V	600	BD-L03A		M	>30.5	C	M	>30.5	C	S	15.5	C
21302P4003M01	AUX FDWTR PP-MOTOR DRIVEN	11E	X4	2AA02	4160V	900	BC-F04A		S	20.5	C	S	20.5	C	2HV-1975 *	20.5	C
21592C7001M01	ESF CHILLER (COMPRESSOR)	11E	X4	2AA02	4160V	400	BG-G02A		P	>30.5	C	P	>30.5	C	2HV-1979 *	>30.5	C
															DISTR PANEL 2AY01		
2180553B04	LOAD CENTER TRANS. 2AB04	11E	X3	2AB04	4160 -480V	1000KVA	BA-D02F	—	—	C	—	0	C	—	0	C
	SECONDARY BREAKER 2AB04	11E	X3	2AB04	480V	—	BA-E02A	—	—	C	S	0.5	C	S	0.5	C
21501A7001M01	CTMT CLG UNIT HIGH SPEED	11E	X4	2AB04	480V	125	BG-B01A	NOTE 10	NA	NA	NOTE 10	NA	NA	S NOTE 15	50.5	C
	CTMT CLG UNIT LOW SPEED	11E	X4	2AB04	480V	62.5	BG-B03F	S	30.5	C	NOTE 10	NA	NA	2HV-5137	30.5	C
21501A7002M01	CTMT CLG UNIT HIGH SPEED	11E	X4	2AB04	480V	125	BG-B01B	NOTE 10	NA	NA	NOTE 10	NA	NA	S NOTE 15	50.5	C
	CTMT CLG UNIT LOW SPEED	11E	X4	2AB04	480V	62.5	BG-B03G	S	30.5	C	NOTE 10	NA	NA	PNL 2AYC1 (NOTE 20)	NA	NA
21501A7005M01	CTMT CLG UNIT HIGH SPEED	11E	X4	2AB04	480V	125	BG-B01E	NOTE 10	NA	NA	NOTE 10	NA	NA	S	30.5	C
	CTMT CLG UNIT LOW SPEED	11E	X4	2AB04	480V	62.5	BG-B03K	S	30.5	C	NOTE 10	NA	NA	21532B7003M01	30.5	C
21501A7006M01	CTMT CLG UNIT HIGH SPEED	11E	X4	2AB04	480V	125	BG-B01F	NOTE 10	NA	NA	NOTE 10	NA	NA	S	30.5	C
	CTMT CLG UNIT LOW SPEED	11E	X4	2AB04	480V	62.5	BG-B03L	S	30.5	C	S	30.5	C	NOTE 10	NA	NA
2180553B05	LOAD CENTER TRANS. 2AB05	11E	X3	2AB05	4.16 -480V	1000KVA	BA-D02L	—	—	C	—	0	C	—	0	C
	SECONDARY BREAKER 2AB05	11E	X3	2AB05	480V	—	BA-E02P	—	—	C	S	0.5	C	S	0.5	C
2156687001M01	DGB VENT FAN	11E	X4	2AB05	480V	50	BG-F01B	I	>30.5	C	I	>30.5	C	I	>30.5	C
2156687003M01	DGB VENT FAN	11E	X4	2AB05	480V	50	BG-F01C	P	>30.5	C	P	>30.5	C	P	>30.5	C
2153\N7001H01	CB CONTROL ROOM HTR	11E	X4	2AB05	480V	118KW	BG-C05P	TK NOTE 11	>30.5	C	TK NOTE 11	>30.5	C	TK NOTE 11	>30.5	C
2153\N7001H01	CB CONTROL ROOM FIL UNIT	11E	X4	2AB05	480V	125	BG-C01E	P	>30.5	C	P	>30.5	C	P	>30.5	C
2151\N7001000	CTB HYDROGEN RECOMBINER	11E	X6	2AB05	480V	75KW	BG-B02U	M	>30.5	C	M	>30.5	C	M	>30.5	C
2180553B15	LOAD CENTER TRANS. 2AB15	11E	X3	2AB15	4.16 -480V	1000KVA	BA-D02K	—	—	C	—	0	C	—	0	C
	SECONDARY BREAKER 2AB15	11E	X3	2AB15	480V	—	BA-E02R	—	—	C	S	0.5	C	S	0.5	C
21202W4001M01	NUC SERV CLG TWR FAN	11E	X4	2AB15	480V	100	BD-K03A	P	>30.5	C	P	>30.5	C	P	>30.5	C
21202W4001M02	NUC SERV CLG TWR FAN	11E	X4	2AB15	480V	100	BD-K03B	P	>30.5	C	P	>30.5	C	P	>30.5	C
21202W4001M03	NUC SERV CLG TWR FAN	11E	X4	2AB15	480V	100	BD-K03C	P	>30.5	C	P	>30.5	C	P	>30.5	C
21202W4001M04	NUC SERV CLG TWR FAN	11E	X4	2AB15	480V	100	BD-K03D	P	>30.5	C	P	>30.5	C	P	>30.5	C
21213P6002M01	SPENT FUEL PIT PUMP	11E	X6	2AB15	480V	100	BD-H01B	M	>30.5	C	M	>30.5	C	2HV-8808C	>30.5	C
21561N7001M01	PIPE PEN RM FILTER/EXH FAN	11E	X4	2AB15	480V	75	BG-D01D	S	15.5	C	S	15.5	C	2HV-19055 *	15.5	C
21561N7001H01	PIPE PEN FLTR/EXH UNIT HTR	11E	X4	2AB15	480V	80KW	BG-D01C	TK NOTE 17	15.5	C	TK NOTE 17	15.5	C	TK NOTE 17	15.5	C
														2HV-19057 *	THERM BARR CW RCP 004	11E	X4	2ABC	480V	0.7	BD-L03N	K	0	C	K	0.5	C
														2HV-8146	CHARG TO REAC COOL SYS ISO	11E	X6	2ABC	480V	0.7	BD-C03T	M	>30.5	10 SEC	M	>30.5	10 SEC
2180553B01	LOAD CENTER TRANS. 2NB01	62E	X3	2NB01	4.16 -480V	1000KVA	BA-D02G	SEE NOTE 3.a	>30.5	C	SEE NOTE 3.a	>30.5	C	—	0	C
	SECONDARY BREAKER	62E	X3	2NB01	480V		BA-E01A	SEE NOTE 3.a	>30.5	C	SEE NOTE 3.a	>30.5	C	S	10.5	C
21511B7001M01	CTB CAV CLG FAN (NON 1E)	62E	X4	2NB01	480V	75	BD-B02G	SEE NOTE 3.a	>30.5	C	SEE NOTE 3.a	>30.5	C	S	10.5	C
2180503PB1	FDR TO HTR PNL 2NBPB1	62E	X3	2NB01	480V	483KW	BD-B01K	SEE NOTE 3.a	>30.5	C	SEE NOTE 3.a	>30.5	C	K	10.5	C
21806B3CN5	BATTERY CHARGER 2ND3ACA	62E	X3	2NB01	480V	150KVA	BA-E03P	SEE NOTE 3.a	>30.5	C	SEE NOTE 3.a	>30.5	C	MM	10.5	C
21806B3CN1	BATTERY CHARGER 2ND1CA	62E	X3	2NB01	480V	100KVA	BA-E03F	SEE NOTE 3.a	>30.5	C	SEE NOTE 3.a	>30.5	C	MM	10.5	C
21806B3CAA	BATTERY CHARGER 2AD1CA	11E	X3	2ABA	480V	100KVA	AA-G01A	N	0	C	N	0.5	C	N	0.5	C
21806B3CCB	BATTERY CHARGER 2C01CB (NOTE 20)	11E	X3	2ABA	480V	75KVA	AA-G01A	N	0	C	N	0.5	C	N	0.5	C
21807Y312	REG XFMR 2ABA07RX	11E	X3	2ABA	480V	7.5KVA	AA-G02A	N	0	C	N	0.5	C	N	0.5	C
21531B7002M01	CBCR CHLR RM VENT FAN	11E	X4	2ABA	480V	5.0	BG-C01B	K	0	C	K	0.5	C	K	0.5	C
21592P7001M01	CB ESF CHILLED WATER PUMP	11E	X4	2ABA	480V	30	BG-G02C	I NOTE 16a	60	C	I NOTE 16a	60.5	C	M	>30.5	C
21531B7005M01	CB CR RETURN AIR FAN (NOTE 18)	11E	X4	2ABA	480V	20	BG-C02Z	NOTE 18	N/A	N/A	NOTE 18	N/A	N/A	NOTE 18	N/A	N/A
21592C7001M02 &M03	CB ESF CHILLER PURGE COMP. B	11E	X4	2ABA	480V	0.75	AA-F16A	TK NOTE 15b,c	>30.5	C	TK NOTE 15b,c	>30.5	C	TK NOTE 15b,c	>30.5	C
21539A7005M01	CB NORMAL AC RM ESF AC FAN MOT (NOTE 20)	11E	X4	2ABA	480V	5.0	BG-C07R	L	0	C	L	0.5	C	K	0.5	C
DISTR PNL 2AYA1	120/240V DISTRIBUTION PANEL	11E	X3	2ABA	480V	15KVA	AA-F18A	N	0	C	N	0.5	C	N	0.5	C
2HV-12\14	MCR AIR INTAKE ISO	11E	X4	2ABA	480V	1.9	BG-C01G	K	0	10SEC	K	0.5	10SEC	K	0.5	10SEC
2HV-12130	CBCR RTRN AIR FAN IN	11E	X4	2ABA	480V	.2	BG-C01X	I NOTE 11	>30.5	10SEC	I NOTE 11	>30.5	10SEC	I NOTE 11	>30.5	10SEC
2HV-12118	CBCR FLTR UNIT IN	11E	X4	2ABA	480V	.2	BG-C01V	M	>30.5	10SEC	M	>30.5	10SEC	M	>30.5	10SEC
2HV-12128	CBCR FLTR UNIT OUT	11E	X4	2ABA	480V	.2	BG-C01X	I NOTE 11	>30.5	10SEC	I NOTE 11	>30.5	10SEC	I NOTE 11	>30.5	10SEC
2HV-11605	NSCW PUMP P4005 OUT	11E	X5	2ABB	480V	1.9	BD-K04Z	I	70.5	15SEC	I	70.5	15SEC	I	70.5	15SEC
2HV-8508A	ALT CHG PUMP MINIFLOW	11E	X6	2ABB	480V	.7	BD-C01T	L	0	10SEC	L	0.5	10SEC	I	0.5	10SEC
2HV-8509A	ALT CHG PUMP MINIFLOW	11E	X6	2ABB	480V	.7	BD-C01V	M	>30.5	10SEC	M	>30.5	10SEC	M	>30.5	10SEC
21807Y3IA11	INVERTER 2AD1I11	11E	X3	2ABB	480V	10KVA	AA-G01A	SEE NOTE 14	0	C	SEE NOTE 14	0.5	C	SEE NOTE 14	0.5	C
21807Y3RX22	REG XFMR 2ABB40RX	11E	X3	2ABB	480V	7.5KVA	AA-G01A	N	0	C	N	0.5	C	N	0.5	C
21202P4008M01	NSCW XFR PUMP	11E	X4	2ABB	480V	30	BG-K01B	M	>30.5	40AYS	M	>30.5	40AYS	M	>30.5	C
21555A7005M01	ELEC MCC ROOM COOLER	11E	X4	2ABB	480V	2.0	BG-D05E	L	0	C	L	0.5	C	K	0.5	C
21555A7011M01	CCWP ROOM COOLER FAN	11E	X4	2ABB	480V	5.0	BG-D05E	L	0	C	L	0.5	C	K	0.5	C
21555A7017M01	SFP PUMP ROOM CLR	11E	X4	2ABB	480V	5.0	BG-D05U	L	0	C	L	0.5	C	K	0.5	C
2HV-11606	NSCW PUMP P4003 OUT	11E	X5	2ABB	480V	2.0	BD-K04Z	I	70.5	15SEC	I	70.5	15SEC	I	70.5	15SEC

NOTES:

1.) LOSS OF OFFSITE POWER (LOP) IS DEFINED AS A LOSS OF VOLTAGE ON THE 4.16KV BUS.

2.)
a. TIME T+0 SECONDS IS DEFINED AS THE TIME THE DIESEL GENERATOR BREAKER IS CLOSED FOR ALL LOP CONDITIONS AND THE TIME THE SI SIGNAL IS RECEIVED IN THE SF SEQUENCER WITH PREFERRED OFFSITE SOURCE AVAILABLE TO THE 4.16KV BUS.

b. DURING LOP CONDITIONS THE MAXIMUM TIME AFTER THE D.G. RECEIVES THE START SIGNAL TO THE FIRST LOAD STEP IS 10 SECONDS. THIS 10 SECOND INTERVAL INCLUDES A MAXIMUM OF 9.5 SECONDS FOR THE DIESEL GENERATOR TO START AND COME UP TO RATED VOLTAGE AND FREQUENCY PLUS 0.5 SECONDS AFTER THE DIESEL GENERATOR BREAKER CLOSES, TO ENERGIZE THE UNLOADED 4.16KV LOADCENTER TRANSFORMERS.

3.)
a. THE 4.16KV BREAKERS SUPPLYING 2NB01 AND 2NB10 WILL BE TRIPPED DURING SI CONDITIONS BY (12) POWER CAN BE RESTORED MANUALLY TO BUSSES 2NB01 AND 2NB10 UNDER ADMINISTRATIVE PROCEDURE AFTER THE SI SIGNAL HAS BEEN MANUALLY OVERRIDDEN FOR THIS 4.16KV BREAKER.

b. DURING LOP CONDITIONS AND SUBSEQUENT SI, ALL BUSSES (EXCEPT 2NB01, 2NB5, 2NB10, 2NBR, 2NBD AND 2NB1) WILL (20.) HAVE POWER AVAILABLE FOR LOADING 0.5 SECONDS AFTER THE DIESEL GENERATOR BREAKER CLOSES.

c. DURING LOP CONDITIONS, ALL BUSSES WILL HAVE POWER AVAILABLE FOR LOADING 0.5 SECONDS AFTER THE DIESEL GENERATOR BREAKER CLOSES.

4.)
a. C MEANS CONTINOUS, S MEANS STANDBY

b. UNLESS OTHERWISE INDICATED MOV'S ARE ASSUMED TO OPERATE FOR 10 SECONDS ONLY.

5.) THIS PUMP WILL BE SEQUENCED ON ONLY IF ONE OR BOTH OF THE PRIMARY PUMPS FAIL TO START ON THE PREVIOUSLY SEQUENCED STEP.

6.) THE SF SEQUENCER WILL ONLY PERMIT THIS LOAD TO START FOR 1 SECOND PERIODS OF TIME AT THE BEGINNING OF THE INDICATED SEQUENCE STEP, ALL SUBSEQUENT STEPS AND AT ANYTIME AFTER 30.5 SECONDS.

7.) THE FIRST LOAD STEP WILL OCCUR (WITH PREFERRED OFFSITE SOURCE AVAILABLE) 0.5 SECONDS AFTER THE SI SIGNAL IS RECEIVED AT THE SF SEQUENCER.

10.) UNDER SI CONDITIONS,THESE LOADS ARE SEQUENCED ON AT LOW SPEED ONLY. FOR LOP CONDITIONS THEY ARE SEQUENCED ON AT HIGH SPEED ONLY.

11.) THESE LOADS ARE INTERLOCKED WITH OTHER LOADS,WHICH ARE BLOCKED BY SEQUENCER,UNTIL 30.5 SECONDS.

12.) TRAIN B SIMILAR EXCEPT WHERE NOTED BY ASTERISK (*) OR IN TRAIN B TABLE (SEE NOTES 18,19,20 & 21)

13.) SI MEANS SAFETY INJECTION SYSTEM SIGNAL ACTUATED.

14.) THIS LOAD FED FROM BREAKER CLOSED ONLY DURING STARTUP OF THE LOAD.

15.) SEQUENCER PROVIDES MOTOR AUTO-START CIRCUIT WITH START SIGNAL AT 30.5 SECOND STEP. HOWEVER MOTOR WILL START AT 50.5 SECONDS DUE TO AN ADDITIONAL TIME DELAY OF 20 SECONDS BY AGASTAT TIME DELAY RELAY IN THE AUTO-START CKT.

16.)
a. SI PROVIDES CB ESF CHILLER WATER PUMP AUTO-START CIRCUIT WITH START SIGNAL. HOWEVER, PUMP MOTOR START WILL BE DELAYED APPROXIMATELY 60 SECONDS DUE TO ADDITION OF AGASTAT TIME DELAY RELAY ON THE AUTO-START CIRCUIT.

b. CB ESF CHILLER PURGE COMP. MTR. IS INTERLOCKED WITH THE USF CHILLER COMPRESSOR WHICH IS BLOCKED BY THE SEQUENCER UNTIL 30.5 SECONDS.

c. CB ESF CHILLER OIL PUMP MOTOR IS INTERLOCKED WITH ESF CHILLER CHILLED WATER FLOW SWITCH. THIS SWITCH OPERATES DUE TO FLOW FROM THE CB ESF CHILLED WATER PUMP APPROXIMATELY 60 SECONDS AFTER THE SIGNAL IS INITIATED.

17.) FAN 2-1561-N7-001 WILL LOAD SHED BUT WILL NOT AUTO-START UNLESS A CVI SIGNAL IS PRESENT. HEATER 2-1561-N7-001-H01 IS INTERLOCKED WITH FAN 2-1561-N7-001 AND IS ALSO PROCESS CONTROLLED.

18.) FAN 2-1531-B7-005 IS DISABLED AND ABANDONED IN PLACE.

19.) THOSE LOADS MARKED BY AN ASTERISK (*) ARE UNIQUE TO THEIR PARTICULAR TRAIN AND DO NOT HAVE AN EQUIVALENT OPPOSITE TRAIN LOAD.

THESE LOADS EITHER FROM THEIR CORRESPONDING OPPOSITE TRAIN LOAD. BOTH LOADS ARE LISTED UNDER THEIR RESPECTIVE TRAIN.

21.) TRAIN A AND B LOADS WHICH DIFFER BY .2 HP OR LESS ARE LISTED ONLY UNDER TRAIN A. ALL OTHER LISTED DATA IS SIMILIAR.

TABLE 1

INITIATING LETTER DESIGNATOR	TIME SEC TO OPERATE AFTER DIESEL GENERATOR BREAKER CLOSES	STARTING INITIATED BY
K	0.5	PROCESS DEVICES (THERMOSTATS, PRESSURE SWITCHES, ETC.)
L	0.5	ACCIDENT SIGNAL (SI) OR LOP
M	>30.5	MANUAL (ASSUME NO MANUAL ACTIONS WILL OCCUR UNTIL SEQUENCE TIMING CYCLE IS COMPLETED)
N	0.5	PANELS & OTHER LOADS FED FROM BRKRS ONLY-WHICH REMAIN CLOSED
P	>30.5 (BLOCKED BY SEQUENCER)	PROCESS DEVICE
S	≤30.5	AUTOMATICALLY BY SEQUENCER CAN BE STOPPED MANUALLY ONLY AFTER 30.5 SEC
MM	0.5	MNL STARTED WITH MEMORY (SLIP ON MAIN START CONT) LOAD WILL RE-START UPON RESTORATION OF POWER
I	VARIOUS	INTERLOCKED WITH OTHER LOADS

NUCLEAR SAFETY RELATED [A] [B]

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NO.	REVISIONS	DATE	DR	CHK	APPV	DTL
	INCORP. PER ABN-36789	11-28-01	CD	EGG	MWD	X

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

DIESEL GENERATOR
2A AND TRAIN "A" AC BUSSES
LOADING TABLE - PAGE 1 OF 2

SCALE: NONE | DRAWING NO. 2X3D-AA-K02A | REV. 12

JOB NO.10604

SIZE E 34x44

EQUIPMENT TAG NUMBER	EQUIPMENT DESCRIPTION	PROJECT CLASSIFICATION (ELECTRICAL SYSTEM)	PROCUREMENT SPECIFICATION	BUS NUMBER	BUS VOLTAGE	NAMEPLATE RATING HP (OR KW)	REFERENCE DWG. 2X3D-	LOSS OF COOLANT ACCIDENT,(SI), PREFERRED OFFSITE SOURCE AVAILABLE: MANUAL PROCESS, ACCI. OR SEQUENCER START SEE TABLE 1	TIME (SECONDS) TO START (NOTE 7)	RUNNING TIME AFTER START (NOTE 4)	LOSS OF OFFSITE POWER & SUBSEQUENT SI: MANUAL PROCESS, ACCI. OR SEQUENCER START SEE TABLE 1	TIME (SEC) TO START IS CLOSED (NOTE 3b)	RUNNING TIME AFTER START (NOTE 4)	LOSS OF OFFSITE POWER: MANUAL PROCESS, ACCI. OR SEQUENCER START SEE TABLE 1	TIME (SEC) TO START AFTER D-G BRKR IS CLOSED (NOTE 3c)	RUNNING TIME AFTER START (NOTE 4)
PNL 2AY01	120/240 DISTRIBUTION PANEL	11E	X3	2ABD	480V	15.0KVA	AA-F11A	N	0	C	N	0.5	C	N	0.5	C
21555A7013M01	CHARGING PUMP ROOM COOLER	11E	X4	2ABD	480V	5.0	BG-D05N	I,K	0	C	I,K	0.5	C	I,K	0.5	C
21555A7015M01	SIS PUMP ROOM COOLER	11E	X4	2ABD	480V	3.0	BG-D05R	I,K	0	C	I,K	0.5	C	I,K	0.5	C
21555A7001M01	MCC ROOM COOLER	11E	X4	2ABD	480V	5.0	BG-D05A	L	0	C	L	0.5	C	K	0.5	C
21555A7003M01	SWGR & MCC ROOM COOLER	11E	X4	2ABD	480V	2.0	BG-D05C	L	0	C	L	0.5	C	K	0.5	C
21208P6006M01	BORIC ACID TRANSFER PUMP	11E	X6	2ABD	480V	20.8	BD-C01F	I	0	C	I	0.5	C	I	0.5	C
21555A7009M01	CTMT SPRAY PUMP COOLER	11E	X4	2ABD	480V	3.0	BG-D05J	I,K	0	C	I,K	0.5	C	I,K	0.5	C
21555A7007M01	RHR PUMP ROOM COOLER	11E	X4	2ABD	480V	5.0	BG-D05G	I,K	0	C	I,K	0.5	C	I,K	0.5	C
21513P5HMAF01	CTMT H2 MONITORING PANEL	11E	X5	2ABD	480V	1.7KVA	AA-F11A	N	0	C	N	0.5	C	N	0.5	C
2HV-2138	RC COOLING COIL OUT ISO	11E	X5	2ABD	480V	.13	BD-K04M	L	0	40SEC	L	0.5	40SEC	M	>30.5	40SEC
2HV-1830	CTMT AIR COOLER CW OUT	11E	X5	2ABD	480V	.13	BD-K04V	L	0	30SEC	L	0.5	30SEC	M	>30.5	30SEC
2LV-0112D	REFUEL W TK TO CHG PUMP	11E	X6	2ABD	480V	1.6	BD-C02H	L	0	15SEC	L	0.5	15SEC	K	0.5	15SEC
2HV-8812A	RHR PUMP INLET	11E	X6	2ABD	480V	7.8	BD-E02E	M	>30.5	20SEC	M	>30.5	20SEC	M	>30.5	20SEC
2HV-8110 •	CHG PUMP TO MINIFLO ISO	11E	X6	2ABD	480V	.7	BD-C03J	L	0	10SEC	L	0.5	10SEC	M	>30.5	10SEC
2HV-8920	SI INLET B MINIFLO ISO	11E	X6	2ABD	480V	.7	BD-D02B	M	>30.5	10SEC	M	>30.5	10SEC	M	>30.5	10SEC
2HV-8821A	RCS HT LEG LP 1/4HDR BYP	11E	X6	2ABD	480V	1.0	BD-D03D	M	>30.5	15SEC	M	>30.5	15SEC	M	>30.5	15SEC
2HV-8802A	RCS HT LEG LP 1/4HDR ISO	11E	X6	2ABD	480V	1.0	BD-D02G	M	>30.5	10SEC	M	>30.5	10SEC	M	>30.5	10SEC
2HV-8923A	SI PUMP INLET	11E	X6	2ABD	480V	1.0	BD-D03F	M	>30.5	15SEC	M	>30.5	15SEC	M	>30.5	15SEC
2HV-8716A	RHR NO. 1 COLD LEG ISO	11E	X6	2ABD	480V	1.6	BD-E02N	M	>30.5	15SEC	M	>30.5	15SEC	M	>30.5	15SEC
2HV-8807A	SI PUMP SUCTION HDR	11E	X6	2ABD	480V	.7	BD-D02P	M	>30.5	15SEC	M	>30.5	15SEC	M	>30.5	15SEC
2HV-1806	CTMT AIR COOLER CW INLET	11E	X5	2ABD	480V	.13	BD-K04N	L		30SEC	L	0.5	30SEC	M	>30.5	30SEC
2HV-8811A	CTMT SUMP ISO	11E	X6	2ABD	480V	7.8	BD-E03F	K,L		20SEC	K,L	0.5	20SEC	M	>30.5	20SEC
2HV-9003A	CTMT SPRAY PUMP SUCTION	11E	X6	2ABD	480V	2.0	BD-J02J	M	>30.5	15SEC	M	>30.5	15SEC	M	>30.5	15SEC
2HV-8814	SI PUMP MINIFLO ISO	11E	X6	2ABD	480V	.7	BD-D03C	M	>30.5	10SEC	M	>30.5	10SEC	M	>30.5	10SEC
2HV-8801A	BORON INJ TK DISC ISO	11E	X6	2ABD	480V	3.2	BD-D02E	L	0	10SEC	L	0.5	10SEC	M	>30.5	10SEC
2FV-0610	RHR PUMP 1 MINIFLO	11E	X6	2ABD	480V	.7	BD-E02C	K	0	10SEC	K	0.5	10SEC	K	0.5	10SEC
2HV-9017A	CSP RFL WTR STOR TK IN	11E	X6	2ABD	480V	1.9	BD-J02G	M	>30.5	15SEC	M	>30.5	15SEC	M	>30.5	15SEC
2HV-8924 •	CHG PUMP HDR TO/FR SIS PA	11E	X6	2ABD	480V	1.0	BD-C04V	M	>30.5	15SEC	M	>30.5	15SEC	M	>30.5	15SEC
2HV-9002A	CSP SUCTION	11E	X6	2ABD	480V	2.0	BD-J02E	M	>30.5	15SEC	M	>30.5	15SEC	M	>30.5	15SEC
2HV-1808	CTMT AIR COOL CW INLET	11E	X5	2ABD	480V	.13	BD-K04U	L	0	30SEC	L	0.5	30SEC	M	>30.5	30SEC
2HV-9001A	CTMT SPRAY HDR ISO	11E	X6	2ABD	480V	1.6	BD-J02C	L	0	15SEC	L	0.5	15SEC	M	>30.5	15SEC
2HV-1822	CTMT AIR COOL CW OUTLET	11E	X5	2ABD	480V	.13	BD-K04P	L	0	30SEC	L	0.5	30SEC	M	>30.5	30SEC
2HV-2134	REACTOR CAVITY CC INLET ISO	11E	X5	2ABD	480V	.13	BD-K04L	L	0	40SEC	L	0.5	40SEC	M	>30.5	40SEC
2HV-8485A	CHARGING PUMP A DISCH	11E	X6	2ABD	480V	1.0	BD-C05A	M	>30.5	10SEC	M	>30.5	10SEC	M	>30.5	10SEC
2HV-8471A	CHARGING PUMP A SUCTION	11E	X6	2ABD	480V	.7	BD-C05C	M	>30.5	10SEC	M	>30.5	10SEC	M	>30.5	10SEC
PNL 2AYE1(NOTE 20)	120/240V DISTRIBUTION PANEL	11E	X3	2ABE	480V	15KVA	AA-F24A	N	0	C	N	0.5	C	N	0.5	C
21516B7001M01	CTB POST LOCA CAV PURGE UNIT	11E	X4	2ABE	480V	5	BG-B02X	L	0	C	L	0.5	C	MM	0.5	C
21806B3CAB	BATTERY CHARGER 2AD1CB	11E	X3	2ABE	480V	100KVA	AA-H01A	N	0	C	N	0.5	C	N	0.5	C
21806B3CCA (NOTE 20)	BATTERY CHARGER 2CD1CA	11E	X3	2ABE	480V	75KVA	AA-H04A	N	0	C	N	0.5	C	N	0.5	C
2HV-2624A	CTB POST LOCA PURGE EXH ISO	11E	X5	2ABE	480V	0.13	BG-B04A	L	0	20SEC	L	0.5	20SEC	M	>30.5	20SEC
2HV-8701A	RHR LOOP 1 INLET ISO	11E	X6	2ABE	480V	2.6	BD-E02G	M	>30.5	120SEC	M	>30.5	120SEC	M	>30.5	120SEC
2HV-8112	RCP SEAL WATER ISO	11E	X6	2ABE	480V	0.70	BD-C03L	L	0	15SEC	L	0.5	15SEC	M	>30.5	15SEC
2HV-9380A	CTMT ATMOS UNIT SERV AIR	11E	X4	2ABE	480V	0.33	BH-R01D	M	>30.5	10SEC	M	>30.5	10SEC	M	>30.5	10SEC
2HV-8808A	ACCUM LOOP 1 ISO	11E	X6	2ABE	480V	10.3	BD-D02R	L	0	10SEC	L	0.5	10SEC	K	0.5	10SEC
2HV-2626A	CTB NORM PURGE SPLY ISO	11E	X5	2ABE	480V	1.33	BG-B04E	L	0	10SEC	L	0.5	10SEC	M	>30.5	10SEC
2HV-2582A	CTB CLR A7001 UNIT (DAMPER)	11E	X4	2ABE	480V	.13	BG-B03V	M	>30.5	10SEC	M	>30.5	10SEC	M	>30.5	10SEC
2HV-2582B	CTB CLR A7002 UNIT (DAMPER)	11E	X4	2ABE	480V	.13	BG-B03V	M	>30.5	10SEC	M	>30.5	10SEC	M	>30.5	10SEC
2HV-19051 •	THERM BARR CW RCP 001	11E	X4	2ABE	480V	0.70	BD-L03K	K	0	10SEC	K	0.5	10SEC	K	0.5	10SEC
2HV-19053 •	THERM BARR CW RCP 002	11E	X4	2ABE	480V	0.70	BD-L03L	K	0	10SEC		0.5	10SEC	K	0.5	10SEC
2HV-8000A	PRESSURIZER POWER RELIEF ISO	11E	X6	2ABE	480V	2.0	BD-B02A	K	0	10SEC	K	0.5	10SEC	K	0.5	10SEC
2AYF1	120/240V DISTRIBUTION PANEL	11E	X3	2ABF	480V	15KVA	AA-F36A	N	0	C	N	0.5	C	N	0.5	C
21593B7001M01	AUX FWP HSE FAN	11E	X4	2ABF	480V	5.0	BG-G03A	K	0	C	K	0.5	C	MM	0.5	C
22403P4001M01	DSL OIL STORAGE TK PUMP	11E	X4	2ABF	480V	1.5	BH-G01X	K	0	C	K	0.5	C	K	0.5	C
22403P4002M01	DSL OIL STORAGE TK PUMP	11E	X4	2ABF	480V	1.5	BH-G01Z	K	0	C	K	0.5	C	K	0.5	C
21540B7001M01	DSL TUNNEL VENT FAN	11E	X4	2ABF	480V	3.0	BG-K01C	K	0	C	K	0.5	C	K	0.5	C
2HV-12051	DGB FAN B7003 DAMPER	11E	X4	2ABF	480V	.2	BG-F01M	I (NOTE 11)	>30.5	10SEC	I (NOTE 11)	>30.5	10SEC	I (NOTE 11)	>30.5	10SEC
2180873114	LTG. ISO XFMR 2ABF13RX	11E	X3	2ABF	480V	15KVA	AA-F36A	N	0	C	N	0.5	C	N	0.5	C
2HV-5119	COND STOR TK TO PUMP	11E	X5	2AGF	480V	.25	BC-F02D	M	>30.5	10SEC	M	>30.5	10SEC	M	>30.5	10SEC
2HV-12006	AUX FWP RM INLET	11E	X4	2ABF	480V	.2	BG-G01M	I (NOTE 11)	0	10SEC	I (NOTE 11)	0.5	10SEC	I (NOTE 11)	0.5	10SEC
2HV-12050	DGB FAN B7001 DAMPER	11E	X4	2ABF	480V	.2	BG-F01M	I (NOTE 11)	>30.5	10SEC	I (NOTE 11)	>30.5	10SEC	I (NOTE 11)	>30.5	10SEC

EQUIPMENT TAG NUMBER	EQUIPMENT DESCRIPTION	PROJECT CLASSIFICATION (ELECTRICAL SYSTEM)	PROCUREMENT SPECIFICATION	BUS NUMBER	BUS VOLTAGE	NAMEPLATE RATING HP (OR KW)	REFERENCE DWG. 2X3D-	LOSS OF COOLANT ACCIDENT,(SI), PREFERRED OFFSITE SOURCE AVAILABLE: MANUAL PROCESS, ACCI. OR SEQUENCER START SEE TABLE 1	TIME (SECONDS) TO START (NOTE 7)	RUNNING TIME AFTER START (NOTE 4)	LOSS OF OFFSITE POWER & SUBSEQUENT SI: MANUAL PROCESS, ACCI. OR SEQUENCER START SEE TABLE 1	TIME (SEC) TO START IS CLOSED (NOTE 3.b)	RUNNING TIME AFTER START (NOTE 4)	LOSS OF OFFSITE POWER: MANUAL PROCESS, ACCI. OR SEQUENCER START SEE TABLE 1	TIME (SEC) TO START AFTER D-G BRKR IS CLOSED (NOTE 3.c)	RUNNING TIME AFTER START (NOTE 4)
21808T3020	LTG XFMR 2NBS47X	62E	X3	2NBS	480V	15KVA	AA-F28A	SEE NOTE 3.a	>30.5	C	SEE NOTE 3.a	>30.5	C	N	10.5	C
21807Y3RX24	REG XFMR 2NBS32RX	62E	X3	2NBS	480V	15KVA	AA-G01B	SEE NOTE 3.a	>30.5	C	SEE NOTE 3.a	>30.5	C	N	10.5	C
PNL 2NYS1	120/240V DISTRIBUTION PANEL	62E	X3	2NBS	480V	10KVA	AA-F28A	SEE NOTE 3.a	>30.5	C	SEE NOTE 3.a	>30.5	C	N	10.5	C
21509B7001M01	CTB CRDM CLG UNIT FAN	62E	X6	2NBS	480V	40.0	BG-B02A	SEE NOTE 3.a	>30.5	C	SEE NOTE 3.a	>30.5	C	M	>30.5	C
21509B7003M01	CTB CRDM CLG UNIT FAN	62E	X6	2NBS	480V	40.0	BG-B02C	SEE NOTE 3.a	>30.5	C	SEE NOTE 3.a	>30.5	C	M	>30.5	C
21807Y3I4	INVERTER 2ND3I4	62E	X3	2NBS	480V	10KVA	AA-F28A	SEE NOTE 3.a	>30.5	C	SEE NOTE 3.a	>30.5	C	N	10.5	C
21807Y3RX27 •	REG TRANSFORMER 2NBS39RX	62E	X3	2NBS	480V	15KVA	AA-F28A	SEE NOTE 3.a	>30.5	S	SEE NOTE 3.a	>30.5	S	N	10.5	S
21512B7001M01	CTB REACTOR SUPPORT COOLING UNIT	62E	X4	2NBS	480V	15.0	BG-B03P	SEE NOTE 3.a	>30.5	C	SEE NOTE 3.a	>30.5	C	MM	10.5	C
21512B7003M01	CTB REACTOR SUPPORT COOLING UNIT	62E	X4	2NBS	480V	15.0	BG-B03R	SEE NOTE 3.a	>30.5	C	SEE NOTE 3.a	>30.5	C	MM	10.5	C
21556B7003M01	RESTRAINT COOLING FAN	62E	X4	2NBS	480V	3.0	BG-D03B	SEE NOTE 3.a	>30.5	C	SEE NOTE 3.a	>30.5	C	MM	10.5	C
21556B7004M01	RESTRAINT COOLING FAN	62E	X4	2NBS	480V	3.0	BG-D03C	SEE NOTE 3.a	>30.5	C	SEE NOTE 3.a	>30.5	C	MM	10.5	C
21556B7007M01	RESTRAINT COOLING FAN	62E	X4	2NBS	480V	3.0	BG-D03F	SEE NOTE 3.a	>30.5	C	SEE NOTE 3.a	>30.5	C	MM	10.5	C
21556B7008M01	RESTRAINT COOLING FAN	62E	X4	2NBS	480V	3.0	BG-D03G	SEE NOTE 3.a	>30.5	C	SEE NOTE 3.a	>30.5	C	MM	10.5	C
21808T3022	LTG TRANSFORMER 2NBS11X	62E	X3	2NBS	480V	15KVA	AA-F28A	SEE NOTE 3.a	>30.5	C	SEE NOTE 3.a	>30.5	C	N	10.5	C
21808T3038	LTG TRANSFORMER 2NBS12X	62E	X3	2NBS	480V	30KVA	AA-F28A	SEE NOTE 3.a	>30.5	C	SEE NOTE 3.a	>30.5	C	N	10.5	C
21808T3053	LTG TRANSFORMER 2NBS14X	62E	X3	2NBS	480V	45KVA	AA-F28A	SEE NOTE 3.a	>30.5	C	SEE NOTE 3.a	>30.5	C	N	10.5	C
21808T3086	LTG TRANSFORMER 2NBS15X	62E	X3	2NBS	480V	45KVA	AA-F28A	SEE NOTE 3.a	>30.5	C	SEE NOTE 3.a	>30.5	C	N	10.5	C
21808T3069	LTG TRANSFORMER 2NBS23X	62E	X3	2NBS	480V	45KVA	BA-E15G	SEE NOTE 3.a	>30.5	C	SEE NOTE 3.a	>30.5	C	N	10.5	C
21807Y3RX16 •	REG TRANSFORMER 2NBS18RX	62E	X3	2NBS	480V	30KVA	AA-G01B	SEE NOTE 3.a	>30.5	C	SEE NOTE 3.a	>30.5	C	N	10.5	C
21807Y3RX18	REG TRANSFORMER 2NBS21RX	62E	X3	2NBS	480V	30KVA	AA-G01B	SEE NOTE 3.a	>30.5	C	SEE NOTE 3.a	>30.5	C	N	10.5	C
21807Y312	INVERTER 2ND312	62E	X6	2NBS	480V	7.5KVA	AA-H07A	SEE NOTE 3.a	>30.5	C	SEE NOTE 3.a	>30.5	C	N	10.5	C
21228P4001M01	REAC MAKE-UP WTR PP 005	62E	X4	2NBS	480V	40.0	BD-L05A	SEE NOTE 3.a	>30.5	C	SEE NOTE 3.a	>30.5	C	K	10.5	C
21817U3007B	CONT H_2 MON HEAT TRACE PNL(STBY)	62E	X3	2NBS	480V	15KW	AA-F28A	SEE NOTE 3.a	>30.5	C	SEE NOTE 3.a	>30.5	C	N	10.5	C
2HV-12646	CTB REACTOR SUPPLY COOLING UNIT	62E	X4	2NBS	480V	.13	BG-B05U	SEE NOTE 3.a	>30.5	10SEC	SEE NOTE 3.a	>30.5	10SEC	I	10.5	10SEC
2HV-12644	CTB REACTOR SUPPLY COOLING UNIT	62E	X4	2NBS	480V	.13	BG-B05S	SEE NOTE 3.a	>30.5	10SEC	SEE NOTE 3.a	>30.5	10SEC	I	10.5	10SEC
2HV-8098 •	EXCESS LTDWN HT EXCH INLET	62E	X6	2NBS	480V	.7	BD-C05R	SEE NOTE 3.a	>30.5	10SEC	SEE NOTE 3.a	>30.5	10SEC	M	>30.5	10SEC
21817U3001A	HEAT TRACE SI SYSTEM	62E	X3	2NBS	480V	15KW	AA-F28A	SEE NOTE 3.a	>30.5	C	SEE NOTE 3.a	>30.5	C	K	10.5	C
21817U3003A	HEAT TRACE (NORMAL) CVCS	62E	X3	2NBS	480V	15KW	AA-F28A	SEE NOTE 3.a	>30.5	C	SEE NOTE 3.a	>30.5	C	K	10.5	C
21817U3005A	HEAT TRACE	62E	X3	2NBS	480V	15KW	AA-F28A	SEE NOTE 3.a	>30.5	C	SEE NOTE 3.a	>30.5	C	K	10.5	C
21817U3002A	AUX HEAT TRACE PNL (LVL A)	62E	X3	2NBS	480V	15KW	AA-F28A	SEE NOTE 3.a	>30.5	C	SEE NOTE 3.a	>30.5	C	K	10.5	C
22403G4001H01	JACKET WTR HTR	62E	X4	2NB1	480V	75KW	AA-F40A	SEE NOTE 3.a	>30.5	C	SEE NOTE 3.a	>30.5	C	K	10.5	C
22403G4001H02	LUBE OIL HTR	62E	X4	2NB1	480V	45KW	AA-F40A	SEE NOTE 3.a	>30.5	C	SEE NOTE 3.a	>30.5	C	K	10.5	C
22403G4001E01	AIR COMPRESSOR AFTERCOOLER	62E	X4	2NB1	480V	.75	AA-F40A	SEE NOTE 3.a	>30.5	C	SEE NOTE 3.a	>30.5	C	K	10.5	C
22403G4001E02	AIR COMPRESSOR AFTERCOOLER	62E	X4	2NB1	480V	.75	AA-F40A	SEE NOTE 3.a	>30.5	C	SEE NOTE 3.a	>30.5	C	K	10.5	C
22403G4001M07	LUBE OIL CIRC PUMP	62E	X4	2NB1	480V	15	AA-F40A	SEE NOTE 3.a	>30.5	C	SEE NOTE 3.a	>30.5	C	K	10.5	C
22403G4001M04	JACKET WTR CIRC PUMP	62E	X4	2NB1	480V	3.0	AA-F40A	SEE NOTE 3.a	>30.5	C	SEE NOTE 3.a	>30.5	C	K	10.5	C
22403G4001M01	AIR START COMPRESSOR	62E	X4	2NB1	480V	30.0	AA-F40A	SEE NOTE 3.a	>30.5	C	SEE NOTE 3.a	>30.5	C	K	10.5	C
22403G4001M02	AIR START COMPRESSOR	62E	X4	2NB1	480V	30.0	AA-F40A	SEE NOTE 3.a	>30.5	C	SEE NOTE 3.a	>30.5	C	K	10.5	C
PNL 2NY11	120/240 V DIST. PNL.	62E	X3	2NB1	480V	10KVA	AA-F40A	SEE NOTE 3.a	>30.5	C	SEE NOTE 3.a	>30.5	C	K	10.5	C
22403G4001G01	GEN. SP. HTR.	62E	X4	2NB1	480V	7.2KW	AA-F40A	SEE NOTE 3.a	>30.5	C	SEE NOTE 3.a	>30.5	C	K	10.5	C

TRAIN B LOADS
(SEE NOTES 19 & 21)

EQUIPMENT TAG NUMBER	EQUIPMENT DESCRIPTION	PROJECT CLASSIFICATION (ELECTRICAL SYSTEM)	PROCUREMENT SPECIFICATION	BUS NUMBER	BUS VOLTAGE	NAMEPLATE RATING HP (OR KW)	REFERENCE DWG. 2X3D-	LOSS OF COOLANT ACCIDENT,(SI), PREFERRED OFFSITE SOURCE AVAILABLE: MANUAL PROCESS, ACCI. OR SEQUENCER START SEE TABLE 1	TIME (SECONDS) TO START (NOTE 7)	RUNNING TIME AFTER START (NOTE 4)	LOSS OF OFFSITE POWER & SUBSEQUENT SI: MANUAL PROCESS, ACCI. OR SEQUENCER START SEE TABLE 1	TIME (SEC) TO START IS CLOSED (NOTE 3.b)	RUNNING TIME AFTER START (NOTE 4)	LOSS OF OFFSITE POWER: MANUAL PROCESS, ACCI. OR SEQUENCER START SEE TABLE 1	TIME (SEC) TO START AFTER D-G BRKR IS CLOSED (NOTE 3.c)	RUNNING TIME AFTER START (NOTE 4)
21539A7002M01 NOTE 20	CB AUX RELAY RM AC UNIT	11E	X4	2BBA	480V	2.0	BG-C07N	L	0	C	L	0.5	C	WM	0.5	C
21806B3CDA NOTE 20	BATTERY CHARGER 2DD1CA	11E	X3	2BBA	480V	50KVA	AA-G01A	N	0	C	N	0.5	C	N	0.5	C
DISTR PNL 2BYA1	120/240 DISTRIBUTION PANEL	11E	X3	2BBA	480V	10KVA	AA-F19A	N	0	C	N	0.5	C	N	0.5	C
2HV-8840 •	RHR HOT LEG INJECT CROSSOVER	11E	X6	2BBB	480V	10.3	BD-D03J	M	>30.5	20SEC	M	>30.5	20SEC	M	>30.5	20SEC
2HV-8111B •	CHG PMP MINI FLOW ISO	11E	X6	2BBB	480V	.7	BD-C05L	L	0	10SEC	L	0.5	10SEC	M	>30.5	10SEC
PNL 2BYC1(NOTE 20)	120V DISTRIBUTION PANEL	11E	X3	2BBC	480V	15KVA	AA-F04A	N	0	C	N	0.5	C	N	0.5	C
2HV-8111A •	CHG PMP MINI FLOW ISO.	11E	X6	2BBD	480V	.7	BD-C03K	L	0	10SEC	L	0.5	10SEC	M	>30.5	10SEC
2HV-8438 •	CHARGING PMP DISCH	11E	X6	2BBD	480V	1.0	BD-C05P	M	>30.5	10SEC	M	>30.5	10SEC	M	>30.5	10SEC
2HV-8103A •	RCP 1 SEAL WATER INLET	11E	X6	2BBD	480V	.7	BD-C04L	M	>30.5	10SEC	M	>30.5	10SEC	M	>30.5	10SEC
2HV-8103B •	RCP 2 SEAL WATER INLET	11E	X6	2BBD	480V	.7	BD-C04H	M	>30.5	10SEC	M	>30.5	10SEC	M	>30.5	10SEC
2180683CDB NOTE 20	BATTERY CHARGER 2DD1CB	11E	X3	2BBE	480V	150KVA 211.8	AA-H05A	N	0	C	N	0.5	C	N	0.5	C
2HV-8103C •	RCP 3 SEAL WATER INLET	11E	X6	2BBE	480V	2.7	BD-C04J	M	>30.5	10SEC	M	>30.5	10SEC	M	>30.5	10SEC
2HV-8103D •	RCP 4 SEAL WATER INLET	11E	X6	2BBE	480V	.66	BD-C04U	M	>30.5	10SEC	M	>30.5	10SEC	M	>30.5	10SEC
2HV-3548 •	RCS HOT LEG SAMPLE	11E	X5	2BBE	480V	.13	BD-N01N	L	0	15SEC	L	0.5	15SEC	M	>30.5	15SEC
2HV-2041 •	THERMAL BARRIER CW RETURN	11E	X4	2BBE	480V	1.0	BD-L03P	K	0	10SEC	K	0.5	10SEC	K	0.5	10SEC
PNL 2BYE1(NOTE 20)	120/240V DISTRIBUTION PANEL	11E	X3	2BBE	480V	10KVA	AA-F25A	N	0	C	N	0.5	C	N	0.5	C
2HV-3500 •	RCS HOT LEG SAMPLE	62E	X5	2NBR	480V	.13	BD-N01A	SEE NOTE 3a	>30.5	15SEC	SEE NOTE 3a	>30.5	15SEC	M	>30.5	15SEC
21808T3003	LTG. TRANSFORMER 2NBR11X	62E	X3	2NBR	480V	30KVA	AA-F27A	SEE NOTE 3a	>30.5	C	SEE NOTE 3a	>30.5	C	N	10.5	C
21806B3CN3	BATTERY CHARGER 2ND2CA	62E	X3	2NBR	480V	45KVA	AA-H06A	SEE NOTE 3a	>30.5	C	SEE NOTE 3a	>30.5	C	N	10.5	C
21541B7001M01	PASS PNL EXH AIR BOOSTER FAN	62E	X4	2NBR	480V	.33	BG-E01A	SEE NOTE 3a	>30.5	C	SEE NOTE 3a	>30.5	C	M	>30.5	C
21807Y313	INVERTER 2ND312	62E	X3	2NBR	480V	7.5KVA	AA-H07A	SEE NOTE 3a	>30.5	C	SEE NOTE 3a	>30.5	C	N	0.5	C
21807Y311	INVERTER 2ND311	62E	X3	2NBR	480V	30KVA	AA-H07B	SEE NOTE 3a	>30.5	C	SEE NOTE 3a	>30.5	C	N	0.5	C
21817U3007A	CTMT H_2 MON HT TRACE PNL	62E	X3	2NBR	480V	30KVA	AA-F27A	SEE NOTE 3a	>30.5	C	SEE NOTE 3a	>30.5	C	K	0.5	C
21301K4001M09 •	TURB GEAR MAIN DRIVE	62E	X4	2NB1G	480V	40	BC-Q52C	SEE NOTE 3a	>30.5	C	SEE NOTE 3a	>30.5	C	M	>30.5	C
21307T4501M03 •	TURB GEAR LO RES TGDP	62E	X4	2NB1G	480V	50	BC-L02A	SEE NOTE 3a	>30.5	C	SEE NOTE 3a	>30.5	C	M	>30.5	C

FOR NOTES & TABLES SEE SHEET 1 DRAWING NO 2X3D-AA-K02A.

NUCLEAR SAFETY RELATED [A] [B] [Q]

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

DIESEL GENERATOR
2A AND TRAIN "A" AC BUSSES
LOADING TABLE - PAGE 2 OF 2

SCALE: NONE

DRAWING NO. 2X3D-AA-K02B

VER. 11.0

JOB NO.10604

SIZE E 34x44

DWG CATEGORY: CRITICAL

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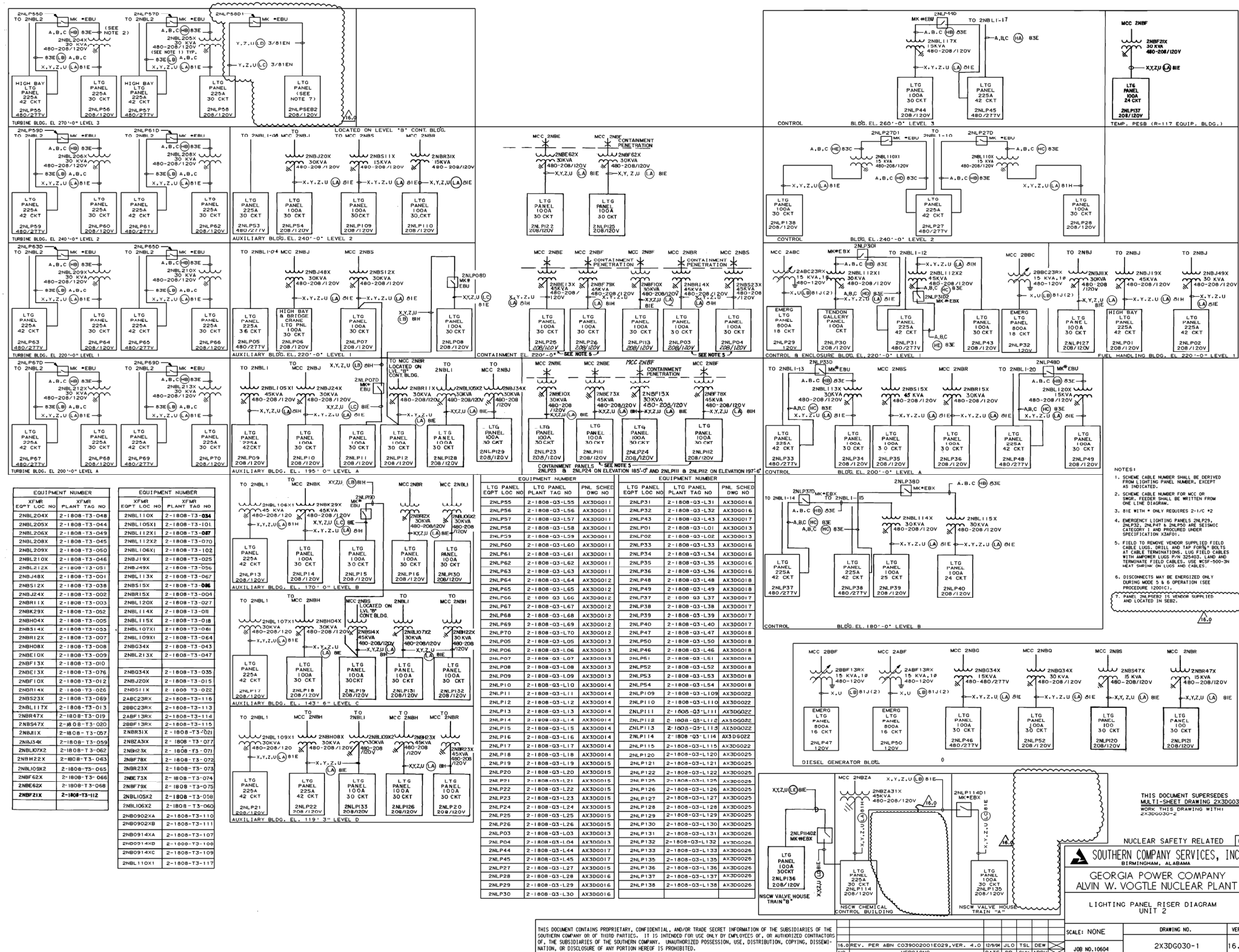
NO.	VERSIONS	DATE	DR	CHK	APPV
11.0	REVISED PER ABN 2081605401E003, VER.1.0	10/10/08	TSL	ELC	JMR



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XFMR EQPT LOC NO	XFMR PLANT TAG NO
2NBL204X	2-1808-T3-048
2NBL205X	2-1808-T3-044
2NBL206X	2-1808-T3-049
2NBL208X	2-1808-T3-045
2NBL209X	2-1808-T3-050
2NBL210X	2-1808-T3-046
2NBL212X	2-1808-T3-051
2NBJ48X	2-1808-T3-001
2NBS12X	2-1808-T3-038
2NBJ24X	2-1808-T3-002
2NBR11X	2-1808-T3-003
2NBK29X	2-1808-T3-052
2NBH04X	2-1808-T3-005
2NBS14X	2-1808-T3-053
2NBR12X	2-1808-T3-007
2NBH08X	2-1808-T3-008
2NBE10X	2-1808-T3-009
2NBF13X	2-1808-T3-010
2NBE13X	2-1808-T3-076
2NBF10X	2-1808-T3-012
2NBR14X	2-1808-T3-026
2NBS23X	2-1808-T3-069
2NBL117X	2-1808-T3-013
2NBR47X	2-1808-T3-019
2NBS47X	2-1808-T3-020
2NBJ11X	2-1808-T3-057
2NBJ34X	2-1808-T3-059
2NBL107X2	2-1808-T3-062
2NBH22X	2-1808-T3-063
2NBL109X2	2-1808-T3-065
2NBF62X	2-1808-T3-066
2NBE62X	2-1808-T3-068
2NBF21X	**2-1808-T3-112**

EQUIPMENT NUMBER	
XFMR EQPT LOC NO	XFMR PLANT TAG NO
2NBL110X	2-1808-T3-**034**
2NBL105X1	2-1808-T3-101
2NBL112X1	2-1808-T3-**047**
2NBL112X2	2-1808-T3-070
2NBL106X1	2-1808-T3-102
2NBJ19X	2-1808-T3-025
2NBJ49X	2-1808-T3-056
2NBL113X	2-1808-T3-067
2NBS15X	2-1808-T3-**086**
2NBR15X	2-1808-T3-004
2NBL120X	2-1808-T3-027
2NBL114X	2-1808-T3-011
2NBL115X	2-1808-T3-018
2NBL107X1	2-1808-T3-061
2NBL109X1	2-1808-T3-064
2NBG34X	2-1808-T3-043
2NBL213X	2-1808-T3-047
2NBG34X	2-1808-T3-035
2NBJ20X	2-1808-T3-015
2NBS11X	2-1808-T3-022
2ABC23RX	2-1808-T3-116
2BBC23RX	2-1808-T3-113
2ABF13RX	2-1808-T3-114
2BBF13RX	2-1808-T3-115
2NBR31X	2-1808-T3-021
2NBZA31X	2-1808-T3-077
2NBH23X	2-1808-T3-071
2NBF78X	2-1808-T3-072
2NBR23X	2-1808-T3-073
2NBE73X	2-1808-T3-074
2NBF79X	2-1808-T3-075
2NBL105X2	2-1808-T3-058
2NBL106X2	2-1808-T3-060
2NB0902XA	2-1808-T3-110
2NB0902XB	2-1808-T3-111
2NB0914XA	2-1808-T3-107
2NB0914XB	2-1808-T3-108
2NB0914XC	2-1808-T3-109
2NBL110X1	2-1808-T3-117

EQUIPMENT NUMBER		
LTG PANEL EQPT LOC NO	LTG PANEL PLANT TAG NO	PNL SCHED DWG NO
2NLP55	2-1808-Q3-L55	AX3DG011
2NLP56	2-1808-Q3-L56	AX3DG011
2NLP57	2-1808-Q3-L57	AX3DG011
2NLP58	2-1808-Q3-L58	AX3DG011
2NLP59	2-1808-Q3-L59	AX3DG011
2NLP60	2-1808-Q3-L60	AX3DG011
2NLP61	2-1808-Q3-L61	AX3DG011
2NLP62	2-1808-Q3-L62	AX3DG011
2NLP63	2-1808-Q3-L63	AX3DG011
2NLP64	2-1808-Q3-L64	AX3DG012
2NLP65	2-1808-Q3-L65	AX3DG012
2NLP66	2-1808-Q3-L66	AX3DG012
2NLP67	2-1808-Q3-L67	AX3DG012
2NLP68	2-1808-Q3-L68	AX3DG012
2NLP69	2-1808-Q3-L69	AX3DG012
2NLP70	2-1808-Q3-L70	AX3DG012
2NLP05	2-1808-Q3-L05	AX3DG013
2NLP06	2-1808-Q3-L06	AX3DG013
2NLP07	2-1808-Q3-L07	AX3DG013
2NLP08	2-1808-Q3-L08	AX3DG013
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2NLP12	2-1808-Q3-L12	AX3DG014
2NLP13	2-1808-Q3-L13	AX3DG014
2NLP14	2-1808-Q3-L14	AX3DG014
2NLP15	2-1808-Q3-L15	AX3DG014
2NLP16	2-1808-Q3-L16	AX3DG014
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2NLP25	2-1808-Q3-L25	AX3DG015
2NLP26	2-1808-Q3-L26	AX3DG015
2NLP03	2-1808-Q3-L03	AX3DG013
2NLP04	2-1808-Q3-L04	AX3DG013
2NLP44	2-1808-Q3-L44	AX3DG017
2NLP45	2-1808-Q3-L45	AX3DG017
2NLP27	2-1808-Q3-L27	AX3DG015
2NLP28	2-1808-Q3-L28	AX3DG016
2NLP29	2-1808-Q3-L29	AX3DG016
2NLP30	2-1808-Q3-L30	AX3DG016

EQUIPMENT NUMBER		
LTG PANEL EQPT LOC NO	LTG PANEL PLANT TAG NO	PNL SCHED DWG NO
2NLP31	2-1808-Q3-L31	AX3DG016
2NLP32	2-1808-Q3-L32	AX3DG016
2NLP43	2-1808-Q3-L43	AX3DG017
2NLP01	2-1808-Q3-L01	AX3DG013
2NLP02	2-1808-Q3-L02	AX3DG013
2NLP33	2-1808-Q3-L33	AX3DG016
2NLP34	2-1808-Q3-L34	AX3DG016
2NLP35	2-1808-Q3-L35	AX3DG016
2NLP36	2-1808-Q3-L36	AX3DG016
2NLP48	2-1808-Q3-L48	AX3DG018
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2NLP37	2-1808-Q3-L37	AX3DG017
2NLP38	2-1808-Q3-L38	AX3DG017
2NLP39	2-1808-Q3-L39	AX3DG017
2NLP40	2-1808-Q3-L40	AX3DG017
2NLP47	2-1808-Q3-L47	AX3DG018
2NLP50	2-1808-Q3-L50	AX3DG018
2NLP46	2-1808-Q3-L46	AX3DG018
2NLP51	2-1808-Q3-L51	AX3DG018
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2NLP121	2-1808-Q3-L121	AX3DG025
2NLP122	2-1808-Q3-L122	AX3DG025
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2NLP129	2-1808-Q3-L129	AX3DG025
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2NLP135	2-1808-Q3-L135	AX3DG026
2NLP136	2-1808-Q3-L136	AX3DG026
2NLP137	2-1808-Q3-L137	AX3DG026
2NLP138	2-1808-Q3-L138	AX3DG026

NOTES:

1. SCHEME CABLE NUMBER SHALL BE DERIVED FROM LIGHTING PANEL NUMBER, EXCEPT AS INDICATED.
2. SCHEME CABLE NUMBER FOR MCC OR SWGR. FEEDER SHALL BE WRITTEN FROM ONE LINE DIAGRAM.
3. 81E WITH * ONLY REQUIRES 2-1/C #2
4. EMERGENCY LIGHTING PANELS 2NLP29, 2NLP32, 2NLP47 & 2NLP50 ARE SEISMIC CATEGORY I AND PROCURED UNDER SPECIFICATION X3AF01.
5. FIELD TO REMOVE VENDOR SUPPLIED FIELD CABLE LUGS. DRILL AND TAP FOR⅜" BOLTS AT CABLE TERMINATIONS. LUG FIELD CABLES WITH AMPOWER LUGS P/N 325403. AND TERMINATE FIELD CABLES. USE WCSF-500-3N HEAT SHRINK ON LUGS AND CABLES.
6. DISCONNECTS MAY BE ENERGIZED ONLY DURING MODE 5 & 6 OPERATION (SEE PROCEDURE 12001C).
7. PANEL 2NLPSEB2 IS VENDOR SUPPLIED AND LOCATED IN SEB2.

THIS DOCUMENT SUPERSEDES MULTI-SHEET DRAWING 2X3DG030
WORK THIS DRAWING WITH:
2X3DG030-2

NUCLEAR SAFETY RELATED Q

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

LIGHTING PANEL RISER DIAGRAM
UNIT 2

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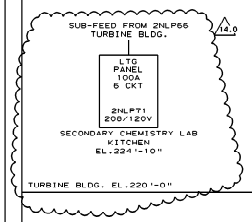
NO.	VERSIONS	DATE	DR	CHK	APPV
16.0	REV. PER ABN CO3900200 1E029, VER. 4.0	12/9/04	JLO	TSL	DEW

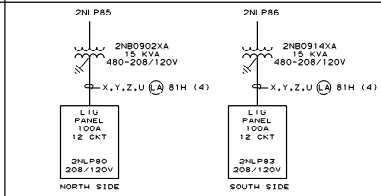
SCALE: NONE

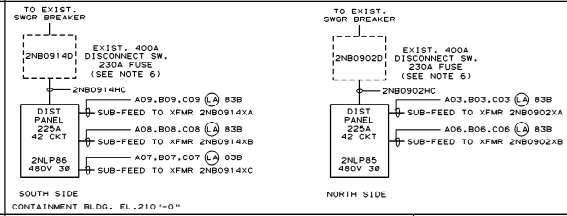
DRAWING NO.	VER.
2X3DG030-1	16.0

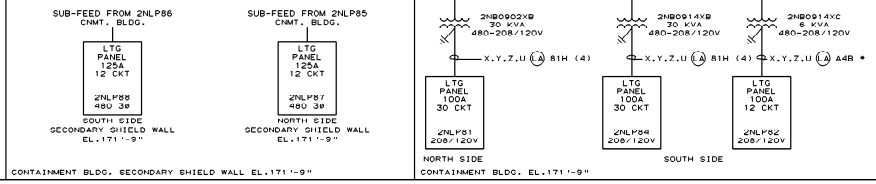
JOB NO.10604

SIZE E 34X44









14.0

EQUIPMENT NUMBER		
LTG PANEL EQPT LOC NO.	LTG PANEL PLANT TAG NO.	PNL SCHED DWG NO
2NLP71	2-1808-03-L71	AX3DG012

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NO.	VERSION	DATE	DR	CHK	APPV	DTL
14.0	REVISED PER ABN-38844	09/03/04	TSL	MDN	DFW	

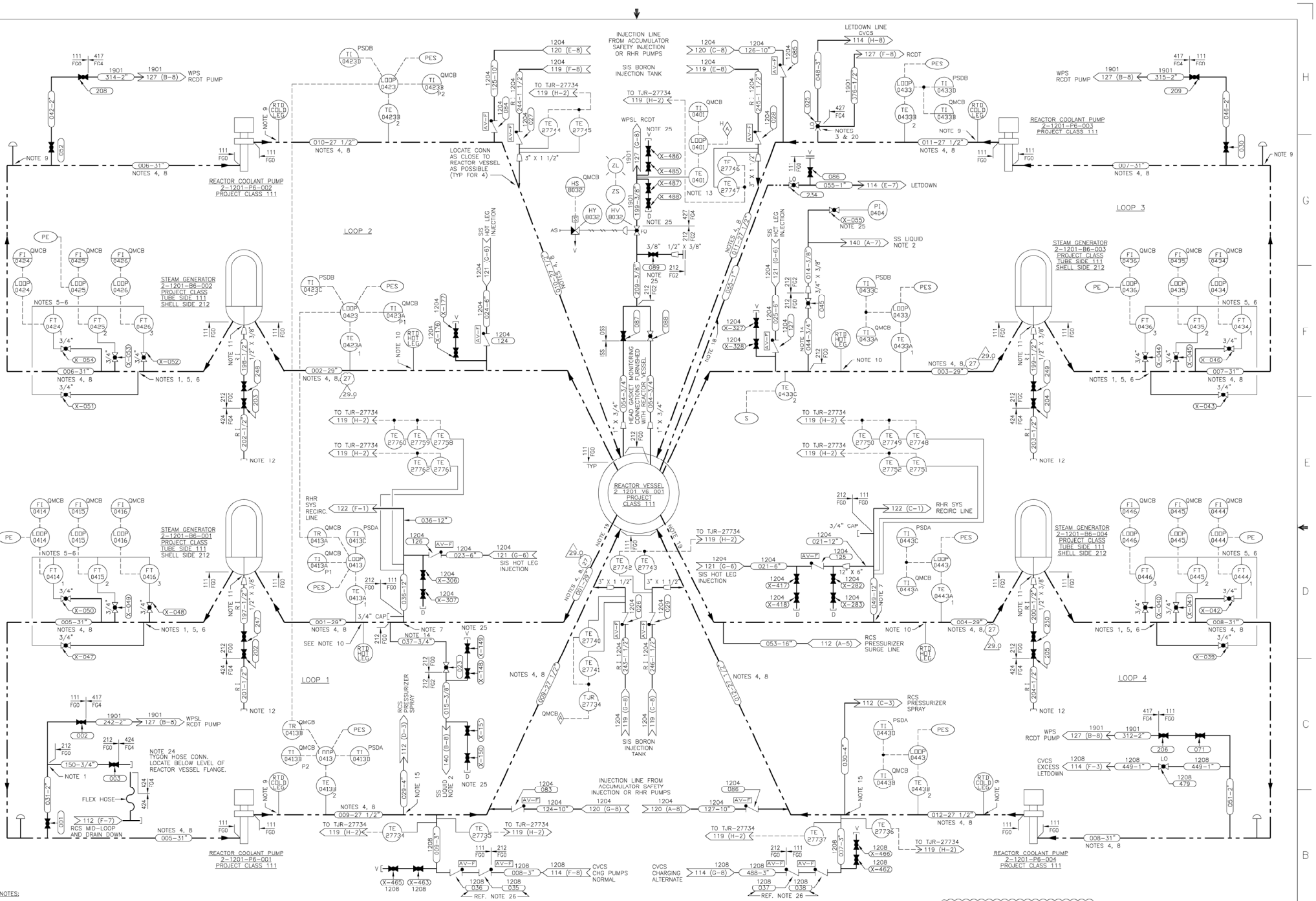
SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

LIGHTING PANEL RISER DIAGRAM
UNIT 2

SCALE: NONE	DRAWING NO.	VER.
JOB NO.10604	2X3DG030-2	14.0

SIZE E 34X44



NOTES:

1. PROVIDE 0.375 ID FLOW RESTRICTION AS SHOWN ON BPC DWG CX4DG001.
2. GROSS FAILED FUEL DETECTOR SYSTEM INSTALLED ON THIS LINE OUTSIDE REACTOR CONTAINMENT. (SEE BPC DWG 2X4DB140)
3. PROVIDE 12" REMOVAL SPOOL.
4. INSIDE DIAMETER.
5. LOCATE FLOWMETER IN 90° ELBOW DOWNSTREAM OF STEAM GENERATORS.
6. ELBOW FLOWMETERS.
7. LOCATE CONNECTION IN BOTTOM HALF OF REACTOR COOLANT PIPING ON 45° ANGLE TO VERTICAL.
8. REACTOR COOLANT LOOP PIPING AND ALL EQUIPMENT FURNISHED BY WESTINGHOUSE. ALL OTHER PIPING AND MANUAL VALVES 2" AND SMALLER FURNISHED BY BECHTEL.
9. FOR CONNECTION DETAILS OF RTD COLD LEG, SEE DETAIL "A" ON BPC DWG 2X4DB113.
10. FOR CONNECTION DETAILS OF RTD HOT LEG, SEE DETAIL "A" ON BPC DGW 2X4DB113.
11. 3/8 INCH ID FLOW RESTRICTOR SUPPLIED WITH STEAM GENERATOR FOR CLASS 1 TO CLASS 2 TRANSITION.
    FOR CLASS 1–CLASS 2 TRANSITION, PROVIDE A 3/8 INCH PIPE TO TUBING TRANSITION PIECE.
12. TYGON HOSE TO BE INSTALLED AT THIS CONNECTION FOR DRAINING BOTTOM OF CHANNEL HEAD TO CONTAINMENT SUMP AFTER REACTOR COOLANT SYSTEM HAS BEEN DRAINED BELOW STEAM GENERATOR NOZZLE.
13. SURFACE MOUNTED TEMPERATURE DETECTOR. LOCATE AT BOTTOM OF PIPE.
14. 0.234 I.D. PIPING.
15. SPRAY LINE SCOOP.
16. FOR REACTOR COOLANT PUMP LUBE OIL SPILL PROTECTION ARRANGEMENT SEE DRAWING 2X4DB112.
17. PORTION OF THE SYSTEM IS SUSPECTED TO CAUSE STRESS CORROSION CRACKING. REFER TO THE APPLICABLE DESIGN DRAWING AND PIPING MATERIAL CLASSIFICATIONS FOR MATERIAL AND HEAT TREATMENT REQUIREMENTS.
18. VENT PIPE FURNISHED WITH REACTOR VESSEL HEAD; REMAINDER OF PIPING AND VALVES TO THE FIRST SPOOL PIECE & FLANGE ALSO PROVIDED BY WESTINGHOUSE, EXCEPT VALVES 234 AND 086.
19. FOR REACTOR VESSEL WATER LEVEL INSTRUMENTATION (RVLIS) SEE P&ID 2X4DB113.
20. FOR TYPICAL DETAIL, SEE STD. DWG. AX4DD000.
21. ALL LINES CLASSIFIED R.V. UNLESS OTHERWISE NOTED.
22. DELETED.
23. DELETED.
24. REMOVE TYGON HOSE CONN., CONNECTED/OPERATIONAL ONLY DURING RCS MIDLOOP AND DRAIN DOWN LEVEL OPERATION.
25. THESE ARE KEROTEST VALVES AND MAY NOT ALLOW REVERSE FLOW.
26. THESE VALVES HAVE A SEAL WELDED BONNET PER NRC COMMITMENT SNC473861.
27. MSIP HAS BEEN APPLIED TO LINE 001, 002, 003, AND 004 NEAR THE REACTOR VESSEL NOZZLE.

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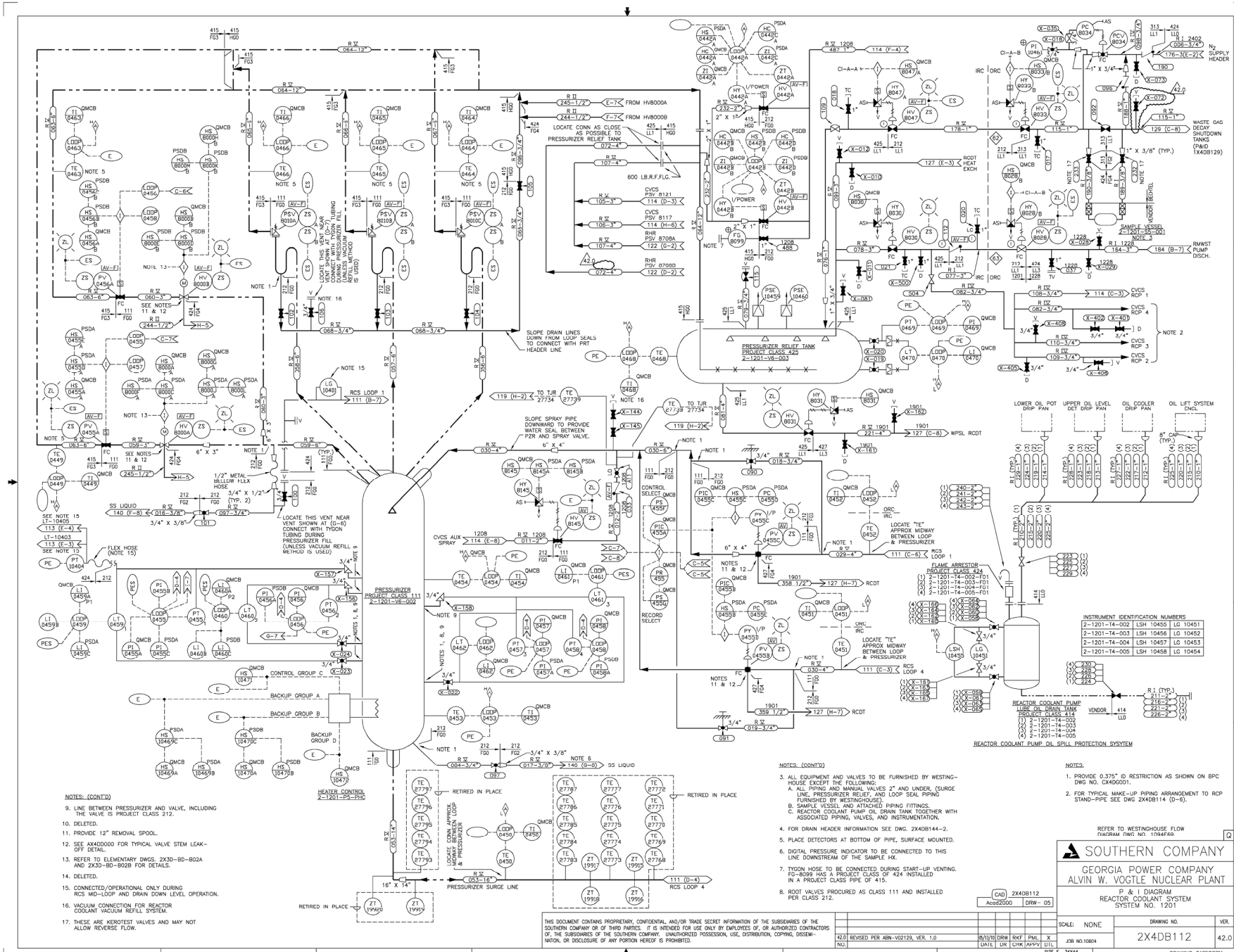
NO.	VERSIONS	DATE	DR	CHK	APPV	DTL
29.0	REVISED PER ABN SNC526823M002, VER. 1.0	10/20/14	LEH	ALM	MDW	

CAD 2X4DB111
ACAD2004 LEH – 05

SOUTHERN COMPANY
GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT
P & I DIAGRAM
REACTOR COOLANT SYSTEM
SYSTEM NO. 1201

SCALE: NONE
JOB NO.10604
DRAWING NO. 2X4DB111
VER. 29.0
SIZE E 34X44
DRAWING CATEGORY: CRITICAL



NOTES: (CONT'D)

9. LINE BETWEEN PRESSURIZER AND VALVE, INCLUDING THE VALVE IS PROJECT CLASS 212.
10. DELETED.
11. PROVIDE 12" REMOVAL SPOOL.
12. SEE AX4DD000 FOR TYPICAL VALVE STEM LEAK-OFF DETAIL.
13. REFER TO ELEMENTARY DWGS. 2X3D-BD-B02A AND 2X3D-BD-B02B FOR DETAILS.
14. DELETED.
15. CONNECTED/OPERATIONAL ONLY DURING RCS MID-LOOP AND DRAIN DOWN LEVEL OPERATION.
16. VACUUM CONNECTION FOR REACTOR COOLANT VACUUM REFILL SYSTEM.
17. THESE ARE KEROTEST VALVES AND MAY NOT ALLOW REVERSE FLOW.

NOTES: (CONT'D)

3. ALL EQUIPMENT AND VALVES TO BE FURNISHED BY WESTINGHOUSE EXCEPT THE FOLLOWING:
   A. ALL PIPING AND MANUAL VALVES 2" AND UNDER, (SURGE LINE, PRESSURIZER RELIEF, AND LOOP SEAL PIPING FURNISHED BY WESTINGHOUSE).
   B. SAMPLE VESSEL AND ATTACHED PIPING FITTINGS.
   C. REACTOR COOLANT PUMP OIL DRAIN TANK TOGETHER WITH ASSOCIATED PIPING, VALVES, AND INSTRUMENTATION.
4. FOR DRAIN HEADER INFORMATION SEE DWG. 2X4DB144-2.
5. PLACE DETECTORS AT BOTTOM OF PIPE, SURFACE MOUNTED.
6. DIGITAL PRESSURE INDICATOR TO BE CONNECTED TO THIS LINE DOWNSTREAM OF THE SAMPLE HX.
7. TYGON HOSE TO BE CONNECTED DURING START-UP VENTING. FC-8099 HAS A PROJECT CLASS OF 424 INSTALLED IN A PROJECT CLASS PIPE OF 415.
8. ROOT VALVES PROCURED AS CLASS 111 AND INSTALLED PER CLASS 212.

NOTES:

1. PROVIDE 0.375" ID RESTRICTION AS SHOWN ON BPC DWG NO. CX4DG001.
2. FOR TYPICAL MAKE-UP PIPING ARRANGEMENT TO RCP STAND-PIPE SEE DWG 2X4DB114 (D-6).

REFER TO WESTINGHOUSE FLOW DIAGRAM DWG NO. 1094F69

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NO.		DATE	DR	CHK	APPV
42.0	REVISED PER ABN-V02129, VER. 1.0	05/13/10	DRW	RKF	PML
					DTL

CAD 2X4DB112
Acad2000 DRW- 05

SOUTHERN COMPANY

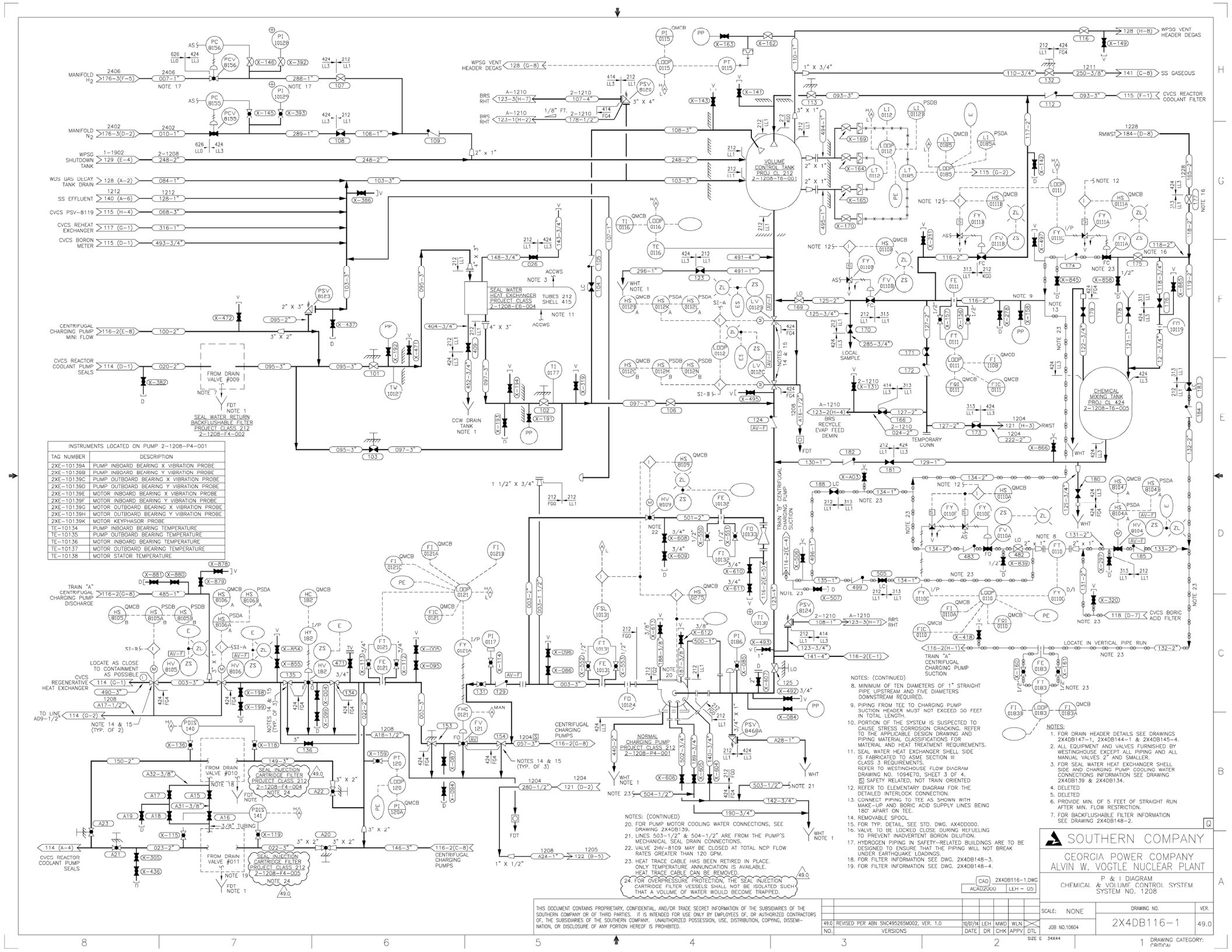
GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
REACTOR COOLANT SYSTEM
SYSTEM NO. 1201

SCALE: NONE	DRAWING NO.	VER.
JOB NO.10604	2X4DB112	42.0

SIZE E 34X44

DRAWING CATEGORY: CRITICAL



INSTRUMENTS LOCATED ON PUMP 2-1208-P4-001	
TAG NUMBER	DESCRIPTION
2XE-10139A	PUMP INBOARD BEARING X VIBRATION PROBE
2XE-10139B	PUMP INBOARD BEARING Y VIBRATION PROBE
2XE-10139C	PUMP OUTBOARD BEARING X VIBRATION PROBE
2XE-10139D	PUMP OUTBOARD BEARING Y VIBRATION PROBE
2XE-10139E	MOTOR INBOARD BEARING X VIBRATION PROBE
2XE-10139F	MOTOR INBOARD BEARING Y VIBRATION PROBE
2XE-10139G	MOTOR OUTBOARD BEARING X VIBRATION PROBE
2XE-10139H	MOTOR OUTBOARD BEARING Y VIBRATION PROBE
2XE-10139K	MOTOR KEYPHASOR PROBE
TE-10134	PUMP INBOARD BEARING TEMPERATURE
TE-10135	PUMP OUTBOARD BEARING TEMPERATURE
TE-10136	MOTOR INBOARD BEARING TEMPERATURE
TE-10137	MOTOR OUTBOARD BEARING TEMPERATURE
TE-10138	MOTOR STATOR TEMPERATURE

NOTES: (CONTINUED)

8. MINIMUM OF TEN DIAMETERS OF 1" STRAIGHT PIPE UPSTREAM AND FIVE DIAMETERS DOWNSTREAM REQUIRED.
9. PIPING FROM TEE TO CHARGING PUMP SUCTION HEADER MUST NOT EXCEED 50 FEET IN TOTAL LENGTH.
10. PORTION OF THE SYSTEM IS SUSPECTED TO CAUSE STRESS CORROSION CRACKING. REFER TO THE APPLICABLE DESIGN DRAWING AND PIPING MATERIAL CLASSIFICATIONS FOR MATERIAL AND HEAT TREATMENT REQUIREMENTS.
11. SEAL WATER HEAT EXCHANGER SHELL SIDE IS FABRICATED TO ASME SECTION III CLASS 3 REQUIREMENTS. REFER TO WESTINGHOUSE FLOW DIAGRAM DRAWING NO. 1094E70, SHEET 3 OF 4. [S] SAFETY RELATED, NOT TRAIN ORIENTED
12. REFER TO ELEMENTARY DIAGRAM FOR THE DETAILED INTERLOCK CONNECTION.
13. CONNECT PIPING TO TEE AS SHOWN WITH MAKE-UP AND BORIC ACID SUPPLY LINES BEING 180° APART ON TEE.
14. REMOVABLE SPOOL.
15. FOR TYP. DETAIL, SEE STD. DWG. AX4DD000.
16. VALVE TO BE LOCKED CLOSE DURING REFUELING TO PREVENT INADVERTENT BORON DILUTION.
17. HYDROGEN PIPING IN SAFETY-RELATED BUILDINGS ARE TO BE DESIGNED TO ENSURE THAT THE PIPING WILL NOT BREAK UNDER EARTHQUAKE LOADINGS.
18. FOR FILTER INFORMATION SEE DWG. 2X4DB148-3.
19. FOR FILTER INFORMATION SEE DWG. 2X4DB148-4.

NOTES: (CONTINUED)

20. FOR PUMP MOTOR COOLING WATER CONNECTIONS, SEE DRAWING 2X4DB139.
21. LINES 503-1/2" & 504-1/2" ARE FROM THE PUMP'S MECHANICAL SEAL DRAIN CONNECTIONS.
22. VALVE 2HV-8109 MAY BE CLOSED AT TOTAL NCP FLOW RATES GREATER THAN 120 GPM.
23. HEAT TRACE CABLE HAS BEEN RETIRED IN PLACE. ONLY TEMPERATURE ANNUNCIATION IS AVAILABLE. HEAT TRACE CABLE CAN BE REMOVED.
24. FOR OVERPRESSURE PROTECTION, THE SEAL INJECTION CARTRIDGE FILTER VESSELS SHALL NOT BE ISOLATED SUCH THAT A VOLUME OF WATER WOULD BECOME TRAPPED.

NOTES:

1. FOR DRAIN HEADER DETAILS SEE DRAWINGS 2X4DB147-1, 2X4DB144-1 & 2X4DB145-4.
2. ALL EQUIPMENT AND VALVES FURNISHED BY WESTINGHOUSE EXCEPT ALL PIPING AND ALL MANUAL VALVES 2" AND SMALLER.
3. FOR SEAL WATER HEAT EXCHANGER SHELL SIDE AND CHARGING PUMP COOLING WATER CONNECTIONS INFORMATION SEE DRAWING 2X4DB139 & 2X4DB134.
4. DELETED
5. DELETED
6. PROVIDE MIN. OF 5 FEET OF STRAIGHT RUN AFTER MIN. FLOW RESTRICTION.
7. FOR BACKFLUSHABLE FILTER INFORMATION SEE DRAWING 2X4DB148-2.

CAD 2X4DB116-1.DWG
ACAD2000 LEH – 05

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NO.	VERSIONS	DATE	DR	CHK	APPV	DTL
49.0	REVISED PER ABN SNC495265M002, VER. 1.0	10/07/14	LEH	MWD	WLN	

SOUTHERN COMPANY

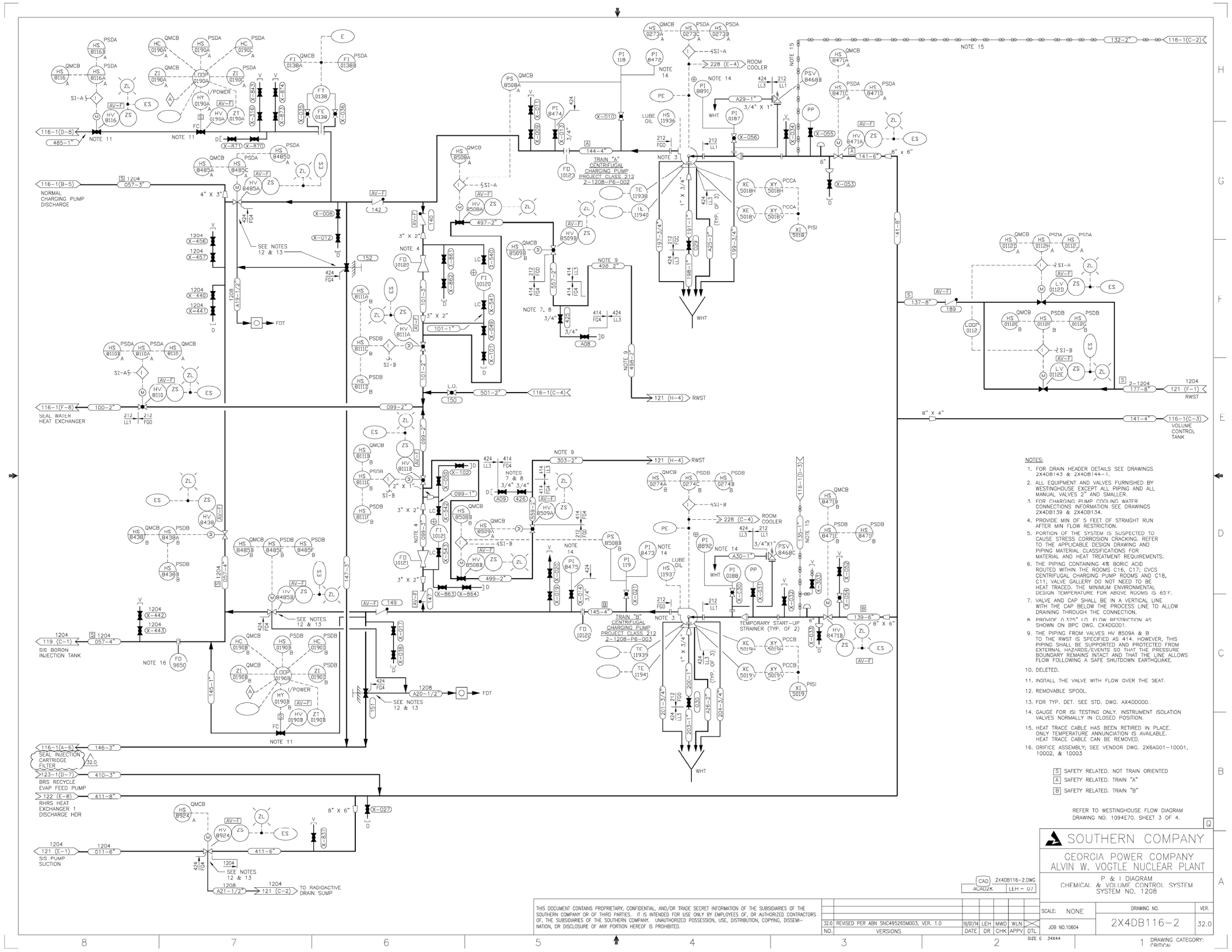
GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
CHEMICAL & VOLUME CONTROL SYSTEM
SYSTEM NO. 1208

SCALE: NONE	DRAWING NO.	VER.
JOB NO.10604	2X4DB116-1	49.0

SIZE E 34X44

DRAWING CATEGORY: CRITICAL



NOTES:

1. FOR DRAIN HEADER DETAILS SEE DRAWINGS 2X4DB143 & 2X4DB144-1.
2. ALL EQUIPMENT AND VALVES FURNISHED BY WESTINGHOUSE EXCEPT ALL PIPING AND ALL MANUAL VALVES 2" AND SMALLER.
3. FOR CHARGING PUMP COOLING WATER CONNECTIONS INFORMATION SEE DRAWINGS 2X4DB139 & 2X4DB134.
4. PROVIDE MIN OF 5 FEET OF STRAIGHT RUN AFTER MIN FLOW RESTRICTION.
5. PORTION OF THE SYSTEM IS SUSPECTED TO CAUSE STRESS CORROSION CRACKING. REFER TO THE APPLICABLE DESIGN DRAWING AND PIPING MATERIAL CLASSIFICATIONS FOR MATERIAL AND HEAT TREATMENT REQUIREMENTS.
6. THE PIPING CONTAINING 4% BORIC ACID ROUTED WITHIN THE ROOMS C16, C17; CVCS CENTRIFUGAL CHARGING PUMP ROOMS AND C18, C11; VALVE GALLERY DO NOT NEED TO BE HEAT TRACED. THE MINIMUM ENVIRONMENTAL DESIGN TEMPERATURE FOR ABOVE ROOMS IS 65°F.
7. VALVE AND CAP SHALL BE IN A VERTICAL LINE WITH THE CAP BELOW THE PROCESS LINE TO ALLOW DRAINING THROUGH THE CONNECTION.
8. PROVIDE 0.375" I.D. FLOW RESTRICTION AS SHOWN ON BPC DWG. CX4DG001.
9. THE PIPING FROM VALVES HV 8509A & B TO THE RWST IS SPECIFIED AS 414. HOWEVER, THIS PIPING SHALL BE SUPPORTED AND PROTECTED FROM EXTERNAL HAZARDS/EVENTS SO THAT THE PRESSURE BOUNDARY REMAINS INTACT AND THAT THE LINE ALLOWS FLOW FOLLOWING A SAFE SHUTDOWN EARTHQUAKE.
10. DELETED.
11. INSTALL THE VALVE WITH FLOW OVER THE SEAT.
12. REMOVABLE SPOOL.
13. FOR TYP. DET. SEE STD. DWG. AX4DD000.
14. GAUGE FOR ISI TESTING ONLY. INSTRUMENT ISOLATION VALVES NORMALLY IN CLOSED POSITION.
15. HEAT TRACE CABLE HAS BEEN RETIRED IN PLACE. ONLY TEMPERATURE ANNUNCIATION IS AVAILABLE. HEAT TRACE CABLE CAN BE REMOVED.
16. ORIFICE ASSEMBLY; SEE VENDOR DWG. 2X6AG01-10001, 10002, & 10003

[S] SAFETY RELATED. NOT TRAIN ORIENTED
[A] SAFETY RELATED. TRAIN "A"
[B] SAFETY RELATED. TRAIN "B"

REFER TO WESTINGHOUSE FLOW DIAGRAM DRAWING NO. 1094E70. SHEET 3 OF 4.

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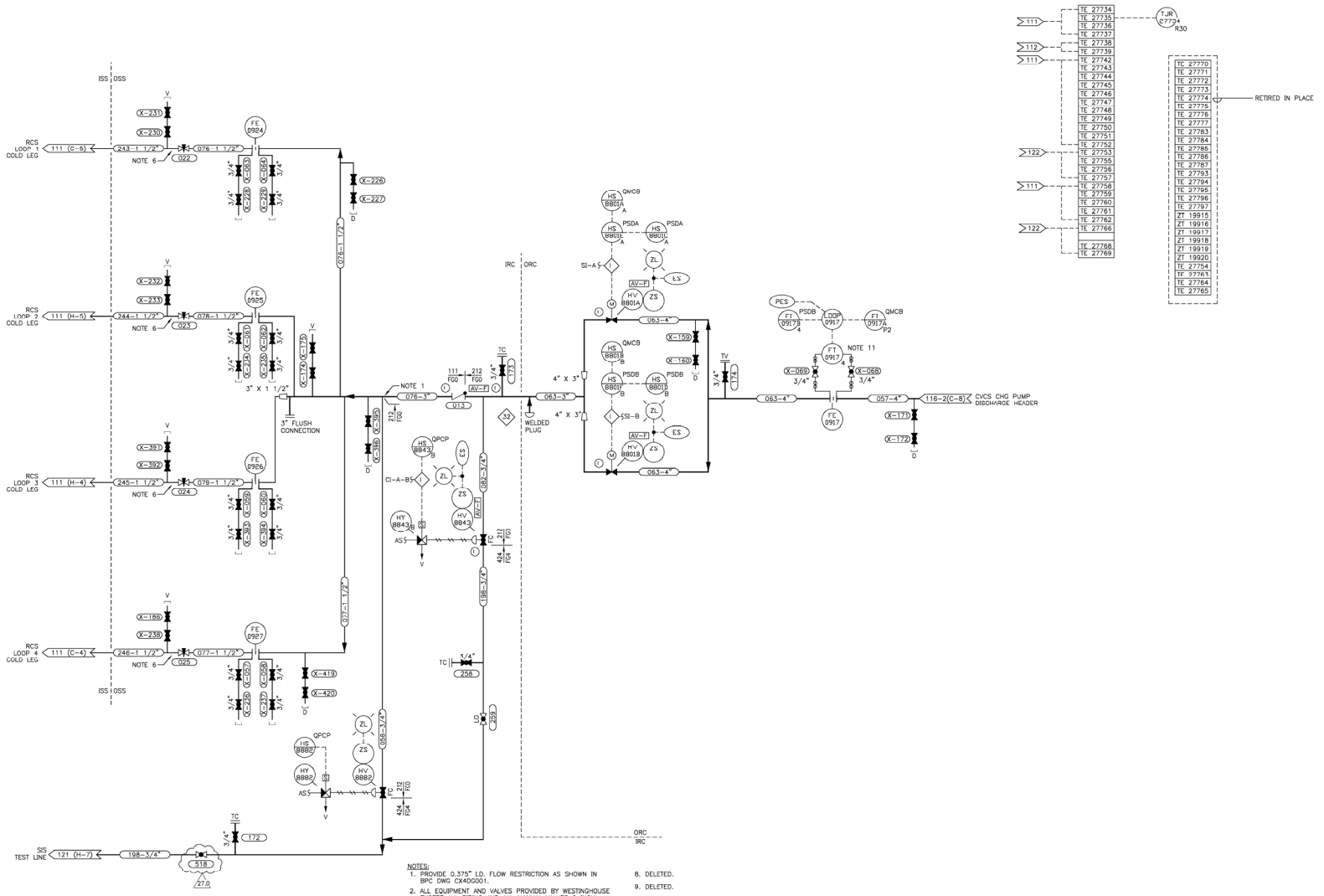
NO.	VERSIONS	DATE	DR	CHK	APPV	DTL
32.0	REVISED PER ABN SNC495265M003, VER. 1.0	10/07/14	LEH	MWD	WLN	

CAD 2X4DB116-2.DWG
ACADZK LEH – 07

SOUTHERN COMPANY
GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
CHEMICAL & VOLUME CONTROL SYSTEM
SYSTEM NO. 1208

SCALE: NONE
JOB NO.10604
DRAWING NO. 2X4DB116-2
VER. 32.0
SIZE E 34X44
DRAWING CATEGORY: CRITICAL



NOTES:

1. PROVIDE 0.375" I.D. FLOW RESTRICTION AS SHOWN IN BPC DWG CX4DS001.
2. ALL EQUIPMENT AND VALVES PROVIDED BY WESTINGHOUSE EXCEPT ALL PIPING AND ALL MANUAL VALVES 2 INCH AND SMALLER, AND EQUIPMENT MARKED ⊕.
3. DELETED.
4. DELETED.
5. DELETED.
6. ADJUST VALVES IN FIELD TO LIMIT CENTRIFUGAL CHARGING PUMP RUNOUT, THEN LOCK IN POSITION. REFER TO SIS START-UP TEST PROCEDURE GAE/GBE-SU-2.3.3. BECHTEL LOG X6AA13-B.
7. PORTION OF THE SYSTEM IS SUSPECTED TO CAUSE STRESS CORROSION CRACKING. REFER TO THE APPLICABLE DESIGN DRAWING AND PIPING MATERIAL CLASSIFICATIONS FOR MATERIAL AND HEAT TREATMENT REQUIREMENTS.
8. DELETED.
9. DELETED.
10. DELETED.
11. HEAT TRACE CABLE HAS BEEN RETIRED IN PLACE. ONLY TEMPERATURE ANNUNCIATION IS AVAILABLE. HEAT TRACE CABLE CAN BE REMOVED.

REFER TO WESTINGHOUSE FLOW DIAGRAM DRAWING NO. 10946E71, SHEET 1 OF 3. Q

SOUTHERN COMPANY

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
SAFETY INJECTION SYSTEM
SYSTEM NO. 1204

CAD 2X4DB119.DWG
ACAD2002 Vep – 02

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NO.	VERSIONS	DATE	DR	CHK	APPV	UTL
27.0	REVISED PER ABN 2071073601M002 VER. 2.0	3-31-09	Vep	ELC	JMR	X

SIZE E 34X44

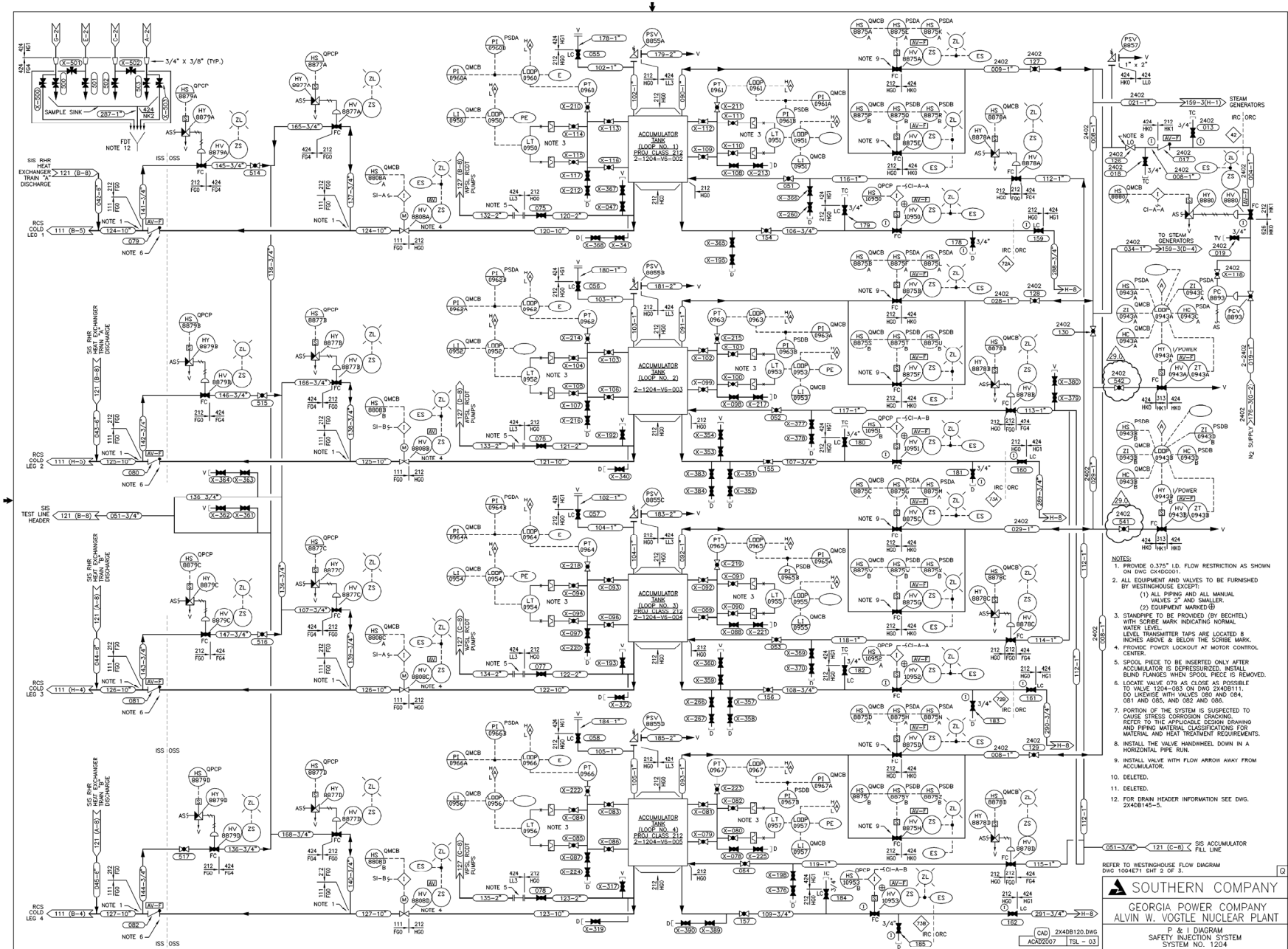
SCALE: NONE

DRAWING NO. 2X4DB119

VER. 27.0

JOB NO.10604

1 DRAWING CATEGORY: CRITICAL



NOTES:

1. PROVIDE 0.375" I.D. FLOW RESTRICTION AS SHOWN ON DWG CX4DC001.
2. ALL EQUIPMENT AND VALVES TO BE FURNISHED BY WESTINGHOUSE EXCEPT:
   (1) ALL PIPING AND ALL MANUAL VALVES 2" AND SMALLER.
   (2) EQUIPMENT MARKED ⊕
3. STANDPIPE TO BE PROVIDED (BY BECHTEL) WITH SCRIBE MARK INDICATING NORMAL WATER LEVEL.
   LEVEL TRANSMITTER TAPS ARE LOCATED 8 INCHES ABOVE & BELOW THE SCRIBE MARK.
4. PROVIDE POWER LOCKOUT AT MOTOR CONTROL CENTER.
5. SPOOL PIECE TO BE INSERTED ONLY AFTER ACCUMULATOR IS DEPRESSURIZED. INSTALL BLIND FLANGES WHEN SPOOL PIECE IS REMOVED.
6. LOCATE VALVE 079 AS CLOSE AS POSSIBLE TO VALVE 1204-083 ON DWG 2X4DB111. DO LIKEWISE WITH VALVES 080 AND 084, 081 AND 085, AND 082 AND 086.
7. PORTION OF THE SYSTEM IS SUSPECTED TO CAUSE STRESS CORROSION CRACKING. REFER TO THE APPLICABLE DESIGN DRAWING AND PIPING MATERIAL CLASSIFICATIONS FOR MATERIAL AND HEAT TREATMENT REQUIREMENTS.
8. INSTALL THE VALVE HANDWHEEL DOWN IN A HORIZONTAL PIPE RUN.
9. INSTALL VALVE WITH FLOW ARROW AWAY FROM ACCUMULATOR.
10. DELETED.
11. DELETED.
12. FOR DRAIN HEADER INFORMATION SEE DWG. 2X4DB145-5.

REFER TO WESTINGHOUSE FLOW DIAGRAM DWG 1094E71 SHT 2 OF 3.

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NO.	VERSIONS	DATE	DR	CHK	APPV	DTL
29.0	REVISED PER ABN 20390013011M016, VER. 1.0	04/01/13	TSL	ELC	JMR	X

CAD 2X4DB120.DWG
ACAD2007 TSL - 03

SOUTHERN COMPANY

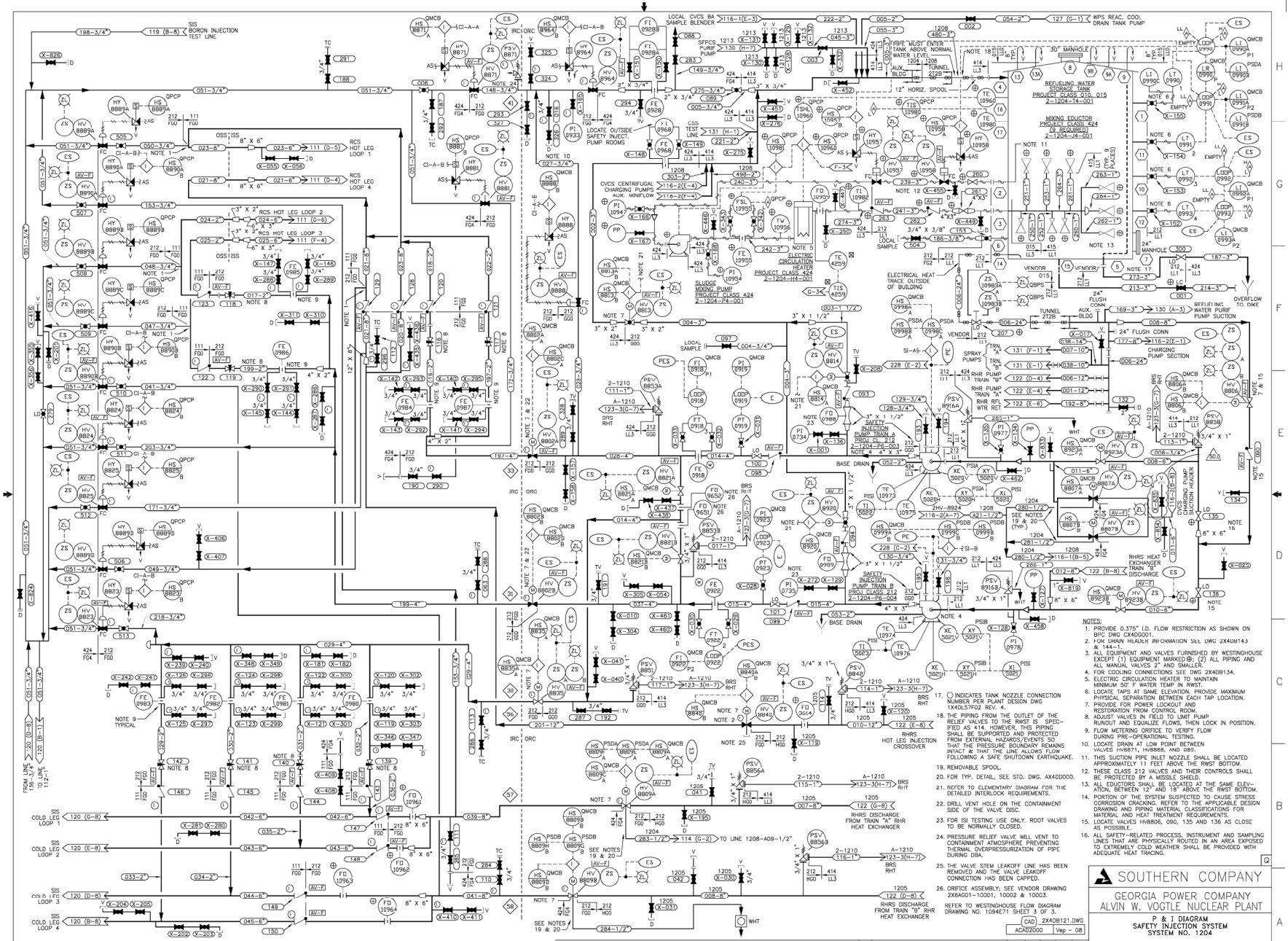
GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
SAFETY INJECTION SYSTEM
SYSTEM NO. 1204

SCALE: NONE	DRAWING NO.	VER.
	2X4DB120	29.0
	JOB NO.10604	

SIZE E 34044

DRAWING CATEGORY: CRITICAL


SIS BORON INJECTION TEST LINE
LOCAL CVCS BA SAMPLE BLENDER
WPS REAC. COOL DRAIN TANK PUMP
REFUELING WATER STORAGE TANK PROJECT CLASS 010, 015 2-1204-T4-001
MIXING EDUCTOR PROJECT CLASS 424 (8 REQUIRED) 2-1204-J4-001
CVCS CENTRIFUGAL CHARGING PUMPS AUX. MINIFLOW
ELECTRIC CIRCULATION HEATER PROJECT CLASS 424 2-1204-H4-001
SLUDGE MIXING PUMP PROJECT CLASS 424 2-1204-P4-001
ELECTRICAL HEAT TRACE OUTSIDE OF BUILDING
SAFETY INJECTION PUMP TRAIN A PROJ CL 212 2-1204-P6-003
SAFETY INJECTION PUMP TRAIN B PROJ CLASS 212 2-1204-P6-004
RHRS HOT LEG INJECTION CROSSOVER
RHRS DISCHARGE FROM TRAIN "A" RHR HEAT EXCHANGER
RHRS DISCHARGE FROM TRAIN "B" RHR HEAT EXCHANGER
CHARGING PUMP SUCTION HEADER
CHARGING PUMP SECTION
REFUELING WATER PURIF PUMP SUCTION
OVERFLOW TO RWST
RCS HOT LEG LOOP 1
RCS HOT LEG LOOP 2
RCS HOT LEG LOOP 3
RCS HOT LEG LOOP 4
SIS COLD LEG LOOP 1
SIS COLD LEG LOOP 2
SIS COLD LEG LOOP 3
SIS COLD LEG LOOP 4
TO LINE 136-3/4"
FROM LINE 136-3/4"
TO LINE 114-1"
TO LINE 1208-A09-1/2"
SEE NOTES 19 & 20
SEE NOTE 9 TYPICAL
BASE DRAIN
VENDOR
TUNNEL 2126 AUX. BLDG
50" MANHOLE
24" MANHOLE
24" FLUSH CONN
NOTES:
1. PROVIDE 0.375" I.D. FLOW RESTRICTION AS SHOWN ON BPC DWG CX40Q001.
2. FOR DRAIN HEADER INFORMATION SEE DWG 2X4DB143 & 144-1.
3. ALL EQUIPMENT AND VALVES FURNISHED BY WESTINGHOUSE EXCEPT (1) EQUIPMENT MARKED Φ; (2) ALL PIPING AND ALL MANUAL VALVES 2" AND SMALLER.
4. FOR COOLING CONNECTIONS SEE DWG 2X4DB134.
5. ELECTRIC CIRCULATION HEATER TO MAINTAIN MINIMUM 50° F WATER TEMP IN RWST.
6. LOCATE TAPS AT SAME ELEVATION, PROVIDE MAXIMUM PHYSICAL SEPARATION BETWEEN EACH TAP LOCATION.
7. PROVIDE FOR POWER LOCKOUT AND RESTORATION FROM CONTROL ROOM.
8. ADJUST VALVES IN FIELD TO LIMIT PUMP RUNOUT AND EQUALIZE FLOWS, THEN LOCK IN POSITION.
9. FLOW METERING ORIFICE TO VERIFY FLOW DURING PRE-OPERATIONAL TESTING.
10. LOCATE DRAIN AT LOW POINT BETWEEN VALVES HV8871, HV8888, AND 080.
11. THIS SUCTION PIPE INLET NOZZLE SHALL BE LOCATED APPROXIMATELY 11 FEET ABOVE THE RWST BOTTOM.
12. THESE CLASS 212 VALVES AND THEIR CONTROLS SHALL BE PROTECTED BY A MISSILE SHIELD.
13. ALL EDUCTORS SHALL BE LOCATED AT THE SAME ELEVATION, BETWEEN 12" AND 18" ABOVE THE RWST BOTTOM.
14. PORTION OF THE SYSTEM SUSPECTED TO CAUSE STRESS CORROSION CRACKING. REFER TO THE APPLICABLE DESIGN DRAWING AND PIPING MATERIAL CLASSIFICATIONS FOR MATERIAL AND HEAT TREATMENT REQUIREMENTS.
15. LOCATE VALVES HV8806, 090, 135 AND 136 AS CLOSE AS POSSIBLE.
16. ALL SAFETY-RELATED PROCESS, INSTRUMENT AND SAMPLING LINES THAT ARE PHYSICALLY ROUTED IN AN AREA EXPOSED TO EXTREMELY COLD WEATHER SHALL BE PROVIDED WITH ADEQUATE HEAT TRACING.
17. ○ INDICATES TANK NOZZLE CONNECTION NUMBER PER PLANT DESIGN DWG 1X4DL57F02 REV. 4.
18. THE PIPING FROM THE OUTLET OF THE RELIEF VALVES TO THE RWST IS SPECIFIED AS 414, HOWEVER, THIS PIPING SHALL BE SUPPORTED AND PROTECTED FROM EXTERNAL HAZARDS/EVENTS SO THAT THE PRESSURE BOUNDARY REMAINS INTACT & THAT THE LINE ALLOWS FLOW FOLLOWING A SAFE SHUTDOWN EARTHQUAKE.
19. REMOVABLE SPOOL.
20. FOR TYP. DETAIL, SEE STD. DWG. 4X4DD000.
21. REFER TO ELEMENTARY DIAGRAM FOR THE DETAILED INTERLOCK REQUIREMENTS.
22. DRILL VENT HOLE ON THE CONTAINMENT SIDE OF THE VALVE DISC.
23. FOR ISI TESTING USE ONLY. ROOT VALVES TO BE NORMALLY CLOSED.
24. PRESSURE RELIEF VALVE WILL VENT TO CONTAINMENT ATMOSPHERE PREVENTING THERMAL OVERPRESSURIZATION OF PIPE DURING DBA.
25. THE VALVE STEM LEAKOFF LINE HAS BEEN REMOVED AND THE VALVE LEAKOFF CONNECTION HAS BEEN CAPPED.
26. ORIFICE ASSEMBLY; SEE VENDOR DRAWING 2X6AG01-10001, 10002 & 10003.
REFER TO WESTINGHOUSE FLOW DIAGRAM DRAWING NO. 1094E71 SHEET 3 OF 3.
SOUTHERN COMPANY
GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT
P & I DIAGRAM
SAFETY INJECTION SYSTEM
SYSTEM NO. 1204
CAD 2X4DB121.DWG
ACAD2000 Vep – 08
SCALE: NONE
DRAWING NO. 2X4DB121
VER. 50.0
JOB NO.10604
SIZE E 34X44
DRAWING CATEGORY: CRITICAL
50.0 REVISED PER ABN 2100547901M003, VER. 1.0 3-25-10 Vep GDM JMR X
NO. VERSIONS DATE DN CHK APPV DTL
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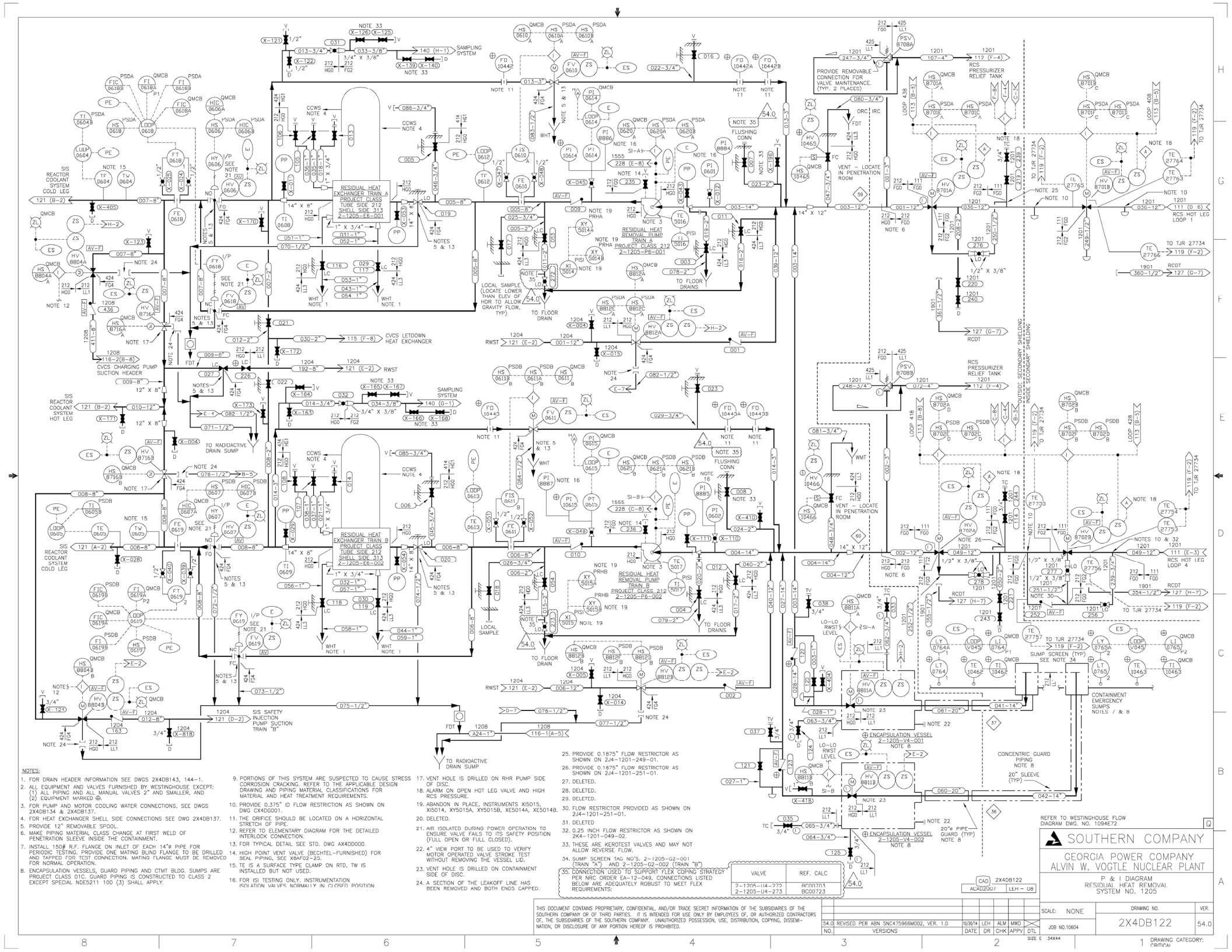

NOTES:

1. FOR DRAIN HEADER INFORMATION SEE DWGS 2X4DB143, 144-1.
2. ALL EQUIPMENT AND VALVES FURNISHED BY WESTINGHOUSE EXCEPT: (1) ALL PIPING AND ALL MANUAL VALVES 2" AND SMALLER, AND (2) EQUIPMENT MARKED Ⓖ.
3. FOR PUMP AND MOTOR COOLING WATER CONNECTIONS, SEE DWGS 2X4DB134 & 2X4DB137.
4. FOR HEAT EXCHANGER SHELL SIDE CONNECTIONS SEE DWG 2X4DB137.
5. PROVIDE 12" REMOVABLE SPOOL.
6. MAKE PIPING MATERIAL CLASS CHANGE AT FIRST WELD OF PENETRATION SLEEVE INSIDE THE CONTAINMENT.
7. INSTALL 150# R.F. FLANGE ON INLET OF EACH 14"ø PIPE FOR PERIODIC TESTING. PROVIDE ONE MATING BLIND FLANGE TO BE DRILLED AND TAPPED FOR TEST CONNECTION. MATING FLANGE MUST BE REMOVED FOR NORMAL OPERATION.
8. ENCAPSULATION VESSELS, GUARD PIPING AND CTMT BLDG. SUMPS ARE PROJECT CLASS 01C. GUARD PIPING IS CONSTRUCTED TO CLASS 2 EXCEPT SPECIAL NDE5211 100 (3) SHALL APPLY.
9. PORTIONS OF THIS SYSTEM ARE SUSPECTED TO CAUSE STRESS CORROSION CRACKING. REFER TO THE APPLICABLE DESIGN DRAWING AND PIPING MATERIAL CLASSIFICATIONS FOR MATERIAL AND HEAT TREATMENT REQUIREMENTS.
10. PROVIDE 0.375" ID FLOW RESTRICTION AS SHOWN ON DWG CX4DG001.
11. THE ORIFICE SHOULD BE LOCATED ON A HORIZONTAL STRETCH OF PIPE.
12. REFER TO ELEMENTARY DIAGRAM FOR THE DETAILED INTERLOCK CONNECTION.
13. FOR TYPICAL DETAIL SEE STD. DWG AX4DD000.
14. HIGH POINT VENT VALVE (BECHTEL–FURNISHED) FOR SEAL PIPING, SEE X8AF02–25.
15. TE IS A SURFACE TYPE CLAMP ON RTD, TW IS INSTALLED BUT NOT USED.
16. FOR ISI TESTING ONLY. INSTRUMENTATION ISOLATION VALVES NORMALLY IN CLOSED POSITION.
17. VENT HOLE IS DRILLED ON RHR PUMP SIDE OF DISC.
18. ALARM ON OPEN HOT LEG VALVE AND HIGH RCS PRESSURE.
19. ABANDON IN PLACE, INSTRUMENTS XI5015, XI5014, XY5015A, XY5015B, XE5014A, XE5014B.
20. DELETED.
21. AIR ISOLATED DURING POWER OPERATION TO ENSURE VALVE FAILS TO ITS SAFETY POSITION (FULL OPEN OR FULL CLOSED).
22. 4" VIEW PORT TO BE USED TO VERIFY MOTOR OPERATED VALVE STROKE TEST WITHOUT REMOVING THE VESSEL LID.
23. VENT HOLE IS DRILLED ON CONTAINMENT SIDE OF DISC.
24. A SECTION OF THE LEAKOFF LINE HAS BEEN REMOVED AND BOTH ENDS CAPPED.
25. PROVIDE 0.1875" FLOW RESTRICTOR AS SHOWN ON 2J4-1201-249-01.
26. PROVIDE 0.1875" FLOW RESTRICTOR AS SHOWN ON 2J4-1201-251-01.
27. DELETED.
28. DELETED.
29. DELETED.
30. FLOW RESTRICTOR PROVIDED AS SHOWN ON 2J4-1201-251-01.
31. DELETED
32. 0.25 INCH FLOW RESTRICTOR AS SHOWN ON 2K4-1201-049-02.
33. THESE ARE KEROTEST VALVES AND MAY NOT ALLOW REVERSE FLOW.
34. SUMP SCREEN TAG NO'S. 2-1205-G2-001 (TRAIN "A") AND 2-1205-G2-002 (TRAIN "B")
35. CONNECTION USED TO SUPPORT FLEX COPING STRATEGY PER NRC ORDER EA-12-049. CONNECTIONS LISTED BELOW ARE ADEQUATELY ROBUST TO MEET FLEX REQUIREMENTS:

VALVE	REF. CALC
2-1205-U4-272	BC00703
2-1205-U4-273	BC00723

REFER TO WESTINGHOUSE FLOW DIAGRAM DWG. NO. 1094E72

SOUTHERN COMPANY
GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT
P & I DIAGRAM
RESIDUAL HEAT REMOVAL
SYSTEM NO. 1205

CAD 2X4DB122
ACADZ007 LEH – 08

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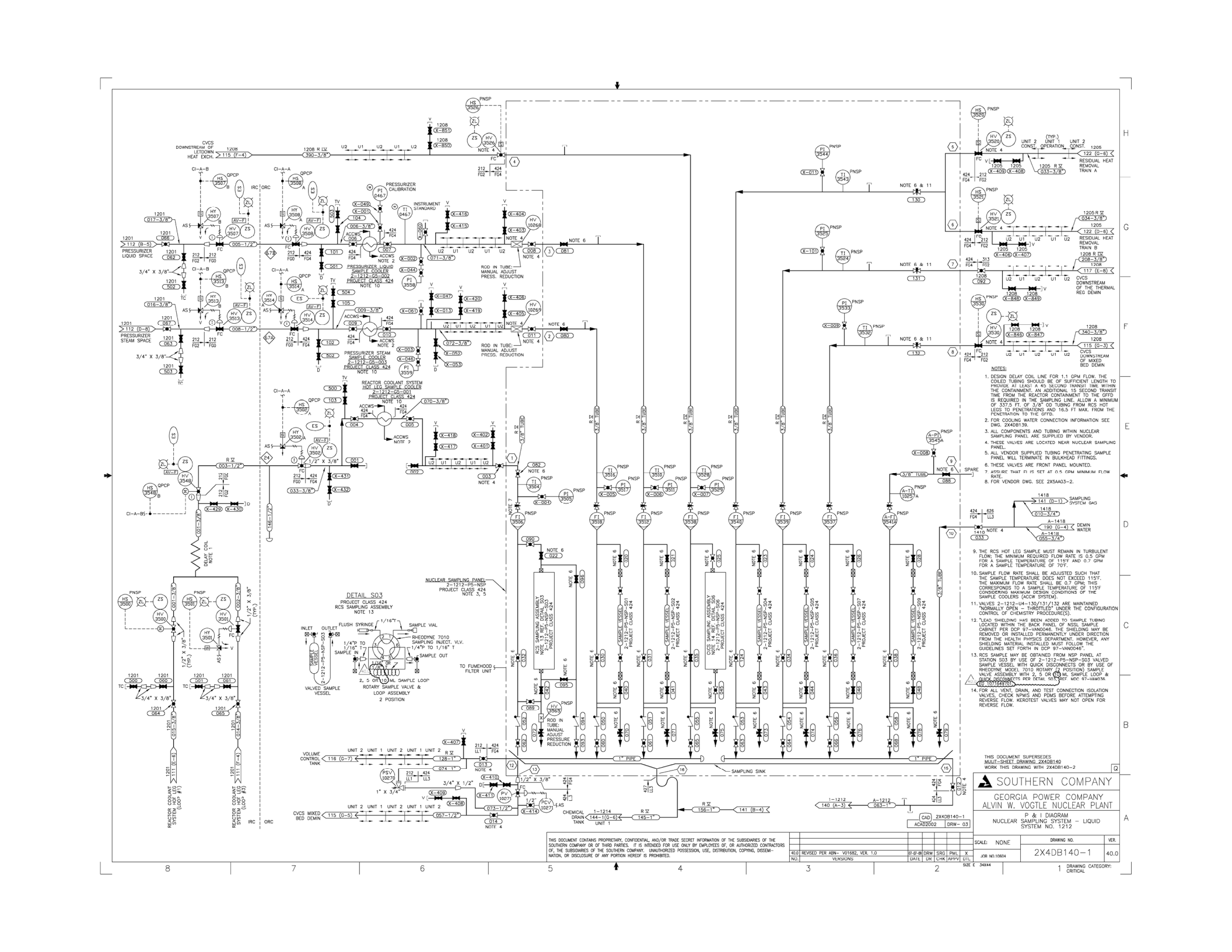
NO.	VERSIONS	DATE	DR	CHK	APPV	DTL
54.0	REVISED PER ABN SNC475966M002, VER. 1.0	10/20/14	LEH	ALM	MWD	

SCALE: NONE
JOB NO.10604
DRAWING NO. 2X4DB122
VER. 54.0
SIZE E 34X44
DRAWING CATEGORY: CRITICAL


PRESSURIZER CALIBRATION
INSTRUMENT STANDARD
PRESSURIZER LIQUID SAMPLE COOLER Z-1212-GS-002 PROJECT CLASS 424 NOTE 10
PRESSURIZER STEAM SAMPLE COOLER Z-1212-GS-003 PROJECT CLASS 424 NOTE 10
REACTOR COOLANT SYSTEM HOT LEG SAMPLE COOLER Z-1212-GS-001 PROJECT CLASS 424 NOTE 10
ROD IN TUBE: MANUAL ADJUST PRESS. REDUCTION
NUCLEAR SAMPLING PANEL Z-1212-PS-NSP PROJECT CLASS 424 NOTE 3, 5
DETAIL S03
PROJECT CLASS 424
RCS SAMPLING ASSEMBLY
NOTE 13
RCS SAMPLING ASSEMBLY NOTE 13 REF. DETAIL S03 Z-1212-PS-NSP-503 PROJECT CLASS 424
CVCS SAMPLING ASSEMBLY NOTE 14 REF. DETAIL S06 Z-1212-PS-NSP-506 PROJECT CLASS 424
SAMPLE VESSEL Z-1212-PS-NSP-501 PROJECT CLASS 424
SAMPLE VESSEL Z-1212-PS-NSP-502 PROJECT CLASS 424
SAMPLE VESSEL Z-1212-PS-NSP-504 PROJECT CLASS 424
SAMPLE VESSEL Z-1212-PS-NSP-505 PROJECT CLASS 424
SAMPLE VESSEL Z-1212-PS-NSP-507 PROJECT CLASS 424
SAMPLE VESSEL Z-1212-PS-NSP-509 PROJECT CLASS 424
INLET
OUTLET
FLUSH SYRINGE
SAMPLE VIAL
RHEODYNE 7010 SAMPLING INJECT. VLV.
SAMPLE IN
SAMPLE OUT
2, 5 OR 10 ML SAMPLE LOOP
ROTARY SAMPLE VALVE & LOOP ASSEMBLY 2 POSITION
VALVED SAMPLE VESSEL
TO FUMEHOOD FILTER UNIT
SAMPLING SINK
1" PIPE
VOLUME CONTROL TANK
CHEMICAL DRAIN TANK
CVCS MIXED BED DEMIN
REACTOR COOLANT SYSTEM HOT LEG (LOOP #1)
REACTOR COOLANT SYSTEM HOT LEG (LOOP #3)
DRAIN COIL NOTE 1
RESIDUAL HEAT REMOVAL TRAIN A
RESIDUAL HEAT REMOVAL TRAIN B
CVCS DOWNSTREAM OF THE THERMAL REG DEMIN
CVCS DOWNSTREAM OF MIXED BED DEMIN
SAMPLING SYSTEM GAS
DEMIN WATER
SPARE
UNIT 2 CONST.
(TYP.) UNIT 1 OPERATION
NOTES:
1. DESIGN DELAY COIL LINE FOR 1.1 GPM FLOW. THE COILED TUBING SHOULD BE OF SUFFICIENT LENGTH TO PROVIDE AT LEAST A 45 SECOND TRANSIT TIME WITHIN THE CONTAINMENT. AN ADDITIONAL 15 SECOND TRANSIT TIME FROM THE REACTOR CONTAINMENT TO THE GFFD IS REQUIRED IN THE SAMPLING LINE. ALLOW A MINIMUM OF 337.5 FT. OF 3/8" OD TUBING FROM RCS HOT LEGS TO PENETRATIONS AND 16.5 FT MAX. FROM THE PENETRATION TO THE GFFD.
2. FOR COOLING WATER CONNECTION INFORMATION SEE DWG. 2X4DB139.
3. ALL COMPONENTS AND TUBING WITHIN NUCLEAR SAMPLING PANEL ARE SUPPLIED BY VENDOR.
4. THESE VALVES ARE LOCATED NEAR NUCLEAR SAMPLING PANEL.
5. ALL VENDOR SUPPLIED TUBING PENETRATING SAMPLE PANEL WILL TERMINATE IN BULKHEAD FITTINGS.
6. THESE VALVES ARE FRONT PANEL MOUNTED.
7. ASSURE THAT FI IS SET AT 0.5 GPM MINIMUM FLOW RATE.
8. FOR VENDOR DWG. SEE 2X5A403-2.
9. THE RCS HOT LEG SAMPLE MUST REMAIN IN TURBULENT FLOW; THE MINIMUM REQUIRED FLOW RATE IS 0.5 GPM FOR A SAMPLE TEMPERATURE OF 115° AND 0.7 GPM FOR A SAMPLE TEMPERATURE OF 70°F.
10. SAMPLE FLOW RATE SHALL BE ADJUSTED SUCH THAT THE SAMPLE TEMPERATURE DOES NOT EXCEED 115°F. THE MAXIMUM FLOW RATE SHALL BE 0.7 GPM; THIS CORRESPONDS TO A SAMPLE TEMPERATURE OF 115°F CONSIDERING MAXIMUM DESIGN CONDITIONS OF THE SAMPLE COOLERS (ACCW SYSTEM).
11. VALVES 2-1212-U4-130/131/132 ARE MAINTAINED "NORMALLY OPEN – THROTTLED" UNDER THE CONFIGURATION CONTROL OF CHEMISTRY PROCEDURE(S).
12. "LEAD SHIELDING HAS BEEN ADDED TO SAMPLE TUBING LOCATED WITHIN THE BACK PANEL OF NSSL SAMPLE CABINET PER DCP 97-VAN0046. THE SHIELDING MAY BE REMOVED OR INSTALLED PERMANENTLY UNDER DIRECTION FROM THE HEALTH PHYSICS DEPARTMENT. HOWEVER, ANY SHIELDING MATERIAL INSTALLED MUST FOLLOW THE GUIDELINES SET FORTH IN DCP 97-VAN0046".
13. RCS SAMPLE MAY BE OBTAINED FROM NSP PANEL AT STATION S03 BY USE OF Z-1212-PS-NSP-503 VALVED SAMPLE VESSEL WITH QUICK DISCONNECTS OR BY USE OF RHEODYNE MODEL 7010 ROTARY (2 POSITION) SAMPLE VALVE ASSEMBLY WITH 2, 5 OR 10 ML SAMPLE LOOP & QUICK DISCONNECTS PER DETAIL S03 REF. MOD 97-VAN0046 (EC 10015420D).
14. FOR ALL VENT, DRAIN, AND TEST CONNECTION ISOLATION VALVES, CHECK NPMIS AND PDMS BEFORE ATTEMPTING REVERSE FLOW. KEROTEST VALVES MAY NOT OPEN FOR REVERSE FLOW.
THIS DOCUMENT SUPERSEDES MULIT-SHEET DRAWING 2X4DB140
WORK THIS DRAWING WITH 2X4DB140-2
SOUTHERN COMPANY
GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT
P & I DIAGRAM
NUCLEAR SAMPLING SYSTEM – LIQUID
SYSTEM NO. 1212
CAD 2X4DB140-1
ACAD2002 DRW- 03
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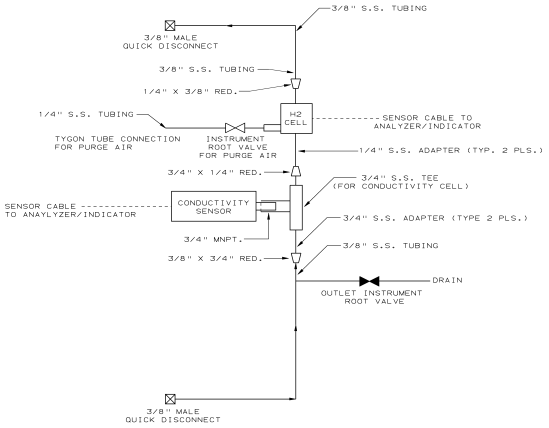
40.0 REVISED PER ABN- V01682, VER. 1.0 01-07-09 DRW SRG PML X
NO. VERSIONS DATE DN CHK APPV DTL
SCALE: NONE
DRAWING NO. 2X4DB140-1
VER. 40.0
JOB NO.10604
SIZE E 34X44
DRAWING CATEGORY: CRITICAL

DETAIL S06

PROJECT CLASS 424
CVCS SAMPLING ASSEMBLY



CONINUATION OF NOTES:

14. IN-LINE HYDROGEN ANALYZERS AND CONDUCTIVITY ELEMENTS HAVE REPLACED THE "SAMPLE BOMB" AND WILL PROVIDE CONTINUOUS SAMPLE CAPABILITY. REFERENCE MOC 98-VAM045.

THIS DOCUMENT SUPERSEDES MULTI-SHEET DRAWING 2X4DB140
WORK THIS DRAWING WITH: 2X4DB140-1
32

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
NUCLEAR SAMPLING SYSTEM-LIQUID
SYSTEM NO. 1212

SCALE: NONE
JOB NO.10604
DRAWING NO. 32 2X4DB140-2
REV. 32

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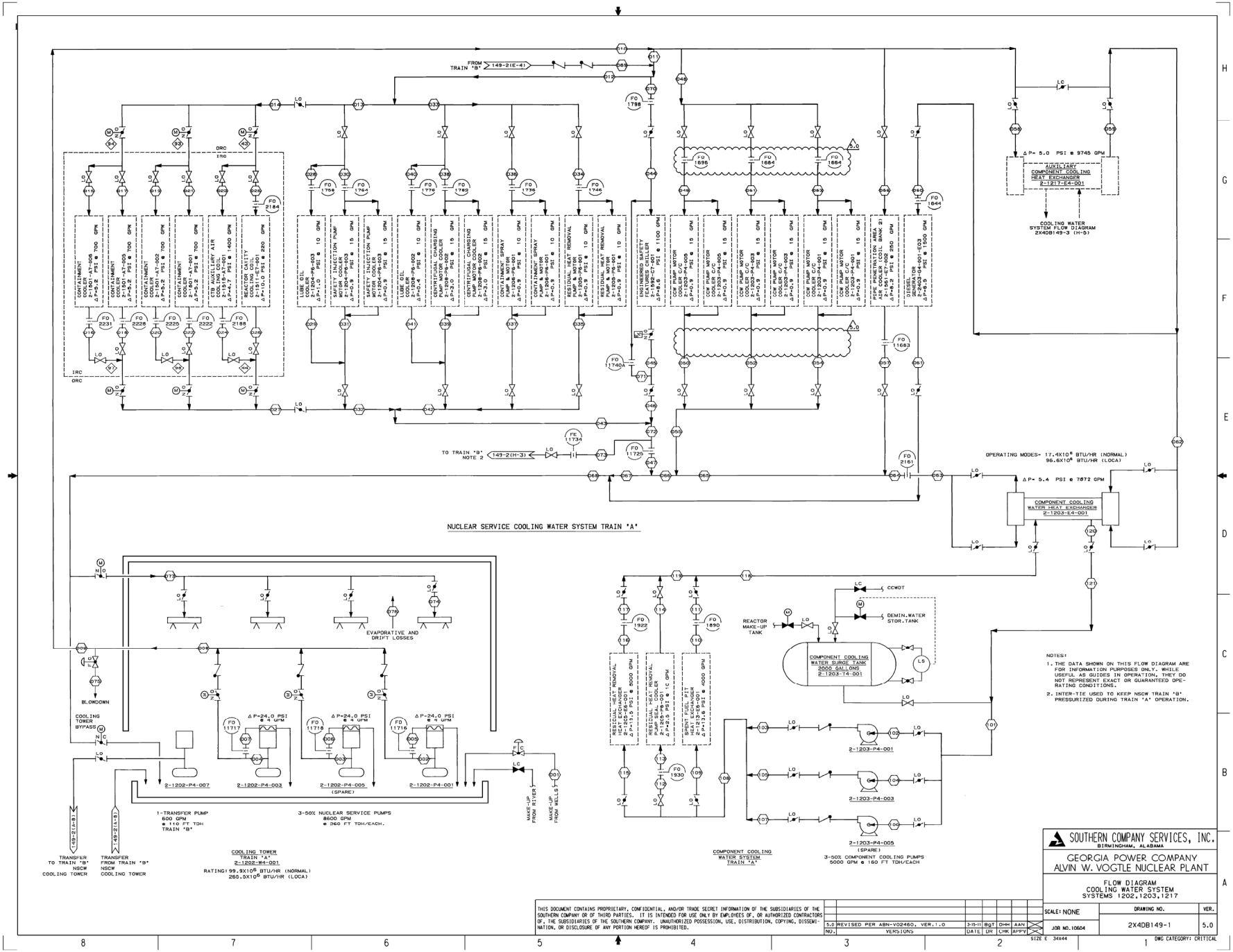
NO.	REVISIONS	DATE	DR	CHK	APPV	DTL
3	INCORPORATED PER REA 01-VAA037	06/04/01	JGM	JRP	PDG	JLM
2	ISSUED FOR CONSTRUCTION	SEE MICROFILM FOR SIGNATURES				
1						
0						

SIZE E 34X44



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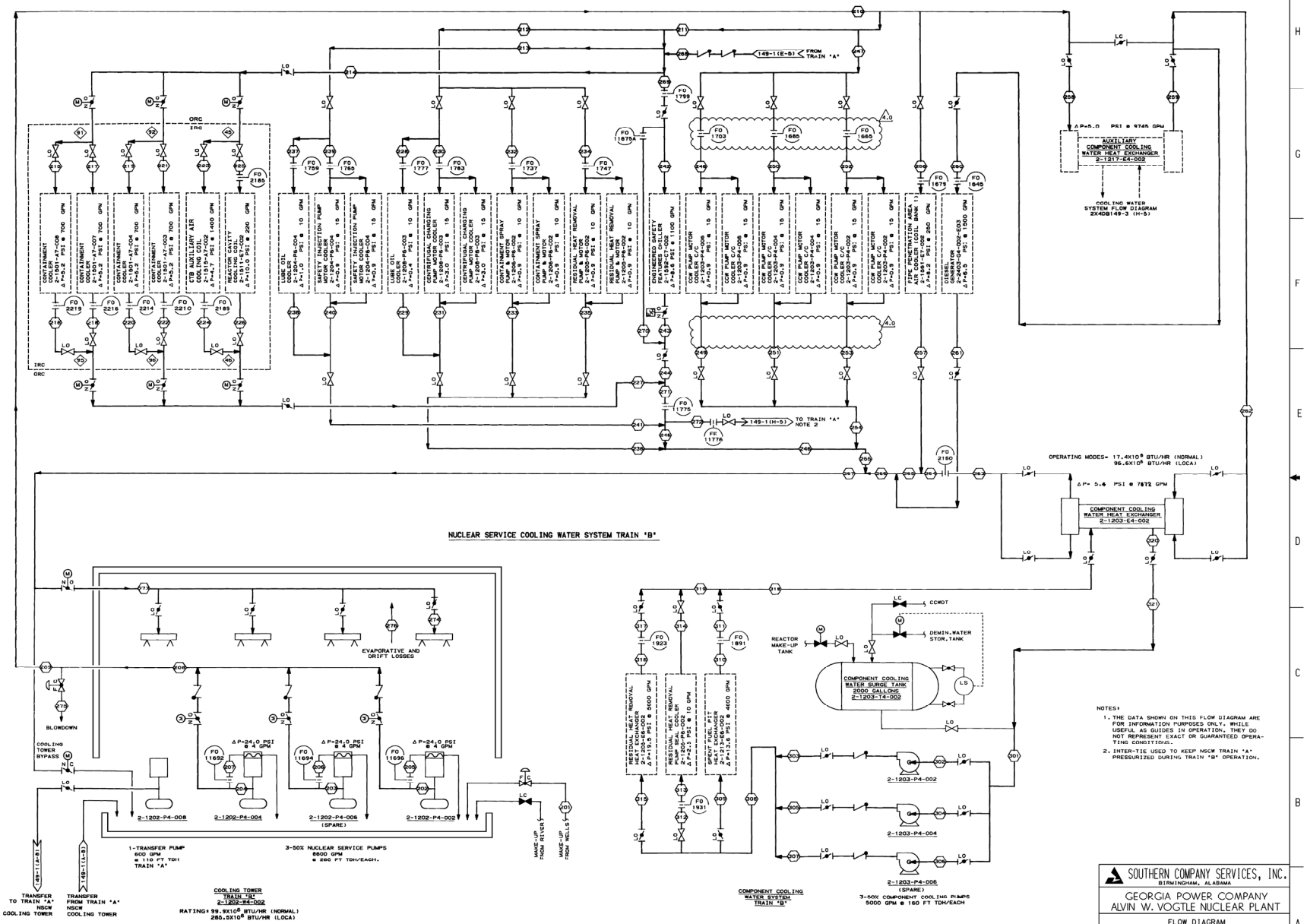
NO.	VERSIONS	DATE	DR	CHK	APPV
5.0	REVISED PER ABN-V02460, VER.1.0	3-15-11	BGT	DHH	AAN

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

FLOW DIAGRAM
COOLING WATER SYSTEM
SYSTEMS 1202,1203,1217

SCALE: NONE
DRAWING NO. 2X4DB149-1
VER. 5.0
JOB NO. 10604
SIZE E 34x44
DWG CATEGORY: CRITICAL



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NO.	VERSIONS	DATE	DR	CHK	APPV
4.0	REVISED PER ABN-V02460, VER.1.0	3-15-11	BGT	DHH	AAN

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

FLOW DIAGRAM
COOLING WATER SYSTEM
SYSTEMS 1202,1203,1217

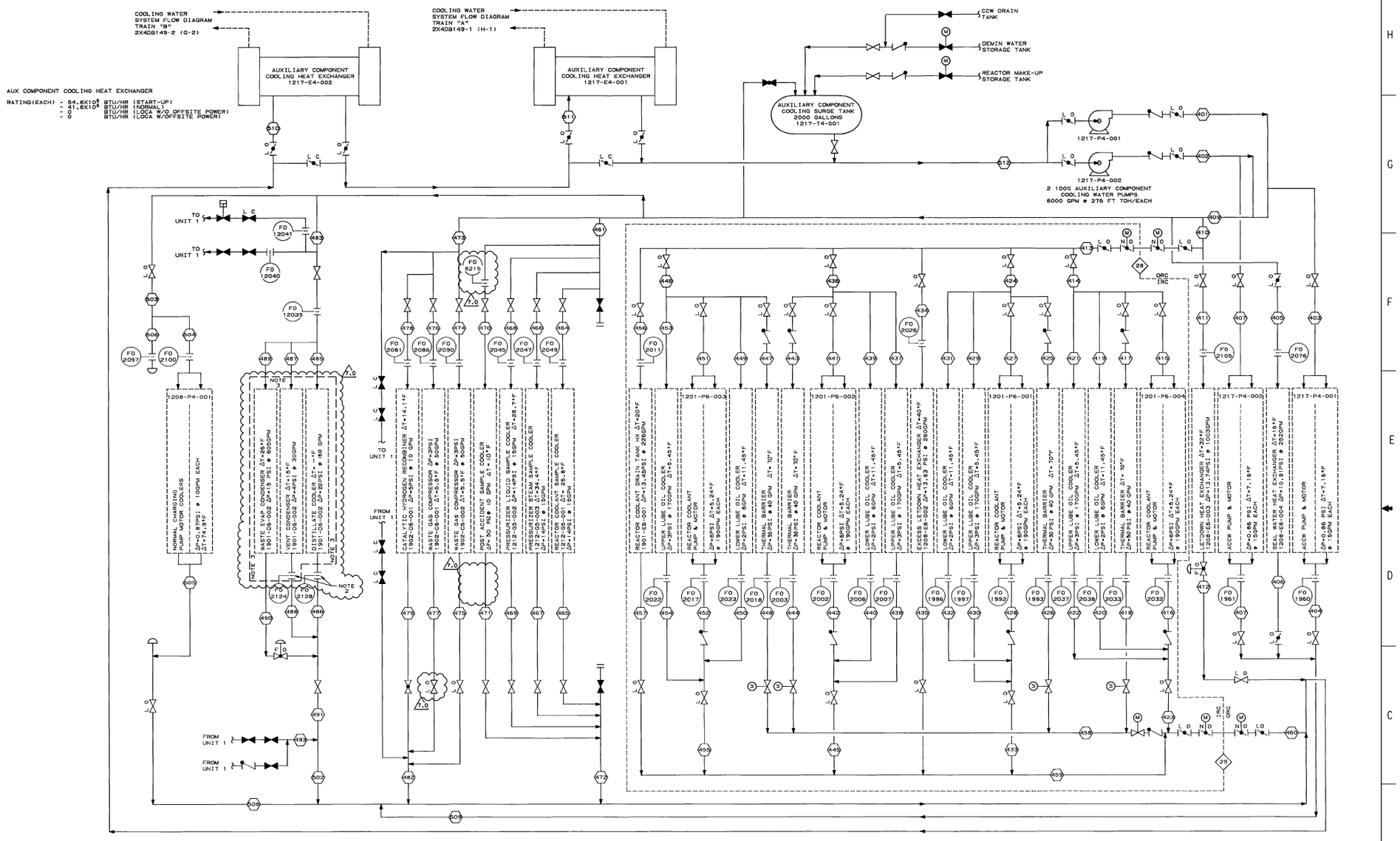
SCALE: NONE

DRAWING NO. 2X4DB149-2 VER. 4.0

JOB NO.10604

SIZE E 34x44

DWG CATEGORY: CRITICAL



AUXILIARY COMPONENT COOLING WATER SYSTEM

NOTE

1. THE DATA SHOWN ON THIS FLOW DIAGRAM ARE FOR INFORMATION PURPOSES ONLY, & WHILE USEFUL AS GUIDES IN OPERATION, DO NOT REPRESENT EXACT OR GUARANTEED OPERATING CONDITIONS.
2. BLANK PLATE
3. "RETIRED IN PLACE"

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

FLOW DIAGRAM
COOLING WATER SYSTEM
SYSTEMS 1202, 1203, 1217

SCALE: NONE

DRAWING NO. 2X4DB149-3

VER. 7.0

JOB NO.10604

SIZE E 34X44

DWG CATEGORY: CRITICAL

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NO.	VERSIONS	DATE	DR	CHK	APPV
7.0	REVISED PER ABN-V02460, VER.1.0	3-15-11	BgT	DHH	AAN

NUCLEAR SERVICE COOLING WATER SYSTEM TRAIN "A" (NOTE 2)

MODES OF OPERATION	PARAMETER		001	002	003	004	005	006	007	008	009	010	011	012	013	014	015	016	017	018	019	020	021	022	023	024	025	026	027	028	029	030	031	032	033	034	035	036	036	037	038	039	040	041	042	043	044	045	046	047	048	049	050	051	052	053	054	055	056
100 % POWER	FLOW	GPM	287	8700	0	8700	5	0	5	17390	17330	17330	6558	4863	4761	4710	746	746	746	746	746	746	746	1491	1491	234	234	4710	13	13	38	38	4760	102	26	26	26	26	26	37	38	13	13	102	4863	1695	1100	1695	6558	114	38	38	38	38	38	38	114	268
	TEMP	°F	70	90	90	90	90	90	90	90	90	90	90	90	90	90	90	97	90	97	90	97	90	97	90	96	90	94	97	90	90	90	90	97	90	90	90	90	90	90	97	90	102	94	97	90	90	90	95	90	90	97	90	97	90	90	95	90
PEAK MSLBA CONDITION (NOTE 4)	FLOW	GPM	0	8200	0	8200	6	0	6	16389	16389	16389	5034	3318	3200	3146	787	787	787	787	787	787	787	0	0	0	0	3146	14	14	40	40	3200	107	27	27	27	27	27	40	40	13	13	107	3307	1727	1100	1727	5034	120	40	40	40	40	40	40	120	283
	TEMP	°F	70	92	92	92	92	92	92	92	92	92	92	92	92	92	92	255	92	255	92	255	92	255	92	92	92	92	255	92	95	92	96	252	92	97	92	98	92	104	104	247	92	101	98	196	92	92	95	92	95	92	95	92	94	92
PEAK LOCA CONDITION (NOTE 5)	FLOW	GPM	0	8200	0	8200	6	0	6	16389	16389	16389	5034	3318	3200	3146	787	787	787	787	787	787	787	0	0	0	0	3146	14	14	40	40	3200	107	27	27	27	27	27	40	40	13	13	107	3307	1727	1100	1727	5034	120	40	40	40	40	40	40	120	283
	TEMP	°F	70	91	91	91	91	91	91	91	91	91	91	91	91	91	91	266	91	266	91	266	91	266	91	91	91	91	266	91	94	91	95	263	91	91	91	118	91	96	91	97	91	102	103	258	91	100	97	202	91	91	94	91	94	91	93	91
RECIRC. MODE POST-LOCA (NOTE 6)	FLOW	GPM	0	8200	0	8200	6	0	6	16389	16389	16389	5034	3318	3200	3146	787	787	787	787	787	787	787	0	0	0	0	3146	14	14	40	40	3200	107	27	27	27	27	27	40	40	13	13	107	3307	1727	1100	1727	5034	120	40	40	40	40	40	40	120	283
	TEMP	°F	70	95	95	95	95	95	95	95	95	95	95	95	95	95	95	242	95	242	95	242	95	242	95	95	95	95	242	95	99	95	99	239	95	95	95	123	95	101	95	102	95	107	107	235	95	105	101	188	95	95	99	95	99	95	98	95

7.0 7.0

NUCLEAR SERVICE COOLING WATER SYSTEM TRAIN "A" (CONTINUED) (NOTE 2)

MODES OF OPERATION	PARAMETER		057	058	059	060	061	062	063	064	065	066	067	068	069	070	071	072	073	074	075	076
100 % POWER	FLOW	GPM	268	10390	10390	1718	1718	8672	8672	8672	8940	9054	5612	17330	0	595	6558	0	17330	4333	65	201
	TEMP	°F	90	90	98	98	98	98	102	102	102	102	99	99	95	90	95	95	99	99	90	90
PEAK MSLBA CONDITION (NOTE 4)	FLOW	GPM	283	10952	10953	1811	1811	9141	9141	9141	9424	9544	14578	16389	0	627	5034	0	16388	4097	0	472
	TEMP	°F	101	92	93	93	112	93	146	146	145	144	133	131	92	92	196	196	131	131	92	104
PEAK LOCA CONDITION (NOTE 5)	FLOW	GPM	283	10952	10953	1811	1811	9141	9141	9141	9424	9544	14578	16389	0	627	5034	0	16389	4097	0	477
	TEMP	°F	100	91	91	91	110	91	95	95	95	95	132	129	91	91	202	202	129	129	91	104
RECIRC. MODE POST-LOCA (NOTE 6)	FLOW	GPM	283	10952	10953	1811	1811	9141	9141	9141	9424	9544	14578	16389	0	627	5034	0	16389	4097	0	432
	TEMP	°F	105	95	95	95	114	95	129	129	128	128	148	144	95	95	188	188	144	144	95	110

7.0 7.0 7.0

COMPONENT COOLING WATER SYSTEM TRAIN "A"

MODES	PARAMETER		101	102	103	104	105	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120	121
100% POWER	FLOW	GPM	8979	4490	4490	4490	4490	0	0	8979	3982	3988	3988	10	10	10	4987	4987	4987	8987	4997	8979	8979
	TEMP	°F	101	101	101	101	101	101	101	101	101	115	115	101	101	101	101	101	101	112	101	101	101
2 TRAIN START OF RECIRC POST-LOCA	FLOW	GPM	8978	4489	4489	4489	4489	0	0	8978	3981	3985	3985	10	10	10	4987	5062	5062	9057	5070	8978	8978
	TEMP	°F	102	102	102	102	102	102	102	102	102	106	106	102	102	107	102	156	156	134	156	102	102
1 TRAIN COOLDOWN, OFFSITE POWER AVAIL.	FLOW	GPM	9000	4500	4500	4500	4500	0	0	9000	3991	4000	4000	10	10	10	4999	5090	5090	9094	5100	9000	9000
	TEMP	°F	120	120	120	120	120	120	120	120	120	133	133	120	120	125	120	189	189	164	189	120	120
SINGLE TRAIN COOLDN W/LOP	FLOW	GPM	9000	4500	4500	4500	4500	0	0	9000	3991	4000	4000	10	10	10	4999	5090	5090	9094	5100	9000	9000
	TEMP	°F	120	120	120	120	120	120	120	120	120	133	133	120	120	126	120	207	207	174	207	120	120

7.0

NUCLEAR SERVICE COOLING WATER SYSTEM TRAIN "B" (NOTE 2)

MODES OF OPERATION	PARAMETER		201	202	203	204	205	206	207	208	209	210	211	212	213	214	215	216	217	218	219	220	221	222	223	224	225	226	227	228	229	230	231	232	233	234	235	236	237	238	239	240	241	242	243	244	245	246	247	248	249	250	251	252	253	254	255	256
100 % POWER	FLOW	GPM	287	8700	0	8700	5	0	5	17390	17330	17330	6558	102	51	4710	746	746	746	746	746	746	746	1491	1491	234	234	4710	13	13	37	37	26	26	26	26	102	13	13	38	38	51	1100	1100	1695	6456	6558	114	38	38	38	38	38	38	114	6672	268
	TEMP	°F	70	90	90	90	90	90	90	90	90	90	90	90	90	90	90	97	90	97	90	97	90	97	90	96	90	94	97	90	102	90	97	90	90	90	90	94	90	90	90	90	90	90	90	95	95	90	90	97	90	97	90	90	95	95	90
PEAK MSLBA CONDITION (NOTE 4)	FLOW	GPM	0	8200	0	8200	6	0	6	16389	16389	16389	5034	107	54	3146	787	787	787	787	787	787	787	0	0	0	0	3146	14	13	40	40	27	27	27	27	107	14	14	40	40	54	1100	1100	1727	4927	5034	120	40	40	40	40	40	40	120	5154	283
	TEMP	°F	70	92	92	92	92	92	92	92	92	92	92	92	92	92	92	255	92	255	92	255	92	255	92	92	92	92	255	92	104	92	98	92	97	92	119	104	92	95	92	96	96	92	101	98	198	196	92	92	95	92	95	92	92	94	193	92
PEAK LOCA CONDITION (NOTE 5)	FLOW	GPM	0	8200	0	8200	6	0	6	16389	16389	16389	5034	107	54	3146	787	787	787	787	787	787	787	0	0	0	0	3146	14	13	40	40	27	27	27	27	107	14	14	40	40	54	1100	1100	1727	4927	5034	120	40	40	40	40	40	40	120	5154	283
	TEMP	°F	70	91	91	91	91	91	91	91	91	91	91	91	91	91	91	266	91	266	91	266	91	266	91	91	91	91	266	91	102	91	97	91	96	91	118	103	91	94	91	95	94	91	100	97	205	202	91	91	94	91	94	91	91	93	200	91
RECIRC. MODE POST-LOCA (NOTE 6)	FLOW	GPM	0	8200	0	8200	6	0	6	16389	16389	16389	5034	107	54	3146	787	787	787	787	787	787	787	0	0	0	0	3146	14	13	40	40	27	27	27	27	107	14	14	40	40	54	1100	1100	1727	4927	5034	120	40	40	40	40	40	40	120	5154	283
	TEMP	°F	70	95	95	95	95	95	95	95	95	95	95	95	95	95	95	242	95	242	95	242	95	242	95	95	95	95	242	95	107	95	102	95	101	95	123	107	95	99	95	99	99	95	105	101	190	188	95	95	99	95	99	95	95	98	186	95

7.0 7.0 7.0

NUCLEAR SERVICE COOLING WATER SYSTEM TRAIN "B" (CONTINUED) (NOTE 2)

MODES OF OPERATION	PARAMETER		257	258	259	260	261	262	263	264	265	266	267	268	269	270	271	272	273	274	275	276
100 % POWER	FLOW	GPM	268	10390	10390	1718	1718	8672	8672	8672	8940	10658	17330	0	1695	595	6405	0	17330	4333	65	201
	TEMP	°F	90	90	98	98	98	102	102	102	101	99	90	90	90	95	95	95	99	99	90	90
PEAK MSLBA CONDITION (NOTE 4)	FLOW	GPM	283	10952	10952	1811	1811	9141	9141	9141	9424	11235	16389	0	1727	627	4873	0	16389	4097	0	472
	TEMP	°F	101	92	93	93	112	93	146	146	145	139	131	92	92	92	199	199	131	131	92	104
PEAK LOCA CONDITION (NOTE 5)	FLOW	GPM	283	10952	10952	1811	1811	9141	9141	9141	9424	11235	16389	0	1727	627	4873	0	16389	4097	0	477
	TEMP	°F	100	91	91	91	110	91	95	95	95	97	129	91	91	91	206	206	129	129	91	104
RECIRC. MODE POST-LOCA (NOTE 6)	FLOW	GPM	283	10952	10952	1811	1811	9141	9141	9141	9424	11235	16389	0	1727	627	4873	0	16389	4097	0	432
	TEMP	°F	105	95	95	95	114	95	129	129	128	125	144	95	95	95	191	191	144	144	95	110

7.0 7.0 7.0

COMPONENT COOLING WATER SYSTEM TRAIN "B"

MODES	PARAMETER		301	302	303	304	305	306	307	308	309	310	311	312	313	314	315	316	317	318	319	320	321
100% POWER	FLOW	GPM	8979	4490	4490	4490	4490	0	0	8979	3982	3988	3988	10	10	10	4987	4987	4987	8987	4997	8979	8979
	TEMP	°F	101	101	101	101	101	101	101	101	101	115	115	101	101	101	101	101	101	112	101	101	101
2 TRAIN START OF RECIRC POST-LOCA	FLOW	GPM	8978	4489	4489	4489	4489	0	0	8978	3981	3985	3985	10	10	10	4987	5062	5062	9057	5070	8978	8979
	TEMP	°F	102	102	102	102	102	102	102	102	102	106	106	102	102	107	102	156	156	134	156	102	102
1 TRAIN COOLDOWN, OFFSITE POWER AVAIL.	FLOW	GPM	9002	4501	4501	4501	4501	0	0	9002	3992	4000	4000	10	10	10	5000	5077	5077	9167	5175	9002	9002
	TEMP	°F	120	120	120	120	120	120	120	120	120	133	133	120	120	125	120	189	189	164	189	120	120
SINGLE TRAIN COOLDN W/LOP	FLOW	GPM	9002	4501	4501	4501	4501	0	0	9002	3992	4000	4000	10	10	10	5000	5077	5077	9167	5175	9002	9002
	TEMP	°F	120	120	120	120	120	120	120	120	120	133	133	120	120	126	120	207	207	174	207	120	120

7.0

AUXILIARY COMPONENT COOLING WATER SYSTEM (NOTE 3)

MODES OF OPERATION	PARAMETER		401	402	403	404	405	406	407	408	409	410	411	412	413	414	415	416	417	418	419	420	421	422	423	424	425	426	427	428	429	430	431	432	433	434	435	436	437	438	439	440	441	442	443	444	445	446	447	448	449	450	451	452	453	454	455	456
100 % POWER	FLOW	GPM	5741	0	35	35	278	278	35	35	5671	3154	262	262	2892	593	352	352	53	53	8	8	180	180	540	596	52	52	355	355	181	181	8	8	544	284	284	583	174	174	8	8	351	351	50	50	533	581	49	49	8	8	350	350	174	174	532	254
	TEMP	°F	105	105	105	105	105	119	105	111	105	105	105	155	105	105	105	111	105	158	105	115	105	110	111	105	105	158	105	111	105	110	105	115	111	105	105	105	105	110	105	115	105	111	105	161	111	105	105	162	105	115	105	111	105	110	111	105
START -UP	FLOW	GPM	6188	0	33	33	264	264	33	33	6122	3738	1003	1003	2735	561	333	333	50	50	8	8	171	171	511	564	49	49	336	336	171	171	8	8	515	269	269	552	165	165	8	8	332	332	47	47	504	550	47	47	8	8	331	331	164	164	503	240
	TEMP	°F	105	105	105	105	105	120	105	112	105	105	105	137	105	105	105	111	105	161	105	115	105	110	111	105	105	161	105	111	105	110	105	115	111	105	105	144	105	105	105	115	105	111	105	164	111	105	105	165	105	115	105	111	105	111	111	105
SHUT DOWN, LOP	FLOW	GPM	6282	0	33	33	260	260	33	33	6216	3862	1162	1162	2700	554	329	329	49	49	8	8	168	168	505	557	49	49	332	332	169	169	8	8	508	265	265	545	163	163	8	8	327	327	47	47	498	543	46	46	8	8	327	327	162	162	497	237
	TEMP	°F	105	105	105	105	105	120	105	112	105	105	105	155	105	105	105	111	105	162	105	105	105	105	105	105	105	162	105	105	105	105	105	105	105	105	105	144	105	105	105	105	105	105	105	165	105	105	105	166	105	105	105	105	105	105	105	105
HOT STANDBY WITH OFFSITE POWER	FLOW	GPM	5741	0	35	35	278	278	35	35	5671	3154	262	262	2892	593	352	352	53	53	8	8	180	180	540	596	52	52	355	355	181	181	8	8	544	284	284	583	174	174	8	8	351	351	50	50	533	581	49	49	8	8	350	350	174	174	532	254
	TEMP	°F	105	105	105	105	105	119	105	111	105	105	105	155	105	105	105	111	105	158	105	105	105	105	105	105	105	158	105	111	105	110	105	115	111	105	105	105	105	110	105	115	105	111	105	161	111	105	105	162	105	115	105	111	105	105	105	105

7.0

CONTINUATION OF AUXILIARY COMPONENT COOLING WATER SYSTEM (NOTE 3)

MODES OF OPERATION	PARAMETER		457	458	459	460	461	464	465	466	467	468	469	470	471	472	473	474	475	476	477	478	479	480 NOTE 7	481 NOTE 7	482	483	484 NOTE 7	485	486	487	488	489	490	491	492 NOTE 7	493 NOTE 7	494 NOTE 7	495 NOTE 7	496 NOTE 7	497 NOTE 7	498 NOTE 7	499 NOTE 7	500 NOTE 7	501 NOTE 7	502	503	504	505	506	508	509	510	511	512
100 % POWER	FLOW	GPM	254	204	2147	2892	102	18	18	19	19	23	23	17	17	102	137	56	56	56	56	12	12	12	12	137	1887	1020	0	0	0	0	1020	1020	1020	38	38	630	123	123	36	36	671	671	867	1887	112	23	23		112	348	5741	5741	5741
	TEMP	°F	123	160	111	115	105	105	126	105	132	105	124	105	105	121	105	105	110	105	105	105	116	105	105	116	109	105	NA	NA	NA	NA	105	105	105	105	121	105	105	118	105	118	105	118	105	126	115	105	105	107		113	116	117	105	105
START -UP	FLOW	GPM	240	193	2031	2735	97	17	17	18	18	22	22	16	16	97	130	53	53	53	53	12	12	12	12	130	1787	966	0	0	0	0	966	966	966	36	36	786	116	116	34	34	635	635	821	1787	107	21	21		107	330	6188	6188	6188
	TEMP	°F	110	163	115	118	105	105	128	105	134	105	125	105	105	122	105	105	110	105	105	105	117	105	117	109	105	NA	NA	NA	NA	NA	105	105	105	105	122	105	105	118	105	118	105	130	127	115	105	105	107		113	117	120	105	105
SHUT DOWN, LOP	FLOW	GPM	237	191	2005	2700	95	17	17	18	18	21	21	16	16	95	128	52	52	52	52	12	12	12	12	128	1765	954	0	0	0	0	954	954	954	35	35	776	115	115	34	34	627	627	811	1765	105	21	21		105	326	6282	6282	6282
	TEMP	°F	105	164	105	109	105	105	105	105	105	105	105	105	105	105	105	105	105	105	105	105	105	105	105	105	105	NA	NA	NA	NA	NA	105	105	105	105	105	105	105	105	105	105	105	105	105	105	105	105	105		105	117	107	105	105
HOT STANDBY WITH OFFSITE POWER	FLOW	GPM	254	204	2147	2892	102	18	18	19	19	23	23	17	17	102	137	56	56	56	56	12	12	12	12	137	1887	1020	0	0	0	0	1020	1020	1020	38	38	630	123	123	36	36	671	671	867	1887	112	23	23		112	348	5741	5741	5741
	TEMP	°F	105	160	106	110	105	105	126	105	132	105	124	105	105	121	105	105	110	105	105	105	116	105	105	108	105	NA	NA	NA	NA	NA	105	105	105	121	105	105	118	105	118	105	128	126	115	105	105	107		113	116	114	105	105

7.0 7.0 7.0

NOTES:

1. THE DATA SHOWN ON THIS FLOW DIAGRAM ARE FOR INFORMATION PURPOSES ONLY. WHILE USEFUL AS GUIDES IN OPERATION, THEY DO NOT REPRESENT EXACT OR GUARANTEED OPERATING CONDITIONS.
2. ALL NSCW DATA SHOWN ARE FOR ONE-TRAIN OPERATION, WITH ALL ACCW HEAT LOADS ON THE OPERATING TRAIN.
3. ACCW DATA ASSUME ACCW HEAT LOADS REJECTED TO TRAIN B NSCW.
4. TEMPERATURES SHOWN HERE ARE THE MAXIMUM TEMPERATURES DURING THE MSLBA TRANSIENT. THE MAXIMUM TEMPERATURE AT EACH NODE MAY NOT OCCUR AT THE SAME TIME DURING THE TRANSIENT. THEREFORE THE INLET AND OUTLET TEMPERATURE DIFFERENCES DO NOT NECESSARILY REPRESENT THE ACTUAL TEMPERATURE RISES ACROSS THE COMPONENTS AT A SPECIFIC TIME. (REFER TO CALC. • X4C1202S18 FOR DETAILS).
5. TEMPERATURES SHOWN HERE ARE THE MAXIMUM TEMPERATURES DURING THE LOCA TRANSIENT. THE MAXIMUM TEMPERATURE AT EACH NODE MAY NOT OCCUR AT THE SAME TIME DURING THE TRANSIENT. THEREFORE THE INLET AND OUTLET TEMPERATURE DIFFERENCES DO NOT NECESSARILY REPRESENT THE ACTUAL TEMPERATURE RISES ACROSS THE COMPONENTS AT A SPECIFIC TIME. (REFER TO CALC. • X4C1202S17 FOR DETAILS).
6. TEMPERATURES SHOWN HERE ARE THE MAXIMUM TEMPERATURES DURING THE POST-LOCA, RECIRCULATION TRANSIENT. THE MAXIMUM TEMPERATURE AT EACH NODE MAY NOT OCCUR AT THE SAME TIME DURING THE TRANSIENT. THEREFORE THE INLET AND OUTLET TEMPERATURE DIFFERENCES DO NOT NECESSARILY REPRESENT THE ACTUAL TEMPERATURE RISES ACROSS THE COMPONENTS AT A SPECIFIC TIME. (REFER TO CALC. • X4C1202C17 FOR DETAILS).
7. ACCW DATA ASSUME THAT UNIT 2 ACCW IS SUPPLIED TO THE COMMON COMPONENTS AS SHOWN ON 1X4DB149-3.

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NO.	VERSIONS	DATE	DR	CHK	APPV
7.0	REVISED PER ABN-V02460, VER.1.0	3-15-11	BgT	DHH	AAN

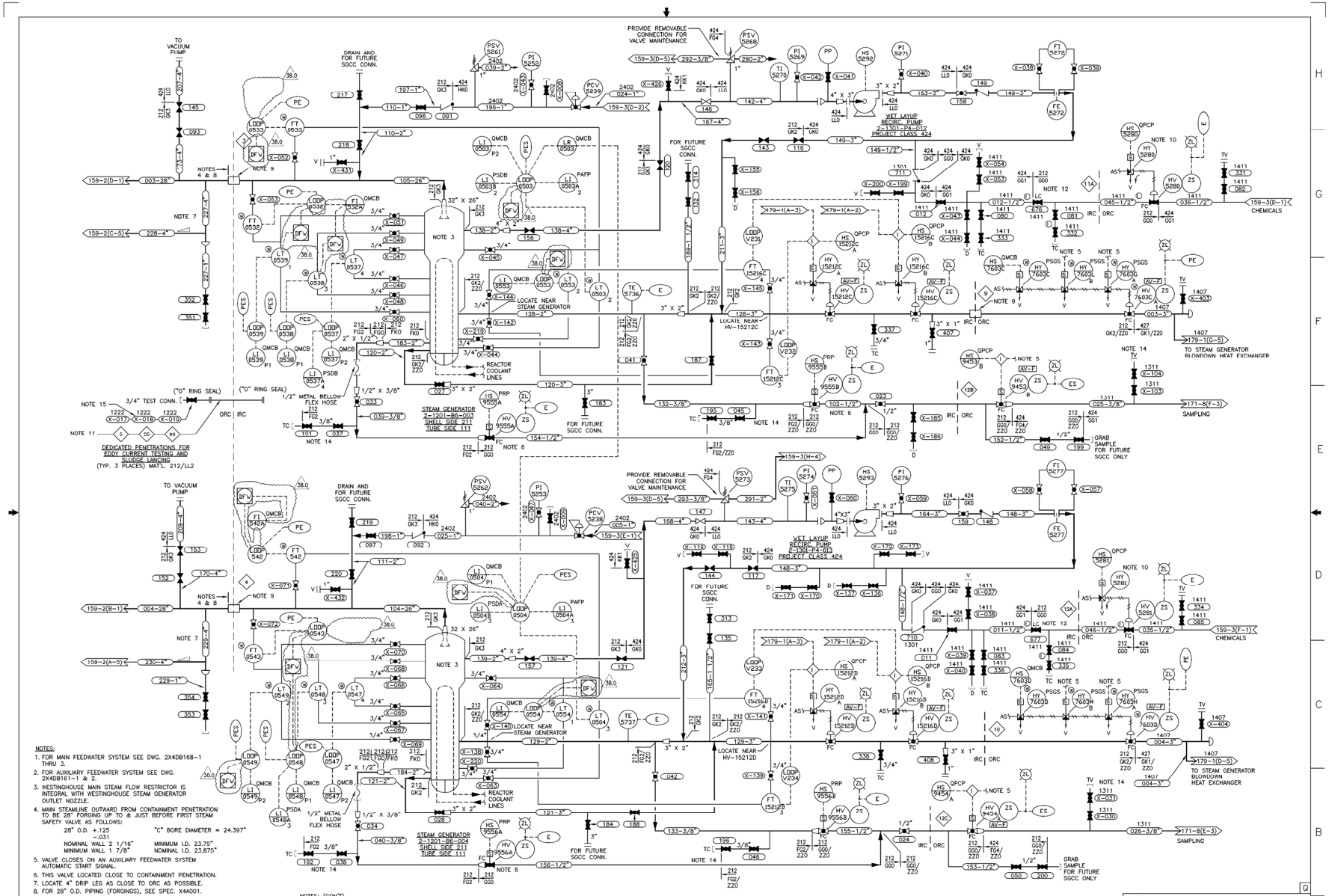
SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

FLOW DIAGRAM
COOLING WATER SYSTEMS
SYSTEMS 1202, 1203, 1217

SCALE: NONE | JOB NO.10604 | DRAWING NO. 2X4DB149-4 | VER. 7.0

SIZE E 34x44 | DWG CATEGORY: CRITICAL



NOTES:

1. FOR MAIN FEEDWATER SYSTEM SEE DWG. 2X4DB168-1 THRU 3.
2. FOR AUXILIARY FEEDWATER SYSTEM SEE DWG. 2X4DB161-1 & 2.
3. WESTINGHOUSE MAIN STEAM FLOW RESTRICTOR IS INTEGRAL WITH WESTINGHOUSE STEAM GENERATOR OUTLET NOZZLE.
4. MAIN STEAMLINE OUTWARD FROM CONTAINMENT PENETRATION TO BE 28" FORGING UP TO & JUST BEFORE FIRST STEAM SAFETY VALVE AS FOLLOWS:

28" O.D. +.125 −.031	"C" BORE DIAMETER = 24.397"
NOMINAL WALL 2 1/16"	MINIMUM I.D. 23.75"
MINIMUM WALL 1 7/8"	NOMINAL I.D. 23.875"

5. VALVE CLOSES ON AN AUXILIARY FEEDWATER SYSTEM AUTOMATIC START SIGNAL.
6. THIS VALVE LOCATED CLOSE TO CONTAINMENT PENETRATION.
7. LOCATE 4" DRIP LEG AS CLOSE TO ORC AS POSSIBLE.
8. FOR 28" O.D. PIPING (FORGINGS), SEE SPEC. X4A001.
9. FOR PIPING FLUID HEAD FITTING OF PENETRATION ASSEMBLY, SEE SPEC. X4A010.
10. POWER TO THE SOLENOID SHALL BE REMOVED DURING ALL MODES OF OPERATION, EXCEPT DURING WET LAYUP OPERATION AND PERIODIC TESTING OF VALVES, BY MOVING THE SLIDE LINKS OF THE STATES TERMINAL BLOCKS. LOCATED IN THE TERMINAL BOX, TO THE OPEN POSITION.
11. DURING MODE 5 (COLD SHUTDOWN) OR MODE 6 (REFUELING), "OUTAGE FIXTURES" MAY BE INSTALLED ON THESE PENETRATIONS TO FACILITATE SLUDGE LANCING AND EDDY CURRENT TESTING INSIDE CONTAINMENT. SEE DRAWING 2X2D01J019 FOR DETAILS OF THE "OUTAGE FIXTURES".
12. THIS VALVE SHALL REMAIN LOCKED CLOSED DURING ALL MODES OF OPERATION, EXCEPT DURING WET LAYUP.
13. DELETED.

NOTES: (CONT'D)

14. THESE ARE KEROTEST VALVES AND MAY NOT ALLOW REVERSE FLOW.
15. VALVE X-018 IS A GATE VALVE.

SOUTHERN COMPANY

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
MAIN STEAM SYSTEM
SYSTEM NO. 1301

CAD 2X4DB159-1
ACAD2007 JLO - 04

SCALE: NONE

DRAWING NO. 2X4DB159-1

VER. 38.0

JOB NO.10604

SIZE E 34044

DRAWING CATEGORY: CRITICAL

NO.	VERSIONS	DATE	DR	CHK	APPV	DTL
38.0	REVISED PER ABN 20390023011174, VER. 1.0	04-15-13	JLO	ELC	JMR	X

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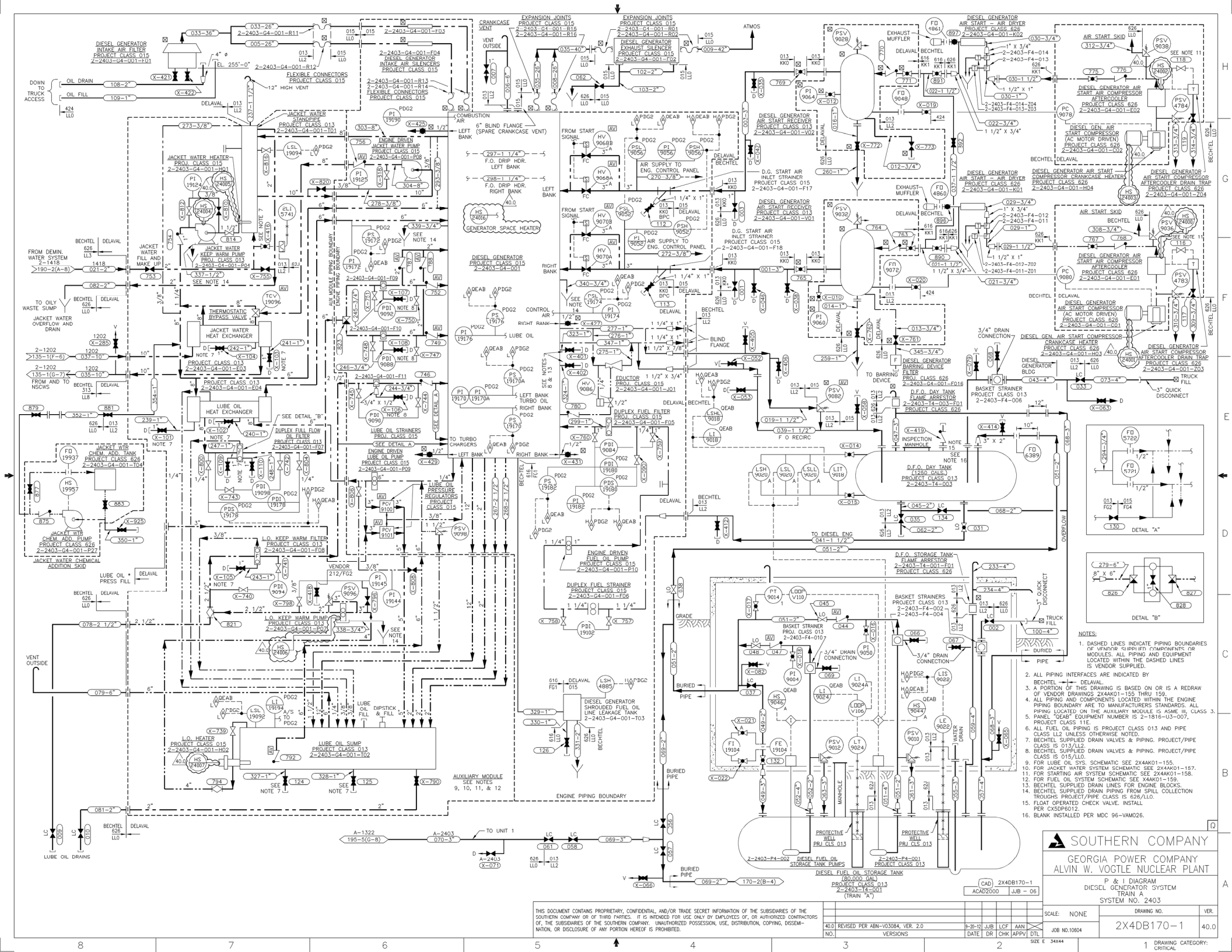

SOUTHERN COMPANY

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
DIESEL GENERATOR SYSTEM
TRAIN A
SYSTEM NO. 2403

CAD ACAD2000 | 2X4DB170-1 | JJB - 06

SCALE: NONE | DRAWING NO. 2X4DB170-1 | VER. 40.0

JOB NO.10604

NO.	VERSIONS	DATE	DR	CHK	APPV	DTL
40.0	REVISED PER ABN-V03084, VER. 2.0	9-20-12	JJB	LCF	AAN	

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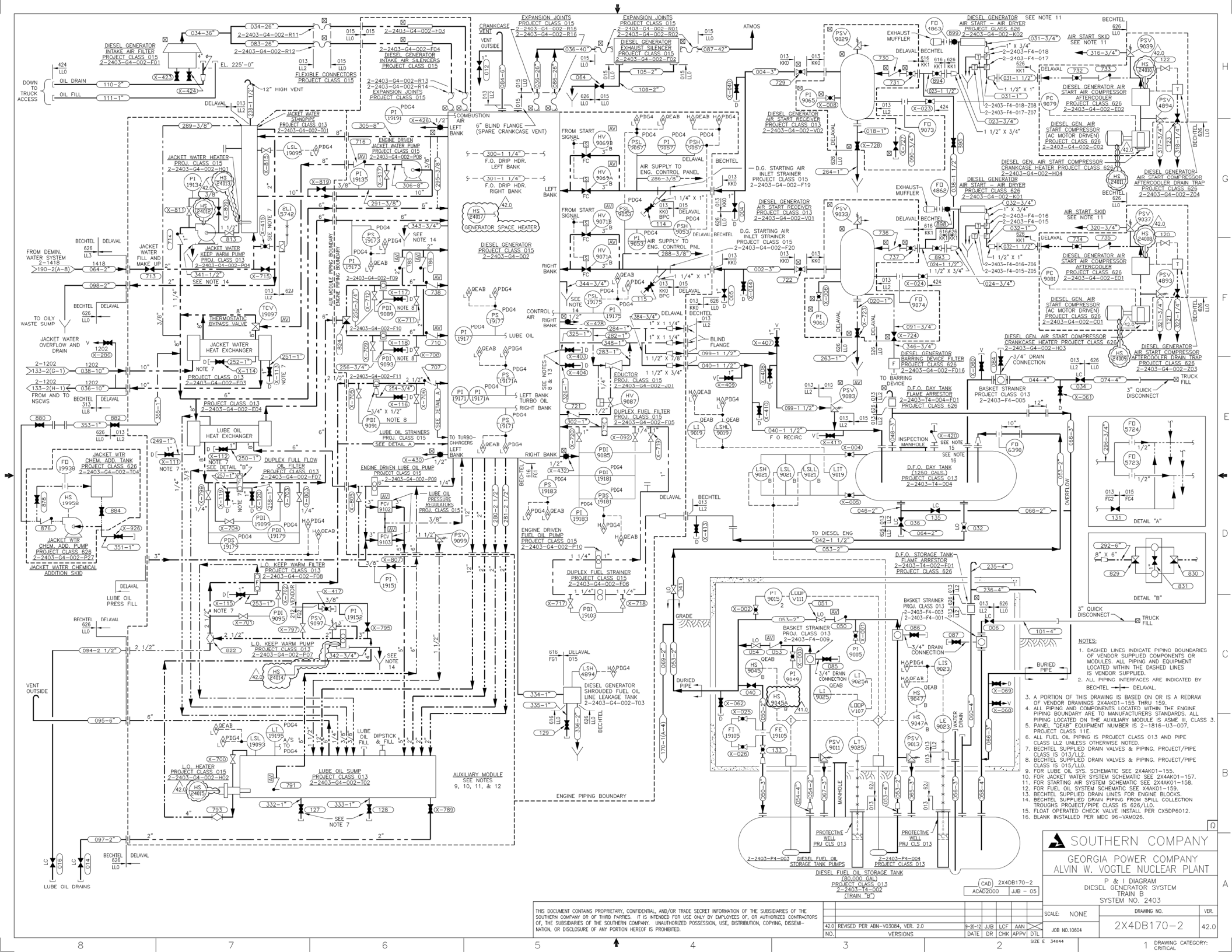
SIZE E 34X44

DRAWING CATEGORY: CRITICAL



NOTES:

1. DASHED LINES INDICATE PIPING BOUNDARIES OF VENDOR SUPPLIED COMPONENTS OR MODULES. ALL PIPING AND EQUIPMENT LOCATED WITHIN THE DASHED LINES IS VENDOR SUPPLIED.
2. ALL PIPING INTERFACES ARE INDICATED BY BECHTEL ⊣ DELAVAL.
3. A PORTION OF THIS DRAWING IS BASED ON OR IS A REDRAW OF VENDOR DRAWINGS 2X4AK01-155 THRU 159.
4. ALL PIPING AND COMPONENTS LOCATED WITHIN THE ENGINE PIPING BOUNDARY ARE TO MANUFACTURERS STANDARDS. ALL PIPING LOCATED ON THE AUXILIARY MODULE IS ASME III, CLASS 3.
5. PANEL LEAK EQUIPMENT NUMBER IS 2-1818-U3-007, PROJECT CLASS 11E.
6. ALL FUEL OIL PIPING IS PROJECT CLASS 013 AND PIPE CLASS LL2 UNLESS OTHERWISE NOTED.
7. BECHTEL SUPPLIED DRAIN VALVES & PIPING. PROJECT/PIPE CLASS IS 013/LL2.
8. BECHTEL SUPPLIED DRAIN VALVES & PIPING. PROJECT/PIPE CLASS IS 015/LL0.
9. FOR LUBE OIL SYS. SCHEMATIC SEE 2X4AK01-155.
10. FOR JACKET WATER SYSTEM SCHEMATIC SEE 2X4AK01-157.
11. FOR STARTING AIR SYSTEM SCHEMATIC SEE 2X4AK01-158.
12. FOR FUEL OIL SYSTEM SCHEMATIC SEE X4AK01-159.
13. BECHTEL SUPPLIED DRAIN LINES FOR ENGINE BLOCKS.
14. BECHTEL SUPPLIED DRAIN PIPING FROM SPILL COLLECTION TROUGHS PROJECT/PIPE CLASS IS 626/LL0.
15. FLOAT OPERATED CHECK VALVE INSTALL PER CX5DP6012.
16. BLANK INSTALLED PER MDC 96-VAM026.

SOUTHERN COMPANY

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
DIESEL GENERATOR SYSTEM
TRAIN B
SYSTEM NO. 2403

CAD 2X4DB170-2
ACAD2000 JJB - 05

NO.	VERSIONS	DATE	DR	CHK	APPV	DTL
42.0	REVISED PER ABN-V03084, VER. 2.0	9-20-12	JJB	LCF	AAN	

SCALE: NONE
DRAWING NO. 2X4DB170-2
VER. 42.0
JOB NO. 10604
SIZE E 34X44
DRAWING CATEGORY: CRITICAL

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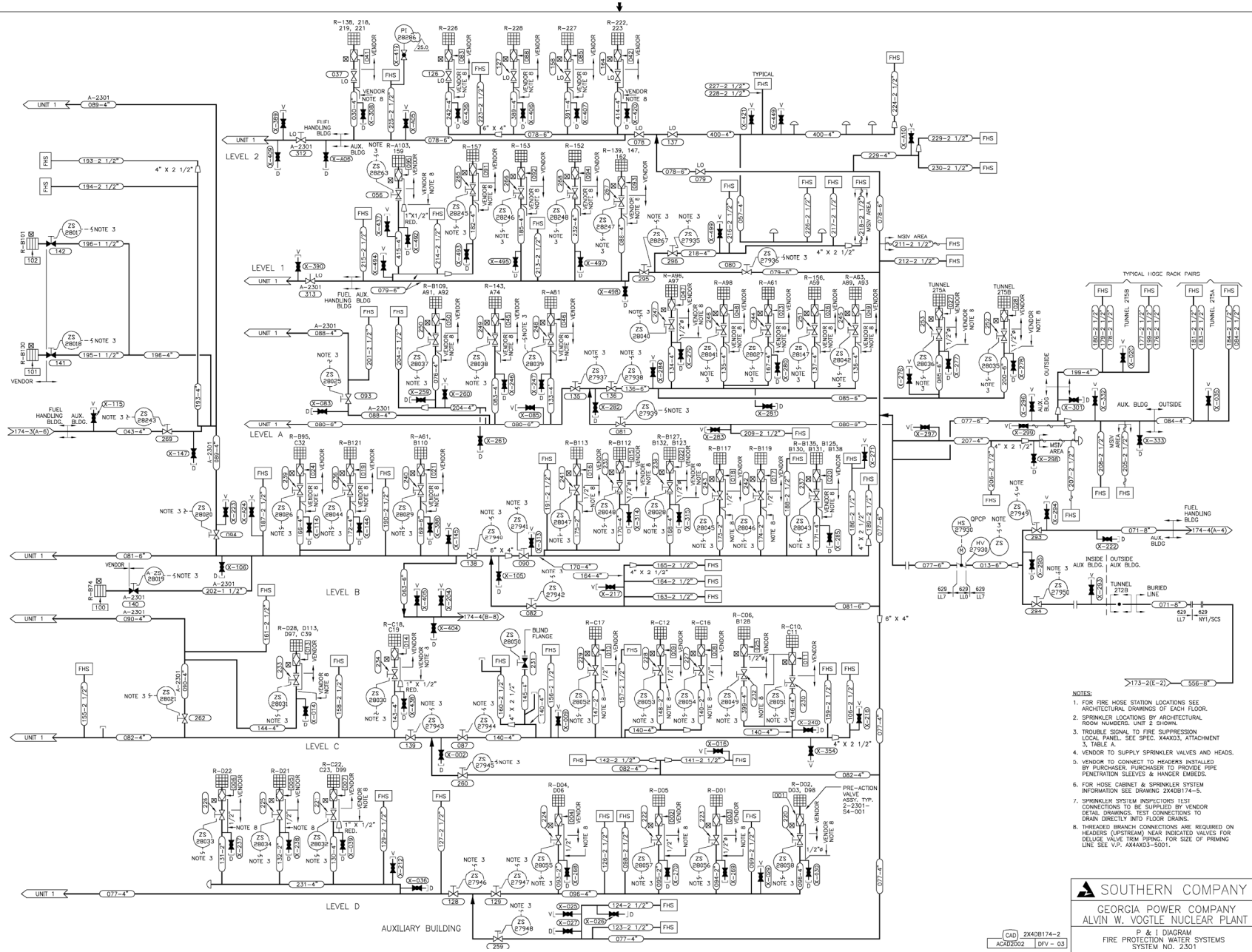
LEVEL 2
LEVEL 1
LEVEL A
LEVEL B
LEVEL C
LEVEL D
AUXILIARY BUILDING
TYPICAL HOSE RACK PAIRS
NOTES:
1. FOR FIRE HOSE STATION LOCATIONS SEE ARCHITECTURAL DRAWINGS OF EACH FLOOR.
2. SPRINKLER LOCATIONS BY ARCHITECTURAL ROOM NUMBERS: UNIT 2 SHOWN.
3. TROUBLE SIGNAL TO FIRE SUPPRESSION LOCAL PANEL. SEE SPEC. X4AX03, ATTACHMENT 3, TABLE A.
4. VENDOR TO SUPPLY SPRINKLER VALVES AND HEADS.
5. VENDOR TO CONNECT TO HEADERS INSTALLED BY PURCHASER. PURCHASER TO PROVIDE PIPE PENETRATION SLEEVES & HANGER EMBEDS.
6. FOR HOSE CABINET & SPRINKLER SYSTEM INFORMATION SEE DRAWING 2X4DB174-5.
7. SPRINKLER SYSTEM INSPECTIONS TEST CONNECTIONS TO BE SUPPLIED BY VENDOR DETAIL DRAWINGS. TEST CONNECTIONS TO DRAIN DIRECTLY INTO FLOOR DRAINS.
8. THREADED BRANCH CONNECTIONS ARE REQUIRED ON HEADERS (UPSTREAM) NEAR INDICATED VALVES FOR DELUGE VALVE TRIM PIPING. FOR SIZE OF PRIMING LINE SEE V.P. AX4AX03-5001.
PRE-ACTION VALVE ASSY. TYP. 2-2301-S4-001
SOUTHERN COMPANY
GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT
P & I DIAGRAM
FIRE PROTECTION WATER SYSTEMS
SYSTEM NO. 2301
CAD 2X4DB174-2
ACAD2002 DFV - 03
SCALE: NONE
DRAWING NO. 2X4DB174-2
VER. 25.0
JOB NO.10604
SIZE E 34X44
DRAWING CATEGORY: CRITICAL
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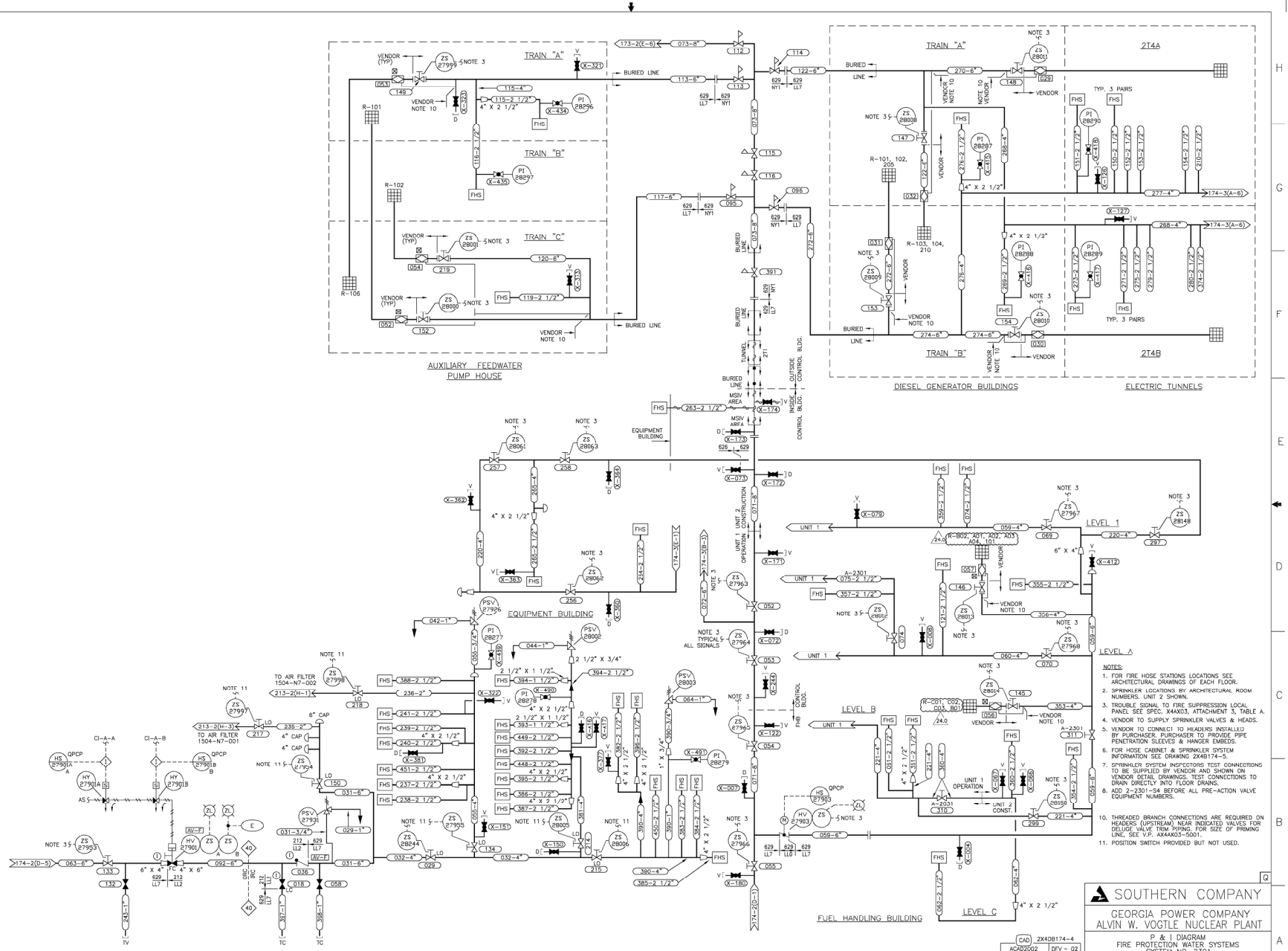
NO. VERSIONS DATE DK CHK APPV DTL
25.0 REVISED PER ABN-V00740, VER. 1.0 5-9-07 DFV WMW WLN X



SOUTHERN COMPANY

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
FIRE PROTECTION WATER SYSTEMS
SYSTEM NO. 2301

CAD 2X4DB174-4
ACAD2002 DFV - 02

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NO.	VERSIONS	DATE	DR	CHK	APPV	DTL
24.0	REVISED PER ABN-V00509, VER. 1.0	10-25-06	DFV	MWD	WLN	X

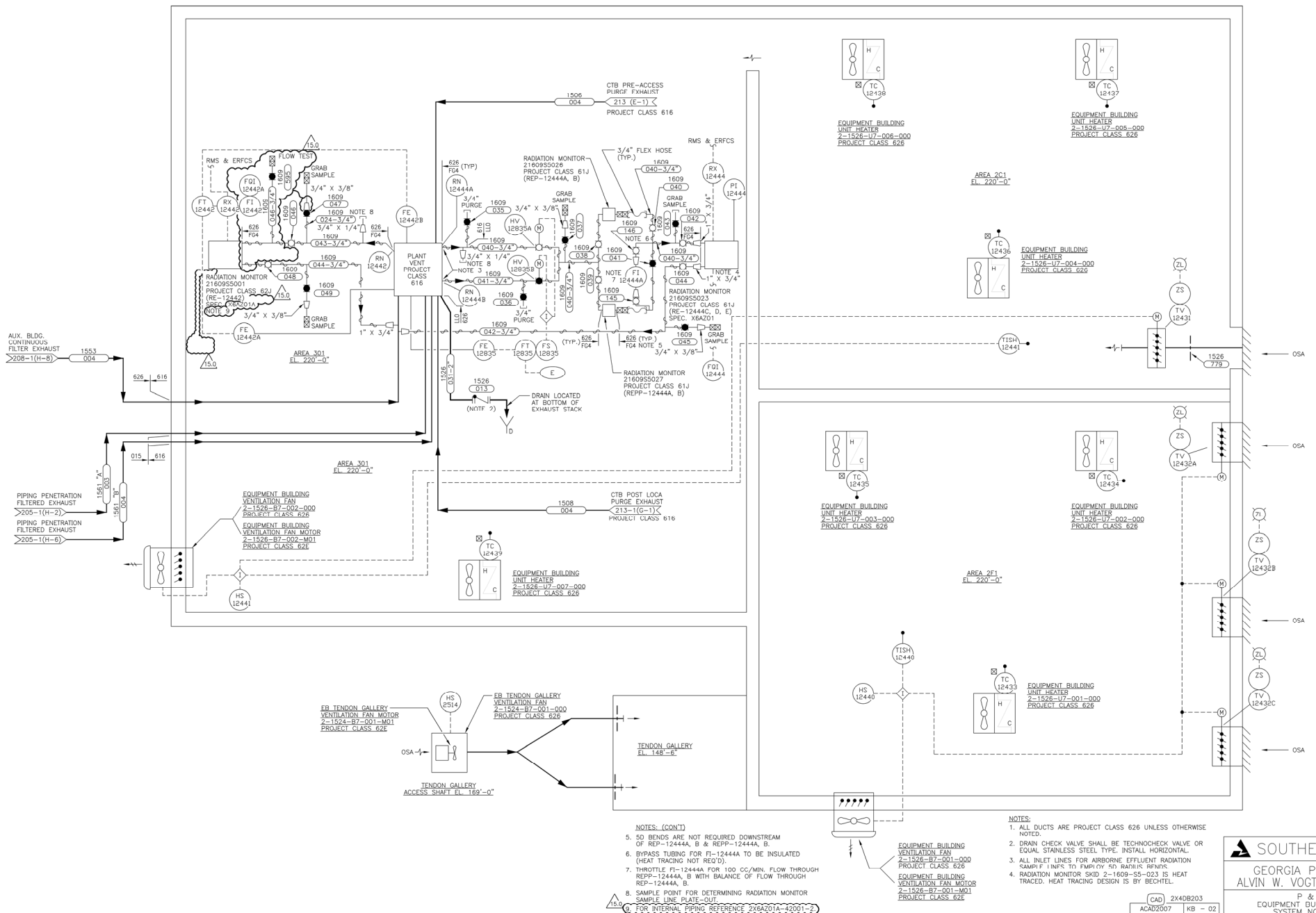
SIZE E 34X44

SCALE: NONE

DRAWING NO. 2X4DB174-4 VER. 24.0

JOB NO.10604

DRAWING CATEGORY: CRITICAL



NOTES: (CON'T)

5. 5D BENDS ARE NOT REQUIRED DOWNSTREAM OF REP-12444A, B & REPP-12444A, B.
6. BYPASS TUBING FOR FI-12444A TO BE INSULATED (HEAT TRACING NOT REQ'D).
7. THROTTLE FI-12444A FOR 100 CC/MIN. FLOW THROUGH REPP-12444A, B WITH BALANCE OF FLOW THROUGH REP-12444A, B.
8. SAMPLE POINT FOR DETERMINING RADIATION MONITOR SAMPLE LINE PLATE-OUT.
15.0 9. FOR INTERNAL PIPING REFERENCE 2X6AZ01A-42001-2.

NOTES:

1. ALL DUCTS ARE PROJECT CLASS 626 UNLESS OTHERWISE NOTED.
2. DRAIN CHECK VALVE SHALL BE TECHNOCHECK VALVE OR EQUAL STAINLESS STEEL TYPE. INSTALL HORIZONTAL.
3. ALL INLET LINES FOR AIRBORNE EFFLUENT RADIATION SAMPLE LINES TO EMPLOY 5D RADIUS BENDS.
4. RADIATION MONITOR SKID 2-1609-S5-023 IS HEAT TRACED. HEAT TRACING DESIGN IS BY BECHTEL.

CAD 2X4DB203
ACAD2007 KB - 02

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NO.	VERSIONS	DATE	DR	CHK	APPV	DTL
15.0	REVISED PER ABN 2080131901J002, VER. 1.0.	03/18/14	KB	RBH	JMR	X

SOUTHERN COMPANY

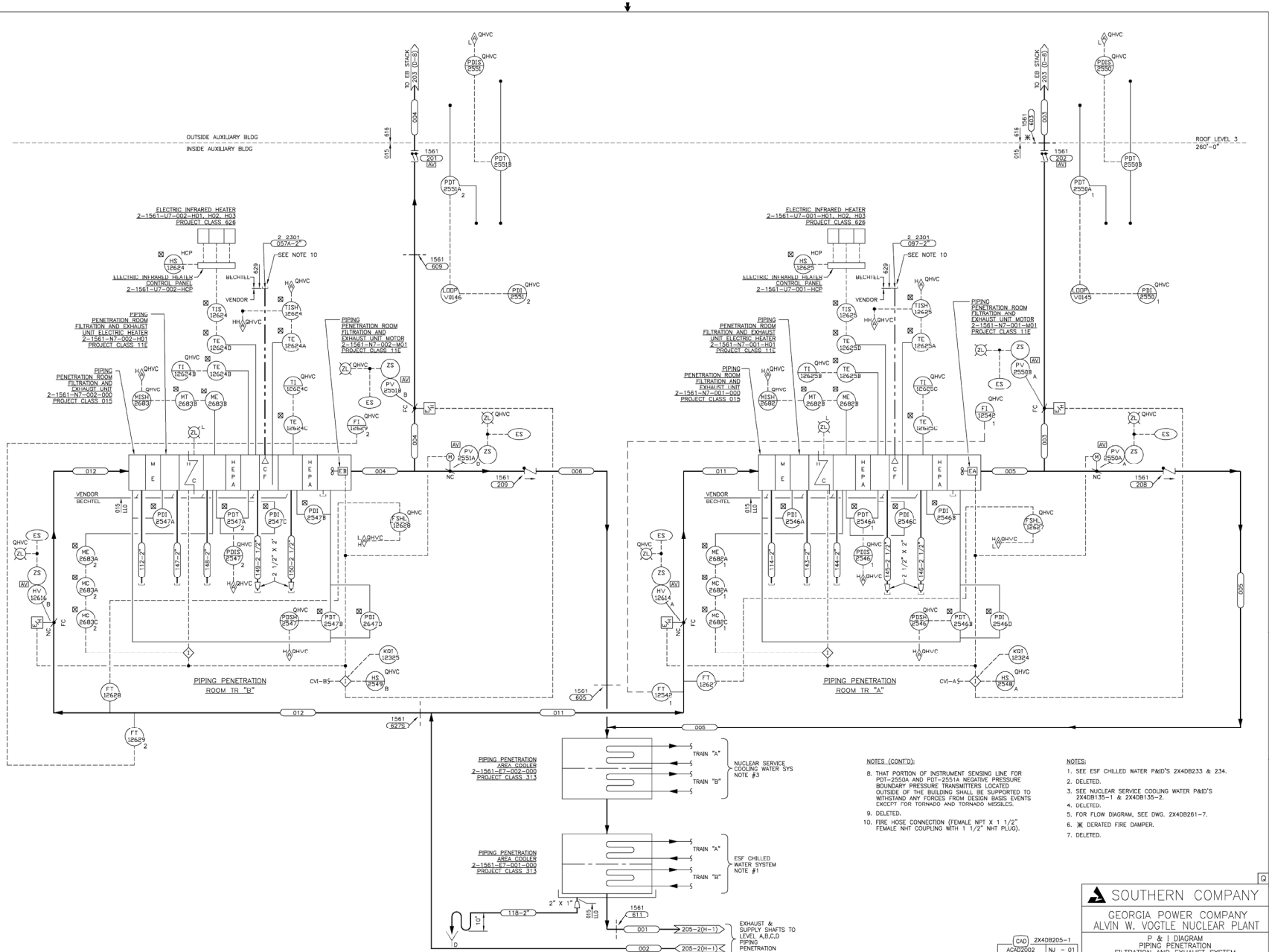
GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
EQUIPMENT BUILDING HVAC SYSTEM
SYSTEM NO. 1526 & 1524

SCALE: NONE	DRAWING NO.	VER.
JOB NO.10604	2X4DB203	15.0

SIZE E 34X44

DRAWING CATEGORY: 1 CRITICAL



NOTES (CONT'D):

8. THAT PORTION OF INSTRUMENT SENSING LINE FOR PDT-2550A AND PDT-2551A NEGATIVE PRESSURE BOUNDARY PRESSURE TRANSMITTERS LOCATED OUTSIDE OF THE BUILDING SHALL BE SUPPORTED TO WITHSTAND ANY FORCES FROM DESIGN BASIS EVENTS EXCEPT FOR TORNADO AND TORNADO MISSILES.
9. DELETED.
10. FIRE HOSE CONNECTION (FEMALE NPT X 1 1/2" FEMALE NHT COUPLING WITH 1 1/2" NHT PLUG).

NOTES:

1. SEE ESF CHILLED WATER P&ID'S 2X4DB233 & 234.
2. DELETED.
3. SEE NUCLEAR SERVICE COOLING WATER P&ID'S 2X4DB135-1 & 2X4DB135-2.
4. DELETED.
5. FOR FLOW DIAGRAM, SEE DWG. 2X4DB261-7.
6. ⋇ DERATED FIRE DAMPER.
7. DELETED.

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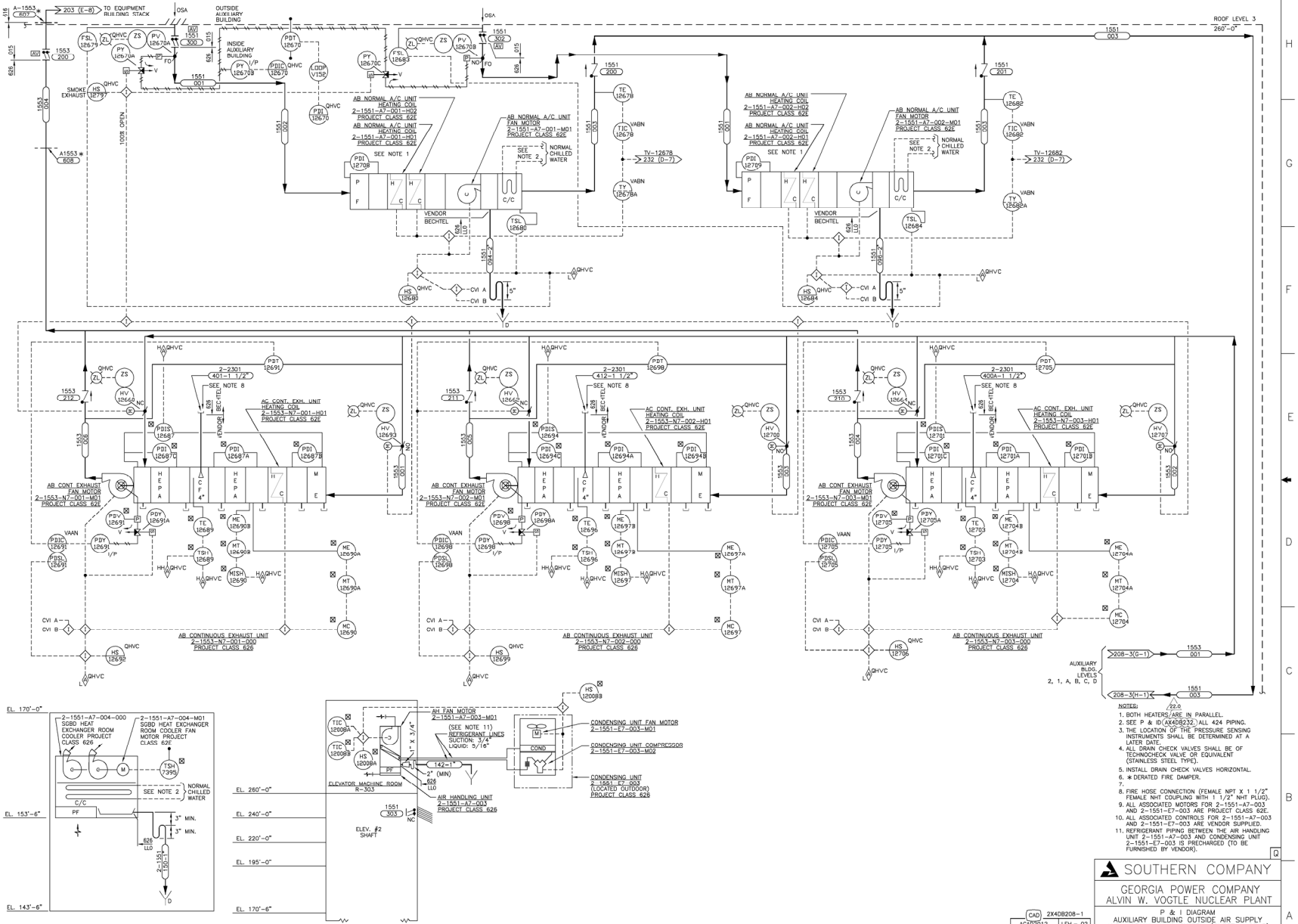
NO.		DATE	DR	CHK	APPV	DTL
19.0	REDRAWN PER ABN-52223, VER.1.0	10-06-05	NJ	LCF	EPD	X

CAD 2X4DB205-1
ACAD2002 NJ - 01

SOUTHERN COMPANY
GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
PIPING PENETRATION
FILTRATION AND EXHAUST SYSTEM
SYSTEM NO. 1561

SCALE: NONE
SIZE E 34X44
DRAWING NO. 2X4DB205-1
VER. 19.0
JOB NO. 10604
DRAWING CATEGORY: CRITICAL



NOTES:

1. BOTH HEATERS ARE IN PARALLEL.
2. SEE P & ID (AX4DB232) ALL 424 PIPING.
3. THE LOCATION OF THE PRESSURE SENSING INSTRUMENTS SHALL BE DETERMINED AT A LATER DATE.
4. ALL DRAIN CHECK VALVES SHALL BE OF TECHNOCHECK VALVE OR EQUIVALENT (STAINLESS STEEL TYPE).
5. INSTALL DRAIN CHECK VALVES HORIZONTAL.
6. * DERATED FIRE DAMPER.
7.
8. FIRE HOSE CONNECTION (FEMALE NPT X 1 1/2" FEMALE NHT COUPLING WITH 1 1/2" NHT PLUG).
9. ALL ASSOCIATED MOTORS FOR 2-1551-A7-003 AND 2-1551-E7-003 ARE PROJECT CLASS 62E.
10. ALL ASSOCIATED CONTROLS FOR 2-1551-A7-003 AND 2-1551-E7-003 ARE VENDOR SUPPLIED.
11. REFRIGERANT PIPING BETWEEN THE AIR HANDLING UNIT 2-1551-A7-003 AND CONDENSING UNIT 2-1551-E7-003 IS PRECHARGED (TO BE FURNISHED BY VENDOR).

SOUTHERN COMPANY

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
AUXILIARY BUILDING OUTSIDE AIR SUPPLY AND NORMAL HVAC SYSTEM (SHEET 1 OF 3)
SYSTEM NO. 1551 & 1553.

CAD 2X4DB208-1
ACAD2012 LEH - 02

SCALE: NONE.
DRAWING NO. 2X4DB208-1
VER. 22.0
JOB NO.10604

NO.		DATE	DR	CHK	APPV	DTL
22.0	REVISED PER ABN-V03987, VER. 1.0	7-1-15	LEH	MWD	WLN	

SIZE E 34844

DRAWING CATEGORY: CRITICAL

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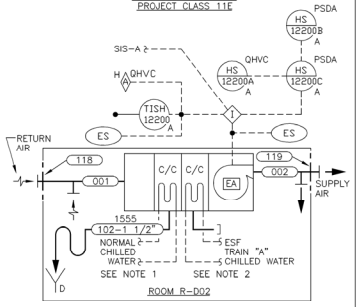
ROOM R-D104

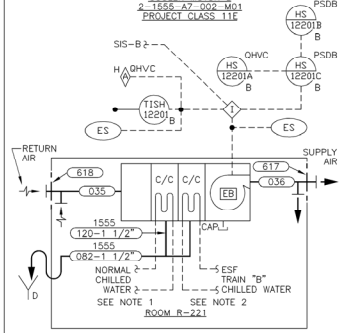
ELECT. SWGR. AND MCC ROOM COOLER MOTOR A 2-1555-A7-001-M01 PROJECT CLASS 11E



ELECT. SWGR. AND MCC ROOM COOLER A EL. 119'-1/4" LEVEL D 2-1555-A7-001-000 PROJECT CLASS 015

ROOM R-Z23

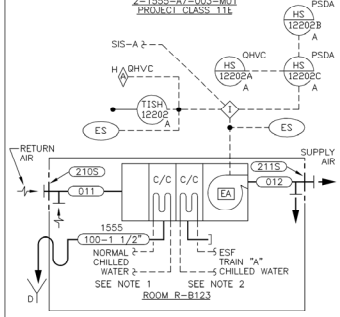
ELECT. SWGR. AND MCC ROOM COOLER MOTOR B 2-1555-A7-002-M01 PROJECT CLASS 11E



ELECT. SWGR. AND MCC ROOM COOLER B EL. 240'-0" LEVEL 2 2-1555-A7-002-000 PROJECT CLASS 015

ROOM R-C07

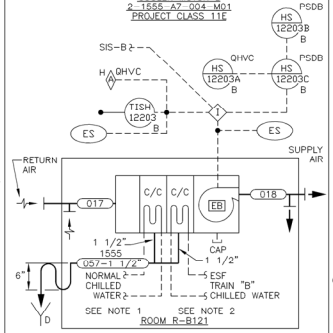
ELECT. SWGR. AND MCC ROOM COOLER MOTOR A 2-1555-A7-003-M01 PROJECT CLASS 11E



ELECT. SWGR. AND MCC ROOM COOLER A EL. 143'-0" LEVEL C 2-1555-A7-003-000 PROJECT CLASS 015

ROOM R-D122

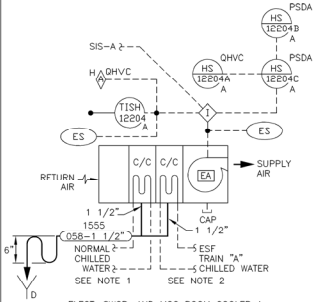
ELECT. SWGR. AND MCC ROOM COOLER MOTOR B 2-1555-A7-004-M01 PROJECT CLASS 11E



ELECT. SWGR. AND MCC ROOM COOLER B EL. 180'-0" LEVEL B 2-1555-A7-004-000 PROJECT CLASS 015

ROOM R-149

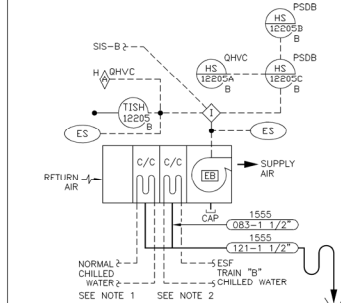
ELECT. SWGR. AND MCC ROOM COOLER MOTOR A 2-1555-A7-005-M01 PROJECT CLASS 11E



ELECT. SWGR. AND MCC ROOM COOLER A EL. 220'-0" LEVEL 1 2-1555-A7-005-000 PROJECT CLASS 015

ROOM R-147

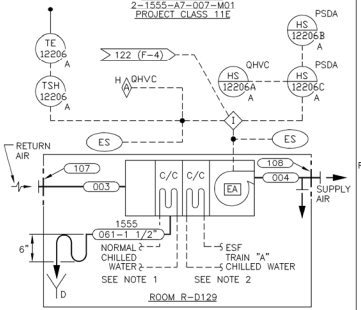
ELECT. SWGR. AND MCC ROOM COOLER MOTOR B 2-1555-A7-006-M01 PROJECT CLASS 11E



ELECT. SWGR. AND MCC ROOM COOLER B EL. 220'-0" LEVEL 1 2-1555-A7-006-000 PROJECT CLASS 015

ROOM R-D22

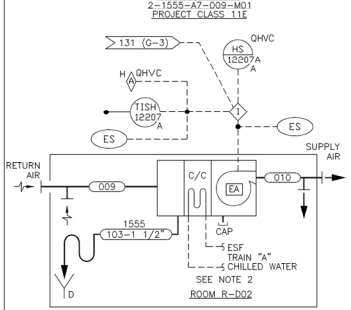
RHR PUMP ROOM COOLER MOTOR A 2-1555-A7-007-M01 PROJECT CLASS 11E



RHR PUMP ROOM COOLER A EL. 119'-1/4" LEVEL D 2-1555-A7-007-000 PROJECT CLASS 015

ROOM R-D05

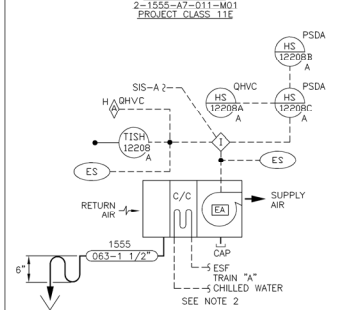
CONTAINMENT SPRAY PUMP ROOM COOLER MOTOR A 2-1555-A7-009-M01 PROJECT CLASS 11E



CONTAINMENT SPRAY PUMP ROOM COOLER A EL. 119'-1/4" LEVEL D 2-1555-A7-009-000 PROJECT CLASS 015

ROOM R-A98

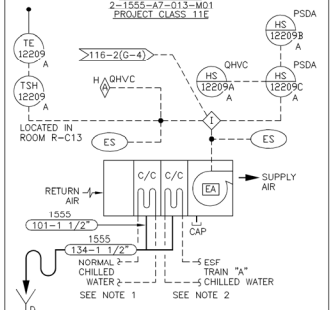
COMPONENT COOLING WATER PUMP ROOM COOLER MOTOR A 2-1555-A7-011-M01 PROJECT CLASS 11E



COMPONENT COOLING WATER PUMP ROOM COOLER A EL. 200'-0" LEVEL A 2-1555-A7-011-000 PROJECT CLASS 015

ROOM R-C16

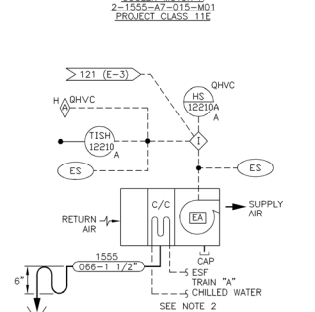
CVCS CHARGING PUMP ROOM COOLER MOTOR A 2-1555-A7-013-M01 PROJECT CLASS 11E



CVCS CHARGING PUMP ROOM COOLER A EL. 143'-0" LEVEL C 2-1555-A7-013-000 PROJECT CLASS 015

ROOM R-B119

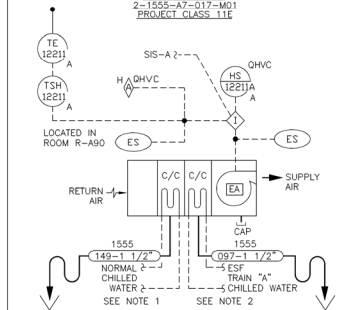
SIS PUMP ROOM COOLER MOTOR A 2-1555-A7-015-M01 PROJECT CLASS 11E



SIS PUMP ROOM COOLER A EL. 160'-0" LEVEL B 2-1555-A7-015-000 PROJECT CLASS 015

ROOM R-A91

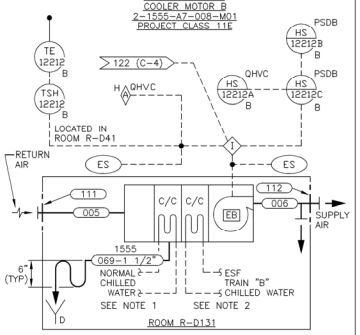
SFP HEAT EXCHANGER AND PUMP ROOM COOLER MOTOR A 2-1555-A7-017-M01 PROJECT CLASS 11E



SFP HEAT EXCHANGER AND PUMP ROOM COOLER A EL. 200'-0" LEVEL A 2-1555-A7-017-000 PROJECT CLASS 015

ROOM R-D21 15.0

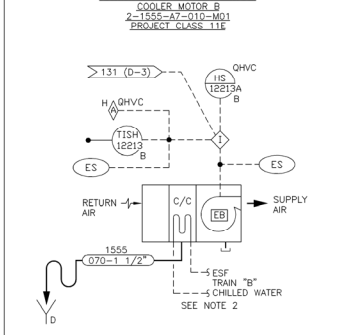
RHR PUMP ROOM COOLER MOTOR B 2-1555-A7-008-M01 PROJECT CLASS 11E



RHR PUMP ROOM COOLER B EL. 119'-1/4" LEVEL D 2-1555-A7-008-000 PROJECT CLASS 015

ROOM R-D04

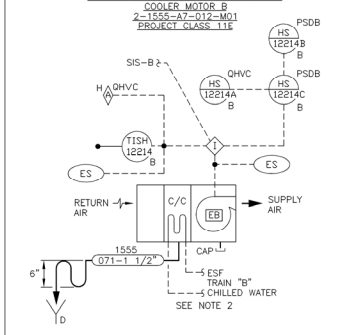
CONTAINMENT SPRAY PUMP ROOM COOLER MOTOR B 2-1555-A7-010-M01 PROJECT CLASS 11E



CONTAINMENT SPRAY PUMP ROOM COOLER B EL. 119'-1/4" LEVEL D 2-1555-A7-010-000 PROJECT CLASS 015

ROOM R-A96

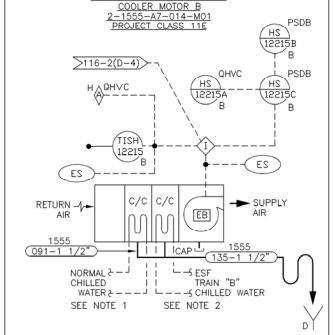
COMPONENT COOLING WATER PUMP ROOM COOLER MOTOR B 2-1555-A7-012-M01 PROJECT CLASS 11E



COMPONENT COOLING WATER PUMP ROOM COOLER B EL. 200'-0" LEVEL A 2-1555-A7-012-000 PROJECT CLASS 015

ROOM R-C17

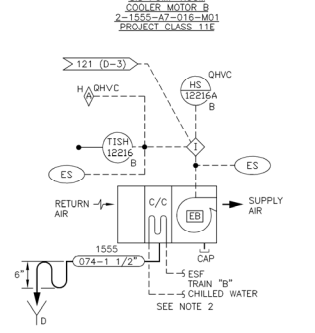
CVCS CHARGING PUMP ROOM COOLER MOTOR B 2-1555-A7-014-M01 PROJECT CLASS 11E



CVCS CHARGING PUMP ROOM COOLER B EL. 143'-0" LEVEL C 2-1555-A7-014-000 PROJECT CLASS 015

ROOM R-B117

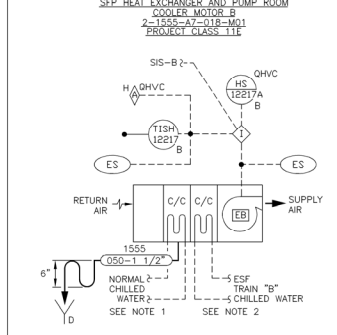
SIS PUMP ROOM COOLER MOTOR B 2-1555-A7-016-M01 PROJECT CLASS 11E



SIS PUMP ROOM COOLER B EL. 180'-0" LEVEL B 2-1555-A7-016-000 PROJECT CLASS 015

ROOM R-A04

SFP HEAT EXCHANGER AND PUMP ROOM COOLER MOTOR B 2-1555-A7-018-M01 PROJECT CLASS 11E



SFP HEAT EXCHANGER AND PUMP ROOM COOLER B EL. 200'-0" LEVEL A 2-1555-A7-018-000 PROJECT CLASS 015

NOTES:
1. SEE P & ID'S AX4DB231 & 232.
2. SEE P & ID'S 2X4DB233 & 234.
3. ALL DUCT IS 015.
4. ALL DRAIN LINES TO BE PROJECT CLASS 626, PIPING CLASS LL0.

CAD 2X4DB228
ACAD2007 JJB - 08

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NO.	VERSIONS	DATE	DR	CHK	APPV	DTL
15.0	REVISED PER ABN-V03256, VER. 1.0	5/30/13	JJB	JWL	RKF	

SOUTHERN COMPANY

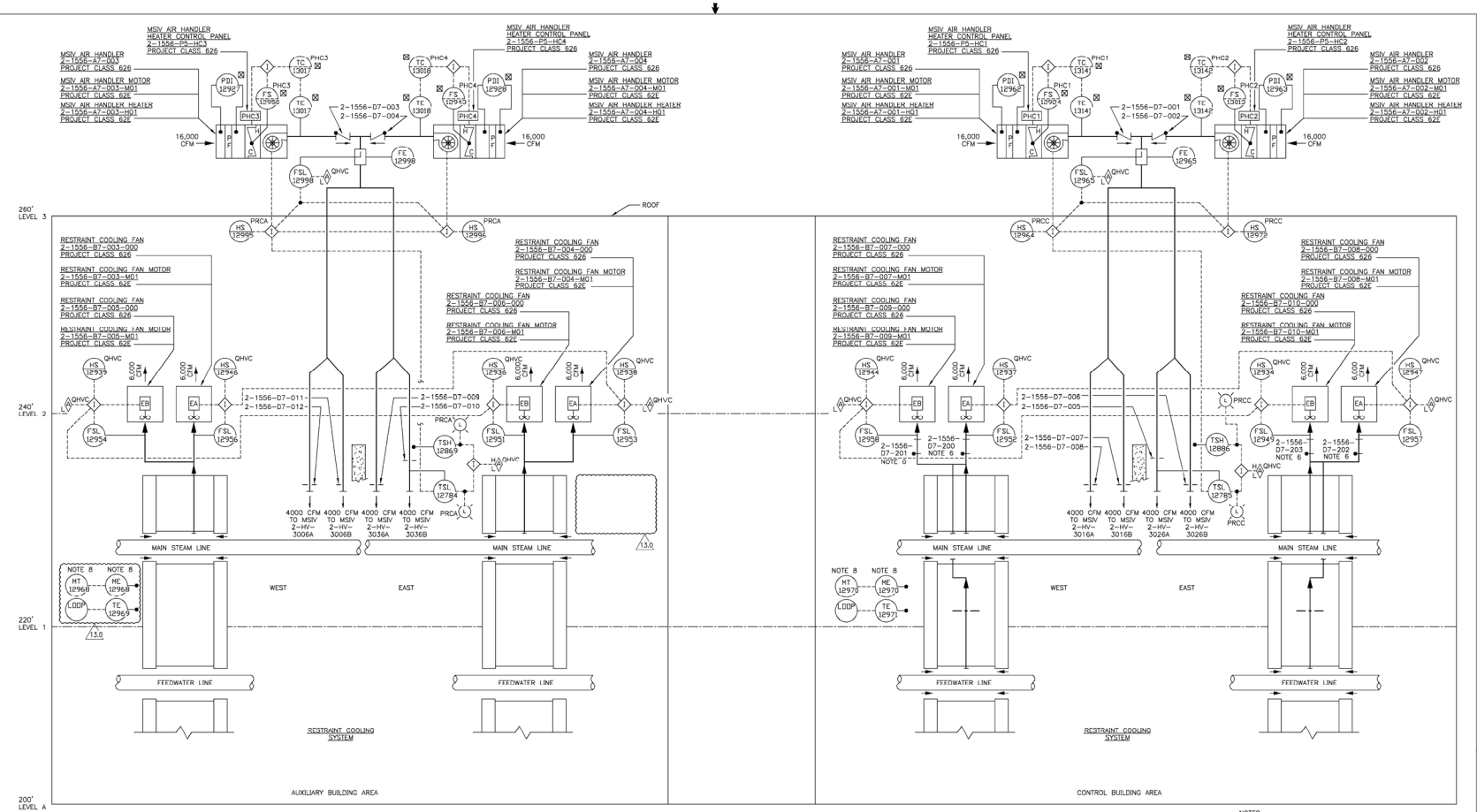
GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
ENGINEERED SAFETY FEATURE
ROOM COOLERS
SYSTEM 1555

SCALE: NONE
JOB NO.10604
DRAWING NO. 2X4DB228
VER. 15.0

SIZE E 34X44

DRAWING CATEGORY: CRITICAL



NOTES:

1. ALL DUCT IS PROJECT CLASS 626 EXCEPT AS NOTED.
2. ALL THE FOUR RESTRAINT COOLING FANS, BOTH IN THE AUXILIARY BUILDING AREA AND THE CONTROL BUILDING AREA, SHALL BE OPERATING TO MAINTAIN CONCRETE TEMPERATURE BELOW 200°F AT THE RESTRAINT AREA.
3. ⊠ FURNISHED WITH ASSOCIATED EQUIPMENT.
4. TEMPERATURE SENSORS TSH12886 (CONTROL BUILDING) AND TSH12869 (AUXILIARY BUILDING) ARE "HIGH TEMP" SENSORS LOCATED IN THE VICINITY OF THE MSIV'S. THESE SENSORS ARE INTENDED TO NOTIFY THE CONTROL ROOM THAT THE TEMPERATURE NEAR THE MSIV ACTUATORS IS APPROACHING THE ENVIRONMENTAL QUALIFICATION UPPER LIMIT FOR NORMAL OPERATION.
5. TEMPERATURE SENSORS TSL12785 (CONTROL BUILDING) AND TSL12784 (AUXILIARY BUILDING) ARE "LOW TEMP" SENSORS LOCATED IN THE AIR HANDLER DISCHARGE AIRSTREAM IN THE VICINITY OF THE MSIV'S. THESE SENSORS ARE INTENDED TO NOTIFY THE CONTROL ROOM THAT THE SUPPLY AIR TEMPERATURE IS APPROACHING THE VALVE ACTUATOR LOW OPERATING TEMPERATURE LIMIT.
6. THE INTERNAL PARTS OF BACK DRAFT DAMPERS 2-1556-D7-200, 201, 202 AND 203 HAVE BEEN REMOVED.
7. TEMPORARY VENTILATION FANS AND DUCTWORK ARE ADDED IN THE MSIV AREA TO PROVIDE LOCALIZED COOLING TO THE MSIV HEAT SENSITIVE COMPONENTS. THIS SYSTEM IS CONTROLLED UNDER THE TEMPORARY MODIFICATION PROCEDURE 00307-C.
8. 2MT-12968, 2ME-12968, 2MT-12970, & 2ME-12970 HAVE BEEN "ABANDONED IN PLACE"

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NO.	VERSIONS	DATE	DR	CHK	APPV	DTL
13.0	REVISED PER ABN-V00917, VER. 2.0.	4-14-08	TJS	SRG	PML	X

CAD 2X4DB245
Acad2000 TJS – 03

SOUTHERN COMPANY

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
PIPING PENETRATION
VENTILATION SYSTEM
SYSTEM NO. 1556

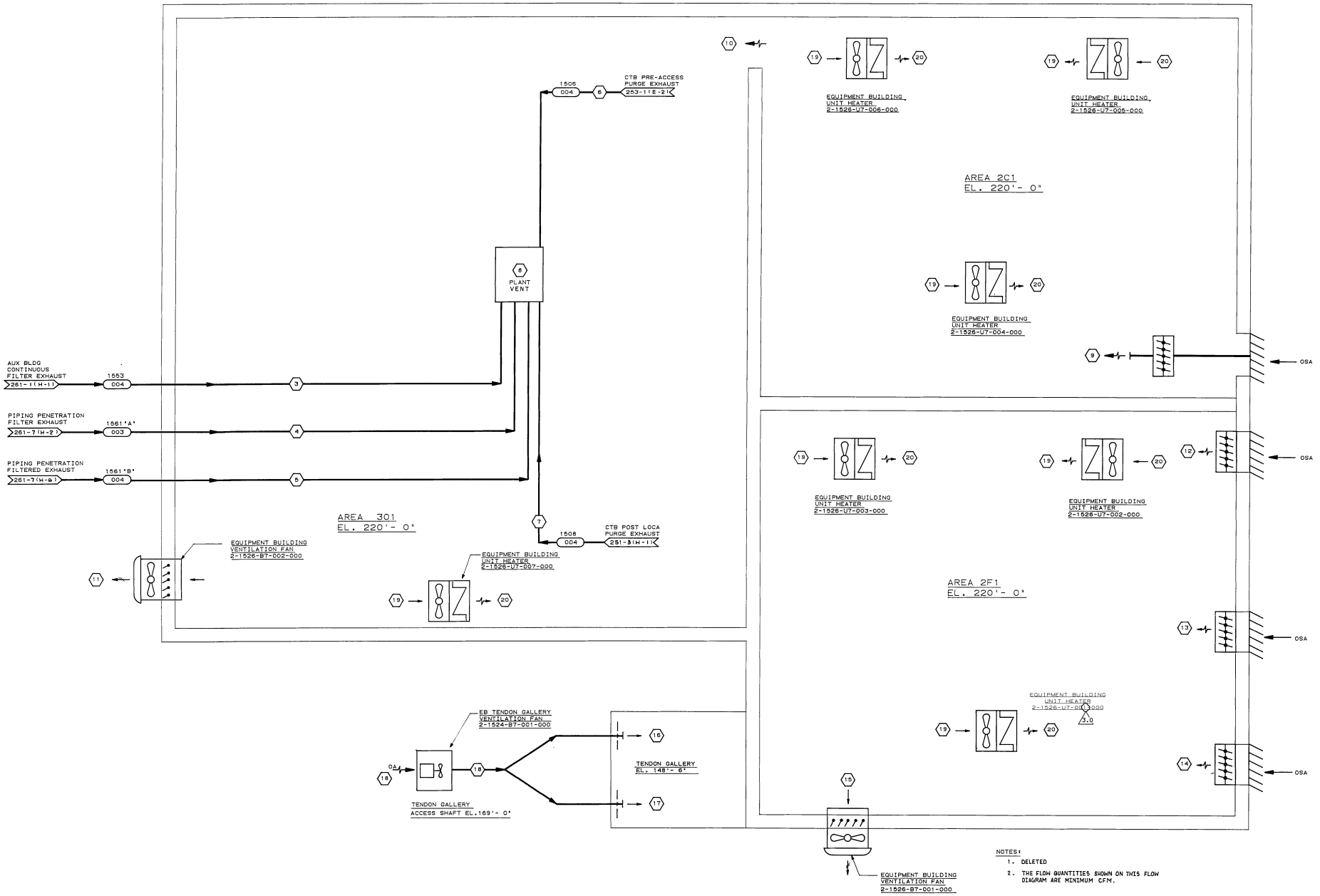
SCALE: NONE

DRAWING NO. 2X4DB245 VER. 13.0

JOB NO. 10604

SIZE E 34X44

DRAWING CATEGORY: CRITICAL



MODE		PARAMETER		⟨1⟩	⟨2⟩	⟨3⟩	⟨4⟩	⟨5⟩	⟨6⟩	⟨7⟩	⟨8⟩	⟨9⟩	⟨10⟩	⟨11⟩	⟨12⟩	⟨13⟩	⟨14⟩	⟨15⟩	⟨16⟩	⟨17⟩	⟨18⟩	⟨19⟩	⟨20⟩
NORMAL	POWER GENERATION	FLOW	CFM	—	—	75395	—	—	5000	—	80395	12250	12250	12,250	450	450	400	1300	2500	2500	5000	500	500
		TEMPERATURE	°F	—	—	104	—	—	120	—	108	98	110	120	98	98	98	120	98	98	98	50	78
		PRESSURE	INCH OF WATER	—	—	0.75	—	—	0.75	—	0.5	—	—	—	—	—	—	-0.5	0.25	0.25	—	—	—
	REFUELING	FLOW	CFM	—	—	75395	—	—	15000	—	90395	12250	12250	12,250	450	450	400	1300	2500	2500	5000	500	500
		TEMPERATURE	°F	—	—	104	—	—	120	—	110	98	110	120	98	98	98	120	98	98	98	50	78
		PRESSURE	INCH OF WATER	—	—	0.75	—	—	0.75	—	0.5	—	—	—	—	—	—	-0.5	0.25	0.25	—	—	—
POST LOCA	WITH OFFSITE POWER	FLOW	CFM	—	—	75395	2700	2700	—	500	75395	12250	12250	12,250	450	450	400	1300	2500	2500	5000	500	500
		TEMPERATURE	°F	—	—	104	126	126	—	120	108	98	110	120	98	98	98	120	98	98	98	50	78
		PRESSURE	INCH OF WATER	—	—	0.75	0.75	0.75	—	0.75	0.5	—	—	—	—	—	—	-0.5	0.25	0.25	—	—	—
	WITHOUT OFFSITE POWER	FLOW	CFM	—	—	—	2700	2700	—	500	2700	—	—	—	—	—	—	—	—	—	—	—	—
		TEMPERATURE	°F	—	—	—	126	126	—	120	126	—	—	—	—	—	—	—	—	—	—	—	—
		PRESSURE	INCH OF WATER	—	—	—	0.75	0.75	—	0.75	0.5	—	—	—	—	—	—	—	—	—	—	—	—

NOTES:
1. DELETED
2. THE FLOW QUANTITIES SHOWN ON THIS FLOW DIAGRAM ARE MINIMUM CFM.

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NO.	VERSIONS	DATE	DR	CHK	APPV
3.0	REVISED PER ABN-81183, VER. 1.0	10-4-05	DFV	WBJ	ASK

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

FLOW DIAGRAM
EQUIPMENT BUILDING HVAC SYSTEM
SYSTEM NO. 1526 & 1524

SCALE: NONE
JOB NO.10604
DRAWING NO. 2X4DB250-1
VER. 3.0
SIZE E 34X44
DRAWING CATEGORY: CRITICAL

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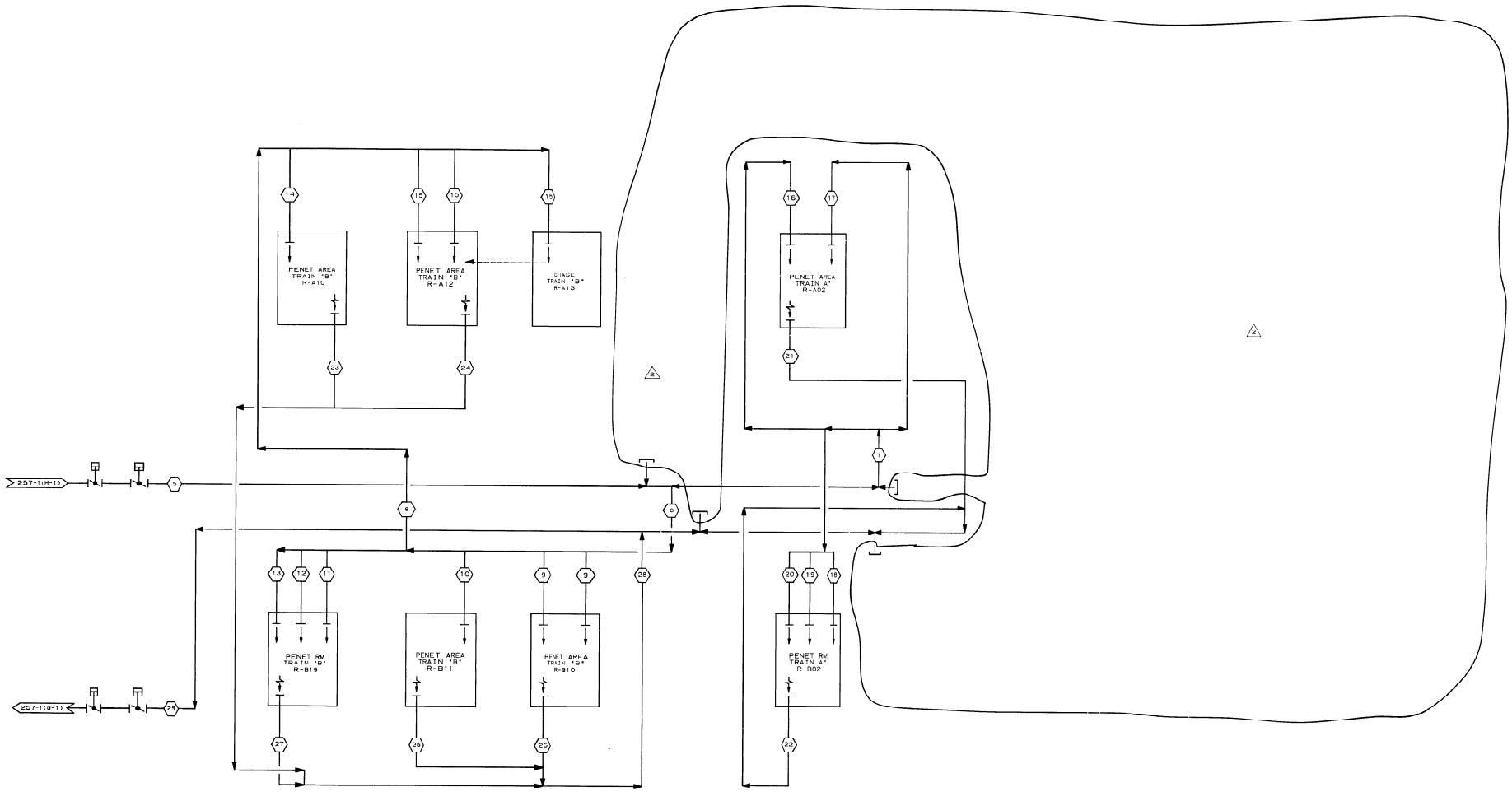


MODE	PARAMETER		⬡	⬡	⬡	⬡	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29
NORMAL	FLOW	CFM					5305	2840	2465	1280	155	85	385	390	390	140	380	605	610	415	420	415	1365	1400	160	1300	100	350	1385	3235	6000
	TEMP	°F					57	57	57	57	59	59	57	59	59	59	59	59	59	59	59	59	83	76	86	89	84	93	82	NA	NA
	PRESSURE	INCH OF WATER																													
POST LOCA	FLOW	CFM																													
	TEMP	°F																													
	PRESSURE	INCH OF WATER												2															2		

NOTES:

1. THE DATA SHOWN ON THIS FLOW DIAGRAM IS FOR INFORMATION PURPOSES ONLY, AND WHILE USEFUL AS GUIDES IN OPERATION, DOES NOT REPRESENT EXACT OR GUARANTEED CONDITIONS.
2. P & ID 2X4DB209.

NO.	REVISIONS	DATE	DR.	CHK.	SUP'V	EGS	OF.E	P.E.	Q.A.E
2	INCORPORATED DCN #2								
1	INCORPORATED DCN #1 & 2								
0	ISSUED FOR CONSTRUCTION								

ENG. MGR. DATE

SEE MICROFILM FOR SIGNATURES

SEE MICROFILM FOR SIGNATURES

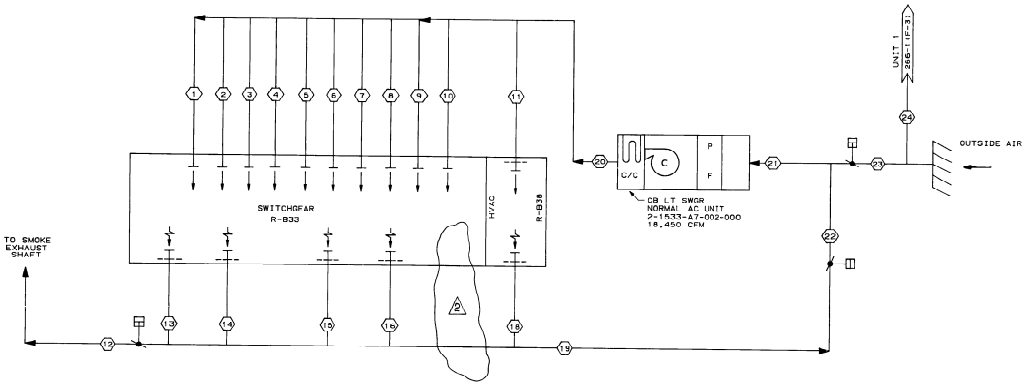
BECHTEL
LOS ANGELES

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

FLOW DIAGRAM
CONTROL BUILDING ELECTRICAL & PIPE PENETRATION ROOM FILTER & EXHAUST SYSTEM
SYSTEM NO. 1562

SCALE: NONE	DRAWING NO.	REV.
JOB NO. 9510	2X4DB259-1	2

SIZE E 34X44



MODE	PARAMETER		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24
SUMMER	FLOW	CFM	1650	1650	1700	1700	1700	1500	1650	1650	1650	1650	1750	0	3710	3710	3710	3720	—	1575	16425	18250	18250	16425	1825	2000
	TEMP	°F	59	59	59	59	59	59	59	59	59	59	59	—	85	85	85	85	—	72	84	57	86	85	98	98
	PRESSURE	IN.OF WATER																	2							
WINTER	FLOW	CFM	1650	1650	1700	1700	1700	1500	1650	1650	1650	1650	1750	0	3710	3710	3710	3720	—	1575	16425	18250	18250	16425	1825	2000
	TEMP	°F	59	69	69	59	59	59	59	59	59	59	59	—	85	85	85	85	—	72	84	57	86	85	17	17
	PRESSURE	IN.OF WATER																								

MODE	PARAMETER		12	22	23	24
SMOKE EXH.	FLOW	CFM	18250	0	18250	20000
	PRESSURE	IN.OF WATER			2	
NORMAL	FLOW	CFM	0	16425	1825	2000
	PRESSURE	IN.OF WATER	—			

NOTES:

1. THE DATA SHOWN ON THIS FLOW DIAGRAM IS FOR INFORMATION PURPOSES ONLY, AND WHILE USEFUL AS GUIDES IN OPERATION, DOES NOT REPRESENT EXACT OR GUARANTEED CONDITIONS.

2. P & ID 2X4DB211.

QA: PLANT VOGTLE CONSTR. MICROFILMING

I CERTIFY THAT THE IMAGE CONTAINED ON THIS FRAME WAS MADE IN THE NORMAL AND REGULAR COURSE OF BUSINESS, ON THE DATE STATED BELOW AND THAT THE IMAGE IS AN ACCURATE REPRODUCTION OF THE RECORD DOCUMENT.

[illegible] 10/15/84
SIGNATURE DATE

[illegible] 10/15/84
CAMERA OPERATOR DATE

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NO.	REVISIONS	DATE	DR.	CHK. SUPV.	EGS	CHF.E	P.E.	Q.A.E.
3	INCORPORATED DCN # 3	10-5-84	C.L.	RJB				
2	INCORPORATED DCN #1 & 2	10-5-84	CAC	SEE MICROFILM FOR SIGNATURES				
1	ISSUED FOR CONSTRUCTION	06-25-84	CAC	SEE MICROFILM FOR SIGNATURES				

ENG. MGR. [illegible] DATE

DATE	DR.	CHK. SUPV.	EGL	EGS	APE	P.E.	Q.A.E.
10-5-84	C.L.	RJB	[illegible]	[illegible]			

BECHTEL
LOS ANGELES

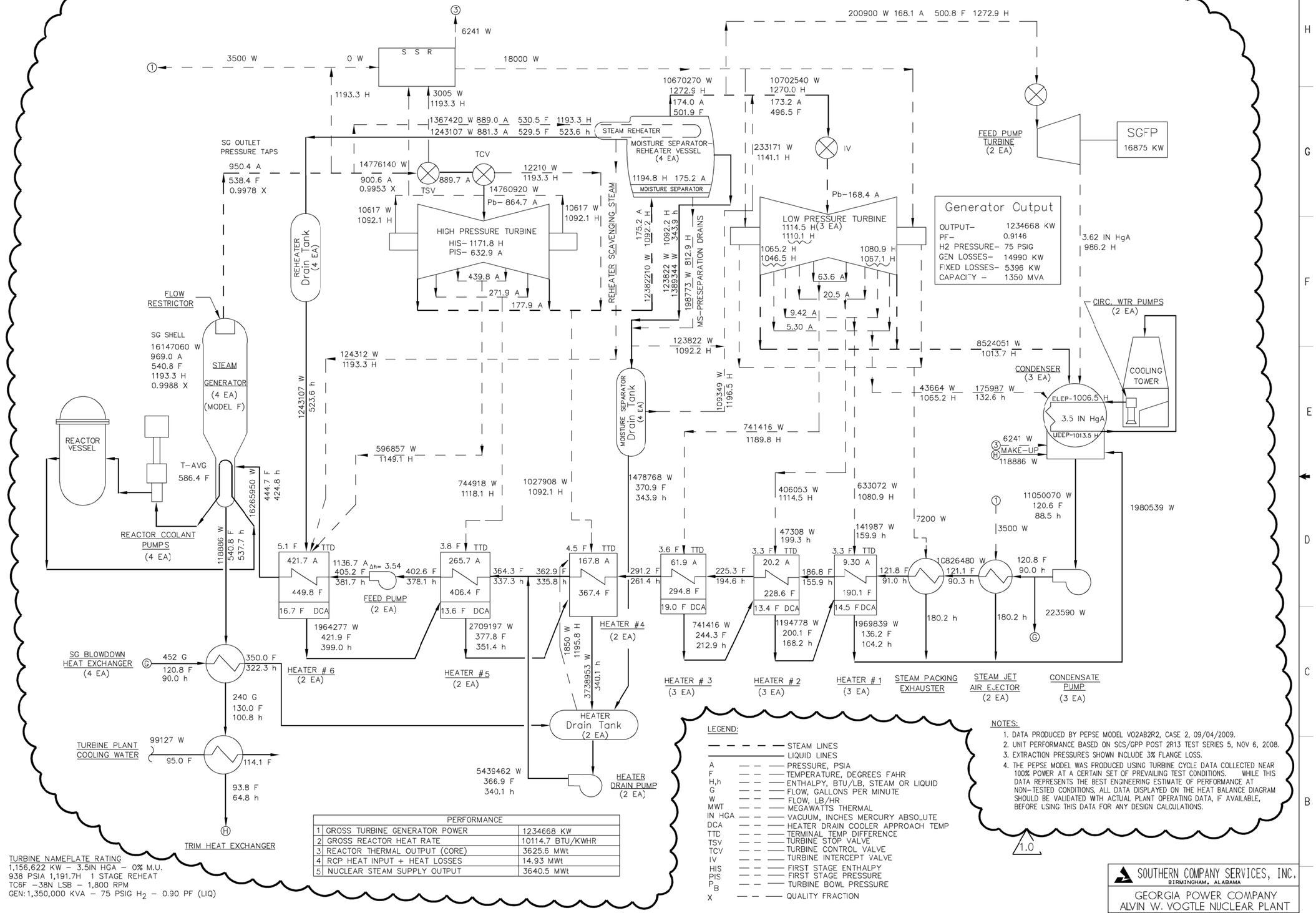
GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

FLOW DIAGRAM
CONTROL BUILDING NORMAL HVAC SYSTEM
LEVEL B
SYSTEM NO. 1533

SCALE: NONE	DRAWING NO.	REV.
JOB NO. 9510	2X4DB266-1	2

"E" SIZE

30X



PERFORMANCE		
1	GROSS TURBINE GENERATOR POWER	1234668 KW
2	GROSS REACTOR HEAT RATE	10114.7 BTU/KWHR
3	REACTOR THERMAL OUTPUT (CORE)	3625.6 MWt
4	RCP HEAT INPUT + HEAT LOSSES	14.93 MWt
5	NUCLEAR STEAM SUPPLY OUTPUT	3640.5 MWt

TURBINE NAMEPLATE RATING
1,156,622 KW - 3.5IN HGA - 0% M.U.
938 PSIA 1,191.7H 1 STAGE REHEAT
TC6F -38N LSB - 1,800 RPM
GEN: 1,350,000 KVA - 75 PSIG H_2 - 0.90 PF (LIQ)

LEGEND:

— — — —	STEAM LINES
————	LIQUID LINES
A	PRESSURE, PSIA
F	TEMPERATURE, DEGREES FAHR
H,h	ENTHALPY, BTU/LB, STEAM OR LIQUID
G	FLOW, GALLONS PER MINUTE
W	FLOW, LB/HR
MWT	MEGAWATTS THERMAL
IN HGA	VACUUM, INCHES MERCURY ABSOLUTE
DCA	HEATER DRAIN COOLER APPROACH TEMP
TTD	TERMINAL TEMP DIFFERENCE
TSV	TURBINE STOP VALVE
TCV	TURBINE CONTROL VALVE
IV	TURBINE INTERCEPT VALVE
HIS	FIRST STAGE ENTHALPY
PIS	FIRST STAGE PRESSURE
P_B	TURBINE BOWL PRESSURE
X	QUALITY FRACTION

NOTES:
1. DATA PRODUCED BY PEPSE MODEL VO2AB2R2, CASE 2, 09/04/2009.
2. UNIT PERFORMANCE BASED ON SCS/GPP POST 2R13 TEST SERIES 5, NOV 6, 2008.
3. EXTRACTION PRESSURES SHOWN INCLUDE 3% FLANGE LOSS.
4. THE PEPSE MODEL WAS PRODUCED USING TURBINE CYCLE DATA COLLECTED NEAR 100% POWER AT A CERTAIN SET OF PREVAILING TEST CONDITIONS. WHILE THIS DATA REPRESENTS THE BEST ENGINEERING ESTIMATE OF PERFORMANCE AT NON-TESTED CONDITIONS, ALL DATA DISPLAYED ON THE HEAT BALANCE DIAGRAM SHOULD BE VALIDATED WITH ACTUAL PLANT OPERATING DATA, IF AVAILABLE, BEFORE USING THIS DATA FOR ANY DESIGN CALCULATIONS.

1.0

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NO.	VERSIONS	DATE	DR	CHK	APPV
1.0	REVISED PER ABN 20538714014005, VER. 1.0.	06/20/13	KB	WLM	JMR

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

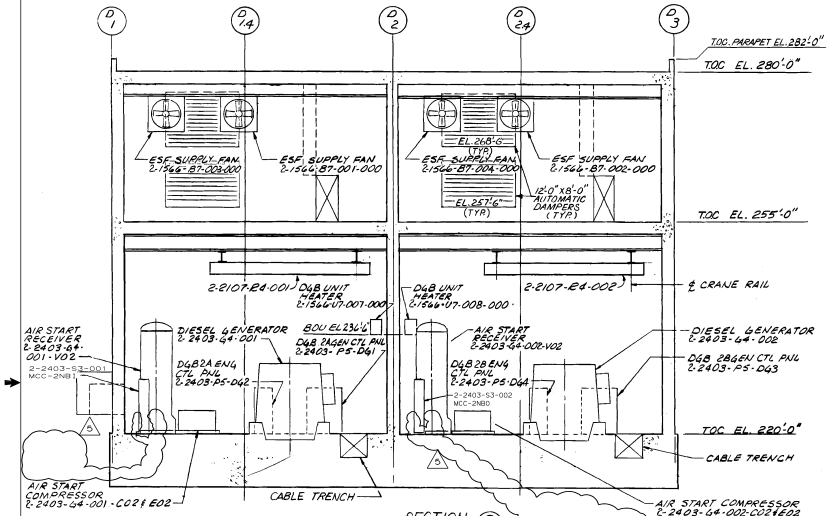
GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

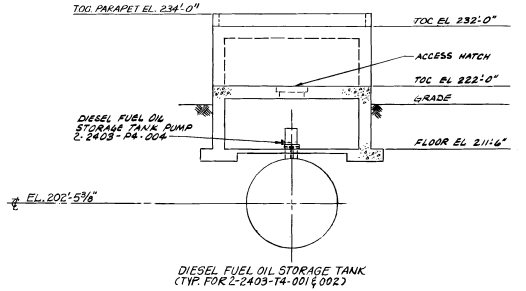
AS BUILT HEAT BALANCE DIAGRAM
(100% REACTOR POWER)

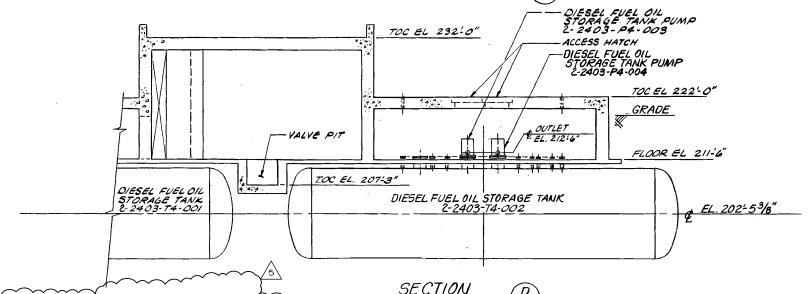
SCALE: NONE	DRAWING NO.	VER.
JOB NO.10604	2X4DC103	1.0

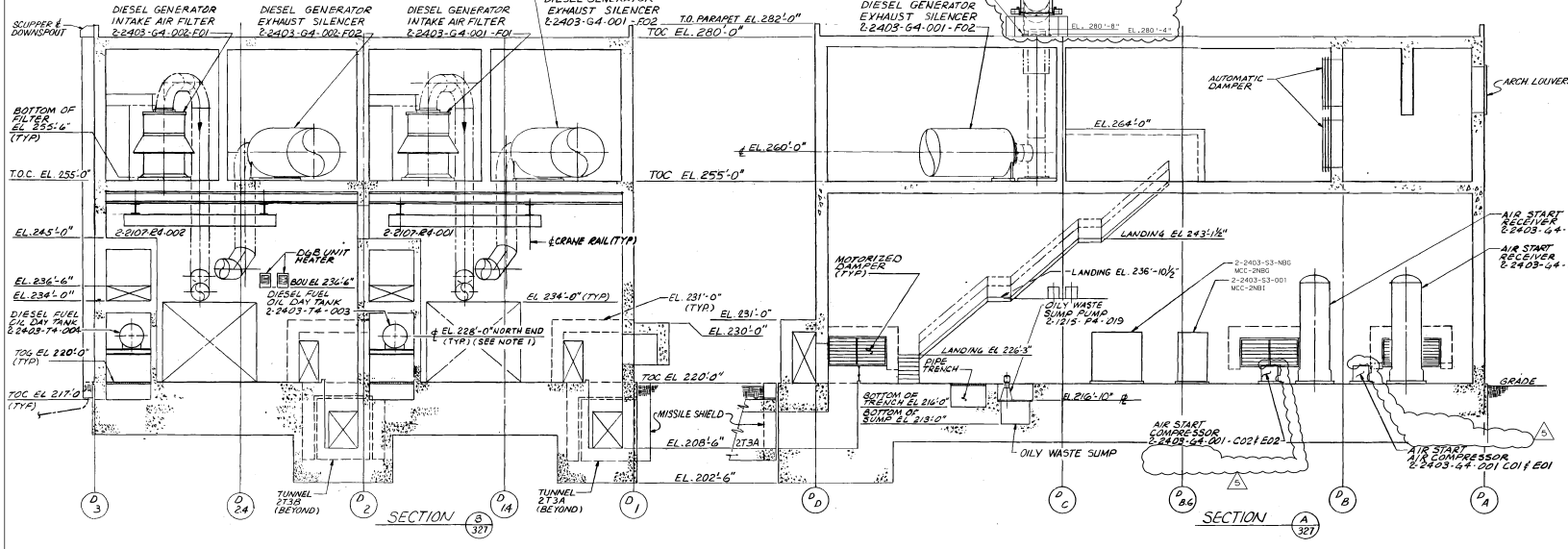
SIZE E 34X44

AUTOCAD NAME: 2X4DC103









NOTE:
1. TANK SLOPES DOWN FROM NORTH END TO SOUTH END.

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

DIESEL GEN. BLDG. UNIT 2
EQUIP. LOCATION LAYOUT SECTIONS

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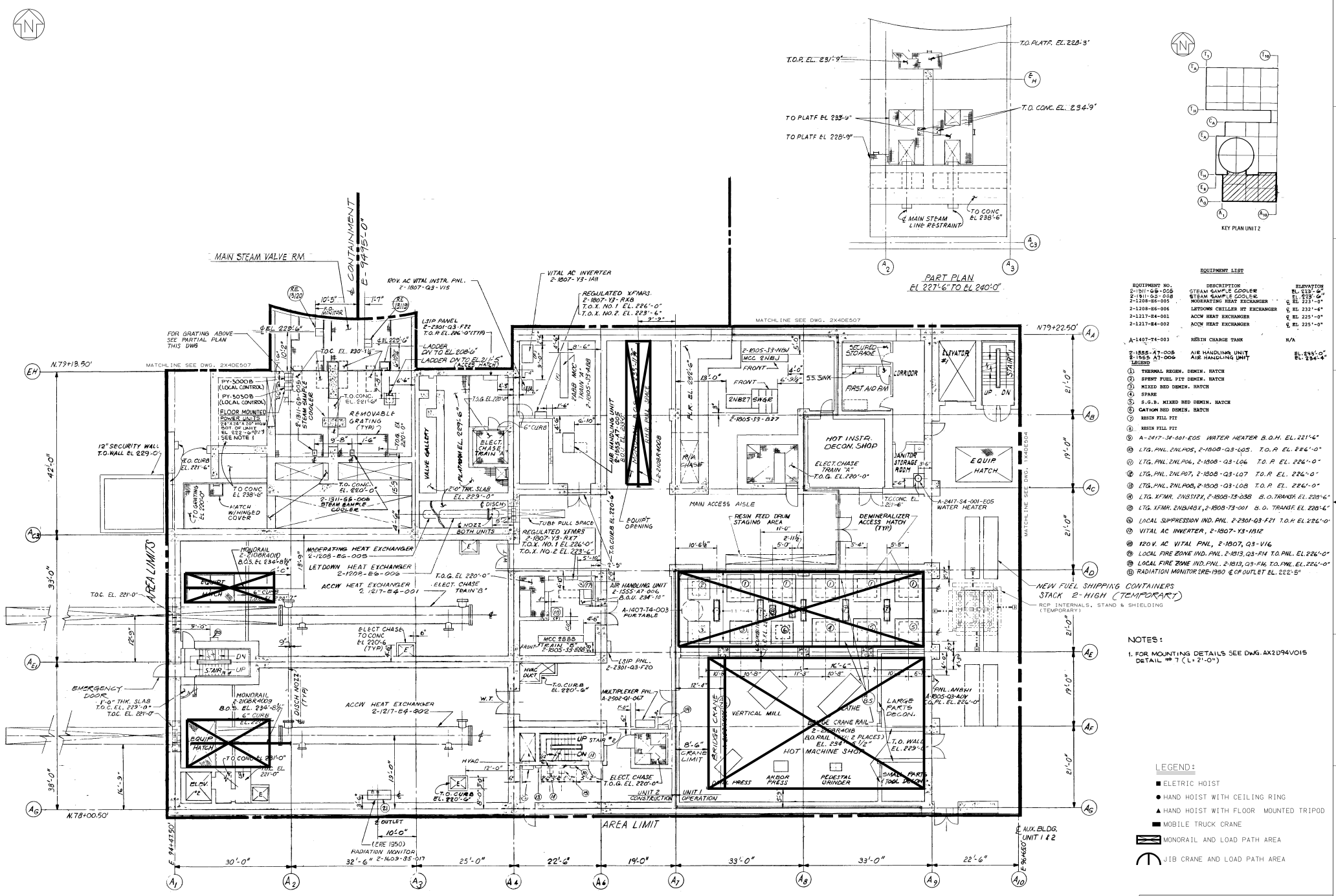
NO.	REVISIONS	DATE	DR	CHK	APPV	DTL
3	INCORPORATED PER DCP 98-V2N0012	12-15-98	JGM	GLB	MWD	
2	AND DCP 95-V2N0071					
1	ISSUED FOR CONSTRUCTION	SEE MICROFILM FOR SIGNATURES				

SCALE: 1/8"=1'-0"
JOB NO. 10604
DRAWING NO. 2X4DE330
REV. 5
SIZE E 34X44
CAD NAME 2X4DE330



EQUIPMENT LIST

EQUIPMENT NO.	DESCRIPTION	ELEVATION
2-1911-G5-005	STEAM SAMPLE COOLER	EL. 223'-6"
2-1911-G5-008	STEAM SAMPLE COOLER	EL. 223'-6"
2-1208-E6-005	MODERATING HEAT EXCHANGER	℄ EL. 223'-0"
2-1208-E6-006	LETDOWN CHILLER HT EXCHANGER	℄ EL. 232'-6"
2-1217-E4-001	ACCW HEAT EXCHANGER	℄ EL. 225'-0"
2-1217-E4-002	ACCW HEAT EXCHANGER	℄ EL. 225'-0"
A-1407-T4-003	RESIN CHARGE TANK	N/A
2-1555-A7-005	AIR HANDLING UNIT	EL. 253'-0"
2-1555-A7-006	AIR HANDLING UNIT	EL. 254'-4"

MARKED

1. THERMAL REGEN. DEMIN. HATCH
2. SPENT FUEL PIT DEMIN. HATCH
3. MIXED BED DEMIN. HATCH
4. SPARE
5. S.G.B. MIXED BED DEMIN. HATCH
6. GATOR BED DEMIN. HATCH
7. RESIN FILL PIT
8. RESIN FILL PIT
9. A-2417-S4-001-E05 WATER HEATER B.O.H. EL. 221'-6"
10. LTG. PNL. 2NLP05, 2-1808-Q3-L05. T.O.P. EL. 226'-0"
11. LTG. PNL. 2NLP06, 2-1808-Q3-L06 T.O.P. EL. 226'-0"
12. LTG. PNL. 2NLP07, 2-1808-Q3-L07 T.O.P. EL. 226'-0"
13. LTG. PNL. 2NLP08, 2-1808-Q3-L08 T.O.P. EL. 226'-0"
14. LTG. XFMR. 2NBS12X, 2-1808-T3-038 B.O. TRANSF. EL. 226'-6"
15. LTG. XFMR. 2NB148X, 2-1808-T3-001 B.O. TRANSF. EL. 226'-6"
16. LOCAL SUPPRESSION IND. PNL. 2-2301-Q3-F21 T.O.P. EL. 226'-0"
17. VITAL AC INVERTER, 2-1807-Y3-1B12
18. 120 V. AC VITAL PNL, 2-1807, Q3-V16
19. LOCAL FIRE ZONE IND. PNL. 2-1813, Q3-F14 T.O. PNL. EL. 226'-0"
20. LOCAL FIRE ZONE IND. PNL. 2-1813, Q3-F16 T.O. PNL. EL. 224'-0"
21. RADIATION MONITOR 2RE-1950 ℄ OF OUTLET EL. 222'-5"

NOTES:

1. FOR MOUNTING DETAILS SEE DWG. AX2D94V015 DETAIL # 7 (L=2'-0")

LEGEND:

- ■ ELETRIC HOIST
- ● HAND HOIST WITH CEILING RING
- ▲ HAND HOIST WITH FLOOR MOUNTED TRIPOD
- ▬ MOBILE TRUCK CRANE
- MONORAIL AND LOAD PATH AREA
- JIB CRANE AND LOAD PATH AREA

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT
HEAVY LOAD HANDLING
AUXILIARY BUILDING LEVEL 1
EL.220'-0" TO 240'-0" (UNIT 2)

SCALE: NONE | DRAWING NO. 2X4DE504 | REV. 0
JOB NO. 10604
SIZE E 34X44
CAD NAME 2X4DE504

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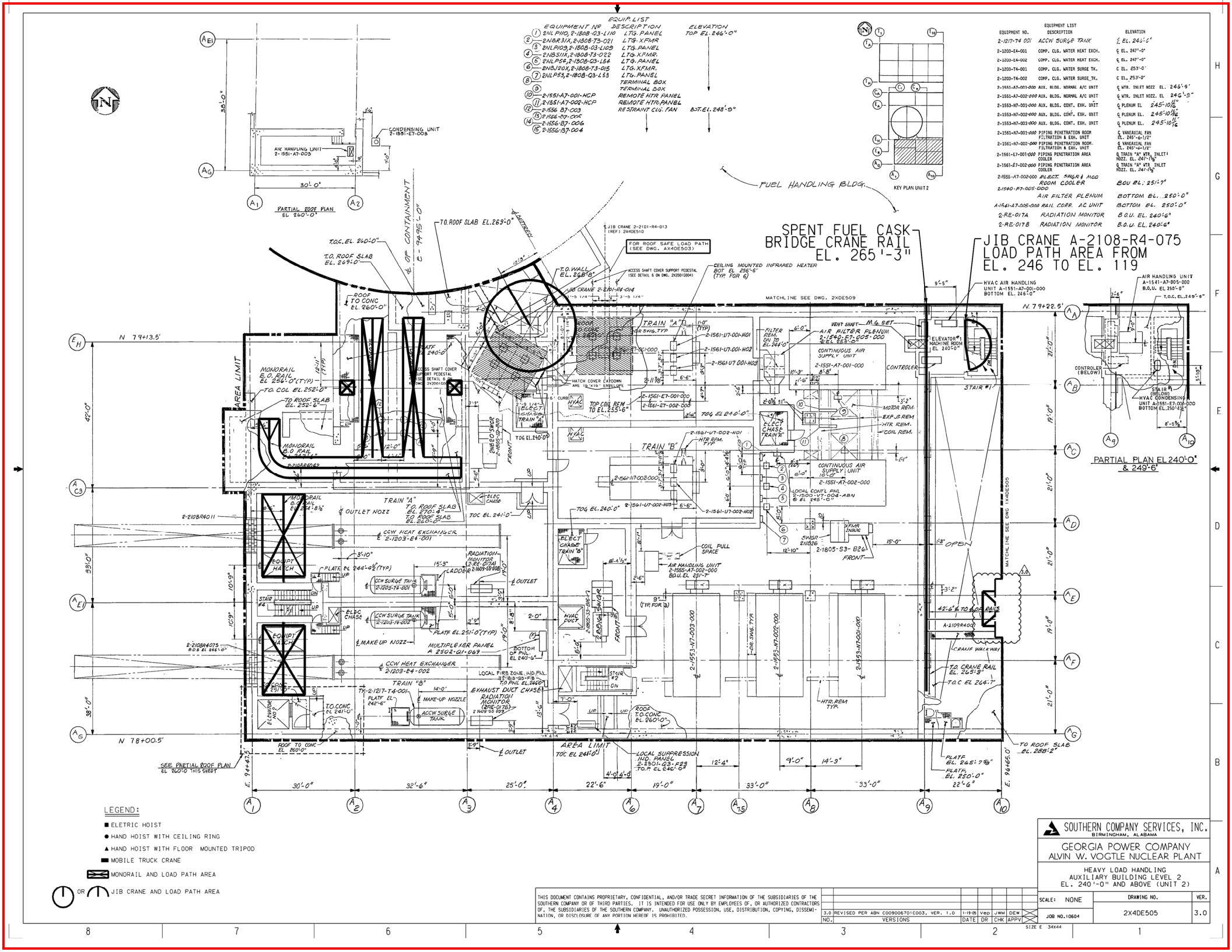
NO.	REVISIONS	DATE	DR	CHK	APPV	DTL
3						
2						
1						
0	ISSUED PER REA 96-VAA054	7-26-96	SRM	RBH	CBH	WFP



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NO.	VERSIONS	DATE	DR	CHK	APPV
3.0	REVISED PER ABN C00900670!C003, VER. 1.0	1-19-05	Vep	JWM	DEW

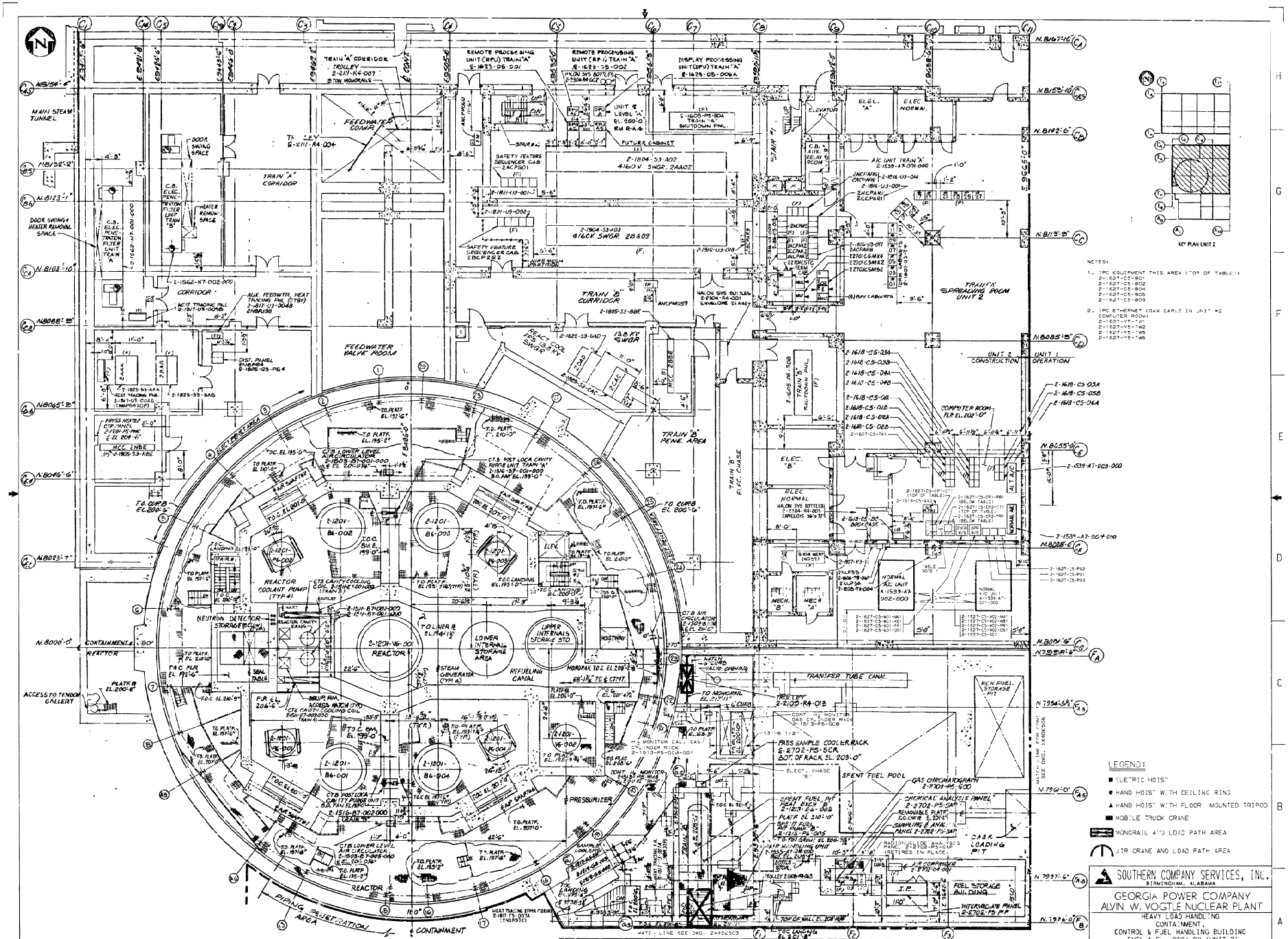
SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

HEAVY LOAD HANDLING
AUXILIARY BUILDING LEVEL 2
EL. 240'-0" AND ABOVE (UNIT 2)

SCALE:	DRAWING NO.	VER.
NONE	2X4DE505	3.0
JOB NO. 10604		

SIZE E 34x44



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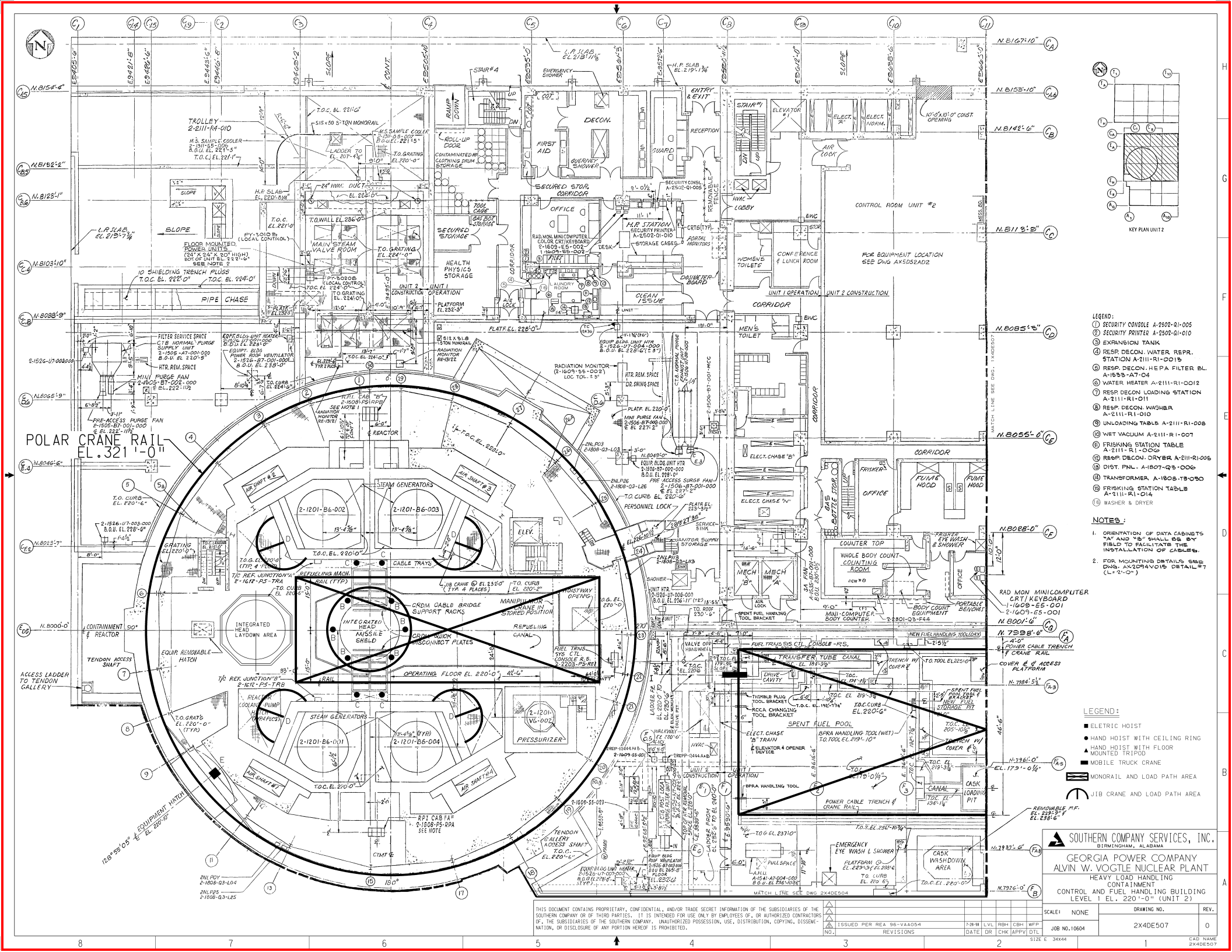
NO.	REVISIONS	DATE	ER	CHK	APPV	D'L
2	ISSUED PER REA 96-VAANO54	7-18-18	LVL	RBH	CBH	AFP

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT
HEAVY LOAD HANDLING CONTAINMENT, CONTROL & FUEL HANDLING BUILDING
LEVEL A EL. 200'-0" (UNIT 2)

SCALE: NONE
DRAWING NO. 2X4DE506
REV. 0
JOB NO. 16604
SIZE E 34X44
CAD NAME 2X4DE506


POLAR CRANE RAIL EL. 321'-0"
REACTOR
STEAM GENERATORS
2-1201-B6-002
2-1201-B6-003
2-1201-B6-001
2-1201-B6-004
PRESSURIZER
2-1201-V6-002
INTEGRATED HEAD MISSILE SHIELD
CONTAINMENT
REACTOR
TENDON ACCESS SHAFT
ACCESS LADDER TO TENDON GALLERY
EQUIP. REMOVABLE HATCH
AIR SHAFT #1
AIR SHAFT #2
AIR SHAFT #3
AIR SHAFT #4
REFUELING CANAL
SPENT FUEL POOL
TRANSFER TUBE CANAL
CASK LOADING PIT
CASK WASHDOWN AREA
CONTROL ROOM UNIT #2
CORRIDOR
DECON.
FIRST AID
GUARD
RECEPTION
ENTRY & EXIT
SECURED STOR. CORRIDOR
SECURED STORAGE
HEALTH PHYSICS STORAGE
MAIN STEAM VALVE ROOM
PIPE CHASE
OFFICE
CONFERENCE & LUNCH ROOM
WOMENS TOILETS
MENS TOILET
LOBBY
AIR LOCK
CLEAN ISSUE
DOSIMETER BOARD
COUNTER TOP
WHOLE BODY COUNT COUNTING ROOM
FRISKER & SHOWER
FUME HOOD
OFFICE
MECH "A"
MECH "B"
ELECT. CHASE "A"
ELECT. CHASE "B"
ELECT. CHASE "N"
GAS BOTTLE STOR.
SPENT FUEL HANDLING TOOL BRACKET
RCCA CHANGING TOOL BRACKET
BPRA HANDLING TOOL
TROLLEY 2-2111-R4-010
M.S. SAMPLE COOLER 2-1911-S5-005
FLOOR MOUNTED POWER UNITS (24" X 24" X 20" HIGH) SEE NOTE 2
10 SHIELDING TRENCH PLUGS
PRE-ACCESS PURGE FAN 2-1505-B7-001-000
MINI PURGE FAN 2-1505-B7-002-000
FILTER SERVICE SPACE C18 NORMAL PURGE SUPPLY UNIT 2-1505-A7-001-000
EQUIP. BLDG. UNIT HEATER 2-1526-B7-001-000
EQUIP. BLDG. POWER ROOF VENTILATOR 2-1526-B7-001-000
RADIATION MONITOR (2-1609-S5-002) LOC TOL. ± 3"
RAD MON MINICOMPUTER CRT/KEYBOARD 2-1609-E5-002 1-1609-S5-003
SECURITY CONSOL A-2502-Q1-005
SECURITY PRINTER A-2502-Q1-010
H.R STATION SECURITY PRINTER-A-2502-Q1-010
CRT'S (TYP.)
DESK
STORAGE CASES
PORTAL MONITORS
REMOVABLE FENCE
10'-0"x10'-0" CONST. OPENING
POE EQUIPMENT LOCATION SEE DWG AX5D52A02
KEY PLAN UNIT 2
MATCH LINE SEE DWG. 1X4DE507
MATCH LINE SEE DWG. 2X4DE504
UNIT 1 OPERATION
UNIT 2 CONSTRUCTION
REFUELING MACH. RAIL (TYP.)
CRDM CABLE BRIDGE SUPPORT RACKS
CRDM QUICK DISCONNECT PLATES
JIB CRANE @ EL 235'0" (TYP. 4 PLACES)
MANIPULATOR CRANE IN STORED POSITION
WIDE WAY OPENING
CABLE TRAYS
FUEL TRANSFER SYS CTL CONSOLE R.S. 2-2203-P5-RE1
FUEL TRANSFER SYS CTL CONSOLE-P.S.
VALVE OPR HANDWHEEL
TROLLEY
DRIVE CAVITY
THIMBLE PLUG TOOL BRACKET
ELECT. CHASE "B" TRAIN
ELEVATOR & OPENER DEVICE
WALKWAY EL 220'-0"
OPERATING FLOOR EL 220'-0"
TENDON GALLERY ACCESS SHAFT
EQUIP. HATCH
RPI CAB "A" 2-1608-P5-RPA SEE NOTE
RPI CAB "B" 2-1608-P5-RPB
RADIATION MONITOR RE-13122
RADIATION MONITOR RE-13121
EQUIP BLDG UNIT HTR 2-1526-U7-004-000
EQUIP BLDG UNIT HTR 2-1526-U7-003-000
EQUIP BLDG UNIT HTR 2-1526-U7-002-000
EQUIP BLDG UNIT HTR 2-1526-U7-001-000
EQUIP BLDG ROOF VENTILATOR 2-1526-B7-001-000
EQUIP BLDG ROOF VENTILATOR 2-1526-B7-002-000
FUEL HANDLING MACHINE A-1541-A7-004-000
PRE ACCESS SURGE FAN 2-1506-B7-001-000
SA FAN A-1535-B7-001-000
2NLP03 2-1808-Q3-L03
2NLP26 2-1808-Q3-L26
2NLP19 2-1808-Q3-L13
2NLP09 2-1808-Q3-L04
2NLP25 2-1608-Q3-L25
2-1609-S5-003
2-1609-E5-001
2-2301-Q3-F4A
2-1526-U7-003-000
2-1526-U7-002-000
2-1526-U7-004-000
2-1526-U7-001-000
2-1505-A7-001-000
2-1505-B7-001-000
2-1505-B7-002-000
2-1506-B7-001-000
2-1506-B7-002-000
2-1506-B7-001-MCC
2-1608-P5-TRA
2-1608-P5-TRB
2-1608-P5-RPA
2-1608-P5-RPB
2-1608-Q3-L03
2-1608-Q3-L04
2-1608-Q3-L25
2-1608-Q3-L26
2-1609-E5-001
2-1609-E5-002
2-1609-S5-002
2-1609-S5-003
2-2203-P5-RE1
2-2301-Q3-F4A
A-1535-B7-001-000
A-1541-A7-004-000
A-2502-Q1-005
A-2502-Q1-010
A-2503-Q1-002
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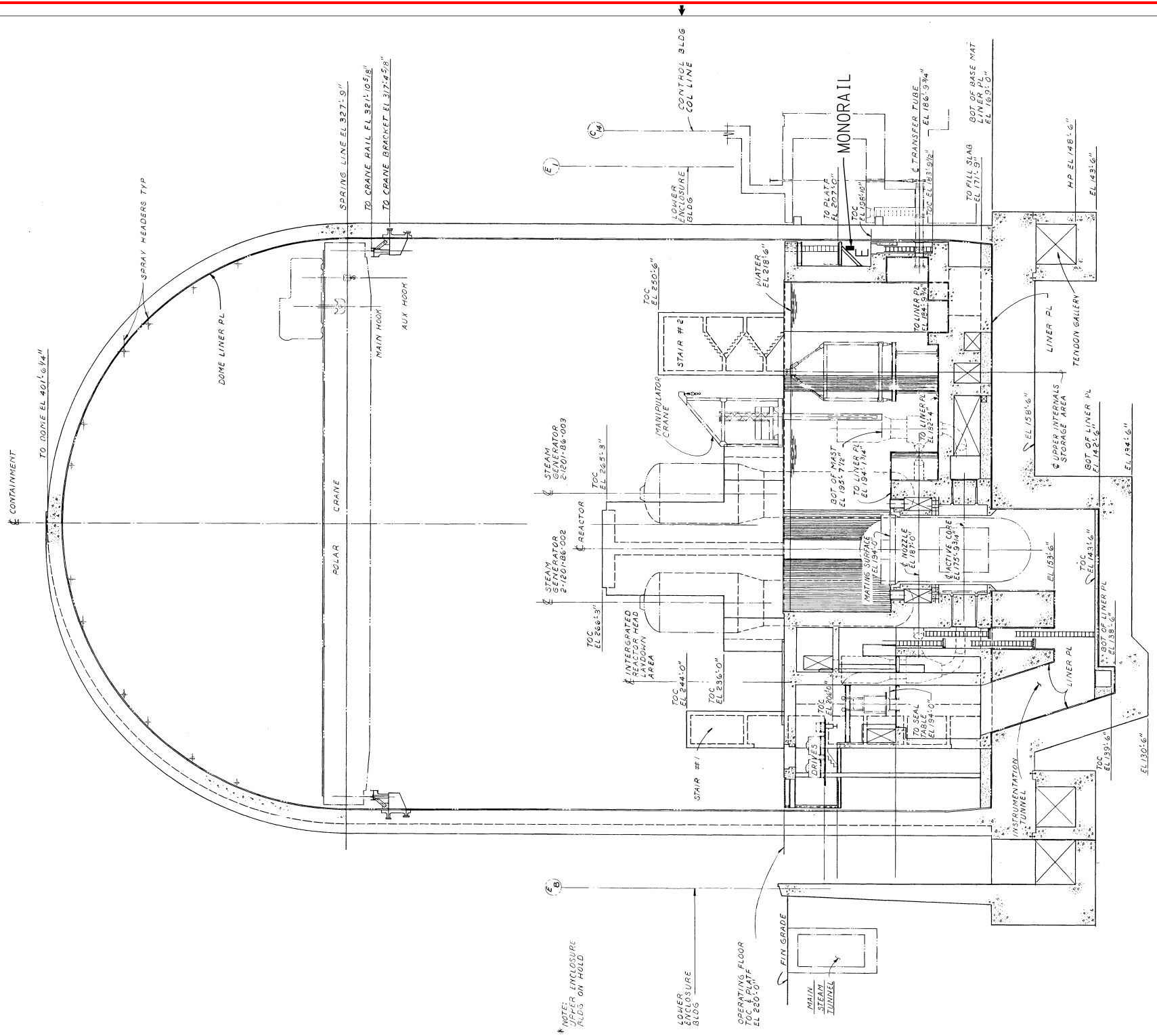
A-2503-Q1-040
A-2503-Q1-041



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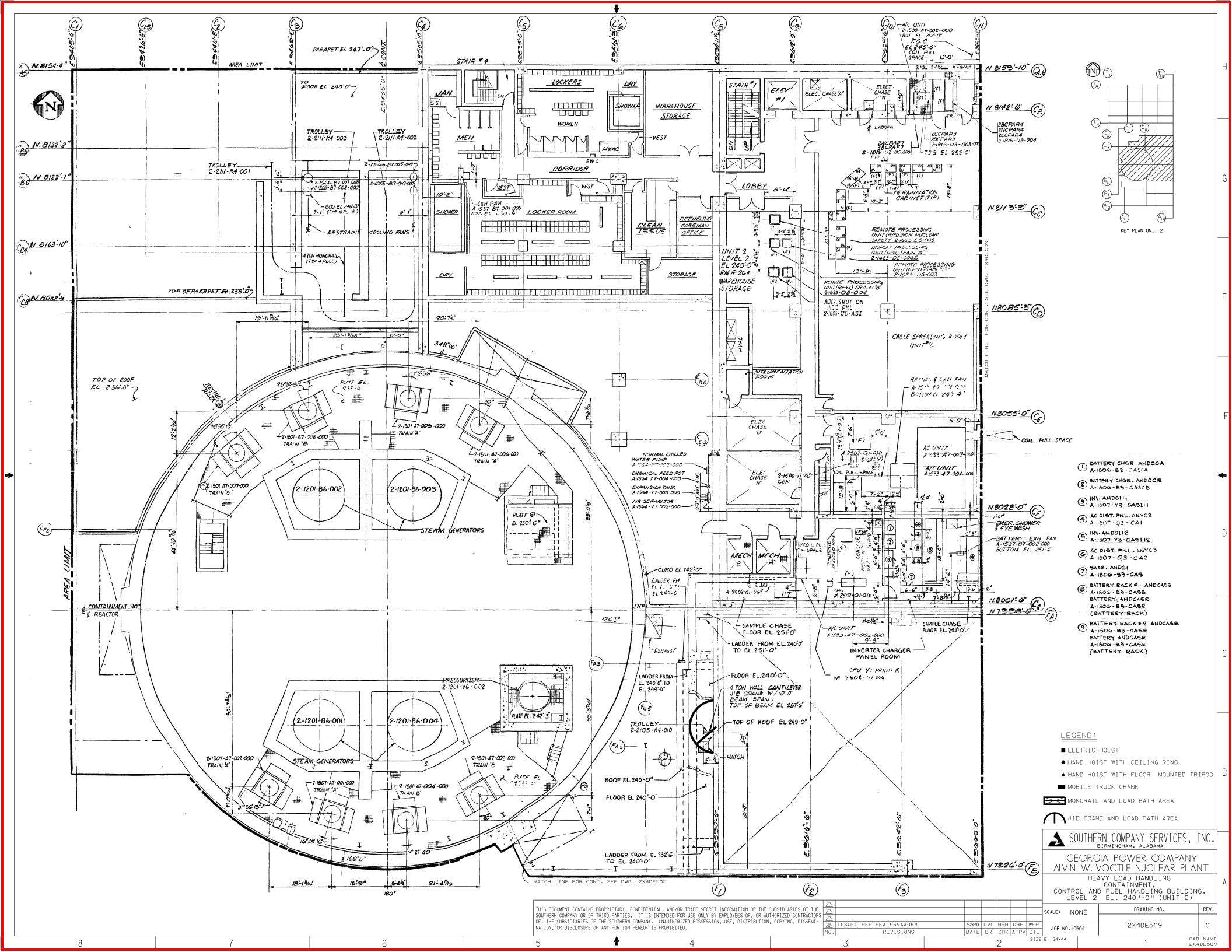
NO.	REVISIONS	DATE	DR	CHK	APPV	DTL
4						
3						
2						
1	ISSUED PER REA 96-VAA054	7-28-98	ELC	RBH	CBH	WFP

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

HEAVY LOAD HANDLING
CONTAINMENT INTERNAL
SECTION LOOKING NORTH

SCALE: NONE
JOB NO. 10604
DRAWING NO. 2X4DE508
REV. 0
SIZE E 34X44
CAD NAME 2X4DE508



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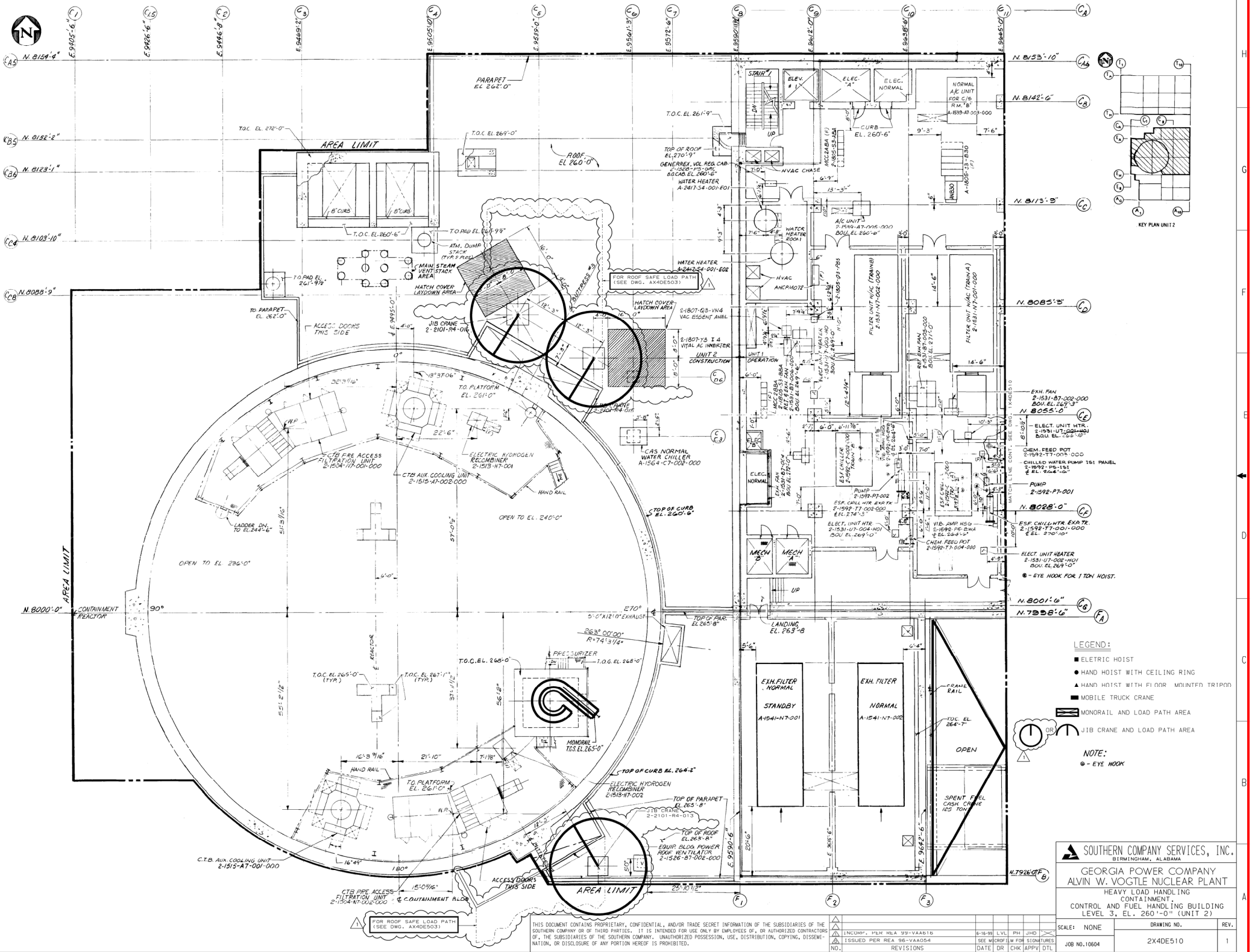
NO.	REVISIONS	DATE	DR	CHK	APPV	DTL
3	ISSUED PER REA 96VAA054	7-26-96	LVL	RBH	CBH	WFP

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT
HEAVY LOAD HANDLING CONTAINMENT, CONTROL AND FUEL HANDLING BUILDING. LEVEL 2 EL. 240'-0" (UNIT 2)

SCALE: NONE	DRAWING NO.	REV.
JOB NO. 10604	2X4DE509	0

SIZE E 34x44
CAD NAME 2X4DE509



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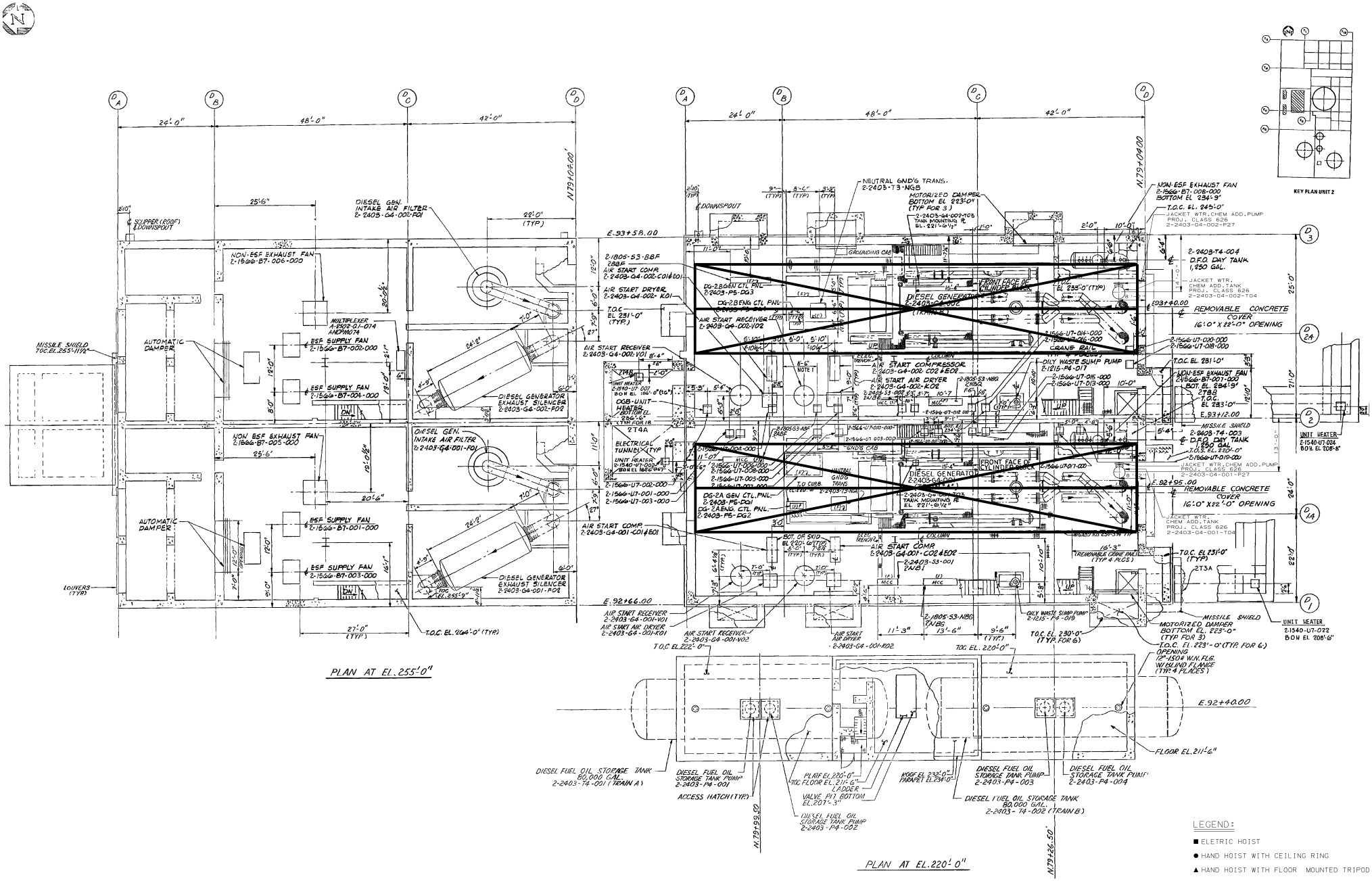
NO.	REVISIONS	DATE	DR	CHK	APPV	DTL
3	INCORP. PER REA 99-VAA616	6-16-99	LVL	PH	JHD	
2	ISSUED PER REA 96-VAA054	SEE MICROFILM FOR SIGNATURES				

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT
HEAVY LOAD HANDLING CONTAINMENT, CONTROL AND FUEL HANDLING BUILDING LEVEL 3, EL. 260'-0" (UNIT 2)

SCALE: NONE	DRAWING NO.	REV.
JOB NO. 10604	2X4DE510	1

SIZE E 34X44
CAD NAME 2X4DE510



SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

HEAVY LOAD HANDLING EQUIPMENT LOCATION LAYOUT DIESEL GENERATOR BUILDING (UNIT 2)

SCALE: NONE | DRAWING NO. 2X4DE511 | REV. 0 | JOB NO. 10604

SIZE E 34X44

CAD NAME 2X4DE511

NO.	REVISIONS	DATE	DR	CHK	APPV	DTL
4						
3						
2						
1	ISSUED PER REA 96-VAA054	7-28-98	JLH	RBH	CBH	WFP

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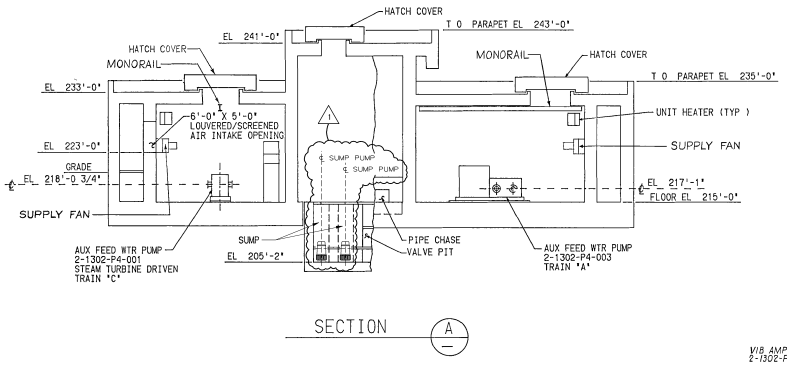
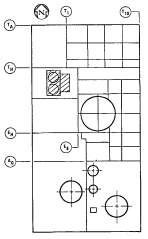
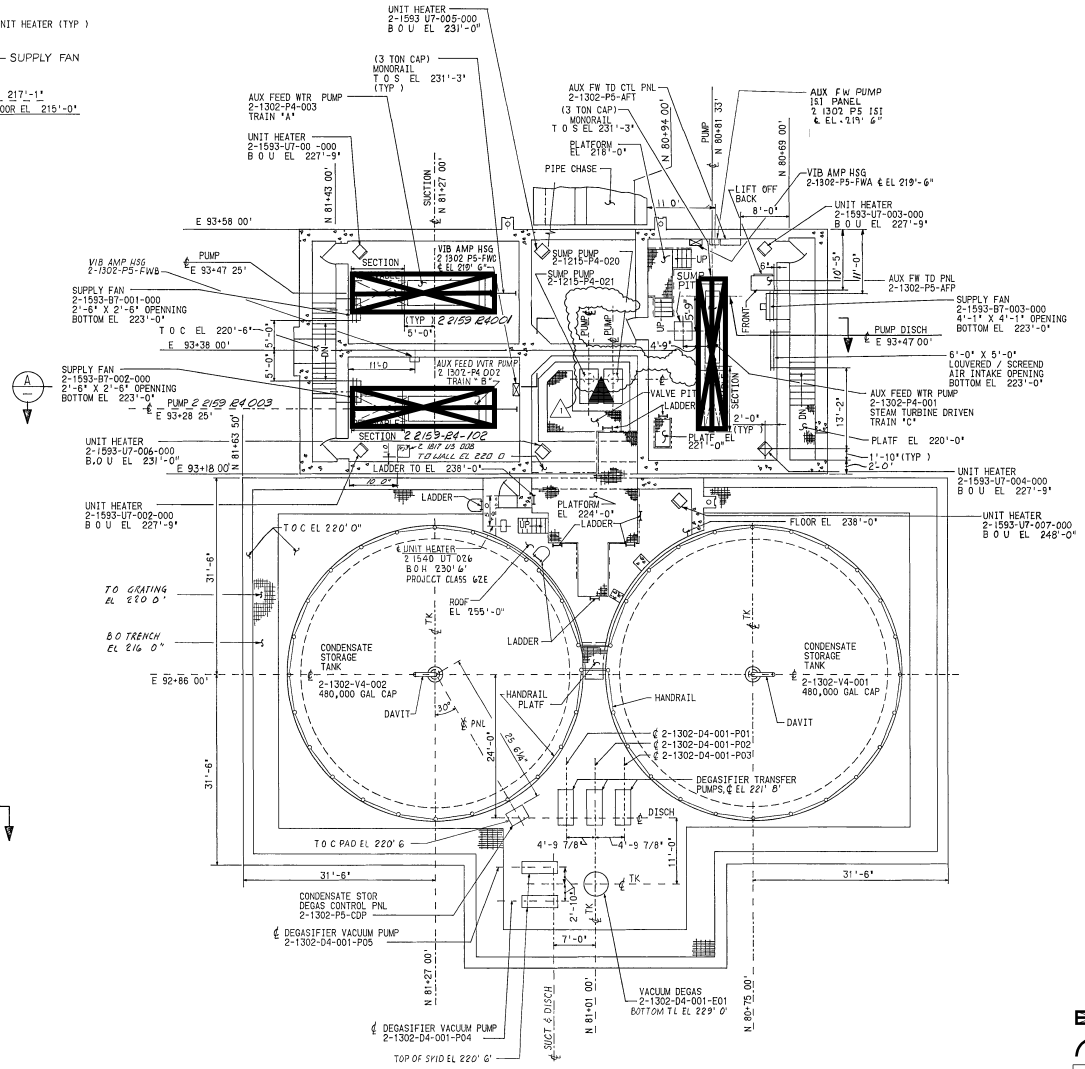
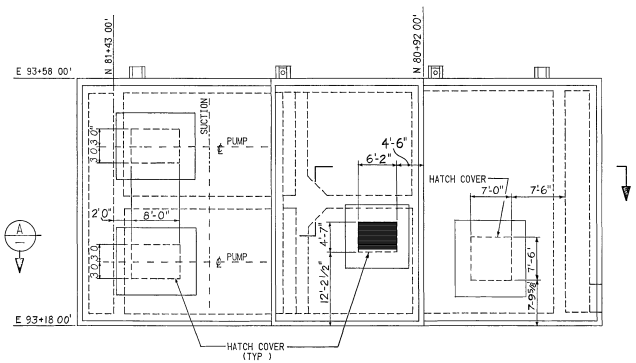


SECTION A





PLAN AT EL. 215'-0" & ABOVE



ROOF PLAN

LEGEND:

- ■ ELETRIC HOIST
- ● HAND HOIST WITH CEILING RING
- ▲ HAND HOIST WITH FLOOR MOUNTED TRIPOD
- MOBILE TRUCK CRANE
- MONORAIL AND LOAD PATH AREA
- JIB CRANE AND LOAD PATH AREA

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

HEAVY LOAD HANDLING
AUXILIARY FEEDWATER PUMP HOUSE AND
CONDENSATE STORAGE TANK

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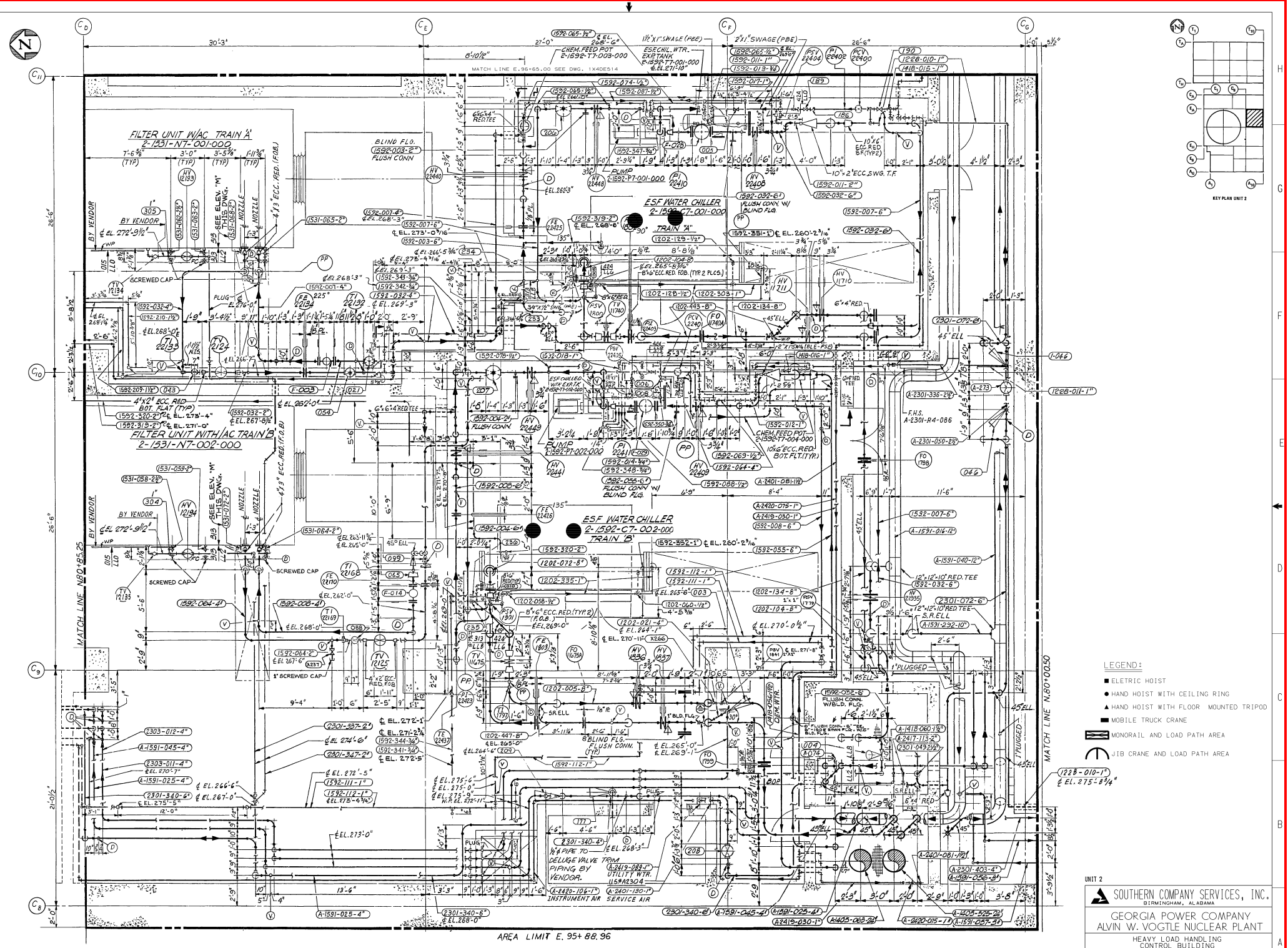
NO.	REVISIONS	DATE	DR	CHK	APPV	DTL
3						
2						
1	INCORPORATED PER DCP 97-VAN0063	01/18/01	ELC	GLB	MWD	
0	ISSUED FOR CONSTRUCTION	SEE MICROFILM FOR SIGNATURES				

SCALE: NONE
JOB NO.10604
DRAWING NO. 2X4DE512
REV. 1
SIZE E 34x44



UNIT 2

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

HEAVY LOAD HANDLING
CONTROL BUILDING
PIPING AREA 2B LEVEL 3
EL. 260'-0" TO 280'-0"

SCALE: NONE | DRAWING NO. 2X4DE514 | REV. 0
JOB NO. 10604

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NO.	REVISIONS	DATE	DR	CHK	APPV	DTL
1	ISSUED PER REA 96-VAA054	7-26-96	ELC	RBH	CBH	WFP

SIZE E 34X44

CAD NAME 2X4DE514

UNIT 2 CONSTRUCTION | UNIT 1 OPERATION

MATCH LINE E.95+14.83

FW VALVE MONORAIL EL. 236'

MSIV/SV MONORAILS EL. 254' (SEE NOTE #1)

AREA LIMIT N.81+69.83'

MATCH LINE E.94+67.66' SEE DWG. 2X4DE516

KEY PLAN UNIT 2

FOR PARTIAL PLAN M.S. BY-PASS PIPING SEE DWG 2X4DL2E02

3 TON MONORAIL

M.S. SAMPLE COOLER 2-1311-65-007

M.S. SAMPLE COOLER 2-1311-65-006

STRESS FORGING (TYP) 2 PLACES

4 TON MONORAIL TYP. FOR 3 (FOR DET. SEE DWG. AXED1GC12)

RADIATION MONITOR RE-13121

RADIATION MONITOR RE-13122

PLATFORM EL. 232'-3"

TO STRUT EL. 240'-0"

LEGEND:

- ELETRIC HOIST
- HAND HOIST WITH CEILING RING
- HAND HOIST WITH FLOOR MOUNTED TRIPOD
- MOBILE TRUCK CRANE
- MONORAIL AND LOAD PATH AREA
- JIB CRANE AND LOAD PATH AREA

NOTES:

1. THE MSIV/SV MONORAIL WITH TROLLEY 2-1211-R4-D01 MAY BE USED FOR LIFTING THE ACTUATOR OF ATMOSPHERIC RELIEF VALVE 2PV-3020 OVER THE MAIN STEAM LINES UPSTREAM OF MSIV, DURING PLANT OPERATION, WITHOUT DOUBLE RIGGING. THE HOIST, THE SLING CONFIGURATION, AND THE CROSBY SHACKLE USED IN THE ACTUATOR LIFT SHALL HAVE A MINIMUM WORKING LOAD LIMIT OF TWO TONS.

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NO.	REVISIONS	DATE	DR	CHK	APPV	DTL
2	ISSUED PER REA 98-V4A054	3-20-03	LRP	HAB	ASK	

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

HEAVY LOAD HANDLING
NORTH MAIN STEAM VALVE ROOM
PIPING AREA 2E
LEVEL 1, EL.220'-0" AND ABOVE

SCALE: NONE	DRAWING NO.	REV.
JOB NO. 10604	2X4DE515	2

SIZE E 34x44

CAD NAME 24DL2E02

FW VALVES MONORAIL EL. 236'

MSIV/SV MONORAILS EL. 254'

4 TON MONORAIL

3 TON MONORAIL

ATMOSPHERIC RELIEF VALVE MONORAIL EL.254'

KEY PLAN UNIT 2

MATCH LINE E. 94+67.67' SEE DWG. 2X4DE515

MATCH LINE E. 94+03.75'

MATCH LINE 80+85.25'

AREA LIMIT N. 81+69.83'

PLATFORM EL 232'-3"

W.P.E EL. 224'-9 9/16"

W.P. EL 225'-8 1/2" (TOP)

W.P. EL 224'-11 1/8" (BELOW)

W.P. ℄ EL 224.9' 221'-3 1/4" (BELOW)

W.P.E EL. 221.0312" (REF.)

T.O.C. EL. 224'-5 1/2"

T.O.C. EL. 224'-4 1/2"

T.O.C. EL. 221'-7"

℄ EL 223'-6"

℄ EL 226'-0"

℄ EL 228'-0"

BOP EL 233'-6"

BOP EL 232'-6"

SLOPE

24"x16"CONC. RED

24"x24"x16" RED TEE

(SEE NOTE #1)

1301-371-2"

1301-014-8"

1302-013-4"

1302-018-4"

1301-224-1"

1301-216-3"

1301-093-10"

1301-012-4"

1302-014-4"

1302-019-4"

1305-081-8"

1511-068-3/8"

1305-070-18"

1511-067 3/8"

1305-135-1 1/2"

1411-050-1/2"

2420-074-1"

2401-080-1"

2419-066-1"

1411-048-1/2"

1511-067-3/8"

1305-135-1 1/2"

1305-070-18"

1411-048-1/2"

1305-061-16"

1305-055-16"

1305-056-42"

1301-004-48"

1301-00736"

2301-011-8"

1411-048-1/2"

2419-066-1"

2401-080-1"

2420-074-1"

1411-050-1/2"

1505-139-1 1/2"

FE 530

FE 520

FE 521

FT 530

FT 520

FT 521

TE 1520

TE 1521

PT 3010

FO 3010

328

324

NOTES:

1. THE MSIV/SV MONORAIL WITH TROLLEY 2-1211-R4-001 MAY BE USED FOR LIFTING THE ACTUATOR OF ATOMOSPHERIC RELIEF VALVE 2PV-3020 OVER THE MAIN STEAM LINES UPSTREAM OF MSIV, DURING PLANT OPERATION, WITHOUT DOUBLE RIDDING. THE HOIST, THE SLING CONFIGURATION, AND THE CROSBY SHACKLE USED IN THE ACTUATOR LIFT SHALL HAVE A MINIMUM WORKING LOAD OF TWO TONS

LEGEND:

- ELETRIC HOIST
- HAND HOIST WITH CEILING RING
- HAND HOIST WITH FLOOR MOUNTED TRIPOD
- MOBILE TRUCK CRANE
- MONORAIL AND LOAD PATH AREA
- JIB CRANE AND LOAD PATH AREA UNIT 2

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

HEAVY LOAD HANDLING
CONTROL BUILDING
PIPING AREA 2F
LEVEL 1, EL.220'-0" AND ABOVE

SCALE: NONE	DRAWING NO.	REV.
JOB NO. 10504	2X4DE516	1

SIZE E 34X44 CAD NAME 2X4DE516

NO.	REVISIONS	DATE	DR	CHK	APPV	DTL
3	INCORP. PER ABN-51837	5-20-03	LRP	HAB	ASK	

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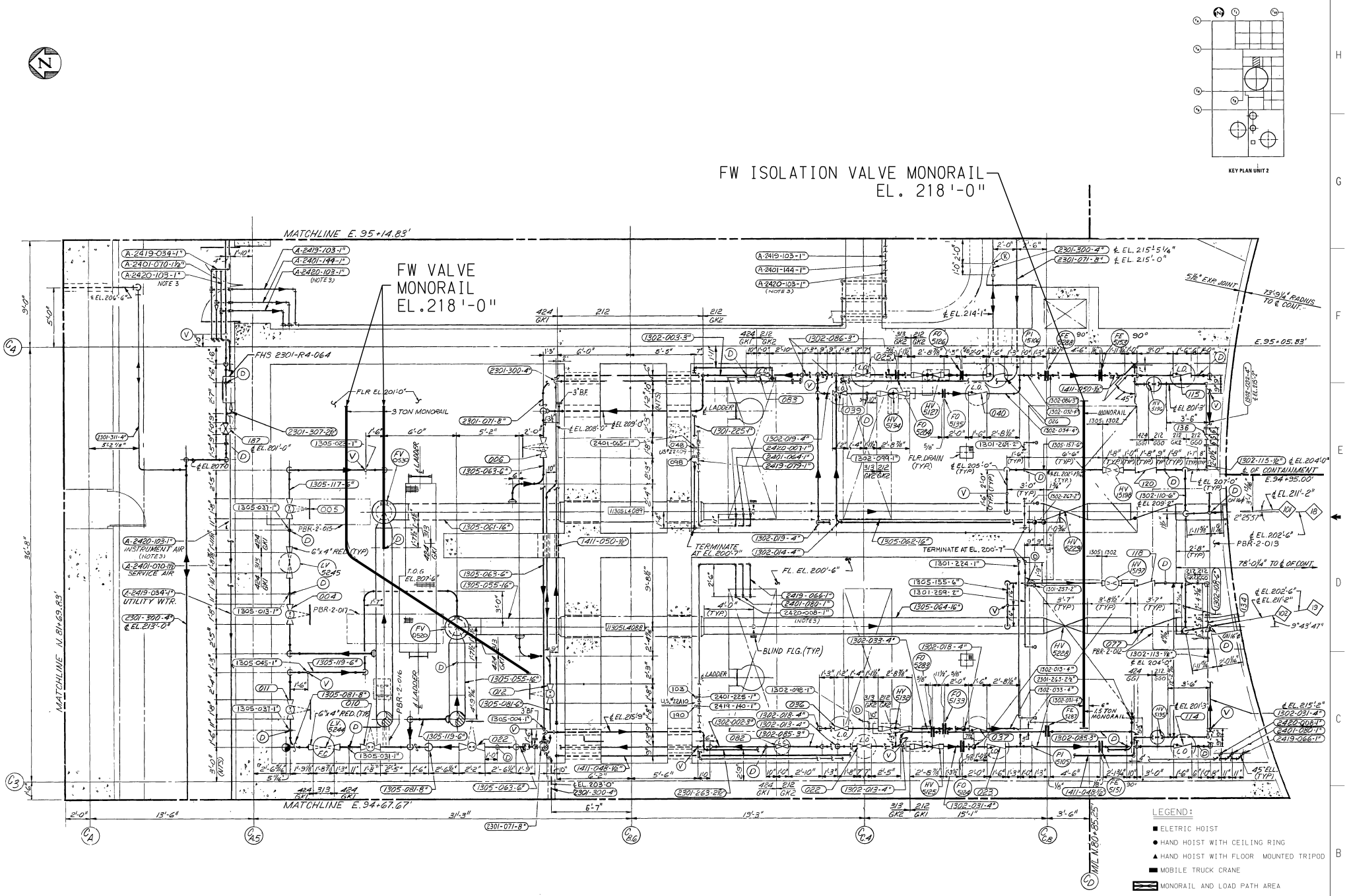

LEGEND:

- ■ ELETRIC HOIST
- ● HAND HOIST WITH CEILING RING
- ▲ HAND HOIST WITH FLOOR MOUNTED TRIPOD
- MOBILE TRUCK CRANE
- MONORAIL AND LOAD PATH AREA
- JIB CRANE AND LOAD PATH AREA

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

HEAVY LOAD HANDLING
NORTH FEEDWATER VALVE
ROOM PIPING AREA 2E LEVEL A
EL. 200'-0" TO EL.220'-0"

SCALE: NONE | DRAWING NO. 2X4DE517 | REV. 0 | JOB NO. 10604

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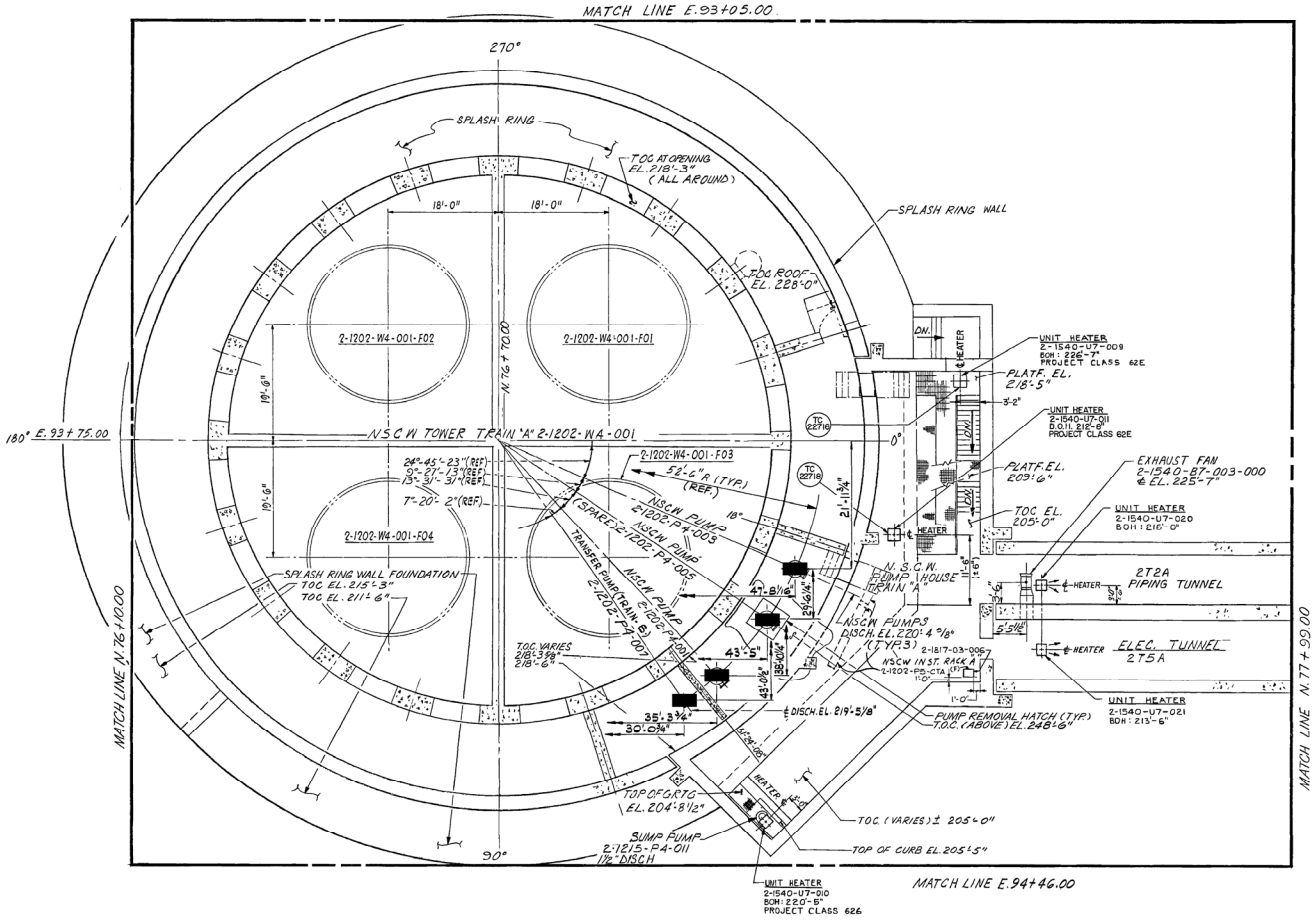
NO.	REVISIONS	DATE	DR	CHK	APPV	DTL
3						
2						
1						
0	ISSUED PER REA 96-VAA054	7-28-98	ELC	RBH	CBH	WFP

SIZE E 34X44

CAD NAME 2X4DE517



LEGEND:

- ■ ELETRIC HOIST
- ● HAND HOIST WITH CEILING RING
- ▲ HAND HOIST WITH FLOOR MOUNTED TRIPOD
- ▬ MOBILE TRUCK CRANE
- MONORAIL AND LOAD PATH AREA
- JIB CRANE AND LOAD PATH AREA

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

HEAVY LOAD HANDLING
TRAIN "A"
NUCLEAR SERVICE COOLING WATER TOWER
LEVEL 1 EL. 220'-0" (UNIT 2)

SCALE: NONE	DRAWING NO.	REV.
JOB NO. 10604	2X4DE518	0

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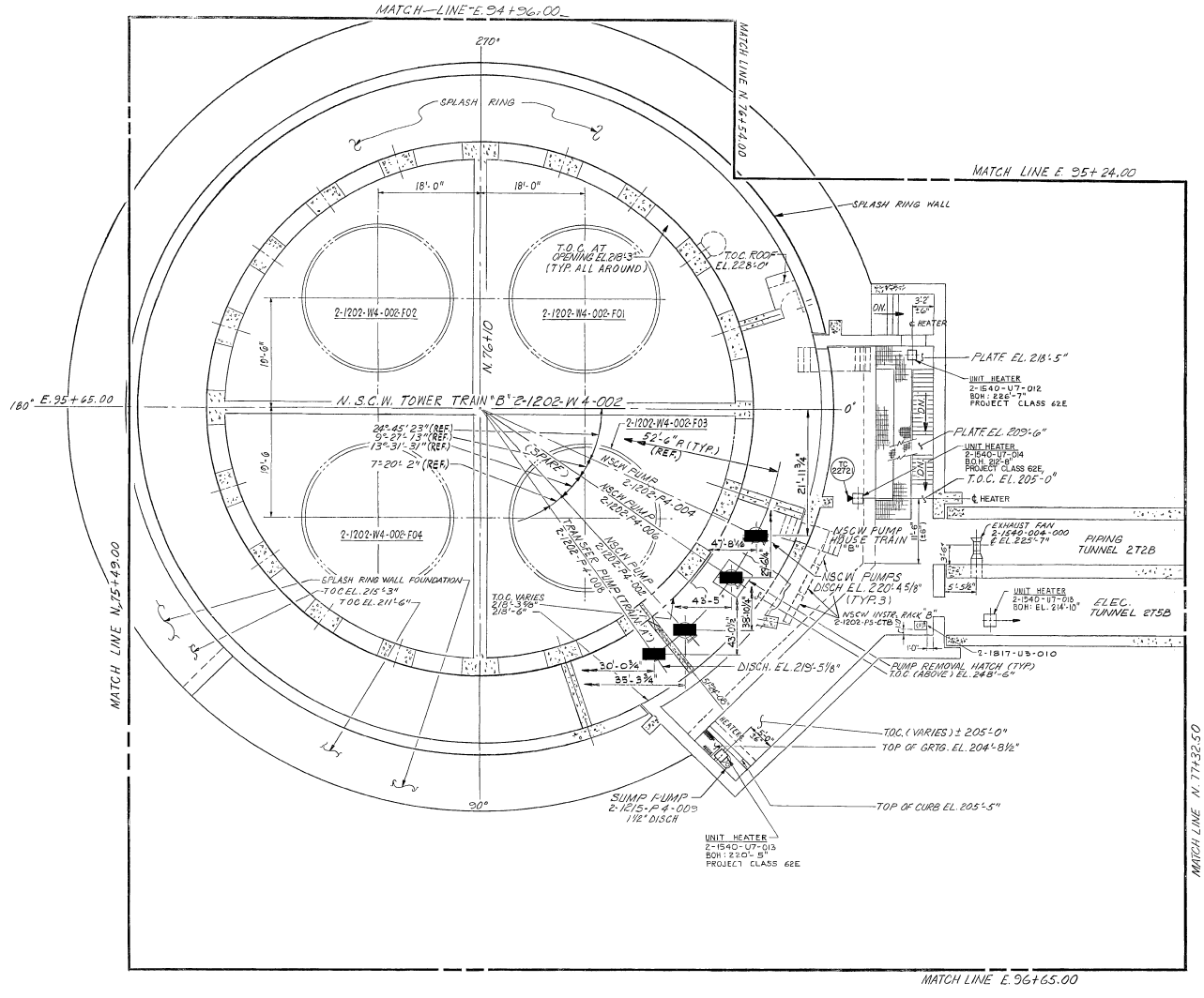
NO.	REVISIONS	DATE	DR	CHK	APPV	DTL
4						
3						
2						
1	ISSUED PER REA 96-VAA054	7-28-98	JLH	RBH	CBH	WFP

SIZE E 34X44

CAD NAME 2X4DE518



LEGEND:

- ■ ELETRIC HOIST
- ● HAND HOIST WITH CEILING RING
- ▲ HAND HOIST WITH FLOOR MOUNTED TRIPOD
- ▬ MOBILE TRUCK CRANE
- MONORAIL AND LOAD PATH AREA
- JIB CRANE AND LOAD PATH AREA

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

HEAVY LOAD HANDLING
TRAIN "B"
NUCLEAR SERVICE COOLING WATER TOWER
LEVEL 1, EL220'-0". (UNIT 2)

SCALE: NONE	DRAWING NO.	REV.
JOB NO.10604	2X4DE519	0

SIZE E 34X44

CAD NAME 2X4DE518

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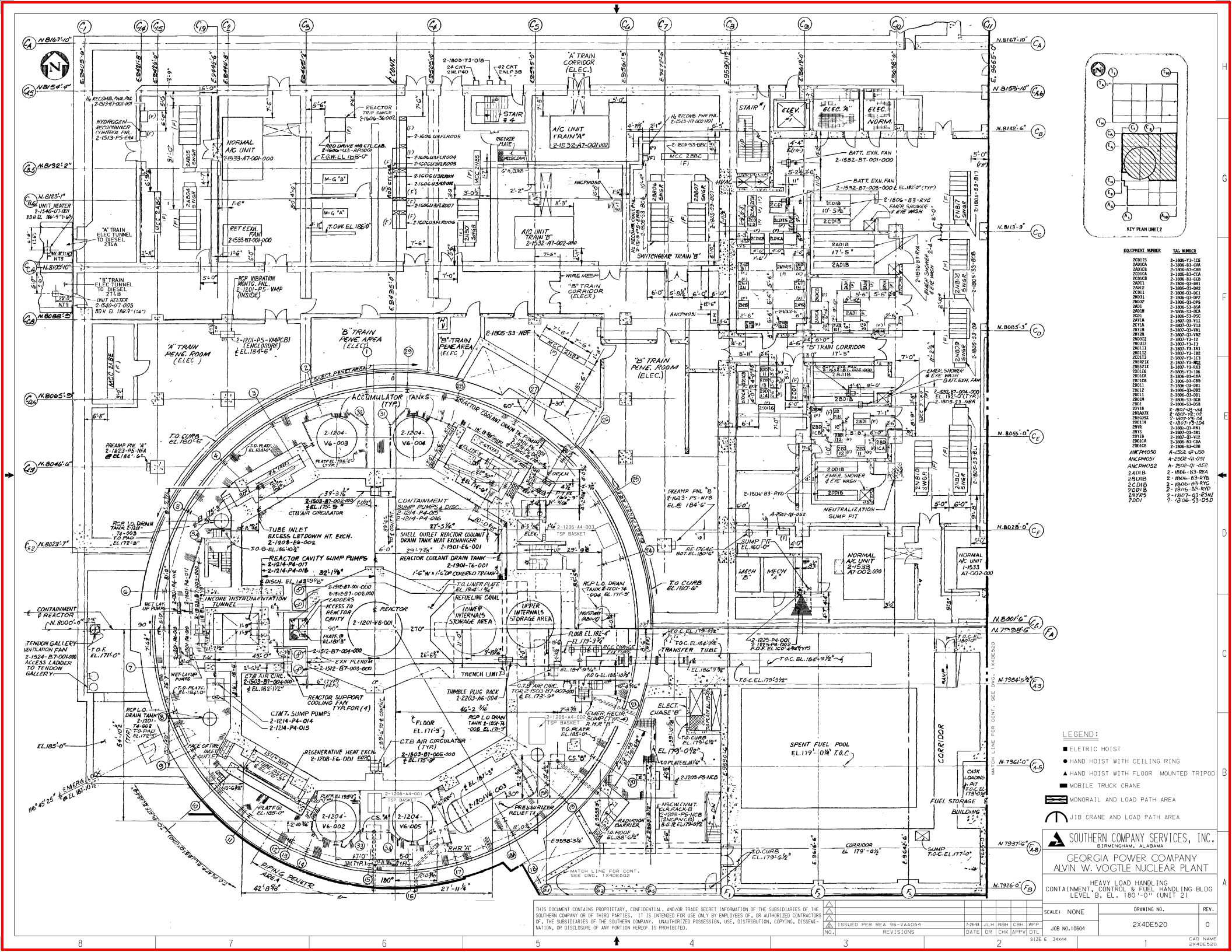
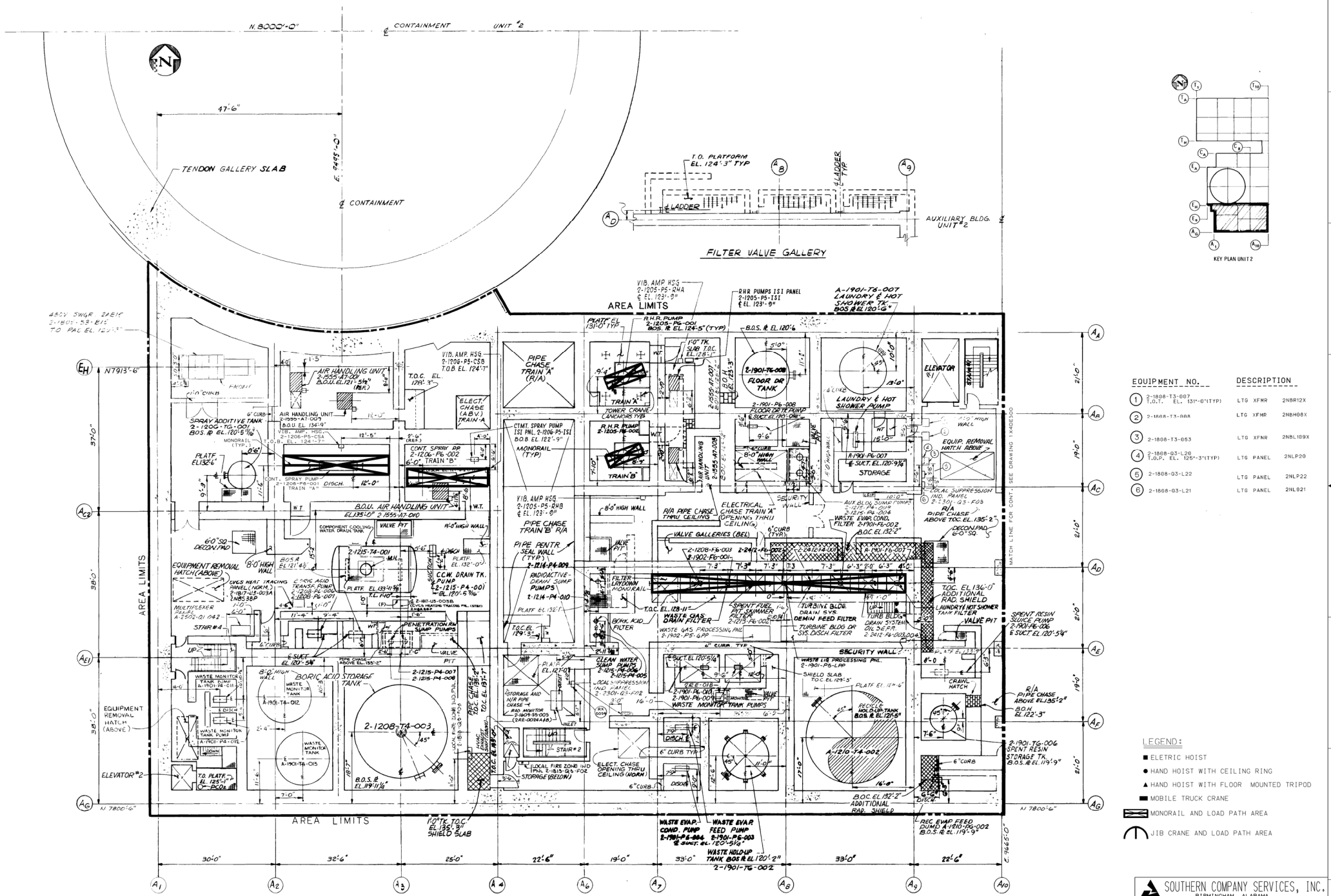
NO.	REVISIONS	DATE	DR	CHK	APPV	DTL
4						
3						
2						
1	ISSUED PER REA 96-VAA054	7-26-96	JLH	RBH	CBH	WFP


ACCUMULATOR TANKS (TYP.)
"A" TRAIN PENE. ROOM (ELEC.)
"B" TRAIN PENE. ROOM (ELEC.)
"A" TRAIN CORRIDOR (ELEC.)
"B" TRAIN CORRIDOR (ELEC.)
"A" TRAIN PENE. AREA (ELEC.)
"B" TRAIN PENE. AREA (ELEC.)
NORMAL A/C UNIT 2-1533-A7-001-000
A/C UNIT TRAIN "A" 2-1532-A7-001-000
A/C UNIT TRAIN "B" 2-1532-A7-002-000
SWITCHGEAR TRAIN "B"
STAIR #1
STAIR #4
ELEV. #1
ELEC. "A"
ELEC. "B"
NORM
BATT. EXH. FAN 2-1532-B7-001-000
BATT. EXH. FAN 2-1532-B7-003-000
RET. EXH. FAN 2-1533-B7-001-000
REACTOR CAVITY SUMP PUMPS 2-1214-P4-017 2-1214-P4-018
REACTOR COOLANT DRAIN TANK 2-1901-T6-001
CONTAINMENT SUMP PUMPS & DISC. 2-1214-P4-005 2-1214-P4-016
TUBE INLET EXCESS LETDOWN HT. EXCH. 2-1208-E6-002
5 SHELL OUTLET REACTOR COOLANT DRAIN TANK HEAT EXCHANGER 2-1901-E6-001
INCORE INSTRUMENTATION TUNNEL
REFUELING CANAL
LOWER INTERNALS STORAGE AREA
UPPER INTERNALS STORAGE AREA
TRANSFER TUBE
SPENT FUEL POOL EL. 179'-0 1/4" T.O.C.
FUEL STORAGE BUILDING
CASK LOADING PIT
CORRIDOR
NEUTRALIZATION SUMP PIT
SUMP PIT EL. 160'-0"
MECH "B"
MECH "A"
NORMAL A/C UNIT 1-1533 A7-002-000
REGENERATIVE HEAT EXCH. 2-1208-E6-001
PRESSURIZER RELIEF TK.
RADIATION BARRIER
TENDON GALLERY VENTILATION FAN 2-1524-B7-001-000
ACCESS LADDER TO TENDON GALLERY
REACTOR SUPPORT COOLING FAN TYP. FOR (4)
CTMT. SUMP PUMPS 2-1214-P4-014 2-1214-P4-015
RCP L.O. DRAIN TANK 2-1201-T4-002
RCP L.O. DRAIN TANK 2-1201-T4-003
RCP L.O. DRAIN TANK 2-1201-T4-004
RCP L.O. DRAIN TANK 2-1201-T4-005
RCP VIBRATION MONTR. PNL. 2-1201-P5-VMP (INSIDE)
"A" TRAIN ELEC TUNNEL TO DIESEL 2TKA
"B" TRAIN ELEC TUNNEL TO DIESEL 2TKB
UNIT HEATER 2-1540-U7-005
PREAMP PNL "A" 2-1623-P5-NFA
PREAMP PNL "B" 2-1623-P5-NFB
NEUTRALIZATION SUMP PIT
EMER. SHOWER & EYE WASH
EMERG. SHOWER & EYE WASH
EMER. SHOWER & EYE WASH
EMERG LOCK
PIPING PENETR. AREA
ELECT. PENETR. AREA
MATCH LINE FOR CONT. SEE DWG. 1X4DE502
MATCH LINE FOR CONT. SEE DWG. 1X4DE520
TO PLANT ACCESS & FUEL HANDLING BLDG
N.8167'-10"
N.8153'-10"
N.8142'-6"
N.8123'-1"
N.8113'-9"
N.8085'-3"
N.8055'-0"
N.8028'-0"
N.8001'-6"
N.7984'-5 1/2"
N.7961'-0"
N.7937'-6"
N.7926'-0"
E.9665'-0"
E.9642'-6"
E.9558'-3 1/4"
E.9530'-1 1/2"
E.9466'-8"
E.9403'-2 1/2"
E.9340'-5"
E.8842'-8"
E.8779'-2"
E.8715'-5"
E.8652'-8"
E.8589'-2"
E.8525'-5"
E.8462'-8"
E.8399'-2"
E.8335'-5"
E.8272'-8"
E.8209'-2"
E.8145'-5"
E.8082'-8"
E.8019'-2"
E.7955'-5"
E.7892'-8"
E.7829'-2"
E.7765'-5"
N.8167'-10"
N.8153'-10"
N.8132'-2"
N.8123'-1"
N.8105'-10"
N.8088'-8"
N.8085'-3"
N.8046'-6"
N.8023'-7"
N.8000'-0"
N.7984'-5 1/2"
N.7961'-0"
N.7937'-6"
N.7926'-0"
EL. 185'-0"
EL. 171'-0"
EL. 160'-0"
EL. 179'-0 1/2"
EL. 179'-6 1/2"
EL. 184'-9 1/2"
EL. 186'-9 3/8"
EL. 171'-9"
EL. 175'-9"
EL. 184'-6"
EL. 195'-6"
EL. 189'-5"
EL. 182'-1 1/2"
EL. 181'-10 1/2"
EL. 184'-0"
EL. 186'-10 3/8"
EL. 173'-9"
EL. 179'-0 1/2"
EL. 185'-0"
EL. 184'-9 3/8"
EL. 179'-3 1/2"
EL. 185'-10 1/2"
EL. 184'-9 1/2"
EL. 179'-6 1/2"
EL. 177'-0"
EL. 179'-0 1/2"
EL. 179'-6 1/2"
EL. 177'-0"
EL. 173'-0 1/2"
EL. 179'-0 1/2"
EL. 188'-0 1/2"
EL. 192'-0" (TYP.)
EL. 192'-0"
EL. 192'-0" (TYP.)
EL. 192'-0"
EL. 192'-0" (TYP.)
EL. 192'-0



EQUIPMENT NO.		DESCRIPTION
1	2-1808-T3-007 T.O.T. EL. 131'-0" (TYP)	LTG XFMR 2NBR12X
2	2-1808-T3-008	LTG XFMR 2NBH08X
3	2-1808-T3-053	LTG XFMR 2NBL109X
4	2-1808-03-L20 T.O.P. EL. 125'-3" (TYP)	LTG PANEL 2NLP20
5	2-1808-03-L22	LTG PANEL 2NLP22
6	2-1808-03-L21	LTG PANEL 2NL921

LEGEND:

- ■ ELETRIC HOIST
- ● HAND HOIST WITH CEILING RING
- ▲ HAND HOIST WITH FLOOR MOUNTED TRIPOD
- ■ MOBILE TRUCK CRANE
- MONORAIL AND LOAD PATH AREA
- JIB CRANE AND LOAD PATH AREA

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NO.	REVISIONS	DATE	DR	CHK	APPV	DTL
A	ISSUED PER REA 96-VAA054	7-28-98	ELC	RBH	CBH	WFP

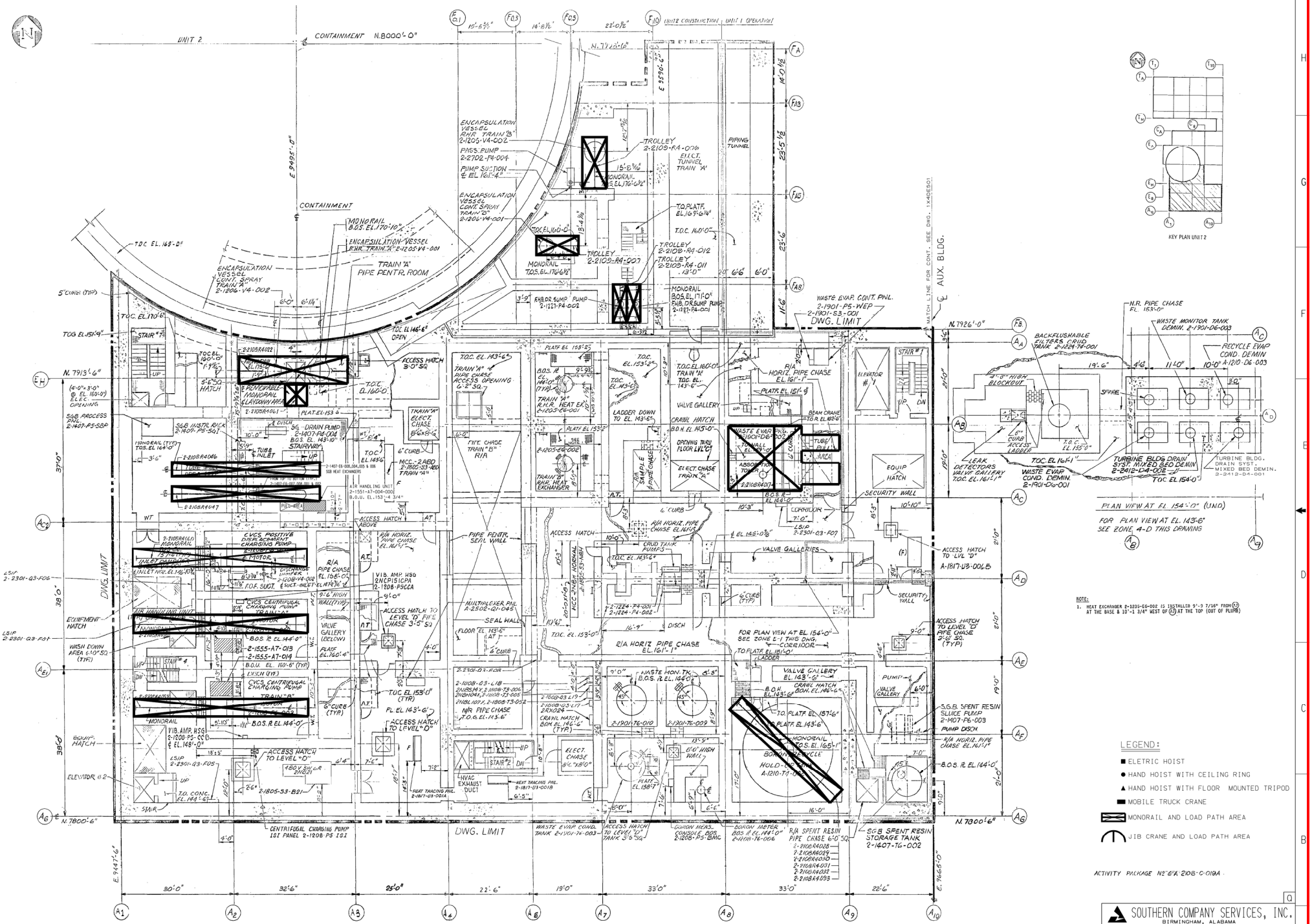
SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

HEAVY LOAD HANDLING
AUXILIARY BUILDING LEVEL D
EL 119'-3" TO 143'3" (UNIT 2)

SCALE: NONE	DRAWING NO.	REV.
JOB NO. 10604	2X4DE521	0

SIZE E 34X44
CAD NAME 2X4DE521



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NO.	REVISIONS	DATE	DR	CHK	APPV	DTL
A	ISSUED PER REA 96-VAA054	7-28-98	ELC	RBH	CBH	WFP

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

HEAVY LOAD HANDLING
AUXILIARY BUILDING LEVEL C
EL. 143'-6" TO 170'-6" (UNIT 2)

SCALE: NONE
JOB NO. 10604
DRAWING NO. 2X4DE522
REV. 0
SIZE E 34X44
CAD NAME 2X4DE522

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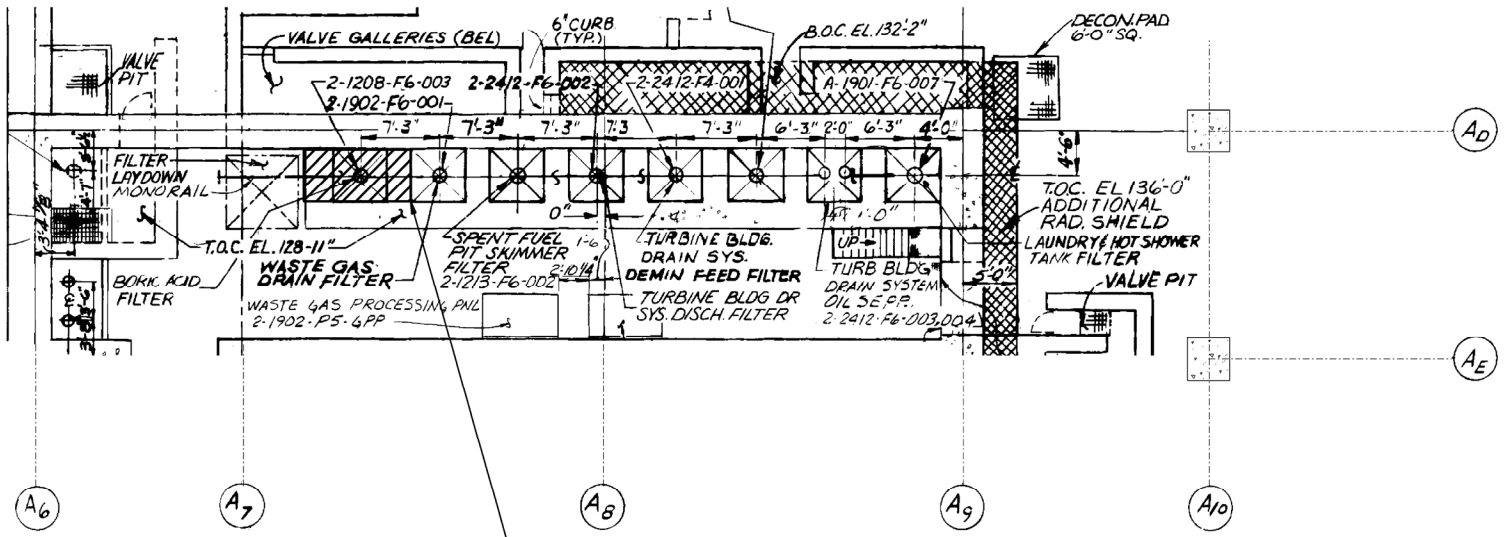


IN THE INDICATED AREA, THE LOAD SHALL NOT BE LIFTED HIGHER THAN 1 FOOT (AS MEASURED FROM THE SLAB TO BOTTOM OF THE LOAD)

NO.	VERSIONS	DATE	DR	CHK	APPV
1.0	REVISED PER ABN-81267, VER. 1.0	01-04-06	KMH	LRP	ASK

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

SAFE LOAD PATH FOR
CARTRIDGE FILTER MONORAIL
AUXILIARY BUILDING LEV. D MONORAIL
2108-R4-064 (UNIT 2)
1.0

SCALE:

JOB NO. 10604

DRAWING NO. 2X4DE601

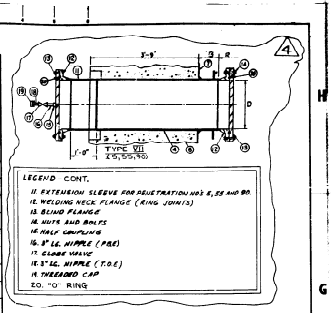
VER. 1.0

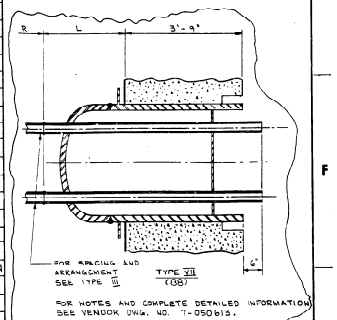
SIZE D 22X34

SCALING RULES

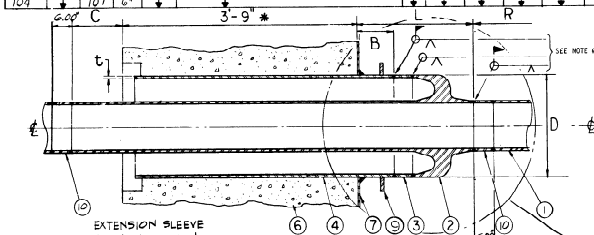
SEE DWG. 2X4DLA13

PENETRATION NO.	LINE NO. SYSTEM	SEQ. NO.	NOM. SIZE	PIPE SPEC.	SYSTEM DESCRIPTION	TYPE	IN-SERVICE INSPECTION (SEE NOTE #7)	NORM. OPERATING PRESS. (PS.I.G.)	NORM. OPERATING TEMP. (°F)	MAX. DESIGN PRESS. (PS.I.G.)	MAX. DESIGN TEMP (°F)	PIPE O.D. (INCH)	PIPE I.D. (INCH)	WALL THK.	MATERIAL (A.S.M.E. SPEC. NO.) (SEE NOTE # 15) ① PIPE REFERENCE	② FLUED HEAD SEE NOTE # 15	③ INTERMEDIATE SLEEVE	④ PENETRATION SLEEVE	PENETRATION ASSEMBLY DIMENSIONS (INCHES) (SEE NOTE # 15) L	B	C	D	t	℄ ELEVATION	℄ AZIMUTH	P. & I.D. DWG. NO.	MECH. DWG. NO.	RADIUS "R"	PENETRATION NO.
70 (A)	1513	002	3/4"	LL1	CONTAINMENT HYDROGEN MONITOR-PS-HMB (OUT)	III	VISUAL	80	340	80	340	1.050	0.824	SCH.40S	SA-312, TP 304L	SA-182, GR F304L	SA-312, TP 304L	SA-333, GR. 6	12.750	6.250	6.000	12.750	0.375	210'-1"	233°30'46"	2X4DB213-2	2X4DL4D03	68'-11½"	70A
(B)		006			(IN)																								B
(C)	1206		1/2"	SEE NOTE #2	CONTAINMENT PRESS INSTR. PT-0934 (NOTE 7)			0	100	60	320	0.840	0.252	SCH XXS	SA-312, TP304L										2X4DB131			C
71 (A)	1513	014	3/4"	LL1	CONTAINMENT HYDROGEN MONITOR-PS-HMA (OUT)			80	340	80	340	1.050	0.824	SCH.40S										189'-6"	153°14'33"	2X4DB213-2	2X4DL4A06		71A
(B)		008			(IN)			80	340																				B
(C)	1206		1½"	SEE NOTE 2	CONTAINMENT PRESS INSTR. PT 0935 (NOTE 7)			0	100	60	320	0.840	0.252	SCH XXS	SA-312, TP304L								2X4DB131			C
72 (A)	1204	106	3/4"	HG0	LOOP 1 ACCUMULATOR V6-002 SAMPLE			700	120	700	300	1.050	0.824	SCH.40S										181'-4½"	177°34'03"	2X4DB120	2X4DL4007		72A
(B)		108			LOOP 3 V6-004																								B
(C)	—	—		SEE NOTE #11	SPARE NECK ASS'Y.			—	—	—	—		0.612	SCH 160	SEE NOTE # 18														C
73 (A)	1204	107		HG0	LOOP 2 ACCUMULATOR V6-003 SAMPLE			700	120	700	300		0.824	SCH.40S	SA-312, TP 304L										180°	2X4DB120			73A
(B)		109			LOOP 4 V6-005																								B
(C)	—	—		SEE NOTE #11	SPARE NECK ASS'Y			—	—	—	—		0.612	SCH 160	SEE NOTE # 18														C
74	—	—	—	—	NOT USED			—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	74
75	2630.3 3431.36	264 66	6A 4B	—	BULK GASES SUPPLY SYSTEM-TEMPORARY	V	N/A	—	—	—	—	—	—	—	—	—	—	SA-333, GR. 6	—	*8.250	—	10.750	0.365	201'-11½"	248°6'28"	AX4DB012 002, 037, 042	2X4DL4D04	—	75
76	—	—	—	—	SPARE PENETRATION SLEEVE	I	—	—	—	—	—	—	—	—	—	—	—		—	8.250	6.0000			189'-6"	182°25'51"	—	2X4DL4D06	—	76
77	1901	022	3"	LL1	RCDT E6001 HT EX DISCH TO WPS WASTE HOLDUP TANK T-002E	II	VISUAL	125	130	150	250	3.500	3.068	SCH 40S	SA-312, TP304L	SA-182, GRF304L	SA 312, TP304L		15.750	6.250	6.000			201'-11½"	245°40'31"	2X4DB127	2X4DL4D04	68'-8½"	77
78	1214	028	2"		CONTAINMENT SUMP PUMPS DISCH. TO WPS, FDT. T-005 & WHT T-002			57	AMB	60	AMB	2.375	2.067											202'-6"	153°14'33"	2X4DB143	2X4DL4A04		78
79	1901	029	3/4"	LL2	RCDT T-001 H2 SUPPLY & WASTE GAS REMOVAL			02	120	150	250	1.050	0.742	SCH.80	SA-106, GR. B	SA-350, LF2	NONE		12.000					201'-11½"	243°14'34"	2X4DB127	2X4DL4D04	69'-0¼"	79
80	7401	050	4"	LL1	SERVICE AIR SUPPLY			120	115	132	115	4.500	4.026	SCH.40S	SA 312, TP304	SA-182, GR.F304L	NONE		16.000			18.000	0.500	189'-6"	181°17'51	2X4DB186-1	2X4DL4D06	68'-8¼"	80
81	2420.	043	2"		INSTRUMENT AIR							2.375	2.067	SCH40S					12.000			10.750	0.365	189'-6"	184°5'54	2X4DB186-2		69'-0¼"	81
82	—	—	—	—	SPARE PENETRATION SLEEVE	V	N/A	—	—	—	—	—	—	—	—	—	—		—	8.250	—	16.000	0.500	210'-1"	245°40'31"	—	2X4DL4D03	—	82
83	1505	003	24"	LL1	CTB NORMAL PROCESS PURGE SUPPLY	SEE NOTE 2	VISUAL	01	98	80	340	24.000	23.250	STD.WT.	SA-312, TP304		NONE		10.000	—	—	24.000		224'-2"	65°40'31"	2X4DB213-1	2X4DL4A03 2X4DL4B03	69'-2¼"	83
84	1506	002			EXHAUST			01	120											—	—			301'37'16"		2X4DL4B23 2X4DL4C02			84
85 (A)	—	—	3/4"	SEE NOTE #11	SPARE NECK ASS'Y.	III	VISUAL	—	—	—	—		1.050	0.612	SCH 160	SEE NOTE # 18	SA-182, GR F304L	SA 312, TP304L		12.750	6.250	6.000	12.750	0.375	182'-3½"	250°32'25"	—	2X4DL4D07	68'-11½"	85A
(B) (A)							VISUAL	—	—	—	—																		B
(C)	1206	—	1½"	SEE NOTE 2	CONTAINMENT PRESSURE INSTR. PT-0936 (NOTE 7)		VISUAL	0	100	60	320	0.840	0.252	SCH XXS	SA-312, TP304L										2X4DB131			C
86 (A)	2702	002	1/2"	LL1	POST ACCIDENT SAMPLING SYSTEM	III		50	300	65	400	0.840	0.622	SCH.40S	SA-312, TP 304L	SA-182, GR F304L	SA 312, TP 304L		23.67	8.250	6.000			235°56'48"	2X4DB110		68'-0 9/16"	86A
(B)	—	—	—	—	SPARE			—	—	—	—	—	—	—	—	—	—		—	—	—				—				B
(C)	2702	005	1"	FG0	POST ACCIDENT SAMPLING SYSTEM			2235	617	2485	650	1.315	0.815	SCH 160	SA-312, TP 304L	SA-182, GRF 304L	SA 312, TP 304L								2X4DB110			C
87	1222	015	8"	LL2	CONTAINMENT LEAK RATE TEST	I	VISUAL	60	100	60	100	8.625	7.981	SCH40	SA-106, GR B	SA 350, LF2	SA 106, GR B	SA-333, GR 6	16.000	6.250	6.000	18.000	0.500	189'-6"	167°50'15"	2X4DB132	2X4DL4A06	68'-8¼"	87
88	1201	4	1½"	FG2	RESERVED FOR RVLIS	VII	VISUAL	2235	120	2485	320	0.840	0.252	XXS	SA-312, TP304L	SA-182, F304L	NONE	4	15.25	8.250	6.000	10.750	0.365	189'-1½"	228°38'50"	2X4DB113	2X4DL4D06	(68'-8") 4	88
89	—	4	26"	4	FUEL TRANSFER TUBE (SEE NOTE #2 & #8)	—	—	—	—	—	—	—	—	—	—	—	—	SA-516, GR 70	—	—	—	—	—	186'-9¾"	—	2X4DB130	2X4DL4D08 2X4DL4D06	SEE NOTE 2	89
90	—	—	—	4	DEDICATED PENETRATION	VII	VISUAL	—	—	—	—	—	—	—	—	—	—	SA-333, GR 6	14.500	8.250	6.000	10.750	0.365	189'-6"	155°40'50"	—	2X4DL4A06	68'-9¾"	90
91	1202	216	8"	LL8	N.S.W. SUPPLY TO CTMT. COOLER	II	VISUAL	145	95	200	280	8.625	7.981	SCH 40S	SA-312, TP 304 L	SA-182, GRF304L	SA312, TP 304L		19.750	6.750	6.000	18.000	0.750	189'-1½"	245°40'31"	2X4DB135-2	2X4DL4A06	68'-4½"	91
92		209			2-1501-A7-0025 (004) TRAIN B																				240°48'37"				92
93		188			2-1501-A7-0026 (001) TRAIN A																			181'-4½"	168°22'39"	2X4DB135-1	2X4DL4A07		93
94		196			2-1501-A7-0064 (004) TRAIN A																			189'-6"			2X4DL4A06		94
95		220			RETURN FM 2-1501-A7-0027 (008) TRAIN B						105													189'-1½"	248°6'28"	2X4DB135-2	2X4DL4D06		95
96		218			2-1501-A7-0063 (003) TRAIN D																				243°14'34"				96
97		001			2-1501-A7-0064 (004) TRAIN A																			189'-6"	150°48'36"	2X4DB135-1	2X4DL4A06		97
98		192			2-1501-A7-0026 (001) TRAIN A																			181'-4½"			2X4DL4A07		98
99	—	—	—	—	NOT USED	—	—	—	—	—	—	—	—	—	—	—	—		—	—	—	—	—	—	—	—	—	—	99
100	1508	001	4"	LL1	CTB POST LOCA PURGE EXHAUST	SEE NOTE 2	VISUAL	03	150	80	340	4.500	4.026	SCH.40S	SA-312-TP304		NONE	SA-333, GR.6	9.250	—	—	4.500	0.237	264'-3"	177°34'-0"	2X4DB213-1	2X4DJ4132 2X4DL4D16	69'-3"	100
101	1302	110	6"	GK2	AUX. FEEDWATER	I	VOLUMETRIC	1150	445	1500	600	6.625	5.501	SCH 120	SA-106, GRB	SA-350, LF2			20.55	6.250	6.000	12.000	0.750	211'-2"	2'-35'-57"	2X4DB163-3	2X4DL4CJ03	68'-3 9/16"	101
102		108	6"																					211'-2"	9'-45'-41"		2X4DJ4B03		102
103		107	6"																					209'-10"	177°34'-0"		2X4DL4A03		103
104		107	6"																					209'-10"	170°16'-12"		2X4DL4A03		104



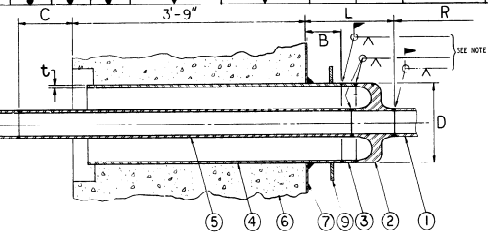


FOR NOTES AND COMPLETE DETAILED INFORMATION SEE VENDOR DWG. NO. T-050813.

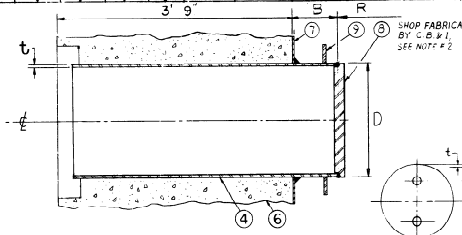


TYPE I

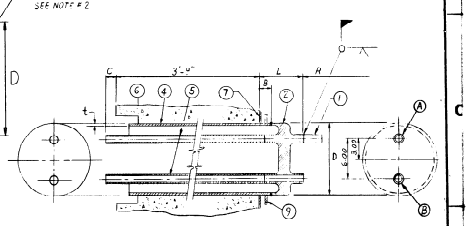
* FOR PENETRATIONS 14 & 2 DIMEN. IS 3'-9 15/16"
FOR PENETRATIONS 18 & 20 DIMEN. IS 3'-9 5/16"



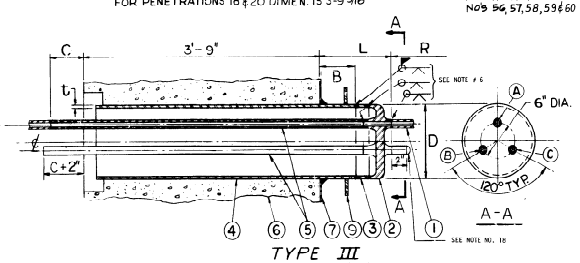
TYPE II



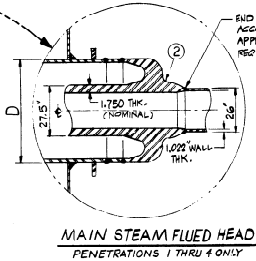
TYPE V
SPARE PENETRATION SLEEVE



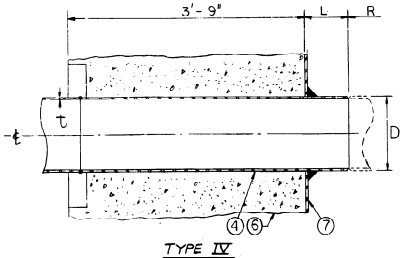
TYPE VI



TYPE III



MAIN STEAM FLUED HEAD
PENETRATIONS 1 THRU 4 ONLY



TYPE IV

LEGEND
1. CONNECTING PIPE OR COMPONENT
2. FLUED HEAD FITTING
3. INTERMEDIATE SLEEVE (NOTE #3 & #13)
4. PENETRATION SLEEVE (NOTE #2)
5. NECK EXTENSION (NOTE #11)
6. CONTAINMENT WALL
7. LINER PLATE (INSIDE CONTAINMENT WALL)
8. PERMANENT END CLOSURE
9. REINFORCING RING PLATE
10. EXTENSION SLEEVE FOR PENETRATION NO'S 56 THRU 60 (NOTE #16)

4 FOR CONTINUATION OF LEGEND SEE ABOVE

THIS DRAWING IS REQUIRED FOR W PROOF OF DESIGN

FOR NOTES & REFERENCES SEE DWG. NO. 2X4DLA13

ACT. PKG. 2-2-2101-L-008A UNIT 2

BECHTEL
LOS ANGELES

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

CONTAINMENT BLDG. UNIT 2
- CONTAINMENT WALL PIPING PENETRATION DESIGN LIST

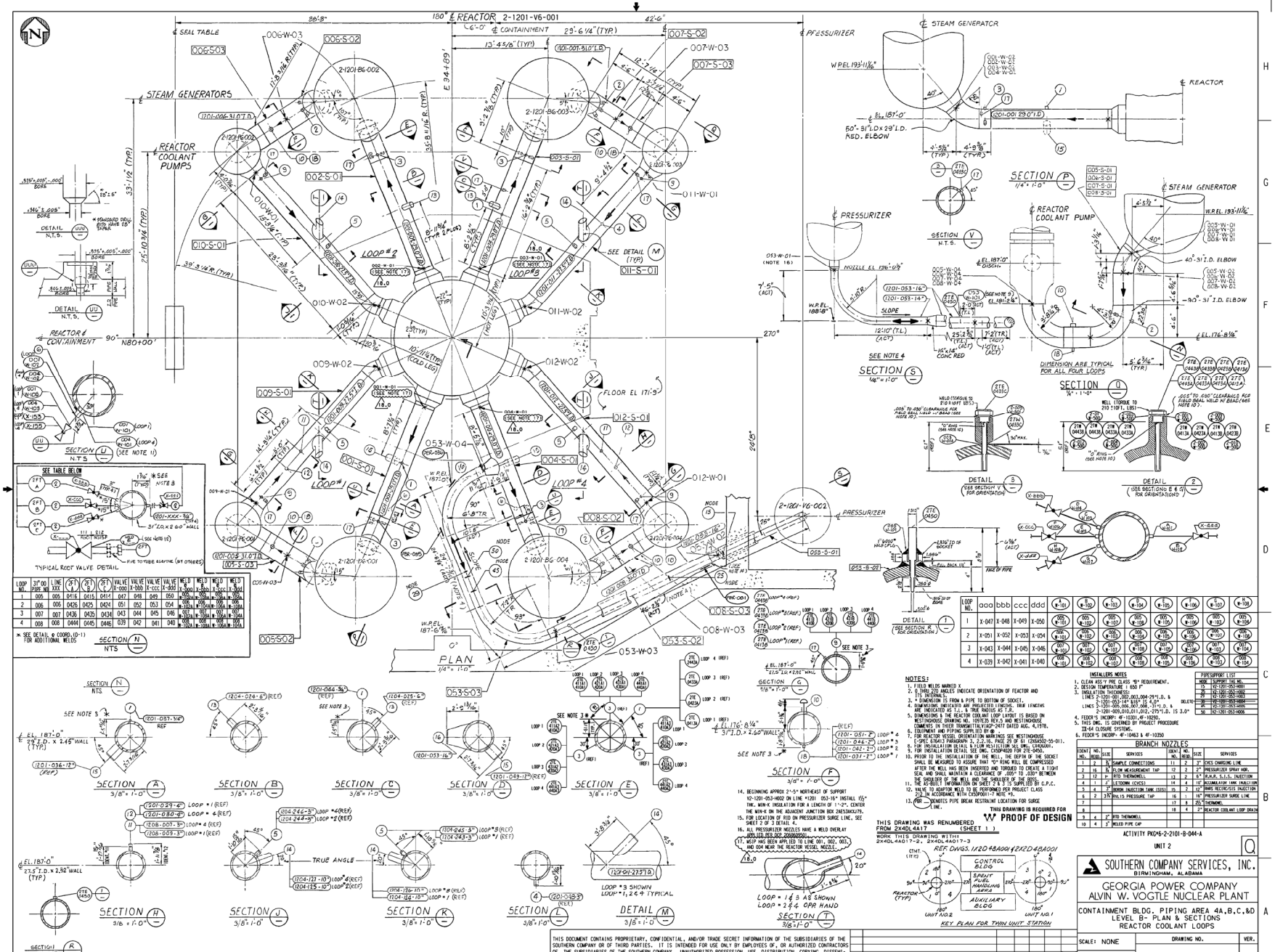
REVISIONS NO.	DATE	DR.	CHK.	SUPV.	EGS	CHF. E	P.E.	Q.A.E.
4								
3	INCORP. DCN'S 2,5&9 AND M-FCAB-19083.	12-90	CCB	Sheet				
2	REVISED PRESS & TEMP PER LATEST LINE LIST REV #22.	3-91	LWL					
1	ISSUED FOR CONSTRUCTION	11-89	AMD					
0	INTERCHANGED PENET #76 & 88 PER LTR. LOG # PSE-5073	6-16-86	DMG					

ENG. MGR. DATE

SCALE: NONE | DRAWING NO. 2X4DLA14 | REV. 4

JOB NO. 9510

SIZE E 34 x 44



LOOP NO.	31"OD PIPE NO.	LINE NO. XXX	24"Ø LINE NO.	2½"Ø LINE NO.	24"Ø LINE NO.	VALVE X-000	VALVE X-100	VALVE X-200	VALVE X-300	WELD NO. X-000	WELD NO. X-100	WELD NO. X-200	WELD NO. X-300
1	005	005	0416	0415	0414	047	048	049	050	005 ℄ 004A	005 ℄ 004A	005 ℄ 004A	005 ℄ 004A
2	006	006	0426	0425	0424	051	052	053	054	006 ℄ 004A	006 ℄ 004A	006 ℄ 004A	006 ℄ 004A
3	007	007	0436	0435	0434	043	044	045	046	007 ℄ 004A	007 ℄ 004A	007 ℄ 004A	007 ℄ 004A
4	008	008	0444	0445	0446	039	042	041	040	008 ℄ 004A	008 ℄ 004A	008 ℄ 004A	008 ℄ 004A

* SEE DETAIL ℄ COORD.(0-1) FOR ADDITIONAL WELDS

LOOP NO.	aaa	bbb	ccc	ddd	(#-101)	(#-102)	(#-103)	(#-104)	(#-105)	(#-106)	(#-107)	(#-108)
1	X-047	X-048	X-049	X-050	005 ℄ 004A	005 ℄ 004A	005 ℄ 004A	005 ℄ 004A	005 ℄ 004A	005 ℄ 004A	005 ℄ 004A	005 ℄ 004A
2	X-051	X-052	X-053	X-054	006 ℄ 004A	006 ℄ 004A	006 ℄ 004A	006 ℄ 004A	006 ℄ 004A	006 ℄ 004A	006 ℄ 004A	006 ℄ 004A
3	X-043	X-044	X-045	X-046	007 ℄ 004A	007 ℄ 004A	007 ℄ 004A	007 ℄ 004A	007 ℄ 004A	007 ℄ 004A	007 ℄ 004A	007 ℄ 004A
4	X-039	X-042	X-041	X-040	008 ℄ 004A	008 ℄ 004A	008 ℄ 004A	008 ℄ 004A	008 ℄ 004A	008 ℄ 004A	008 ℄ 004A	008 ℄ 004A

NOTES:
1. FIELD WELDS MARKED X.
2. ALL 90° ELBOWS INDICATE ORIENTATION OF REACTOR AND PRESSURIZER NOZZLES.
3. * DIMENSION IS FROM ℄ PIPE TO BOTTOM OF SOCKET.
4. DIMENSIONS INDICATED ARE PROJECTED LENGTHS. TRUE LENGTHS ARE INDICATED AS T.L. & TRUE RADIUS AS T.R.
5. DIMENSIONS & THE REACTOR COOLANT LOOP LAYOUT IS BASED ON WESTINGHOUSE DRAWING NO. 1099E21 REV.5 AND WESTINGHOUSE COMMENTS IN THEIR TRANSMITTAL(NQP-2477 DATED AUG. 4, 1978.
6. EQUIPMENT AND PIPING SUPPLIED BY ⊕.
7. FOR REACTOR VESSEL ORIENTATION MARKINGS SEE WESTINGHOUSE DRAWING NO. 1099E21 PARAGRAPH. 3.2.16. PAGE 29 OF SI 1210A502-55-011.
8. FOR INSTALLATION DETAIL & FLOW RESTRICTION SEE ORIFICE SCHEDULE.
9. FOR INSTALLATION DETAIL SEE ENG. CASEMODS FOR 216-0400.
10. PRIOR TO THE INSTALLATION OF THE WELL, THE DEPTH OF THE SOCKET SHALL BE MEASURED TO ASSURE THAT 10" RING WILL BE COMPRESSED AFTER THE WELL HAS BEEN INSERTED AND TORQUED TO CREATE A TIGHT SEAL AND SHALL MAINTAIN A CLEARANCE OF .005" TO .030" BETWEEN THE SHOULDER OF THE WELL AND THE SHOULDER OF THE BORE.
11. THE AS-BUILT INFORMATION ON SHEET 2 & 3 IS SUPPLIED BY G.P.C.
12. VALVE TO ADAPTOR WELD TO BE PERFORMED PER PROJECT CLASS 2¼ IN ACCORDANCE WITH CASEMOD#1-7 NOTE #8.
13. (PSR) DENOTES PIPE BREAK RESTRAINT LOCATION FOR SURGE LINE.
14. BEGINNING APPROX 2'-5" NORTHEAST OF SUPPORT V2-1201-053-H002 ON LINE #1201 053-16" INSTALL 1½" THK. MIN-K INSULATION FOR A LENGTH OF 1'-2". CENTER THE MIN-K ON THE ADJACENT JUNCTION BOX 2HE53MA0179.
15. FOR LOCATION OF RID ON PRESSURIZER SURGE LINE, SEE SHEET 2 OF 3 DETAIL A.
16. ALL PRESSURIZER NOZZLES HAVE A WELD OVERLAY APPLIED PER WCAP 8768(DRAFT).
17. NCIP HAS BEEN APPLIED TO LINE 001, 002, 003, AND 004 NEAR THE REACTOR VESSEL NOZZLE.

INSTALLERS NOTES:
1. CLEAN ASS'Y PRE CLASS "B" REQUIREMENT.
2. DESIGN TEMPERATURE = 650 F°
3. INSULATION THICKNESS:
LINES P-1201-001,002,003,004-29"I.D. & P-1201-005-14" &16" I.D. = 4.5"
LINES P-1201-009,010,011,012,-275"I.D. IS 3.0"
P-1201-009,010,011,012,-275"I.D. IS 3.0"
4. FEDCR'S INCORP: 4F-10301, 4F-10290.
5. THIS DWG. IS GOVERNED BY PROJECT PROCEDURE ZI-64 CLOSURE SYSTEMS.
6. FEDCR'S INCORP: 4F-10463 & 4F-10350

PIPESUPPORT LIST		
WDEK	SUPPORT TAG NO.	
10	V2-1201-053-H001	
29	V2-1201-053-H002	
30	V2-1201-053-H003	DELETE
28	V2-1201-053-H004	
49	V2-1201-053-H005	
50	V2-1201-053-H006	

BRANCH NOZZLES							
IDENT NO.	NO. REQD.	SIZE	SERVICES	IDENT NO.	NO. REQD.	SIZE	SERVICES
1	2	¾	SAMPLE CONNECTIONS	11	2	3"	CVCS CHARGING LINE
2	16	¾	PLUM MEASUREMENT TAP	12	2	2"	PRESSURIZER SPRAY NOR.
3	12	¾	RTD THERMOWELL	13	2	6"	R.H.R. S.I.S. INJECTION
4	1	3"	LETDOWN (CVCS)	14	4	10"	ACCUMULATOR TANK INJECTION
5	4	3"	BORON INJECTION TANK (SIS)	15	2	12"	EMRG RECIRC/SIS INJECTION
6	2	3/4"	PRESSURE TAP	16	1	16"	PRESSURIZER SURGE LINE
7				17	8	2½	THERMOWELL
8				18	4	2"	REACTOR COOLANT LOOP DRAIN
9	4	2"	RTD THERMOWELL				
10	4	2"	WELED PIPE CAP				

THIS DRAWING IS REQUIRED FOR W PROOF OF DESIGN

THIS DRAWING WAS RENUMBERED FROM 2X4DL4A17 (SHEET 1)
WORK THIS DRAWING WITH: 2X4DL4A017-2, 2X4DL4A017-3
REF. DWGS. 1X2D48A001 & 2X2D48A001

ACTIVITY PKG#5-2-2101-B-044-A
UNIT 2

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

CONTAINMENT BLDG. PIPING AREA 4A,B,C,&D
LEVEL B- PLAN & SECTIONS
REACTOR COOLANT LOOPS

SCALE: NONE
DRAWING NO. 2X4DL4A017-1
VER. 18.0
JOB NO. 10604
SIZE E 34X44

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NO.	VERSIONS	DATE	DR	CHK	APPV
18.0	REVISED PER ABN SNC526823W003, VER 1.0	1/4/15	LCH	ALM	MHD

PLAN DIM. FROM REF. DWG.

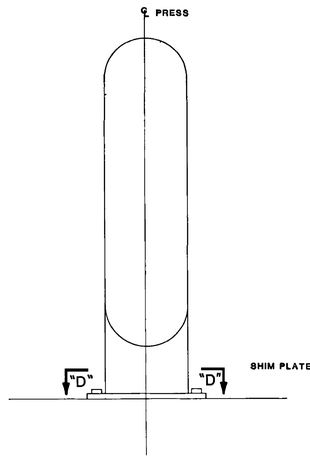
N

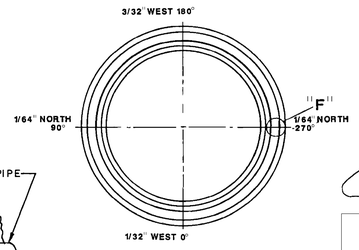
REATOR COOLANT PUMP CASING ELEV. & LOC.					
STA	PLAN	LOOP1	LOOP2	LOOP3	LOOP4
PLAN	N/A	189'-7.500"	189'-7.500"	189'-7.500"	189'-7.500"
A	N/A	189'-7.308"	189'-7.584"	189'-7.524"	189'-7.548"
A1	N/A	N/A	N/A	N/A	N/A
B	N/A	189'-7.308"	189'-7.572"	189'-7.524"	189'-7.560"
B1	N/A	N/A	N/A	N/A	N/A
C	N/A	189'-7.308"	189'-7.572"	189'-7.524"	189'-7.524"
C1	N/A	N/A	N/A	N/A	N/A
HORZ. LOC. NORTH SOUTH	25'-10⅞"	-25'-11⅜"	+25'-10⅞"	+25'-10⅝"	-25'-11⅞"
HORZ. LOC. EAST WEST	29'-6"	-29'-5⅞"	+29'-5⅜"	+29'-5¼"	+29'-5⅝"
AZ	48° 42'-42"	48° 40'-21"	131° 18'-14"	228° 40-25"	311° 22'-13"
FIELD BOOK	———	SC2-30-68	SC2-30-60	SC2-30-35	SC2-30-44
FIELD BOOK		SC2-30-49	SC2-30-50	SC2-30-37	SC2-30-45

+ NORTH / EAST FROM ℄ R.V.
- SOUTH / WEST FROM ℄ R.V.

STEAM GENERATOR ELEV. LOC. PLUM.					
STA	PLAN	LOOP1	LOOP2	LOOP3	LOOP4
PLAN	N/A				
W	N/A	195'-5.496"	195'-5.172"	195'-5.011"	195'-5.220"
X	N/A	195'-5.496"	195'-5.184"	195'-4.996"	195'-5.184"
Y	N/A	195'-5.568"		195'-5.002"	195'-5.256"
Z	N/A	195'-5.568"	195'-5.124"	195'-5.052"	195'-5.520"
HORZ. LOC. NORTH SOUTH	33'-1½"	-33'-1⅜"	-33'-1"	+33'-1"	-33'-1⅜"
HORZ. LOC. EAST WEST	13'-4⅝	-13'-4⅜"	-13'-5⅝"	+13'-4⅞"	+13'-4⅜"
PLUM NORTH SOUTH	0	-1⅜"	-1/8"	+3/8"	+3/8"
PLUM EAST WEST	0	-2⅜"	+3/4"	-3/16"	+5/8"
AZ	22° 00'-00"	22° 01'-23"	157° 53'-12"	202° 00'-17"	337° 59'-16"
FIELD BOOK	———	SC2-30-64	SC2-30-55	SC2-30-25	SC2-30-42
				SC2-30-38	SC2-30-48

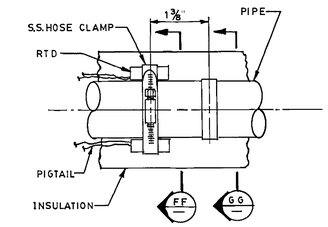
+ NORTH / EAST FROM ℄ R.V.
- SOUTH / WEST FROM ℄ R.V.

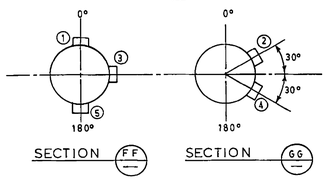


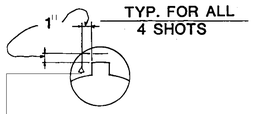


@ CORE SUPPORT LEDGE

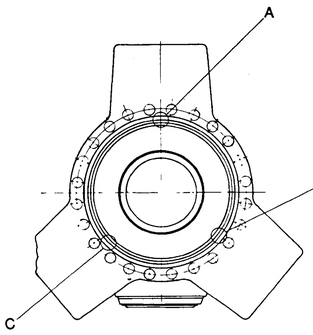
SECTION "A"-"A"



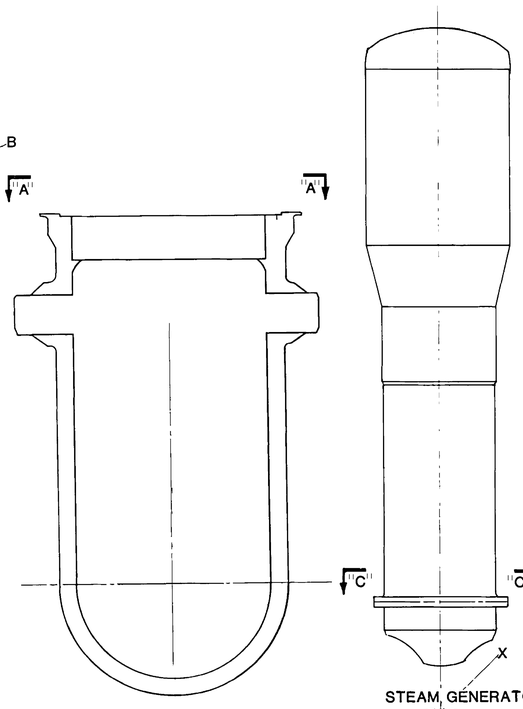




ELEVATIONS
0° - 192" - 11.037"
90° - 192' - 11.052"
180° - 192' - 11.041"
270° - 192' - 11.051"
AVG. PLAN - 192' - 11.013"

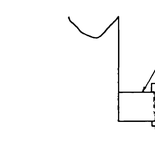


SECTION "B"-"B"



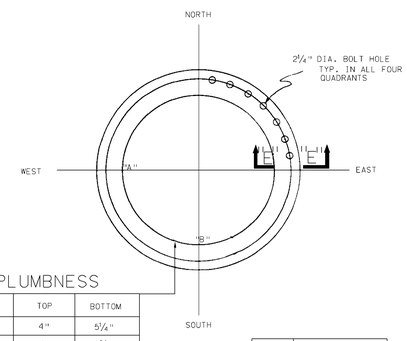
REACTOR VESSEL

SECTION "C"-"C"



SECTION "E"-"E"

DETAIL 1
NOTE SHOTS WERE TAKEN ON CENTER LINE OF BOLTS. ON TOP OF S.G. SUPPORT RING



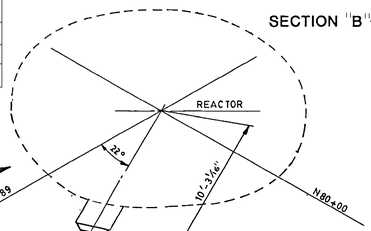
PLUMBNESS

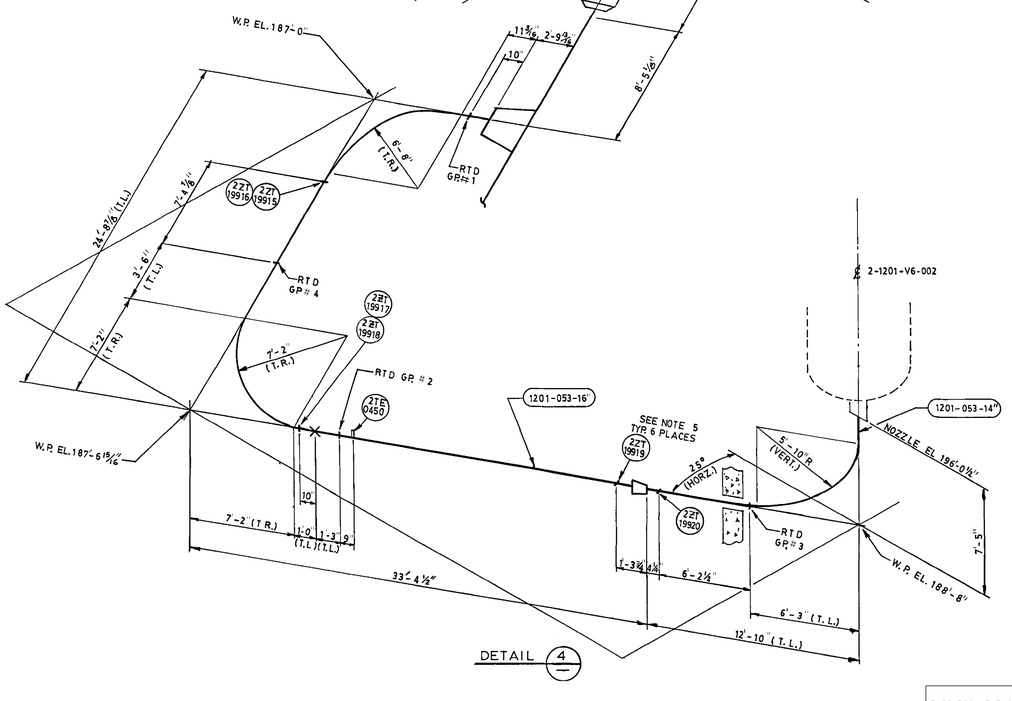
STA.	TOP	BOTTOM
"A"	4"	5¼"
"B"	4"	4⅛"
ERROR	WEST 1¼"	SOUTH ⅞"

PLAN	- 195.958
NORTH	- 195.948
WEST	- 195.931
SOUTH	- 195.932
EAST	- 195.948

SECTION "D"-"D"

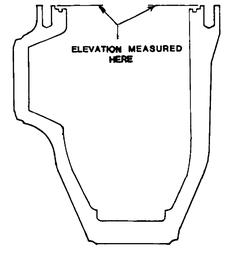
T.E. # FOR PRESSURIZER SURGE LINE					
RTD LOCATION GROUP#	1	2	3	4	5
1	2TE 27768	2TE 27769	2TE 27770	2TE 27771	2TE 27772
2	2TE 27783	2TE 27784	2TE 27785	2TE 27786	2TE 27787
3	2TE 27793	2TE 27794	2TE 27795	2TE 27796	2TE 27797
5	2TE 27773	2TE 27774	2TE 27775	2TE 27776	2TE 27777





DETAIL 4

ELEVATION CHECK



REACTOR COOLING PUMP CASING
TYP. 3 PLACES

NOTES. 1) DETAIL 2 IS TYPICAL FOR ALL FOUR (4) B TEAM GENERATORS.
2) DETAIL 3 IS TYPICAL FOR ALL FOUR (4) REACTOR COOLANT PUMP CASING.
3) RTD GROUP # FOR REFERENCE ONLY.
4) RTD INSTALLATION TOLERANCE ±6".
5) LANYARDS FOR 2ZT-19915, 2ZT-19916, 2ZT-19917, 2ZT-19918, 2ZT-19919, & 2ZT-19920 MANUFACTURE BY GENISCO TECHNOLOGY CORPORATION MODEL: PT-10A-100G (WITH TERMINAL STRIPS).

THIS DRAWING WAS RENUMBERED FROM 2X4DL4A17 (SHEET 2)
WORK THIS DRAWING WITH: 2X4DL4A017-1, 2X4DL4A017-3

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NO.	REVISIONS	DATE	DR	CHK	APPV	DTL
4	INCORP. PER ABN-38664	04-28-03	BgT	LRP	ASK	X

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

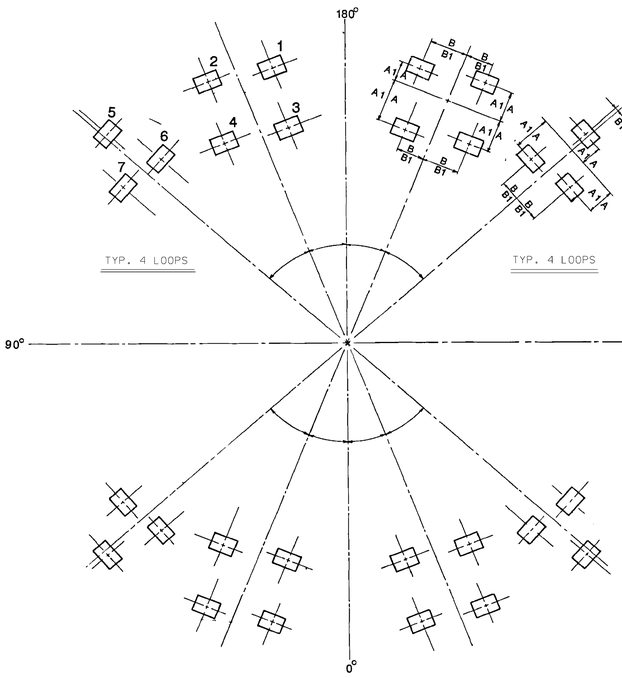
GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

CONTAINMENT BLDG.
PIPING AREA 4A,B,C,D, LEVEL B
PLAN & SECTION REACTOR COOLANT LOOP

SCALE: NONE	DRAWING NO.	REV.
JOB NO.10604	2X4DL4A017-2	16

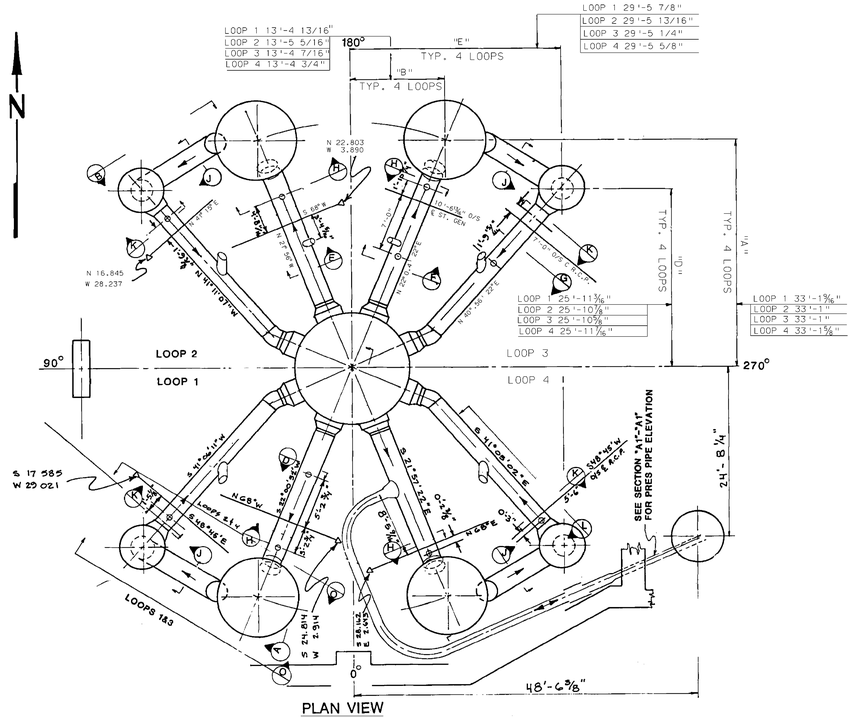
SIZE E 34X44

CAD NAME 2X4DL4A0172

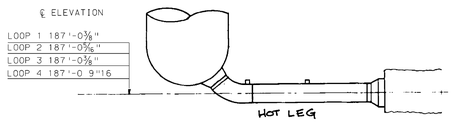


BASE PLATE GENERAL ARRANGEMENT

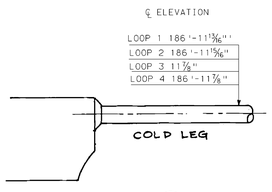
FOR BASE PLATE DETAILS SEE DWGS.



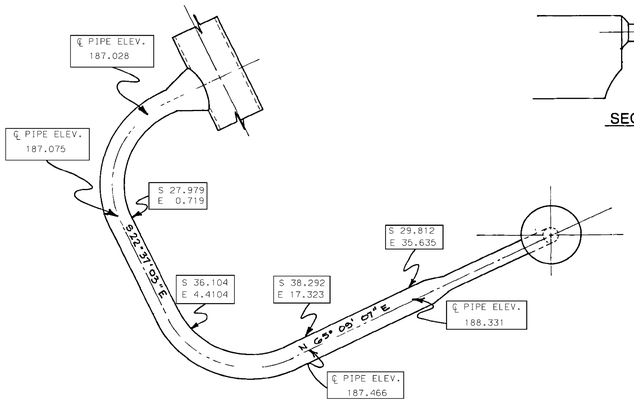
PLAN VIEW

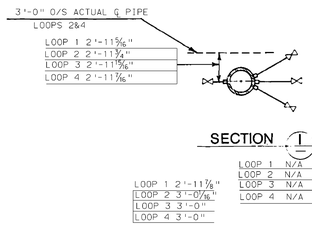


SECTION A

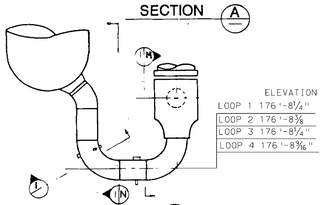


SECTION L

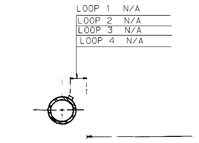




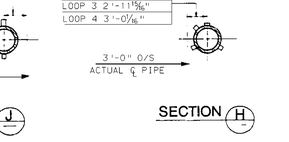
SECTION I



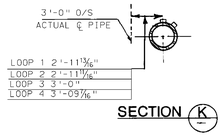
SECTION B



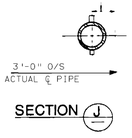
SECTION C



SECTION G

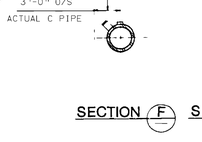


SECTION K

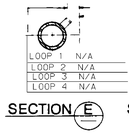


SECTION J

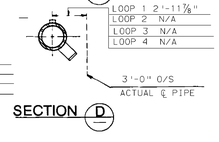
SECTION H



SECTION F

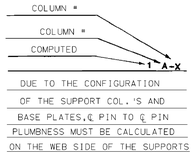


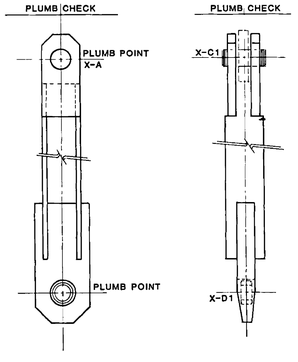
SECTION E



SECTION D

SECTION A1

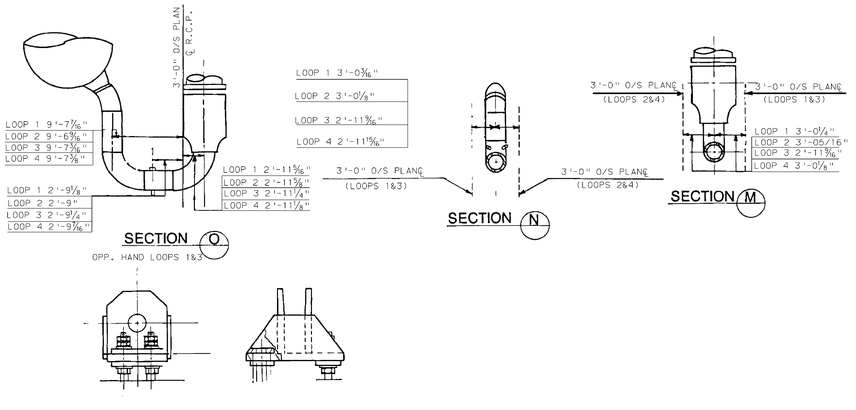




VERTICAL COLUMNS

S.G. – 4 REQD. PER LOOP

R.C. PUMP – 3 REQD. PER LOOP



SECTION Q

OPP. HAND LOOPS 1&3

SECTION N

SECTION M

R.C. PUMP & S.G COLUMN BASE

7 BASE REQD. PER LOOP

BASE PLATES LOCATION & EL.

LOOP #	PLATE #	ACTUAL "A"	PLAN "A"	ACTUAL "B"	PLAN "B"	ACTUAL EL.	PLAN EL.	FIELD BOOK
1	1	5'-9"	3'-9"	5'-0 5/16"	5'-0 3/16"	172.704	172.667	SC2-30
1	2	5'-0 5/16"	5'-0 3/16"	3'-8 7/8"	3'-9"	172.687		
1	3	5'-1 1/16"	5'-0 3/16"	3'-9 1/8"	3'-9"	172.741		
1	4	3'-9"	3'-9"	5'-0 5/16"	5'-0 3/16"	172.719		
1	5	4'-2 11/16"	4'-2 9/16"	0'-6 1/16"	0'-6 1/16"	172.660		
1	6	3'-9 3/4"	3'-7 9/16"	2'-9 13/16"	2'-9 7/8"	172.664		
1	7	2'-3 7/8"	2'-3 5/8"	3'-7"	3'-7"	172.684		
2	1	3'-9 7/8"	3'-9"	5'-0 13/16"	5'-0 13/16"	172.728		
2	2	5'-1 1/16"	5'-0 3/16"	3'-9 3/8"	3'-9"	172.744		
2	3	5'-0 5/16"	5'-0 7/16"	3'-9"	3'-9"	172.768		
2	4	3'-9"	3'-9"	5'-0 11/16"	5'-0 13/16"	172.692		
2	5	4'-2 9/16"	4'-2 7/16"	0'-6 3/4"	0'-6 11/16"	172.677		
2	6	3'-9 9/16"	3'-7 5/16"	2'-9 5/8"	2'-9 7/8"	172.678		
2	7	2'-3 1/16"	2'-3 1/8"	3'-6 13/16"	3'-7"	172.670		
3	1	3'-9 7/16"	3'-9"	5'-0 15/16"	5'-0 13/16"	172.712		
3	2	5'-0 15/16"	5'-0 13/16"	3'-9"	3'-9"	172.712		
3	3	5'-0 9/16"	5'-0 13/16"	3'-8 13/16"	3'-9"	172.697		
3	4	3'-8 11/16"	3'-9"	5'-0 5/8"	5'-0 13/16"	172.706		
3	5	4'-2 1/16"	4'-2 9/16"	0'-6 11/16"	0'-6 11/16"	172.669		
3	6	3'-10 1/2"	3'-7 9/16"	2'-9 7/8"	2'-9 7/8"	172.672		
3	7	2'-3 1/4"	2'-3 3/8"	3'-7 1/16"	3'-7"	172.689		
4	1	3'-8 13/16"	3'-9"	5'-0 11/16"	5'-0 13/16"	172.715		
4	2	5'-0 13/16"	5'-0 11/16"	3'-9 1/16"	3'-9"	172.734		
4	3	5'-0 13/16"	5'-0 11/16"	3'-9 1/16"	3'-9"	172.722		
4	4	3'-9 3/16"	3'-9"	5'-0 5/8"	5'-0 13/16"	172.722		
4	5	4'-1 13/16"	4'-2 9/16"	0'-6 13/16"	0'-6 11/16"	172.673		
4	6	3'-9 13/16"	3'-7 9/16"	2'-9 13/16"	2'-9 7/8"	172.697		
4	7	2'-3 7/8"	2'-3 3/8"	3'-6 15/16"	3'-7"	172.682	▼	▼

SUPPORT-LOCATION & PLUMBNESS

LOOP #	COL. #	ACTUAL A1	PLAN A1	ACTUAL B1	PLAN B1	DIST TOP TWRD. VES.	DIST TOP FROM VES.	DIST TOP EAST	DIST TOP WEST	FIELD BOOK
1	1	N/A	N/A	N/A	N/A	1 11/16"			7/16"	SC2-30
1	2					2 1/8"			15/16"	
1	3					1 5/8"			1/8"	
1	4					2 5/16"			3/8"	
1	5					2"			1/2"	
1	6						3 1/2"	3/4"		
1	7					1 13/16"			5/16"	
2	1					2 1/16"			13/16"	
2	2					2 3/16"		13/32"		
2	3					2 13/16"			13/32"	
2	4					1 11/16"			13/32"	
2	5					2 1/16"			7/16"	
2	6						3 3/16"		5/8"	
2	7					1 1/2"			1/16"	
3	1					2 1/2"		1/8"		
3	2					2 7/8"		5/8"		
3	3					2 5/32"			3/16"	
3	4					3 1/8"		1/2"		
3	5					2"			3/8"	
3	6						3 1/2"		5/8"	
3	7					1 3/4"		1/2"		
4	1					1 1/2"			1/2"	
4	2					1 5/16"		3/8"		
4	3					1 1/2"			1/4"	
4	4					1 11/16"		3/16"		
4	5					3/4"			1/2"	
4	6						3"		1"	
4	7	▼	▼	▼	▼	1 3/4"		0	—	▼

NOTE:

1. DENOTES ASBUILT DIMENSION.
2. A-MEASURED TO BASE PLATE A1-MAESURED TO COL.
3. FROM VESSEL OR TOWARD VESSEL, REFERS TO REACTOR VESSEL.
4. N/A: NOT APPLICABLE
5. O/S: OFF SET

THIS DRAWING WAS RENUMBERED
FROM 2X4DL4A17 (SHEET 3)

WORK THIS DRAWING WITH:
2X4DL4A017-1, 2X4DL4A017-2

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NO.	REVISIONS	DATE	DR	CHK	APPV	DTL
4	INCORP. PER ABN-38664	04-28-03	BgT	LRP	ASK	X

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

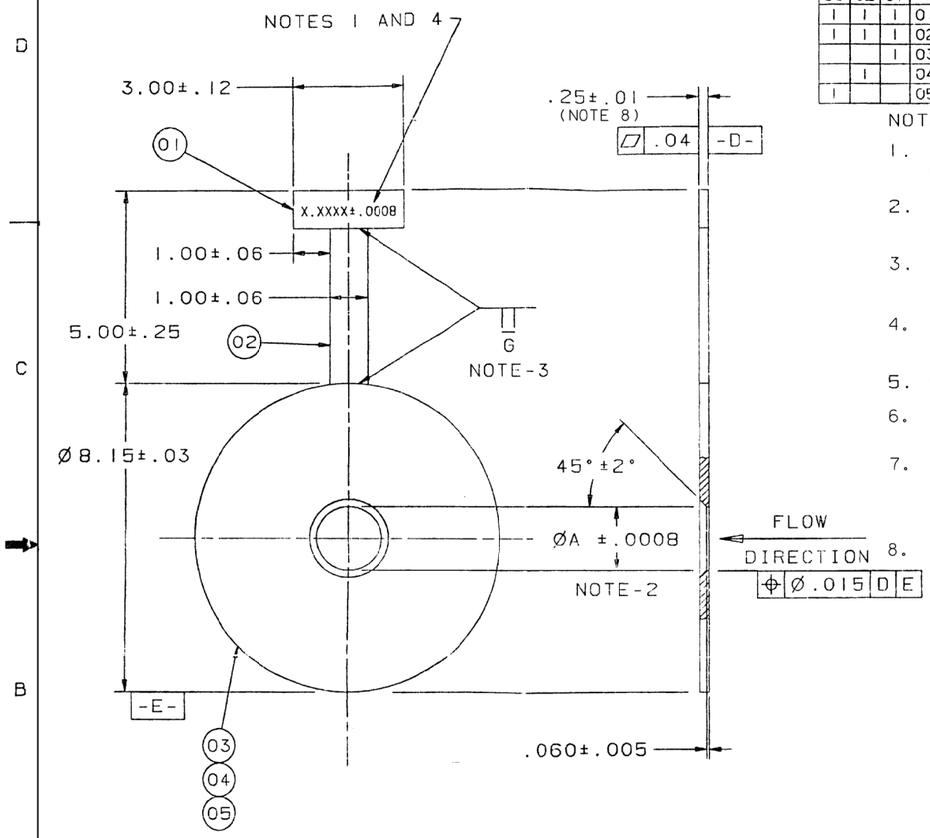
GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

CONTAINMENT BLDG.
PIPING AREA 4A,B,C,D, LEVEL B
PLAN & SECTION REACTOR COOLANT LOOP

SCALE: NONE	DRAWING NO.	REV.
JOB NO.10604	2X4DL4A017-3	16

SIZE E 34X44

CAD NAME 2X4DL4A0173



IDENT CLASS							
GROUP NOTE							
REQD 03	REQD 02	REQD 01	ITEM	NOTE IDENT	PART NAME	(SIZE) REFERENCE INFORMATION	DEF	MATL SIZE CODE PART NUMBER OR REF DWG
1	1	1	01		NAME PLATE	ASME SA 240 304-316 SST.		
1	1	1	02		HANDLE	ASME SA 240 304-316 SST.		
		1	03		DISK	ASME SA 240 304-316 SST.		
	1		04		DISK	ASME SA 240 304-316 SST.		
1			05		DISK	ASME SA 240 304-316 SST.		

BILL OF MATERIAL

NOTES:

1. IDENTIFICATION CHARACTERS MAY BE STAMPED OR ETCHED APPROX. .25 HIGH.
2. INLET EDGE AT DIAMETER "A" MUST BE SQUARE, SHARP, AND FREE FROM BURRS, NICKS OR ROUNDING.
3. OTHER MEANS OF ATTACHMENT MAY BE USED PROVIDED PRIOR W APPROVAL IS GIVEN.
4. NAMEPLATE SHALL IDENTIFY ORIFICE SIZE, ØA INCLUDING TOLERANCE, GRADE, SERIAL NUMBER, AND HEAT NUMBER. (2.0)
5. WESTINGHOUSE SPIN NO. - CSORCP (GAE).
6. FOR GROUP B1, B2, B3 AND B4 USE ITEMS 1, 2 AND 3. THESE GROUPS MANUFACTURED ON SITE.
7. EITHER A SERRATED CONCENTRIC OR SERRATED SPIRAL FINISH HAVING A RESULTANT SURFACE FINISH FROM 125 μIN. TO 250 μIN. AVERAGE ROUGHNESS SHALL BE FURNISHED FOR THE GASKET SEATING SURFACES.
8. FOR 2FO-10123 THE PLATE THICKNESS IS 0.625" ± 0.035"

GROUP	A
01	1.4523
02	1.4826
03	1.5167
04	1.2931
05	1.2627
06	1.2356
B1	1.3986
B2	1.3674
B3	1.3383
B4	3.4380 W/O TAPER

INSTRUMENT TAG NUMBER	GROUP
FO 10123	02
FO 10122	B4

SO. GAE-205
D. 3DS090-01
IT.
REVISION

TOLERANCE & MACHINE NOTES
(UNLESS OTHERWISE SPECIFIED)

DRAWING PRACTICES, GEOMETRIC SYMBOLS, DIMENSIONING, CLEARANCING & INTERPRETATION BASED ON ANSI Y14 SERIES STANDARD & PS 80400HA.

DIMENSIONS IN INCHES BASED ON 68°F(20°C)
RADIUS OR CHAMFER ALL EDGES-----.005-.030
FILLET RADII-----.03±.01
ANGULARITY: CHAMFERS-----±2°

MAXIMUM SURFACE ROUGHNESS-250μIN. (√)AA

SPIN: DWG. REF.
NEXT ASSY.
THIRD ANGLE PROJECTION

WESTINGHOUSE PROPRIETARY DATA

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DFTM MILLSLAGLE 10/29/91
CHKR
DES ENGR
MFG ENGR
QA ENG
APPD
APP
APP
APP
APP
DFTG MGR 10-29-91
SCALE
WEIGHT-LBS.

INTERACTIVE GRAPHICS
A-MILL-8251C26-01-001-102991

Westinghouse Electric Corporation
WATER REACTOR DIVISIONS - MONROEVILLE, PA. U.S.A.

TITLE
GAE CHARGING PUMP
FLAT PLATE ORIFICES
ORIFICE ASSEMBLY

DIMENSIONS IN INCHES
SIZE | DSM NO. C | DWG NO. 8251C26 | REV. D
SHT 1 OF 1 SHTS

THIS DWG. IS REFERENCED IN VENDOR MANUAL: N/A
TAB/SECT. N/A
PAGE N/A
FIGURE N/A

VERSION 2.0 | DATE 01/13/14

REVISED BY SNC PER ABN-V03188, VER 1.0

SEE MICROFILM FOR PREVIOUS REV. SIGNATURES.

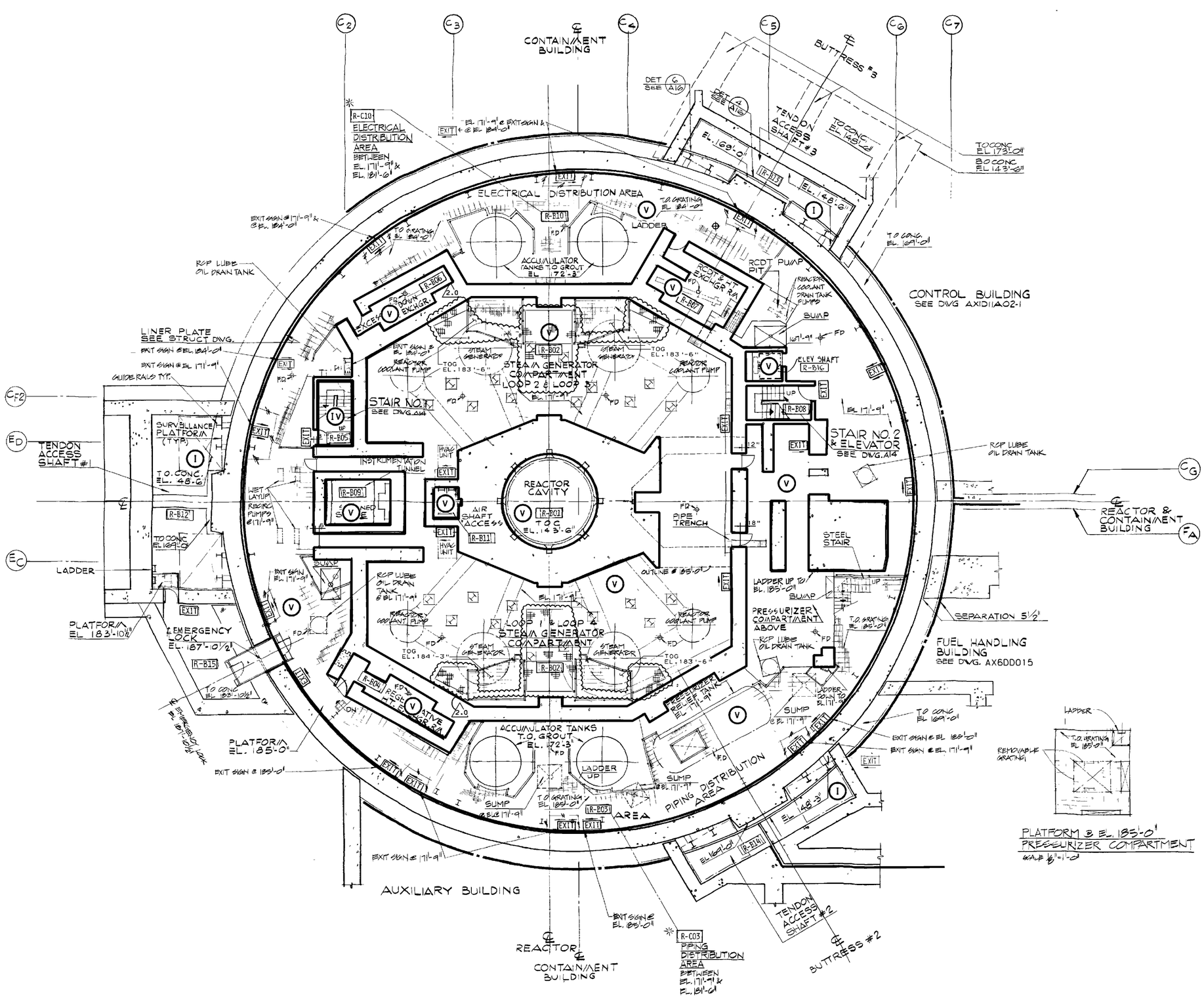
BY	CHK'D	APPR.1	APPR.2
DCP	ELC	JMR	

CAD MICROSTATION DCP/01
2X6AH02-30000.DGN

Southern Nuclear Operating Company, Inc.
FOR
VOGTLE ELECTRIC GENERATING PLANT
UNIT NO. 2

TITLE:
GAE CHARGING PUMP
FLAT PLATE ORIFICES
ORIFICE ASSEMBLY

VENDOR: WESTINGHOUSE | P.O.#: PAV2-2
DRAWING NO. 2X6AH02-30000
SIZE D



ACCESSIBILITY ZONES DURING OPERATIONS		
ZONE	EXPOSURE MREM/HR	OCCUPANCY HRS/WK
I	< 0.25	UNLIMITED
II	0.25-2.50	168-40
III	2.50-15	40-6
IV	15-100	6-1
V	> 100	<1
		ZONE BOUNDARY

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

RAD ZONES ACCESSIBILITY DURING OPERATIONS
CONTAINMENT BUILDING
FLOOR PLAN EL 171'-9"
(UNIT 2)

SCALE: NONE

DRAWING NO. 2X6DD001

VER. 2.0

JOB NO. 10604

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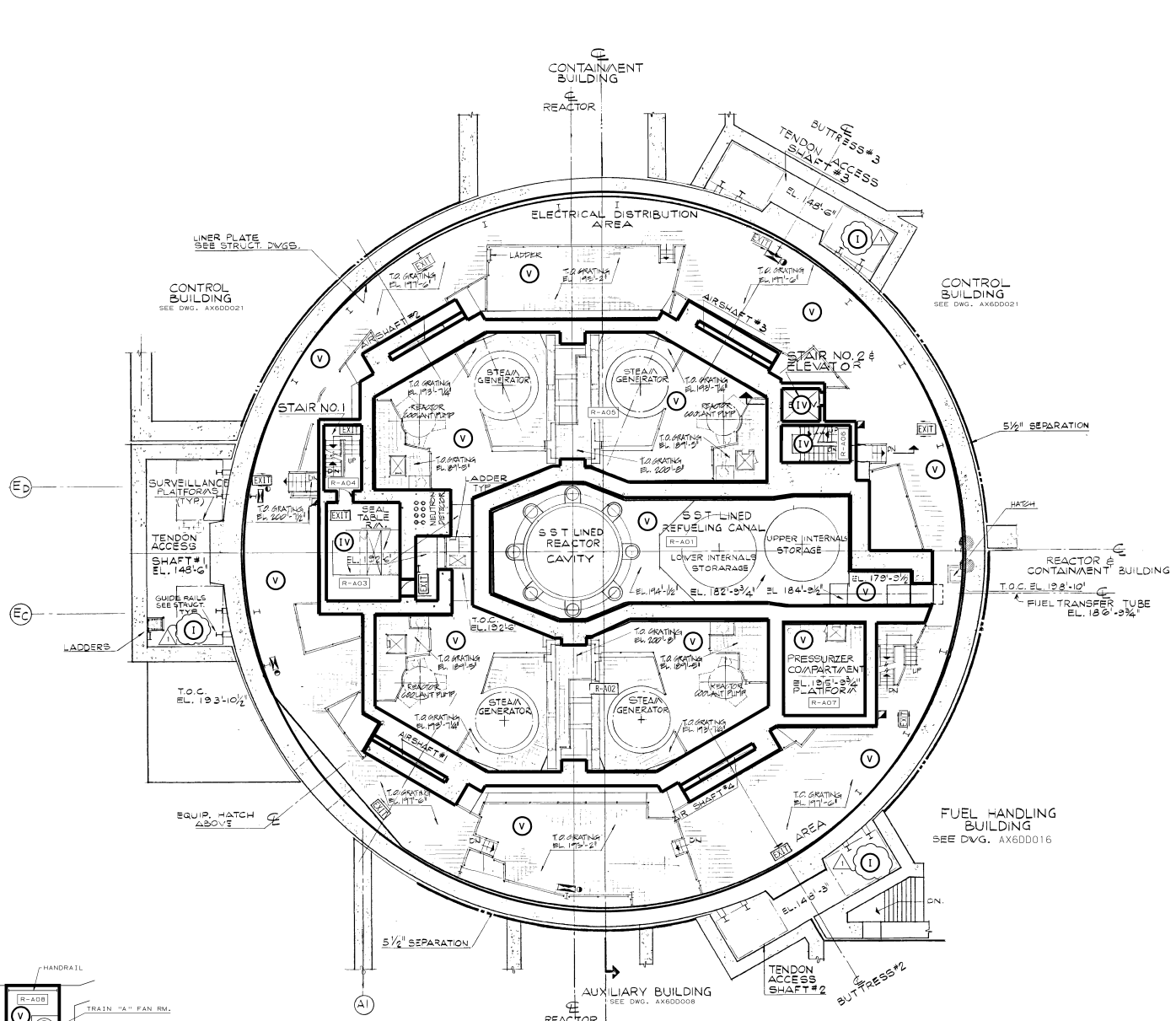
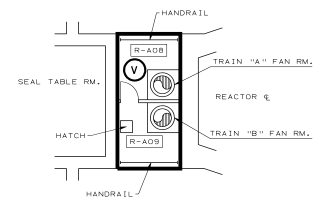
NO.	VERSIONS	DATE	DR	CHK	APPV
2.0	REVISED PER ABN 20290013O1C012, VER.	11-18-05	vep	CFC	DEW

SIZE E 34x44





FAN ROOM PLAN @ EL. 206'-0" (1)

ACCESSIBILITY ZONES DURING OPERATIONS		
ZONE	EXPOSURE MREM/HR	OCCUPANCY HRS/WK
I	< 0.25	UNLIMITED
II	0.25-2.50	168-40
III	2.50-15	40-6
IV	15-100	6-1
V	> 100	<1
		ZONE BOUNDARY

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

RAD ZONES ACCESSIBILITY DURING OPERATIONS CONTAINMENT BUILDING FLOOR PLAN EL. 195 FT.-0 IN. (UNIT 2)

SCALE: NONE | DRAWING NO. 2X6DD002 | REV. 1

JOB NO. 10604

NO.	REVISIONS	DATE	DR	CHK	APPV	DTL
3						
2						
1	REVISED PER REA 96-VAA054	10-1-98	GS	RBH	CBH	
0	ISSUED PER REA 96-VAA054	SEE M	CROFIL	M FOR	SIGNATURES	

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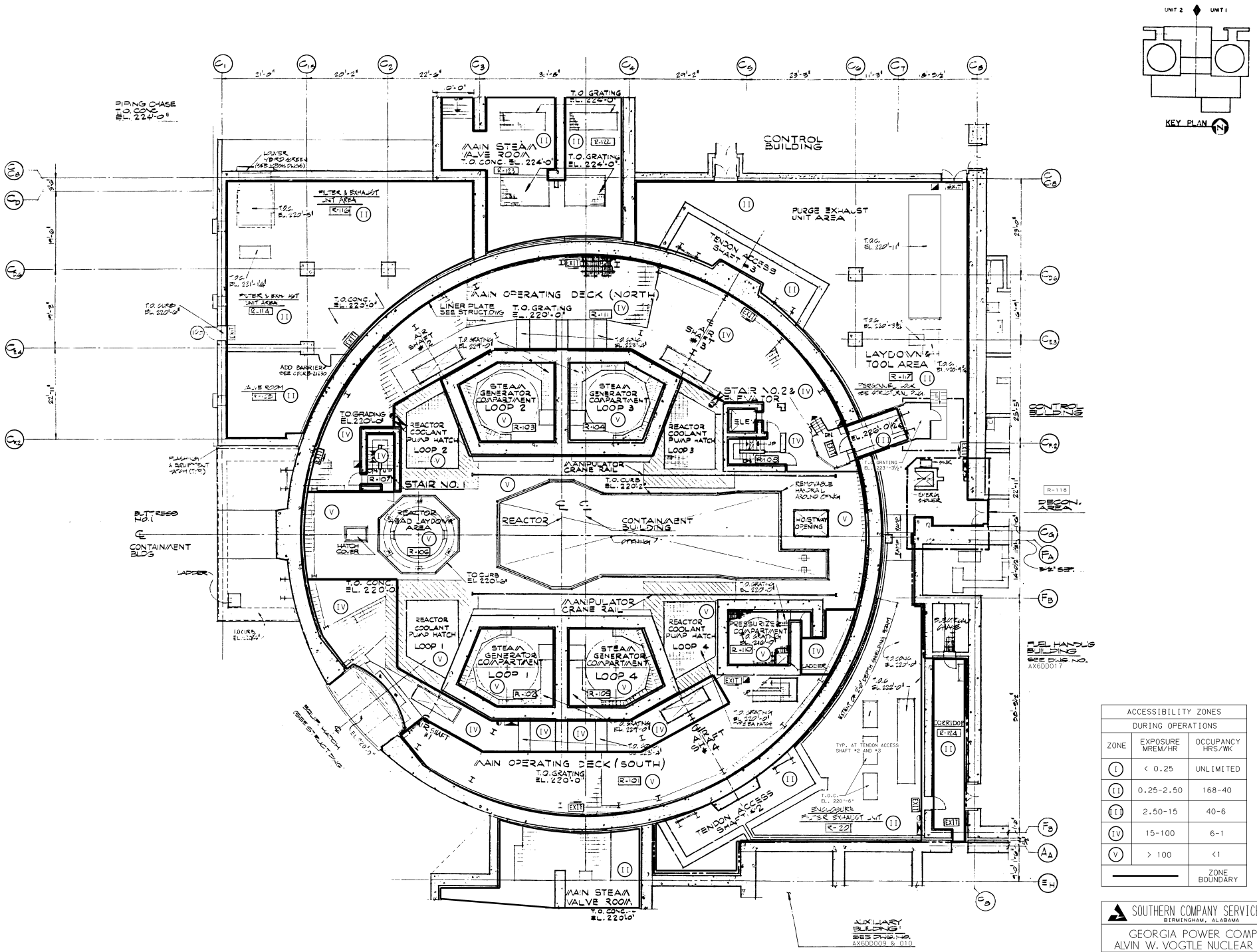
SIZE E 34X44 | CAD NAME 2X6DD002



ACCESSIBILITY ZONES DURING OPERATIONS		
ZONE	EXPOSURE MREM/HR	OCCUPANCY HRS/WK
(I)	< 0.25	UNLIMITED
(II)	0.25-2.50	168-40
(III)	2.50-15	40-6
(IV)	15-100	6-1
(V)	> 100	<1
———		ZONE BOUNDARY

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

RAD ZONES ACCESSIBILITY DURING OPERATIONS
CONTAINMENT BUILDING & EQUIPMENT BUILDING
FLOOR PLAN EL. 220 FT.-0 IN.

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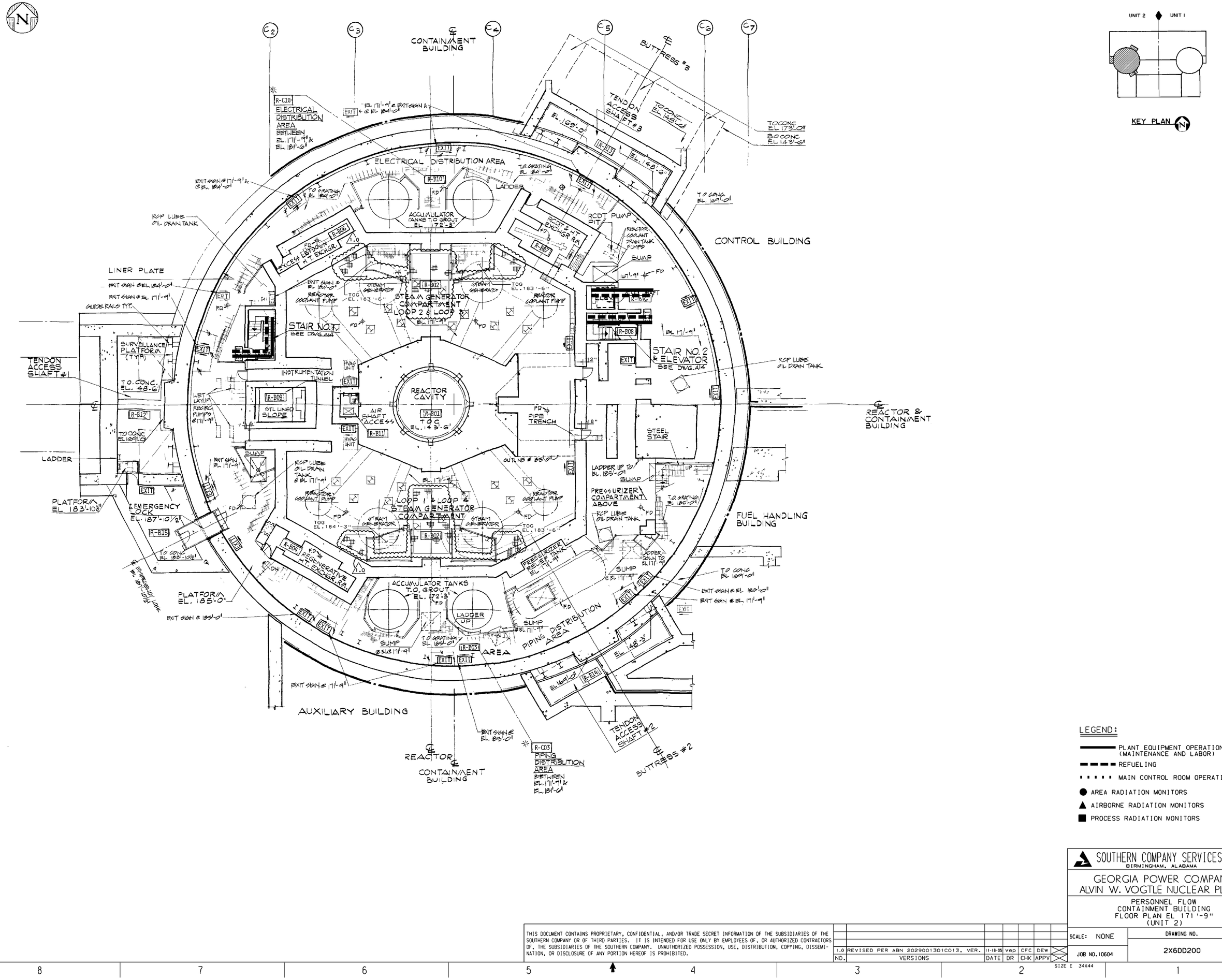
NO.	REVISIONS	DATE	DR	CHK	APPV	DTL
4						
3						
2						
1	ISSUED PER REA 96-VAA054	7-26-96	ELC	RBH	CBH	WFP

SCALE: NONE
JOB NO. 10604
DRAWING NO. 2X6DD003
REV. 0
SIZE E 34X44
CAD NAME 2X6DD003



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1.0	REVISED PER ABN 20290013010013. VER.	11-18-05	vep	CFC	DEW

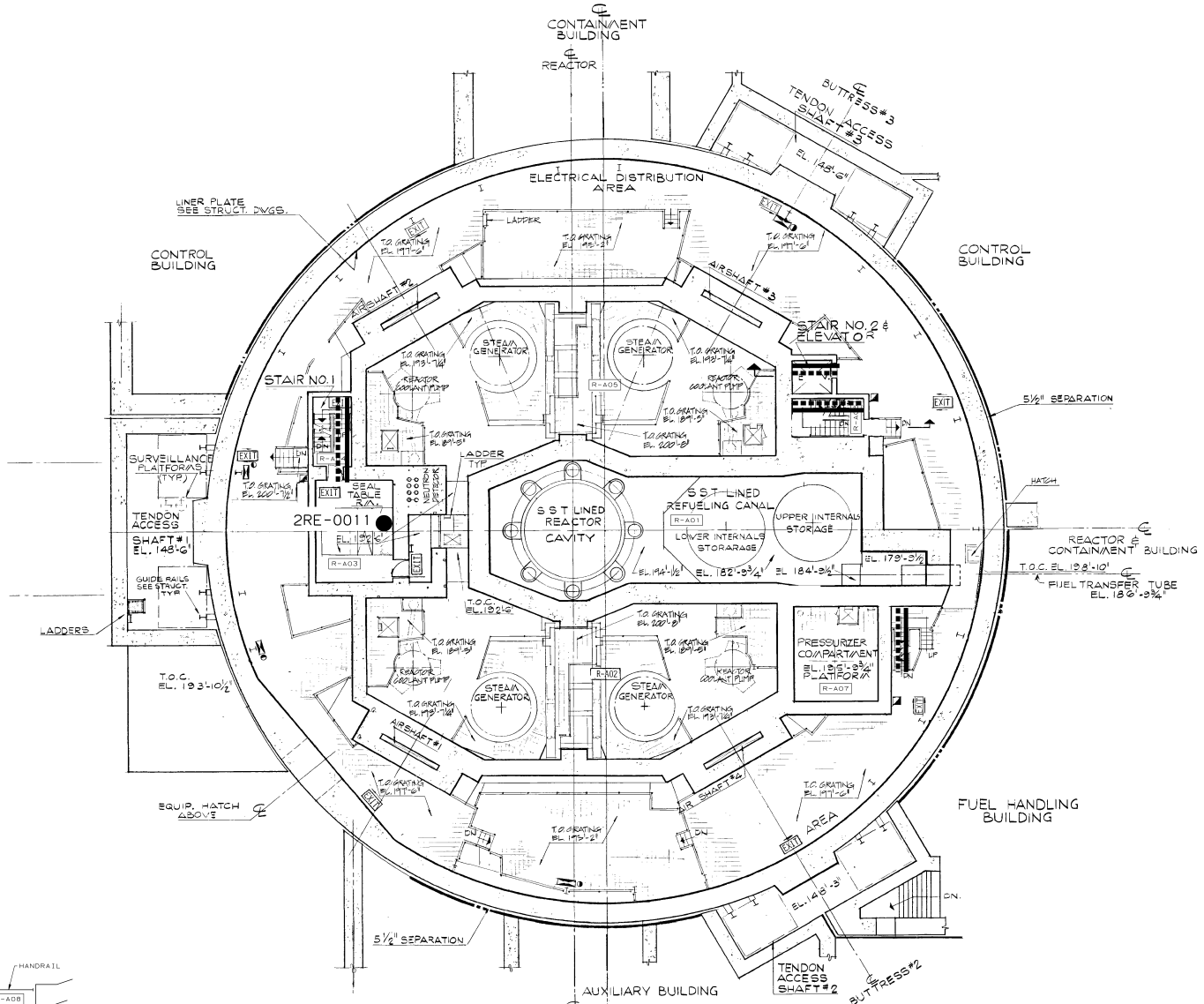
SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

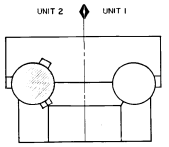
GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

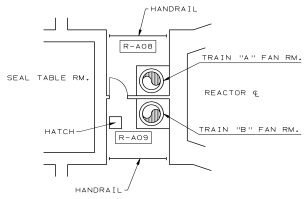
PERSONNEL FLOW
CONTAINMENT BUILDING
FLOOR PLAN EL 171'-9"
(UNIT 2)

SCALE: NONE	DRAWING NO.	VER.
JOB NO.10604	2X6DD200	1.0

SIZE E 34x44







FAN ROOM PLAN @ EL. 206'-0" (1)

LEGEND:

- PLANT EQUIPMENT OPERATIONS (MAINTENANCE AND LABOR)
- REFUELING
- MAIN CONTROL ROOM OPERATIONS
- ● AREA RADIATION MONITORS
- ▲ AIRBORNE RADIATION MONITORS
- ■ PROCESS RADIATION MONITORS

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NO.	REVISIONS	DATE	DR	CHK	APPV	DTL
4						
3						
2						
1	ISSUED PER REA 96-VAA054	7-26-96	JLH	RBH	CBH	WFP

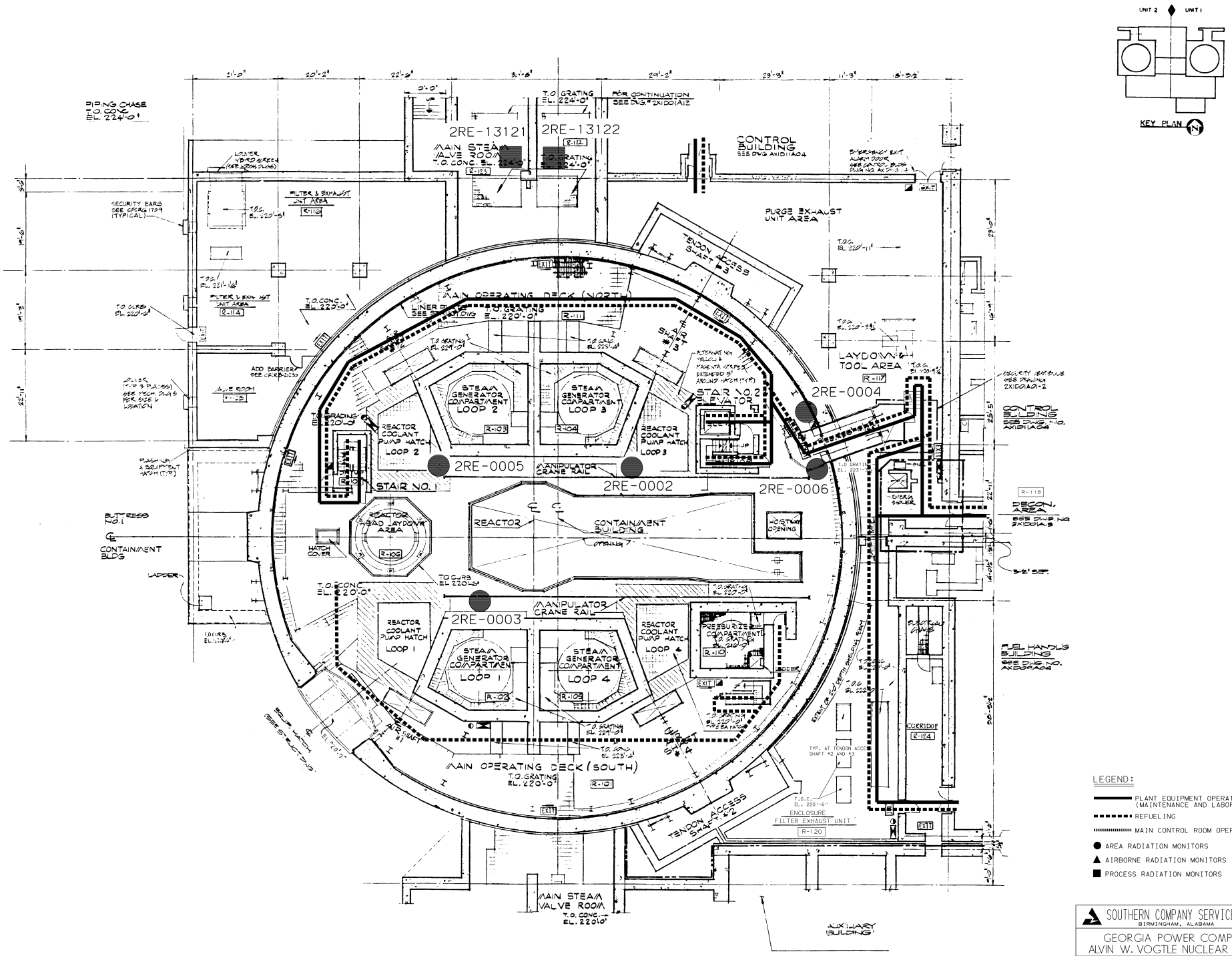
SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

PERSONNEL FLOW
CONTAINMENT BUILDING FLOOR PLAN
EL. 195' - 0" (UNIT 2)

SCALE: NONE	DRAWING NO.	REV.
JOB NO. 10604	2X6DD201	0

SIZE E 34X44
CAD NAME 2X6DD201



SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

PERSONNEL FLOW
CONTAINMENT BUILDING AND EQUIPMENT BUILDING FLOOR PLAN
EL. 220'-0" UNIT 2

SCALE: NONE	DRAWING NO.	REV.
JOB NO.10604	2X6DD202	0

SIZE E 34X44

CAD NAME 2X6DD202

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NO.	REVISIONS	DATE	DR	CHK	APPV	DTL
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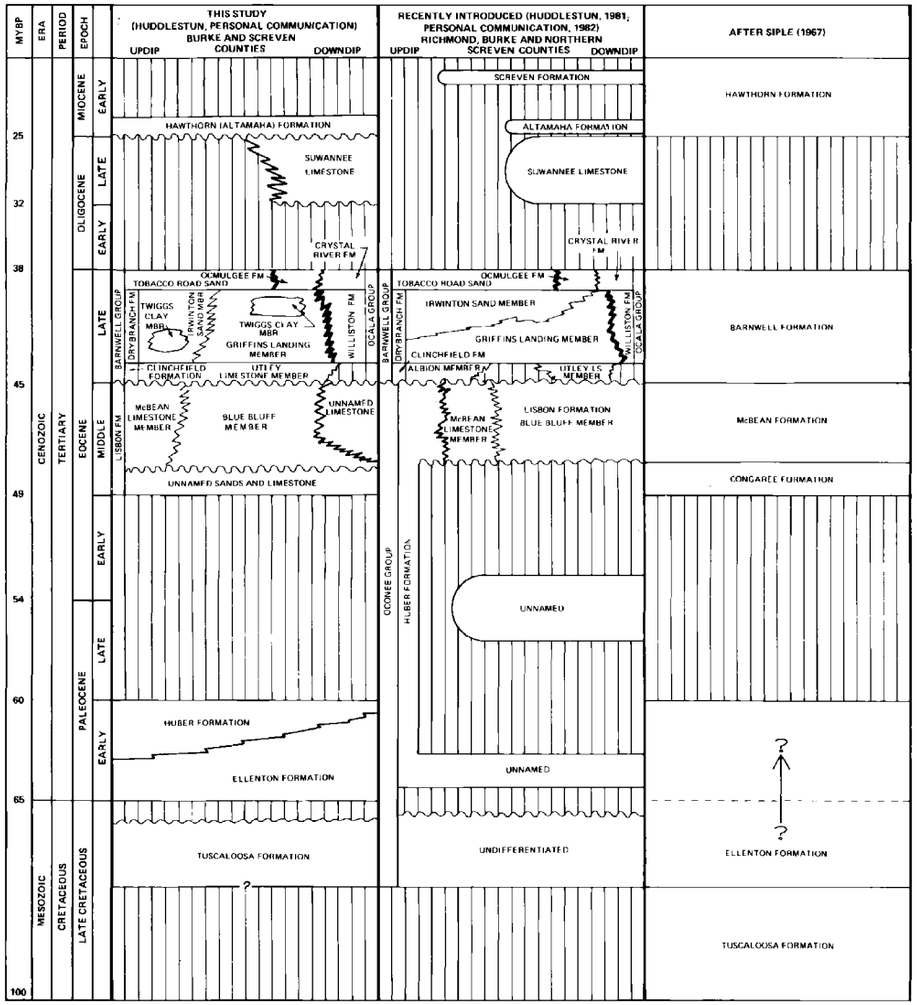
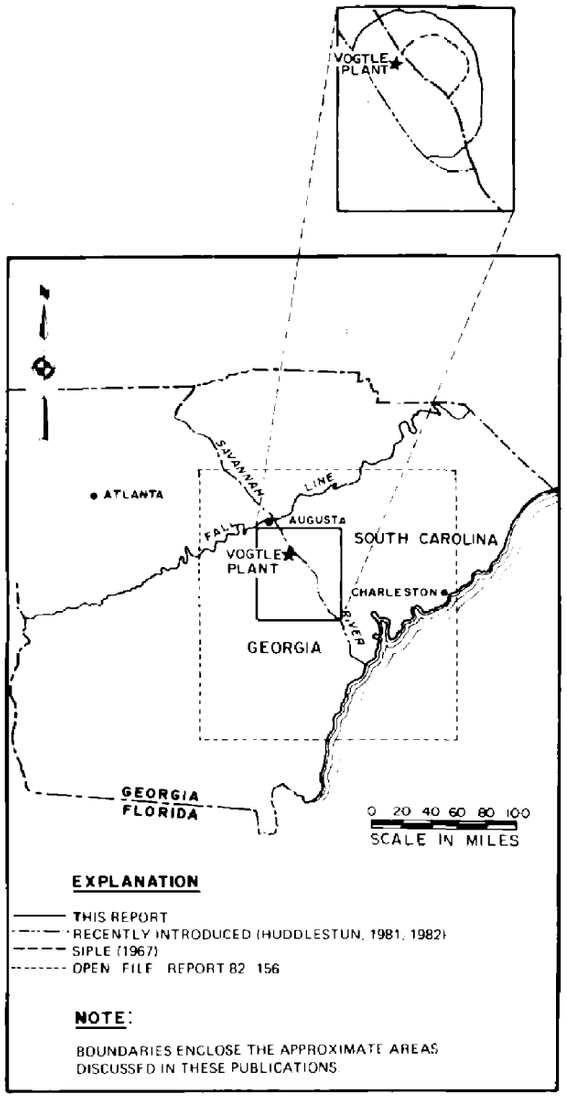
3						
2						
1	ISSUED PER REA 96-VAA054	7-26-96	JLH	RBH	CBH	WFP





REFERENCES

UFSAR SECTION 2.5.1
REF. 28,34,36,61 AND 62

EXPLANATION

- UNCONFORMITY
- AGE OF FORMATION IN QUESTION AT TIME OF REPORT
- ? ACTUAL AGE OF BOUNDARY IN QUESTION ?
- EROSION OR NON DEPOSITION
- CONTACT BETWEEN MEMBERS
- CONTACT BETWEEN FORMATIONS
- CONTACT BETWEEN GROUPS

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

STRATIGRAPHIC CORRELATION CHART

NO.	REVISIONS	DATE	DR	CHK	APPV	DTL
0	ISSUED PER REA 96-VAA054	9-30-98	GS	RBH	CBH	WFP

SCALE: NONE | DRAWING NO. AX6DD339 | REV 0

JOB NO. 10604

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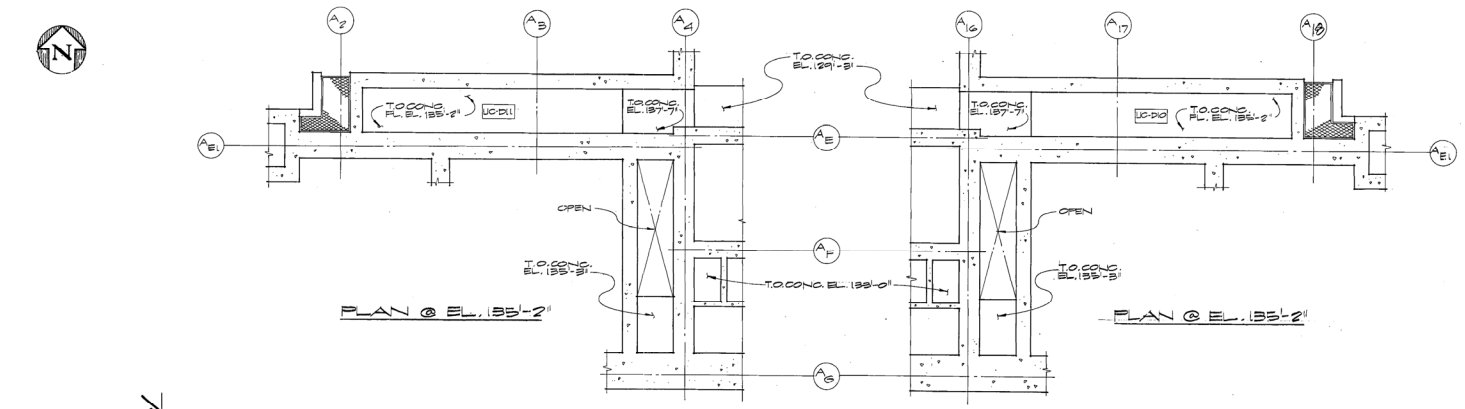
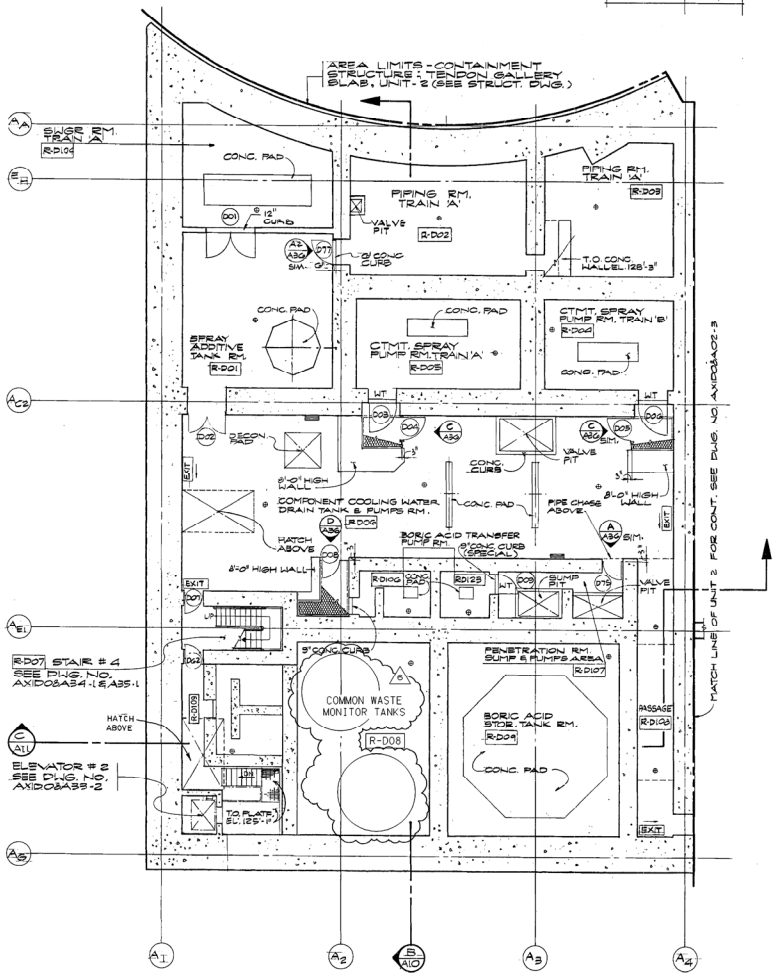
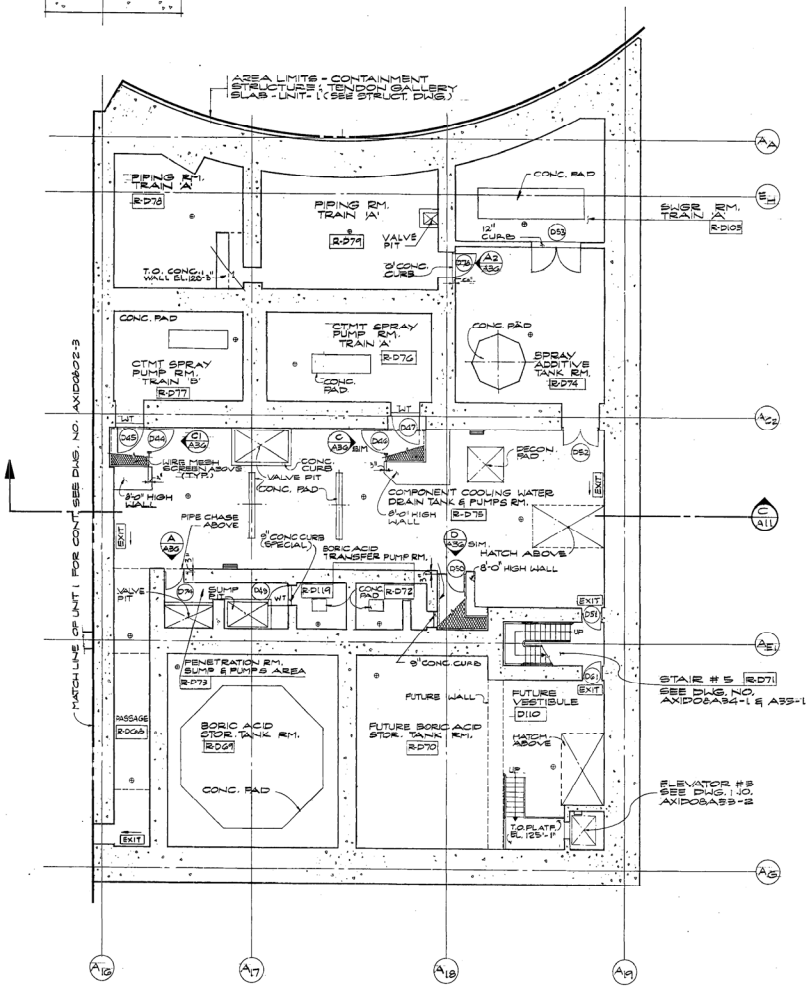
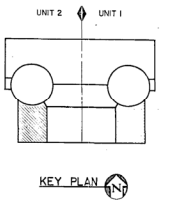
CAD NAME AX6DD339 SIZE B 11x17









NOTES

1. FOR CONC. PAD - SEE STRUCTURAL DWG'S.
2. FOR SERVICE STAIRS AND LADDERS SEE STRUCTURAL DWG'S.
3. FOR WALL DIMENSIONS SEE STRUCT. DRAWINGS
4. FOR GENERAL NOTES SEE DWG. AX1D08A01-1

LEGEND

- UTILITY STATION
- EXIT — EXIT SIGN LOCATION
- KNOCK OUT PANEL
- FLOOR DRAIN

REFERENCE DRAWINGS

AX1D08A02-3	FLOOR PLAN
AX1D08A10	BUILDING SECTIONS
AX1D08A11	BUILDING SECTION
AX1D08A14	FINISH SCHEDULE
AX1D08A15	FINISH SCHEDULE
AX1D08A15-1	FINISH SCHEDULE
AX1D08A24-1	DOOR SCHEDULE
AX1D08A24-2	DOOR SCHEDULE
AX1D08A33-2	ELEVATOR 2&3 PLANS SECTIONS & DETAILS
AX1D08A55-1	STAIR DETAILS

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NO.	REVISIONS	DATE	DR	CHK	APPV	DTL
4	INCORP. PER ABN-32149	8-3-97	SRM	RWB	MWD	
3	ISSUED FOR CONSTRUCTION	SEE MICROFILM FOR SIGNATURES				

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

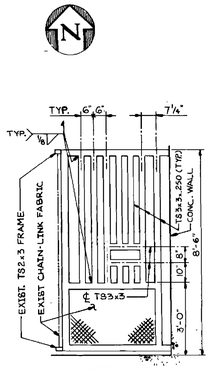
AUXILIARY BUILDING
FLOOR PLAN EL. 119FT - 3IN
LEVEL D

SCALE:	DRAWING NO.	REV.
NONE	AX1D08A02-2	6

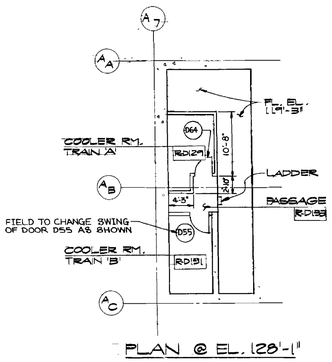
JOB NO. 10604

SIZE E 34x44

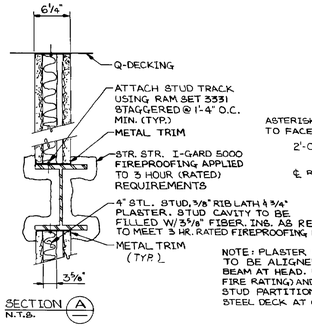
CAD NAME A10BA022



DETAIL 2
N.T.B.



PLAN @ EL. 128'-1"



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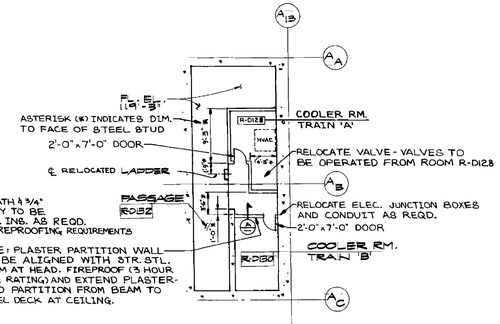
N.T.B.

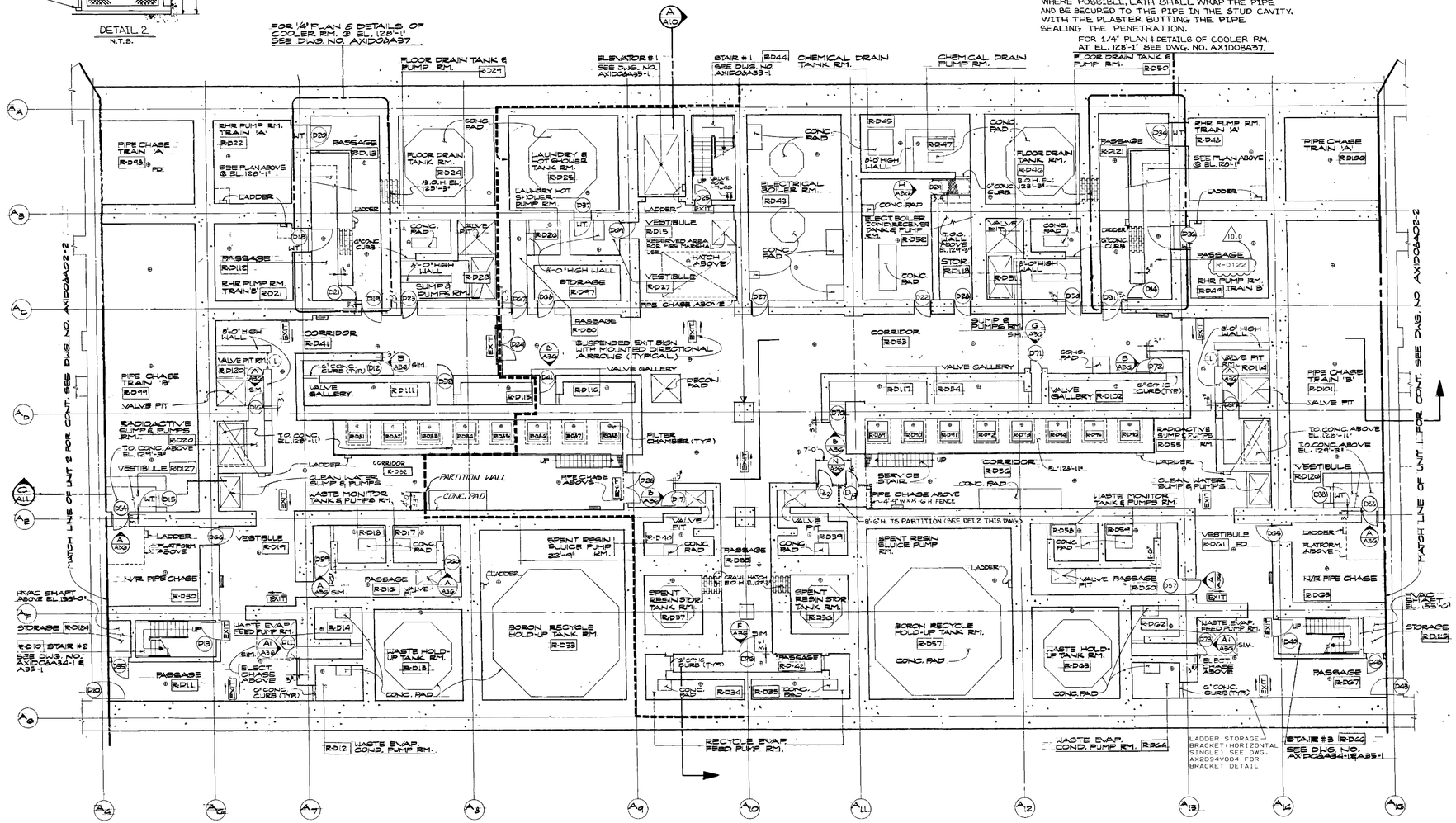


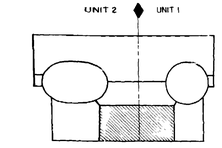
PLAN @ EL. 128'-1"

NOTE: ALL PIPE AND CONDUIT PENETRATING PLASTER WALLS SHALL BE SEALED AS REQD. WHERE POSSIBLE, LATH SHALL WRAP THE PIPE AND BE SECURED TO THE PIPE IN THE STUD CAVITY. WITH THE PLASTER BUTTING THE PIPE SEALING THE PENETRATION.

FOR 1/4" PLAN & DETAILS OF COOLER RM. AT EL. 128'-1" SEE DWG. NO. AX1D08A37.

FOR ½" PLAN & DETAILS OF COOLER RM @ EL. 128'-1" SEE DWG. NO. AX1D08A37





KEY PLAN

NOTES

1. FOR CONC. PAD - SEE STRUCTURAL DWG'S.
2. FOR SERVICE STAIRS AND LADDERS SEE STRUCTURAL DWG'S.
3. FOR WALL DIMENSIONS SEE STRUCT. DRAWINGS
4. FOR GENERAL NOTES SEE DWG. AX1D08A01-1

LEGEND

- UTILITY STATION;
- EXIT — EXIT SIGN LOCATION
- KNOCK OUT PANEL
- WIRE MESH ENCLOSURE
- FLOOR DRAIN
- (1) CHAIN: "CAUTION RADIATION AREA" SIGN
- UNIT 1 OPERATION/UNIT 2 CONSTRUCTION BOUNDARY

REFERENCE DRAWINGS

AX1D08A02-2	FLOOR PLAN
AX1D08A10	BUILDING SECTIONS
AX1D08A11	BUILDING SECTION
AX1D08A14	FINISH SCHEDULE
AX1D08A15	FINISH SCHEDULE
AX1D08A15-1	FINISH SCHEDULE
AX1D08A24-1	DOOR SCHEDULE
AX1D08A24-2	DOOR SCHEDULE
AX1D08A53-1	ELEVATOR & STAIR #1 PLANS, SECTIONS & DETAILS
AX1D08A34-1 & AX1D08A35-1	STAIR DETAILS
AX1D08A37	MISCELLANEOUS DETAILS

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

AUXILIARY BUILDING
FLOOR PLAN EL. 119FT-3IN
LEVEL D

SCALE: 1/8"=1'-0" | DRAWING NO. AX1D08A02-3 | VER. 10.0
JOB NO. 10604
SIZE E 34x44

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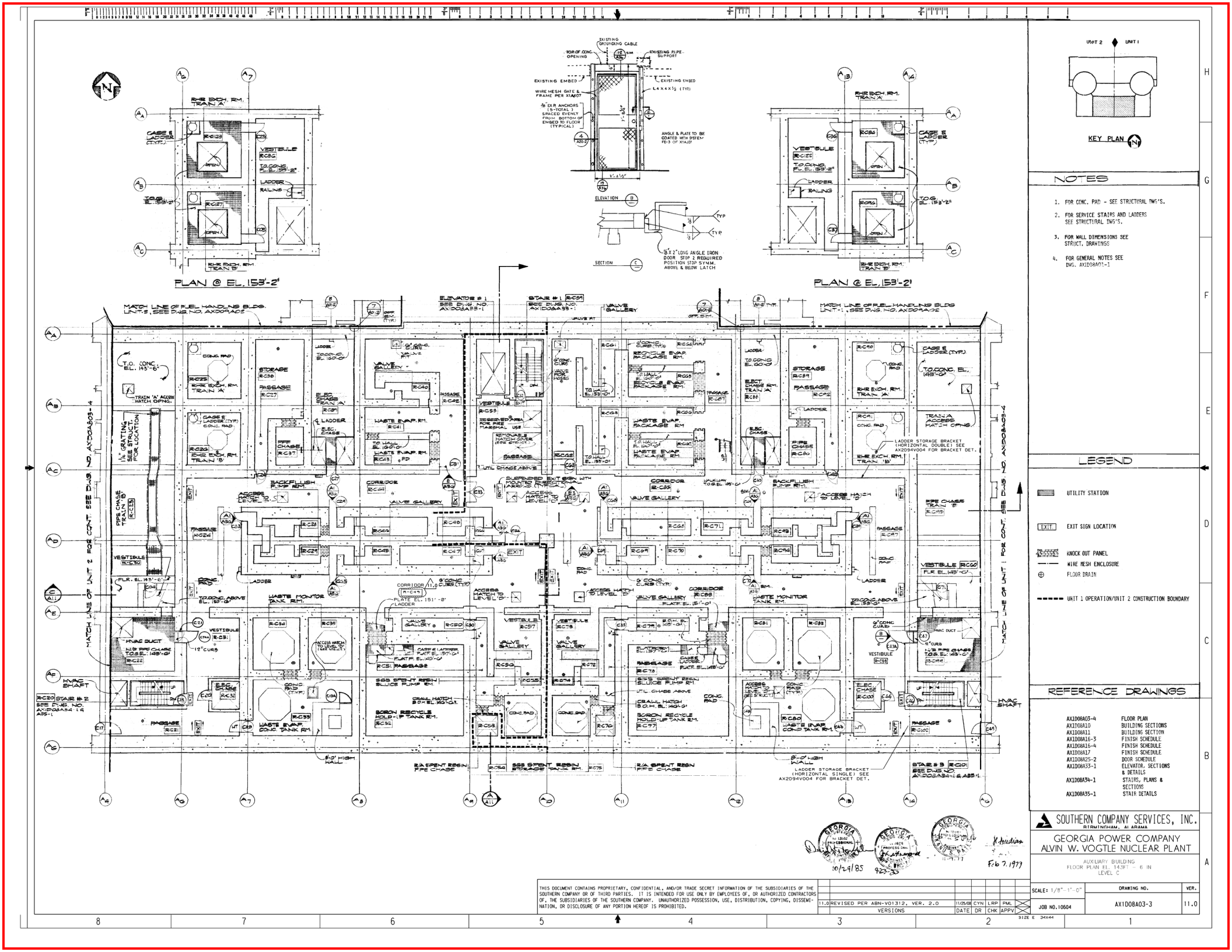
VERSIONS	DATE	DR	CHK	APPV
10.0 REVISED PER ABN-V01312, VER. 2.0	11/24/08	CYN	LRP	PML


PLAN @ EL. 153'-2"
PLAN @ EL. 153'-2"
ELEVATION
SECTION
KEY PLAN
UNIT 2
UNIT 1
NOTES
1. FOR CONC. PAD - SEE STRUCTURAL DWG'S.
2. FOR SERVICE STAIRS AND LADDERS SEE STRUCTURAL DWG'S.
3. FOR WALL DIMENSIONS SEE STRUCT. DRAWINGS
4. FOR GENERAL NOTES SEE DWG. AX1D08A01-1
LEGEND
UTILITY STATION
EXIT SIGN LOCATION
KNOCK OUT PANEL
WIRE MESH ENCLOSURE
FLOOR DRAIN
UNIT 1 OPERATION/UNIT 2 CONSTRUCTION BOUNDARY
REFERENCE DRAWINGS
AX1D08A03-4 FLOOR PLAN
AX1D08A10 BUILDING SECTIONS
AX1D08A11 BUILDING SECTION
AX1D08A16-3 FINISH SCHEDULE
AX1D08A16-4 FINISH SCHEDULE
AX1D08A17 FINISH SCHEDULE
AX1D08A25-2 DOOR SCHEDULE
AX1D08A33-1 ELEVATOR SECTIONS & DETAILS
AX1D08A34-1 STAIRS, PLANS & SECTIONS
AX1D08A35-1 STAIR DETAILS
MATCH LINE OF UNIT 2 FOR CONT. SEE DWG. NO. AX1D08A03-4
MATCH LINE OF UNIT 1 FOR CONT. SEE DWG. NO. AX1D08A03-4
MATCH LINE OF F.H.L. HANDLING BLDG. UNIT-2, SEE DWG. NO. AX1D08A02
MATCH LINE OF F.H.L. HANDLING BLDG. UNIT-1, SEE DWG. NO. AX1D08A02
ELEVATOR # 1 SEE DWG. NO. AX1D08A35-1
STAIR # 1 SEE DWG. NO. AX1D08A34-1 & A35-1
STAIR # 2 SEE DWG. NO. AX1D08A34-1 & A35-1
STAIR # 3 SEE DWG. NO. AX1D08A34-1 & A35-1
SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA
GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT
AUXILIARY BUILDING
FLOOR PLAN EL. 143FT - 6 IN
LEVEL C
SCALE: 1/8"-1'-0"
DRAWING NO. AX1D08A03-3
VER. 11.0
JOB NO. 10604
SIZE E 34X44
11.0 REVISED PER ABN-V01312, VER. 2.0 11/25/08 CYN LRP PML
VERSIONS DATE DR CHK APPV
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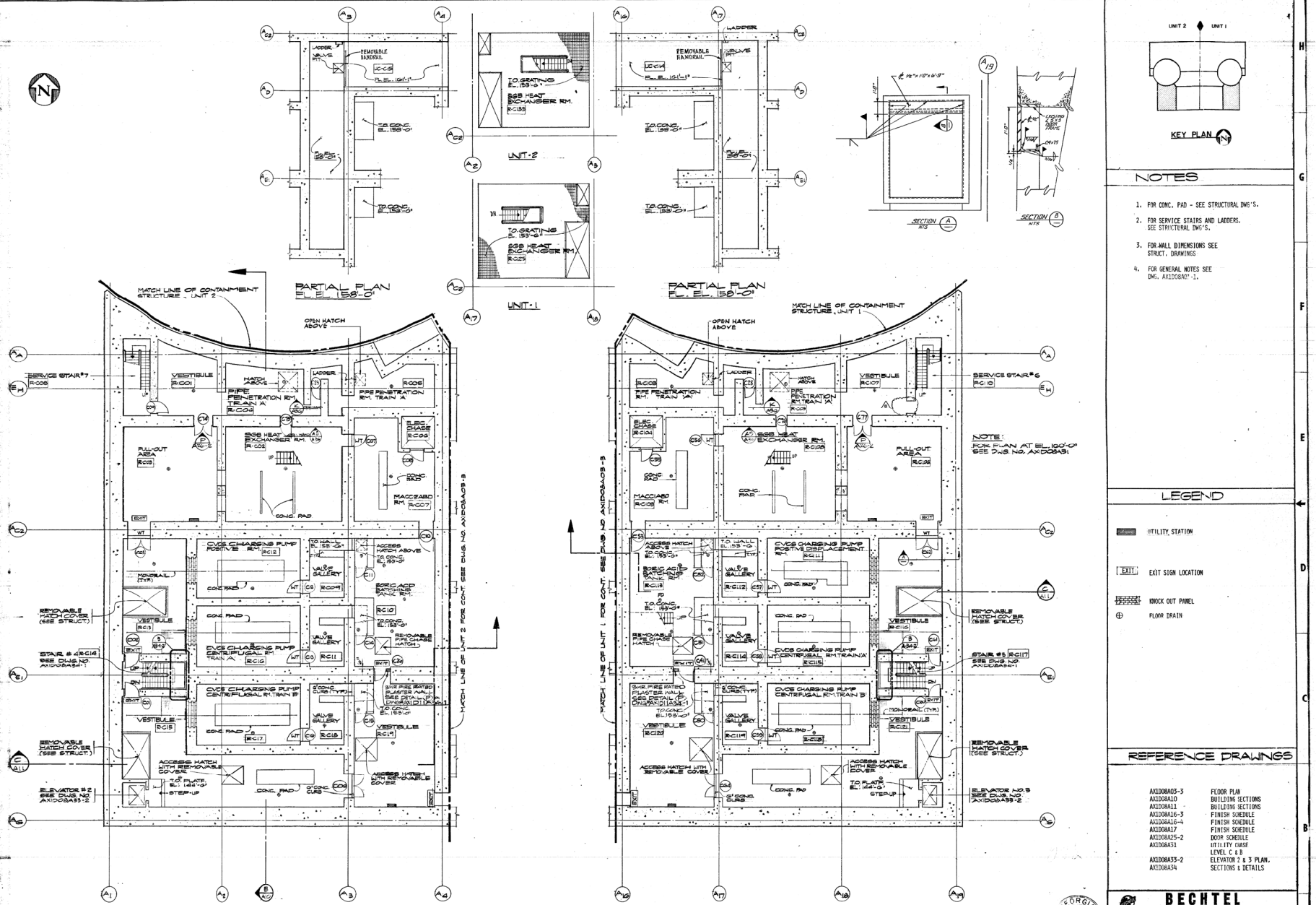
10/29/85
9-23-83
F.b 7. 1977



KEY PLAN

NOTES

1. FOR CONC. PAD - SEE STRUCTURAL DWG'S.
2. FOR SERVICE STAIRS AND LADDERS, SEE STRUCTURAL DWG'S.
3. FOR WALL DIMENSIONS SEE STRUCT. DRAWINGS
4. FOR GENERAL NOTES SEE DWG. AX1D08A01-1.

LEGEND

- UTILITY STATION
- EXIT — EXIT SIGN LOCATION
- KNOCK OUT PANEL
- FLOOR DRAIN

REFERENCE DRAWINGS

AX1D08A03-3	FLOOR PLAN
AX1D08A10	BUILDING SECTIONS
AX1D08A11	BUILDING SECTIONS
AX1D08A16-3	FINISH SCHEDULE
AX1D08A16-4	FINISH SCHEDULE
AX1D08A17	FINISH SCHEDULE
AX1D08A25-2	DOOR SCHEDULE
AX1D08A31	UTILITY CHASE LEVEL C & B
AX1D08A33-2	ELEVATOR 2 & 3 PLAN,
AX1D08A34	SECTIONS & DETAILS

BECHTEL
LOS ANGELES

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

AUXILIARY BUILDING
FLOOR PLAN EL. 141'FT - 6IN
LEVEL C

SCALE: 1/8"=1'-0" | DRAWING NO. AX1D08A03-4 | REV. 8
JOB NO. 9510 - 001

Feb 7, 1979

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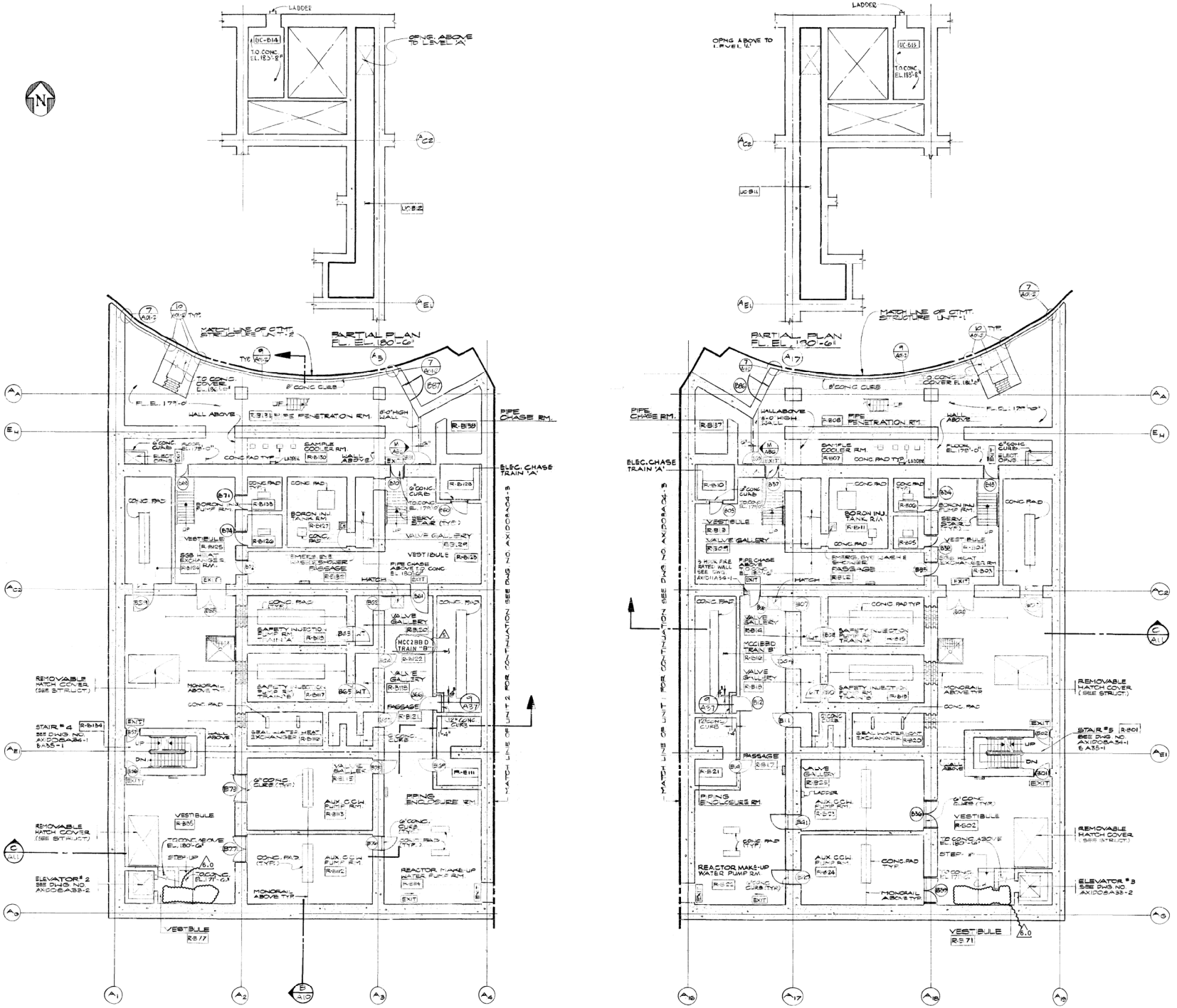
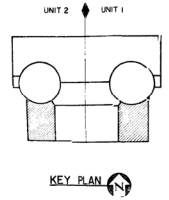
REVISIONS

NO	DESCRIPTION	DATE	BY
A	INCORP. ABN 87-VIE0047A001	10/17/78	W.V.G.
B	INCORP CCR#5-11780 DCN No. 2	9-13-78	
A	INCORP FIRE PROTECTION REV. CCR#5 NO. C&S 5-C41	11-2-78	
B	REVISED AS SHOWN	11-30-78	
C	INCORP DCN. NO. 1	2-14-79	
A	ADDED NEW DOOR	2-13-79	
B	REVISED AS SHOWN	1-15-79	
C	DOORS & PARTIAL WALLS ADDED	2-7-79	
A	REVISED FOR CONSTRUCTION	3-1-79	

ENG. MGR. — DATE

SIZE E 34X44





NOTES

1. FOR CONC. PAD - SEE STRUCTURAL DWG'S.
2. FOR SERVICE STAIRS AND LADDERS SEE STRUCTURAL DWG'S.
3. FOR WALL DIMENSIONS SEE STRUCT. DRAWINGS
4. FOR GENERAL NOTES SEE DWG. AX1D08A01-1

LEGEND

- UTILITY STATION
- EXIT — EXIT SIGN LOCATION
- KNOCK OUT PANEL
- FLOOR DRAIN

REFERENCE DRAWINGS

AX1D08A04-3	FLOOR PLAN
AX1D08A10	BUILDING SECTIONS
AX1D08A11	BUILDING SECTION
AX1D08A18-1	FINISH SCHEDULE
AX1D08A18-2	FINISH SCHEDULE
AX1D08A19	FINISH SCHEDULE
AX1D08A26-1	DOOR SCHEDULE
AX1D08A26-2	DOOR SCHEDULE
AX1D08A33-2	ELEVATOR 2&3 PLANS, SECTIONS
AX1D08A34-1	STAIRS, PLANS & SECTIONS
AX1D08A35-1	STAIR DETAILS

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

AUXILIARY BUILDING
FLOOR PLAN EL. 170'-6"
LEVEL B

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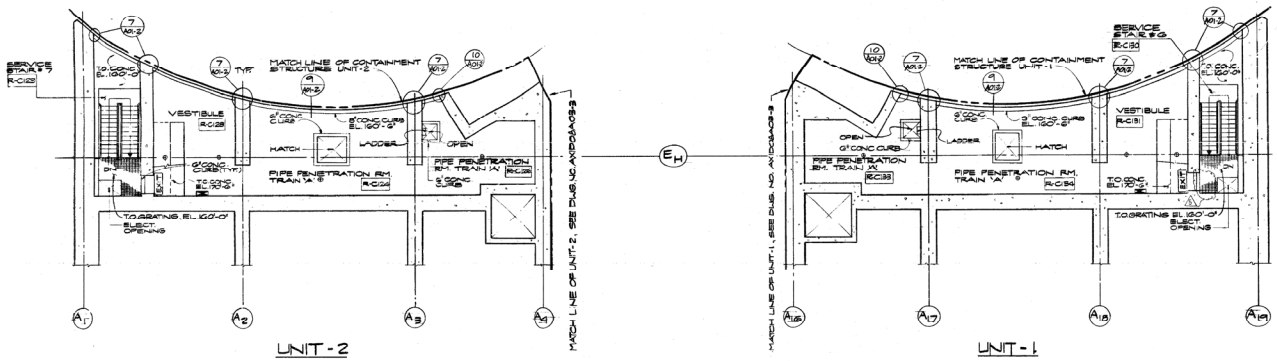
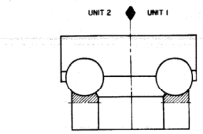
NO.	VERSIONS	DATE	DR	CHK	APPV
6.0	REVISED PER ABN-V70066, VER. 1.0	08-25-06	KMH	LRP	ASK

SCALE: 1/8" = 1'-0"
JOB NO.10604
DRAWING NO. AX1D08A04-4
VER. 6.0
SIZE E 34X44



PARTIAL PLAN - LEVEL 'C', EL. 160'-0"



GENERAL NOTES

1. FOR CONC. PAD - SEE STRUCTURAL DWG'S.
2. FOR SERVICE STAIRS AND LADDERS SEE STRUCTURAL DWG'S.
3. FOR WALL DIMENSIONS SEE STRUCT. DRAWINGS
4. FOR GENERAL NOTES SEE DWG. AX1D08A01-1.

LEGEND

- FIRE HOSE CABINET
- EXIT — EXIT SIGN LOCATION
- FLOOR DRAIN

REFERENCE DRAWING

AX1D08A03-4	FLOOR PLANS
AX1D08A10	BUILDING SECTIONS
AX1D08A11	BUILDING SECTION
AX1D08A17	FINISH SCHEDULE
AX1D08A25-2	DOOR SCHEDULE

3/24/83 — 11-7-77 — Feb. 7, 1979

NO.	REVISIONS	DATE	DR.	CHK.	SUPV.	EGS	CHF.E	P.E.	Q.A.E.

NO.	REVISIONS	DATE	DR.	CHK.	SUPV.	EGS	CHF.E	P.E.	Q.A.E.

NO.	REVISIONS	DATE	DR.	CHK.	SUPV.	EGS	CHF.E	P.E.	K.P.E.
3	INCORP. DCN # 1	11/78	D.D.						
2	PLAN @ EL. 160'-0" OMITTED/ADD. DETAIL-A, L, J, B	11-7-79							
1	PLAN AT EL. 158'-0" OMITTED	2-27-79							
0	ISSUED FOR CONSTRUCTION @ LEVEL 'C'	2-7-79							

NO.	REVISIONS	DATE	DR.	CHK.	SUPV.	EGS	CHF.E	P.E.	K.P.E.
	INCORP. ASB 87-VTE0047A001	10/7/81	W.V.G.	10/8	MBK	TH			
	REVISIONS	DATE	DR.	CHK.	RE	DM	DM	DM	PDM
	INCORP. ORDR-23351	11/19/81	SM	11/23					

ENG. MGR. — DATE

BECHTEL
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GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

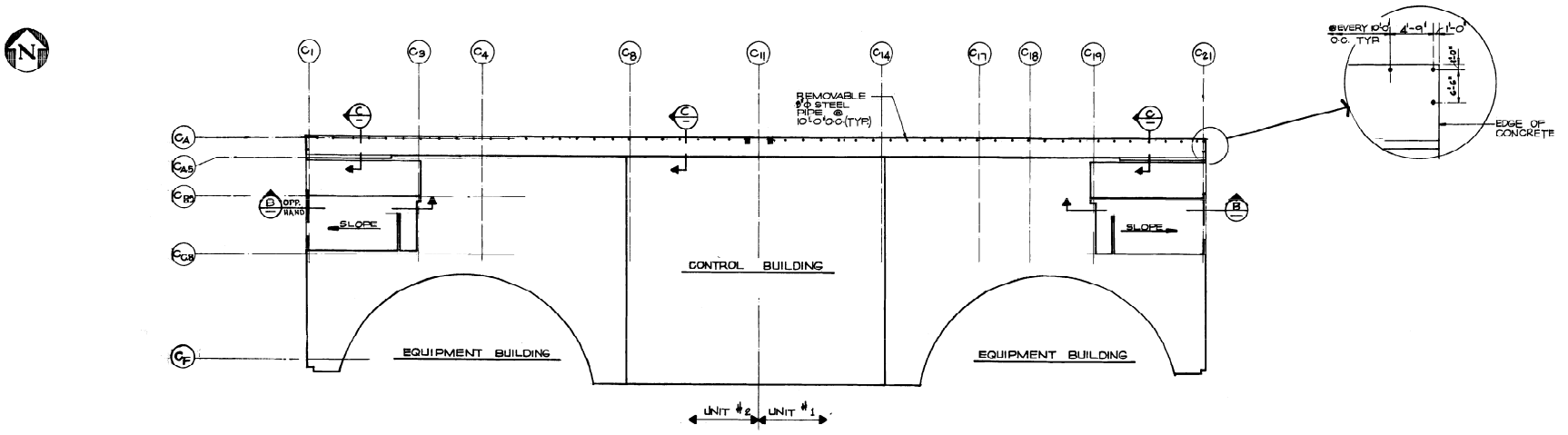
AUXILIARY BUILDING
PARTIAL PLANS
UTILITY CHASE LEVEL C4B

SCALE: 1/8"=1'-0" — DRAWING NO. AX1D08A31 — REV. 5

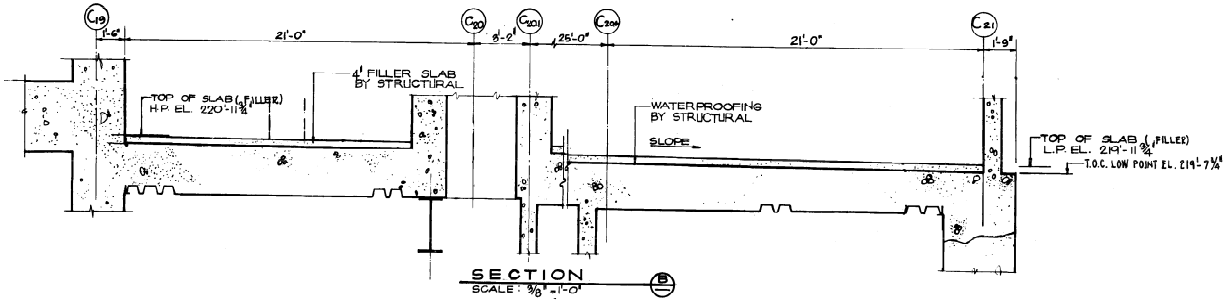
JOB NO. 9610-001

SIZE E 36X44

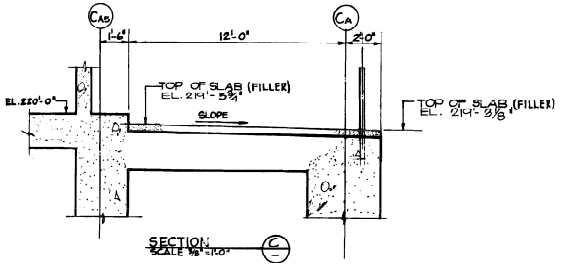
8 7 6 5 4 3 2 1



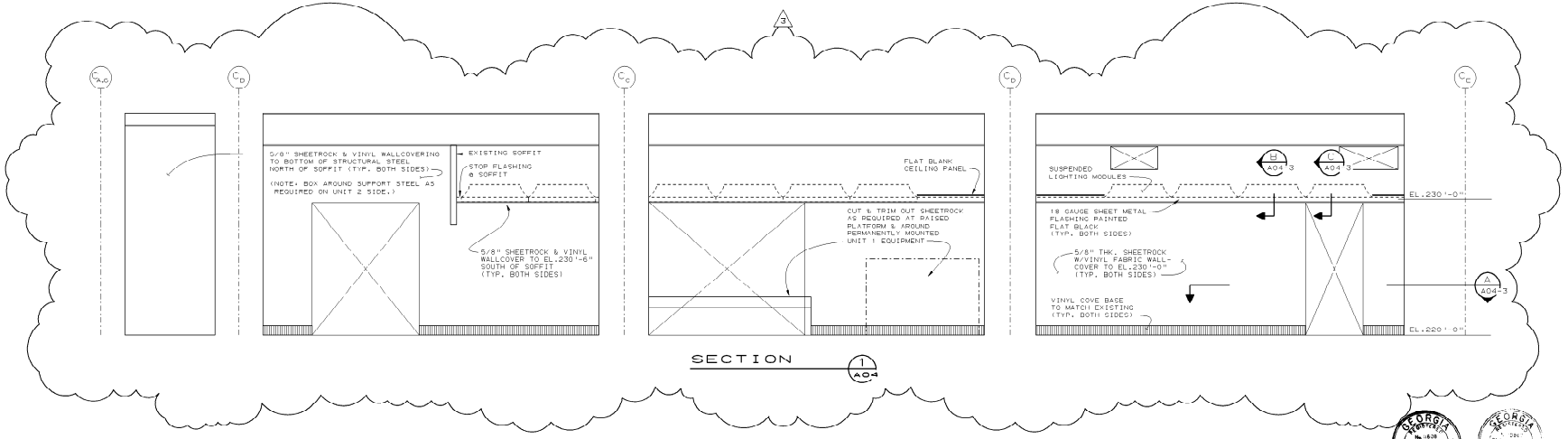
PLAN @ EL. 220'-0"
SCALE: 1" = 25'-0"



SECTION
SCALE: ⅜" = 1'-0"



SECTION
SCALE: ⅜" = 1'-0"



SECTION
1 A04

NOTES

1. FOR GENERAL NOTES SEE DWG. AX1D11A01-1

REFERENCE DRAWINGS

ARCHITECTURAL DRAWINGS
AX1D11A01-1
AX1D11A04
AX1D11A04-3
1X1D01A13
2X1D01A13

STRUCTURAL DRAWINGS
AX2D11A01
AX2D11A018
AX2D11A021
AX12D11A023

GEORGIA REGISTERED No. 16128 PROFESSIONAL ENGINEER K. C. WHISENANT 1/8/92

GEORGIA REGISTERED No. 20154 PROFESSIONAL ENGINEER C. H. ROBERTS, JR. 1C-18-79

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

CONTROL BUILDING
FILLER SLAB PLAN, SECTIONS & DETAILS
LEVEL 1

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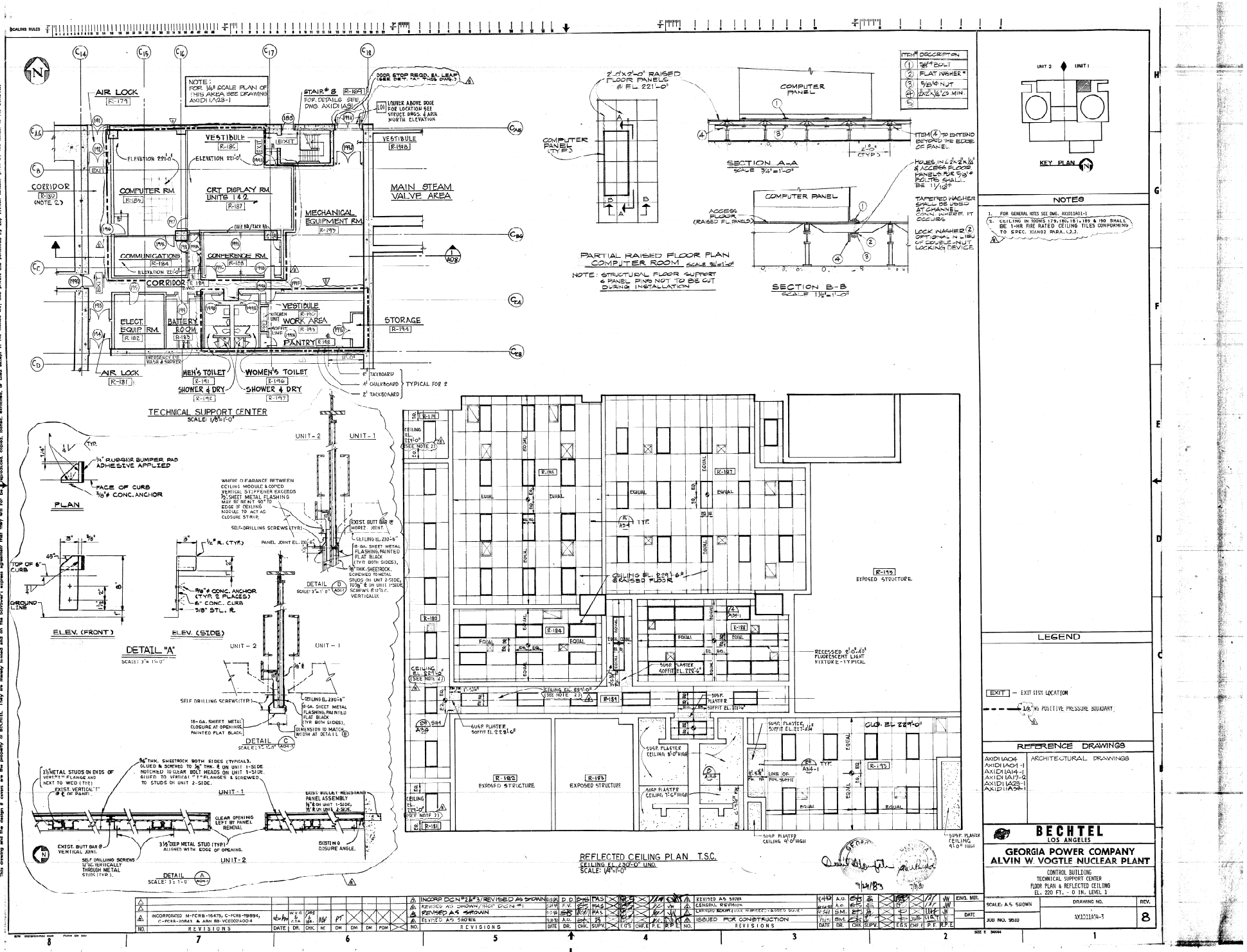
NO.	REVISIONS	DATE	DR	CHK	APPV	DTL
2	INCORP. PER ABN 88VLE021A003	7/15/98	SMM	KWB	SM	
1	ISSUED FOR CONSTRUCTION	SEE MYLAR	M FOR	SIGNATURES		

SCALE: AS NOTED
JOB NO. 10604
DRAWING NO. AX1D11A04-1
REV. 3
SIZE E 34x44
CAD NAME A1D1A041


AIR LOCK R-179
NOTE: FOR 1/4" SCALE PLAN OF THIS AREA SEE DRAWING AXIDI1A23-1
STAIR # 8 R-164
FOR DETAILS SEE DWG. AXIDI1A9
DOOR STOP REQD. EL. LEAF SEE DET. 4A-7-1-08 DWG-7
LOUVER ABOVE DOOR FOR LOCATION SEE STRUCT. DWGS. & ARCH NORTH ELEVATION
VESTIBULE R-180
VESTIBULE R-180A
ELEVATION 229'-0"
ELEVATION 221'-0"
CORRIDOR R-190 (NOTE 2)
COMPUTER RM. R-189
CRT DISPLAY RM. UNITS 1 & 2 R-187
MAIN STEAM VALVE AREA
MECHANICAL EQUIPMENT RM. R-199
COMMUNICATIONS R-184
CONFERENCE RM. R-188
CORRIDOR R-184
ELECT. EQUIP. RM. R-182
BATTERY ROOM R-183
VESTIBULE R-180
KITCHEN UNIT WORK AREA
PANTRY R-198
STORAGE R-194
EMERGENCY EYE WASH & SHOWER
AIR LOCK R-181
MEN'S TOILET R-191
SHOWER & DRY R-192
WOMEN'S TOILET R-196
SHOWER & DRY R-197
6' TACKBOARD
4' CHALKBOARD
2' TACKBOARD
TYPICAL FOR 2
TECHNICAL SUPPORT CENTER
SCALE: 1/8"=1'-0"
2'-0"x2'-0" RAISED FLOOR PANELS @ EL 221'-0"
COMPUTER PANEL (TYP.)
PARTIAL RAISED FLOOR PLAN COMPUTER ROOM
SCALE 3/4"=1'-0"
NOTE: STRUCTURAL FLOOR SUPPORT & PANEL PINS NOT TO BE CUT DURING INSTALLATION
COMPUTER PANEL
SECTION A-A
SCALE 3/4"=1'-0"
ITEM (4) TO EXTEND BEYOND THE EDGE OF PANEL
ACCESS FLOOR (RAISED FL. PANEL)
HOLES IN 2'-0"x2'-0" ½" ALUMINUM ACCESS FLOOR PANELS FOR 5/8" BOLTS SHALL BE 1 1/16" ø
TAPERED WASHER SHALL BE USED AT CHANNEL CONN. WHERE IT OCCURS
LOCK WASHER (2) OPTIONAL IN LIEU OF DOUBLE NUT LOCKING DEVICE
SECTION B-B
SCALE 1½"=1'-0"
ITEM DESCRIPTION
1 3/8" BOLT
2 FLAT WASHER *
3 5/8"-11 NUT
4 2"x2"x1/4"x25 MIN.
5
UNIT 2
UNIT 1
KEY PLAN
NOTES
1. FOR GENERAL NOTES SEE DWG. AXIDI1A01-1
2. CEILING IN ROOMS 179, 180, 181, 189 & 190 SHALL BE 1-HR FIRE RATED CEILING TILES CONFORMING TO SPEC. X1AN02 PARA. 1.2.2.
PLAN
1/4" TYP.
½" RUBBER BUMPER PAD ADHESIVE APPLIED
FACE OF CURB
⅜"ø CONC. ANCHOR
ELEV. (FRONT)
ELEV. (SIDE)
DETAIL "A"
SCALE: 3"=1'-0"
TOP OF 6" CURB
45°
GROUND LINE
3"
⅜"
½" R. (TYP.)
⅞"ø CONC. ANCHOR (TYP. 2 PLACES)
6" CONC. CURB
⅞" ST. L. R
UNIT-2
UNIT-1
WHERE CLEARANCE BETWEEN CEILING MODULE & CONCD. VERTICAL STIFFENER EXCEEDS ½" SHEET METAL FLASHING MAY BE RENT SO TO EDGE OF CEILING MODULE TO ACT AS CLOSURE STRIP.
SELF-DRILLING SCREWS (TYP.)
EXIST. BUTT BAR @ HORIZ. JOINT
CEILING EL. 230'-6"
18-GA. SHEET METAL FLASHING, PAINTED FLAT BLACK (TYP. BOTH SIDES).
⅝" THK. SHEETROCK, SCREWED TO METAL STUDS ON UNIT 2-SIDE, 10⅜" ø ON UNIT 1-SIDE SCREWS @ 12" O.C. VERTICALLY.
PANEL JOINT EL. 230'-6"
DETAIL D
SCALE: 3"=1'-0"
DETAIL C
SCALE: 3"=1'-0"
⅝" THK. SHEETROCK BOTH SIDES (TYPICAL). GLUED & SCREWED TO ⅜" THK. ø ON UNIT 1-SIDE NOTCHED TO CLEAR BOLT HEADS ON UNIT 1-SIDE GLUED TO VERTICAL "T" FLANGES & SCREWED TO STUDS ON UNIT 2-SIDE.
3½" METAL STUDS IN ENDS OF VERTICAL "T" FLANGE AND NEXT TO WCD (TYP.)
EXIST. VERTICAL "T" ø OF PANEL
CLEAR OPENING LEFT BY PANEL REMOVAL
3½" DEEP METAL STUD (TYP.) ALIGNED WITH EDGE OF OPENING.
EXIST. BUTT BAR @ VERTICAL JOINT.
SELF DRILLING SCREWS 12"ø VERTICALLY THROUGH METAL STUDS (TYP.).
EXISTING CLOSURE ANGLE.
BASIC BULLET RESISTANT PANEL ASSEMBLY ⅜" ø ON UNIT 1-SIDE, ⅝" ø ON UNIT 2 SIDE.
DETAIL A
SCALE: 3"=1'-0"
REFLECTED CEILING PLAN T.S.C.
CEILING EL. 230'-0" UNO.
SCALE: 1/4"=1'-0"
R-179 CEILING EL. 221'-0" (SEE NOTE 2)
R-186 EQUAL
R-187 EQUAL
R-199 EXPOSED STRUCTURE
R-180 EQUAL
R-184 EQUAL
R-182 EXPOSED STRUCTURE
R-183 EXPOSED STRUCTURE
R-181 CEILING EL. 221'-6" (SEE NOTE 2)
CEILING EL. 229'-6" RAISED FLOOR
CEILING EL. 221'-6" (SEE NOTE 2)
CEILING EL. 221'-6" (SEE NOTE 2)
CEILING EL. 221'-6" (SEE NOTE 2)

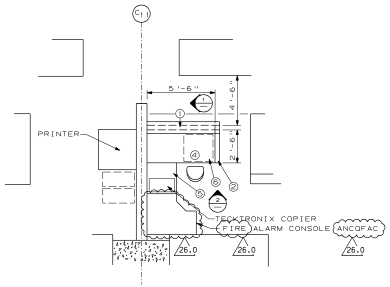
① 5'-6" LONG x 48" HIGH PARTITION WITH 15" WIDE x 5'-6" LONG COUNTER

② 2'-0" LONG x 48" HIGH PARTITION

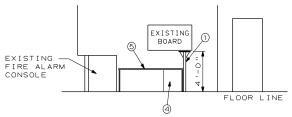
④ 2'-6" WIDE x 5'-6" LONG PANEL- MOUNTED WORK SURFACE

⑤ 2'-6" WIDE x 3'-0" LONG FREE STANDING DESK

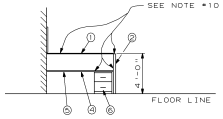
⑥ 2'-0" WIDE x 18" DEEP x 27¾" HIGH LATERAL FILE



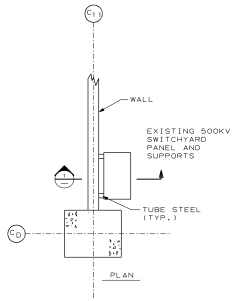
DETAIL A



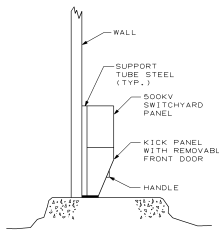
ELEVATION 1



ELEVATION 2

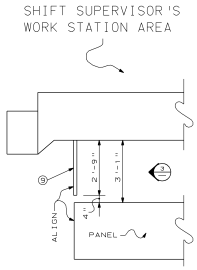


DETAIL B

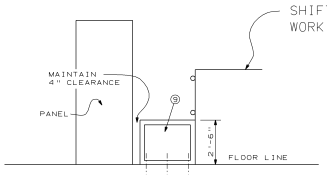


SECTION 1

⑨ REMOVE EXISTING AND REPLACE WITH: 2'-9" WIDE x 2'-6" HIGH x 2" DEEP DEMOUNTABLE PARTITION BOLTED DIRECTLY TO CONC. FLOOR w/3 ⅜" Ø CONC. EXP. ANCHORS.



DETAIL C



ELEVATION 3

THIS DRAWING WAS RENUMBERED FROM AX1D11A04 (SHEET 2)

WORK THIS DRAWING WITH: AX1D11A04-4

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NO.	VERSIONS	DATE	DR	CHK	APPV
26.0	REVISED PER ABN C0522678001E050, VER. 1.0	2-20-07	JLO	ELC	JMR

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

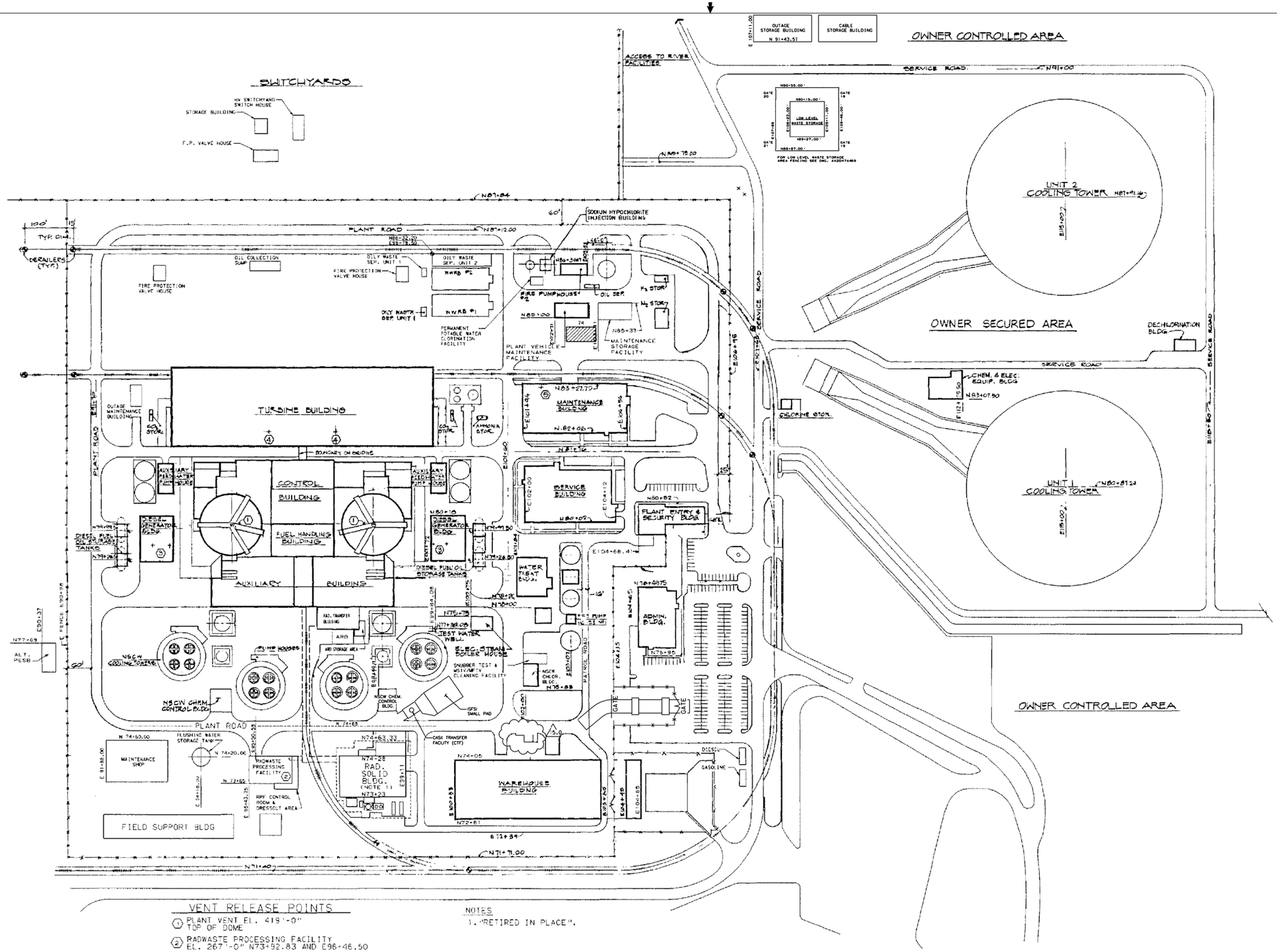
CONTROL BUILDING
FLOOR PLAN EL.220'-0"
LEVEL 1 SHEET 2 OF 2

SCALE: NONE

JOB NO.10604

DRAWING NO.	VER.
AX1D11A04-5	26.0

SIZE E 34X44



VENT RELEASE POINTS

1. PLANT VENT EL. 419'-0" TOP OF DOME
2. RADWASTE PROCESSING FACILITY EL. 267'-0" N73+92.83 AND E96+46.50
3. DIESEL GENERATOR EXHAUST EL. 283'-6"
4. TURBINE BUILDING - SEE SCS
5. AUXILIARY BOILER - SEE SCS

NOTES

1. "RETIRED IN PLACE".

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NO.	VERSIONS	DATE	DR	CHK	APPV
15.0	REVISED PER ABN C10086790IC016, VER. 2.0	08/22/16	LEH	MWD	WLN

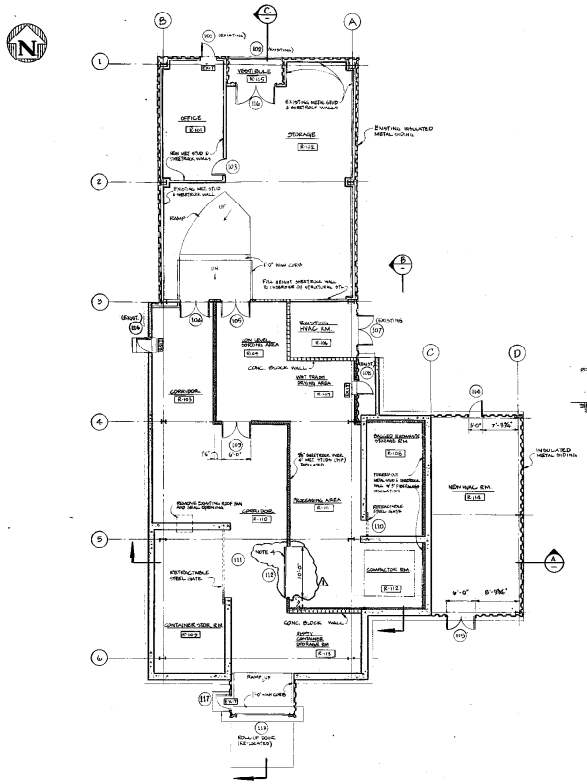
SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

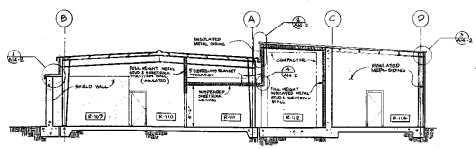
SITE PLOT PLAN

SCALE: NONE	DRAWING NO.	VER.
JOB NO.10604	AX1D45A01	15.0

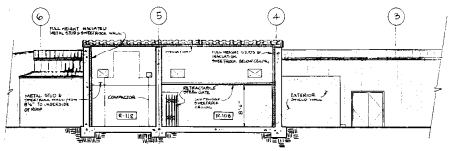
SIZE E 34x44



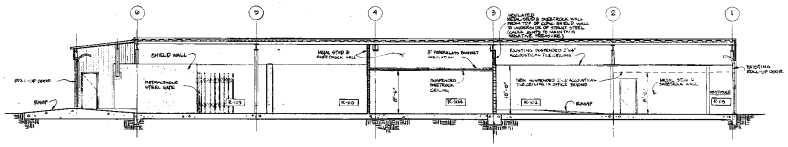
DRY ACTIVE WASTE PROCESSING BUILDING PLAN
SCALE: 1/8"=1'-0"



SECTION (A)
SCALE: 1/8"=1'-0"



SECTION (B)
SCALE: 1/8"=1'-0"



SECTION (C)
SCALE: 1/8"=1'-0"

GENERAL NOTES

1. NEW METAL SIDING SHALL BE PAINTED TAN TO MATCH THE EXISTING FINISH.
2. FASCIA PANELS, GUTTERS & DOWNSPOUTS SHALL BE PAINTED A CONTRASTING DARK BROWN COLOR.
3. FIELD TO CHECK ALL DETAILS OF THE PROPOSED MATERIAL FOR SIDING AND ROOF PANELS FOR NRC/OSHA ROUTE W & APPROVAL.
4. DOOR 112 IS A FABRICATED, SLIDING DOOR, FIELD FITTED TO NOMINAL DIMENSIONS, 10'WIDE x 10' HIGH. MOUNTING HARDWARE INSTALLED ON WALL SIDE OF R.O. AIR TIGHT SEAL AROUND DOOR FRAME & NEOPRENE STRIP FRAMES ALONG THE BOTTOM OF THE DOOR. DOOR COATING AS SPECIFIED. FIELD TO FURNISH MOUNTING DETAILS, HARDWARE ETC. △

FINISH SCHEDULE

NOTES:
1. COATING SYSTEMS PER SPECIFICATION X1A107
2. FINISH COLOR SHALL BE AT PURCHASER'S OPTION
3. ALL HOLLOW METAL DOORS TO BE COATED WITH FD-12.

ABBREVIATIONS:
AT — ACOUSTICAL TILE
CMU — CONCRETE MASONRY UNITS
EX — EXPOSED STRUCTURE
GWB — INTERIOR WALLBOARD
P — PAINT

ROOM NUMBER	SPACE TITLE	FLOORS MATL.	FINISH	SYSTEM	NORTH WALL MATL.	FINISH	SYSTEM	EAST WALL MATL.	FINISH	SYSTEM	SOUTH WALL MATL.	FINISH	SYSTEM	WEST WALL MATL.	FINISH	SYSTEM	CEILING MATL.	FINISH	SYSTEM	REMARKS
R-101	OFFICE	C	P	FC-19	GWB	P	FC-25	GWB	P	FC-25	GWB	P	FC-25	GWB	P	FC-25	AT	–	–	
R-102	STORAGE AREA	C	P	FD-19	GWB	P	FC-25	GWB	P	FC-25	GWB	P	FC-25	GWB	P	FC-25	A1	+	+	* FINISHES AS REQUIRED.
R-103	CORRIDOR	C	P	FD-19	GWB	P	FC-24	GWB	P	FC-24	C	P	FD-14	C	P	FD-14	ES	+	+	+ LEAVE EXISTING FINISHED
R-104	LOW LEVEL SORTING AREA	C	P	FD-19	CMU	P	FC-24	CMU	P	FD-14	GWB	P	FC-24	GWB	P	FC-24	CMU	P	FC-24	
R-105	SPARE	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	
R-106	EXISTING HVAC RM.	C	+	–	PLYWOOD	+	ES	–			PLYWOOD	+	+	PLYWOOD	+	ES	+	+	+	LEAVE EXISTING FINISHED
R-107	WET TRASH DRYING AREA	C	P	FD-19	CMU	P	FD-14	GWB	P	FC-24	–	–	–	–	–	–	GWB	P	FC-24	
R-108	BAGGED RADWASTE STORAGE	C	P	FD-19	GWB	P	FC-24	GWB	P	FC-24	C	P	FD-14	C	P	FD-14	GWB	P	FC-24	
R-109	CONTAINER WITHDRAW RM.	C	P	FD-19	C	P	FD-14	C	P	FD-14	C	P	FD-14	C	P	FD-14	ES	+	+	+ LEAVE EXISTING FINISHED
R-110	CORRIDOR	C	P	FD-19	GWB	P	FC-24	GWB	P	FC-24	ES	+	+	C	P	FD-14	ES	+	+	+ LEAVE EXISTING FINISHED
R-111	PROCESSING AREA	C	P	FD-19	–	–	–	C	P	FD-14	GWB	P	FC-24	GWB	P	FC-24	GWB	P	FC-24	
R-112	COMPACTOR RM.	C	P	FD-19	C	P	FD-14	GWB	P	FC-24	GWB	P	FC-24	GWB	P	FC-24	GWB	P	FC-24	
R-113	EMPTY CONTAINER STOR. RM.	C	P	FD-19	CMU	P	FD-14	C	P	FD-14	C	P	FD-19	–	–	–	ES	+	+	* LEAVE EXISTING FINISHES
R-114	NEW HVAC RM.	C	P	FC-19	ES	P	FC-19	ES	P	FC-19	ES	P	FC-19	C	P	FC-19	ES	P	FC-8	
R-115	VESTIBULE	C	P	FD-19	ES	–	–	GWB	P	FC-24	–	–	–	GWB	P	FC-24	GWB	P	FC-24	

DOOR SCHEDULE

NOTES:
1. SEE MECHANICAL DRAWINGS FOR SPECIFICATIONS OF SPECIFICATION X1A-101.
2. FINISH HARDWARE PER REQUIREMENTS OF SPECIFICATION X1A-210.

TAG NUMBER	ROOM NO.	ROOM TITLE	DOOR MATERIAL	DOOR SIZE	DOOR THICKNESS	FRAME MATERIAL	HARDWARE SET NO.	FIRE RATING	REMARKS
VA2165U 101	R-101	OFFICE	"	"	"	"	"	NO	* USE EXISTING FRAME, DOOR & HARDWARE IN PLACE
102	R-115	VESTIBULE	"	"	"	"	"	NO	* USE EXISTING ROLL UP DOOR IN PLACE
103	R-101	OFFICE	HOLLOW METAL	3'-0" x 7'-0"	1 3/4"	PRESSED STEEL	33	NO	–
104	R-103	CORRIDOR	HOLLOW METAL	PR. 3'-0" x 7'-0"	1 3/4"	PRESSED STEEL	9	NO	AIRTIGHT
105	R-104	LOW LEVEL SORTING AREA	HOLLOW METAL	PR. 3'-0" x 7'-0"	1 3/4"	PRESSED STEEL	9	NO	AIRTIGHT
106	R-103	CORRIDOR	"	"	"	"	"	NO	* USE EXISTING FRAME, DOOR, HARDWARE, RAISE ELEVATION
107	R-106	EXISTING HVAC RM.	"	"	"	"	"	NO	* USE EXISTING FRAME, DOOR & HARDWARE IN PLACE
108	R-107	WET TRASH DRYING AREA	"	"	"	"	"	NO	* USE EXISTING FRAME, DOOR & HARDWARE; ADD AIRTIGHT GASKET, PAINT ENTIRE
109	R-104	LOW LEVEL SORTING AREA	HOLLOW METAL	PR. 3'-0" x 7'-0"	1 3/4"	PRESSED STEEL	9	NO	AIRTIGHT
110	R-105	BAGGED RADWASTE STORAGE RM.	STEEL	8'-0" x 7'-0"	–	–	RAILROAD	NO	RETRACTABLE STEEL GATE (CRITERION LT 3150 OR EQUAL)
111	R-109	CONTAINER STORAGE RM.	STEEL	13'-2" x 8'-0"	–	–	RAILROAD	NO	RETRACTABLE STEEL GATE (CRITERION LT 3150 OR EQUAL)
112	R-111	PROCESSING AREA	HOLLOW METAL	(NOW USED AS)	1 3/4"	(FAB. HOOD)	NOTE 4	NO	AIRTIGHT (NOTE 4, SLIDING DOOR) △
113	R-110	CORRIDOR	"	"	"	"	"	NO	RAISE EXISTING ROLL UP DOOR, RELOCATE
114	R-114	NEW HVAC RM.	HOLLOW METAL	3'-0" x 7'-0"	1 3/4"	PRESSED STEEL	55	NO	–
115	R-114	NEW HVAC RM.	HOLLOW METAL	PR. 3'-0" x 7'-0"	1 3/4"	PRESSED STEEL	9	NO	–
116	R-102	STORAGE	"	"	"	"	"	NO	* USE EXISTING FRAME, DOOR & HARDWARE IN PLACE
117	R-110	CORRIDOR	HOLLOW METAL	8'-0" x 7'-0"	1 3/4"	PRESSED STEEL	14	NO	AIRTIGHT

NO.	REVISIONS	DATE	DR.	CHK.	SUPV.	EGS	CHF.E	P.E.	Q.A.E.
△	REVISED PER RER 84-0135	3-1-85	PLS						
△	ISSUED FOR CONSTRUCTION	4-1-85	PLS						

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

DRY ACTIVE WASTE PROCESSING BUILDING
ARCHITECTURAL FLOOR PLAN
SECTIONS, SCHEDULES

ENG. MGR.	
DATE	
SCALE: AS NOTED	DRAWING NO. AX1D65A14-1
JOB NO. 17555	REV. 1

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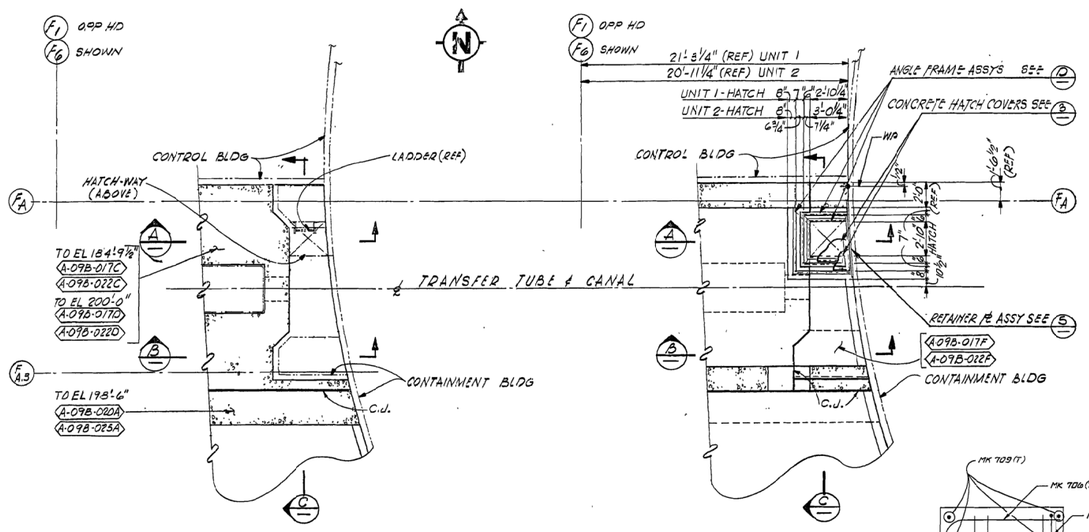
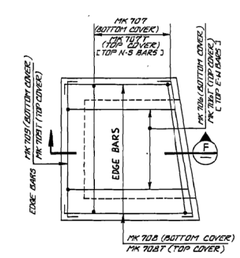
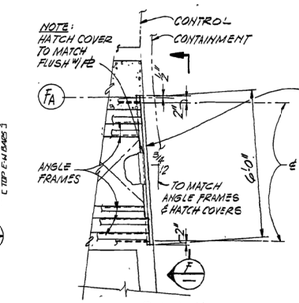
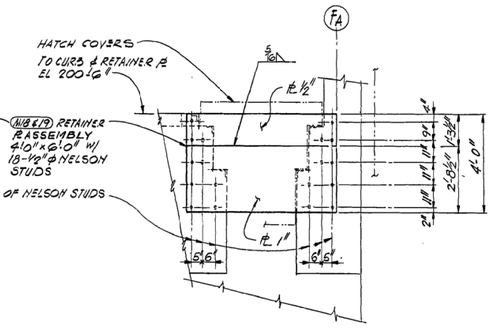


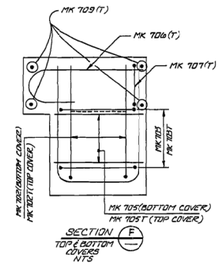
TRANSFER TUBE SHIELDING AND HATCHWAY

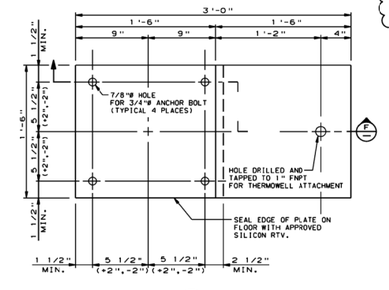
WORK WITH DWGS. AX2D09A005, 09A006, 09A008 & 09A009

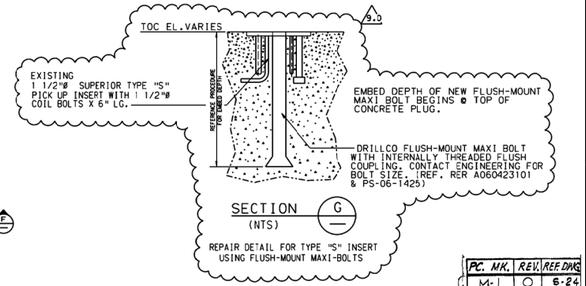












PC. MK.	REV.	REF DWG
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M-2	0	S-24
M-3	0	S-25
M-4	0	S-25
M-5	0	S-24
M-6	0	S-26
M-7	0	S-25
M-8	0	S-26
M-9	1	S-24
M-10	1	S-24
M-11	1	S-24
M-12	1	S-25
M-13	1	S-25
M-14	1	S-25
M-15	1	S-25
M-16	1	S-29
M-17	1	S-29
M-18	2	S-33
M-19	2	S-33

UNIT-1 1-REQ'D EA
2-REQ'D -
UNIT-1 1-REQ'D EA
16-REQ'D -
2-REQ'D -
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1-REQ'D -
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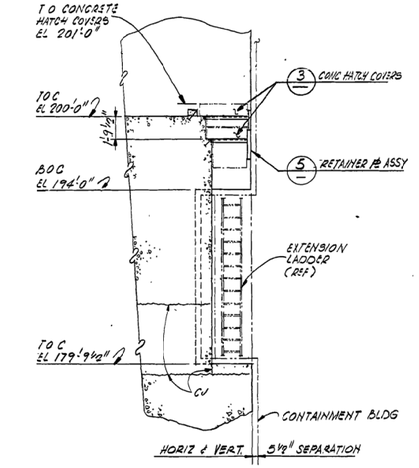
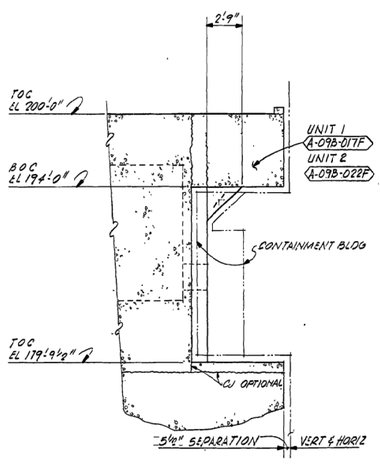
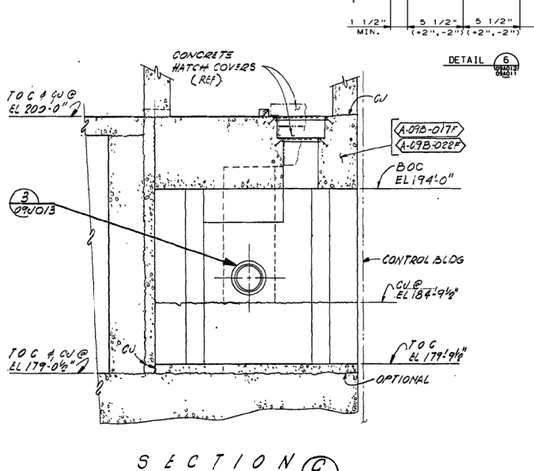
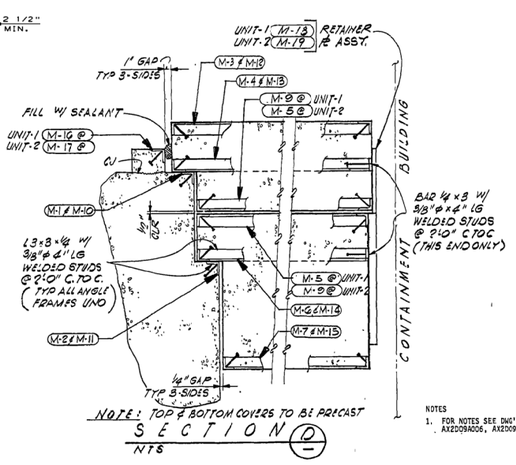
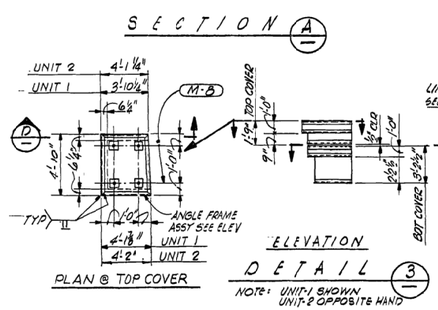


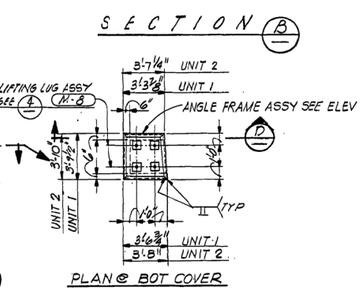


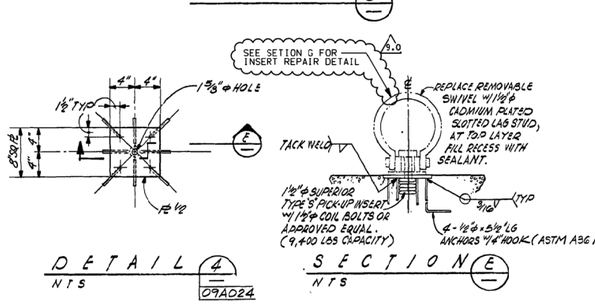


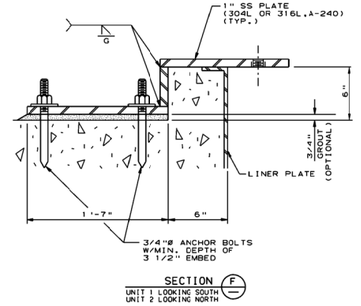


NOTES
1. FOR NOTES SEE DWG'S AX2D09A005, AX2D09A006, AX2D09A008 & AX2D09A009.









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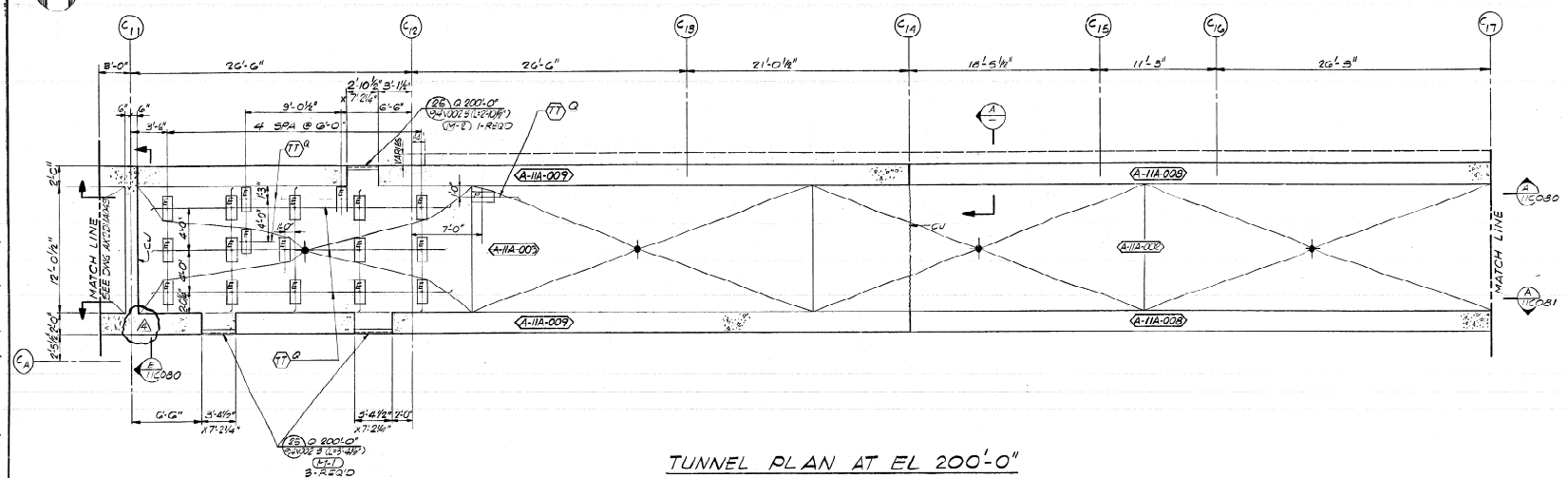
NO.	VERSIONS	DATE	DR	CHK	APPV
9.0	REVISED PER ABN-V03214, VER. 1.0	6-13-13	Vep	JWM	JMR

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

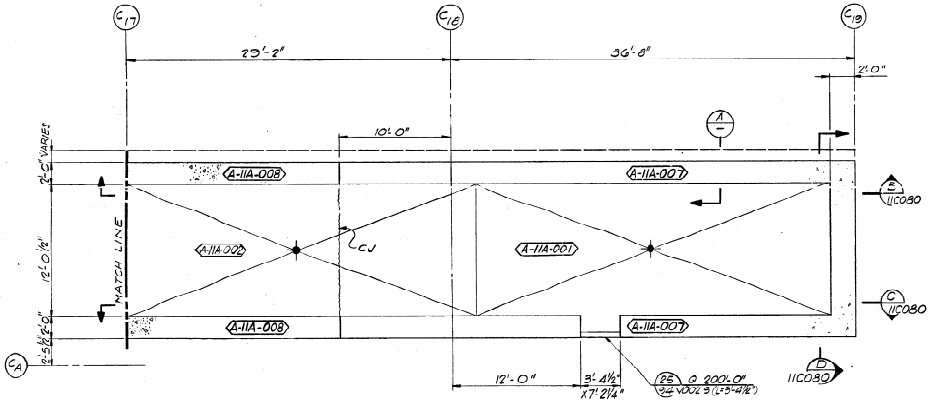
GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

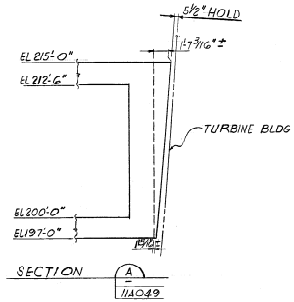
FUEL HANDLING BUILDING
PARTIAL PLANS SHT. 1

SCALE: NONE
JOB NO. 10604
DRAWING NO. AX2D09A017
VER. 9.0
SIZE E 34x44

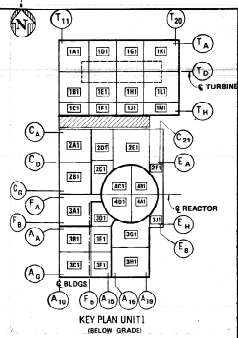


TUNNEL PLAN AT EL 200'-0"





TUNNEL PLAN AT EL 200'-0"



PC. MK.	REV.	REF DWG.
M1	0	S-2
M2	0	S-2

NOTES

1. FOR NOTES SEE DWG. AX2D11A001.
2. HIGH POINT OF FINISH FLOOR @ EL. 200'-0" SLOPES FLOORS TO DRAINS. (TOP OF DRAIN EL. 199'-10" UNLESS NOTED OTHERWISE).
3. TOP ELEVATION OF INSERT PLATES SHALL BE AT EL 200'-0".

ITEM	TYPE	QUAN	REF	ITEM	TYPE	QUAN	REV	ITEM	TYPE	QUAN	REF

EM NO. BU-11-A048

NO.	REVISIONS	DATE	DR.	CHK.	SUPV.	EGS	CHF.E	P.E	Q.A.E.
1	INCORP DCN #1	7-19	TN	BSP	BD	JP	NW		
2	ADD INSERT PLATES	8/20	AC	BSP	BD	JP	NW		
3	ADD PLATE & MISC DETAIL	9/30	AC	BSP	BD	JP	NW		
4	REV PLAN & ADD SECTION 'A'	11/10	AC	BSP	GO	JP	NW		
5	ISSUED FOR CONSTRUCTION	7/11	ZEE	BSP	GO	JP	NW		

NO.	REVISIONS	DATE	DR.	CHK.	SUPV.	EGS	CHF.E	P.E	Q.A.E.

ENG. MGR. _____ DATE _____

BECHTEL
LOS ANGELES

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

CONTROL BUILDING
FORMING PLAN AT EL. 200'-0"
ELECTRICAL TUNNEL UNIT 1

SCALE 1/4" = 1'-0" | DRAWING NO. AX2D11A048 | REV. 4

JOB NO. 9510

SIZE E 36X44

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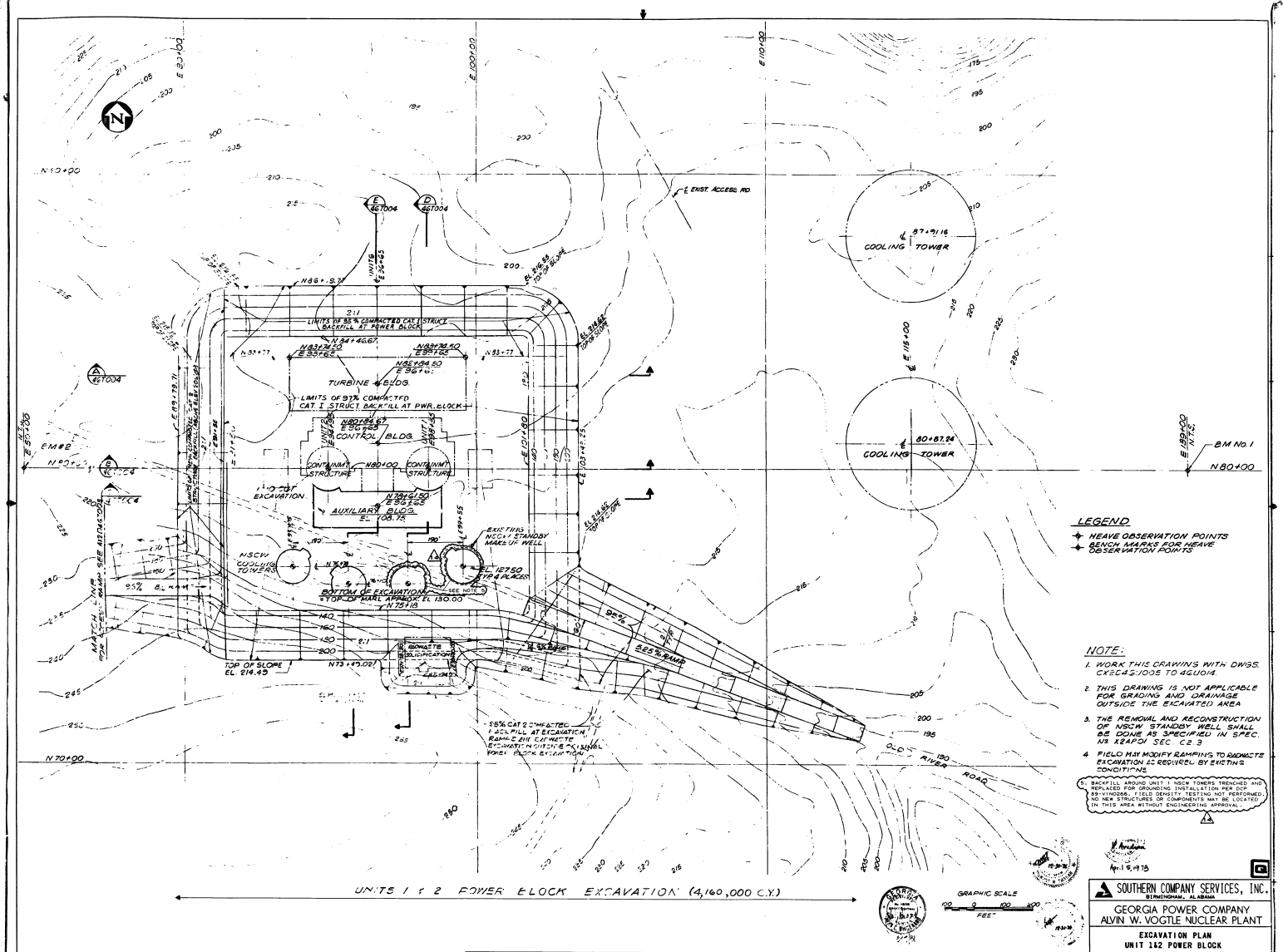
AX2D11A048



LEGEND

- HEAVE OBSERVATION POINTS
- BENCH MARKS FOR HEAVE OBSERVATION POINTS

NOTE:

1. WORK THIS DRAWING WITH DWGS. CXEC62/1005 TO 4GUOI4.
2. THIS DRAWING IS NOT APPLICABLE FOR GRADING AND DRAINAGE OUTSIDE THE EXCAVATED AREA
3. THE REMOVAL AND RECONSTRUCTION OF NSCW STANDBY WELL SHALL BE DONE AS SPECIFIED IN SPEC. NO. X2AP01 SEC. C2.3
4. FIELD MAY MODIFY RAMPING TO REDUCE EXCAVATION AS REQUIRED BY EXISTING CONDITIONS.
5. BACKFILL AROUND UNIT 1 NSCW TOWERS TRENCHED AND REPLACED FOR GROUNDING INSTALLATION PER DCP 89-V1N0286. FIELD DENSITY TESTING NOT PERFORMED. NO NEW STRUCTURES OR COMPONENTS MAY BE LOCATED IN THIS AREA WITHOUT ENGINEERING APPROVAL.

Apr-15-1998

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

EXCAVATION PLAN
UNIT 1&2 POWER BLOCK

SCALE: 1"=100'

DRAWING NO. AX2D46T001 REV. 14

JOB NO. 10604

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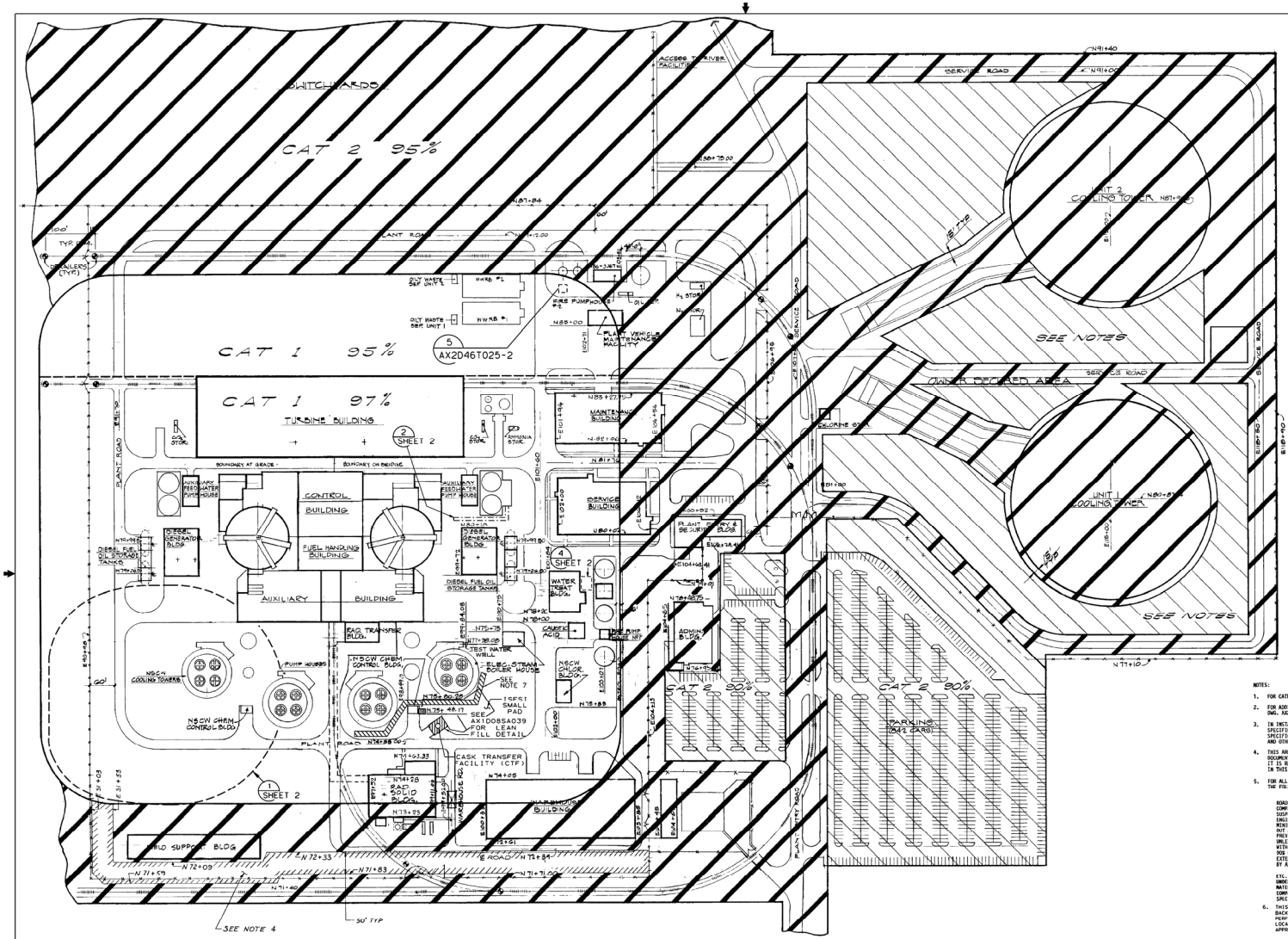
NO.	REVISIONS	DATE	DR	CHK	APPV	DTI
3	INCORP. ABN-24564	7/98	MMD			
2	REVISIONS	DATE	DR	CHK	APPV	DTI
1	ISSUED FOR CONSTRUCTION	SEE MICROFILM FOR SIGNATURES				
NO.	REVISIONS	DATE	DR			

NO. MOR. DATE

SIZE E 34x44



NOTES:

1. FOR CATEGORY 1 BACKFILL REQUIREMENTS, SEE DWGS AX2D46T001, T004.
2. FOR ADDITIONAL NON-CATEGORY 1 BACKFILL REQUIREMENTS, SEE DWG. AX2D46T026.
3. IN INSTANCES WHERE A CONFLICT EXISTS BETWEEN THIS DWG AND THE SPECIFICATIONS, THE SPECIFICATIONS SHALL GOVERN. REFERENCE SPECIFICATION NO. X2AP01, SECTIONS C2.2, C2.4, C2.5, C2.8 & C11.1 AND OTHER RELEVANT SPECIFICATIONS.
4. THIS AREA BACKFILLED WITH AREA FILL FROM ELEVATION 199.00 FT. NO DOCUMENTATION WAS PROVIDED AND NO MINIMUM COMPACTION WAS VERIFIED. IT IS RECOMMENDED THAT NO STRUCTURES, PIPING, DUCTS ETC. BE PLACED IN THIS AREA WITHOUT REPLACING AREA FILL WITH COMPACTED BACKFILL.
5. FOR ALL NON-CATEGORY 1 AREAS DESIGNATED AS 95% OR 90% COMPACTION, THE FOLLOWING SHALL APPLY:

   IN NON-CATEGORY 1 BACKFILL AREAS WHERE NO STRUCTURES, ROADS, UTILITIES, ETC EXIST OR ARE CONTEMPLATED, THE BACKFILL COMPACTION REQUIREMENTS MAY BE REDUCED OR ELIMINATED AND TESTING SUSPENDED AT THE DIRECTION AND DISCRETION OF THE GPC CIVIL AREA ENGINEER. IN SUCH AREAS, 90% COMPACTION SHALL BE PROVIDED A MINIMUM OF ONE FOOT BEYOND THE FOOT PRINT OF STRUCTURES AND ROADS AND 5 FEET BEYOND THE TOP OF SLOPE OF PREVIOUSLY PLACED NON-CATEGORY 1 BACKFILL AT 95% COMPACTION. UNLESS OTHERWISE SHOWN OR SPECIFIED, FILL SHALL BE PLACED WITHIN 10 FEET HORIZONTALLY OF A STRUCTURE AT A MINIMUM OF 90% COMPACTION. GEORGIA POWER COMPANY SHALL DOCUMENT THE EXTENT OF FILL PLACED UNDER THESE DISCRETIONARY PROVISIONS BY AS BUILT DRAWINGS.

   FUTURE STRUCTURES, RAILROADS, ROADS, PIPELINES, UTILITIES, ETC. THAT ARE TO BE LOCATED IN AREAS CONTAINING MATERIAL PLACED UNDER THE ABOVE DEFINED PROVISIONS SHALL REQUIRE REMOVAL OF FILL MATERIAL SO PLACED. SUCH MATERIAL SHALL BE REPLACED WITH MATERIAL COMPACTED IN ACCORDANCE WITH THE DRAWINGS AND THE PROVISIONS OF SPECIFICATION X2AP01 PERTINENT TO THE SPECIFIC APPLICATION.
6. THIS AREA CONTAINS APPROXIMATELY 6' OF UNDOCUMENTED BACKFILL AT GRADE LEVEL (FIELD DENSITY TESTING NOT PERFORMED). NO NEW STRUCTURES OR COMPONENTS MAY BE LOCATED ABOVE THIS BACKFILL WITHOUT ENGINEERING APPROVAL.
7. IT IS ALLOWABLE TO BACKFILL THE EXCAVATIONS AROUND THE MOPE CONDUITS WITH LEAN CONCRETE (MIN 2000 PSI). REFER TO BACKFILL SPECIFICATION X2AP01-C2.4.
8. OVERFLOW PARKING LOT AND FLEX EQUIPTMENT STORAGE BUILDING NOT SHOWN ON THIS DRAWING. FOR BACKFILL REQUIREMENTS OF AFOREMENTIONED AREA, SEE DRAWING AX2AP01-S0264-S ISSUED PER NCR SNC528260.

1.0

THIS DRAWING WAS RENUMBERED FROM AX2D46T025 (SHEET 1)
WORK THIS DRAWING WITH: AX2D46T025-2

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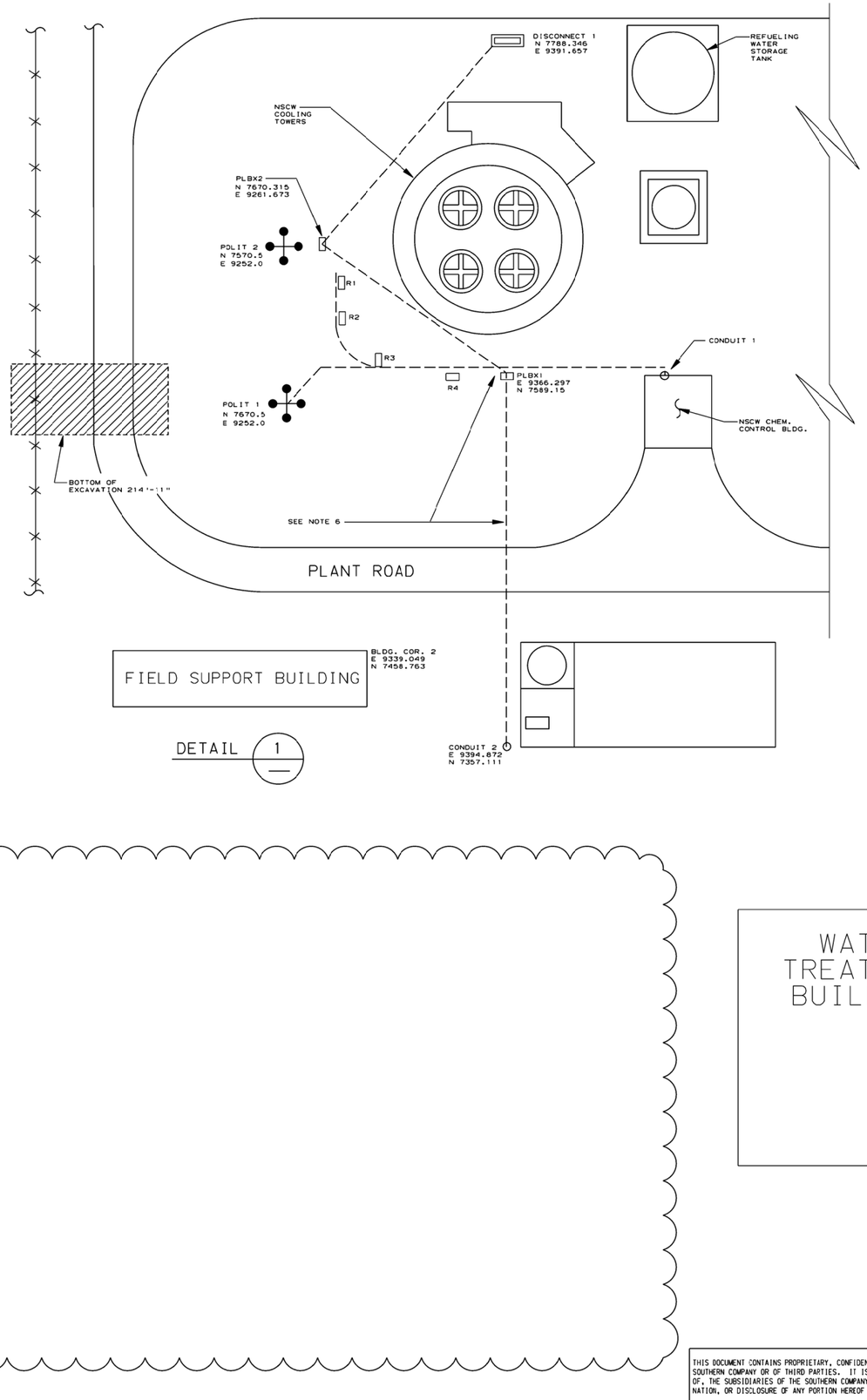
NO.	VERSIONS	DATE	DR	CHK	APPV
1.0	REVISED PER ADN SNC528260C014, VER. 1.0	02/05/16	TDA	PTL	WLN

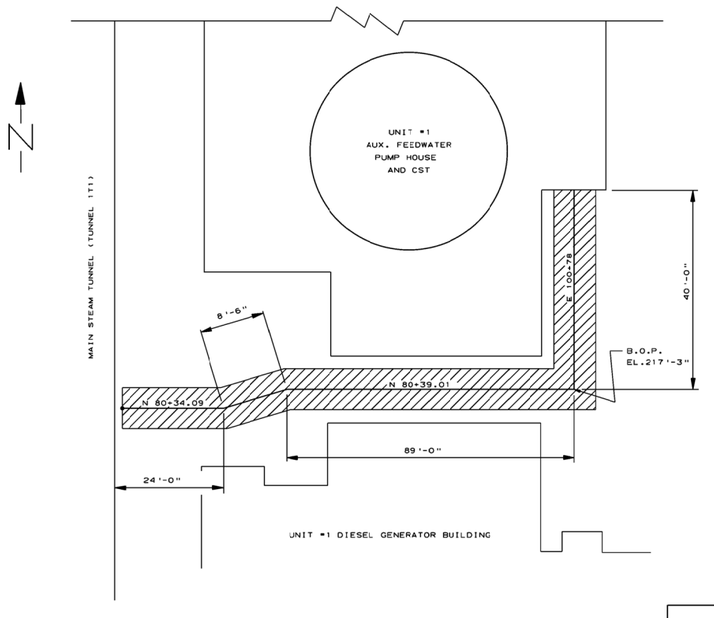
SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

NON-CATEGORY 1
BACKFILL REQUIREMENTS
YARD AREA
(SHEET 1 OF 2)

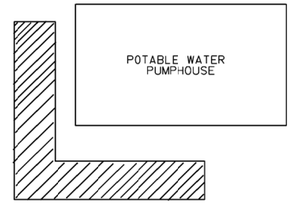
SCALE: 1"=80'
JOB NO.10604
DRAWING NO. AX2D46T025-1
VER. 1.0
SIZE E 34x44



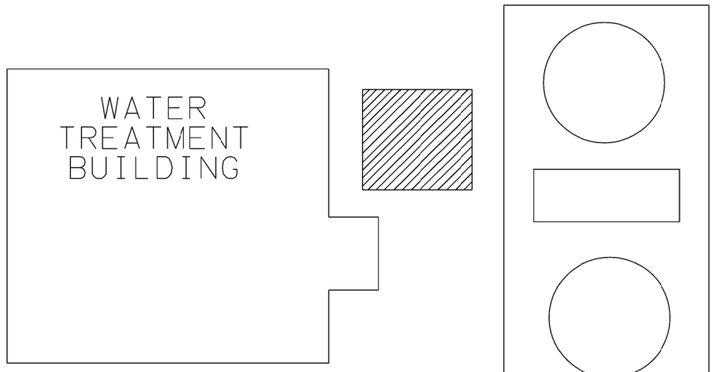


DETAIL 1

DETAIL 2



DETAIL 5



DETAIL 4

LEGEND

--- UNDOCUMENTED BACKFILL

R1 = EAST 9278.401 NORTH 7643.797
R2 = EAST 9283.337 NORTH 7612.469
R3 = EAST 9290.515 NORTH 7575.917
R4 = EAST 9358.851 NORTH 7586.894

UNDOCUMENTED COMPACTED BACKFILL

THIS DRAWING WAS RENUMBERED FROM AX2D46T025 (SHEET 2)

WORK THIS DRAWING WITH:
AX2D46T025-1

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NO.	VERSIONS	DATE	DR	CHK	APPV
7.0	REVISED PER ABN C100868001C014, VER 1.0	8-22-13	Vap	JWM	JMR

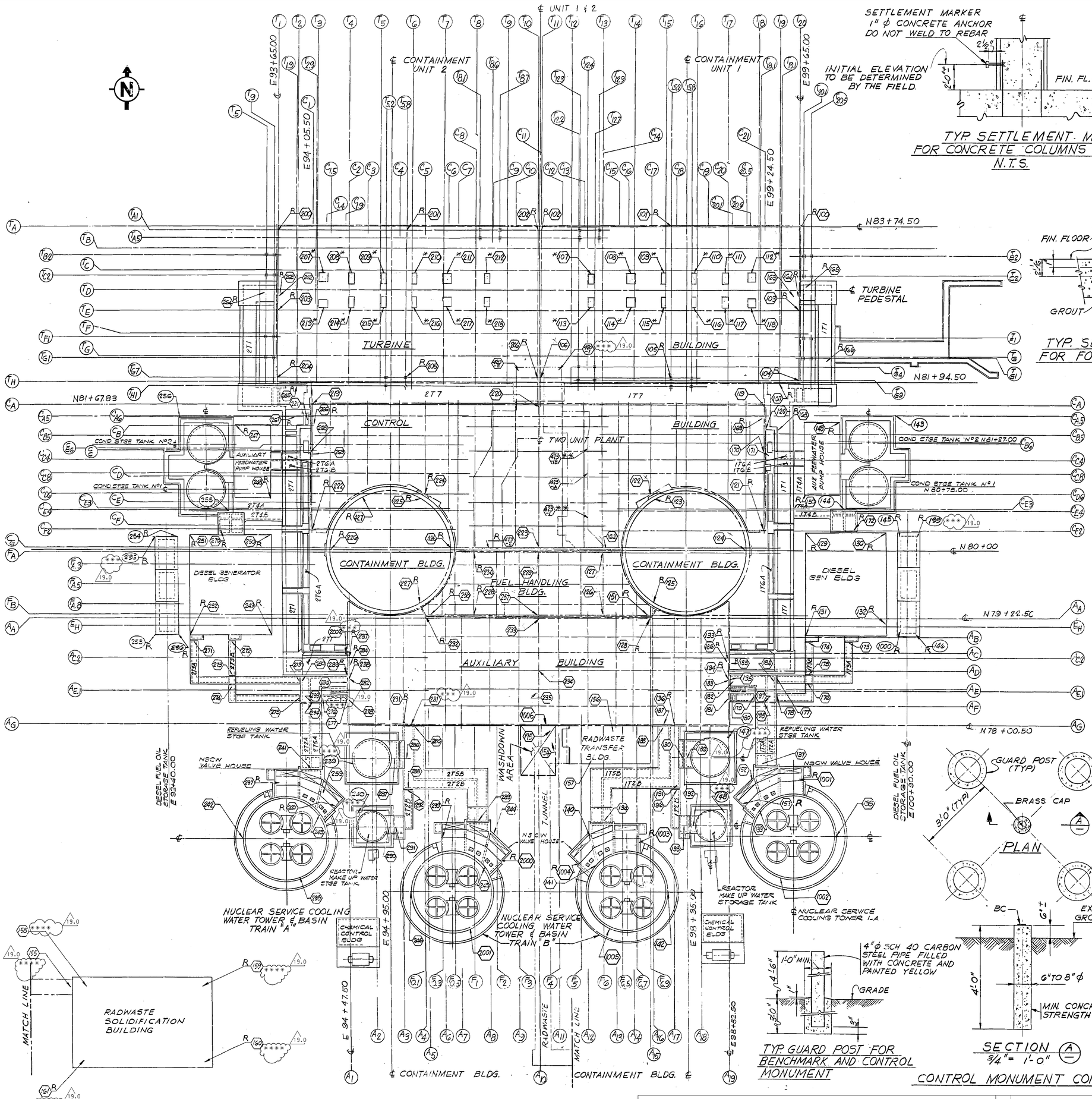
SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

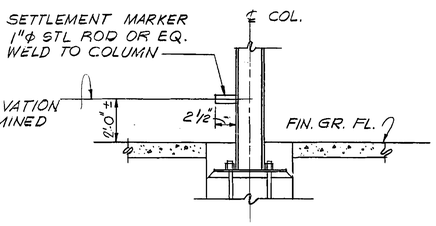
NON-CATEGORY 1
BACKFILL REQUIREMENTS
YARD AREA
(SHEET 2 OF 2)

SCALE: NONE	DRAWING NO.	VER.
JOB NO.10604	AX2D46T025-2	7.0

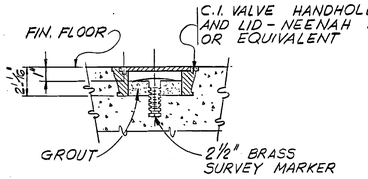
SIZE E 34x44



TYP. SETTLEMENT MARKER FOR CONCRETE COLUMNS OR WALLS
N.T.S.



TYP. SETTLEMENT MARKER FOR STEEL COLUMNS
N.T.S.



TYP. SETTLEMENT MARKER FOR FOUNDATION SLABS
N.T.S.

LEGEND

⬡ - INDICATES LOCATION AND NUMBER OF SETTLEMENT OBSERVATION MARKERS ON STRUCTURES.

⬭ - INDICATES LOCATION AND NUMBER OF SETTLEMENT OBSERVATION MARKER ON CATEGORY 1 TANKS.

NOTES.

1. CONSTRUCTION SHALL INSTALL SETTLEMENT MARKERS SHOWN ON THE LOCATION DRAWING AS SOON AS THE STEEL COLUMNS ARE ERECTED OR THE FORMS ARE STRIPPED FROM CONCRETE COLUMNS, WALLS, AND FOUNDATIONS (BEGINNING AT LEVEL "D" (BASEMAT) AUX. BLDG.
2. SETTLEMENT NUMBERS SHALL BE STENCILED OR OTHERWISE CLEARLY MARKED ON A FLAT SURFACE ADJACENT TO THE MARKER.
3. CONSTRUCTION SHALL LOCATE THE SETTLEMENT MARKERS ON OR NEAR THE SPECIFIED COLUMNS, ETC., SO THAT A CLEAR LINE OF SIGHT IS AVAILABLE FROM CENTRALIZED INSTRUMENT STATIONS WHERE POSSIBLE.
4. USING THE SETTLEMENT MARKER LOCATION DRAWING AND THE PLANT GENERAL ARRANGEMENT DRAWINGS THE LOCATIONS FOR THE INSTRUMENT STATIONS CAN BE ESTABLISHED.
5. ANY MARKER WHICH IS DESTROYED OR BECOMES INACCESSIBLE SHALL BE IMMEDIATELY REPLACED AS NEAR AS POSSIBLE TO THE ORIGINAL LOCATION AND ELEVATION. CONSTRUCTION WILL TIE THE ELEVATION OF A REPLACED OR RELOCATED MARKER TO THAT OF THE OLD MARKER. AN ADJUSTED INITIAL ELEVATION SHALL BE ESTABLISHED FOR THE NEW MARKER SO THAT AN UNINTERRUPTED SETTLEMENT OBSERVATION OF THE MARKER CAN BE MAINTAINED.
6. FOR CONVERSION TO GEORGIA STATE GRID COORDINATE SYSTEM SEE DWG. NO. CX2D45V002.
7. THE LOCATIONS OF BENCHMARKS ARE SHOWN ON DWG. AX2D46T001.
8. THE MONITORING AND RECORDING OF SETTLEMENT READINGS SHALL CONFORM TO THE PROCEDURES GIVEN IN SPECIFICATION NO. X2AP01 DIVISION C10 SECTION C10.1.
9. CONSTRUCTION OF CONTROL MONUMENTS SHALL BE AS SOON AS PRACTICAL AND AS SHOWN ON DETAIL DRAWING AX2D55V001 AND SHALL CONFORM TO SPECIFICATION NO. X2AP01 DIVISION C10 SECTION C10.1.48.
10. GPC FIELD TO FURNISH BECHTEL ENGINEERING WITH PROPOSED LOCATION FOR CONTROL MONUMENTS FOR INCLUSION INTO THE DRAWING.

* 11. READINGS DISCONTINUED 1/15/84.

12. CONCRETE ANCHORS MAY BE SUBSTITUTED FOR SETTLEMENT MARKERS 162 THROUGT 194.

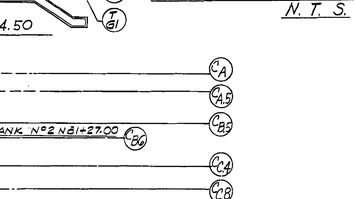
** 13. READINGS DISCONTINUED 10/20/87.

14. FOR TURBINE GENERATOR PEDESTAL SURVEY MARKERS REFERENCE DWGS. 1X2D51C001, 1X2D51C002, 1X2D51C004, 2X2D51C001, 2X2D51C002, 2X2D51C004.
15. FOR TURBINE BUILDING SHEETPILE DEFLECTION MARKERS AND SETTLEMENT MARKERS, SEE DWG. NO. AX2D10B008
16. FOR LOCATION OF SURVEY MARKERS USED DURING BACKFILL DEWATERING SEE DWG. NO. AX2D55V014
17. MARKERS RELOCATED TO IMPROVE ACCESS BEGINNING 10/20/87. MARKER AT ORIGINAL LOCATION TO BE PRESERVED FOR POSSIBLE FUTURE READINGS.

*** 18. READINGS DISCONTINUED 4/8/2005.

**** 19. READINGS DISCONTINUED 10/23/1991.

TYP GUARD POST FOR BENCHMARK AND CONTROL MONUMENT



SECTION (A)
3/4" = 1'-0"

CONTROL MONUMENT CONSTRUCTION DETAIL

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NO.	VERSIONS	DATE	DR	CHK	APPV
19.0	REVISED PER ABN-52167, VER.1.0	6-30-05	BgT	LRP	ASK

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

SETTLEMENT OBSERVATION MARKERS
LOCATION & DETAIL

SCALE: 1"=40'-0" | JOB NO.10604 | SIZE E 34X44

DRAWING NO.	VER.
AX2D55V001	19.0

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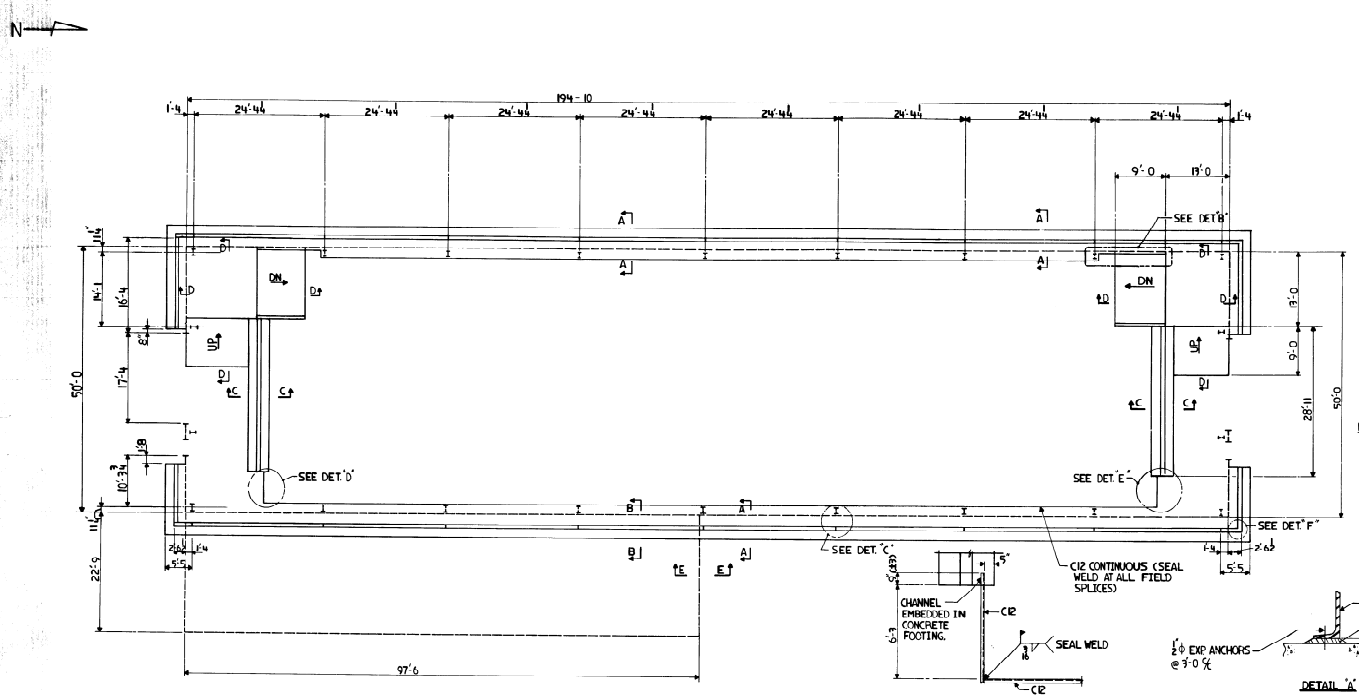
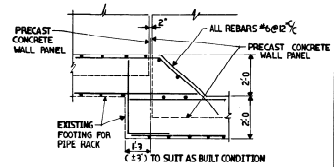
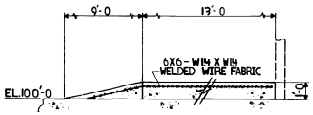


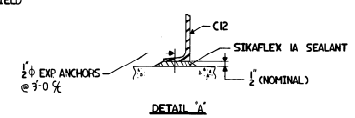


SECTION E-E
SCALE: ½"=1'-0



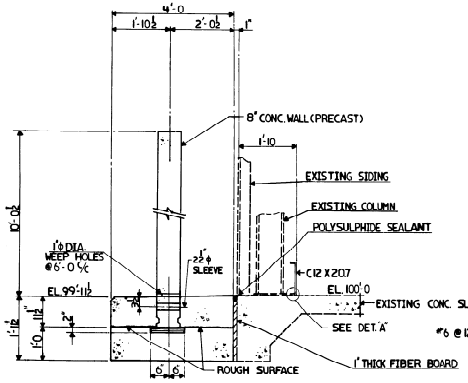
SECTION D-D
SCALE: NONE

DETAIL 'D' (SHOWN)
DETAIL 'E' (OPP HAND)
SCALE: ⅜"=1'-0

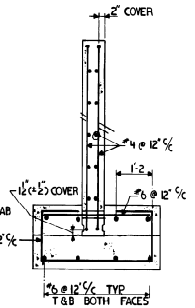


DETAIL 'A'

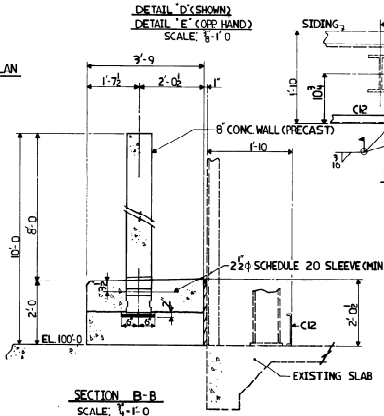
DRY ACTIVE WASTE STORAGE BLDG. PLAN
SCALE: ⅛"=1'-0



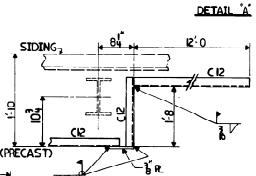
SECTION A-A
SCALE: ¼"=1'-0



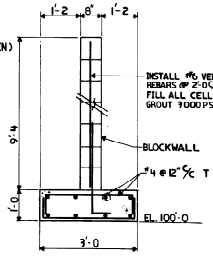
TYPICAL REINFORCING SECT.



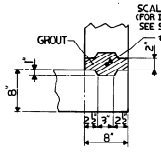
SECTION B-B
SCALE: ¼"=1'-0
(FOR INFORMATION NOT SHOWN SEE SECT. A-A)



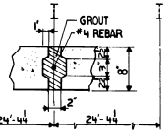
DETAIL 'B'



SECTION C-C
SCALE: ¾"=1'-0



DETAIL 'F'



DETAIL 'C'
SCALE: 1½"=1'-0

NOTES

1) FOR GENERAL NOTES SEE DWG. AX2D64901Ø
2) FOR REBAR LAP SPLICE AND DEVELOPMENT LENGTHS SEE DWG. AX2D64ØØ1Ø
3) MINIMUM CONCRETE STRENGTH AT 28 DAYS:
   i. FOOTING: 3000 PSI
   ii. PRECAST WALL UNITS: 4000 PSI
   iii. MUD MAT: 2000 PSI
   MAX. AGGREGATE SIZE FOR ALL MIXES TO BE 3/4"
4) CEMENT GROUT SHALL HAVE 4000 PSI (MIN.) COMPRESSIVE STRENGTH AT 28 DAYS (UNO)
5) ALL CIVIL WORK SHALL BE PERFORMED PER SPECIFICATION X2AP01
6) EXISTING FLOOR SLAB OF THE BLDG. SHALL BE COATED WITH SYSTEM FD-19 PER SPECIFICATION X1AJ03
7) LIFTING ARRANGEMENT FOR PRECAST WALL UNITS INCLUDING LOCATION OF LIFTING POINTS SHALL BE DETERMINED BY CPC'S CIVIL ENGINEER AND APPROVED BY NPSC
8) INSTALLATION SEQUENCE FOR PRECAST CONCRETE WALL UNITS:
   i. POUR LOWER PORTION OF THE WALL FOOTING
   ii. LIFT PRECAST WALL UNITS IN PLACE
   iii. LEVEL AND PLUMB THE WALL UNITS BY PROVIDING SHIMS AS REQUIRED
   iv. PROVIDE TEMPORARY HORIZONTAL SUPPORTS FOR THE WALL UNITS
   v. GROUT UNDER THE WALL UNITS
   vi. INSTALL REBARS FOR SECOND STAGE CONCRETE OF FOOTING
   vii. POUR SECOND STAGE CONCRETE FOR FOOTING, AND ENSURE THAT CONCRETE IS PLACED IN THE 2 1/2" DIA. SLEEVES WITHOUT ANY VOIDS
   viii. REMOVE TEMPORARY HORIZONTAL SUPPORTS AFTER FOOTING CONCRETE STRENGTH HAS ATTAINED A STRENGTH OF 3000 PSI
   ix. FILL GROUT BETWEEN ADJACENT WALL UNITS
9) THIS DRAWING SHOWS THE DETAILS OF SHIELDING WALLS AND OTHER MISCELLANEOUS ITEMS WHICH WILL BE ADDED TO EXISTING PIPING WARehouse TO MAKE IT SUITABLE FOR DRY ACTIVE WASTE STORAGE. ALL EXISTING DIMENSIONS AND DETAILS SHOWN ON THIS DRAWING ARE BASED UPON FIELD MEASUREMENT AND SURVEY.
10) TO SHOW RELATIVE ELEVATION DIFFERENCES BETWEEN DIFFERENT PARTS OF SHIELDING WALLS AND FOOTINGS, THE ELEVATION OF EXISTING CONCRETE SLAB IS DESIGNATED AS 100'-0". THIS IS NOT THE TRUE ELEVATION OF THE SLAB.

NO.	REVISIONS	DATE	DR.	CHK.	SUPV.	EGS	CHF.E	P.E.	Q.A.E.
3	ISSUED FOR CONSTRUCTION	8-3-88	NP	WJB-S					

ENGL. MGR. DATE

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

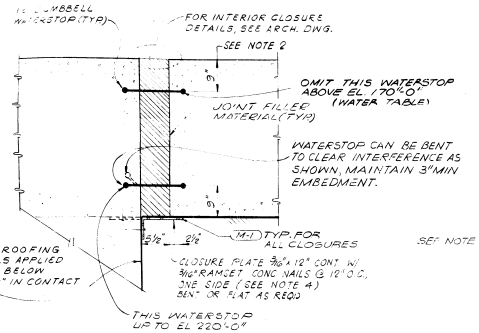
DRY ACTIVE WASTE STORAGE BLDG.
PLAN AND SECTIONS

SCALE: AS NOTED	DRAWING NO.	REV.
	AX2D65A400	0

"E" SIZE

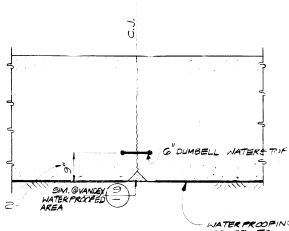
SCALING RULES

PIECE MARK	REV	DWG NO
M-1	1	—



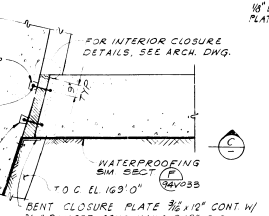
DETAIL 1 — N.T.S. — 94V030

(TYP. VERTICAL SEPARATION JOINT @ WALLS BELOW GRADE IN CONTACT W/ SOIL)



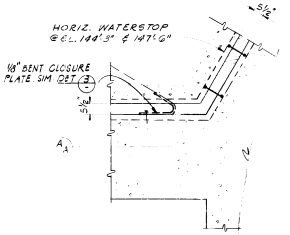
DETAIL 2 — N.T.S. — 94V030

(TYP. VERTICAL CONSTRUCTION JOINT @ WALLS BELOW GRADE IN CONTACT W/ SOIL)



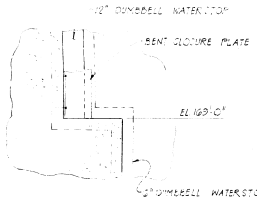
DETAIL 3 — 1/2" = 1'-0" — 94V030

PLAN ABOVE EL 169'-0"

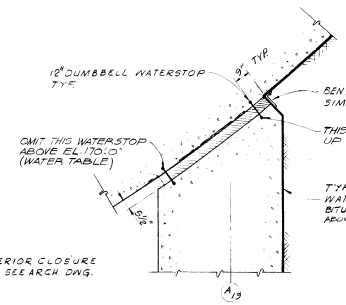


DETAIL 4 — 1/2" = 1'-0" — 94V030

PLAN @ EL. 148'-6"

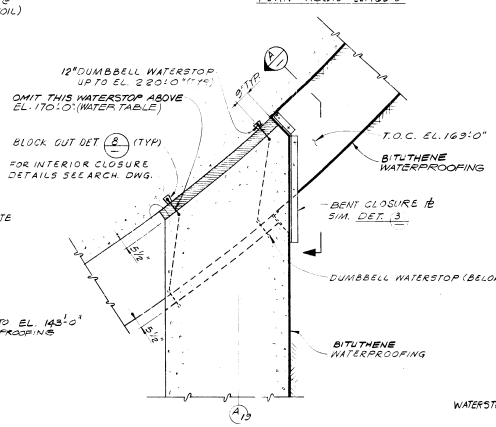


SECTION C — N.T.S.



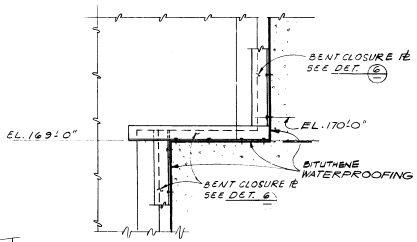
DETAIL 6 — 1/2" = 1'-0" — 94V030

PLAN BELOW EL. 169'-0"

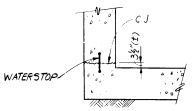


DETAIL 7 — 1/2" = 1'-0" — 94V030

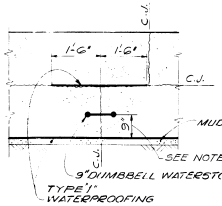
PLAN @ EL. 169'-0" & ABOVE



SECTION A — 1/2" = 1'-0"

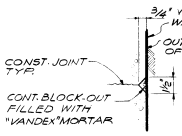


DETAIL 12

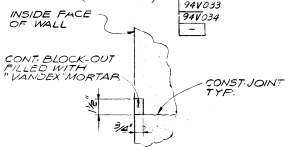


SECTION B — N.T.S. — 94V030

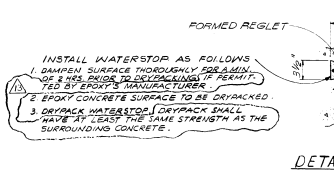
(TYP. HORIZ. CONSTRUCTION JOINT @ BASE MATS)



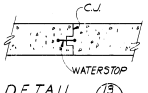
DETAIL 9 (V. SST. "B") — 94V032, 94V033, 94V034



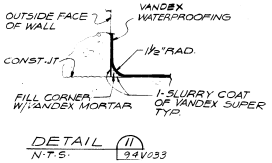
DETAIL 10 (V. SST. "A") — 94V032



DETAIL 8



DETAIL 13



DETAIL 11 — N.T.S. — 94V033

NOTES:

1. FOR LEGEND & NOTES SEE DWG AX2D94V030
2. WATERSTOP LOCATIONS MAY BE CHANGED WITH PRIOR APPROVAL BY THE ENGINEER TO AVOID LOCAL INTERFERENCE WITH REINFORCING BARS
3. FOR NOTES ON WATERPROOFING SEE DWG AX2D94V033
4. CLOSURE PLATE SHALL BE A36 STEEL, PROCURED UNDER SPEC X2AP01 SECTION C9.2. CLOSURE PLATE SHALL BE SAND-BLASTED AND COATED WITH ONE OF THE FOLLOWING MATERIALS:
   a. TNEMEC ASPHALTUM OR TNEMEC-TAR413 HB
   b. KOPPERS SUPER SERVICE BLACK OR GEL TYPE BITUMASTIC 300 HB
   c. RILEY BITUMASTIC
   d. CARBOLINE CARBOMASTIC #14
5. CLOSURE PLATES SHALL BE PROVIDED AT SEISMIC SEPARATION JOINTS BELOW EL 220'-0" IN CONTACT WITH SOIL
6. IN CASE OF REBAR INTERFERENCE OR CONGESTION ON THE EAST, WEST & SOUTH EXTERIOR WALLS OF THE AUXILIARY BUILDING, THE WATERSTOP AT THE BOTTOM OF THE LEVEL TO SLAB MAY BE LOWERED 4" 9" MIN. TO 24" MAX. DISTANCE FROM THE EXTERIOR FACE OF THE WALL.
7. WATERSTOPS MAY BE BENT TO ALLOW FOR REBAR INSTALLATION AND ALLOW FOR PROPER REBAR COVERAGE.

NOTES: (CONT'D)

8. CLOSURE PLATE MAY BE FABRICATED FROM 1/8" THICK MATERIAL. CERTIFIED MILL TEST REPORTS ARE NOT REQUIRED FOR 1/8" THICK CLOSURE PLATE.
9. DETAILS 12 & 13, ALONG WITH NOTE 7, CAN BE USED AS AN OPTION FOR REBAR AND WATERSTOP INTERFERENCE.

M. Arvelo
June 6, 1978

NO.	REVISIONS	DATE	DR.	CHK.	SUPV.	EGL	EGS	CHF. E	P.E.	Q.A.E.
14	DCN #14	8-25-80	RW	WN	GO					
13	ADDED: BITUTHENE, DCN 12 & 13	6-16-80			GO					
12	ISSUED FOR CHIEF ENG. APPROVAL	12/1980	CJ	DP	GO					
11	DCN #9, 10, 11	11-20-79	RW	EA	GO					
10	DCN #7, REV. NOTE 4	7-20-79	RW	EA	GO					
9	DCN #3, 4, 5 & 6	3-30-79	RW	EA	GO					
8	DCN #2, DWG. NO. WAS AX2D03H002	9-5-78	LOGS	EA	GO					
7	ISSUED FOR CONSTRUCTION	5-12-78	LOGS	EA	GO					
6	INCORP. DCN EQ(F-FCR8-14853).	12-13-77	RG	ST	ST					
5	DCN-19 W/MODIFICATION	2-21-77	FW	MA	JW					
4	DCN #16, 17 & 18. DCN #16 MODIFIED	11-24-76	TAD	DP	JW					
3	ISSUED FOR CONSTRUCTION	5-27-76	LOGS	EA	GO	HA				
2										
1										

ENG. MGR. DATE

SCALE: NOTED

DRAWING NO. 13

JOB NO. 9510

AX2D94V031

BECHTEL LOS ANGELES

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

WATERSTOPS & WATERPROOFING SECTIONS AND DETAILS SHEET 1

AX2D94V031

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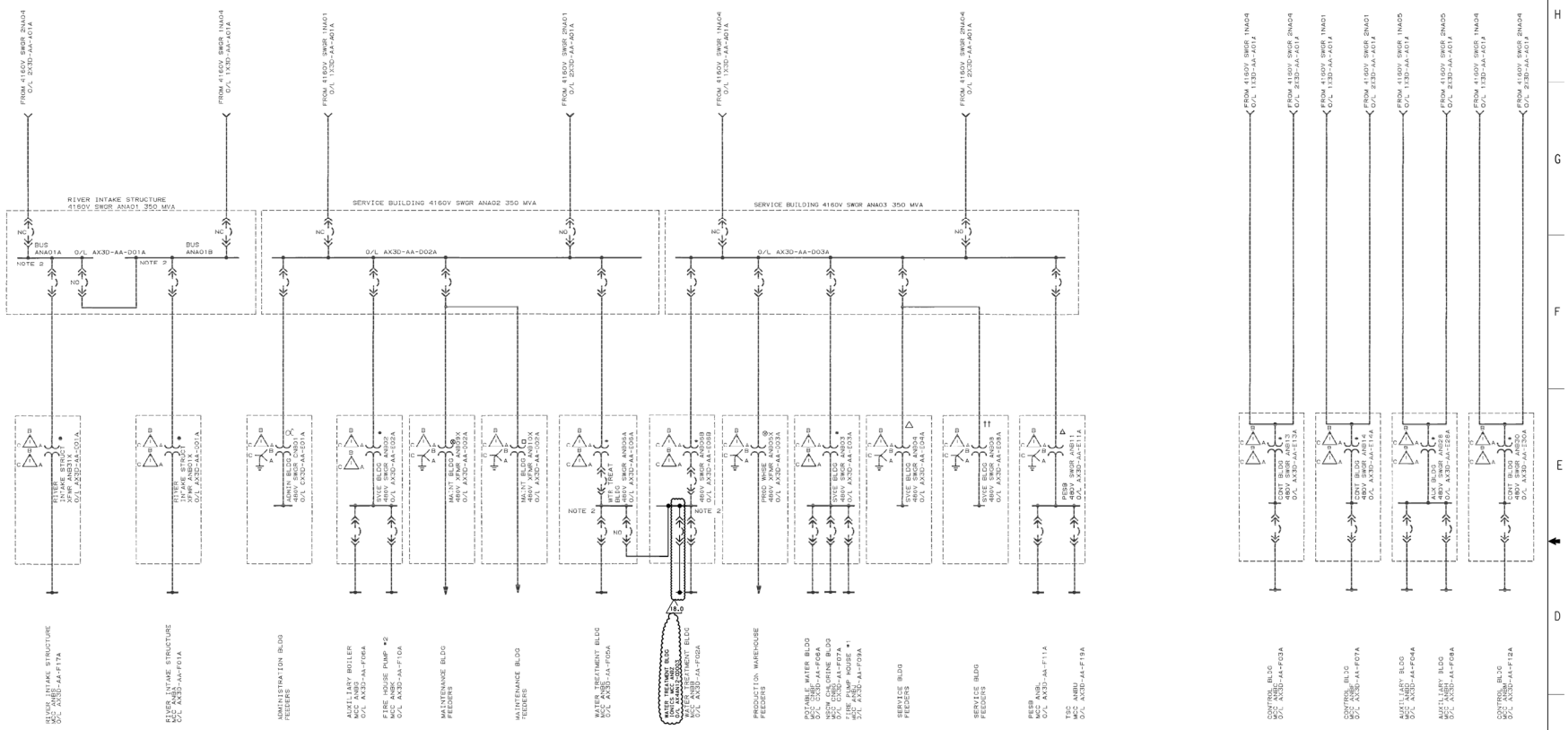

NOTES:

1. FOR DEVICE TABLE, GENERAL NOTES, REFERENCE DWGS AND LEGEND SEE DWGS 1X3D-AA-A00A THRU 1X3D-AA-A00E.
2. INCOMING BREAKERS SHALL BE INTERLOCKED SUCH THAT ONLY ONE FEEDER CAN SUPPLY THE BUS.
3. * LOAD CENTER TRANSFORMERS 4160-480V 1000/1333 KVA AA/FA 115°C RISE, Z=5.75%.
4. • LOAD CENTER TRANSFORMERS 4160-480V 300 KVA AA,115°C RISE, Z=2.7%.
5. ○ LOAD CENTER TRANSFORMERS 4160-480Y/277V 1000 KVA AA,115°C RISE,Z=8.0%.
6. + LOAD CENTER TRANSFORMERS 4160-480Y/277V 750 KVA AA,115°C RISE,Z=8.0%.
7. ⊕ LOAD CENTER TRANSFORMERS 4160-480Y/277V 750 KVA AA,115°C RISE,Z=5.75%.
8. □ LOAD CENTER TRANSFORMERS 4160-480Y/277V, 750 KVA AA, 80°C RISE, Z=5.75%
9. DELETED
10. △ LOAD CENTER TRANSFORMER 4160-480Y-277V 1000/1333KVA AA/FA 115°C RISE. Z=5.75%
11. FOR SWITCHING/LOAD DISPATCH EQUIP. NUMBERS SEE DWG 1X3D-AA-A01B.
12. †† LOAD CENTER TRANSFORMERS 4160-480Y/277V 1000/1333 KVA, AA/FA 115 DEG C RISE. Z=8.0%.

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REVISIONS						
18.0	REV PER ABN-V02086, VER. 1.0	4-19-10	DRW	JEM	PML	
0	ISSUED FOR CONSTRUCTION	SEE MICROFILM FOR SIGNATURES				
		DATE	DR	CHK	APPV	DTL

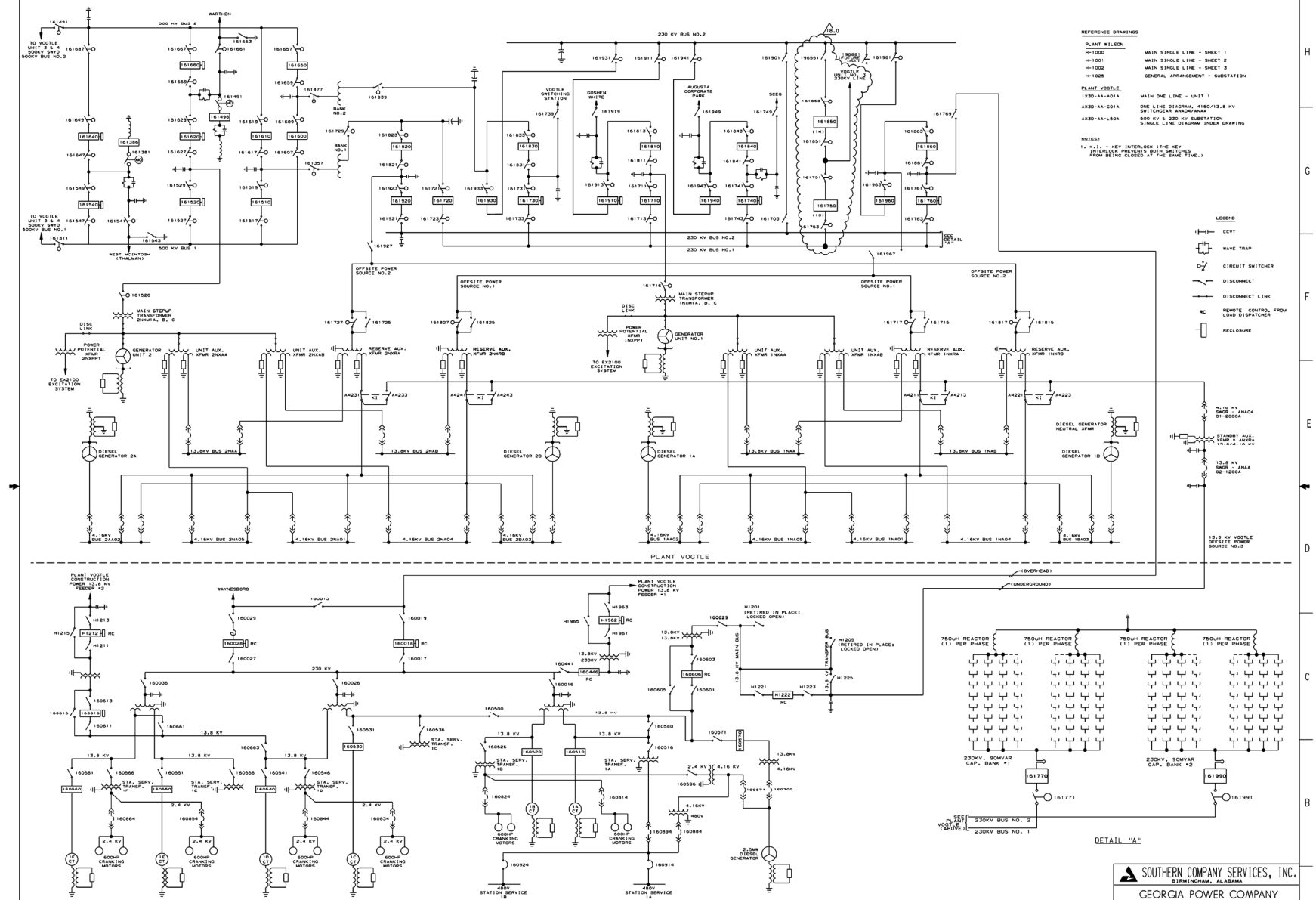
SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

MAIN ONE LINE
COMMON UNITS 1 AND 2

SCALE: NONE | DRAWING NO. AX3D-AA-A01A | VER. 18.0 | JOB NO. 10604

SIZE E 34x44 | DNG CATEGORY: CRITICAL



SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

VOGTLE - WILSON
MASTER ONE LINE DIAGRAM

SCALE: NONE

DRAWING NO. AX3D-AA-A03A

VER. 18.0

JOB NO.10604

NO.	VERSIONS	DATE	DR	CHK	APPV
18.0	REVISED PER ABN SNC445003E003, VER. 1.0	10/02/15	TDA	ALM	WLN

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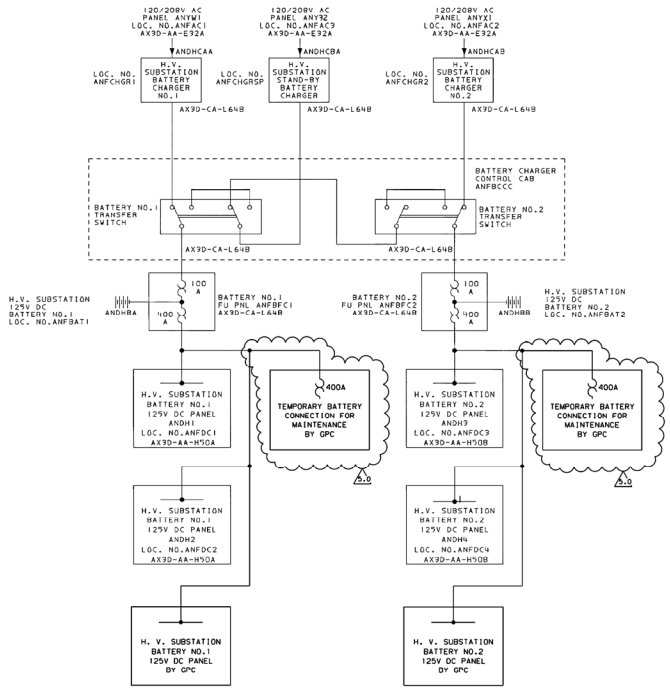
512E E 34x44

DMC CATEGORY: CRITICAL



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NO.	VERSIONS	DATE	DR	CHK	APPV
5.0	REVISED PER ABN SNC527044E005, VER. 1.0	7/20/15	LEH	ALM	WLN

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

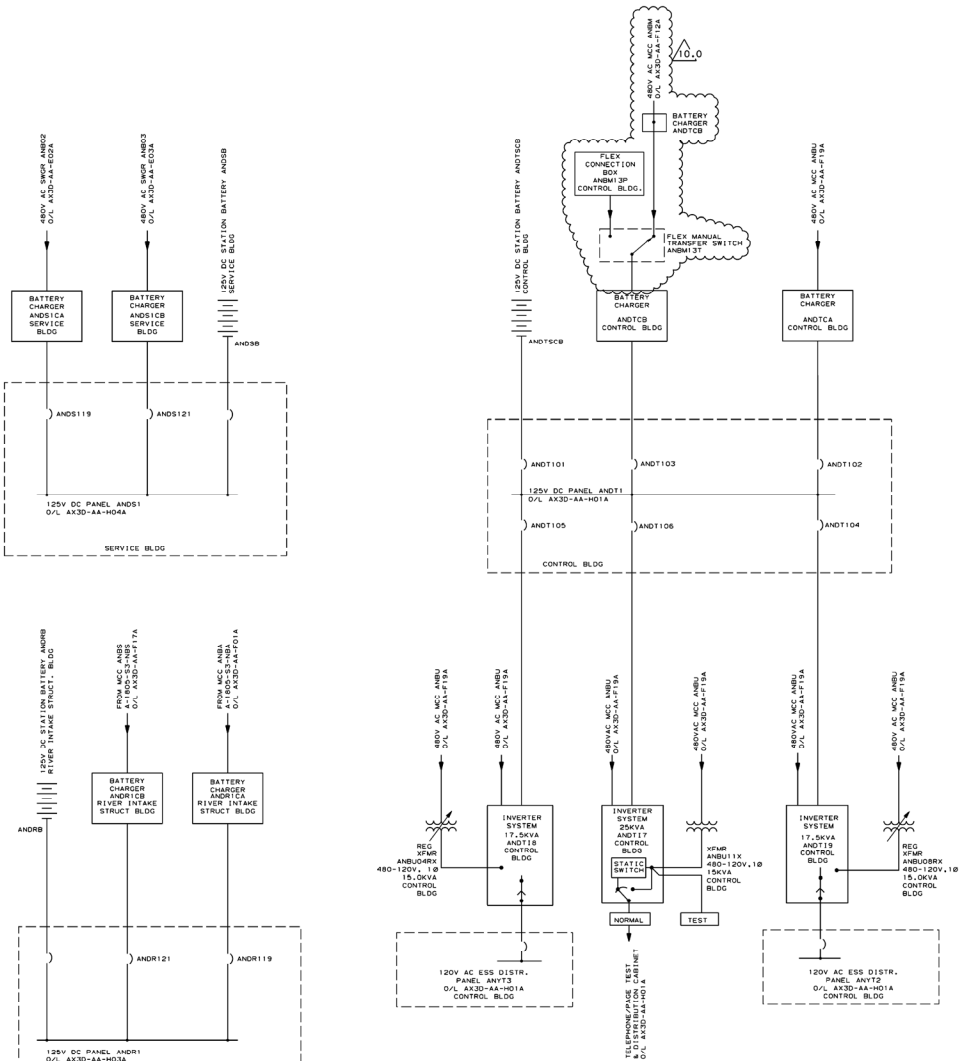
GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

MAIN ONE LINE
125V DC
HIGH VOLTAGE SUBSTATION

SCALE: NONE	DRAWING NO.	VER.
JOB NO.10604	AX3D-AA-G02A	5.0

SIZE E 34X44

DWG CATEGORY: CRITICAL



NOTE:
1. MECHANICAL INTERLOCKS SHALL BE PROVIDED SUCH THAT ONLY ONE BREAKER OF THE TWO INTERLOCKED IS CLOSED.

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

MAIN ONE LINE
NON-CLASS 1E 125V DC AND
120V ESSENTIAL AC SYSTEM (COMMON)

SCALE: NONE

DRAWING NO. AX3D-AA-G02B VER. 10.0

JOB NO. 10604

SIZE E 34x44

OWG CATEGORY: CRITICAL

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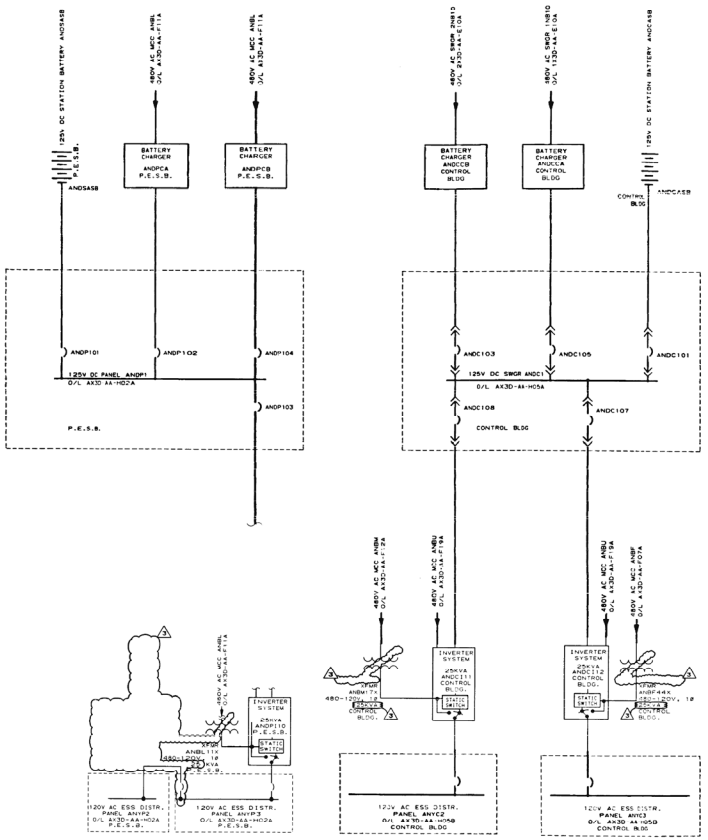
NO.	VERSIONS	DATE	DR	CHK	APP'V
10.0	REVISED PER ABN SNC475965E013, VER.1.0	4/16/15	TDA	SO	WLN



NOTE:
1. FOR HIGH VOLTAGE SWITCHYARD 125V DC SYSTEM MAIN ONE LINE SEE DWG AX3D-AA-G02A & AX3D-AA-G02B.

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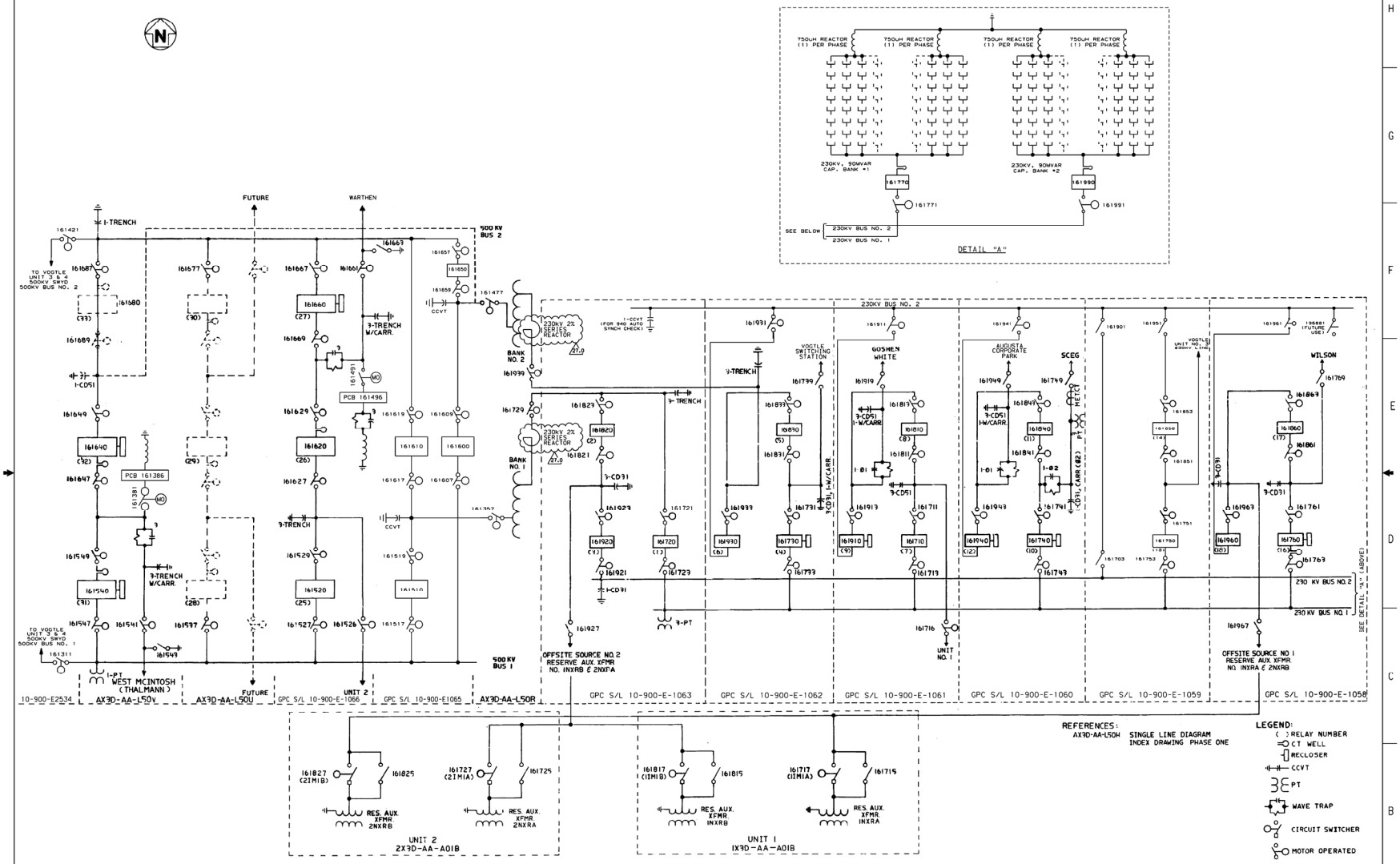
SOUTHERN COMPANY SERVICES, INC. BIRMINGHAM, ALABAMA	
GEORGIA POWER COMPANY ALVIN W. VOGTLE NUCLEAR PLANT	
MAIN ONE LINE CAS & SAS DISTRIBUTION SYSTEM	
SCALE: NONE	DRAWING NO. AX3D-AA-G02C
JOB NO. 10604	REV. 3

NO.	REVISIONS	DATE	DR	CHK	RE	DM	QM	DM	PDM
	INCORP. ABN-19774.								
	ISSUED FOR CONSTRUCTION.				SEE MICROFILM FOR SIGNATURES				
NO.	REVISIONS	DATE	DR	CHK	RE	DM	QM	DM	PDM

ENG. MGR. DATE

SIZE E 31x44



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27.0	REVISED PER ABN SNC444999E003, VER. 1.0	05/11/16	SMD	SO	WLN

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

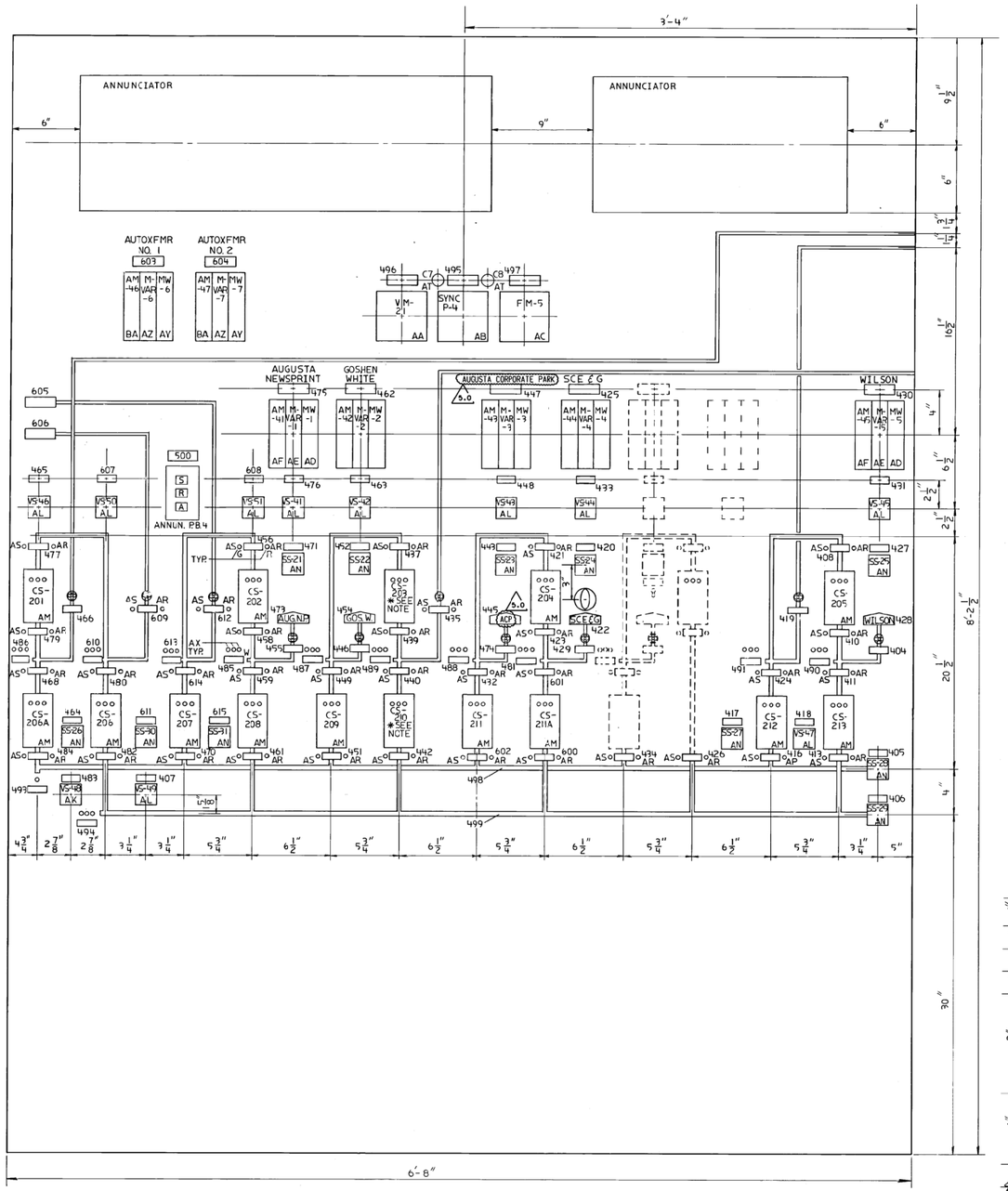
GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

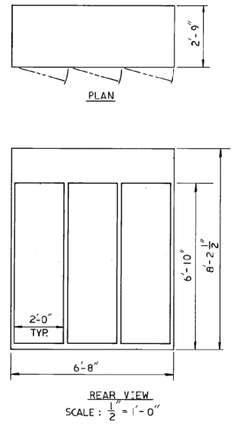
500KV & 230KV SUBSTATION
SINGLE LINE DIAGRAM
INDEX DRAWING

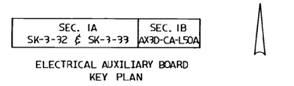
SCALE: NONE
JOB NO.10604
SIZE E 34X44

DRAWING NO. AX3D-AA-L50A
VER. 27.0

DWG CATEGORY: CRITICAL



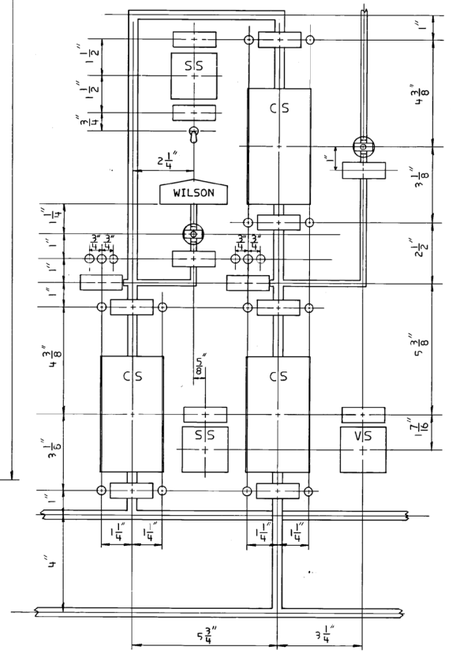




N

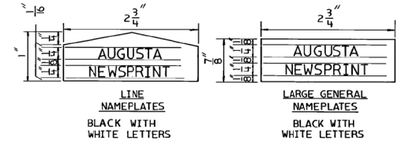
TYPICAL ARRANGEMENT

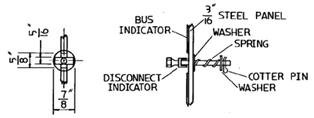
MAIN MIMIC BUS TO BE 3/8" WIDE, RED IN COLOR.
MIMIC BUS TO LINES, UNITS & AUX. TRANSFORMER TO BE 1/4" WIDE, RED IN COLOR.
* BLANK PLATE SIMILAR IN MATERIAL & CONSTRUCTION TO CONTROL SWITCH. DRILL FOR LIGHTS ONLY.



REFERENCE DRAWINGS

AX3D-AA-L50A THRU L50G - SINGLE LINE DIAGRAM
X3AE01 ATTACHMENT 24





NOTES

1. ADDED GOSHEN LINES 2 & 3, VOLT SWITCHES FOR BUS 1 & 2, TRAIN A & B. EXTENDED PANEL HEIGHT REMOVED UNIT NO.1 HIGH SIDE CONTROL, VOLT & SYN. SWITCHES ADDED NOTES INDICATED BY *
2. REARRANGED BOARD TO SHOW NEW LOCATION OF OFFSITE SOURCES AND 230KV LINES AND THE ADDITION OF AUTO-XFMR NOS. 1 & 2.
3. RELIANCE TO PUNCH AND DRILL HOLES FOR ALL FUTURE DEVICES (SHOWN DOTTED) AND COVER HOLES WITH BLANK PLATES.
4. WILSON LINE WAS WAYNESBORO, FUT. SCE & G LINE WAS WILSON, GOSHEN BLACK WAS GOSHEN NO.3, GOSHEN WHITE WAS GOSHEN NO. 2, AUGUSTA NEWSPRINT WAS GOSHEN NO. 1. DELETED NAMEPLATES 401,403,414,415 & 492. RELOCATED N.P. NOS. 427, 428, 430, 431 & 490. REVISED N.P. NOS. 419, 443, 445, 447, 448, 452, 454, 455, 462, 463, 471, 473, 475, 476, 485, 486, 487, 488 & 491.
5. ADDED N.P. NOS. 420, 422, 425, 433, 481 & 617 FOR SCE & G 230KV LINE. ADDED RECLS21. DELETED 2-MK. AD, 2-MK AE, 2-MK. AF, 1-MK AG, 1-MK AH & 1-MK AJ. REVISED GOS. WHITE, GOS. BLACK & SCE & G METERING, AM-42,43,44 & MVAR-2, 3,4 & MW-2,3,4.

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NO.	VERSIONS	DATE	DR	CHK	APPV
5.0	REVISED PER ABN-V03668. VER. 3.0	10/14/14	LEH	ALM	WLN

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

UNIT NO. 1
ELECTRICAL AUXILIARY BOARD, SEC. 1B
230 KV SUBSTATION CONTROL

SCALE: NONE	DRAWING NO.	VER.
JOB NO. 10604	AX3D-CA-L50A	5.0

SIZE E 34x44

LIGHTING SYMBOLS

- 2(2) 200VA; 4'-6" M 5 (PC1) — FIXTURE TYPE "2(2)" SHOWN ON FIXTURE SCHED. DWG. 200W, DENOTES WATTAGE OF FIXTURE, 5 DENOTES QUANTITY OF FIXTURES, 4'-6" DENOTES ACTUAL MOUNTING HEIGHT, TO BOTTOM OF FIXTURE M DENOTES MERC. VAPOR; i.e. I INCANDESCENT, F FLUORESCENT, MH METAL HALIDE, S HI-PRESS. SODIUM. (LED LIGHT EMITTING DIODE) (PC1) INDICATES FIXTURE CONTROLLED BY PHOTOCELL SCHEDULE OUTLET NO. PC1 (SEE AX3DG031) DETAIL "A" AS SHOWN ON DWG AX33DG032
- DET. A 3312 SYM. — POWER & LIGHTING DISTRIBUTION PANEL 120/208 V., 3Ø, 4W, NORMAL-
- ESSENTIAL LIGHTING DISTRIBUTION PANEL (VOLTAGE AS SHOWN ON LIGHTING RISER DIAGRAMS
- LIGHTING DISTRIBUTION PANEL ABOVE(TYP.) - 3Ø, 4W.
- INCANDESCENT, MERCURY METAL HALIDE FOR HI-PRESS. SODIUM FIXTURE "a" INDICATES THE FIXTURE WILL BE SWITCHED BY SWITCH 3A. THE UNLESS-1A INDICATES THE PANEL & CIRCUIT NUMBER TO WHICH IT IS CON-NECTED. I.E. PANEL 1NLP30 CIRCUIT NUMBER 14.
- a NLP30-14
- DC MODULAR LIGHTING UNIT - EACH ARROW INDICATES ONE LAMP UNIT
- FLUORESCENT FIXTURE.
- INDICATES FIXTURE FEED FROM EMERGENCY PANEL (REQUIRED FOR SAFE SHUTDOWN)
- HANDRAIL MOUNTED FIXTURE
- WALL MOUNTED FIXTURE
- ESSENTIAL LIGHTING FIXTURE
- EXIT LIGHT WITH DIRECTIONAL ARROW
- ET — TRITIUM EXIT SIGN. TO BE USED FOR NON RAD AREAS ONLY.
- DT — DIRECTIONAL TRITIUM EXIT SIGN. TO BE USED FOR NON RAD AREAS ONLY.
- E — INDICATES FIXTURE FED FROM EMERGENCY PANEL (REQUIRED FOR SAFE SHUTDOWN)
- SEI — ESSENTIAL LIGHTING FIXTURE
- FLOODLIGHT
- FLOODLIGHT ON EDG CIRCUIT
- S E R — EMERGENCY LIGHTING UNIT-EACH ARROW INDICATES ONE LAMP UNIT R-RECEPTACLE, TWIST LOCK, INDICATES EMERGENCY BATTERY PACK THAT MUST BE CARRIED TO LOCATION INDICATED AND PLUGGED IN, FOR REFUELING OR OTHER CONTAINMENT ACCESS. S(E) INDICATES SEISMIC EMERGENCY BATTERY PACK (MARK# FEX) REQUIRED FOR SAFE SHUTDOWN RECEPTACLE TWIST-LOCK "CO (20 V) 20A "I" OR "D" WITH COVER "EMI, PILM"EXT (20 V) WITH 3/4" M TYPE. "SC" CORD (10'C) OR "FM-A" CORD (10'C)
- S_D — SWITCH, SINGLE POLE, 20 AMP., 125 V.A.C. LETTER INDICATES FIXTURES CONTROLLED
- S_{D2} — SWITCH, DOUBLE POLE, 20 AMP., 125 V.A.C.
- S_3 — SWITCH, 3 WAY, 20 AMP., 125 V.A.C.
- T — THERMOSTAT
- SINGLE CONVENIENCE RECEPTACLE, 3 WIRE 2P., GROUNDED, 20A-125V AC/DC. TWIST-LOCK HUBBELL CAT. #23CN10 OR APPROVED EQUAL. (MARK #LRL)
- SINGLE CLOCK RECEPTACLE, 3 WIRE, 15A-125V AC, 2P & ONE POLE GROUNDED CLOCK & PAN HANGER WITH STAINLESS STEEL PLATE. SIERRA CAT. #2723 OR APPROVED EQUAL (MARK #RPH)
- DUPLEX CONVENIENCE RECEPTACLES, 3 WIRE, 2 POLE 15A, 250V W/I POLE GROUNDED, HUBBELL CAT. #5662 OR APPROVED EQUAL. STRAIGHT BLADE (MARK #RDP & EDG)
- DUPLEX CONVENIENCE RECEPTACLE, 3 WIRE, GROUNDED TYPE, 20A- 250 HUBBELL CAT. #5362 OR APPROVED EQUAL. STRAIGHT BLADE. (MARK # EDG)
- GF — DUPLEX GROUND FAULT RECEPTACLE, 3 WIRE, 20A-125V HUBBELL CAT.#GF-5362 OR APPROVED EQUAL, STRAIGHT BLADE (MARK#ECH)
- WP — DEADTED RECEPTACLE MOUNTED IN APPLETON FS-1-TS OUTLET BOX W/WEATHERPROOF COVER APPLETON CAT.#SE-HYDE SURFACE MTD. OR APPROVED EQUAL
- DC — SINGE CONVENIENCE OUTLET, 3 WIRE, 2 POLE, 20A 125V DC, TWIST LOCK, HUBBELL CAT.#23CN10 OR APPROVED EQUAL. (SEE NOTE 26) (MARK#EDG).
- 30° ANGLE REFLECTOR FIXTURE - ARROW INDICATES LIGHT AIMING
- JUNCTION BOX
- CONDULET
- CONDUIT RUN-EXPOSED OR ABOVE CEILING, UNLESS NOTED OTHERWISE.
- CONDUIT RUN-CONCEALED IN FLOOR SLAB, UNDERGROUND OR UNDER PLATFORMS.
- CONDUIT UP
- CONDUIT DOWN
- CONDUIT RUN-CHANGE IN ELEVATION
- S_P — SURFACE MOUNTED LIGHTING PRESSWITCH SINGLE POLE, 20 A., 120 V.-HUBBELL CAT. #1361-W/WPLATE WEATHERPROOF YELLOW PLATE & 125 V. PILOT LIGHT - HUBBELL #1NCMB5 MOUNTED IN A MALLEABLE IRON "FS" BOX, APPLETON CAT, #FS-1-75, MOUNTED AT 4'-6" FROM FLOOR.
- 2NLP10-17 — LIGHTING CIRCUIT IDENTIFICATION: UNIT #2-LIGHTING PANEL 2NLP10 BREAKER #17
- 7 — LIGHTING TRANSFORMER 480 V. 3Ø.3W. PRIMARY, 208Y/120V.3Ø.4W SECONDARY, KVA SIZE PER RISER DIAGRAMS
- 113 — DENOTES ROOM OR AREA NUMBER. SEE ARCHITECTURAL FINISH AND DOOR SCHEDULES FOR COMPLETE INFORMATION.
- S_{PL} — SWITCH, SINGLE POLE, 20A WITH PILOT LIGHT 4'-6"
- S_{2PL} — SWITCH, 2-POLE, 20A WITH PILOT LIGHT 4'-6"
- (S) — DENOTES SIESMIC CATEGORY I SYSTEMS (REF LIGHTING NOTE 46)

LIGHTING NOTES

1. THE LIGHTING SYSTEM SHALL BE INSTALLED IN ACCORDANCE WITH THE LATEST ISSUE OF THE NATIONAL ELECTRIC CODE
2. CONDUITS, ALS AND WIRES FOR LIGHTING PANEL BRANCH CIRCUITS ARE NOT SHOWN ON LIGHTING PLAN DRAWINGS. FIELD TO REFER TO LIGHTING RISER DRAWINGS (SK-5-) FOR BRANCH CIRCUIT REQUIREMENTS.
3. ALL EMBEDDED LIGHTING CONDUITS WITHIN THE PLANT SHALL BE GALVANIZED RIGID STEEL WITH WATERTIGHT CONNECTORS.
4. CONDUIT RUNS IN WALLS AND FLOOR SLAB SHALL BE 3/4" UNLESS OTHERWISE NOTED. CONDUITS IN CONCRETE SLAB ABOVE GRADE SHALL BE MINIMUM 3/4" UNLESS OTHERWISE NOTED.
5. DIRECT BURIAL CABLE SHALL BE USED FOR MULTIPLE ROADWAY LIGHTING UNDERGROUND COPPER CABLE, TYPE "USE", 600V.
6. EXPOSED CONDUIT SHALL BE RUN EITHER PERPENDICULAR OR PARALLEL TO ADJACENT STRUCTURAL STEEL IN WORKMAN-LIKE MANNER.
7. FIXTURE STEMS (PENDANTS) SHALL BE 3/4" RIGID STEEL CONDUIT GALV. UNLESS OTHERWISE NOTED.
8. ALUMINUM SHEATHED TYPE (ALS) CABLE MAY BE USED FOR RECESSED LIGHTING IN SUSPENDED CEILINGS.
9. ALUMINUM SHEATHED TYPE (ALS) CABLE SHALL BE USED FOR LIGHTING WHERE POSSIBLE. (SEE NOTE 24)
10. FIXTURE STEMS SHALL NOT EXCEED 96"
11. FIXTURES WITHOUT AN ELEVATION NOTED SHALL BE SURFACE OR FLUSH MTD ON STEEL OR CEILING. FIXTURES AND RECEPTACLES SHALL BE INSTALLED AS SHOWN ON DWGS.
12. FIXTURES SHOWN IN BLACK ARE ON ESSENTIAL LIGHTING CIRCUITS.
13. ALL SURFACE MOUNTED RECEPTACLES SHALL BE 3'-0" ABOVE FINISHED FLOOR, UNLESS OTHERWISE NOTED. FLUSH MOUNTED RECEPTACLES SHALL BE 12". CLOCK RECEPTACLES SHALL BE 7'-0" ABOVE FINISHED FLOOR U.N.O.
14. LOCAL SWITCHES FOR LIGHTING CIRCUITS IN OFFICES AND ROOMS WITH HOLLOW PARTITIONED WALLS SHALL BE FLUSH MOUNTED IN FINISH WALL OR EQUAL WITH SWITCH PLATE HUBBELL CATH2017 OR EQUAL. AT 4'-0" ABOVE FL.
15. LOCAL SWITCHES IN ALL OTHER INDOOR LOCATIONS SHALL BE SURFACE MOUNTED IN FD CONDULETS WITH SWEET COVERS.
16. LOCAL SWITCHES IN LOCATIONS EXPOSED TO THE WEATHER SHALL BE FLUSH MOUNTED IN FD CONDULETS WITH WEATHER-PROOF COVERS.
17. DELETED
18. LIGHTING BRANCH CIRCUITS OR HOME RUNS SHALL NOT BE INSTALLED IN CABLE TRAYS.
19. LIGHTING CIRCUITS FOR FIXTURES IN ALL AREAS BUT SMALL ROOMS ARE TO BE DISTRIBUTED SO THE FAILURE OF ONE CIRCUIT WILL NOT COMPLETELY BLACK OUT AN AREA.
20. CABLE SPLICES SHALL BE MADE WITH SOLDERLESS TYPE CONNECTORS (WIRE NUTS OR APPROVED EQUAL).
21. ALL LIGHTING BRANCH CIRCUIT RUNS IN UNDERGROUND DUCTS SHALL BE 600 VOLT TYPE THN AND SHALL RUN IN BOTTS TERRATHERM DUCTS.
22. CONDUIT INSTALLED IN CONCRETE SLAB.
    (A) CONDUITS SHALL BE INSTALLED IN A MANNER AS TO INTERFERE WITH STEEL REINFORCING BARS.
    (B) NO FIXTURE BOXES OR CONDUIT SHALL BE INSTALLED WITH-IN A 3 FT. RADIUS OF DECK DRAIN HOLES. ANY EXCEPTIONS MUST BE APPROVED BY THE FIELD ENGINEER.
    (C) FIXTURE BOXES SHALL NOT INTERFERE WITH ANY STRUCTURAL STEEL
    (D) CONDUITS SHALL BE RUN AS CLOSE TO BEAMS AS PRACTICAL.
    (E) ALL CONDUITS SHALL BE SECURED TO ANCHORED CHAIR TO PREVENT ANY FLOATING OR FALLING.
23. MATERIAL CONTAINING ALUMINUM OR TEFLON SHALL NOT BE USED INSIDE THE CONTAINMENT BUILDING.
24. EXPOSED RACEWAY SYSTEM SHALL BE AS SHOWN BELOW FOR THE BLDG./AREA AS INDICATED.
    A. CONTROL BLDG: SERVICE BLDG: WATER TREATMENT BLDG. AUXILIARY BLDG. DIESEL GENERATOR BLDG.
       (a) ALUMINUM-SHEATHED CABLE (ALS)
       (b) GALV. RIGID STEEL CONDUIT
    B. TURBINE BLDG.
       (a) ALUMINUM-SHEATHED CABLE (ALS)
       (b) ALUMINUM CONDUIT
    C. INTAKE STRUCTURE AND TUNNELS
       (a) GALV. RIGID STEEL CONDUIT OR ALUMINUM-SHEATHED CABLE (ALS) IN PROTECTED AREAS
    D. CONTAINMENT BLDG. & FUEL HANDLING (POOL AREA)
       (a) GALV. RIGID STEEL CONDUIT WITH MALLEABLE IRON FITTINGS OR "MC" INTERLOCK CABLE AND FITTINGS.
    E. OUTSIDE AREAS
       (a) DUCT RUNS - PVC
       (b) EXPOSED CONDENSATE STORAGE TANKS 1 & 2 - PVC OR GALV. RIGID STEEL CONDUIT.
    F. COOLING TOWERS (NEW TOWERS)
       (a) PVC CONDUIT
       (b) RIGID STEEL CONDUIT, COATED EPOXY WILL BE APPLIED TO ALL STEEL SURFACES (SYSTEM PC20) OF SPEC. 81A207 INCLUDING SUPPORTS
       (c) ALS CABLE: COATED EPOXY WILL BE APPLIED OVER THE SURFACE OF CABLE IN AREAS SUBJECT TO DEBRIS COLLECTION.
    G. RADWASTE BLDG.
       (a) GALVANIZED RIGID STEEL CONDUIT
       (b) "MC" INTERLOCK CABLE & FITTINGS
       (c) ALUMINUM SHEATH CABLE (ALS)
25. DC RECEPTACLES SHALL BE TAGGED IN FIELD TO CLEARLY IDENTIFY THEM FROM AC RECEPTACLES.
26. ALUMINUM-SHEATHED CABLE SHALL NOT BE USED IN AREAS WHERE IT WILL BE SUBJECTED TO SEVERE PHYSICAL DAMAGE.
27. ALL LIGHTING CIRCUITS CAPABLE OF BEING FED FROM CLASS 1E SYSTEM SHALL USE THE SAME COLOR CODING FOR THE WIRE AND CONDUIT AS THE COLOR CODING USED FOR THE TRAIN THEY ARE ASSOCIATED WITH.
28. WHERE NO STEEL BEAMS ARE AVAILABLE EMBED PLATES ARE PROVIDED FOR LIGHTING FIXTURE SUPPORT. ALS OR CDS (CROSS - COPPER SHEATHED CABLE) MAY BE CINCH ANCHORED TO CONCRETE WALLS OR CEILINGS.
29. CABLES FROM THE THREE LIGHTING SUBSYSTEMS: NORMAL, ESSENTIAL AND EMERGENCY, SHALL BE ROUTED IN SEPARATE RACEWAYS. HOWEVER, THERE ARE NO REQUIRED SEPARATION DISTANCES BETWEEN THESE RACEWAYS, OR ALS CABLE WHICHEVER IS USED.
30. GALV. SHEET STEEL BOXES MAY BE USED FOR LTG. CKTS EXCEPT WHEN EXPOSED, SHOWN OTHERWISE ON LTG DETAILS.
31. LIGHTING FIXTURE LOCATIONS WHERE SHOWN ON DRAWINGS ARE FOR FIELD GUIDANCE ONLY. FINAL LOCATIONS TO BE DETERMINED BY FIELD.
32. MOUNTING DETAILS WHEN SHOWN ON PLAN DRAWINGS ARE FOR FIELD GUIDANCE ONLY. FIELD MAY SELECT THE APPLICABLE MOUNTING DETAIL FROM THE MOUNTING DETAIL DRAWING(S) EXCEPT FOR DETAIL "B" ON DWG AX3DG032. THIS NOTE DOES NOT APPLY FOR SEISMIC CATEGORY I SYSTEMS (SEE NOTE 46).
33. WHERE ASTM-A307 BOLTS ARE SPECIFIED, NELSON STUDS OF EQUAL SIZE MAY BE USED. BOLTS OR STUDS SHALL BE INSTALLED WITH STANDARD WASHER AND NUT.
34. ALL FIXTURES MOUNTED TO CONCRETE WALLS, FIELD MAY INSTALL 1/4" SPACERS.
35. WHERE NO EMBED PLATE IS AVAILABLE FOR LIGHTING FIXTURE SUPPORT AND 1/4" OR 1/2" NELSON STUDS OR UNISTRUT-1/2" CAPSCREWS ARE CALLED FOR, A CONCRETE ANCHOR OR EQUIVALENT SIZE MAY BE USED. THE ANCHOR BOLT SPACING WILL BE DETERMINED BY THE SPACING OF THE HOLES IN THE DESIGNED COUPLER.
36. FIELD MAY USE TYPE XHHW CABLE FOR LIGHTING HOME RUNS TO THE LIGHTING PANELS FROM THE FIRST JUNCTION BOX. THE CABLES SHALL BE RUN IN RIGID STEEL CONDUIT SIZED PER NEC.
37. ALL PHOTOCELLS SHALL BE MOUNTED ABOVE AND SHIELDED FROM ADJACENT FIXTURES.
38. A CONDULET MAY BE USED AS A SPLICE POINT IN THE STEM OR TO THE BASE OF PENDANT OR WALL MOUNTED INCANDESCENT FIXTURES, IN ORDER TO CONNECT 8#2 OR 5#4 #10 HIGH TEMP. WIRE TO THE S.O.C. CABLE FOR FIXTURES REQUIRING HIGH TEMP. WIRE (ABOVE 90°C).
39. LIGHTING PANEL DISCONNECT SWITCH WILL BE APPROXIMATELY 5'-0" (TOP OF SWITCH) ABOVE FINISH FLOOR.
40. THE MOUNTING HT OF EMERGENCY FIXTURES SHALL BE 7'-0" MIN. & 10'-0" MAX FROM FINISHED FLOOR TO FIXTURES LAMP HEADS.
41. FIELD MAY USE 2-3/4" TOGGLE BOLT FOR ATTACHING LIGHTING OUTLET BOX (4"SQ. GALVANIZED STEEL, MAY BE 2") & ALS CABLE TO PLASTER WALL ONLY.
42. MULTIOUTLET ASSEMBLIES OF LIGHTS SHALL BE ATTACHED TO WALLS WITH EMBEDDED PLASTIC SCREW ANCHORS 1" LONG AND #8 OR #10 SELF TAPPING SCREWS 1" LONG. (MATERIAL SHALL BE FROM FIELD STOCK)
43. FIXTURES LOCATED ON FIELD WALLS MAY BE MOUNTED PER DETAIL B1 DRAWING AX3DG040, SUBSTITUTE 3/8" BOLT 12" LONG (MIN) FOR 1/4 X 1 3/4" BOLT AS REQUIRED. QUANTITY TO BE DETERMINED BY FIXTURES TYPE.

COMMUNICATION SYMBOLS

MK #	PABX SYSTEM	
	PABX TELEPHONE	M — MODEM (SUPPLIED AND WIRED BY NUCLEAR OPERATIONS)
	PABX TELEPHONE JACK	CT — COMPUTER TERMINAL (SUPPLIED, WIRED AND MOUNTED BY NUCLEAR OPERATIONS)
a	ACOUSTIC TELEPHONE ENCLOSURE (GAI-TRONICS MODEL NO. RFP 5803-001/ADCO MODEL NO. B-D 45L SERIES). FULL LENGTH "HEAR-HERE BOOTH". BACK WALL MOUNTED. WITH INTERNAL FLUORESCENT LIGHT FIXTURE/BALLAST AND ALUMINUM WIRE-WAY COVER PLATE REMOVED. MK #TPO	
	AT&T MERLIN TELEPHONE, TELEPHONE JACK	S — STATUS (SUPPLIED AND WIRED BY NUCLEAR OPERATIONS)
P	PABX TELEPHONE PULL BOX MOUNTED IN FINISH WALL	P — PRINTER (SUPPLIED AND WIRED BY NUCLEAR OPERATIONS)
SC	GTM PABX TELEPHONE TERMINAL BOX	SC — SIGNAL CONVERTER (SUPPLIED AND WIRED BY NUCLEAR OPERATIONS)
	FLUSH MOUNTED STANDARD WALL RECEPTABLE FOR USE WITH INTER-COMM TYPE RS 42 A/U CONTROL CABLE INSTALLATION; NUCLEAR PLANT MANAGEMENT INFORMATION SYSTEM (NPMIS)	

TELEPHONE/PAGE SYSTEM

1	TEK, L	SPEAKER (G.T.C. 13302) W/MTG BRACKET.
2	TEM	SPEAKER: (G.T.C. 13307-002)
3	TEO	SPEAKER: (G.T.C 13320) OR (G.T.C HP 15-8) OR UNIVERSITY SOUND (8AB OR 2BCB ED C090179501 APPROVED USE OF ELECTRO-VOICE MODEL PAGE SPEAKER.
4	TFA, B	SPEAKER: UNIVERSITY MODEL CF1015-8 (TFA) CF1032-8 (REF. ED 99-V40094). FOR USE IN CONTAINMENT.
5	TES	SPEAKER: CONE TYPE, W/CIRC. STYRENE GRILL RADIOACTIVE AREA.
C 1	TET	SPEAKER: CEILING MOUNTED BI-DIRECTIONAL SPEAKER.
C 2	TEQ	SPEAKER: CEILING MOUNTED WITH DISPERSION DEFLECTOR (G.T.C. 812)
C 3	TER	SPEAKER: CEILING MOUNTED (G.T.C. 804) OR (G.T.C. B650-B)
	TEU	SPEAKER: BI-DIRECTIONAL W/MOUNTING BRACKET (G.T.C. 13309) DRIVER MK# TFI (GTC-13312-101)
		(SPEAKER LOCATED IN RADIOACTIVE AREAS)
W	TEP	SPEAKER, WALL MOUNTED
WP	TEC	WALL HANDSET/SPEAKER AMPLIFIER STATION (WEATHERPROOF) (G.T.C. 7305-104)
R	TEA	WALL HANDSET/SPEAKER AMPLIFIER STATION (G.T.C. 7005-102-G B40811).
WP	TED	WALL HANDSET FLUSH MOUNTED WITH REMOTE AMPLIFIER ASSEMBLY (G.T.C. 7100-102)
ELEV	TEE	PANEL HANDSET/SPEAKER AMPLIFIER STATION (G.T.C. 7105-102) (FOR ELEVATOR CABS)
DM	TFD	DESK TOP HANDSET WITH ZONE MERGING BUTTON (G.T.C. 7205-102-FB5078)
D	TEF	DESK TOP HANDSET/SPEAKER AMPLIFIER (G.T.C. 7205-102)
E	TEG	DESK EDGE HANDSET/SPEAKER AMPLIFIER STATION (G.T.C. 7105-102)
W	TEB	WALL HANDSET/SPEAKER AMPLIFIER STATION IN CONTAINMENT (G.T.C. 7005-102-FB5076)
A	TEH	SPEAKER AMPLIFIER, (G.T.C. 650-001 OR 760-401) OR 751-001) FOR RETROFIT APPLICATIONS USE HARNESS KIT (G.T.C. 12586-002.
	TEI	SPEAKER AMPLIFIER IN CONTAINMENT (G.T.C. 650 001 FB5077 OR 750-401-FB5077) RETROFIT IS NOT POSSIBLE; ENTIRE ASSEMBLY (ENCLOSURE AND AMPLIFIER) MUST BE REPLACED.

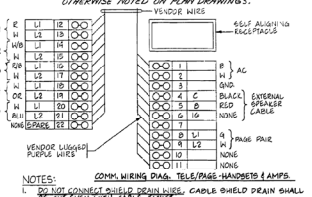
J	TEW	PORTABLE HANDSET RECEPTACLE (TOMSON #TC-1604-SPA) HOLD
U	TEX	PORTABLE HANDSET RECEPTACLE IN CONTAINMENT (THOMSON #TC-1606 U)
A WP	TEJ	SPEAKER AMPLIFIER, (G.T.C. 660-001 OR 760-401) (WEATHERPROOF) FOR RETROFIT APPLICATIONS USE HARNESS KIT G.T.C. 12586-003.
J	TEY	JUNCTION BOX (G.T.C. 10434-002). IN UNIT 2 C.W. THOMSON TC-1607-SP OR MODIFIED "EFA" TERMINAL BOX (OR EQUAL) WITH A MINIMUM OF 22 TERMINAL POINTS MAY BE USED.
J	GFY	JUNCTION BOX, NEMA 4, FOR CONTAINMENT USE
R	TFC	REMOTE PLANT ALARM SWITCH BOX
	GBC	WARNING LIGHT, 120 VAC, 100 WATT, RED, 60 FLASHES PER MINUTE, FEDERAL SIGNAL CORP. MODEL #371A, (MOUNTING KIT MARK # GBG)
10A	GBI	TONE RECEIVER ASSEMBLY, GAI-TRONICS CORP. MODEL NO. GB4060-1161 (10 MA" RATED CONTACTS) OR MODEL NO. GB4060-10 (WITH 10 AMP RATED CONTACTS). VOLUME CONTROL GAITRONICS CAT #1050 001

SOUND POWER SYSTEM

A	TGA	SOUND POWER SYSTEM JACK: MAINTENANCE SYSTEM
R/F	TGB	SOUND POWER SYSTEM JACK: REFUEL SYSTEM
S/D	TGC	SOUND POWER SYSTEM JACK: SHUTDOWN SYSTEM
J		JUNCTION BOX (SIZE AS REQUIRED)
P		PULL BOX (SIZE AS REQUIRED)

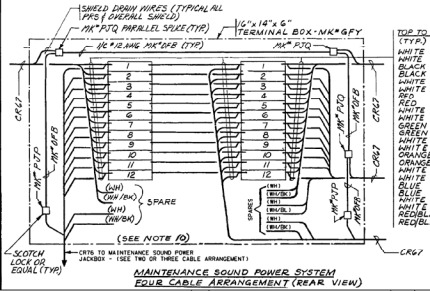
COMMUNICATION NOTES

1. LOCATIONS FOR TELEPHONES, JACKS, SPEAKERS AND BOXES SHOWN ON THE DRAWINGS ARE FOR FIELD GUIDANCE ONLY. FINAL LOCATIONS ARE TO BE DETERMINED BY FIELD.
2. RACEWAY SIZING & CABLE ROUTING SHALL BE DETERMINED FROM THE RISER DIAGRAMS.
3. ALL EMBEDDED CONDUITS FOR COMMUNICATIONS, SHALL BE GALVANIZED RIGID STEEL CONDUIT WITH WATERTIGHT CONNECTORS, OR AS DETERMINED BY FIELD ENGINEER.
4. EMBEDDED CONDUIT IS SHOWN SCHEMATICALLY ONLY. FIELD TO DETERMINE ROUTING AT TIME OF INSTALLATION.
5. EXPOSED RACEWAY SYSTEM SHALL BE THE SAME AS FOR LIGHTING EXCEPT "ALS" SHALL BE USED ONLY FOR AMPLIFIER-TO-SPEAKER CONNECTIONS IN NON-CONTAINMENT BLDG AREAS. MC (INTERLOCK CABLE) SHALL BE USED FOR THESE CONNECTIONS INSIDE THE CONTAINMENT BLDG.
6. RACEWAYS FOR THE THREE COMMUNICATIONS SUB SYSTEMS (PABX, TEL/PAGE, SOUND POWER) SHALL BE INDEPENDENT. THE RACEWAYS FOR THE TWO SEPARATE SOUND POWER PHONE NETWORKS SHALL BE INDEPENDENT UNLESS NOTED OTHERWISE.
7. MOUNTING DETAILS WHEN SHOWN ON PLAN DRAWING ARE FOR FIELD GUIDANCE ONLY. FIELD MAY SELECT THE APPLICABLE MOUNTING DETAIL FROM THE MOUNTING DETAIL DRAWING
8. ALL COMMUNICATIONS EQUIPMENT MOUNTED TO CONCRETE WALLS, FIELD MAY INSTALL 1/4" SPACERS.
9. PABX TELEPHONE TERMINAL BOXES SHALL BE PROVIDED WITH FIELD INSTALLED 3/4" PLYWOOD BACKBOARDS OR TROUGHS TO FIT THE INTERIOR OF THE BOXES. TERMINAL BLOCKS WILL BE PROVIDED BY PABX TELEPHONE SYSTEM SUPPLIER.
10. OVERALL SHIELD AND PAIR SHIELD DRAIN WIRES IN EACH CABLE SHALL BE CONNECTED FROM CABLE TO CABLE AT EACH SOUND POWERED SHUTDOWN AND REFUEL JACK STATION AND AT EACH SOUND POWERED TERMINAL BOX. SHIELDS SHALL BE CONNECTED AS SHOWN IN THE WIRING DIAGRAMS. SHIELDS ARE TO BE INSULATED FROM GROUND AND FROM EACH OTHER AT ALL LOCATIONS NOT SHOWN AS SHIELD GROUND POINTS (SEE NOTE #11).
11. GROUND ALL SHIELD DRAIN WIRE WITH #12AWG, TO JUNCTION BOX. USE #6AWG WIRE TO GROUND JUNCTION BOX TO THE NEAREST GROUND PAD. (SEE RISER DIAGRAM FOR LOCATION OF BOXES TO BE GROUNDED)
12. MARK NUMBERS "TEL" & "TEN" (PREVIOUSLY DESIGNATED FOR USE IN RADIACTIVE AREAS) CAN BE USED INTERCHANGEABLY WITH MARK NUMBERS "TEK" & "TEM" RESPECTIVELY, AT FIELD ENGINEERS DISCRETION.
13. REMOUNT THE 2 JACKS SUPPLIED ON DAVID CLARK PLATE(MK#TSB) TO STANDARD FD BOX COVER PLATE MK# EJA. SEE NOTE 14.
14. FIELD MAY USE CAST BOXES (SUCH AS MK# J3J, CUT TO 4") AS PROVIDED FOR CONNECTION TO JACK BOXES. SEE RISER DIAGRAM FOR MINIMUM CONDUIT SIZES.
15. FOR MK #TGA, MOUNT 6 JACKS SUPPLIED BY DAVID CLARK CO. ONTO COVER OF JUNCTION BOX MK #GEM OR EQUAL.
16. COVER PLATES FOR SHUTDOWN/REFUEL MK#EJA SHALL BE COLOR CODED. SHUTDOWN TO BE - RED REFUEL TO BE - ORANGE
17. RECOMMENDED MOUNTING HEIGHT FOR WALL MOUNTED SPEAKERS SHALL BE APPROXIMATELY 10'-0" ABOVE FINISHED FLOOR, UNLESS OTHERWISE NOTED ON PLAN DRAWINGS.

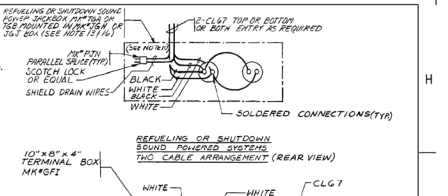


COMM. WIRING DIAG. TELE/PAGE-HANDSETS & AMPS.

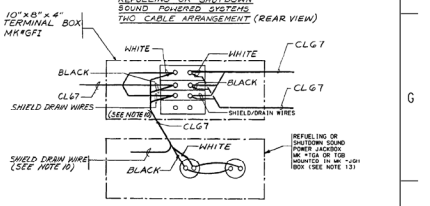
NOTES:
1. DO NOT CONNECT SHIELD DRAIN WIRE. CABLE SHIELD DRAIN SHALL BE CUT EVEN WITH CABLE JACKET.
2. TO ELIMINATE SPEAKER MUTING, MOVE VENDOR LUGGED PURPLE WIRE FROM TERMINAL "1" TO TERMINAL "8".



MAINTENANCE SOUND POWER SYSTEM
FOUR CABLE ARRANGEMENT (REAR VIEW)



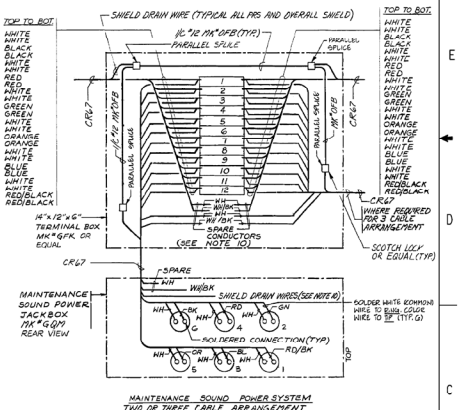
REFUELING OR SHUTDOWN SOUND POWERED SYSTEMS
TWO CABLE ARRANGEMENT (REAR VIEW)



REFUELING & SHUTDOWN SYSTEMS
THREE CABLE ARRANGEMENT (REAR VIEW)

COMMUNICATION NOTES (CONTD)

18. ELECTRICAL METALLIC TUBING (EMT) WITH XHHW WIRE MAY BE USED AS AN OPTION TO ALS AND METAL CLAD (MC) CABLE FOR TELEPHONE PAGE CIRCUITS. FOR OTHER REQUIREMENTS OF EMT SEE LIGHTING NOTES #47 THRU #52 THIS DRAWING.
19. ROUTING FOR PABX OR PABX TELEPHONE SYSTEM RISER DIAGRAM TAKES PRECEDENCE OVER PLAN DRAWINGS FOR UNIT 2 ONLY.
20. TYPICAL JO(AX2D0490502 SERIES) TYPE SUPPORTS MAY BE USED TO SUPPORT COMMUNICATION DEVICES (HAND STATIONS, AMPS, JACK BOXES, ECT.) PROVIDED SUPPORT IS FOR COMMUNICATION DEVICE ONLY AND DOES NOT SUPPORT CONDUIT. ALL INSTALLATIONS SHALL MEET THE BENDING REQUIREMENTS OF AX2D0490502-A1 AND THE INDUSTRIAL SUPPORT TYPES. (ERT MAYBE ROUTED TO THE DEVICE BUT MAY NOT BE ATTACHED TO JO TYPE SUPPORT)
21. FOR UNIT 2 ONLY, NUMBERS LOCATED ADJACENT TO PABX SYMBOL OR LTG. & COMM. DRAWINGS ARE FOR LOCATION IDENTIFICATION ONLY. GPC COMMUNICATIONS DEPT. WILL ASSIGN ACTUAL PABX TELEPHONE NUMBERS.
22. FOR UNIT 2 ONLY, LIGHTING AND COMMUNICATION PLAN DRAWINGS DO NOT REFLECT AS-BUILT CONDITION OF PABX SYSTEM. SEE PABX RISER.



MAINTENANCE SOUND POWER SYSTEM
TWO OR THREE CABLE ARRANGEMENT

REFERENCE DRAWINGS:

AX3DG031	LIGHTING FIXTURE SCHEDULE
AX3DG032	LIGHTING & COMMUNICATIONS MOUNTING DETAILS
AX3DG033	LIGHTING & COMMUNICATIONS MOUNTING DETAILS
AX3DG034	LIGHTING & COMMUNICATIONS MOUNTING DETAILS
AX3DG039	LIGHTING & COMMUNICATIONS MOUNTING DETAILS
AX3DG040	LIGHTING & COMMUNICATIONS MOUNTING DETAILS

NO.	VERSIONS	DATE	DR	CHK	APPV
38.	REVISED PER ABN SNC62360TE004, VER. 1.0	9/01/15	TDA	ALM	WLN

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SOUTHERN COMPANY SERVICES, INC

LIGHTING NOTES
(CONT'D FROM SHEET 1)

44. WHEN CONDUCTORS IN MC CABLE ARE NOT COLOR CODED BLACK, RED, BLUE, & WHITE ALL CONDUCTORS SHALL BE COLOR AS STATED

EXAMPLE *1		EXAMPLE *2	
FROM	TO	FROM	TO
BLACK-1	BLACK	BLACK	BLACK
BLACK-2	RED	RED	RED
BLACK-3	BLUE	GREEN	BLUE
BLACK-4	WHITE	WHITE	WHITE

45. FIELD MAY SUBSTITUTE UNISTRUT P1001 FOR P1001 OR P1000 FOR P1000 FOR USE WITH LIGHTING & COMMUNICATIONS MOUNTING DETAILS.

46. ONLY SUPPORT DETAILS 8, 13, 17, 18, 20, 24(OPTIONS 1, 2, 3), 26, 27, 28, 30, 45, AND 78(ALTERNATE 1) SHALL BE USED FOR SEISMIC CATEGORY I SYSTEMS.

47. ELECTRICAL METALLIC TUBING (EMT) WITH 90MM WIRE MAY BE USED AS AN OPTION TO #18 AWG METAL CLAD (MC) CABLE FOR LIGHTING FIXTURES. THE SIZE OF EMT SHALL BE 1/2" MINIMUM AND 1" MAXIMUM. EMT SHALL BE SECURELY FASTENED PER INDUSTRY STANDARDS AT LEAST ONCE EVERY 10 FEET AND WITHIN 3 FEET OF EACH OUTLET BOX, JUNCTION BOX, CABINET, OR FITTING. TUBING AND FITTINGS SHALL BE ALUMINUM FREE. THE TUBING AND FITTINGS SHALL BE OF THE RAINTIGHT TYPE. EMT SHALL NOT BE THREADED. THIS SUBSTITUTION OF EMT IS NOT ALLOWED FOR EMERGENCY LIGHTING CIRCUITS. FOR SIZING OF EMT SEE LIGHTING NOTE *48.

48. THE MAXIMUM NUMBER OF CONDUCTORS IN EMT SHALL BE AS FOLLOWS

CONDUCTOR SIZE	MIN. SIZE EMT	MAX. NO. COND.
#14	1/2"	13
#12	1/2"	10
#10	1/2"	5
#8	1/2"	2
#8	3/4"	12
#6	1/2"	1
#6	3/4"	5
#4	1/2"	1
#4	3/4"	12
#3	1"	7
#2	1"	5
#1	1"	3

49. A RUN OF ELECTRICAL METALLIC TUBING BETWEEN OUTLET AND OUTLET, FITTING AND FITTING, OR OUTLET AND FITTING SHALL NOT CONTAIN MORE THAN THE EQUIVALENT OF FOUR 90 DEGREE BENDS (360 DEGREES TOTAL) INCLUDING THOSE BENDS LOCATED IMMEDIATELY AT THE OUTLET OR FITTING.

50. BENDS IN THE TUBING SHALL BE SO MADE THAT THE TUBING WILL NOT BE DAMAGED AND THE INTERNAL DIAMETER OF THE TUBING WILL NOT BE EFFECTIVELY REDUCED. THE RADIUS OF THE CENTER LINE OF THE HAND BEND OF ANY STEEL BEND SHALL BE THE SAME AS RIGID STEEL CONDUIT.

51. CUT ENDS OF ELECTRICAL METALLIC TUBING SHALL BE REAMED OR OTHERWISE FINISHED TO REMOVE ROUGH EDGES.

52. FOR SEPERATION REQUIREMENTS OF EMT SEE CONSTRUCTION SPECIFICATION X3MW01 EB ATTACHMENT B.

53. UNIVERSAL CLAMP MAY BE USED TO ATTACH EMT TO UNISTRUT AS FOLLOWS

SIZE OF EMT	CLAMP
1/2"	P1211 OR EQUAL
3/4"	P1212 OR EQUAL
1"	P1213 OR EQUAL

54. SUPPORT EMT WITHIN 3 FEET OF CONDULET BY ONE SIDE AND WITHIN 5 FEET ON THE OTHER SIDE ON CONTAINMENT LINER (SEE NOTE *47 FOR OTHER LOCATIONS).

55. ONE-HOLE CLAMP BACKS MAY BE USED TO ATTACH EMT CONDUITS TO CONCRETE WALLS AS FOLLOWS:

	APPLETON
1/2" EMT - CLB 50M OR EQUAL	
3/4" EMT - CLB 75M OR EQUAL	
1" EMT - CLB 100M OR EQUAL	

CLAMP BACKS TO BE USED WITH RESPECTIVE SIZE 1 HOLE CLAMPS FOR EMT.

56. FIXTURE TYPES FAY AND FAZ ARE INTERCHANGEABLE, EXCEPT WHEN MOUNTED FROM HANGRAIL WHERE TYPE FAZ MUST BE USED.

57. WHEN MOUNTING FIXTURES FROM CONDUITS STEMS PER DETAIL 18 OR 17 ON DWG. 2X3D-8032, FIXTURE LOOPS AND HOOKS MAY BE OMITTED AND STEM THREADED DIRECTLY INTO FIXTURE PROVIDED HOOKS AND LOOPS ARE USED AT THE TOP OF STEM.

58. TYPICAL CONDUIT SUPPORTS MAY BE USED FOR EMT IS THE MAXIMUM OF RIGID CONDUITS AS SPECIFIED ON THE INDIVIDUAL SUPPORT TYPES FOR ZONE 1 OR ZONE 11 AS APPLICABLE. U.N.O. A SMALLER SIZE EMT MAY BE SUBSTITUTED PROVIDED THE ALLOWABLE LOAD BY WEIGHT IS NOT EXCEEDED (FOR MC SUPPORTS EQUIVALENT EMT WEIGHT MAY BE USED). FOR WEIGHTS OF RIGID AND EMT, SEE THE TABLE BELOW FOR THE EQUIVALENT WEIGHT.

EMT SIZE	EQUIVALENT WEIGHT (LB/FT)	TYPE OF STRAP	TYPE OF CLAMP
1/2"	1.1	P-2558-S	P-1213 AC-100
3/4"	2.2	P-2558-S	P-1213 AC-100
1"	2.2	P-2558-S	P-1213 AC-100

UNISTRUT PIPE STRAP → UNISTRUT UNIVERSAL → APPLETON EMT ONE HOLE

(* FOR CONSTRUCTION EQUIVALENT USE ONLY)

C) FOR ATTACHMENT OF EMT TO SUPPORTS (OR BETWEEN SUPPORTS) WHERE THE SPAN EXCEEDS ABOVE OR EQUAL SHALL BE USED.

D) EMT MAY BE ATTACHED TO EXISTING RIGID CONDUIT SUPPORTS AS ALLOWED BY NOTE (E) BUT MAY NOT BE ATTACHED TO OTHER EXISTING SUPPORTS.

F) EXISTING RIGID CONDUIT SUPPORTS MAY BE USED TO SUPPORT EMT AS LONG AS THE FOLLOWING RESTRICTIONS ARE MET:

1. WHEN COMBINING EMT AND RIGID CONDUITS ON THE SAME SUPPORT, THE SUM OF ALL OF THE RIGID CONDUIT WEIGHTS PLUS ALL OF THE EMT WEIGHTS (REF. DWG. AX3D94X0552-3-1 SHEET 21 AND DWG. AX3D94X0552-3-1 TABLE 1) SHALL NOT EXCEED THE TOTAL WEIGHT OF THE MAXIMUM NUMBER OF RIGID CONDUITS AS SPECIFIED ON THE INDIVIDUAL RIGID CONDUIT SUPPORT TYPES.
2. RESTRICTIONS FOR SUPPORT OF EMT AS SPECIFIED IN NOTES A THRU F SHALL APPLY.

F) MAXIMUM SPACING FOR EMT SUPPORTS SHALL BE 10'-0".

LIGHTING NOTES
(CONT'D)

59. FABRICATION OF A LIGHTING FIXTURE OF A GIVEN MARK NO. FROM PARTS COMPONENTS OF OTHER FIXTURE TYPES (MARK NOS. IS ACCEPTABLE, PROVIDED ALL COMPONENT RATING, REFLECTOR/REFRACTORS, AND OTHER HARDWARE ARE EQUAL TO THAT OF THE FIXTURE TYPE BEING FABRICATED. THE HOUSING MUST BE RELABELED WITH THE APPROPRIATE FIXTURE MARK NUMBER TO PROVIDE MAINTENANCE PERSONNEL (FOR UNIT 2 ONLY)

63. EMERGENCY LIGHTING UNIT (EMERGENCY BATTERY PACK FIXTURE) MK *F2K AND MK *F2N ARE INTERCHANGEABLE WITH THE EXCEPTION OF WHERE SYMBOL "SEIS" IS ADJACENT TO EMERGENCY LIGHTING SYMBOL ONLY MK *F2K SHOULD BE USED.

61. FOR UNIT 2 OPTION PENDANT MOUNTED IN-CANDESCENT FIXTURES.

A) REPLACE BROWN CONNECTION BLOCKS WITH BLACK CONNECTION BLOCKS (NEW GREEN BLOCKS ARE ALSO ACCEPTABLE) IN PENDANT MOUNTED INCANDESCENT FIXTURE (MK *FAW).

B) REMOVE EXISTING GLOBES, GLOBES AND FIXTURES GUARDS FROM SEDW PENDANT MOUNTED INCANDESCENT LIGHTING FIXTURE (MK *FAW) IN ALL AREAS EXCEPT CONTAINMENT BLDG. AND FUEL HANDLING BLDG. REPLACE REFLECTORS AS NECESSARY AS IMPROVEMENT IF REQUIRED

(C) EXISTING 300W, 120V STANDARD LAMPS IN PENDANT INCANDESCENT FIXTURES (MK *FAW) SHALL NOT BE REPLACED BY USE REGULAR OR SMATTERPROOF LAMPS AS THE NEED REQUIRES.

62. PHOTOCELLS MAY BE USED TO CONTROL THE DUSK TO DAWN OPERATION OF EXTERIOR LIGHT FIXTURES WHERE REQUIRED.

63. PHOTOCELL CONTROL MAY BE USED IN LIEU OF HARD-WIRED OR SWITCHED CONTROL UNDER THE FOLLOWING CONDITIONS:

A) LIGHT FIXTURES THAT ARE REQUIRED FOR CONTINUOUS OPERATION SHALL BE CONTROLLED BY THE SWITCH OR SHALL BE HARD WIRED "ON" TO MAINTAIN ADEQUATE LIGHTING LEVELS.

B) LIGHT FIXTURES THAT ARE LOCATED IN AREAS WHERE AVAILABLE NATURAL LIGHTING IS SUFFICIENT TO MAINTAIN ADEQUATE LIGHTING LEVELS MAY BE CONTROLLED BY PHOTOCELLS. LIGHT FIXTURES CONTROLLED BY PHOTOCELLS SHALL NOT BE SWITCHED.

C) WHEN THE SWITCH ONLY CONTROLS EXTERIOR LIGHTING FIXTURES, THE SWITCH MAY BE REMOVED AND A PHOTOCELL ADDED TO CONTROL THE FIXTURES.

64. CORRUGATED ALUMINUM SHEATH CABLE (MC CABLE) MAY BE USED IN LIEU OF ALUMINUM SHEATH CABLE OF THE SAME CONDUCTOR SIZE AND NUMBER OF CONDUCTORS.

65. ENSURE THAT ONLY ONE MANUAL TRANSFER SWITCH IS IN THE ON POSITION TO PREVENT PARALLELING OF VOLTAGE SOURCES.

COMMUNICATION SYMBOLS

RADIAX ANTENNA SYSTEM

- SPLITTER, SUPPLIED BY GPC
- CONNECTOR, SUPPLIED BY GPC
- 7/8" HELIAX, SUPPLIED BY GPC
- 7/8" RADIAX, SUPPLIED BY GPC
- 1/2" RADIAX, SUPPLIED BY GPC
- REPEATER, SUPPLIED BY GPC
- ANTENNA, SUPPLIED BY GPC

MK= PABX SYSTEM (CONTINUED FROM SHEET 1)

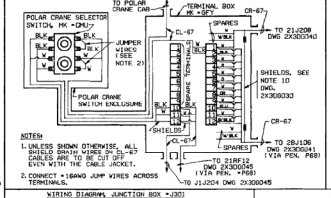
PABX TELEPHONE EXTENSION WITH ONE OR MORE PABX TELEPHONE PARTY LINES CONNECTED TO THE ASSIGNED EXTENSION. EXTENSION NO. IS FOR INFO. ONLY. ACTUAL TELEPHONE EXTENSION NUMBERS ARE ASSIGNED BY GPC TELECOMMUNICATIONS DEPT.

PABX TELEPHONE PARTY LINE CONNECTED TO THE INDICATED PABX TELEPHONE EXTENSION.

TELEPHONE/PAGE SYSTEM (CONTINUED FROM SHEET 1)

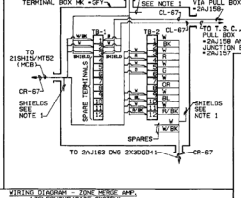
TELEPHONE/PAGE EXTENSION. HANDSET STATION (TYPE XX.MK.* XXX.AS REQUIRED) CONNECTED TO THE INDICATED HANDSET/SPEAKER AMPLIFIER STATION (OPT. NUMBER=HXXXX); WITH INTERNAL SPEAKER AMPLIFIER WHERE NO ADDITIONAL SPEAKER COVERAGE IS REQUIRED.

WIRING DIAGRAM
POLAR CRANE SELECTOR SWITCH AND TERMINAL BOX *21/28S
(CTMT BLDG, LEVEL 1, ROOM *111, MAINT./REFUEL SYSTEM)



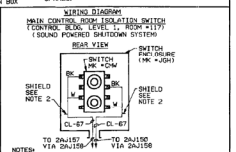
NOTES:
1. UNLESS SHOWN OTHERWISE, ALL SHIELD DRAIN WIRES ON CL-67 CABLES ARE TO BE CUT OFF EVEN WITH THE CABLE JACKET.
2. CONNECT *18AWG JUMP WIRES ACROSS TERMINALS.

WIRING DIAGRAM
MAINTENANCE/SHUTDOWN BOX *2AJ152
(CONTROL BLDG. LEVEL A, ROOM *A231)



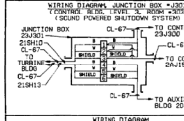
NOTES:
1. SHIELD DRAIN WIRES CONDUCTORS ON *TB-1* ARE TO BE CONNECTED AS SHOWN. SHIELD DRAIN WIRES FOR CONDUCTORS ON TERMINALS 1-4, 19-5, AND CABLE SHIELD DRAIN WIRES ARE TO BE CONNECTED AS STATED FOR NOTE 1D ON DWG 2X3D0030. THE REMAINING SHIELD DRAIN WIRES ARE TO BE CUT OFF EVEN WITH CABLE JACKET.
2. ALL CONDUCTORS NOT SHOWN ON TB-1 AND TB-2 ARE TO TAPED AND TAGGED AS SPARES.

WIRING DIAGRAM
MAIN CONTROL ROOM ISOLATION SWITCH (CONTROL BLDG. LEVEL 1, ROOM *117)
(SOUND POWERED SHUTDOWN SYSTEM)

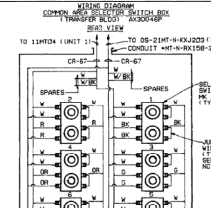


NOTES:
1. CONNECT WIRES TO *HC-* TERMINALS ON ISOLATION SWITCH.
2. SHIELD DRAIN WIRES ARE TO BE CUT OFF EVEN WITH CABLE JACKET.

WIRING DIAGRAM
JUNCTION BOX *2DJ01 (TURBINE BLDG. LEVEL -4, ROOM *201)
(SOUND POWERED SHUTDOWN SYSTEM)

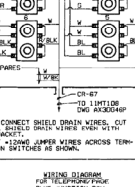


WIRING DIAGRAM
COMMON AREA SELECTOR SWITCH
(TRANSFER BLDG.) AX3D046P



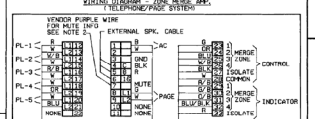
NOTES:
1. DO NOT CONNECT SHIELD DRAIN WIRES. CUT OFF ALL SHIELD DRAIN WIRES EVEN WITH CABLE JACKET.
2. CONNECT *12WG JUMPER WIRES ACROSS TERMINALS ON SWITCHES AS SHOWN.

WIRING DIAGRAM
FOR TELEPHONE/PAGE
BLUE JUNCTION BOX
THOMSON



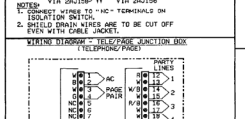
NOTE:
1/2W NOT CONNECT SHIELD DRAIN WIRE. CABLE SHIELD DRAIN WIRE SHALL BE CUT EVEN WITH CABLE JACKET.

WIRING DIAGRAM - TONE MERGE AMP.
(TELEPHONE/PAGE SYSTEM)



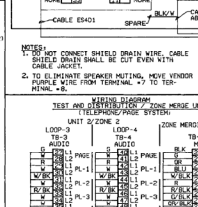
NOTES:
1. DO NOT CONNECT SHIELD DRAIN WIRE. CABLE SHIELD DRAIN SHALL BE CUT EVEN WITH CABLE JACKET.
2. TO ELIMINATE SPEAKER MUTING, MOVE VENDOR PURPLE WIRE FROM TERMINAL *7 TO TERMINAL *8.

WIRING DIAGRAM - TELEPAGE JUNCTION BOX
(TELEPHONE/PAGE)



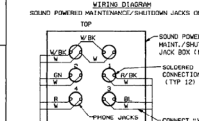
NOTE:
1. DO NOT CONNECT SHIELD DRAIN WIRE. CABLE SHIELD DRAIN WIRE SHALL BE CUT EVEN WITH CABLE JACKET.

WIRING DIAGRAM
TEST AND DISTRIBUTION / ZONE MERGE UNIT
(TELEPHONE/PAGE SYSTEM)



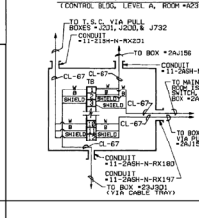
NOTE:
1. DO NOT CONNECT SHIELD DRAIN WIRES ON CABLE *ES401. SHIELD DRAIN WIRES SHALL BE CUT OFF EVEN WITH CABLE JACKET.

WIRING DIAGRAM
SOUND POWERED MAINTENANCE/SHUTDOWN JACKS ON MCB

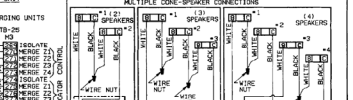


NOTES:
1. ALL CONDUCTORS NOT SHOWN ARE SPARES.
2. ALL SHIELDS AND SHIELD DRAIN WIRES ARE TO BE CUT OFF EVEN WITH CABLE JACKET AND TAPED.

WIRING DIAGRAM, SHUTDOWN BOX *2AJ157
(CONTROL BLDG, LEVEL A, ROOM *A231)

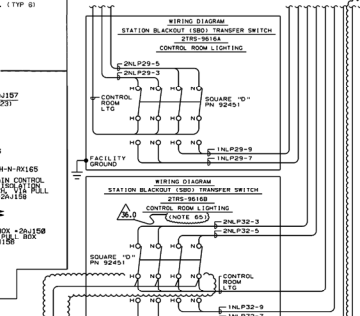


MULTIPLE SPEAKER CONNECTIONS
MULTIPLE CONE-SPEAKER CONNECTIONS

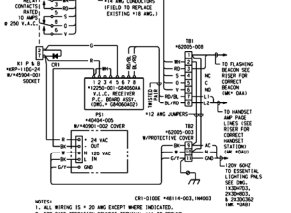


NOTES:
1. THE HORN AND CONE-TYPE SPEAKER ASSEMBLIES SUPPLIED BY DAITRONICS CORPORATION HAVE EITHER 8 OR 16 OHM IMPEDANCE.
2. ALL SPEAKERS IN A SYSTEM SHOULD BE PHASED SO THAT ALL DIAPHRAGMS MOVE SIMULTANEOUSLY IN THE SAME DIRECTION.
3. NORMALLY, ONLY ONE OF THE LARGE HORN TYPE SPEAKERS IS CONNECTED TO AN AMPLIFIER. IF TWO 16 OHM SPEAKERS MUST BE OPERATED FROM ONE AMPLIFIER, THEY SHOULD BE WIRED IN PARALLEL AND CONNECTED TO THE 8 OHM OUTPUT.
4. SEVERAL OF THE CONE-TYPE OR SMALLER HORN-TYPE SPEAKERS CAN BE DRIVEN FROM ONE AMPLIFIER IN MANY SITUATIONS. THE DIAGRAM BELOW SHOWS HOW TWO, THREE, OR FOUR 8 OHM UNITS MAY BE WIRED TO ONE AMPLIFIER.
5. WHEN MORE THAN ONE SPEAKER IS TO BE WIRED TO ONLY ONE AMPLIFIER, SPEAKERS ARE TO BE WIRED AS SHOWN IN THE DIAGRAM ABOVE.
6. NO CHANGE IS REQUIRED WHEN ONLY ONE SPEAKER IS WIRED TO ONE AMPLIFIER.
7. WHEN BOX AND RED CONDUCTORS ARE USED, CONNECT "BLACK" TO COMMON (-C) AND "RED" TO B.A. OR 16 A. SEE DIAGRAM ABOVE.

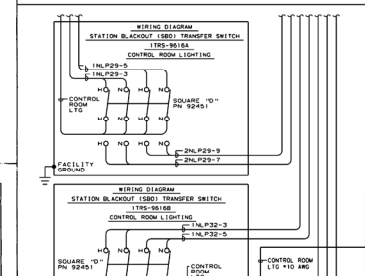
WIRING DIAGRAM
STATION BLACKOUT (SBO) TRANSFER SWITCH
21RS-9616A
CONTROL ROOM LIGHTING
(NOTE 65)



CAITRONICS TONE RECEIVER ASSEMBLY
CODE GB4060-10 CONNECTION DIAGRAM



WIRING DIAGRAM
STATION BLACKOUT (SBO) TRANSFER SWITCH
11RS-9616A
CONTROL ROOM LIGHTING



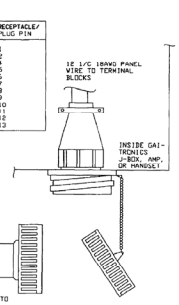
WIRING DIAGRAM
STATION BLACKOUT (SBO) TRANSFER SWITCH
11RS-9616B
CONTROL ROOM LIGHTING

WIRING DIAGRAM
UNIT 1 MCR TRAIN B
FLEX MANUAL TRANSFER SWITCH 1DY1B13T (1-1808-L3-13T) (NOTE 65)

CABLE PAIR	WIRE COLOR	RECEPTACLE/ PLUG PIN
PARTY LINE 1	RED	1
PARTY LINE 1	WHITE	4
PARTY LINE 2	WHITE/BLACK	5
PARTY LINE 2	WHITE	6
PARTY LINE 3	RED/BLACK	7
PARTY LINE 3	WHITE	8
PARTY LINE 4	WHITE	9
PARTY LINE 4	ORANGE	10
PARTY LINE *5	WHITE	11
PAGE PAIR	GREEN	12
PAGE PAIR	WHITE	13

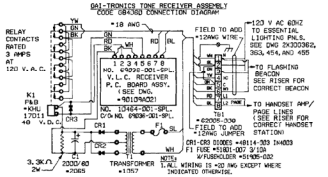
PARTS LIST

DESCRIPTION	AMP P/N
PLUG	208486-3
CABLE CLAMP (PLUG)	208945-7
PINS (CRIMP)	66099-2
PINS (SOLDER)	66180-1
RECEPTACLE	208487-1
CABLE CLAMP (RECPT)	206302-1
SOCKETS (CRIMP)	66101-2
SOCKETS (SOLDER)	66181-1
CAP	211903-1



WIRING DETAIL FOR TELE/PAGE (GAI-TRONICS) AUDIO ONLY RECPTACLE AND PLUG.

GAI-TRONICS TONE RECEIVER ASSEMBLY
CODE GB4360 CONNECTION DIAGRAM*



1. ALL WIRING IS *20 AWG EXCEPT WHERE INDICATED OTHERWISE.

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VERSIONS	DATE	DR	CHK	APPV
36.0 REVISED PER ABN SNC475965E006, VER. 2.0	12/16/15	EJ	ALM	WLN

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

LIGHTING & COMMUNICATIONS GENERAL NOTES & SYMBOLS

SCALE: NONE	DRAWING NO.	VER.
JOB NO.N/A	AX3DG030-2	36.0

SIZE E 34x44

Replacing amplifier 651-001 with amplifier 751-001 will require a retrofit of the enclosure. Use kit numbers shown below.

INDOOR ENCLOSURE (G.T.C. 12586-002)

1. Remove and discard the harness assembly (TB1), the connector (J1), and the associated hardware, see Figure 1. Note: Do not remove TB2.
2. Install harness assembly plate by aligning the plate with the holes used to mount the previous TB1. Secure using the two screws included in the kit.

WEATHERPROOF ENCLOSURE (G.T.C. 12586-003)

1. Remove label on the front cover of the enclosure and discard.
2. Remove and discard the harness assembly (TB1), the connector (J1), and the associated hardware, see figure 2. Note: Do not remove TB2.
3. Install harness assembly plate by aligning the plate with the holes used to mount the previous TB1. Secure using the two screws included in the kit.

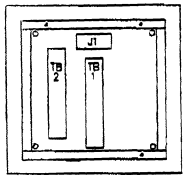


Figure 1, Indoor Enclosure

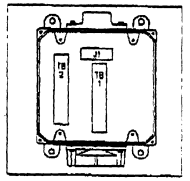
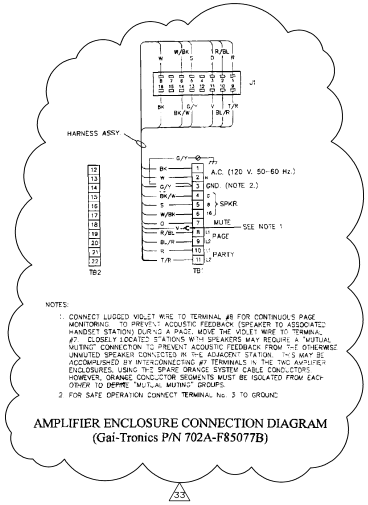


Figure 2, Weatherproof Enclosure

AMPLIFIER RETROFIT KIT



AMPLIFIER ENCLOSURE CONNECTION DIAGRAM
(Gai-Tronics P/N 702A-F85077B)

33

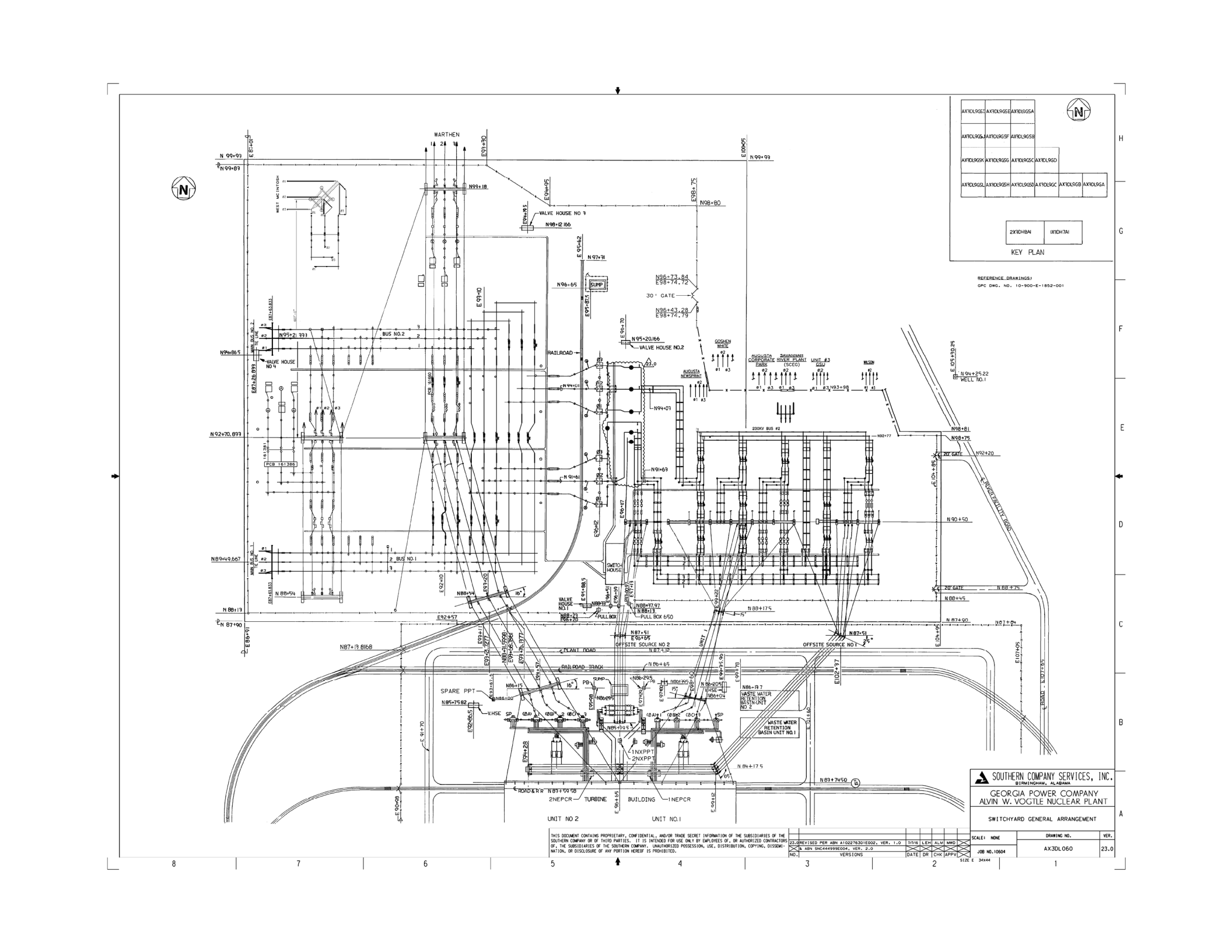
SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

LIGHTING & COMMUNICATIONS
GENERAL NOTES
& SYMBOLS

SCALE: NONE	DRAWING NO.	REV.
JOB NO.10604	AX3DG030 SHEET 3 OF 3	33

SIZE E 34X44


WARTHEN
VALVE HOUSE NO 3
N98+12.166
VALVE HOUSE NO 4
VALVE HOUSE NO.2
N95+20.166
BUS NO.2
BUS NO.1
RAILROAD
SUMP
30' GATE
N96+73.84
E98+74.72
N96+43.28
E98+74.79
GOSHEN WHITE
AUGUSTA CORPORATE PARK
SAVANNAH RIVER PLANT (SCCO)
UNIT #3 GSU
AUGUSTA NEWSPRINT
WILSON
N94+25.22
WELL NO.1
230KV BUS #2
20' GATE
RIVER BEND TIE LINE
SWITCH HOUSE
VALVE HOUSE NO.1
PULL BOX
PULL BOX 650
OFFSITE SOURCE NO.2
OFFSITE SOURCE NO.1
PLANT ROAD
RAIL ROAD TRACK
SPARE PPT
V.HSE
WASTE WATER RETENTION BASIN-UNIT NO.2
WASTE WATER RETENTION BASIN UNIT NO.1
1NXPPT
2NXPPT
ROAD & R.R
2NEPCR
TURBINE
BUILDING
1NEPCR
UNIT NO.2
UNIT NO.1
N 99+93
N 99+83
N94+86.5
N92+70.893
N89+49.667
N 88+13
N 87+90
N87+19.8868
N87+51
N 84+17.5
N 83+74.50
E 86+91
E 90+98
E 91+70
E 94+28
E 102+97
E 107+85
E 107+95
E 104+85
E 104+95
E 105+30.25
E 93+30
E 100+95
N98+80
E98+.75
E 95+62
N 97+31
N96+65
E95+67.5
E 93+40
E 90+70
N94+03
N91+63
E 90+87
E 90+12
E 92+10
E 92+57
N88+54
E93+20
N88+5
N88+17.5
N 90+50
N92+20
N98+81
N98+75
N88+75
E 93+22
E 95+88.5
E 96+96
E 96+51
N88+37.97
N88+13
N87+51
E95+55
N 87.+13
N 86+65
N86-295
N86+20
N86+13.7
N86+15
N85+39.5
N85+7582
E92+85.5
E92+08
E 91+70
E 90+98
E 90+65
E 90+12
E 99+35.96
E 99+78.3
E 97+60
E 97+73
E 97+13
E 93+27
E 93+11
E 93+02.27
E 93+01.31
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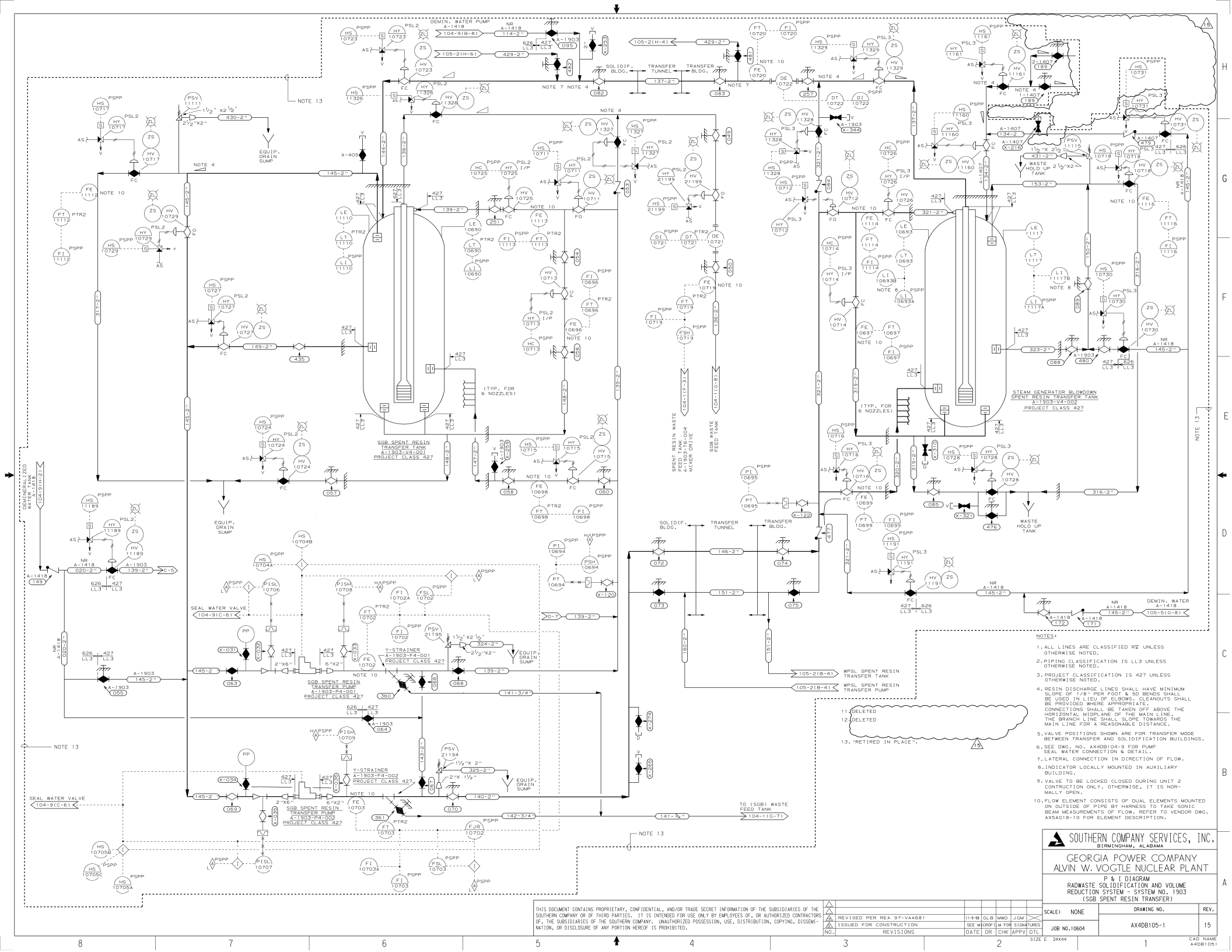
E 93+00.86
E 93+00.86



NOTES:

1. ALL LINES ARE CLASSIFIED R% UNLESS OTHERWISE NOTED.
2. PIPING CLASSIFICATION IS LL3 UNLESS OTHERWISE NOTED.
3. PROJECT CLASSIFICATION IS 427 UNLESS OTHERWISE NOTED.
4. RESIN DISCHARGE LINES SHALL HAVE MINIMUM SLOPE OF 1/8" PER FOOT & 5D BENDS SHALL BE USED IN LIEU OF ELBOWS. CLEANOUTS SHALL BE PROVIDED WHERE APPROPRIATE. CONNECTIONS SHALL BE TAKEN OFF ABOVE THE HORIZONTAL MIDPLANE OF THE MAIN LINE. THE BRANCH LINE SHALL SLOPE TOWARDS THE MAIN LINE FOR A REASONABLE DISTANCE.
5. VALVE POSITIONS SHOWN ARE FOR TRANSFER MODE BETWEEN TRANSFER AND SOLIDIFICATION BUILDINGS.
6. SEE DWG. NO. AX4DB104-9 FOR PUMP SEAL WATER CONNECTION & DETAIL.
7. LATERAL CONNECTION IN DIRECTION OF FLOW.
8. INDICATOR LOCATED LOCALLY MOUNTED IN AUXILIARY BUILDING.
9. VALVE TO BE LOCKED CLOSED DURING UNIT 2 CONTRUCTION ONLY. OTHERWISE, IT IS NORMALLY OPEN.
10. FLOW ELEMENT CONSISTS OF DUAL ELEMENTS MOUNTED ON OUTSIDE OF PIPE BY HARNESS TO TAKE SONIC BEAM MEASUREMENTS OF FLOW. REFER TO VENDOR DWG. AX5AG18-10 FOR ELEMENT DESCRIPTION.

11. DELETED
12. DELETED
13. "RETIRED IN PLACE".

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NO.	REVISIONS	DATE	DR	CHK	APPV	DTL
19	REVISED PER REA 97-VAA681	11-13-98	GLB	MWD	JGM	
18	ISSUED FOR CONSTRUCTION	SEE MICROFILM FOR SIGNATURES				

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
RADWASTE SOLIDIFICATION AND VOLUME REDUCTION SYSTEM - SYSTEM NO. 1903
(SGB SPENT RESIN TRANSFER)

SCALE: NONE	DRAWING NO.	REV.
JOB NO.10604	AX4DB105-1	19

SIZE E 34X44

CAD NAME A4DB1051

NOTES (CONTINUED)

12. "RETIRED IN PLACE"

NOTES

1. ALL LINES ARE CLASSIFIED RW UNLESS OTHERWISE NOTED.
2. PIPING CLASSIFICATION IS LL3 UNLESS OTHERWISE NOTED.
3. PROJECT CLASSIFICATION IS 427 UNLESS OTHERWISE NOTED.
4. RESIN DISCHARGE LINES SHALL HAVE MINIMUM SLOPE OF 1/8" PER FOOT & 50 BENDS SHALL BE USED IN LIEU OF ELBOWS. CLEANOUTS SHALL BE PROVIDED WHERE APPROPRIATE.
   CONNECTIONS SHALL BE TAKEN OFF ABOVE THE HORIZONTAL MIDPLANE OF THE MAIN LINE. THE BRANCH LINE SHALL SLOPE TOWARDS THE MAIN LINE FOR A REASONABLE DISTANCE.
5. VALVE POSITIONS SHOWN ARE FOR TRANSFER MODE BETWEEN TRANSFER AND SOLIDIFICATION BLDGS.
6. SEE DWG. NO. AX4DB104-9 FOR PUMP SEAL WATER CONNECTION & DETAIL.
7. LATERAL CONNECTION IN DIRECTION OF FLOW.
8. INDICATOR LOCALLY MOUNTED IN AUXILIARY BLDG.
9. FLOW ELEMENT CONSISTS OF DUAL ELEMENTS MOUNTED ON OUTSIDE OF PIPE BY HARNESS, TO TAKE SONIC BEAM MEASUREMENTS OF FLOW. REFER TO VENDOR DWG. AX5AG18-10 FOR ELEMENT DESCRIPTION.
10. DELETED
11. DELETED

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NO.	REVISIONS	DATE	DR	CHK	APPV	DTL
3	REVISED PER DCP 95-VAN0026	1-12-99	LVL	GLB	MWD	
2	ISSUED FOR CONSTRUCTION	SEE MICROFILM FOR SIGNATURES				
1						

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

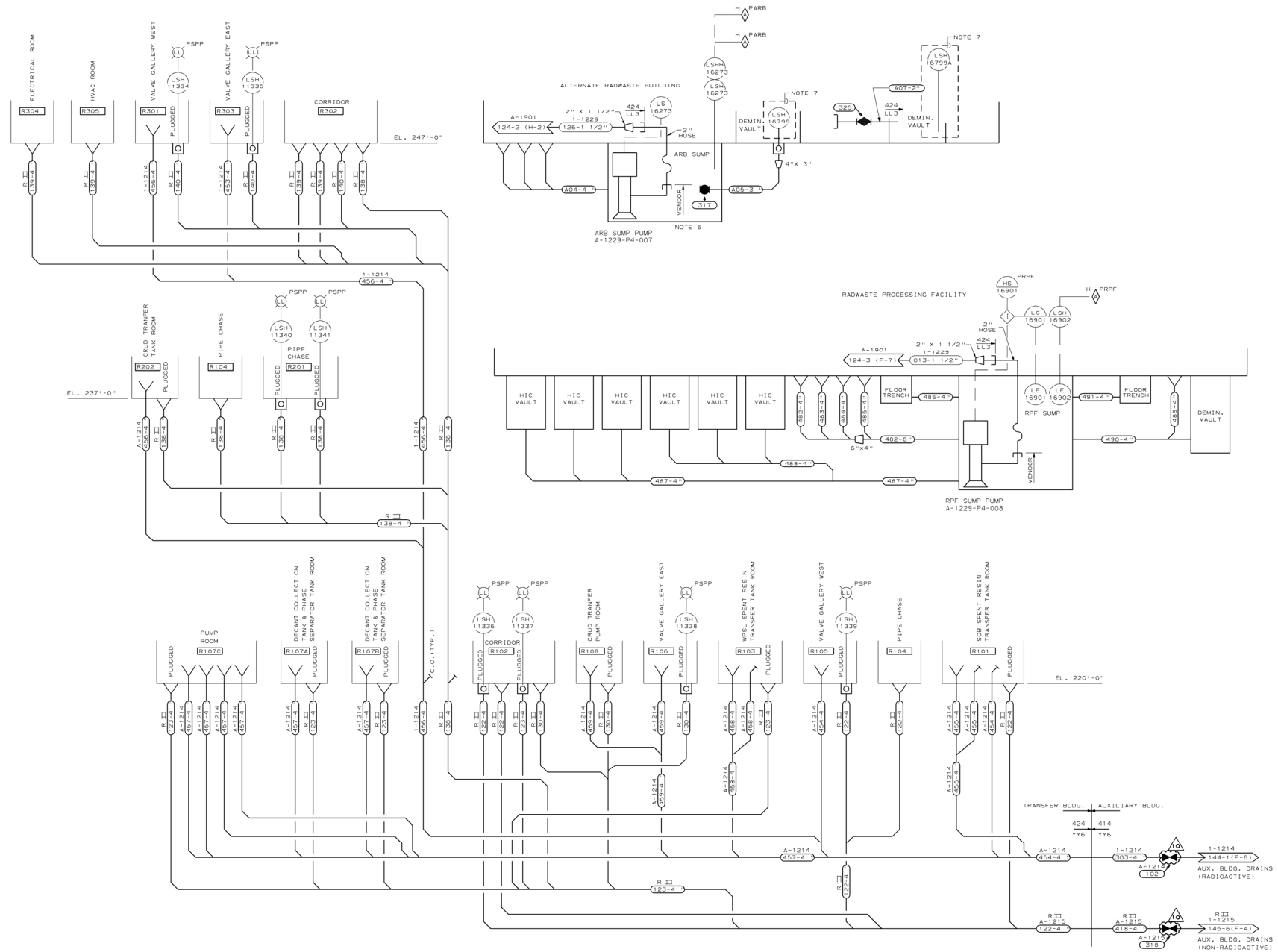
GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
RADWASTE SOLIDIFICATION AND VOLUME REDUCTION SYSTEM - SYSTEM NO. 1903
(WPSL SPENT RESIN TRANSFER)

SCALE: NONE	DRAWING NO.	REV.
JOB NO.10604	AX4DB105-2	13

SIZE E 34X44

CAD NAME A4DB1052



NOTES:

1. ALL VALVES AND LINES ARE PROJECT CLASS 424 UNLESS OTHERWISE NOTED.
2. ALL DRAIN PIPING SPECIFICATION ARE YY6 UNLESS OTHERWISE NOTED.
3. REFER TO DRAWING NOS. AX4DH0001 & AX4DH0002 FOR DRAIN AND CLEANOUT DESCRIPTIONS.
4. DRAIN FIXTURES CONNECTING TO 424 DRAINS ARE PROJECT CLASS 626 MANUFACTURER'S COMMERCIAL STANDARD UNLESS OTHERWISE NOTED.
5. LINES ARE CLASSIFIED RW UNLESS OTHERWISE NOTED.
6. SUMP CONTENT TO BE PUMPED BY A PORTABLE PUMP TO EITHER WASTE HOLDUP TANK OR FLOOR DRAIN TANK IN AUX. BLDG.
7. "RETIRED IN PLACE."

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NO.	REVISIONS	DATE	DR	CHK	APPV	DTL
3						
2						
1	INCORP. PER ABN-39481	12-11-03	DFV	MWD	DLL	X

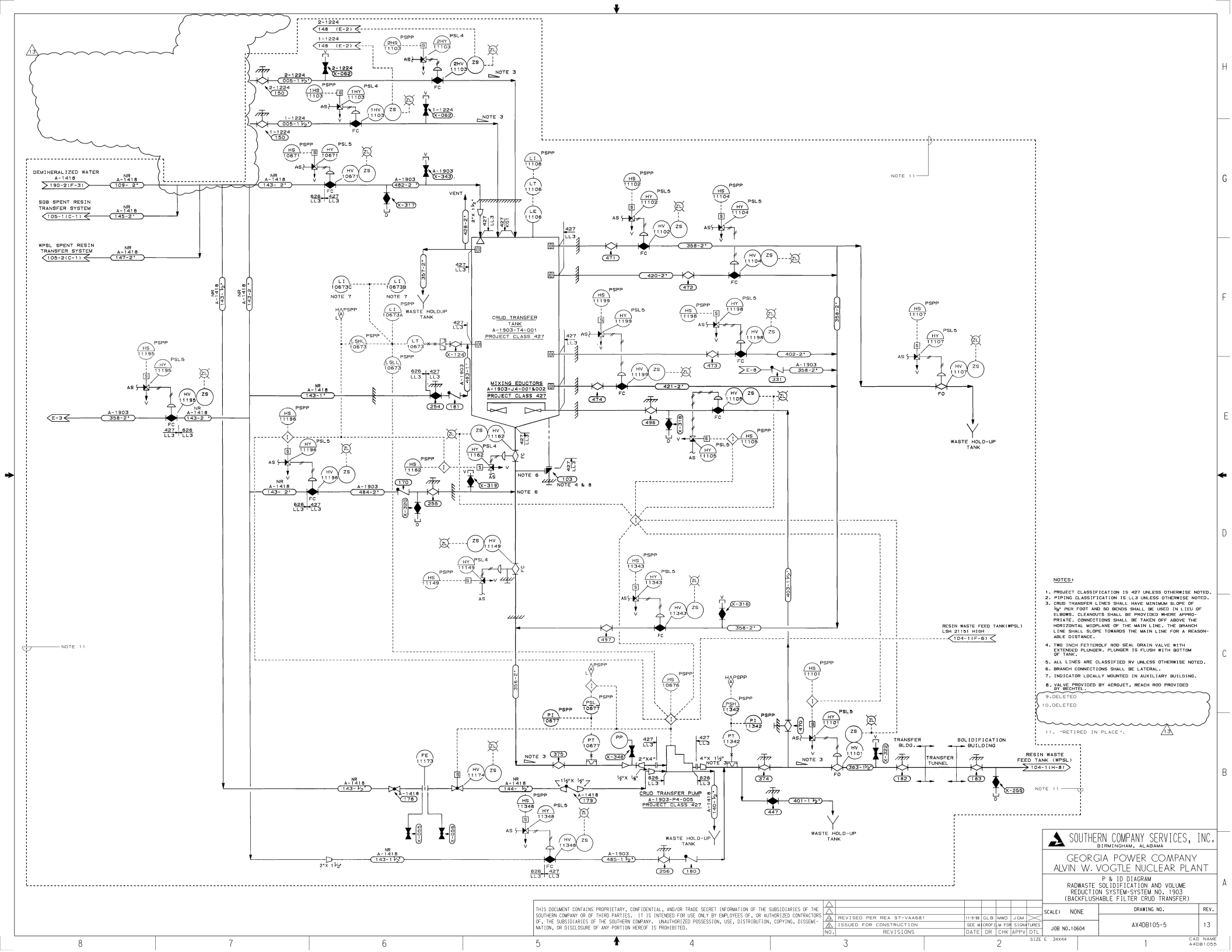
SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAMS
AUXILIARY BUILDING & MISC. DRAINS
SYSTEM NO. 1215
(TRANSFER BUILDING)

SCALE: NONE	DRAWING NO.	REV.
JOB NO.10604	AX4DB105-4	10

SIZE E 34X44



NOTES:

1. PROJECT CLASSIFICATION IS 427 UNLESS OTHERWISE NOTED.
2. PIPING CLASSIFICATION IS LL3 UNLESS OTHERWISE NOTED.
3. CRUD TRANSFER LINES SHALL HAVE MINIMUM SLOPE OF 1/8" PER FOOT AND 5D BENDS SHALL BE USED IN LIEU OF ELBOWS. CLEANOUTS SHALL BE PROVIDED WHERE APPROPRIATE. CONNECTIONS SHALL BE TAKEN OFF ABOVE THE HORIZONTAL MIDPLANE OF THE MAIN LINE. THE BRANCH LINE SHALL SLOPE TOWARDS THE MAIN LINE FOR A REASONABLE DISTANCE.
4. TWO INCH FETTEROLF ROD SEAL DRAIN VALVE WITH EXTENDED PLUNGER. PLUNGER IS FLUSH WITH BOTTOM OF TANK.
5. ALL LINES ARE CLASSIFIED RV UNLESS OTHERWISE NOTED.
6. BRANCH CONNECTIONS SHALL BE LATERAL.
7. INDICATOR LOCALLY MOUNTED IN AUXILIARY BUILDING.
8. VALVE PROVIDED BY AEROJET, REACH ROD PROVIDED BY BECHTEL.
9. DELETED
10. DELETED
11. "RETIRED IN PLACE".

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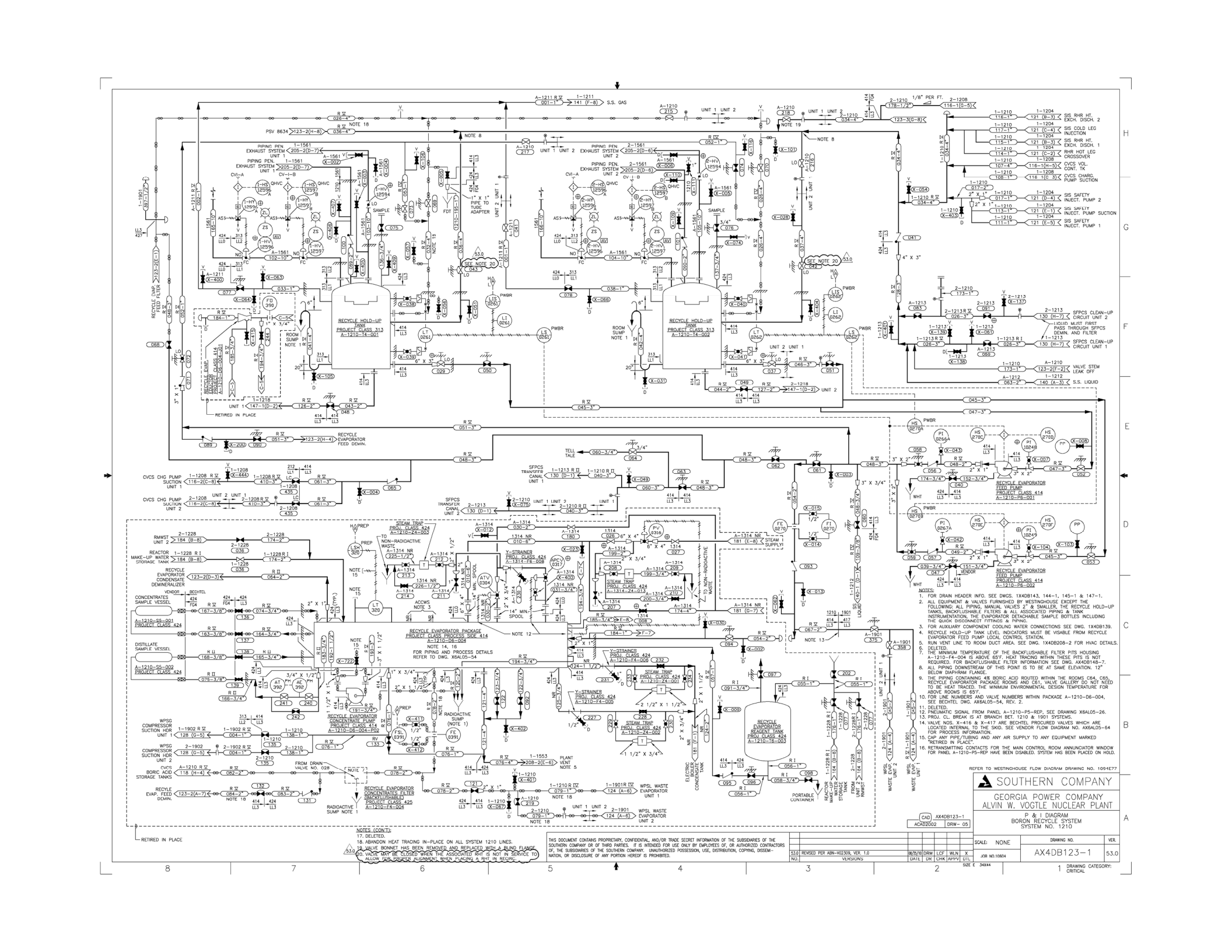
NO.	REVISIONS	DATE	DR	CHK	APPV	DTL
3						
2						
1	REVISED PER REA 97-VAA681	11-5-98	GLB	MWD	JGM	
0	ISSUED FOR CONSTRUCTION	SEE MICROFILM FOR SIGNATURES				

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & ID DIAGRAM
RADWASTE SOLIDIFICATION AND VOLUME REDUCTION SYSTEM-SYSTEM NO. 1903
(BACKFLUSHABLE FILTER CRUD TRANSFER)

SCALE: NONE
JOB NO.10604
DRAWING NO. AX4DB105-5
REV. 13
SIZE E 34X44
CAD NAME A4DB1055


RECYCLE HOLD-UP TANK PROJECT CLASS 313 A-1210-T4-001
RECYCLE HOLD-UP TANK PROJECT CLASS 313 A-1210-T4-002
RECYCLE EVAPORATOR PACKAGE PROJECT CLASS PROCESS SIDE 414 A-1210-D6-004
RECYCLE EVAPORATOR FEED PUMP PROJECT CLASS 414 A-1210-PB-001
RECYCLE EVAPORATOR FEED PUMP PROJECT CLASS 414 A-1210-PB-002
RECYCLE EVAPORATOR CONCENTRATES FILTER (BACKFLUSHABLE) PROJECT CLASS 425 A-1210-F4-004
RECYCLE EVAPORATOR REAGENT TANK A-1210-T6-003
RECYCLE EVAPORATOR CONCENTRATE PUMP PROJECT CLASS 414 A-1210-D6-004-P02
RECYCLE EVAPORATOR CONDENSATE DEMINERALIZER
RECYCLE EVAP. FEED DEMIN.
RECYCLE EVAPORATOR FEED DEMIN.
RECYCLE EVAPORATOR CONCENTRATES SAMPLE VESSEL A-1210-S5-001 PROJECT CLASS 424
DISTILLATE SAMPLE VESSEL A-1210-S5-002 PROJECT CLASS 424
BORIC ACID STORAGE TANKS
REACTOR MAKE-UP WATER STORAGE TANK
CVCS CHG PUMP SUCTION UNIT 1
CVCS CHG PUMP SUCTION UNIT 2
SFPCS TRANSFER CANAL UNIT 1
SFPCS TRANSFER CANAL UNIT 2
RMWST UNIT 2
PORTABLE CONTAINER
ELECTRIC CONDENSATE TANK
RADIOACTIVE SUMP NOTE 1
RADIOACTIVE SUMP (NOTE 1)
TO NON-RADIOACTIVE WASTE
TO RADIOACTIVE WASTE
ACCWS NOTE 3
FROM DRAIN VALVE NO. 028
PLANT VENT NOTE 5
WPSG COMPRESSOR SUCTION HDR UNIT 1
WPSG COMPRESSOR SUCTION HDR UNIT 2
WPSL WASTE EVAPORATOR UNIT 1
WPSL WASTE EVAPORATOR UNIT 2
WPSL WASTE EVAPORATOR UNIT 1
WPSL WASTE EVAPORATOR UNIT 2
S.S. GAS
S.S. LIQUID
VALVE STEM LEAK OFF
SIS RHR HT. EXCH. DISCH. 2
SIS COLD LEG INJECTION
SIS RHR HT. EXCH. DISCH. 1
RHR HOT LEG CROSSOVER
CVCS VOL. CONT. TK.
CVCS CHARG. PUMP SUCTION
SIS SAFETY INJECT. PUMP 2
SIS SAFETY INJECT. PUMP SUCTION
SIS SAFETY INJECT. PUMP 1
SFPCS CLEAN-UP CIRCUIT UNIT 2
SFPCS CLEAN-UP CIRCUIT UNIT 1
LIQUID MUST FIRST PASS THROUGH SFPCS DEMIN. AND FILTER
RETIRED IN PLACE
NOTE 8
NOTE 12
NOTE 13
NOTE 15
NOTE 18
NOTE 19
SEE NOTE 20
ROOM SUMP NOTE 1
PWBR
PSV 8634
PIPING PEN. EXHAUST SYSTEM UNIT 1 1-1561 205-2(D-7)
PIPING PEN. EXHAUST SYSTEM UNIT 2 2-1561 205-2(D-6)
PIPE TO TUBE ADAPTER 1" X 1"
STEAM TRAP PROJ. CLASS 424 A-1210-Z4-001
STEAM TRAP PROJ. CLASS 424 A-1210-Z4-002
STEAM TRAP PROJ. CLASS 424 A-1314-Z4-013
Y-STRAINER PROJ. CLASS 424 A-1314-F4-008
Y-STRAINER PROJ. CLASS 424 A-1210-F4-005
Y-STRAINER PROJ. CLASS 424 A-1210-F4-008
14" MIN. SPOOL
3/4" X 1/2" REDUCER
1/2" X 3/4" REDUCER
1" X 3/4" REDUCER
1" X 1/2" REDUCER
2" X 1" REDUCER
2" X 1 1/2" REDUCER
3" X 1 1/2" REDUCER
3" X 2" REDUCER
4" X 3" REDUCER
6" X 4" REDUCER
6" X 3" REDUCER
1/8" PER FT.
UNIT 1 UNIT 2
UNIT 2 UNIT 1
HS 0270A
HS 0270C
HS 0270D
HS 0270E
HS 0270F
PI 0264A
PI 0267A
PI 10248
PI 10249
FE 0270
FT 0270
PV 0316
LT 320
LT 0261
LT 0262
LS 0261
LS 0262
LI 0261
LI 0262
FD 390
AT 392
AE 392
FSL 0391
FE 0391
SSM 320
APCV 0317
ATV 0304
HV 12596
HV 12597
HV 12594
NOTES:
1. FOR DRAIN HEADER INFO. SEE DWGS. 1X4DB143, 144-1, 145-1 & 147-1.
2. ALL EQUIPMENT & VALVES FURNISHED BY WESTINGHOUSE EXCEPT THE FOLLOWING: ALL PIPING, MANUAL VALVES 2" & SMALLER, THE RECYCLE HOLD-UP TANKS, BACKFLUSHABLE FILTERS & ALL ASSOCIATED PIPING & TANK INSTRUMENTATION, THE EVAPORATOR DETACHABLE SAMPLE BOTTLES INCLUDING THE QUICK DISCONNECT FITTINGS & PIPING.
3. FOR AUXILIARY COMPONENT COOLING WATER CONNECTIONS SEE DWG. 1X4DB139.
4. RECYCLE HOLD-UP TANK LEVEL INDICATORS MUST BE VISABLE FROM RECYCLE EVAPORATOR FEED PUMP LOCAL CONTROL STATION.
5. RUN VENT LINE TO ROOM DUCT AREA. SEE DWG. 1X4DB208-2 FOR HVAC DETAILS.
6. DELETED.
7. THE MINIMUM TEMPERATURE OF THE BACKFLUSHABLE FILTER PITS HOUSING A-1210-F4-004 IS ABOVE 65°F. HEAT TRACING WITHIN THESE PITS IS NOT REQUIRED. FOR BACKFLUSHABLE FILTER INFORMATION SEE DWG. AX4DB148-7.
8. ALL PIPING DOWNSTREAM OF THIS POINT IS TO BE AT SAME ELEVATION. 12" BELOW DIAPHRAM FLANGE.
9. THE PIPING CONTAINING 4% BORIC ACID ROUTED WITHIN THE ROOMS C84, C85, RECYCLE EVAPORATOR PACKAGE ROOMS AND C61, VALVE GALLERY DO NOT NEED TO BE HEAT TRACED. THE MINIMUM ENVIRONMENTAL DESIGN TEMPERATURE FOR ABOVE ROOMS IS 65°F.
10. FOR LINE NUMBERS AND VALVE NUMBERS WITHIN PACKAGE A-1210-D6-004, SEE BECHTEL DWG. AX6AL05-54, REV. 2.
11. DELETED.
12. PNEUMATIC SIGNAL FROM PANEL A-1210-P5-REP, SEE DRAWING X6AL05-26.
13. PROJ. CL. BREAK IS AT BRANCH BET. 1210 & 1901 SYSTEMS.
14. VALVE NOS. X-416 & X-417 ARE BECHTEL PROCURED VALVES WHICH ARE LOCATED INTERNAL TO THE SKID. SEE VENDOR FLOW DIAGRAM NO. AX6AL05-64 FOR PROCESS INFORMATION.
15. CAP ANY PIPE/TUBING AND ANY AIR SUPPLY TO ANY EQUIPMENT MARKED "RETIRED IN PLACE".
16. RETRANSMITTING CONTACTS FOR THE MAIN CONTROL ROOM ANNUNCIATOR WINDOW FOR PANEL A-1210-P5-REP HAVE BEEN DISABLED. SYSTEM HAS BEEN PLACED ON HOLD.
NOTES (CONT):
17. DELETED.
18. ABANDON HEAT TRACING IN-PLACE ON ALL SYSTEM 1210 LINES.
19. VALVE BONNET HAS BEEN REMOVED AND REPLACED WITH A BLIND FLANGE. NO VALVE MAY BE CLOSED WHEN THE ASSOCIATED RHT IS NOT IN SERVICE TO ALLOW FOR PROPER ALIGNMENT WHEN PLACING A RHT IN SERVICE.
REFER TO WESTINGHOUSE FLOW DIAGRAM DRAWING NO. 1091E77
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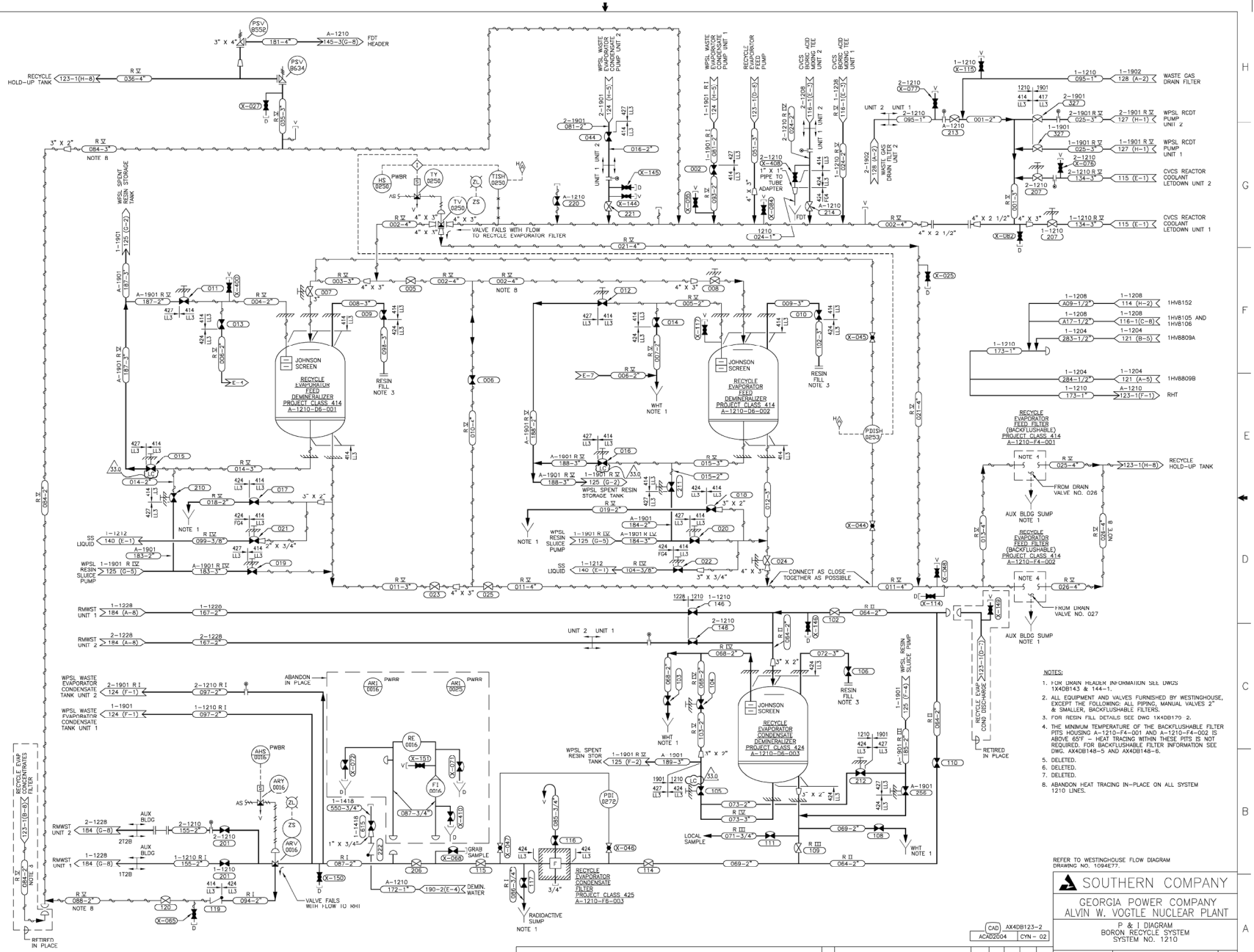
53.0 REVISED PER ABN-W02309, VER. 1.0 08/15/16 DRW LCF WLN X
NO. VERSIONS DATE DR CHK APPV DTL
SOUTHERN COMPANY
GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT
P & I DIAGRAM
BORON RECYCLE SYSTEM
SYSTEM NO. 1210
CAD AX4DB123-1
ACAD2002 DRW- 05
SCALE: NONE
DRAWING NO. AX4DB123-1
VER. 53.0
JOB NO.10604
SIZE E 34X44
DRAWING CATEGORY: CRITICAL



NOTES:

1. FOR DRAIN HEADER INFORMATION SEE DWGS 1X4DB143 & 144-1.
2. ALL EQUIPMENT AND VALVES FURNISHED BY WESTINGHOUSE, EXCEPT THE FOLLOWING: ALL PIPING, MANUAL VALVES 2" & SMALLER, BACKFLUSHABLE FILTERS.
3. FOR RESIN FILL DETAILS SEE DWG 1X4DB179 2.
4. THE MINIMUM TEMPERATURE OF THE BACKFLUSHABLE FILTER PITS HOUSING A-1210-F4-001 AND A-1210-F4-002 IS ABOVE 65F - HEAT TRACING WITHIN THESE PITS IS NOT REQUIRED. FOR BACKFLUSHABLE FILTER INFORMATION SEE DWG. AX4DB148-5 AND AX4DB148-6.
5. DELETED.
6. DELETED.
7. DELETED.
8. ABANDON HEAT TRACING IN-PLACE ON ALL SYSTEM 1210 LINES.

REFER TO WESTINGHOUSE FLOW DIAGRAM DRAWING NO. 1094E77.

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NO.	VERSIONS	DATE	DR	CHK	APPV	DTL
33.0	REVISED PER ABN-V01041, VER.1.0	10-31-07	BgT	MWD	WLN	X

CAD AX4DB123-2
ACAD2004 CYN- 02

SOUTHERN COMPANY

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
BORON RECYCLE SYSTEM
SYSTEM NO. 1210

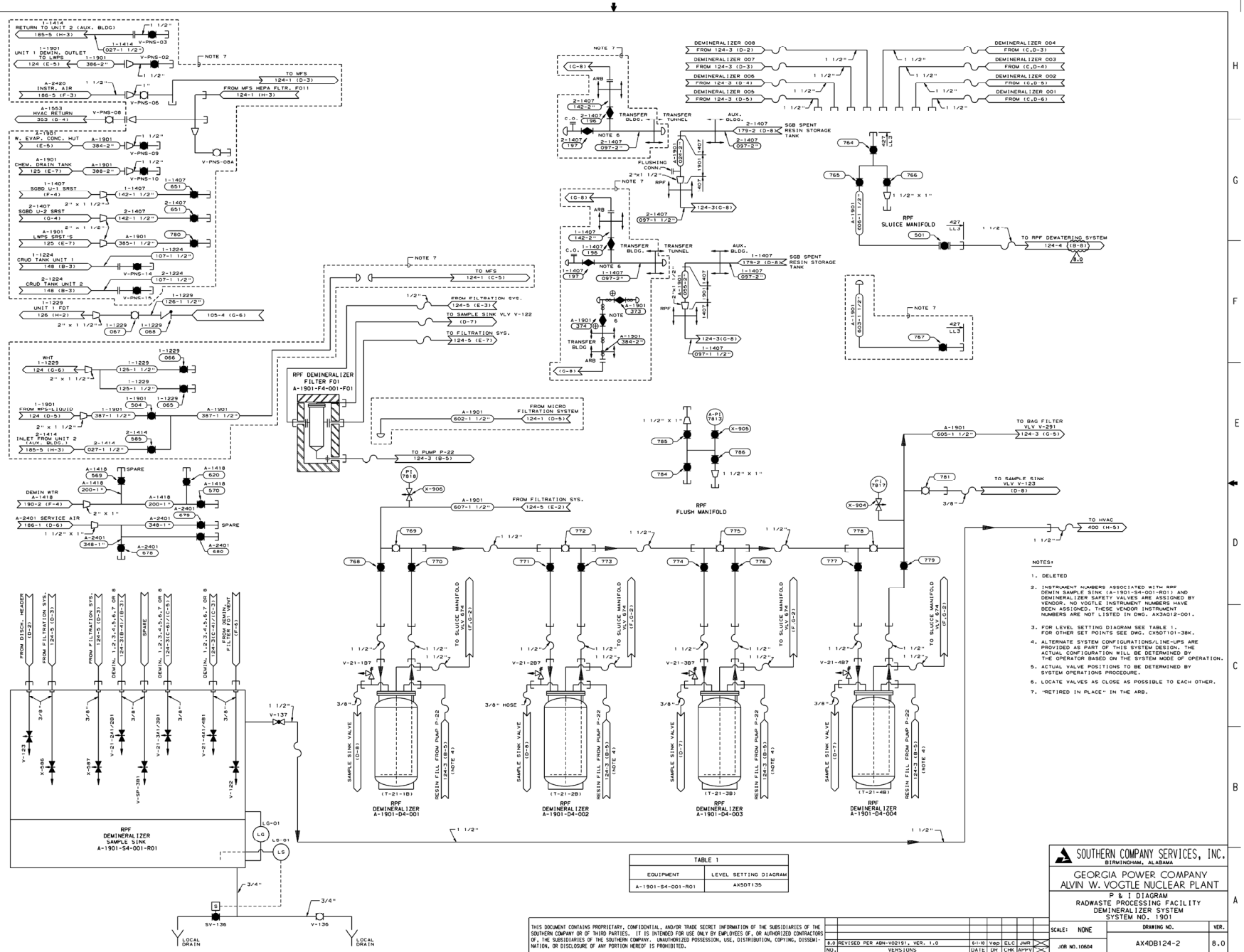
SCALE: NONE

DRAWING NO. AX4DB123-2 VER. 33.0

JOB NO.10604

SIZE E 34X44

DRAWING CATEGORY: CRITICAL



SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
RADWASTE PROCESSING FACILITY
DEMINERALIZER SYSTEM
SYSTEM NO. 1901

SCALE: NONE	DRAWING NO.	VER.
JOB NO. 10604	AX4DB124-2	8.0

SIZE E 34x44

DWG CATEGORY: CRITICAL

NO.	VERSIONS	DATE	DR	CHK	APPV
8.0	REVISED PER ABN-V02191, VER. 1.0	6-11-10	Vap	ELC	JMR

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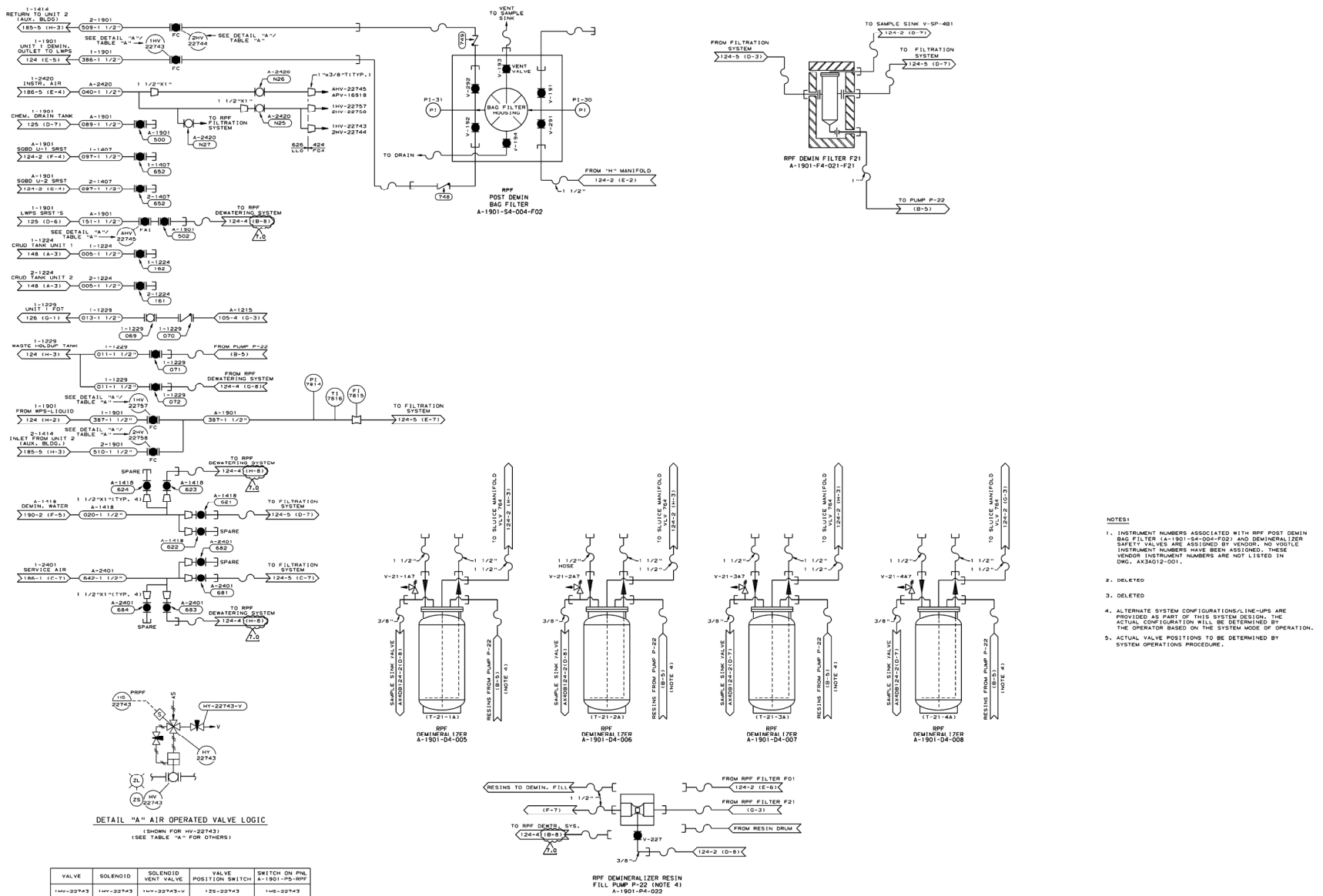

VALVE	SOLENOID	SOLENOID VENT VALVE	VALVE POSITION SWITCH	SWITCH ON PNL A-1901-PS-RPF
1HV-22743	1HY-22743	1HY-22743-V	1ZS-22743	1HS-22743
2HV-22744	2HY-22744	2HY-22744-V	2ZS-22744	2HS-22744
AHV-22745	AHY-22745	AHY-22745-V	AZS-22745	AHS-22745
1HV-22757	1HY-22757	1HY-22757-V	1ZS-22757	1HS-22757
2HV-22758	2HY-22758	2HY-22758-V	2ZS-22758	2HS-22758

TABLE "A"

NOTES:

1. INSTRUMENT NUMBERS ASSOCIATED WITH RPF POST DEMIN BAG FILTER (A-1901-S4-004-F02) AND DEMINERALIZER SAFETY VALVES ARE ASSIGNED BY VENDOR. NO VOGTLE INSTRUMENT NUMBERS HAVE BEEN ASSIGNED. THESE VENDOR INSTRUMENT NUMBERS ARE NOT LISTED IN DWG. AX3A012-001.
2. DELETED
3. DELETED
4. ALTERNATE SYSTEM CONFIGURATIONS/LINE-UPS ARE PROVIDED AS PART OF THIS SYSTEM DESIGN. THE ACTUAL CONFIGURATION WILL BE DETERMINED BY THE OPERATOR BASED ON THE SYSTEM MODE OF OPERATION.
5. ACTUAL VALVE POSITIONS TO BE DETERMINED BY SYSTEM OPERATIONS PROCEDURE.

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NO.	VERSIONS	DATE	DR	CHK	APPV
7.0	REVISED PER ABN-V02191, VER. 1.0	6-11-10	Vap	ELC	JMR

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
RADWASTE PROCESSING FACILITY
DEMINERALIZER SYSTEM
SYSTEM NO. 1901

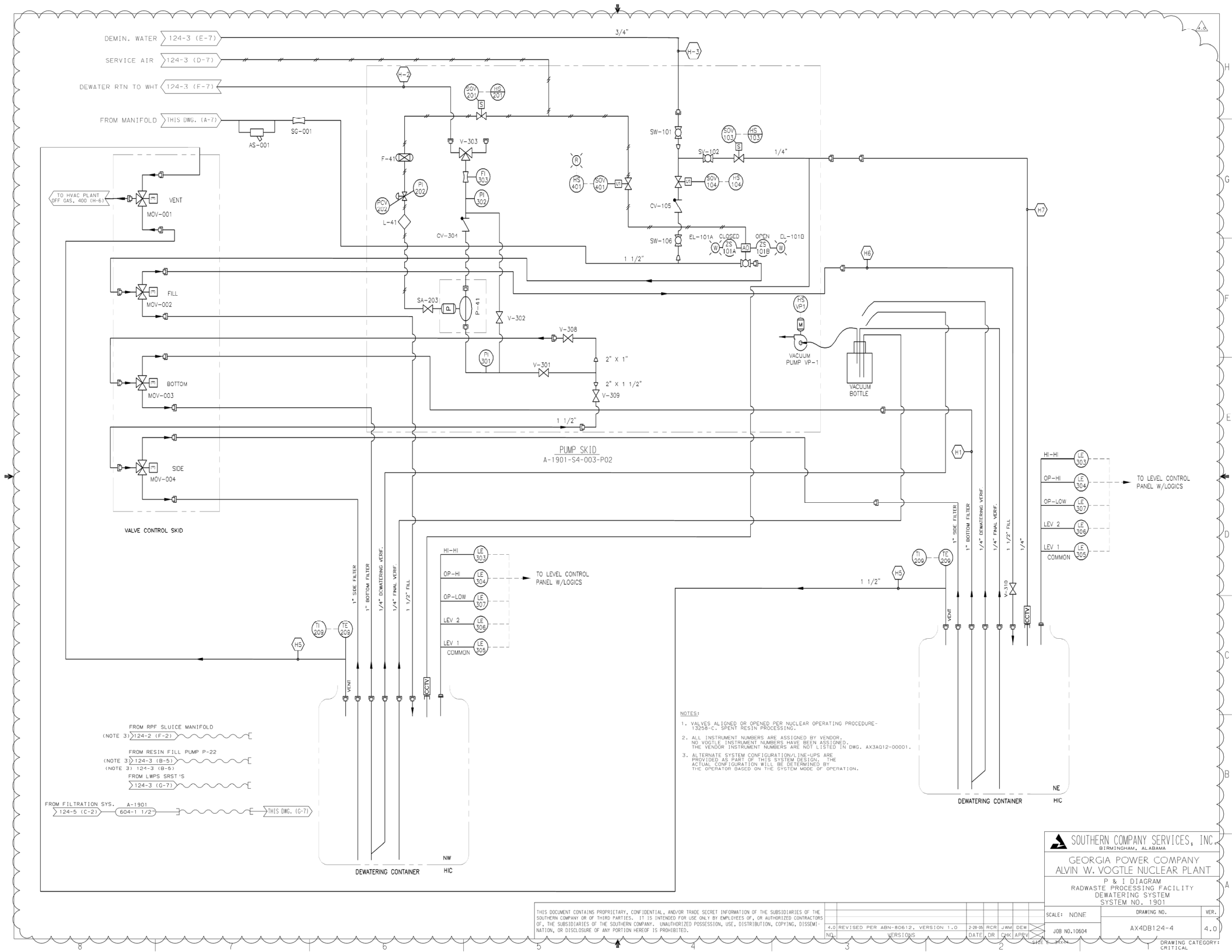
SCALE: NONE

DRAWING NO.	VER.
AX4DB124-3	7.0

JOB NO. 10604

SIZE E 34x44

DWG CATEGORY: CRITICAL



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NO.	VERSIONS	DATE	DR	CHK	APPV
4.0	REVISED PER ABN-80612, VERSION 1.0	2-28-05	RCR	JWM	DEW

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

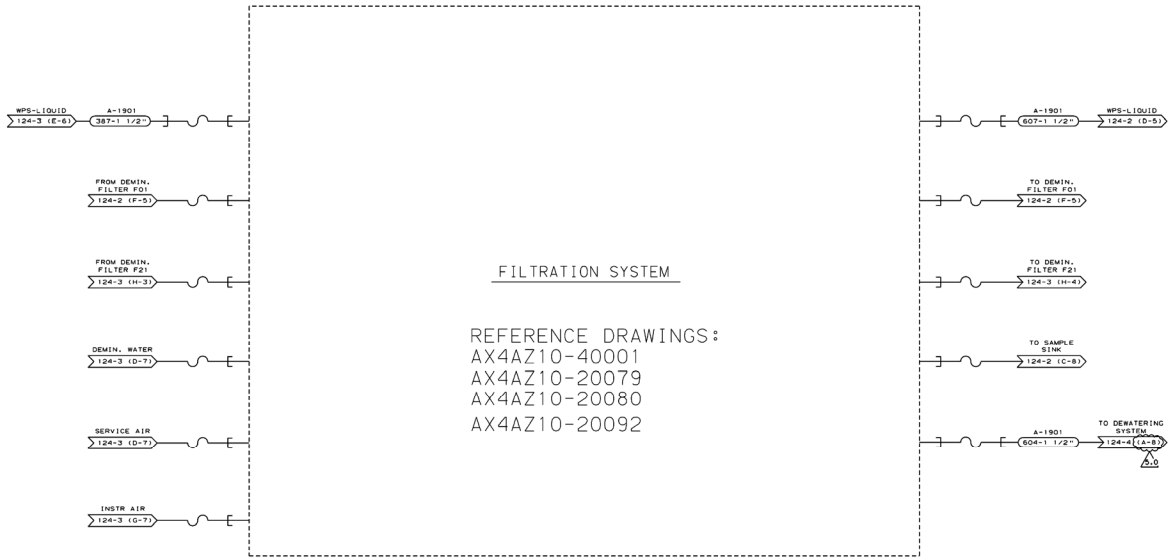
GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
RADWASTE PROCESSING FACILITY
DEWATERING SYSTEM
SYSTEM NO. 1901

SCALE: NONE	DRAWING NO.	VER.
JOB NO.10604	AX4DB124-4	4.0

SIZE E 30X42

DRAWING CATEGORY: CRITICAL



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NO.	VERSIONS	DATE	DR	CHK	APPV
5.0	REVISED PER ABN-V02191, VER. 1.0	6-11-10	Vap	ELC	JMR

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

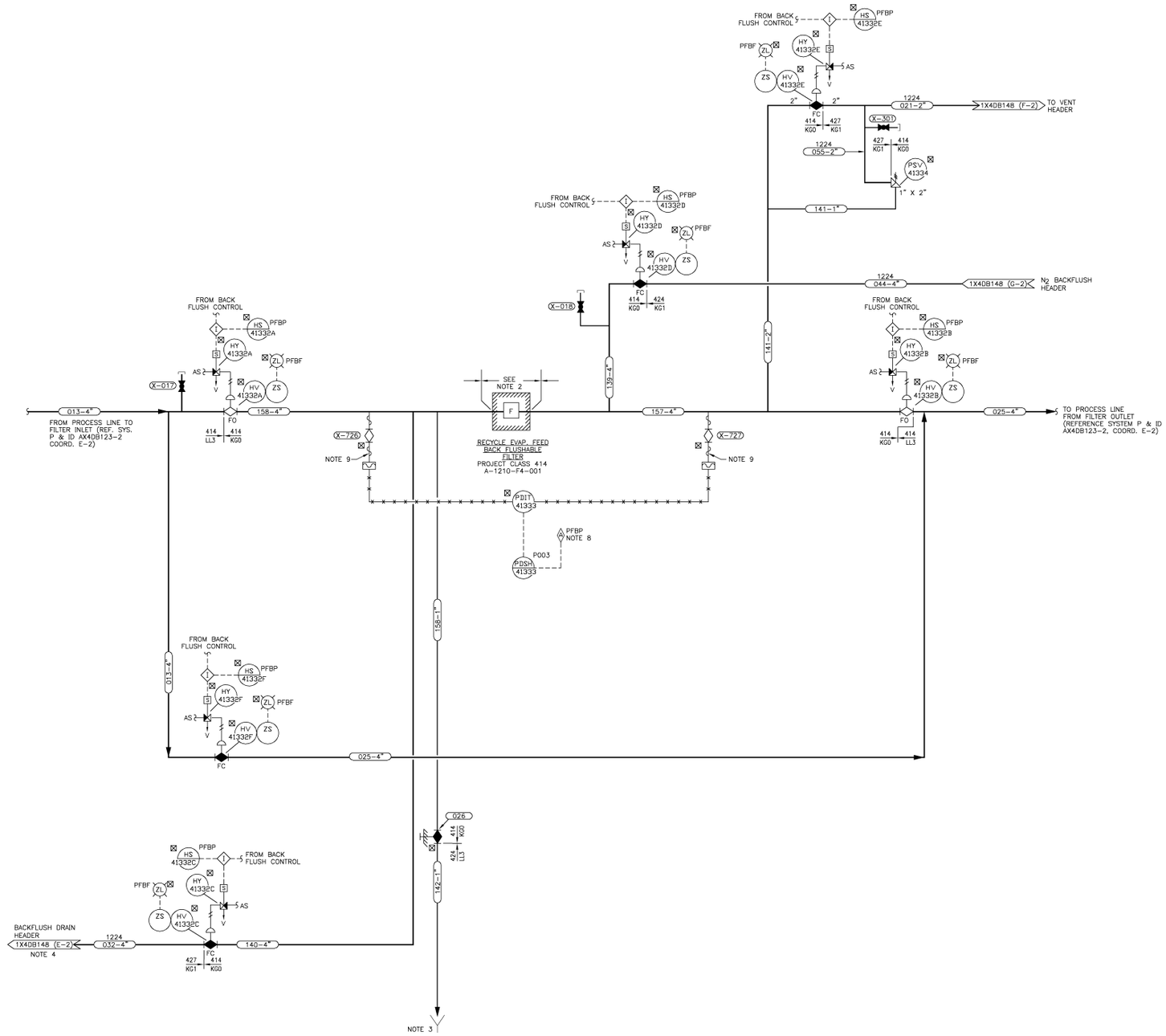
GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
RADWASTE PROCESSING FACILITY
FILTRATION SYSTEM
SYSTEM NO. 1901

SCALE: NONE	DRAWING NO.	VER.
JOB NO. 10604	AX4DB124-5	5.0

SIZE E 34x44

DWG CATEGORY: CRITICAL



NOTES:

1. FOR DRAIN HEADER INFORMATION SEE 1X4DB143 AND 1X4DB144-1.
2. ALL EQUIPMENT, VALVES AND INSTRUMENTATION SHALL BE SUPPLIED BY THE VENDOR. ALL INTERCONNECTING PIPING SHALL BE FURNISHED BY BECHTEL.
3. SEE SYSTEM P & I DIAGRAM AX4DB123-2 FOR DRAINAGE SYSTEM.
4. BACKFLUSH DRAIN LINES WILL SLOPE TO THE BACKFLUSHABLE FILTER CRUD TANK WITH NO POCKETS.
5. RADIATION ZONE CLASSIFICATION FOR THE EQUIPMENT IN THIS SYSTEM IS RV.
6. ALL LINES AND VALVES, REGARDLESS OF SIZE, SHALL BE BUTT-WELDED IN ORDER TO MINIMIZE CRUD TRAPS.
7. SEE VENDOR DRAWINGS FOR DETAILS OF VALVES SHOWN WITH DIAPHRAGM ACTUATORS.
8. THIS ALARM HAS A COMMON ANNUNCIATOR IN THE MAIN CONTROL ROOM.
9. ABANDON HEAT TRACING IN PLACE ON ALL SYSTEM 1210 LINES.
10. BACKFLUSHABLE FILTER A-1210-F4-001 HAS BEEN GUTTED AND USED FOR PIPING.

CAD AX4DB148-5
ACAD2004 CYN - 01

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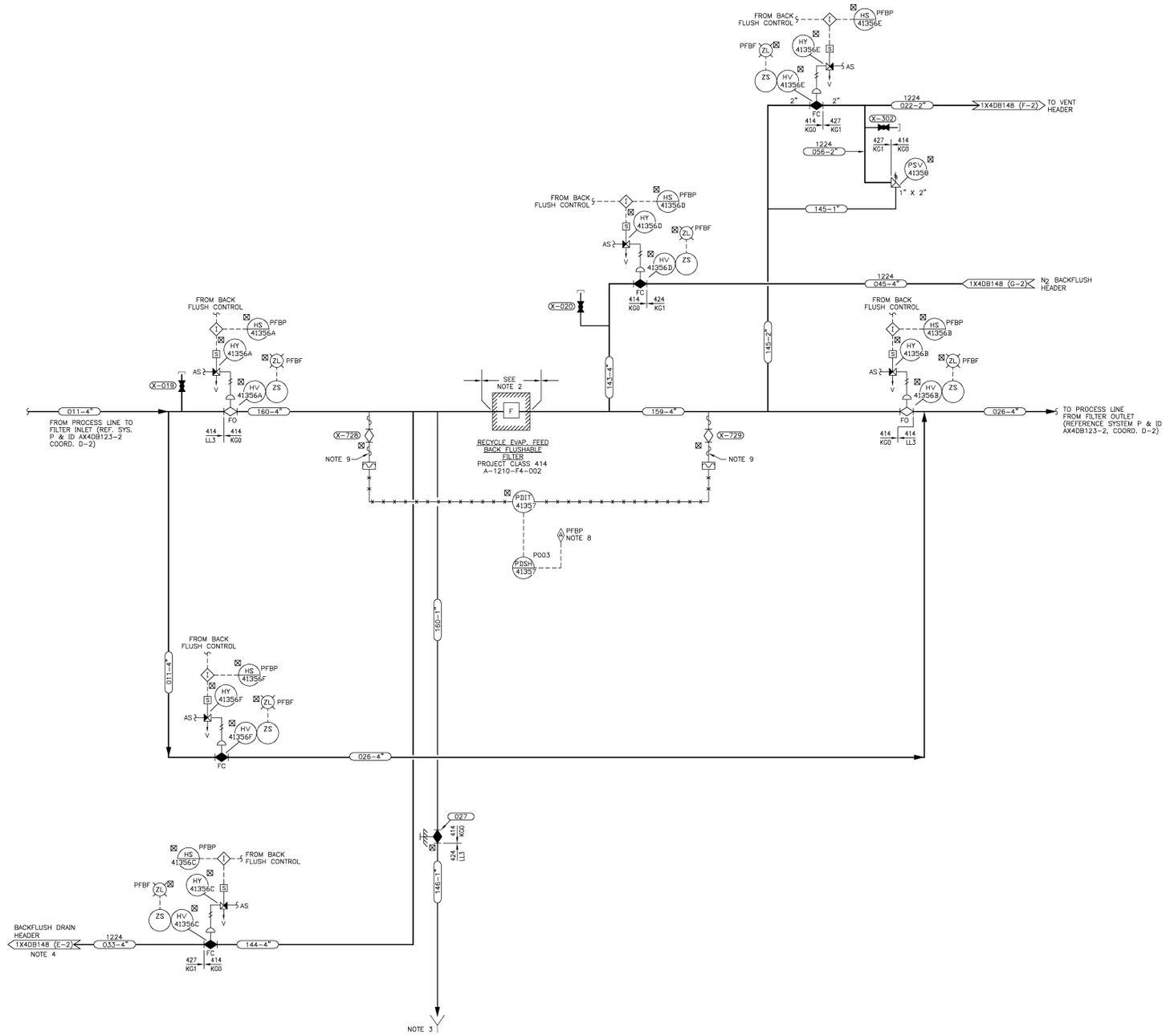
NO.	VERSIONS	DATE	DR	CHK	APPV	DTL
1.0	REDRAWN PER ABN-V70141, VER. 1.0	10-27-06	CYN	LCF	SHS	X

SOUTHERN COMPANY

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
RECYCLE EVAPORATOR FEED BACKFLUSHABLE FILTER SYSTEM SYSTEM NO. 1210

SCALE: NONE
JOB NO.10604
DRAWING NO. AX4DB148-5
VER. 1.0
SIZE E 34X44
DRAWING CATEGORY: CRITICAL



NOTES:

1. FOR DRAIN HEADER INFORMATION SEE 1X4DB143 AND 1X4DB144-1.
2. ALL EQUIPMENT, VALVES AND INSTRUMENTATION SHALL BE SUPPLIED BY THE VENDOR. ALL INTERCONNECTING PIPING SHALL BE FURNISHED BY BECHTEL.
3. SEE SYSTEM P & I DIAGRAM AX4DB123-2 FOR DRAINAGE SYSTEM.
4. BACKFLUSH DRAIN LINES WILL SLOPE TO THE BACKFLUSHABLE FILTER CRUD TANK WITH NO POCKETS.
5. RADIATION ZONE CLASSIFICATION FOR THE EQUIPMENT IN THIS SYSTEM IS RV.
6. ALL LINES AND VALVES, REGARDLESS OF SIZE, SHALL BE BUTT-WELDED IN ORDER TO MINIMIZE CRUD TRAPS.
7. SEE VENDOR DRAWINGS FOR DETAILS OF VALVES SHOWN WITH DIAPHRAGM ACTUATORS.
8. THIS ALARM HAS A COMMON ANNUNCIATOR IN THE MAIN CONTROL ROOM.
9. ABANDON HEAT TRACING IN PLACE ON ALL SYSTEM 1210 LINES.
10. BACKFLUSHABLE FILTER A-1210-F4-002 HAS BEEN GUTTED AND USED FOR PIPING.

CAD AX4DB148-6
ACAD2004 CYN - 01

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NO.	VERSIONS	DATE	DR	CHK	APPV	DTL
1.0	REDRAWN PER ABN-V70141, VER. 1.0	10-27-06	CYN	LCF	SHS	X

SOUTHERN COMPANY

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

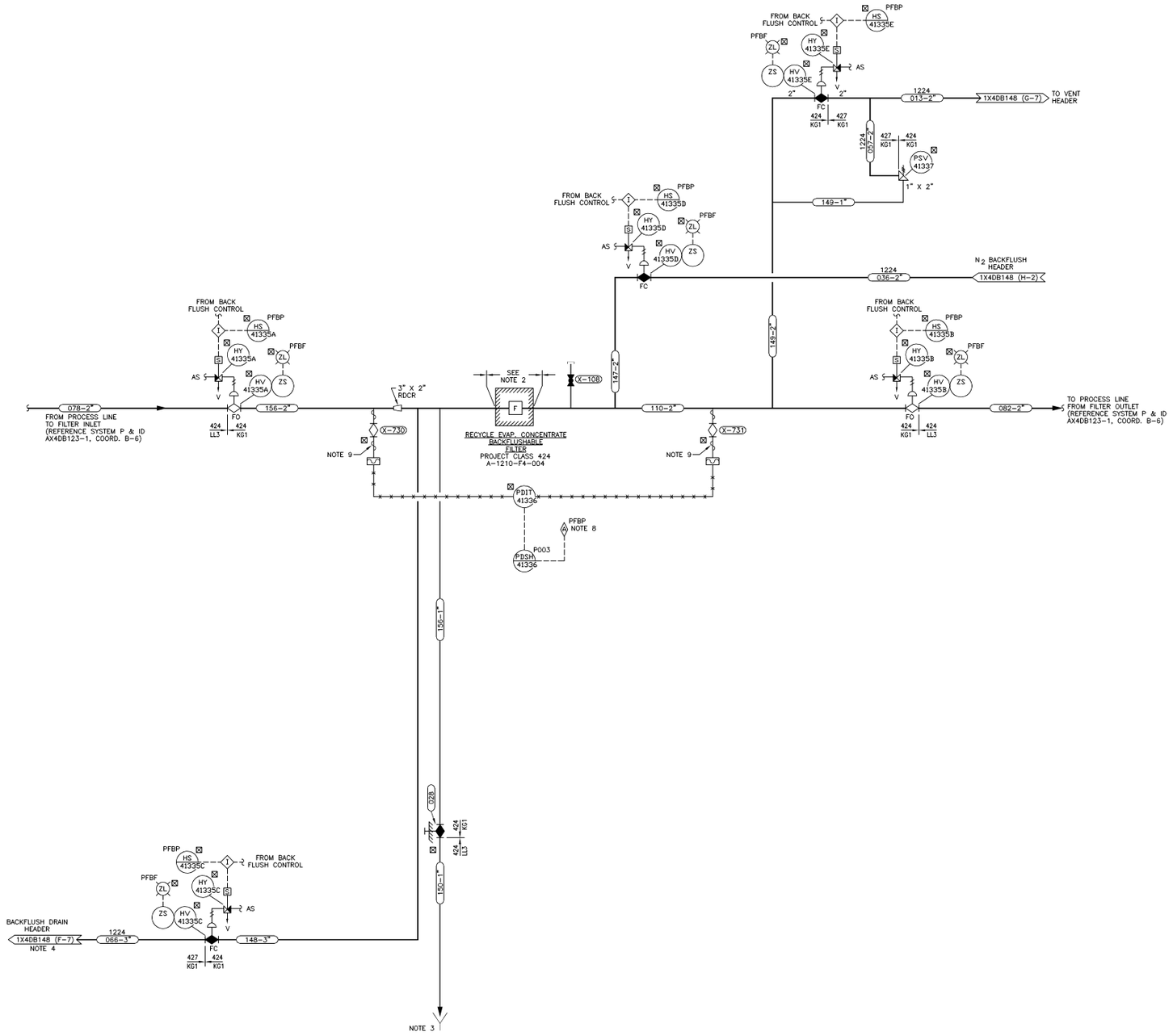
P & I DIAGRAM
RECYCLE EVAPORATOR FEED BACKFLUSHABLE FILTER SYSTEM SYSTEM NO. 1210

SCALE: NONE
JOB NO.10604

DRAWING NO. AX4DB148-6
VER. 1.0

SIZE E 34X44

DRAWING CATEGORY: CRITICAL



NOTES:

1. FOR DRAIN HEADER INFORMATION SEE DWG. 1X4DB143 AND 1X4DB144-1.
2. ALL EQUIPMENT, VALVES, AND INSTRUMENTATION SHALL BE SUPPLIED BY THE VENDOR. ALL INTERCONNECTING PIPING SHALL BE FURNISHED BY BECHTEL.
3. SEE SYSTEM P & I DIAGRAM AX4DB123-1 FOR DRAINAGE SYSTEM.
4. BACKFLUSH DRAIN LINES WILL SLOPE TO THE BACKFLUSHABLE FILTER CRUD TANK WITH NO POCKETS.
5. RADIATION ZONE CLASSIFICATION FOR THE EQUIPMENT IN THIS SYSTEM IS RV.
6. ALL LINES AND VALVES, REGARDLESS OF SIZE, SHALL BE BUTT-WELDED IN ORDER TO MINIMIZE CRUD TRAPS.
7. SEE VENDOR DRAWINGS FOR DETAILS OF VALVES SHOWN WITH DIAPHRAGM ACTUATORS.
8. THIS ALARM HAS A COMMON ANNUNCIATOR IN THE MAIN CONTROL ROOM.
9. ABANDON HEAT TRACING IN PLACE ON ALL SYSTEM 1210 LINES.
10. FILTER IS RETIRED IN PLACE.

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NO.	VERSIONS	DATE	DR	CHK	APPV	DTL
1.0	REDRAWN PER ABN-V70114, VER. 1.0	08/21/06	NJ	LCF	SHS	X

CAD AX4DB148-7
ACAD2004 NJ - 01

SOUTHERN COMPANY

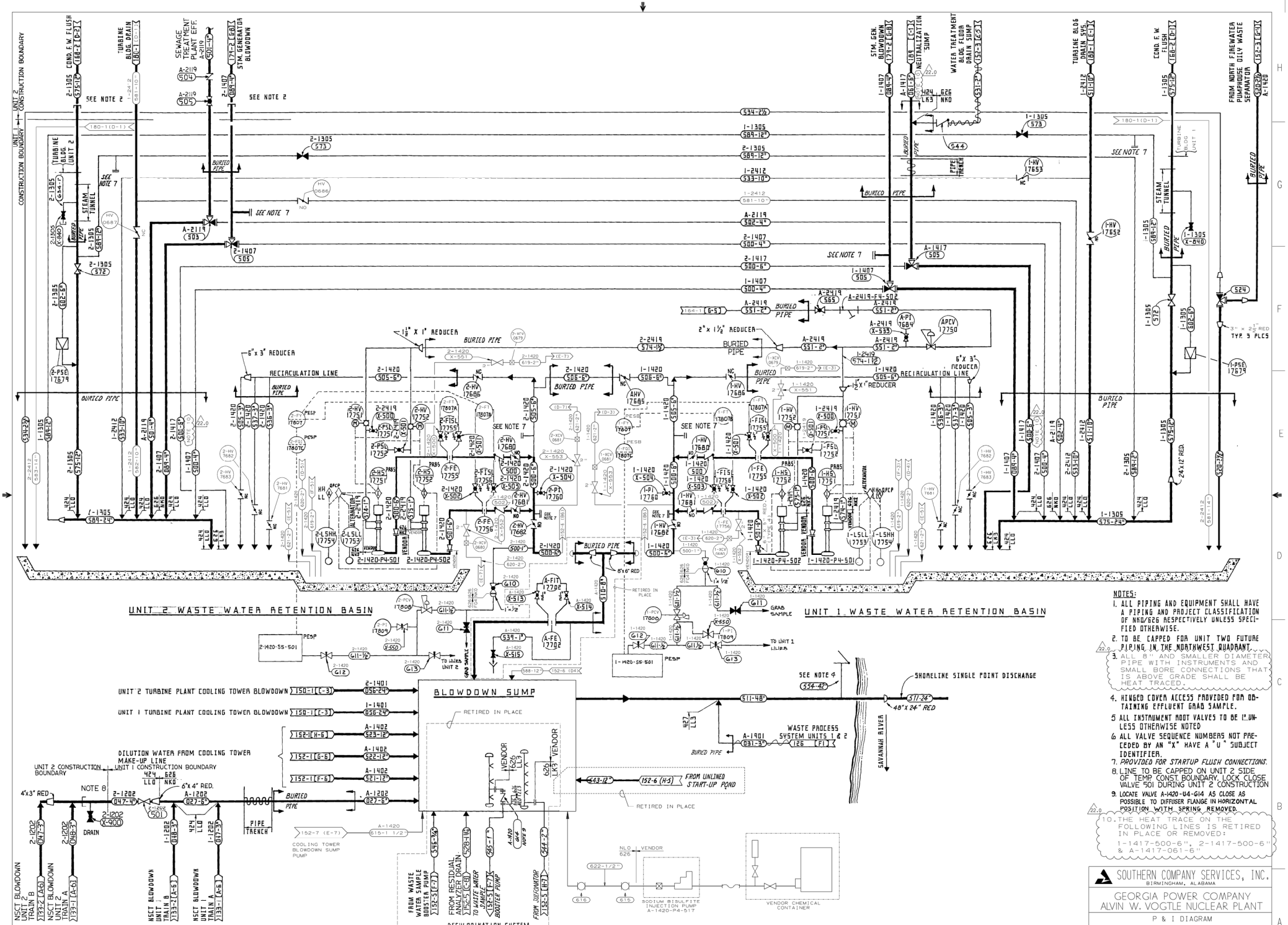
GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
RECYCLE EVAPORATOR CONCENTRATE
BACKFLUSHABLE FILTERS SYSTEM
SYSTEM NO. 1210

SCALE: NONE	DRAWING NO.	VER.
JOB NO.10604	AX4DB148-7	1.0

SIZE E 34X44

DRAWING CATEGORY: CRITICAL



NOTES:

1. ALL PIPING AND EQUIPMENT SHALL HAVE A PIPING AND PROJECT CLASSIFICATION OF NNO/626 RESPECTIVELY UNLESS SPECIFIED OTHERWISE.
2. TO BE CAPPED FOR UNIT TWO FUTURE PIPING IN THE NORTHWEST QUADRANT.
22.0 3. ALL 8" AND SMALLER DIAMETER PIPE WITH INSTRUMENTS AND SMALL BORE CONNECTIONS THAT IS ABOVE GRADE SHALL BE HEAT TRACED.
4. HINGED COVER ACCESS PROVIDED FOR OBTAINING EFFLUENT GRAB SAMPLE.
5. ALL INSTRUMENT ROOT VALVES TO BE 1" UNLESS OTHERWISE NOTED
6. ALL VALVE SEQUENCE NUMBERS NOT PRECEDED BY AN "X" HAVE A "U" SUBJECT IDENTIFIER.
7. *PROVIDED FOR STARTUP FLUSH CONNECTIONS.*
8. LINE TO BE CAPPED ON UNIT 2 SIDE OF TEMP CONST. BOUNDARY. LOCK CLOSE VALVE 501 DURING UNIT 2 CONSTRUCTION
9. LOCATE VALVE A-1420-U4-G14 AS CLOSE AS POSSIBLE TO DIFFUSER FLANGE IN HORIZONTAL POSITION WITH SPRING REMOVED.
22.0 10. THE HEAT TRACE ON THE FOLLOWING LINES IS RETIRED IN PLACE OR REMOVED:
1-1417-500-6", 2-1417-500-6" & A-1417-061-6"

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NO.	VERSIONS	DATE	DR	CHK	APPV
22.0	REVISED PER ABN C99900120IM007, VER. 1.0	6-23-08	JLO	TSL	JMR

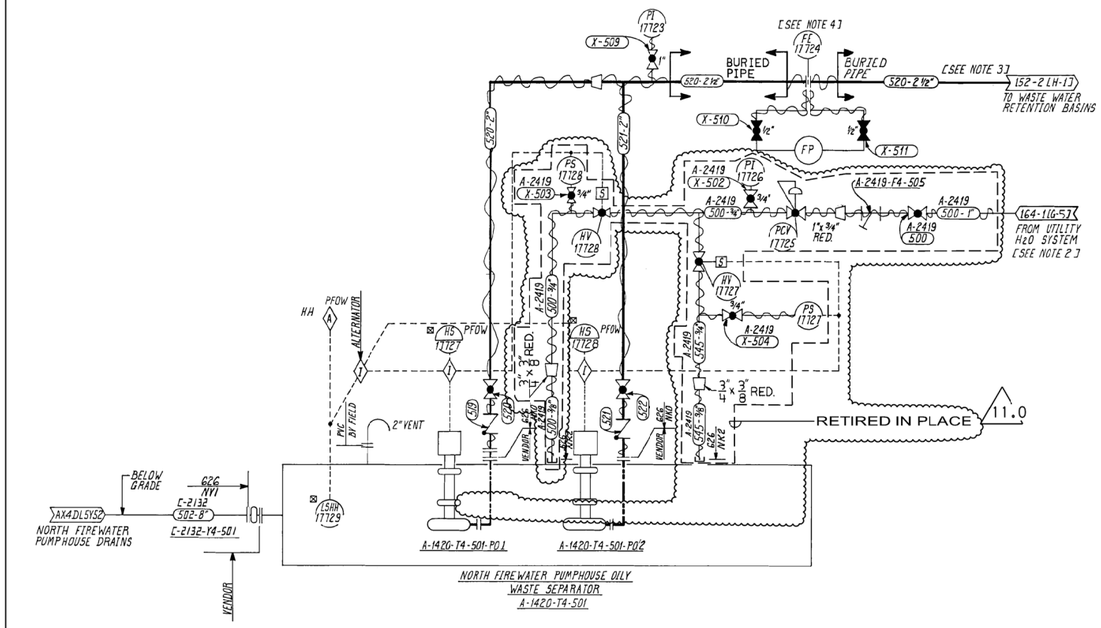
SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

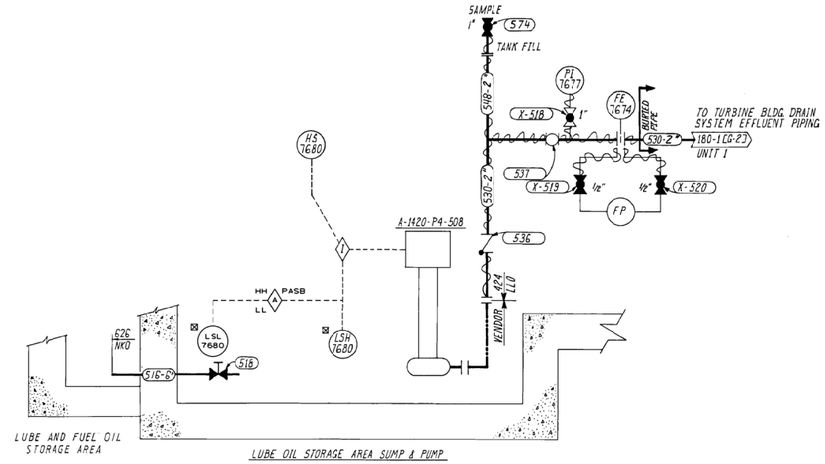
GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

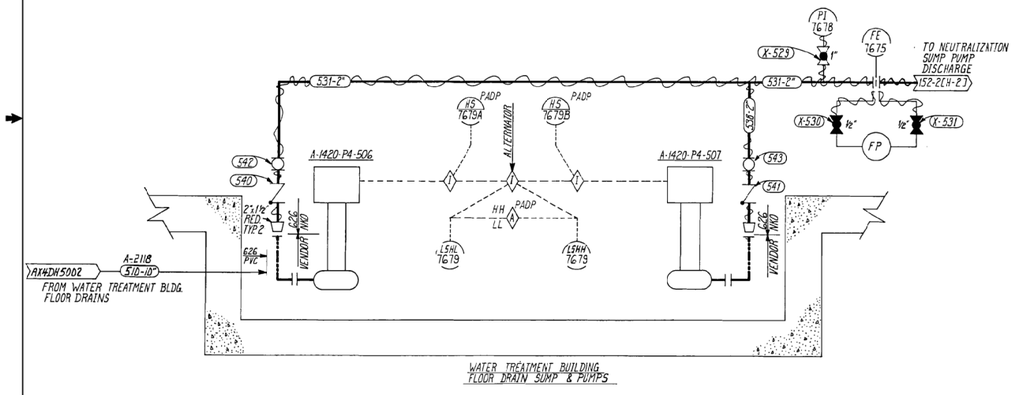
P & I DIAGRAM
WASTE WATER EFFLUENT SYSTEM
SYSTEM NO. 1420

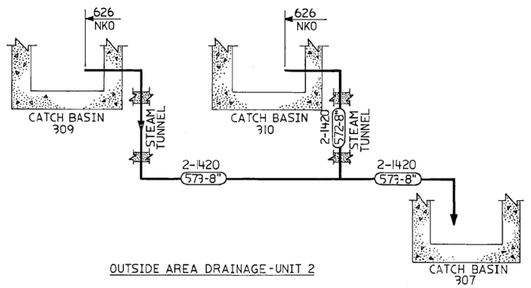
SCALE: NONE	DRAWING NO.	VER.
JOB NO.10604	AX4DB152-2	22.0

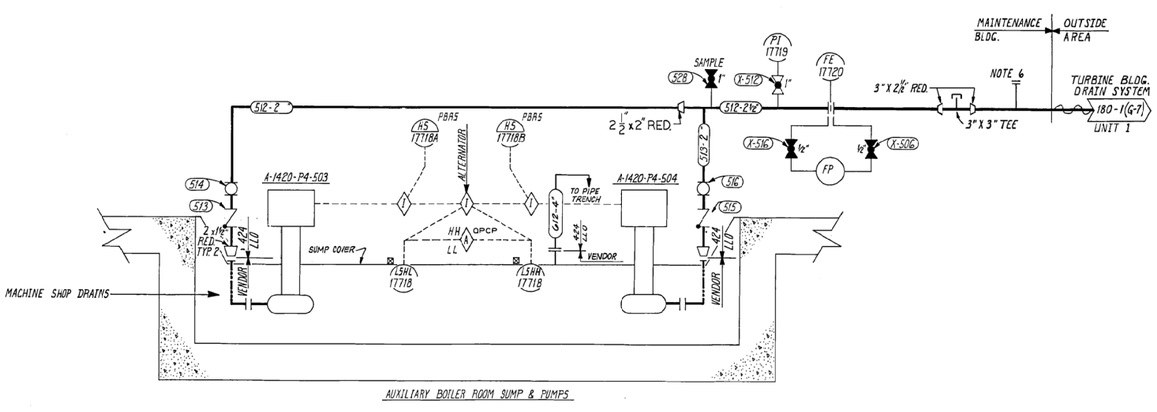
SIZE E 34x44 DWG CATEGORY: CRITICAL

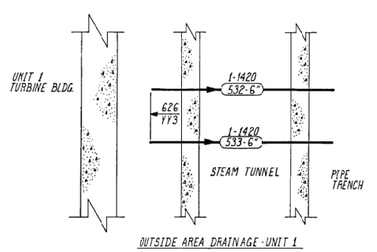












NOTES:

1. ALL VALVE SEQUENCE NUMBERS NOT PRECEDED BY AN "X" HAVE A "U" IDENTIFIER.
2. CONSTRUCTION WATER SUPPLIED FOR TEMP. SERVICE ONLY. PERMANENT SUPPLY WILL BE FROM UTILITY WATER SYSTEM.
3. DISCHARGE SHALL BE TO DITCH DURING CONSTRUCTION. PERMANENT DISCHARGE WILL BE TO WASTE WATER RETENTION BASIN.
4. A FLOW RESTRICTING ORIFICE (D=0.5") SHALL BE UTILIZED DURING CONSTRUCTION.
5. ALL 8" AND SMALLER DIA. PIPE, EXCLUDING DRAIN LINES, THAT IS ABOVE GRADE SHALL BE HEAT TRACED.
6. TEMPORARY CONNECTION TO START-UP POND.

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NO.	VERSIONS	DATE	DR	CHK	APPV
11.0	REVISED PER ABN SNC367406M006, VER. 1.0	07-02-14	PLB	SO	JCG

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

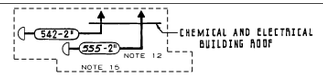
GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

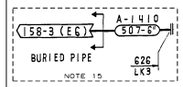
P & I DIAGRAM
WASTE WATER EFFLUENT SYSTEM 11.0
SYSTEM NO. 1420

SCALE: NONE	DRAWING NO.	VER.
JOB NO. 10604	AX4DB152-3	11.0

SIZE E 34x44

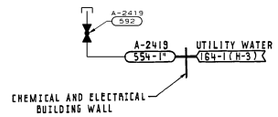
DWG CATEGORY: CRITICAL





GENERAL NOTES

1. DELETED
2. DELETED
3. DELETED
4. DELETED
5. DELETED
6. DELETED
7. DELETED
8. DELETED
9. DELETED
10. DELETED
11. DELETED
12. FIELD TO INSTALL INSECT SCREENS ON ALL VENTS.
13. DELETED
14. DELETED
15. "RETIRED IN PLACE"



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VERSIONS	DATE	DR	CHK	APPV
12.0 REV. PER ABN C999001201M005, VER.2.0	6/24/08	JLO	TSL	JMR

SIZE E 34X44

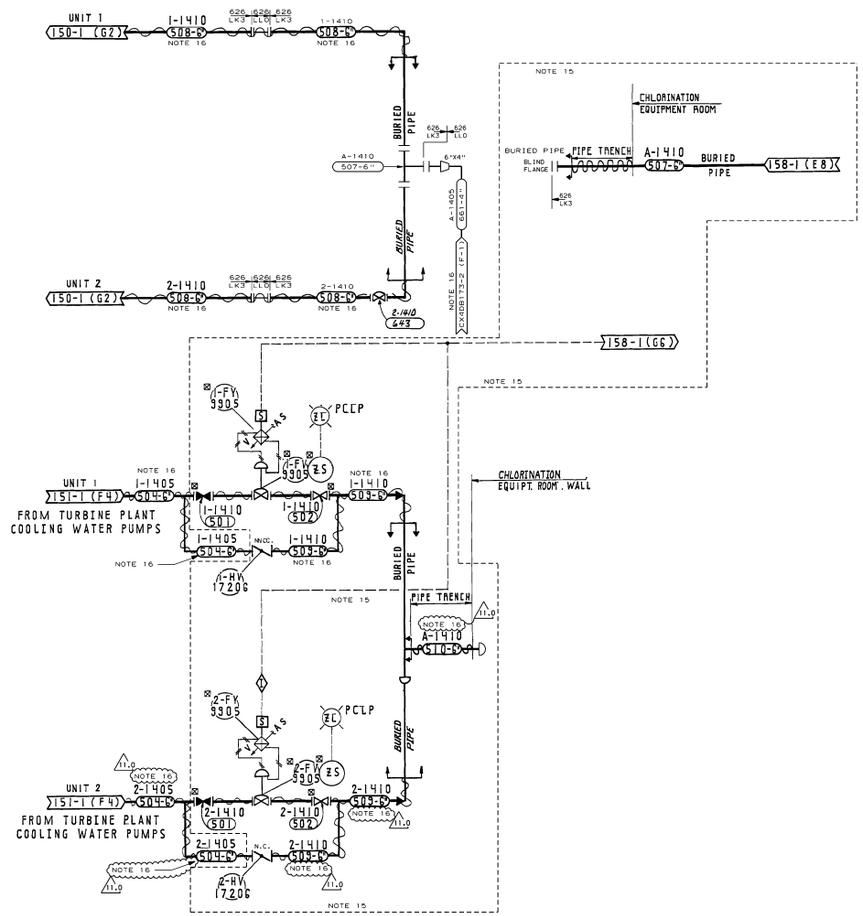
SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
CIRC. WATER CHEMICAL INJ. SYS.
SYSTEM NO. 1410

SCALE: NONE	DRAWING NO.	VER.
JOB NO.10604	AX4DB158-1	12.0

DRAWING CATEGORY: CRITICAL



SEE DWG. NO. AX4DB158-1 FOR NOTES

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NO.	VERSIONS	DATE	DR	CHK	APPV
11.0	REVISED PER ABN C999001201M006, VER. 2.0	6-23-08	JLO	TSL	JMR

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

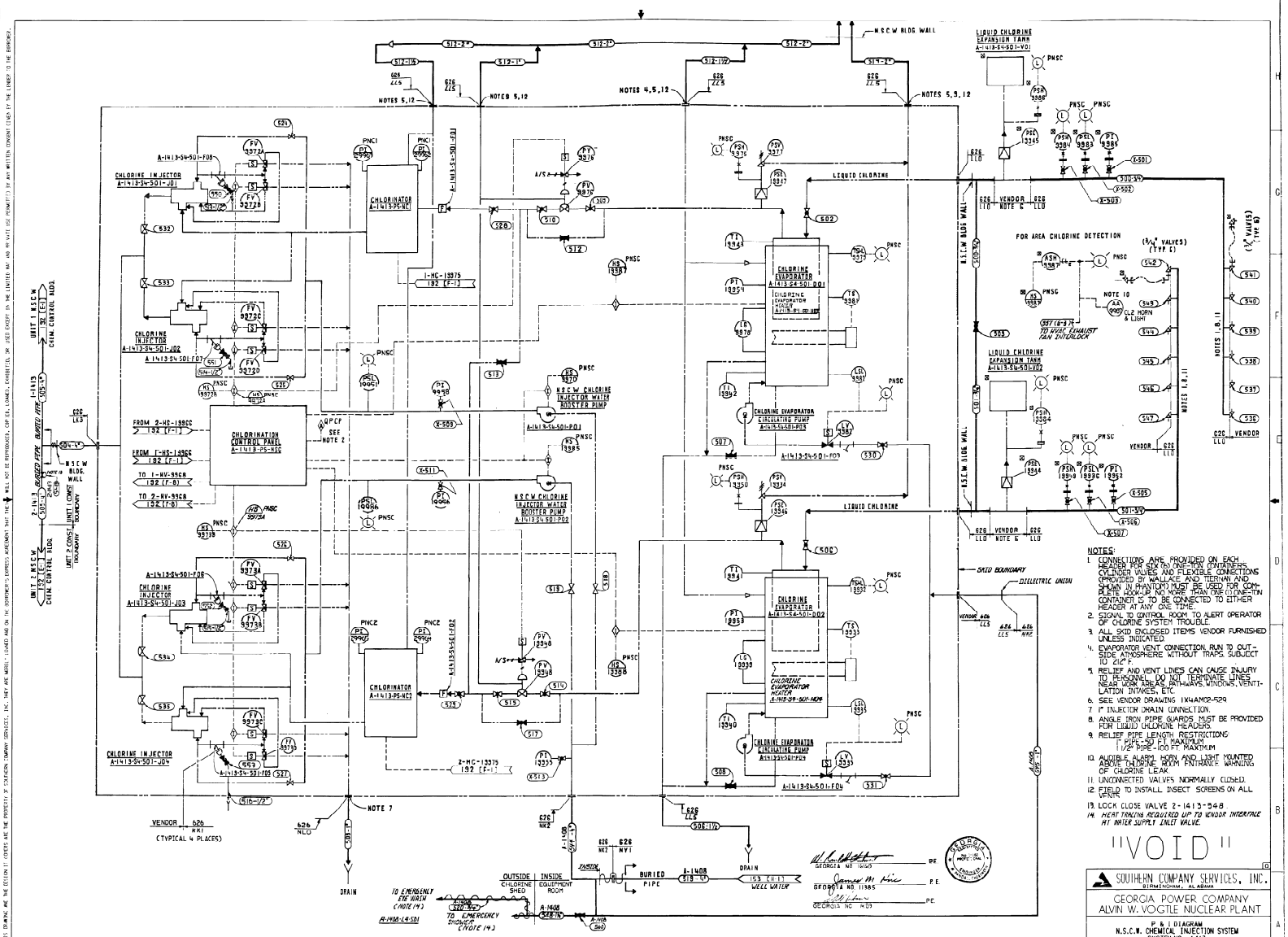
GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
CIRC. WATER CHEMICAL INJ. SYS.
SYSTEM NO. 1410

SCALE: NONE	DRAWING NO.	VER.
JOB NO.10604	AX4DB158-3	11.0

SIZE E 34X44

DWG. CATEGORY: CRITICAL



NOTES:

1. CONNECTIONS ARE PROVIDED ON EACH HEADER FOR SIX (6) ONE-TON CONTAINERS. CYLINDER VALVES AND FLEXIBLE CONNECTIONS PROVIDED BY WALLACE AND TIERNAN AND SHOWN IN PHANTOM) MUST BE USED FOR COMPLETE HOOK-UP. NO MORE THAN ONE CONNECTION CONTAINER IS TO BE CONNECTED TO EITHER HEADER AT ANY ONE TIME.
2. SIGNAL TO CONTROL ROOM TO ALERT OPERATOR OF CHLORINE SYSTEM TROUBLE.
3. ALL SKID ENCLOSED ITEMS VENDOR FURNISHED UNLESS INDICATED.
4. EVAPORATOR VENT CONNECTION. RUN TO OUTSIDE ATMOSPHERE WITHOUT TRAPS, SUBJECT TO 212° F.
5. RELIEF AND VENT LINES CAN CAUSE INJURY TO PERSONNEL. DO NOT TERMINATE LINES NEAR WORK AREAS, PATHWAYS, WINDOWS, VENTILATION INTAKES, ETC.
6. SEE VENDOR DRAWING 1X4AM20-529.
7. 1" INJECTION DRAIN CONNECTION.
8. ANGLE IRON PIPE GUARDS MUST BE PROVIDED FOR LIQUID CHLORINE HEADERS.
9. RELIEF PIPE LENGTH RESTRICTIONS:
   1" PIPE - 50 FT. MAXIMUM.
   1 1/2" PIPE - 100 FT. MAXIMUM.
10. AUDIBLE ALARM, HORN AND LIGHT MOUNTED ABOVE CHLORINE ROOM ENTRANCE WARNING OF CHLORINE LEAK.
11. UNCONNECTED VALVES NORMALLY CLOSED.
12. FIELD TO INSTALL INSECT SCREENS ON ALL VENTS.
13. LOCK CLOSE VALVE 2-1413-548.
14. *HEAT TRACING REQUIRED UP TO VENDOR INTERFACE AT WATER SUPPLY INLET VALVE.*

"VOID"

REVISIONS												
NO.	DESCRIPTION	DATE	DR	CHK	PE	DM	DM	DM	PEM	ENG. MGR.		
0	ISSUED FOR CONSTRUCTION	SEE MICROFILM FOR SIGNATURES								DATE		
NO.	REVISIONS	DATE	DR	CHK	SUPT	EGI	EGS	MPE	PE	DAE		

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
N.S.C.W. CHEMICAL INJECTION SYSTEM
SYSTEM NO. 1413

SCALE: NONE	DRAWING NO.	REV.
JOB NO. 10604	AX4DB165	9

SIZE E 32x44

GEORGIA NO. 16160 PE
GEORGIA NO. 11585 P.E.
GEORGIA NO. 14109 PE.

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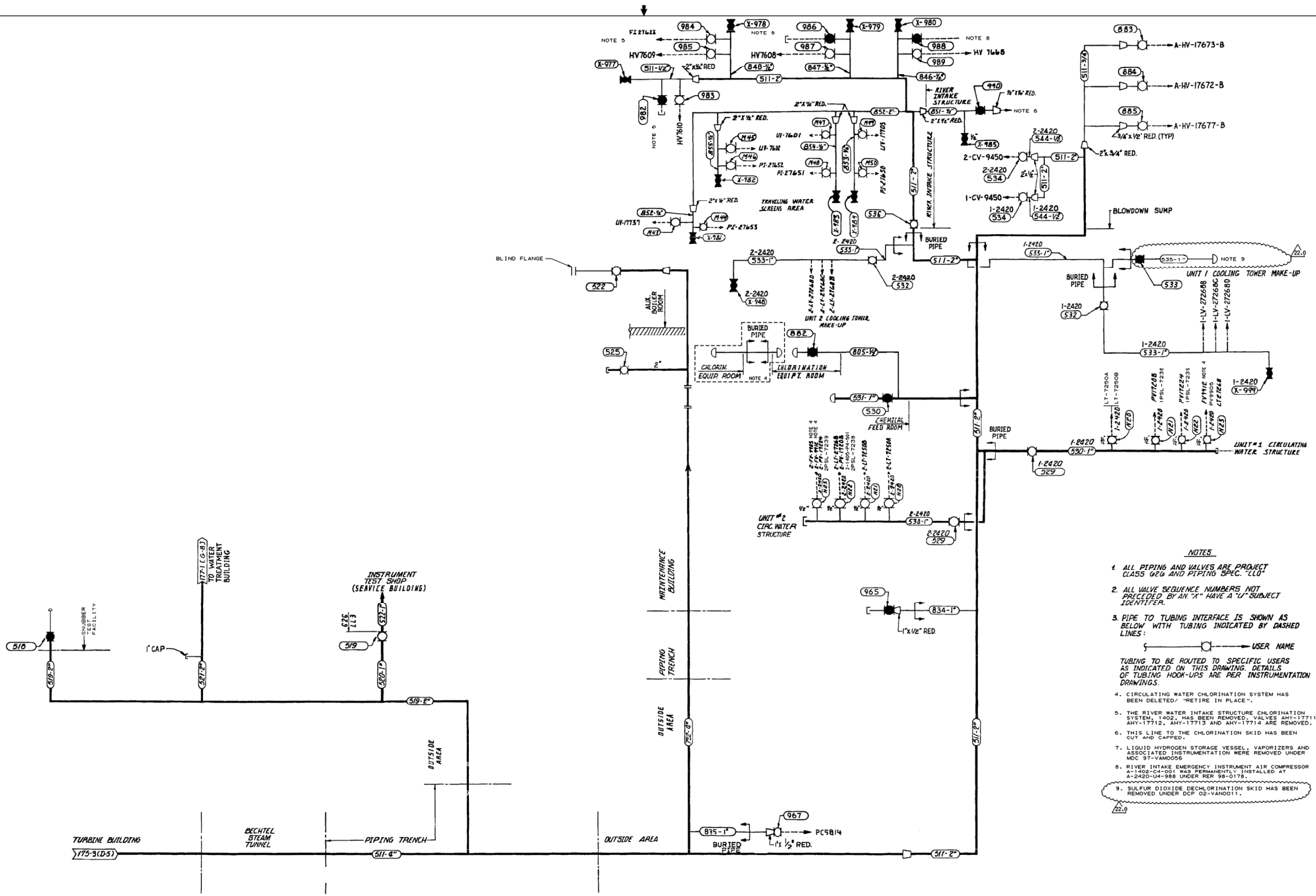

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NO.	VERSION	DATE	DR	CHK	APPV	DTL
22.0	REVISED PER CO29001101M003, VER.1.0	11/17/04	TSL	RBH	DEW	X

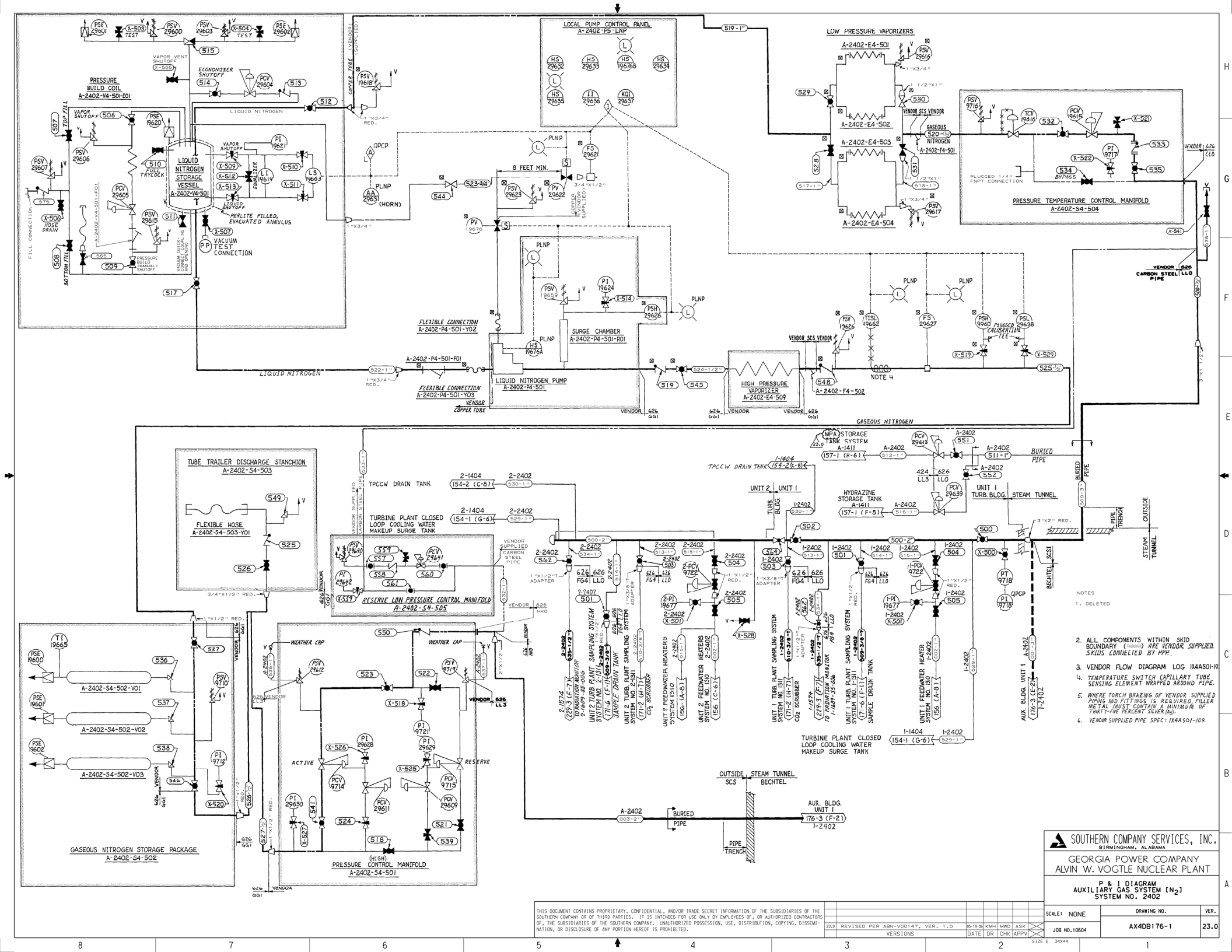
SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
INSTRUMENT AIR
SYSTEM NO. 2420
SHEET 5 OF 5

SCALE: NONE	DRAWING NO.	VER.
JOB NO. 10604	AX4DB175-5	22.0

SIZE E 34X44



SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
AUXILIARY GAS SYSTEM (N_2)
SYSTEM NO. 2402

SCALE: NONE
DRAWING NO. AX4DB176-1
VER. 23.0
JOB NO. 10604
SIZE E 34X44

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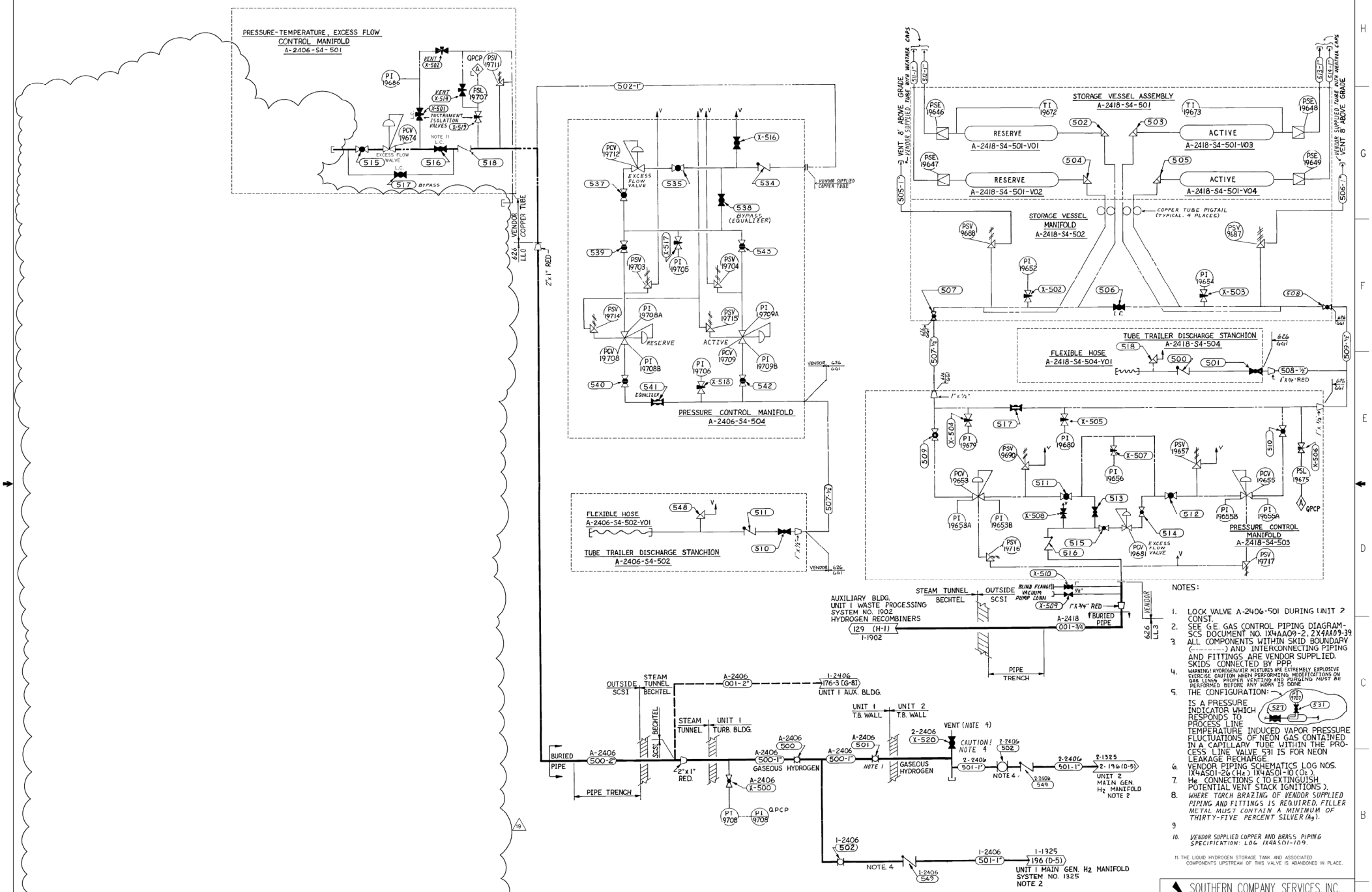
VERSIONS	DATE	DR	CHK	APPV
23.0 REVISED PER ABN-V00147, VER. 1.0	05-19-06	KMH	MWD	ASK



NOTES:

1. LOCK VALVE A-2406-501 DURING UNIT 2 CONST.
2. SEE G.E. GAS CONTROL PIPING DIAGRAM-SCS DOCUMENT NO. 1X4AA09-2, 2X4AA09-39
3. ALL COMPONENTS WITHIN SKID BOUNDARY (--------) AND INTERCONNECTING PIPING AND FITTINGS ARE VENDOR SUPPLIED. SKIDS CONNECTED BY PPP.
4. WARNING! HYDROGEN/AIR MIXTURES ARE EXTREMELY EXPLOSIVE EXERCISE CAUTION WHEN PERFORMING MODIFICATIONS ON GAS LINES. PROPER VENTING AND PURGING MUST BE PERFORMED BEFORE ANY WORK IS DONE
5. THE CONFIGURATION: (PI 9707, 527, 531) IS A PRESSURE INDICATOR WHICH RESPONDS TO PROCESS LINE TEMPERATURE INDUCED VAPOR PRESSURE FLUCTUATIONS OF NEON GAS CONTAINED IN A CAPILLARY TUBE WITHIN THE PROCESS LINE. VALVE 531 IS FOR NEON LEAKAGE RECHARGE.
6. VENDOR PIPING SCHEMATICS LOG NOS. 1X4AS01-26 (H_2) 1X4AS01-10 (O_2).
7. H_2 CONNECTIONS (TO EXTINGUISH POTENTIAL VENT STACK IGNITIONS).
8. *WHERE TORCH BRAZING OF VENDOR SUPPLIED PIPING AND FITTINGS IS REQUIRED, FILLER METAL MUST CONTAIN A MINIMUM OF THIRTY-FIVE PERCENT SILVER (Ag).*
9.
10. *VENDOR SUPPLIED COPPER AND BRASS PIPING SPECIFICATION: L06 1X4AS01-109.*

11. THE LIQUID HYDROGEN STORAGE TANK AND ASSOCIATED COMPONENTS UPSTREAM OF THIS VALVE IS ABANDONED IN PLACE.

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NO.	REVISIONS	DATE	DR	CHK	APPV	DTL
4	INCORP. PER ABN-32708	12-8-97	GS	GLB	WFP	
3	ISSUED FOR CONSTRUCTION	SEE MICROFILM FOR SIGNATURES				

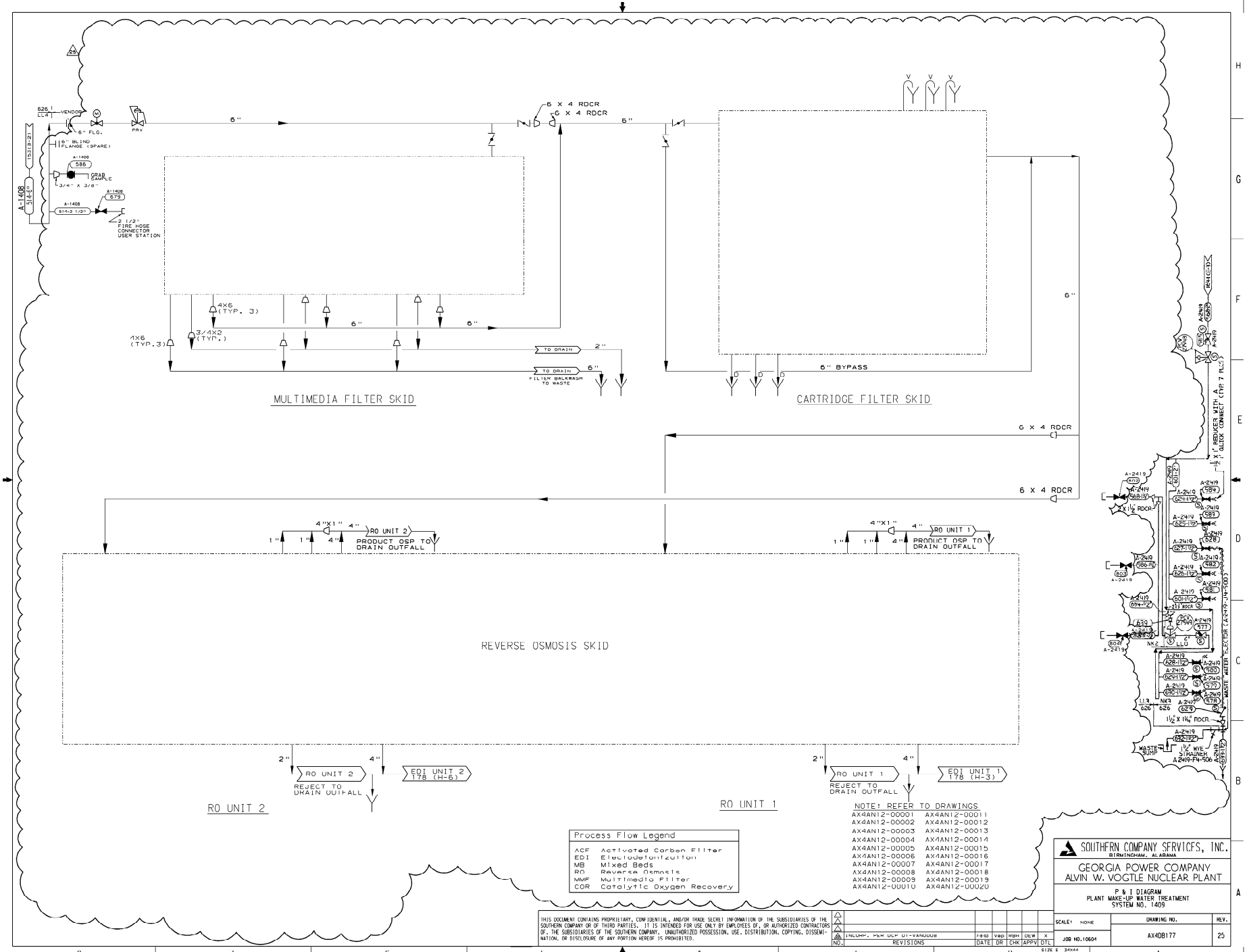
SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
AUXILIARY GAS SYSTEMS (H & O)
SYSTEM NOS. 2406 & 2418

SCALE: NONE
JOB NO. 10604
DRAWING NO. AX4DB176-2
REV. 19

SIZE E 34X44
CAD NAME A4DB1762



Process Flow Legend	
ACF	Activated Carbon Filter
EDI	Electrodeionization
MB	Mixed Beds
RO	Reverse Osmosis
MMF	Multimedia Filter
COR	Catalytic Oxygen Recovery

NOTE: REFER TO DRAWINGS

AX4AN12-00001	AX4AN12-00011
AX4AN12-00002	AX4AN12-00012
AX4AN12-00003	AX4AN12-00013
AX4AN12-00004	AX4AN12-00014
AX4AN12-00005	AX4AN12-00015
AX4AN12-00006	AX4AN12-00016
AX4AN12-00007	AX4AN12-00017
AX4AN12-00008	AX4AN12-00018
AX4AN12-00009	AX4AN12-00019
AX4AN12-00010	AX4AN12-00020

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NO.	REVISIONS	DATE	DR	CHK	APPV	DTL
A	INCORP. PER DCP 01-VENDOR	7-9-03	VBP	RBH	DEW	X

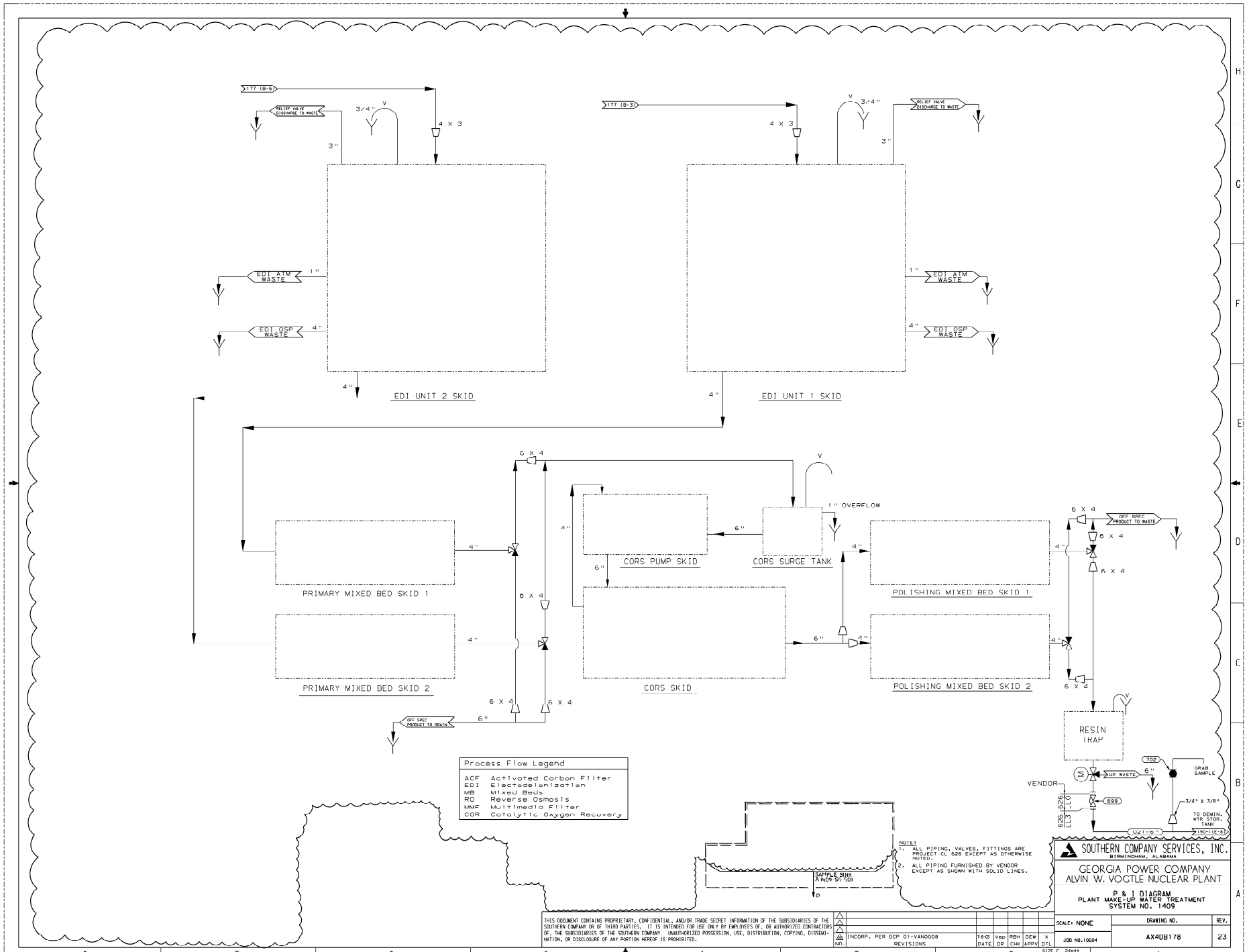
SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
PLANT MAKE-UP WATER TREATMENT
SYSTEM NO. 1409

SCALE: NONE	DRAWING NO.	REV.
JOB NO. 10604	AX4DB177	25

SIZE E 34X44



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NO.	REVISIONS	DATE	DR	CHK	APPV
1	INCORP. PER DCP 01-VAN0008	7-8-03	Vap	RBH	DEW

DTL X

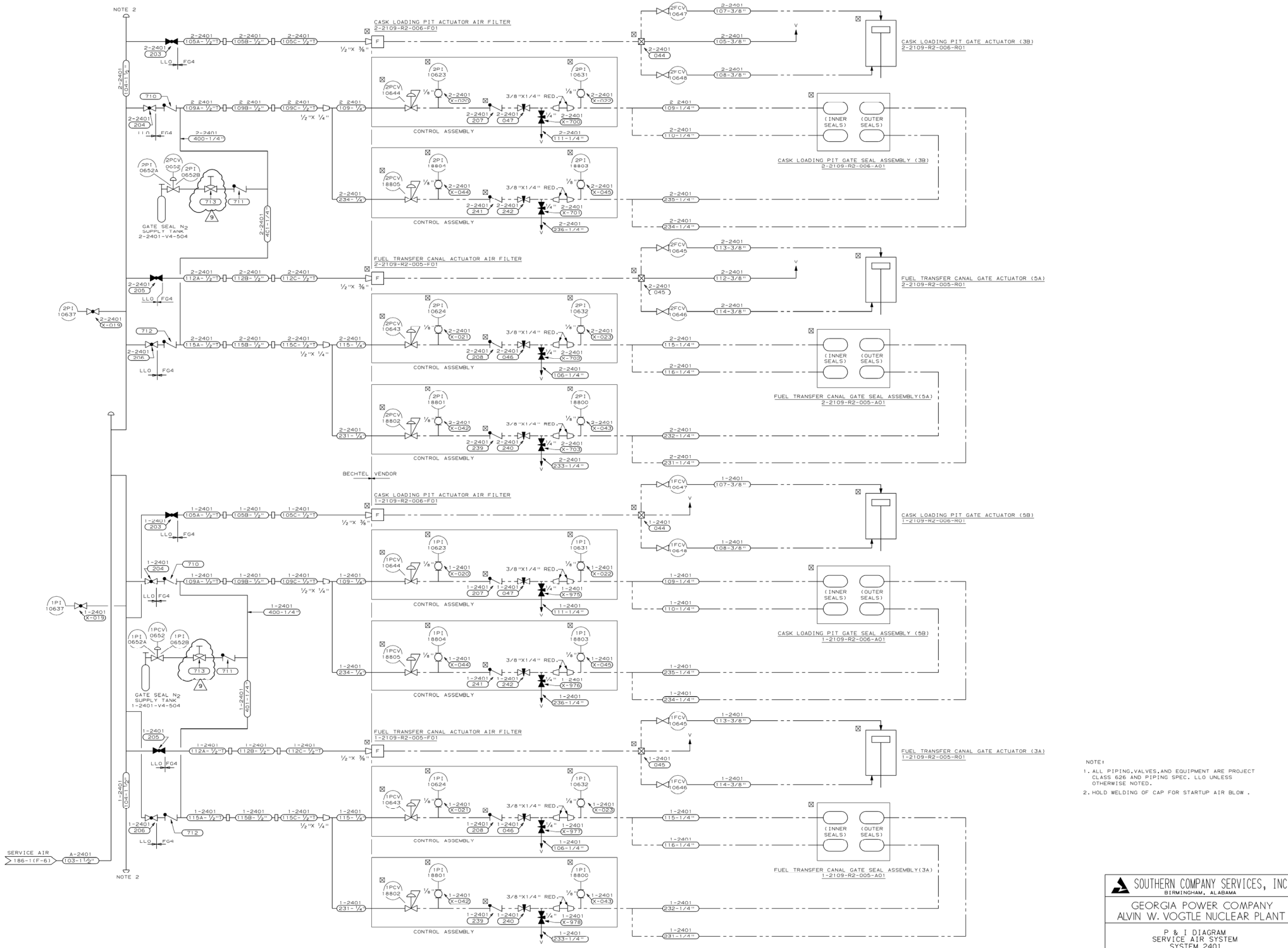
SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
PLANT MAKE-UP WATER TREATMENT
SYSTEM NO. 1409

SCALE: NONE	DRAWING NO.	REV.
JOB NO. 10604	AX4DB178	23

SIZE E 34x44



NOTE:

1. ALL PIPING,VALVES,AND EQUIPMENT ARE PROJECT CLASS 626 AND PIPING SPEC. LLO UNLESS OTHERWISE NOTED.
2. HOLD WELDING OF CAP FOR STARTUP AIR BLOW .

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NO.	REVISIONS	DATE	DR	CHK	APPV	DTL
1	INCORP. PER ABN-37561	07/03/02	ELC	GLB	MWD	X

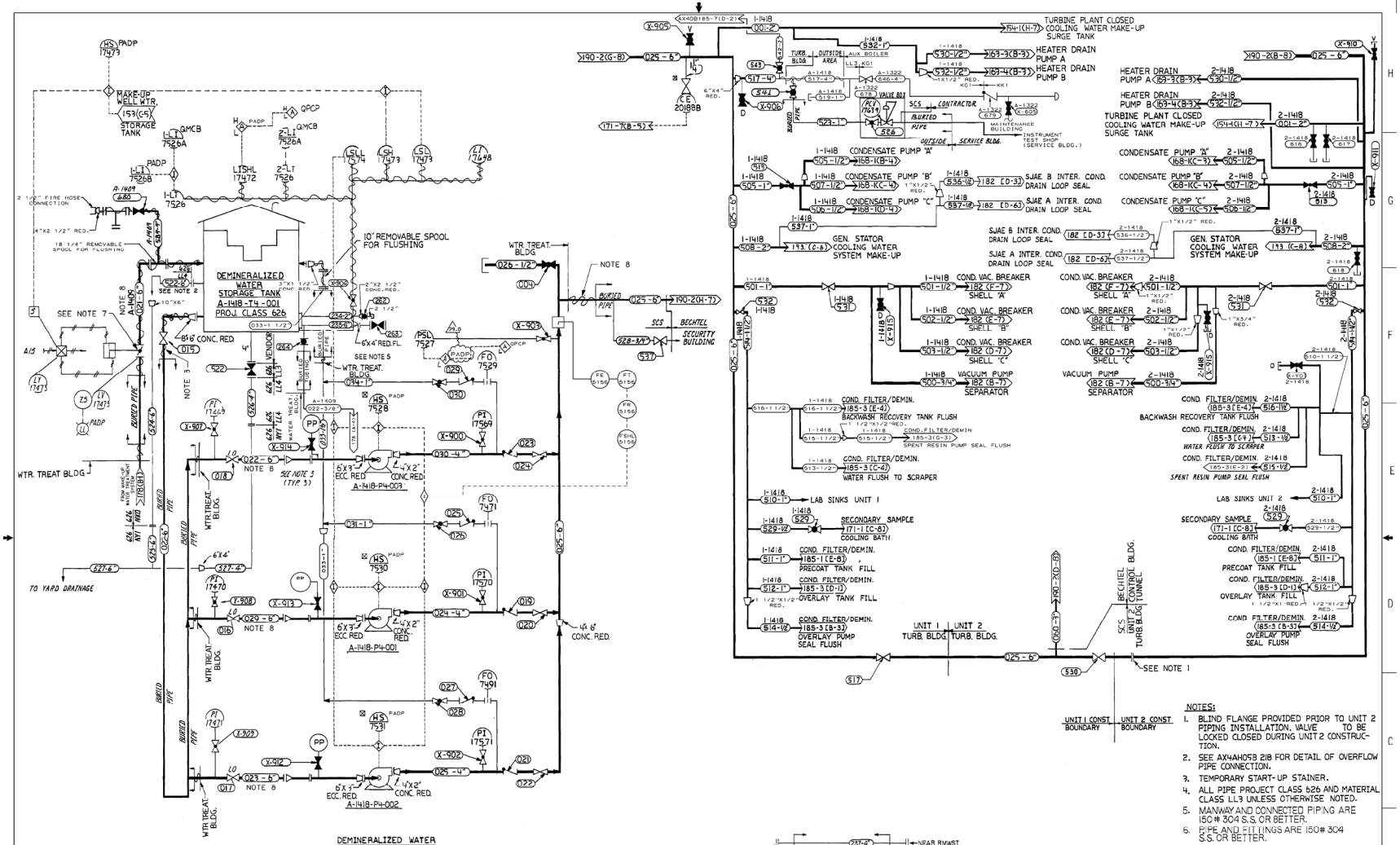
SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
SERVICE AIR SYSTEM
SYSTEM 2401

SCALE: NONE	DRAWING NO.	REV.
JOB NO.10604	AX4DB186-3	9

SIZE E 34X44



NOTES:

1. BLIND FLANGE PROVIDED PRIOR TO UNIT 2 PIPING INSTALLATION. VALVE TO BE LOCKED CLOSED DURING UNIT 2 CONSTRUCTION.
2. SEE AX4AH059 218 FOR DETAIL OF OVERFLOW PIPE CONNECTION.
3. TEMPORARY START-UP STAINER.
4. ALL PIPE PROJECT CLASS 626 AND MATERIAL CLASS LL3 UNLESS OTHERWISE NOTED.
5. MANWAY AND CONNECTED PIPING ARE 150# 304 S.S. OR BETTER.
6. PIPE AND FITTINGS ARE 150# 304 S.S. OR BETTER.
7. AVL-17473 IS NORMALLY IN MANUAL AND OPEN WITH AIR SUPPLY ISOLATED.
0. THE HEAT TRACE ON THE FOLLOWING LINES IS RETIRED IN PLACE OR REMOVED:

   A-1418-022-6", A-1418-023-6", A-1418-025-6", A-1418-029-6" & A-1409-021-6"

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
PLANT DEMINERALIZED WATER
SYSTEM NO. 1418 SHEET 1

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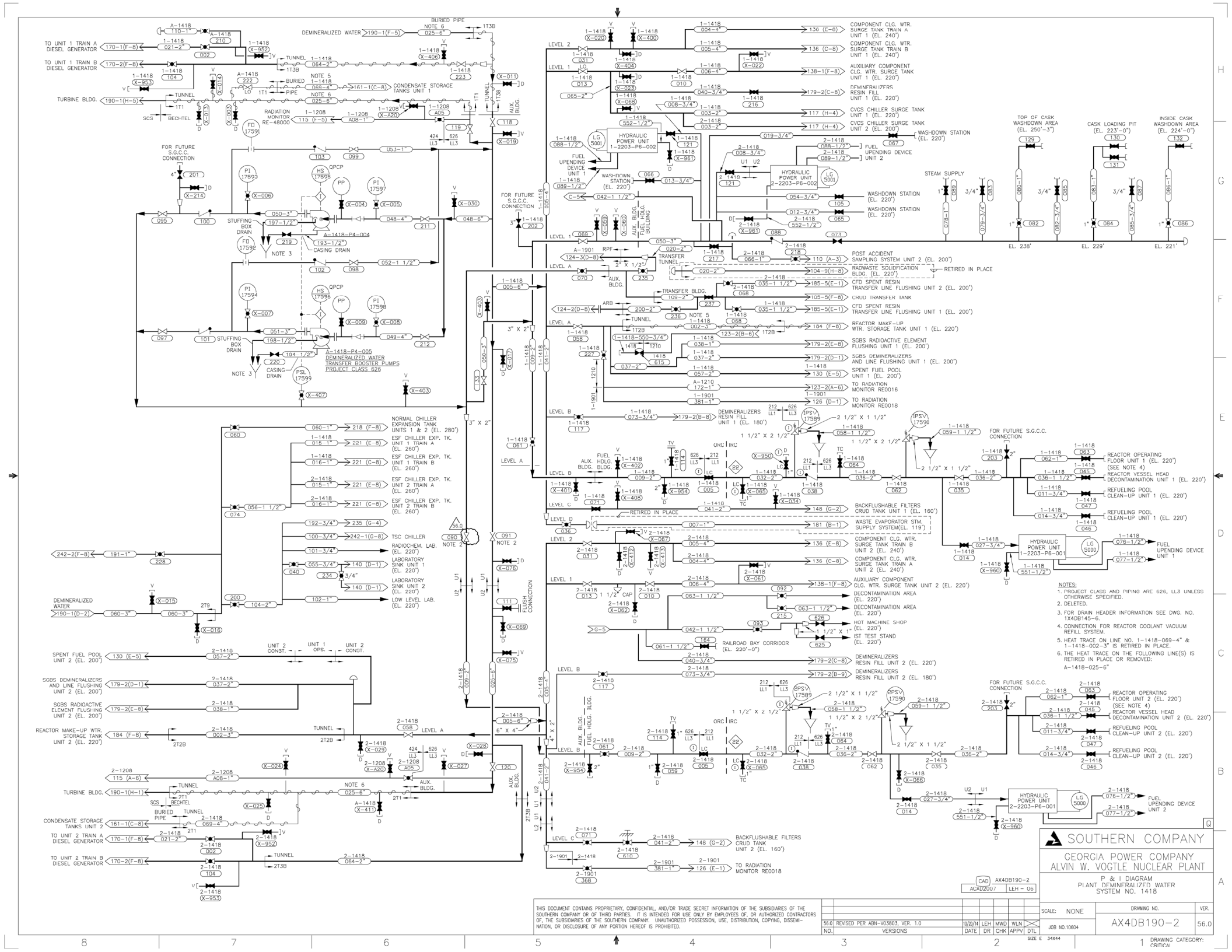
NO.	VERSIONS	DATE	DR	CHK	APPV
29.0	REVISED PER ABN DCP 01-VANDOOR	3-30-01	JLO	RBH	JMR

SCALE: NONE
JOB NO. 10604
DRAWING NO. AX4DB190-1
VER. 29.0
SIZE E 34X44
DRAWING CATEGORY: CRITICAL



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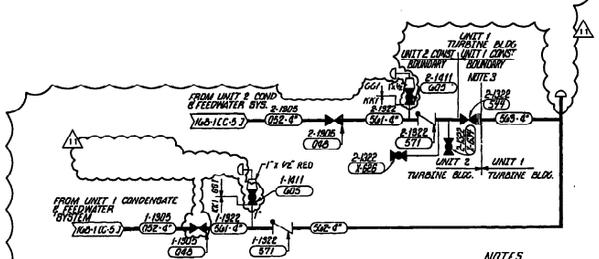
NO.	VERSIONS	DATE	DR	CHK	APPV	DTL
56.0	REVISED PER ABN-V03803, VER. 1.0	10/20/14	LEH	MWD	WLN	

CAD AX4DB190-2
ACAD2007 LEH – 06

SOUTHERN COMPANY
GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT
P & I DIAGRAM
PLANT DEMINERALIZED WATER SYSTEM NO. 1418

SCALE: NONE
JOB NO.10604
SIZE E 34X44
DRAWING NO. AX4DB190-2
VER. 56.0
DRAWING CATEGORY: CRITICAL


UNIT 1 TURBINE BLDG.
UNIT 2 CONST. BOUNDARY
UNIT 1 CONST. BOUNDARY
NOTE 3
2-1411
GO3
2-1332
571
2-1322
571
2-1322
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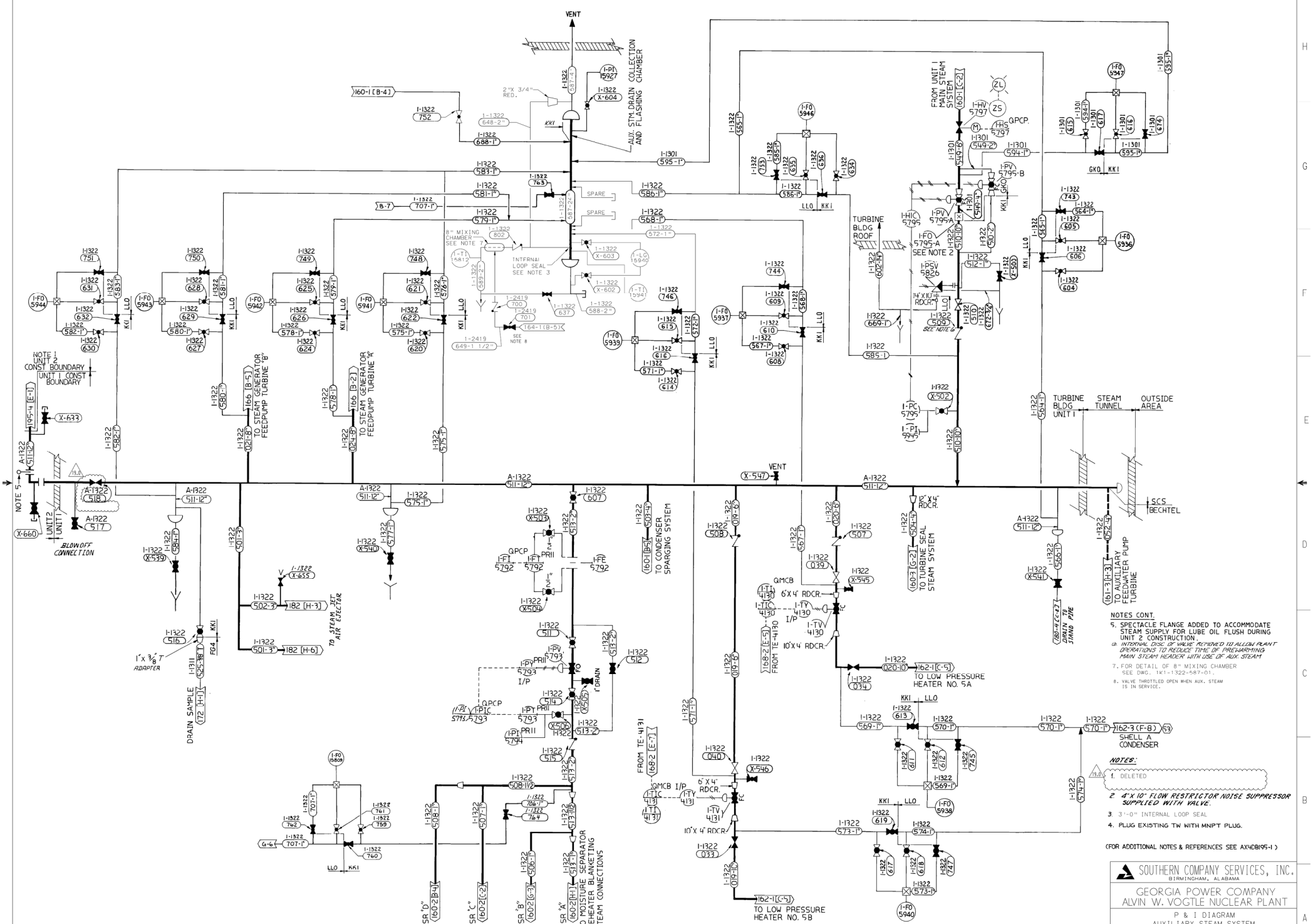
2-1322



NOTES CONT.

5. SPECTACLE FLANGE ADDED TO ACCOMMODATE STEAM SUPPLY FOR LUBE OIL FLUSH DURING UNIT 2 CONSTRUCTION.
   - 5a. INTERNAL DISC OF VALVE REMOVED TO ALLOW PLANT OPERATIONS TO REDUCE TIME OF PREWARMING MAIN STEAM HEADER WITH USE OF AUX. STEAM.
7. FOR DETAIL OF 8" MIXING CHAMBER SEE DWG. 1K1-1322-687-01.
8. VALVE THROTTLED OPEN WHEN AUX. STEAM IS IN SERVICE.

NOTES:

1. DELETED
2. 4"X 10" FLOW RESTRICTOR NOISE SUPPRESSOR SUPPLIED WITH VALVE.
3. 3'-0" INTERNAL LOOP SEAL
4. PLUG EXISTING TW WITH MNPT PLUG.

(FOR ADDITIONAL NOTES & REFERENCES SEE AX4DB195-1)

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NO.	VERSIONS	DATE	DR	CHK	APPV	DTL
19.0	REVISED PER ABN-80470, VERSION 1.0	01/05/04	TSL	MWD	ASK	X

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

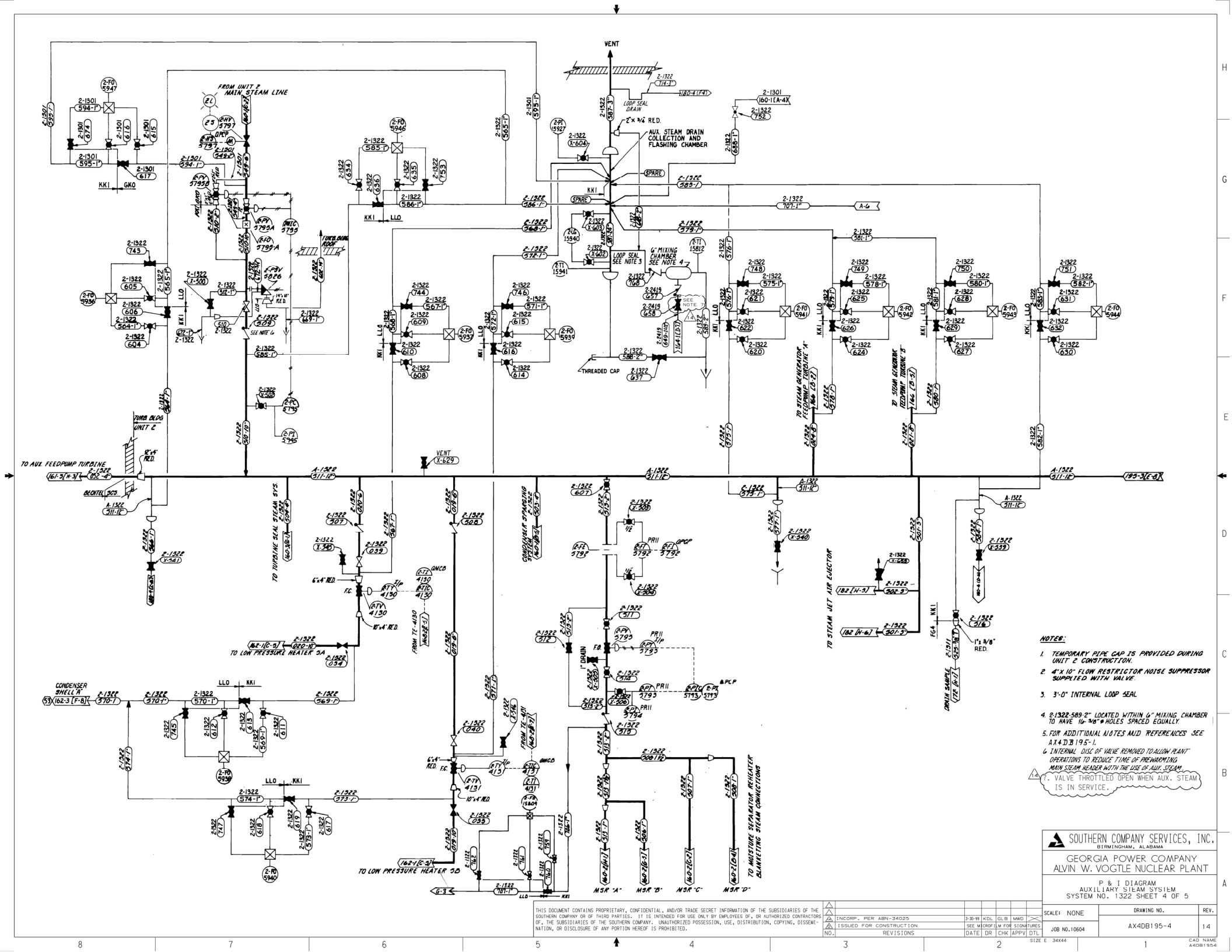
GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
AUXILIARY STEAM SYSTEM
SYSTEM NO. 1322 SHEET 3 OF 5

SCALE: NONE	DRAWING NO.	VER.
JOB NO. 10604	AX4DB195-3	19.0

SIZE E 34x44

CRITICAL DOCUMENT



NOTES:

1. TEMPORARY PIPE CAP IS PROVIDED DURING UNIT 2 CONSTRUCTION.
2. 4"X 10" FLOW RESTRICTOR NOISE SUPPRESSOR SUPPLIED WITH VALVE.
3. 3'-0" INTERNAL LOOP SEAL
4. 2-1322-589-2" LOCATED WITHIN 6" MIXING CHAMBER TO HAVE 16-3/8"ø HOLES SPACED EQUALLY.
5. FOR ADDITIONAL NOTES AND REFERENCES SEE AX4DB195-1.
6. INTERNAL DISC OF VALVE REMOVED TO ALLOW PLANT OPERATIONS TO REDUCE TIME OF PREWARMING MAIN STEAM HEADER WITH THE USE OF AUX. STEAM.
7. VALVE THROTTLED OPEN WHEN AUX. STEAM IS IN SERVICE.

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NO.	REVISIONS	DATE	DR	CHK	APPV	DTL
3	INCORP. PER ABN-34025	3-30-99	KDL	GLB	MWD	
2	ISSUED FOR CONSTRUCTION	SEE MICROFILM FOR SIGNATURES				

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

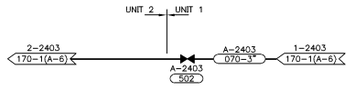
GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

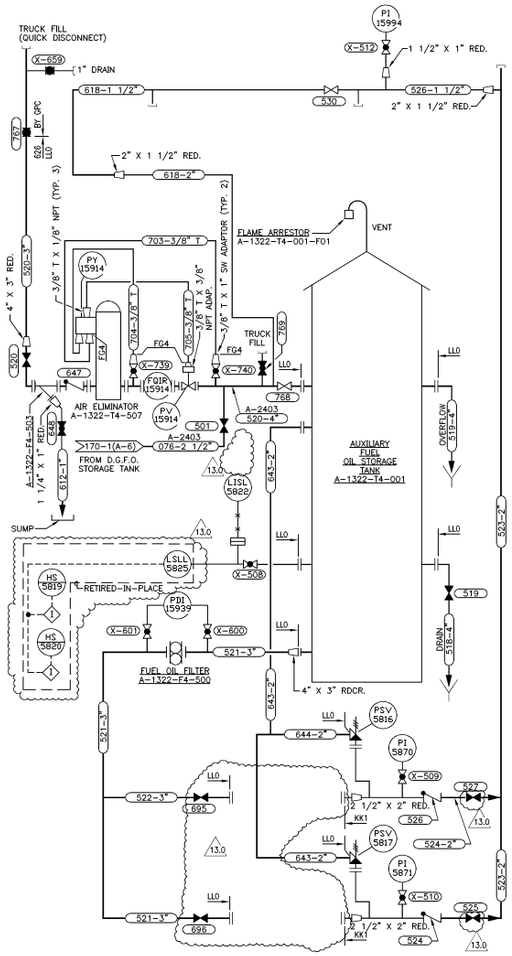
P & I DIAGRAM
AUXILIARY STEAM SYSTEM
SYSTEM NO. 1322 SHEET 4 OF 5

SCALE: NONE	DRAWING NO.	REV.
JOB NO. 10604	AX4DB195-4	14

SIZE E 34X44

CAD NAME AX4DB195-4





NOTES:

1. ALL PIPING AND EQUIPMENT ARE PROJECT CLASS 626 UNLESS OTHERWISE NOTED.
2. DELETED.
3. ALL INSTRUMENT ROOT VALVES SUPPLIED BY SCS ARE 1" UNLESS OTHERWISE NOTED.
4. ALL VALVE SEQUENCE NUMBERS NOT PRECEDED BY AN "X" HAVE A "U" SUBJECT IDENTIFIER.

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NO.	VERSIONS	DATE	DR	CHK	APPV	DTL
13.0	REVISED PER ABN C081591601M002, VER. 1.0	04/21/12	DCP	ELC	JMR	X

CAD AX4DB195-5
2007 DCP- 02

SOUTHERN COMPANY

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
AUXILIARY STEAM SYSTEM
SYSTEM NO. 1322

SCALE: NONE

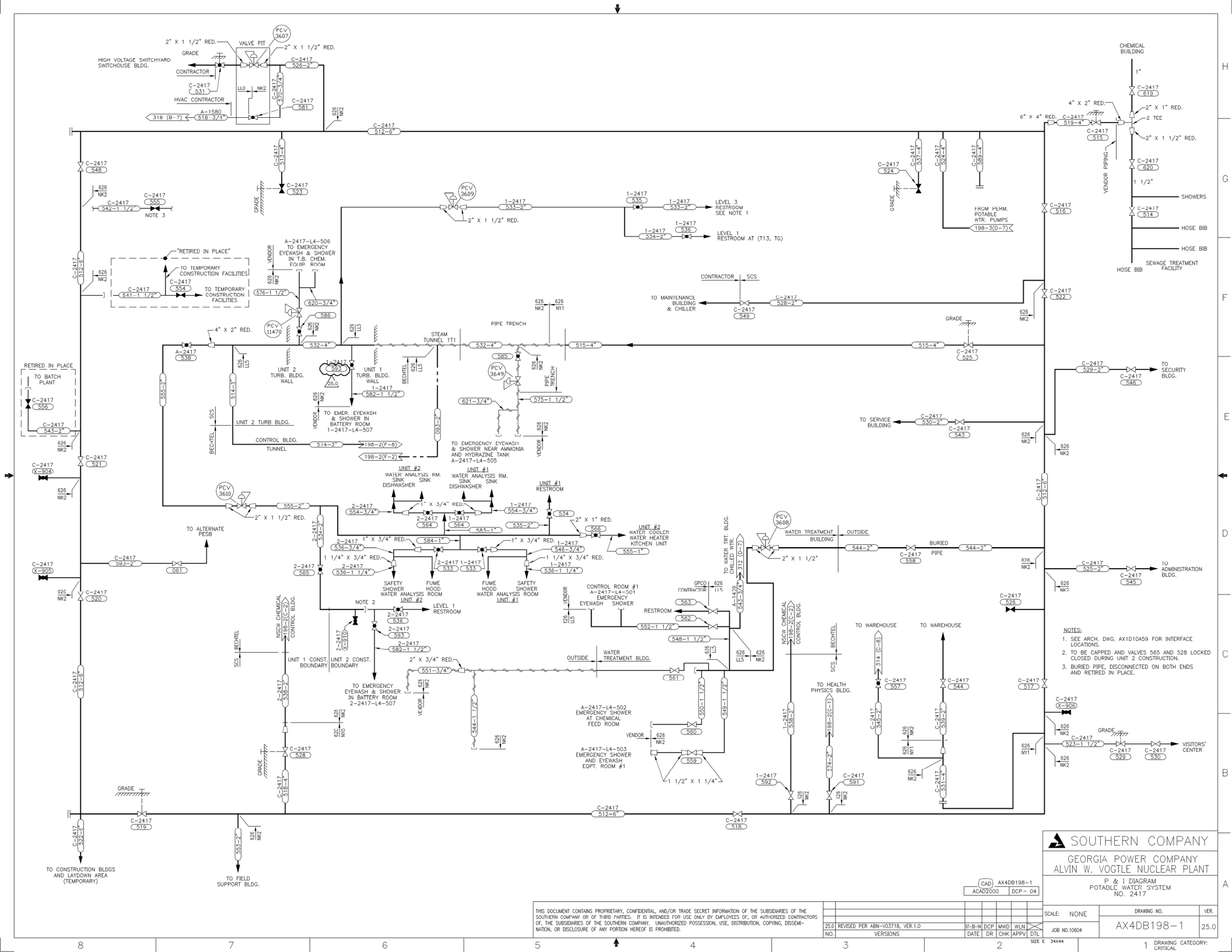
DRAWING NO. AX4DB195-5

VER. 13.0

JOB NO.10604

SIZE E 34X44

DRAWING CATEGORY: CRITICAL



SOUTHERN COMPANY

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
POTABLE WATER SYSTEM
NO. 2417

CAD AX4DB198-1
ACAD2000 DCP- 04

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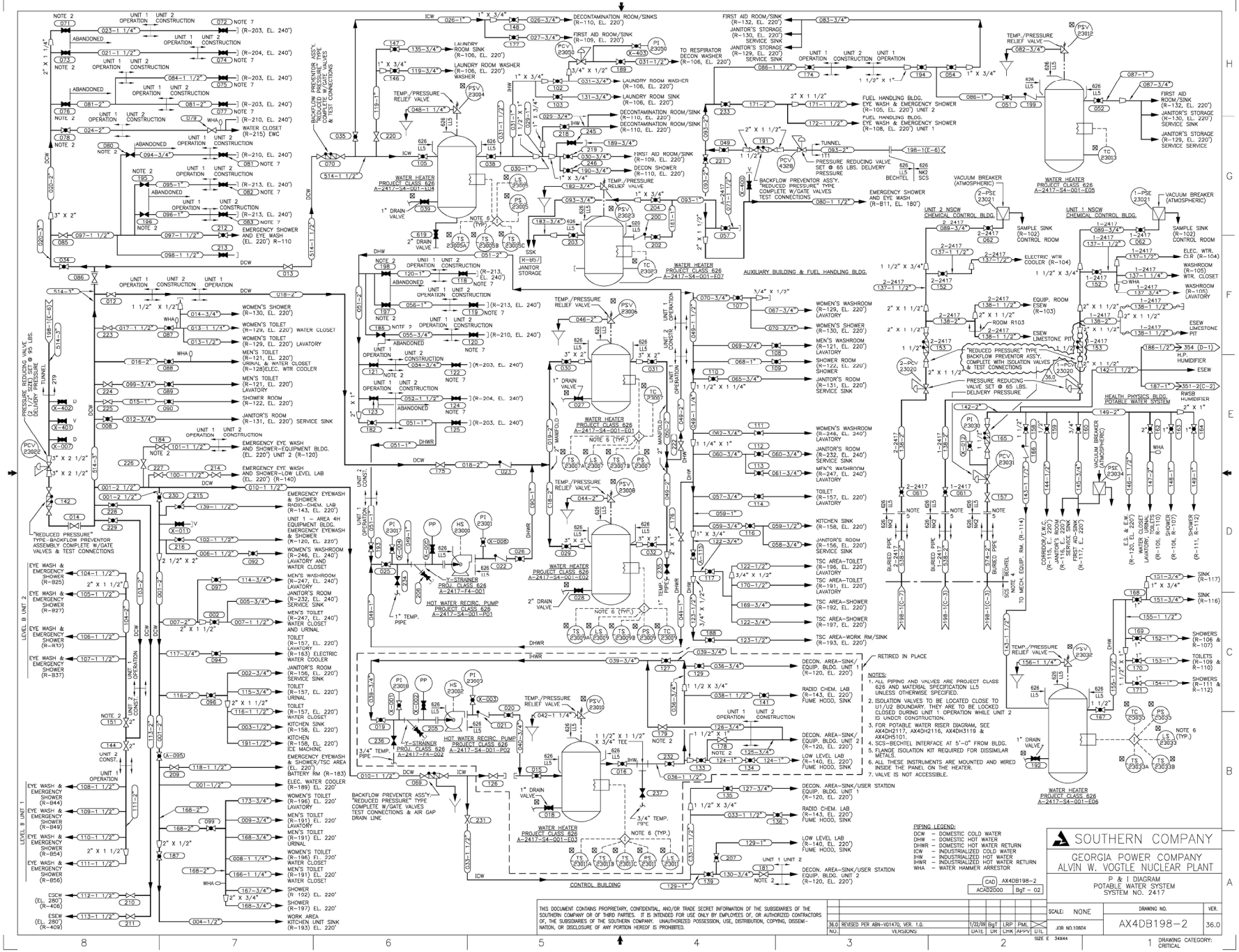
NO.	VERSIONS	DATE	DR	CHK	APPV	DTL
25.0	REVISED PER ABN-V03718, VER.1.0	07-30-14	DCP	MWD	WLN	

SCALE: NONE
JOB NO.10604
DRAWING NO. AX4DB198-1
VER. 25.0
SIZE E 34X44
DRAWING CATEGORY: CRITICAL


DECONTAMINATION ROOM/SINKS (R-110, EL. 220')
FIRST AID ROOM/SINK (R-109, EL. 220')
LAUNDRY ROOM SINK (R-106, EL. 220')
LAUNDRY ROOM WASHER (R-106, EL. 220') WASHER
TO RESPIRATOR DECON WASHER (R-106, EL. 220')
FIRST AID ROOM/SINK (R-132, EL. 220')
JANITOR'S STORAGE SERVICE SINK (R-130, EL. 220')
JANITOR'S STORAGE SERVICE SINK (R-129, EL. 220')
FUEL HANDLING BLDG. EYE WASH & EMERGENCY SHOWER (R-105, EL. 220') UNIT 2
FUEL HANDLING BLDG. EYE WASH & EMERGENCY SHOWER (R-108, EL. 220') UNIT 1
FIRST AID ROOM/SINK (R-132, EL. 220')
JANITOR'S STORAGE (R-130, EL. 220') SERVICE SINK
JANITOR'S STORAGE (R-129, EL. 220') SERVICE SINK
WATER HEATER PROJECT CLASS 626 A-2417-S4-001-L04
WATER HEATER PROJECT CLASS 626 A-2417-S4-001-E07
WATER HEATER PROJECT CLASS 626 A-2417-S4-001-E05
WATER HEATER PROJECT CLASS 626 A-2417-S4-001-E02
WATER HEATER PROJECT CLASS 626 A-2417-S4-001-E01
WATER HEATER PROJECT CLASS 626 A-2417-S4-001-E06
HOT WATER RECIRC. PUMP PROJECT CLASS 626 A-2417-S4-001-P01
HOT WATER RECIRC. PUMP PROJECT CLASS 626 A-2417-S4-001-P02
Y-STRAINER PROJ. CLASS 626 A-2417-F4-001
Y-STRAINER PROJ. CLASS 626 A-2417-F4-002
AUXILIARY BUILDING & FUEL HANDLING BLDG.
UNIT 2 NSCW CHEMICAL CONTROL BLDG.
UNIT 1 NSCW CHEMICAL CONTROL BLDG.
HEALTH PHYSICS BLDG. POTABLE WATER SYSTEM
CONTROL BUILDING
BACKFLOW PREVENTER ASS'Y. "REDUCED PRESSURE" TYPE COMPLETE W/GATE VALVES TEST CONNECTIONS & AIR GAP DRAIN LINE
"REDUCED PRESSURE" TYPE-BACKFLOW PREVENTOR ASSEMBLY COMPLETE W/GATE VALVES & TEST CONNECTIONS
PRESSURE REDUCING VALVE SET @ 85 LBS. DELIVERY PRESSURE - TUNNEL
PRESSURE REDUCING VALVE SET @ 65 LBS. DELIVERY PRESSURE
BECHTEL
TO MECH. EQUIP. RM. (R-114)
RETIRED IN PLACE
NOTES:
1. ALL PIPING AND VALVES ARE PROJECT CLASS 626 AND MATERIAL SPECIFICATION LL5 UNLESS OTHERWISE SPECIFIED.
2. ISOLATION VALVES TO BE LOCATED CLOSE TO U1/U2 BOUNDARY. THEY ARE TO BE LOCKED CLOSED DURING UNIT 1 OPERATION WHILE UNIT 2 IS UNDER CONSTRUCTION.
3. FOR POTABLE WATER RISER DIAGRAM, SEE AX4DH2117, AX4DH2116, AX4DH3119 & AX4DH5101.
4. SCS-BECHTEL INTERFACE AT 5'-0" FROM BLDG.
5. FLANGE ISOLATION KIT REQUIRED FOR DISSIMILAR METALS.
6. ALL THESE INSTRUMENTS ARE MOUNTED AND WIRED INSIDE THE PANEL ON THE HEATER.
7. VALVE IS NOT ACCESSIBLE.
PIPING LEGEND:
DCW – DOMESTIC COLD WATER
DHW – DOMESTIC HOT WATER
DHWR – DOMESTIC HOT WATER RETURN
ICW – INDUSTRIALIZED COLD WATER
IHW – INDUSTRIALIZED HOT WATER
IHWR – INDUSTRIALIZED HOT WATER RETURN
WHA – WATER HAMMER ARRESTOR
SOUTHERN COMPANY
GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT
P & I DIAGRAM
POTABLE WATER SYSTEM
SYSTEM NO. 2417
CAD AX4DB198-2
ACAD2000 BgT - 02
SCALE: NONE
DRAWING NO. AX4DB198-2
VER. 36.0
JOB NO.10604
SIZE E 34X44
DRAWING CATEGORY: CRITICAL
36.0 REVISED PER ABN-W1470, VER. 1.0. 1/22/09 BgT LRP PML
NO. REVISIONS DATE DN CHK APPV UTL
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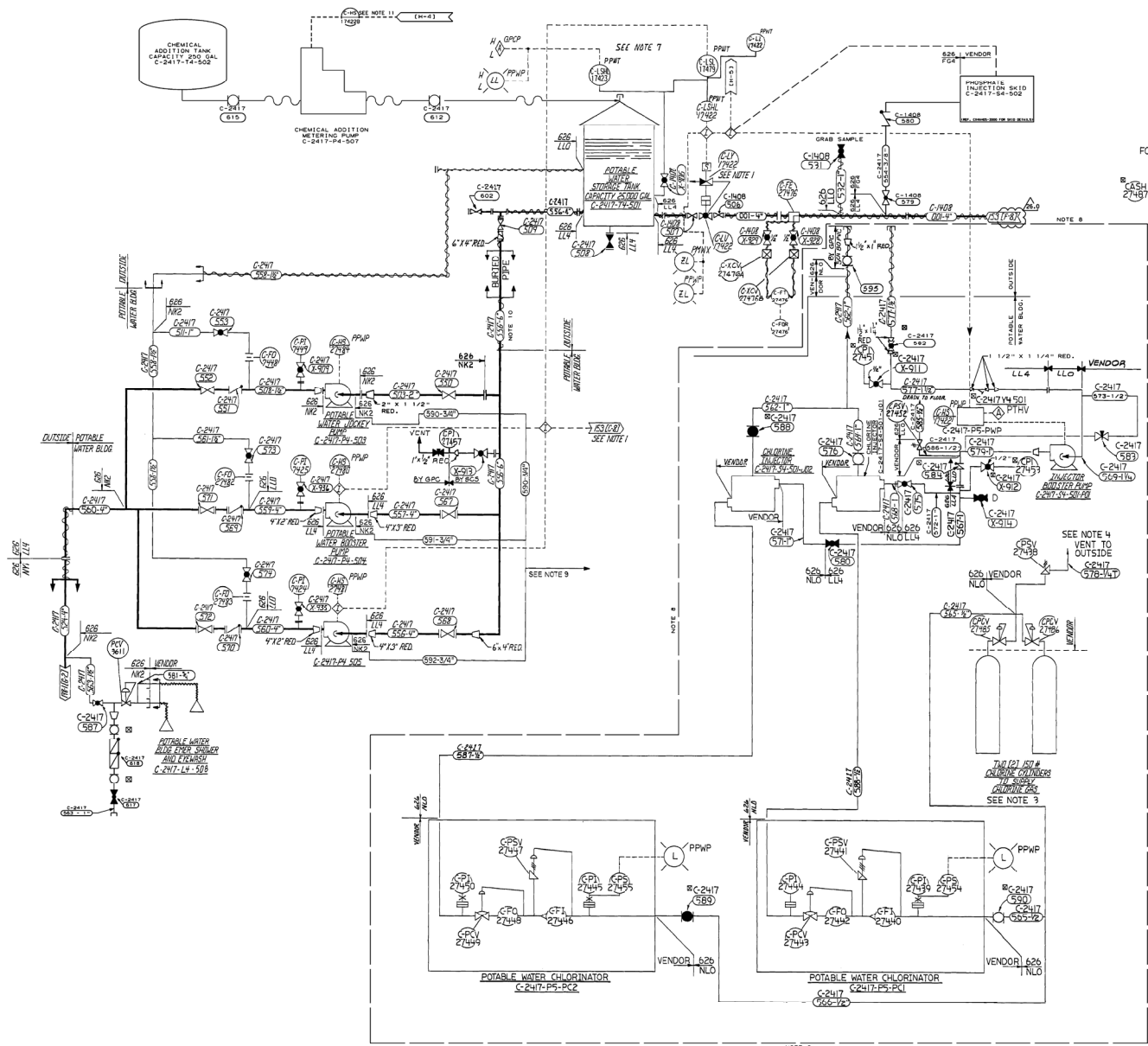

NOTES:

1. NOTE DELETED
2. AUDIBLE ALARM HORN AND LIGHT MOUNTED ABOVE CHLORINE ROOM ENTRANCE WARNING OF CHLORINE LEAK
3. GPCO TO SUPPLY CHLORINE CYLINDERS
4. VENT LINE MUST TERMINATE IN AN AREA WHERE GAS FUMES CANNOT CAUSE INJURY TO PERSONNEL.
5. SOLUTION LINE IS SCH.80 NLO.
6. NOTE DELETED
7. PANEL PPWT LOCATED IN A HEATED CABINET ON THE POTABLE WATER STORAGE TANK THEREFORE HEAT TRACING OF INSTRUMENTS NOT REQUIRED
8. GASEOUS CHLORINATION EQUIPMENT TO BE ABANDONED IN PLACE. SYSTEM DENOTED BY DASH (— — — —) LINE. PANEL C-2417-P5-PWP WILL REMAIN OPERABLE.
9. PUMP BASE DRAIN SHALL HAVE A DOWNWARD SLOPE AND TERMINATE OUTSIDE THE POTABLE WATER BLDG. SCHEDULE 40 PIPE MAY BE USED FOR LINE NOS. C-2417-L4-590, C-2417-L4-591 AND C-2417-L4-592.
10. THE HEAT TRACE ON THE FOLLOWING LINE(S) IS RETIRED IN PLACE OR REMOVED:
    C-2417-556-6"
11. CHS17228 IS LOCATED IN THE POTABLE WATER SODIUM HYPOCHLORITE BLDG., PANEL C2417P5PWH.

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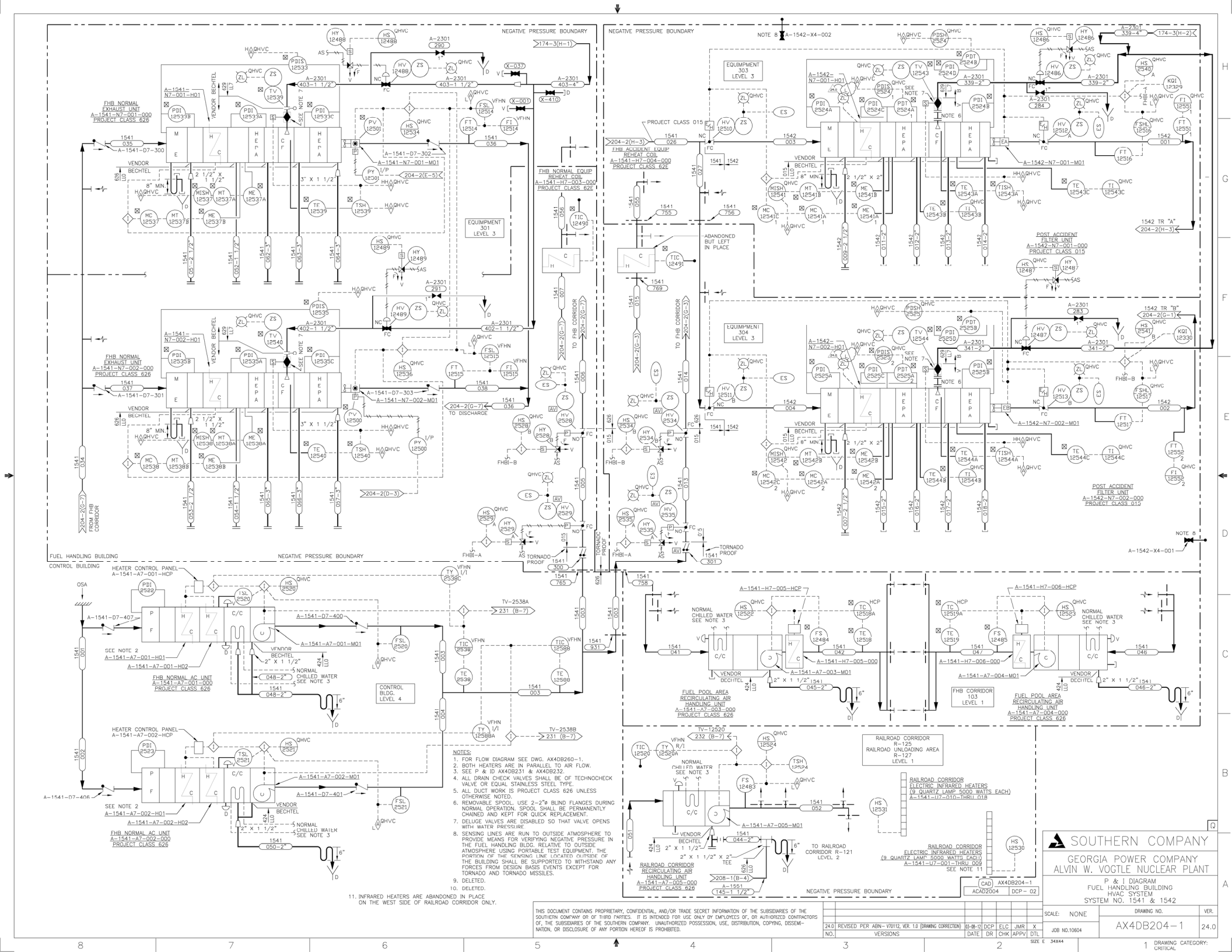
NO.	VERSIONS	DATE	DR	CHK	APPV
26.0	REVISED PER ABN-V02537, VER.1.0	3-19-11	BGT	LRP	AAN

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
POTABLE WATER SYSTEM
NO. 2417

SCALE: NONE
JOB NO. 10604
DRAWING NO. AX4DB198-3
VER. 26.0
SIZE E 34x44
DRAWING CATEGORY: CRITICAL



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NO.	VERSIONS	DATE	DR	CHK	APPV	DTL
24.0	REVISED PER ABN- V70112, VER. 1.0 (DRAWING CORRECTION)	03-08-12	DCP	ELC	JMR	X

SOUTHERN COMPANY

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

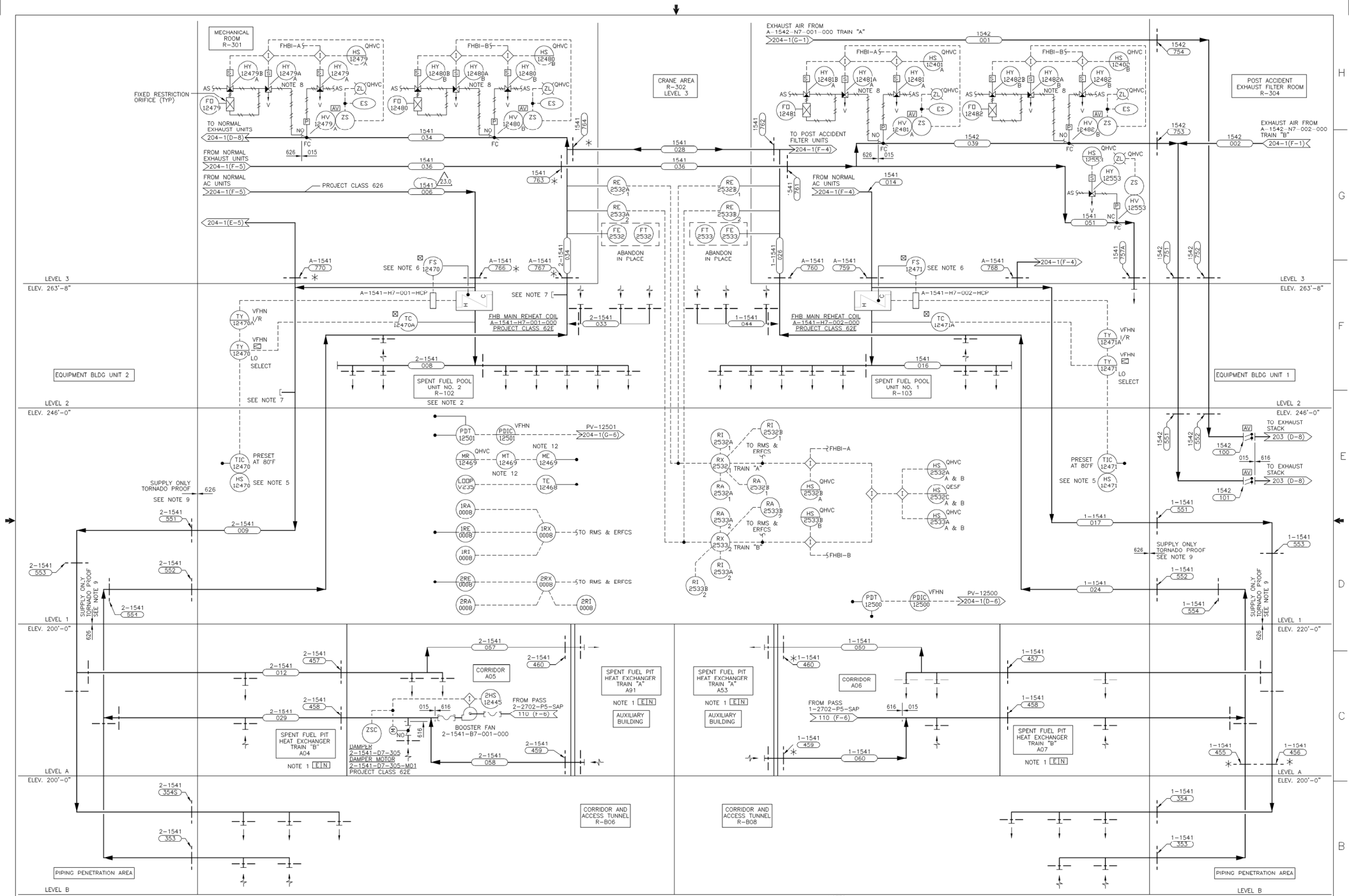
P & I DIAGRAM
FUEL HANDLING BUILDING
HVAC SYSTEM
SYSTEM NO. 1541 & 1542

SCALE: NONE	DRAWING NO.	VER.
JOB NO.10604	AX4DB204-1	24.0

CAD AX4DB204-1
ACAD2004 DCP- 02

SIZE E 34X44

DRAWING CATEGORY: CRITICAL



NOTES: (CONTINUED)

7. FORMER DUCT TO AND FROM HOT MACHINE SHOP CAPPED PER DCP-88-VIE0099.
8. CONTROL AIR TO INFLATABLE SEAL ON BUBBLE-TIGHT DAMPER.
9. DUCTWORK DESIGNATED AS TORNADO PROOF SHALL BE CONSIDERED AS PROJECT CLASS 015. THIS DUCTWORK SHALL BE CONSTRUCTED IN ACCORDANCE WITH THE REQUIREMENTS OF AX2D67W001.
10. ✳ DERATED FIRE DAMPER.
11. DELETED.
12. AMT-12469 & AME-12469 HAVE BEEN "ABANDONED IN PLACE".

NOTES:

1. SPACE IS ALSO COOLED BY SYSTEM 1555, SEE P & ID 1X4DB228.
   E EMERGENCY ESF COIL
   N NORMAL COOL
2. EQUIPMENT SHOWN ON P & ID AX4DB204-1.
3. ALL DUCTWORK IS PROJECT CLASS 015 UNLESS OTHERWISE NOTED.
4. ALL DUCTWORK IS SYSTEM 1541 UNLESS OTHERWISE NOTED.
5. HS-12470 AND HS-12471 ARE NORMALLY IN "OFF" POSITION AND MANUALLY SWITCHED TO "ON" DURING REFUELING OPERATIONS ONLY.
6. INSTRUMENT IS FACTORY WIRED. WIRING CONNECTION IS SHOWN ONLY TO CLARIFY SYSTEM CONTROL INTERLOCK.

CAD AX4DB204-2
ACAD2004 PVK - 02

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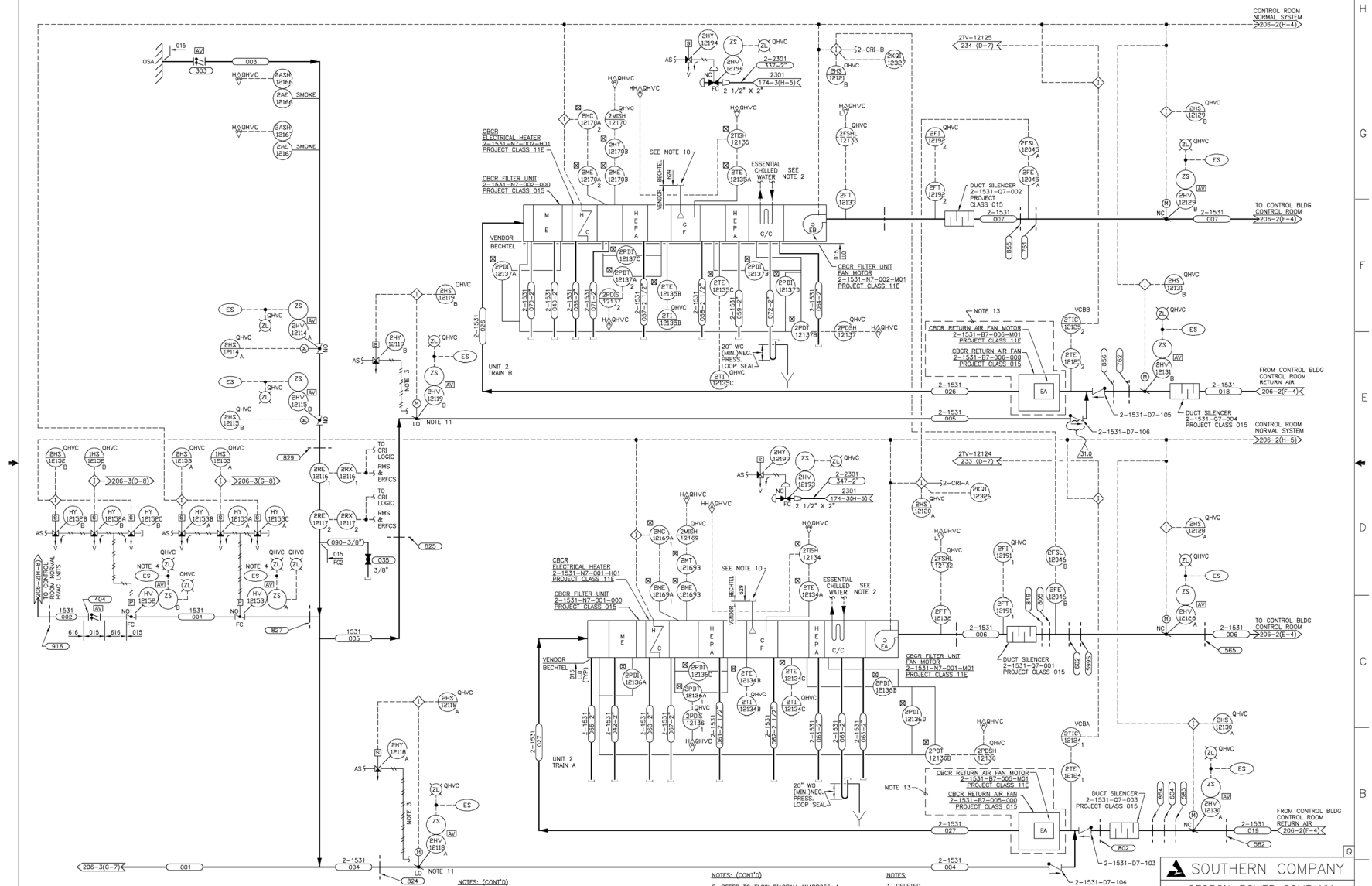
NO.	VERSIONS	DATE	DR	CHK	APPV	DTL
23.0	REVISED PER ABN-V03440, VER. 1.0	10/29/13	PVK	CJK	RKF	X

SOUTHERN COMPANY

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
FUEL HANDLING BUILDING
HVAC SYSTEM
SYSTEM NO. 1541 & 1542

SCALE: NONE
JOB NO.10604
DRAWING NO. AX4DB204-2
VER. 23.0
SIZE E 34X44
DRAWING CATEGORY: CRITICAL



NOTES: (CONT'D)

11. PROVIDE FOR POWER LOCKOUT AND RESTORATION FROM CONTROL ROOM.
12. DELETED
13. THE RETURN AIR FANS 2-1531-B7-005 & 2-1531-B7-006 ARE RETIRED IN PLACE WITH THE BLADES REMOVED.

NOTES: (CONT'D)

5. REFER TO FLOW DIAGRAM AX4DB255-1.
6. DELETED
7. DELETED
8. DELETED
9. DELETED
10. FIRE HOSE CONNECTION (FEMALE NPT X 1 1/2" FEMALE NHT COUPLING WITH 1 1/2" NHT PLUG.)

NOTES:

1. DELETED
2. FOR ESSENTIAL CHILLED WATER TRAIN "A" SEE DWG 2X4DB233. FOR TRAIN "B" SEE DWG 2X4DB234.
3. CONTROL AIR TO INFLATABLE SEAL ON BUBBLETIGHT DAMPER.
4. POSITION INDICATION OF THESE COMMON VALVES IS INPUT TO ONLY THE UNIT 1 COMPUTER.

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NO.		DATE	DN	CHK	APPV	DTL
31.0	REVISED PER ABN-V00745, VER.1.0	06-13-07	BgT	LRP	WLN	

CAD AX4DB206-1
ACAD2002 BgT - 02

SOUTHERN COMPANY

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
CONTROL BUILDING CONTROL ROOM HVAC SYSTEM LEVEL 1 SYSTEM NO. 15.31

SCALE: NONE

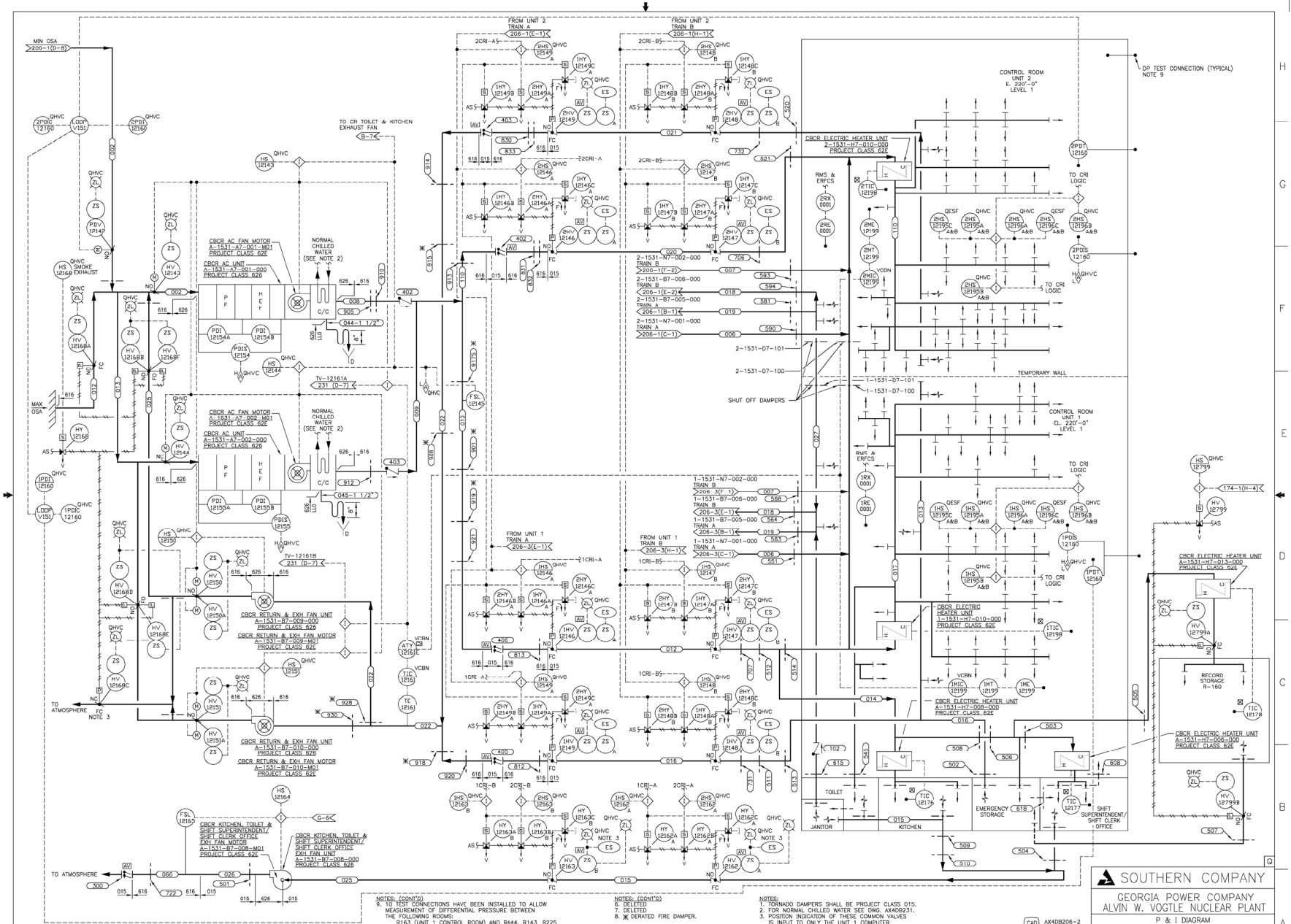
DRAWING NO. AX4DB206-1

VER. 31.0

JOB NO.10604

SIZE E 34X44

DRAWING CATEGORY: CRITICAL



NOTES: (CONT'D)
9. TO TEST CONNECTIONS HAVE BEEN INSTALLED TO ALLOW MEASUREMENT OF DIFFERENTIAL PRESSURE BETWEEN THE FOLLOWING ROOMS:
R163 (UNIT 1 CONTROL ROOM) AND RA44, R143, R225
R164 (UNIT 2 CONTROL ROOM) AND R117, RA23, R141, R224
R155 AND R158
R161 AND R199
R180 AND R160.
EACH CONNECTION IS CAPPED TO MAINTAIN CONTROL ROOM PRESSURE BOUNDARY.

NOTES: (CONT'D)
6. DELETED
7. DELETED
8. ✱ DERATED FIRE DAMPER.

NOTES:
1. TORNADO DAMPERS SHALL BE PROJECT CLASS 01S.
2. FOR NORMAL CHILLED WATER SEE DWG. AX4D9231.
3. POSITION INDICATION OF THESE COMMON VALVES IS INPUT TO ONLY THE UNIT 1 COMPUTER.
4. DELETED
5. REFER TO FLOW DIAGRAM AX4DB255-1.

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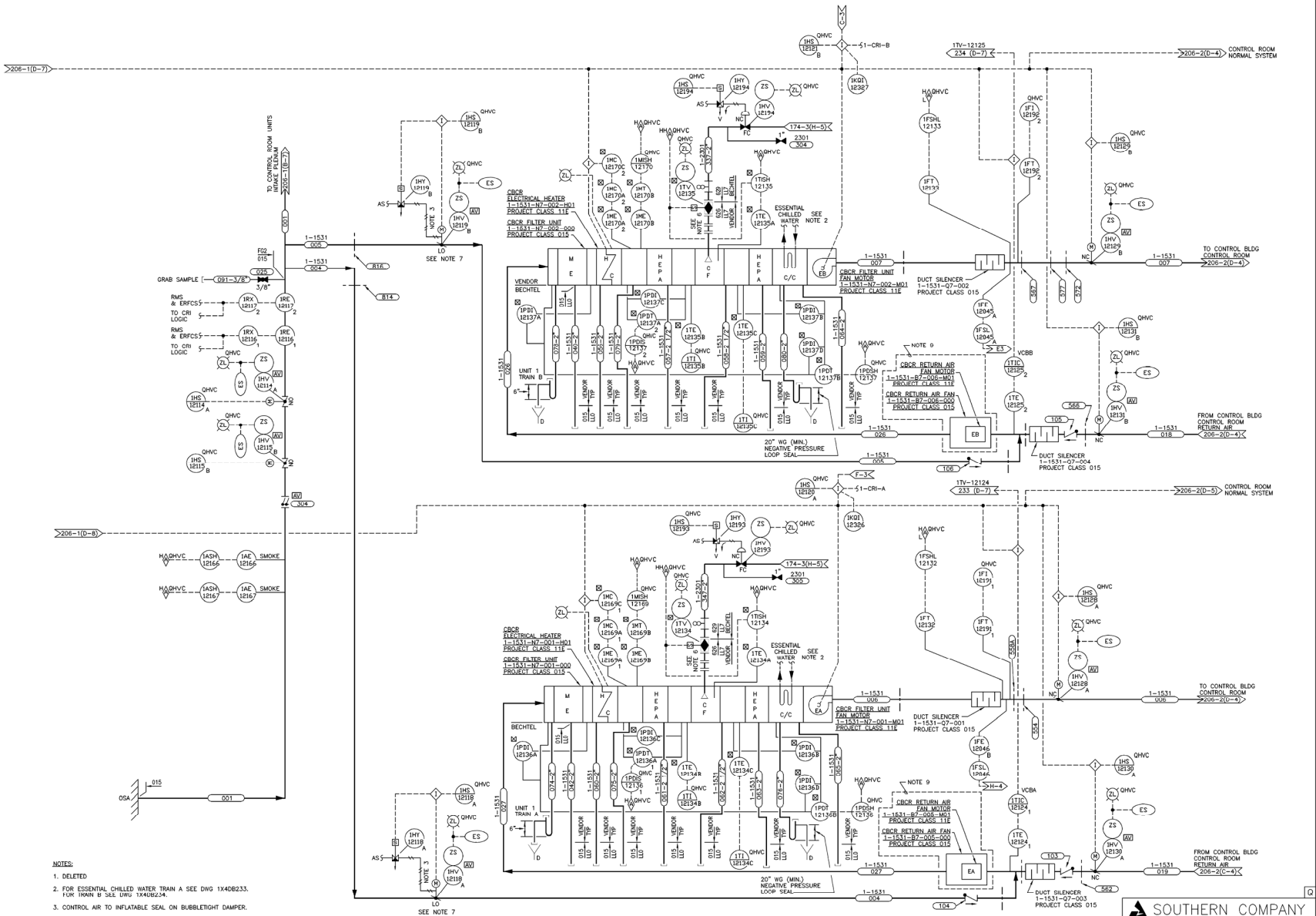
NO.	VERSIONS	DATE	DR	CHK	APPV	DTL
23.0	REDRAWN PER ABN-V70112, VER. 1.0	08-21-06	NJ	LCF	EPD	X

CAD AX4DB206-2
ACAD2002 NJ – 01

SOUTHERN COMPANY
GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT
P & I DIAGRAM
CONTROL BUILDING CONTROL ROOM HVAC SYSTEM LEVEL 1 SYSTEM NO. 15.51

SCALE: NONE
DRAWING NO. AX4DB206-2
VER. 23.0
JOB NO.10604
SIZE E 34X44
DRAWING CATEGORY: CRITICAL



NOTES:

1. DELETED
2. FOR ESSENTIAL CHILLED WATER TRAIN A SEE DWG 1X4DB233. FOR TRAIN B SEE DWG 1X4DB234.
3. CONTROL AIR TO INFLATABLE SEAL ON BUBBLETIGHT DAMPER.
4. DELETED
5. REFER TO FLOW DIAGRAM AX4DB250-1.
6. DELUGE VALVES ARE DISABLED SO THAT VALVE OPENS WITH WATER PRESSURE. REMOVABLE SPOOL PIECE IS REMOVED FOR NORMAL OPERATION AND INSTALLED TO DELUGE CARBON FILTER.
7. PROVIDE FOR POWER LOCKOUT AND RESTORATION FROM CONTROL ROOM.
8. DELETED
9. RETURN AIR FANS 1-1531-B7-005 AND 1-1531-B7-006 ARE RETIRED IN PLACE WITH THE BLADES REMOVED.

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NO.	VERSIONS	DATE	DR	CHK	APPV	DTL
29.0	REDRAWN PER ABN- V70113, VER. 1.0	08/21/06	NJ	LCF	EPD	X

SIZE E 34X44

CAD AX4DB206-3
ACAD2002 NJ - 01

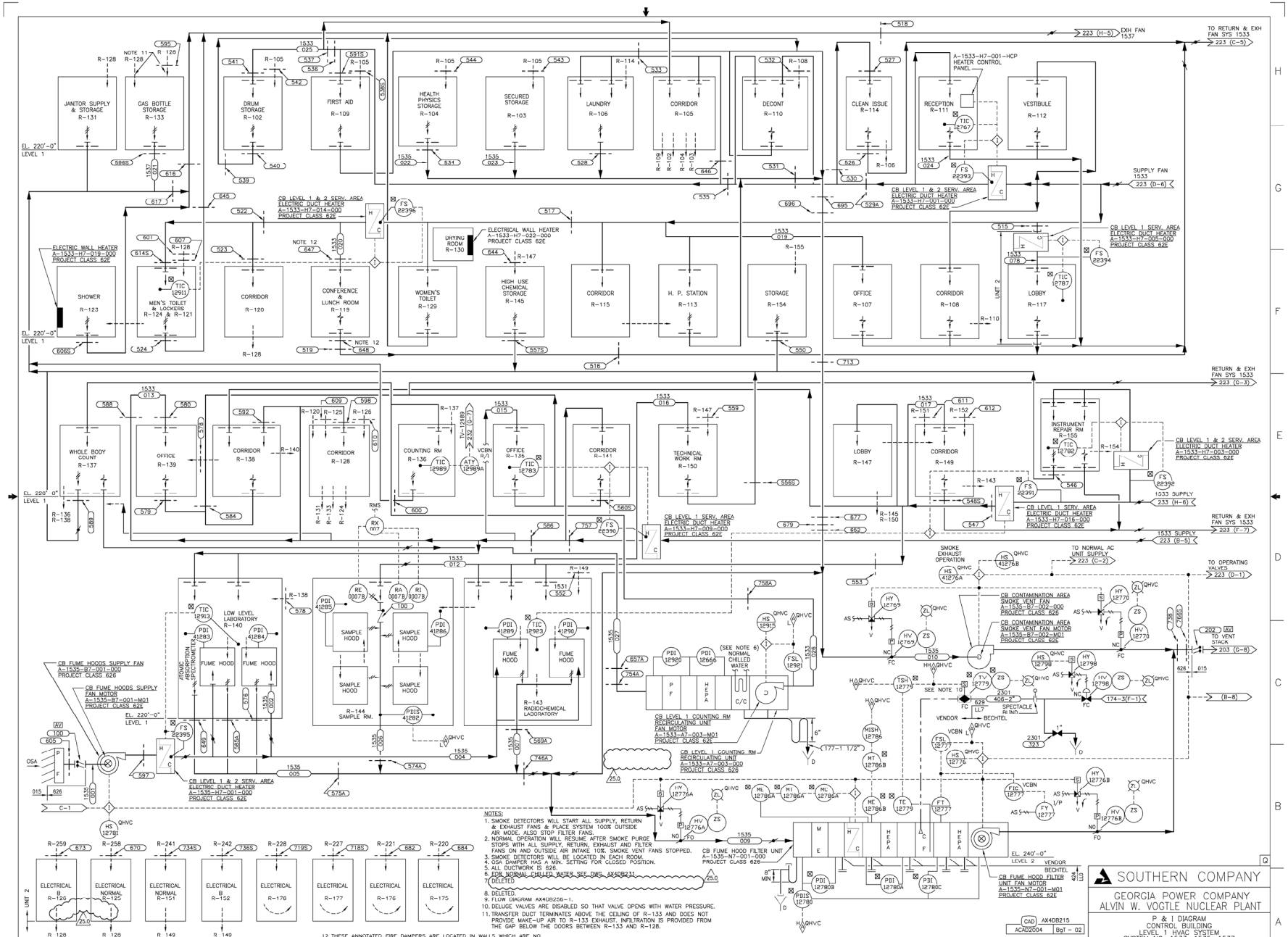
SOUTHERN COMPANY

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
CONTROL BUILDING CONTROL ROOM
HVAC SYSTEM LEVEL 1
SYSTEM NO. 1531

SCALE: NONE	DRAWING NO.	VER.
JOB NO.10604	AX4DB206-3	29.0

1 DRAWING CATEGORY: CRITICAL



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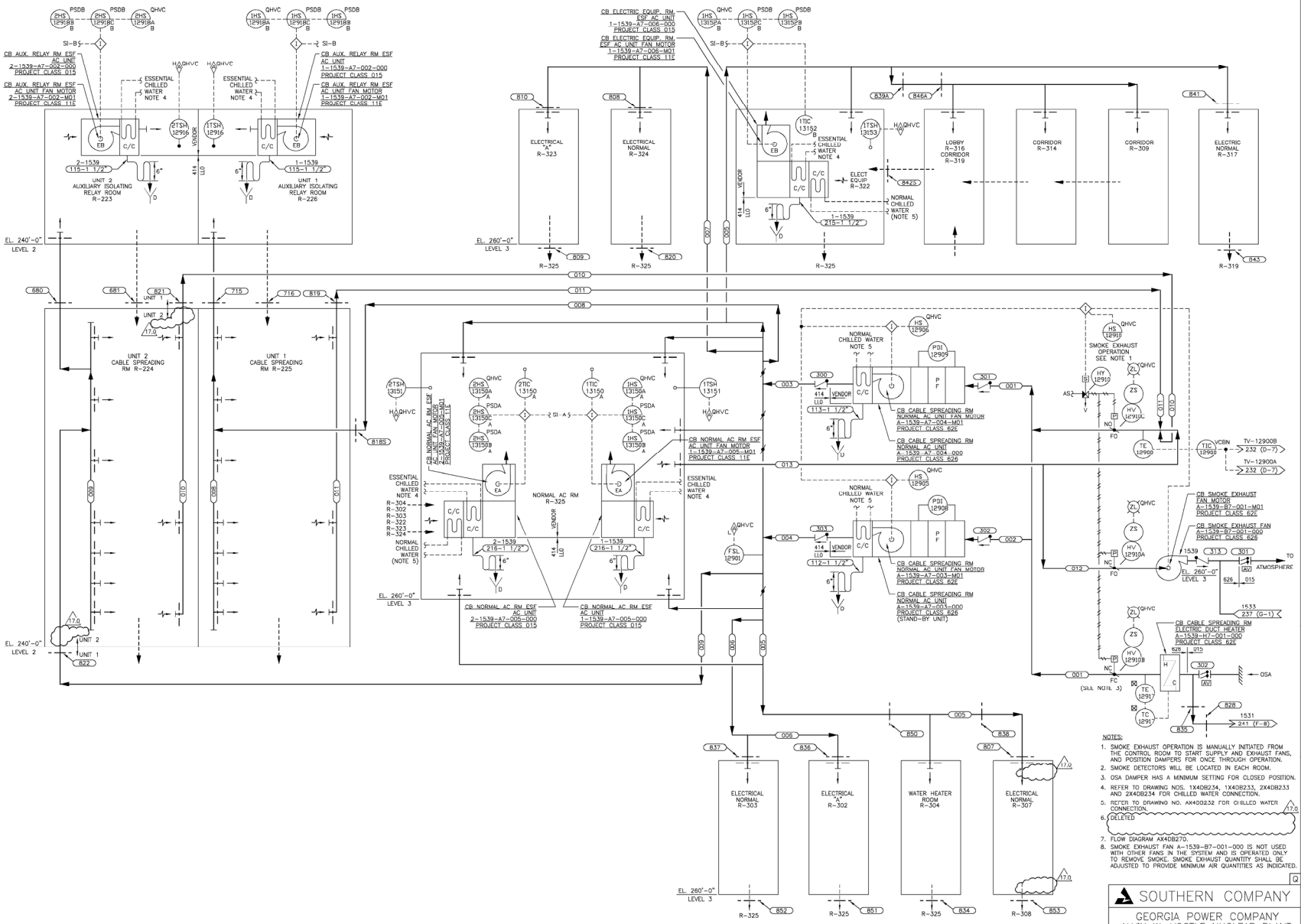
NO.	VERSIONS	DATE	DR	CHK	APPV	UTL
25.0	REVISED PER ABN-V00466, VER.1.0	1-19-07	BgT	MWD	WLN	

CAD AX4DB215
ACAD2004 BgT – 02

SOUTHERN COMPANY
GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT
P & I DIAGRAM
CONTROL BUILDING
LEVEL 1 HVAC SYSTEM
SYSTEM NO. 1533, 1535, 1537

SCALE: NONE
DRAWING NO. AX4DB215
VER. 25.0
JOB NO. 10604
SIZE E 34X44
DRAWING CATEGORY: CRITICAL



NOTES:

1. SMOKE EXHAUST OPERATION IS MANUALLY INITIATED FROM THE CONTROL ROOM TO START SUPPLY AND EXHAUST FANS, AND POSITION DAMPERS FOR ONCE THROUGH OPERATION.
2. SMOKE DETECTORS WILL BE LOCATED IN EACH ROOM.
3. OSA DAMPER HAS A MINIMUM SETTING FOR CLOSED POSITION.
4. REFER TO DRAWING NOS. 1X4DB234, 1X4DB233, 2X4DB233 AND 2X4DB234 FOR CHILLED WATER CONNECTION.
5. REFER TO DRAWING NO. AX4DB232 FOR CHILLED WATER CONNECTION. 17.0
6. DELETED 17.0
7. FLOW DIAGRAM AX4DB270.
8. SMOKE EXHAUST FAN A-1539-B7-001-000 IS NOT USED WITH OTHER FANS IN THE SYSTEM AND IS OPERATED ONLY TO REMOVE SMOKE. SMOKE EXHAUST QUANTITY SHALL BE ADJUSTED TO PROVIDE MINIMUM AIR QUANTITIES AS INDICATED.

Q

SOUTHERN COMPANY

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
CONTROL BLDG. CABLE SPREADING ROOM
LEVEL 2 & 3 HVAC
SYSTEM NO. 15.39

CAD AX4DB216
Acad2002 BgT - 03

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NO.	VERSIONS	DATE	DR	CHK	APPV	UTL
17.0	REVISED PER ABN-V00466, VER.1.0	1-19-07	BgT	MWD	WLN	

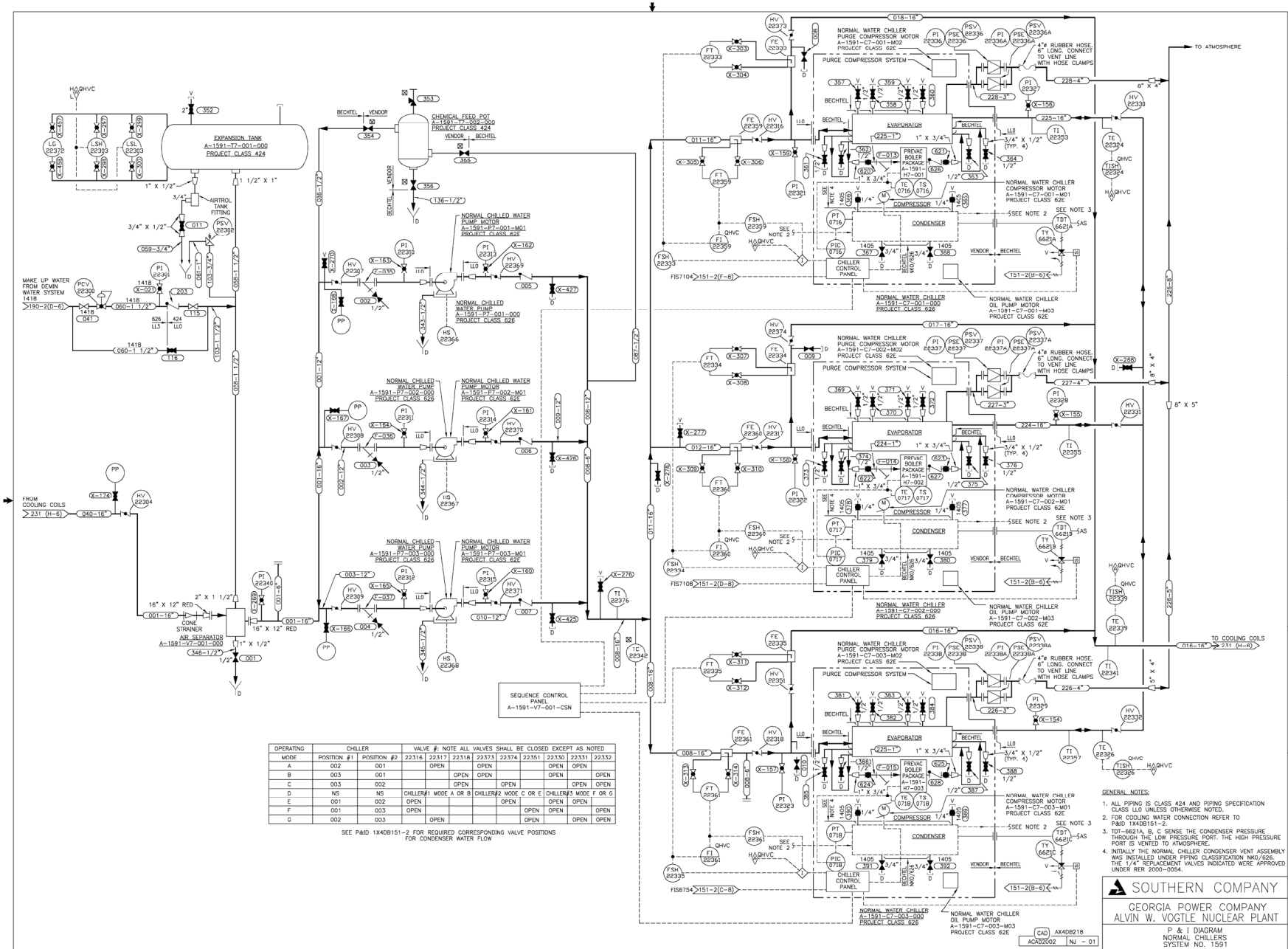
SCALE: NONE

DRAWING NO. AX4DB216 VER. 17.0

JOB NO.10604

SIZE E 34X44

DRAWING CATEGORY: CRITICAL



OPERATING	CHILLER		VALVE #: NOTE ALL VALVES SHALL BE CLOSED EXCEPT AS NOTED								
MODE	POSITION #1	POSITION #2	22316	22317	22318	22373	22374	22351	22330	22331	22332
A	002	001		OPEN		OPEN			OPEN	OPEN	
B	003	001			OPEN	OPEN			OPEN		OPEN
C	003	002			OPEN		OPEN			OPEN	OPEN
D	NS	NS	CHILLER #1 MODE A OR B			CHILLER #2 MODE C OR E			CHILLER #3 MODE F OR G		
E	001	002	OPEN				OPEN		OPEN	OPEN	
F	001	003	OPEN					OPEN	OPEN		OPEN
G	002	003		OPEN				OPEN		OPEN	OPEN

SEE P&ID 1X4DB151-2 FOR REQUIRED CORRESPONDING VALVE POSITIONS FOR CONDENSER WATER FLOW

GENERAL NOTES:

1. ALL PIPING IS CLASS 424 AND PIPING SPECIFICATION CLASS LLO UNLESS OTHERWISE NOTED.
2. FOR COOLING WATER CONNECTION REFER TO P&ID 1X4DB151-2.
3. TDT-6621A, B, C SENSE THE CONDENSER PRESSURE THROUGH THE LOW PRESSURE PORT. THE HIGH PRESSURE PORT IS VENTED TO ATMOSPHERE.
4. INITIALLY THE NORMAL CHILLER CONDENSER VENT ASSEMBLY WAS INSTALLED UNDER PIPING CLASSIFICATION NKO/626. THE 1/4" REPLACEMENT VALVES INDICATED WERE APPROVED UNDER RER 2000-0054.

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NO.	VERSION	DATE	DN	CHK	AP'VL	DTL
22.0	REDRAWN PER ABN-V70079, VER. 1.0	7-11-06	NJ	LCF	SHS	X

CAD AX4DB218
ACAD2002 | NJ – 01

SOUTHERN COMPANY

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
NORMAL CHILLERS
SYSTEM NO. 1591

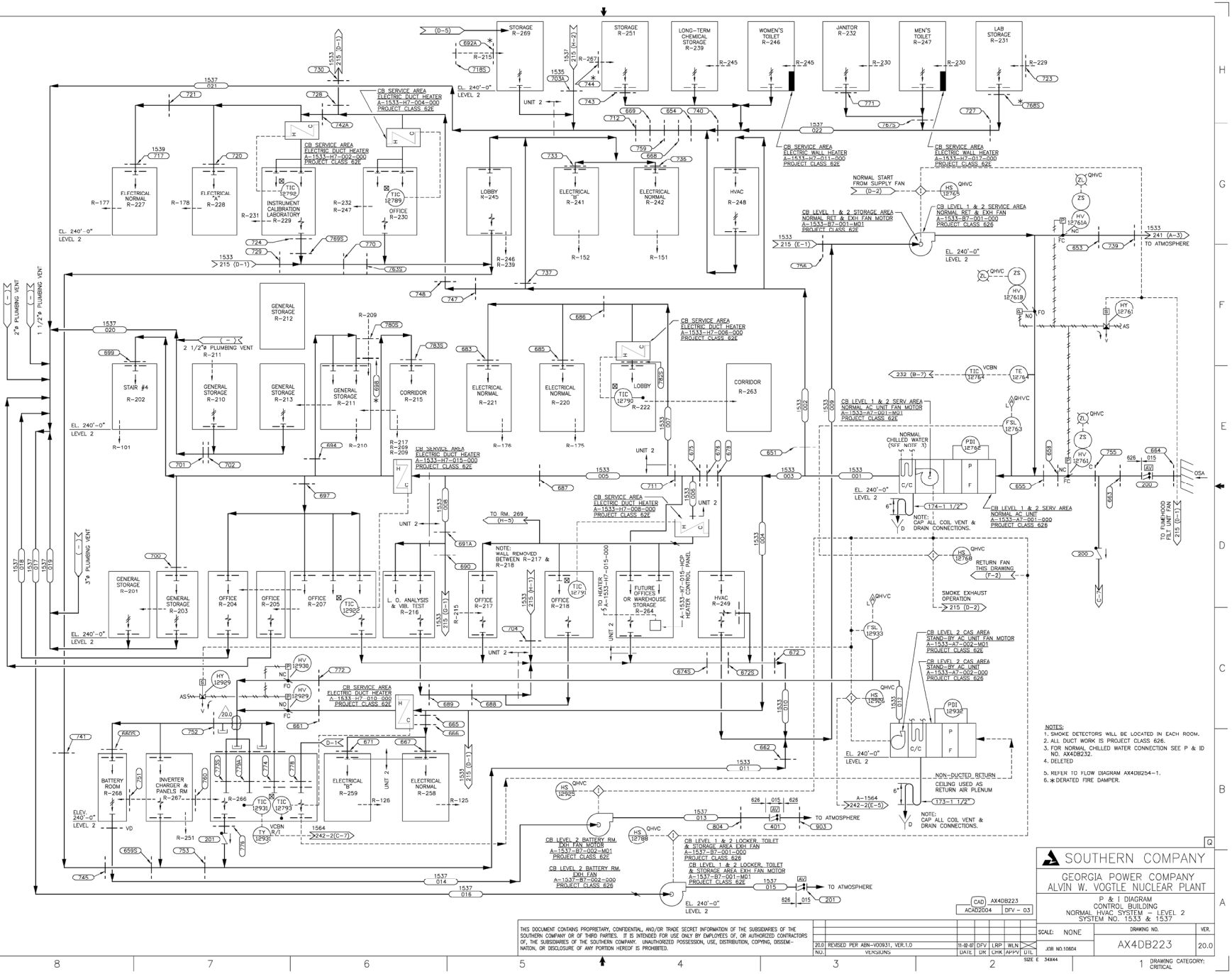
SCALE: NONE

DRAWING NO. AX4DB218 VER. 22.0

JOB NO.10604

SIZE E 34X44

1 DRAWING CATEGORY: CRITICAL


STORAGE R-269
STORAGE R-251
LONG-TERM CHEMICAL STORAGE R-239
WOMEN'S TOILET R-246
JANITOR R-232
MEN'S TOILET R-247
LAB STORAGE R-231
UNIT 2
EL. 240'-0" LEVEL 2
CB SERVICE AREA ELECTRIC DUCT HEATER A-1533-H7-004-000 PROJECT CLASS 62E
CB SERVICE AREA ELECTRIC DUCT HEATER A-1533-H7-009-000 PROJECT CLASS 62E
CB SERVICE AREA ELECTRIC WALL HEATER A-1533-H7-011-000 PROJECT CLASS 62E
CB SERVICE AREA ELECTRIC WALL HEATER A-1533-H7-017-000 PROJECT CLASS 62E
ELECTRICAL NORMAL R-227
ELECTRICAL "A" R-228
INSTRUMENT CALIBRATION LABORATORY R-229
OFFICE R-230
LOBBY R-245
ELECTRICAL "B" R-241
ELECTRICAL NORMAL R-242
HVAC R-248
NORMAL START FROM SUPPLY FAN (D-2)
CB LEVEL 1 & 2 STORAGE AREA NORMAL RET & EXH FAN MOTOR A-1533-B7-001-M01 PROJECT CLASS 62E
CB LEVEL 1 & 2 SERVICE AREA NORMAL RET & EXH FAN A-1533-B7-001-000 PROJECT CLASS 62E
TO ATMOSPHERE
GENERAL STORAGE R-212
GENERAL STORAGE R-211
GENERAL STORAGE R-210
GENERAL STORAGE R-213
CORRIDOR R-215
ELECTRICAL NORMAL R-221
ELECTRICAL NORMAL R-220
LOBBY R-222
CORRIDOR R-263
STAIR #4 R-202
CB SERVICE AREA ELECTRIC DUCT HEATER A-1533-H7-006-000 PROJECT CLASS 62E
CB LEVEL 1 & 2 SERV AREA NORMAL AC UNIT FAN MOTOR A-1533-A7-001-M01 PROJECT CLASS 62E
NORMAL CHILLED WATER (SEE NOTE 3)
CB LEVEL 1 & 2 SERV AREA NORMAL AC UNIT A-1533-A7-001-000 PROJECT CLASS 62E
TO PUMPHOUSE FILT UNIT FAN
CB SERVICE AREA ELECTRIC DUCT HEATER A-1533-H7-008-000 PROJECT CLASS 62E
CB SERVICE AREA ELECTRIC DUCT HEATER A-1533-H7-010-000 PROJECT CLASS 62E
CB SERVICE AREA ELECTRIC DUCT HEATER A-1533-H7-015-000 PROJECT CLASS 62E
TO RM. 269 (H-5)
NOTE: WALL REMOVED BETWEEN R-217 & R-218
A-1533-H7-015-HCP HEATER CONTROL PANEL
GENERAL STORAGE R-201
GENERAL STORAGE R-203
OFFICE R-204
OFFICE R-205
OFFICE R-207
L. O. ANALYSIS & VIB. TEST R-216
OFFICE R-217
OFFICE R-218
FUTURE OFFICES OR WAREHOUSE STORAGE R-264
HVAC R-249
RETURN FAN THIS DRAWING (F-2)
SMOKE EXHAUST OPERATION 215 (D-2)
CB LEVEL 2 CAS AREA STAND-BY AC UNIT FAN MOTOR A-1533-A7-002-M01 PROJECT CLASS 62E
CB LEVEL 2 CAS AREA STAND-BY AC UNIT A-1533-A7-002-000 PROJECT CLASS 62E
NON-DUCTED RETURN CEILING USED AS RETURN AIR PLENUM
NOTE: CAP ALL COIL VENT & DRAIN CONNECTIONS.
BATTERY ROOM R-268
INVERTER CHARGER & PANELS RM R-267
ELECTRICAL "B" R-259
ELECTRICAL NORMAL R-258
CB LEVEL 2 BATTERY RM. EXH FAN MOTOR A-1537-B7-002-M01 PROJECT CLASS 62E
CB LEVEL 2 BATTERY RM. EXH FAN A-1537-B7-002-000 PROJECT CLASS 62E
CB LEVEL 1 & 2 LOCKER TOILET & STORAGE AREA EXH FAN A-1537-B7-001-000 PROJECT CLASS 62E
CB LEVEL 1 & 2 LOCKER TOILET & STORAGE AREA EXH FAN MOTOR A-1537-B7-001-M01 PROJECT CLASS 62E
2"ø PLUMBING VENT
1 1/2"ø PLUMBING VENT
2 1/2"ø PLUMBING VENT
3"ø PLUMBING VENT
NOTES:
1. SMOKE DETECTORS WILL BE LOCATED IN EACH ROOM.
2. ALL DUCT WORK IS PROJECT CLASS 62E.
3. FOR NORMAL CHILLED WATER CONNECTION SEE P & ID NO. AX4DB232.
4. DELETED
5. REFER TO FLOW DIAGRAM AX4DB254-1.
6. ✱ DERATED FIRE DAMPER.
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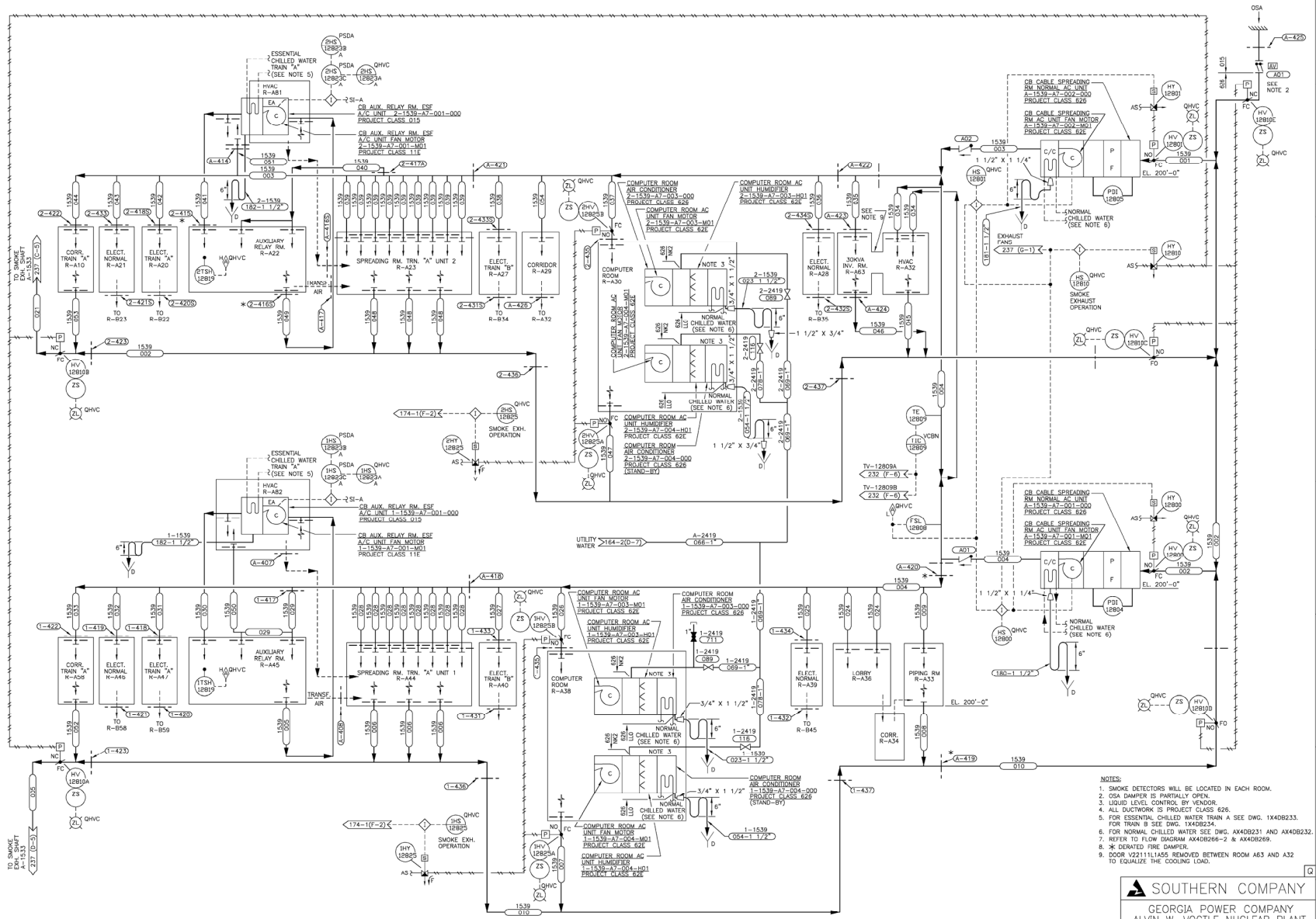
20.0 REVISED PER ABN-V00931, VER.1.0 11-02-07 DFV LRP WLN
NO. VERSIONS DATE DR CHK APPV UTL
SOUTHERN COMPANY
GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT
P & I DIAGRAM
CONTROL BUILDING
NORMAL HVAC SYSTEM — LEVEL 2
SYSTEM NO. 1533 & 1537
CAD AX4DB223
ACAD2004 DFV – 03
SCALE: NONE
DRAWING NO. AX4DB223
VER. 20.0
JOB NO.10604
SIZE E 34X44
DRAWING CATEGORY: CRITICAL



SOUTHERN COMPANY

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
CONTROL BUILDING CABLE SPREADING ROOM
LEVEL A HVAC SYSTEM
SYSTEM NO. 15.39

CAD AX4DB225
ACAD2002 CYN- 01

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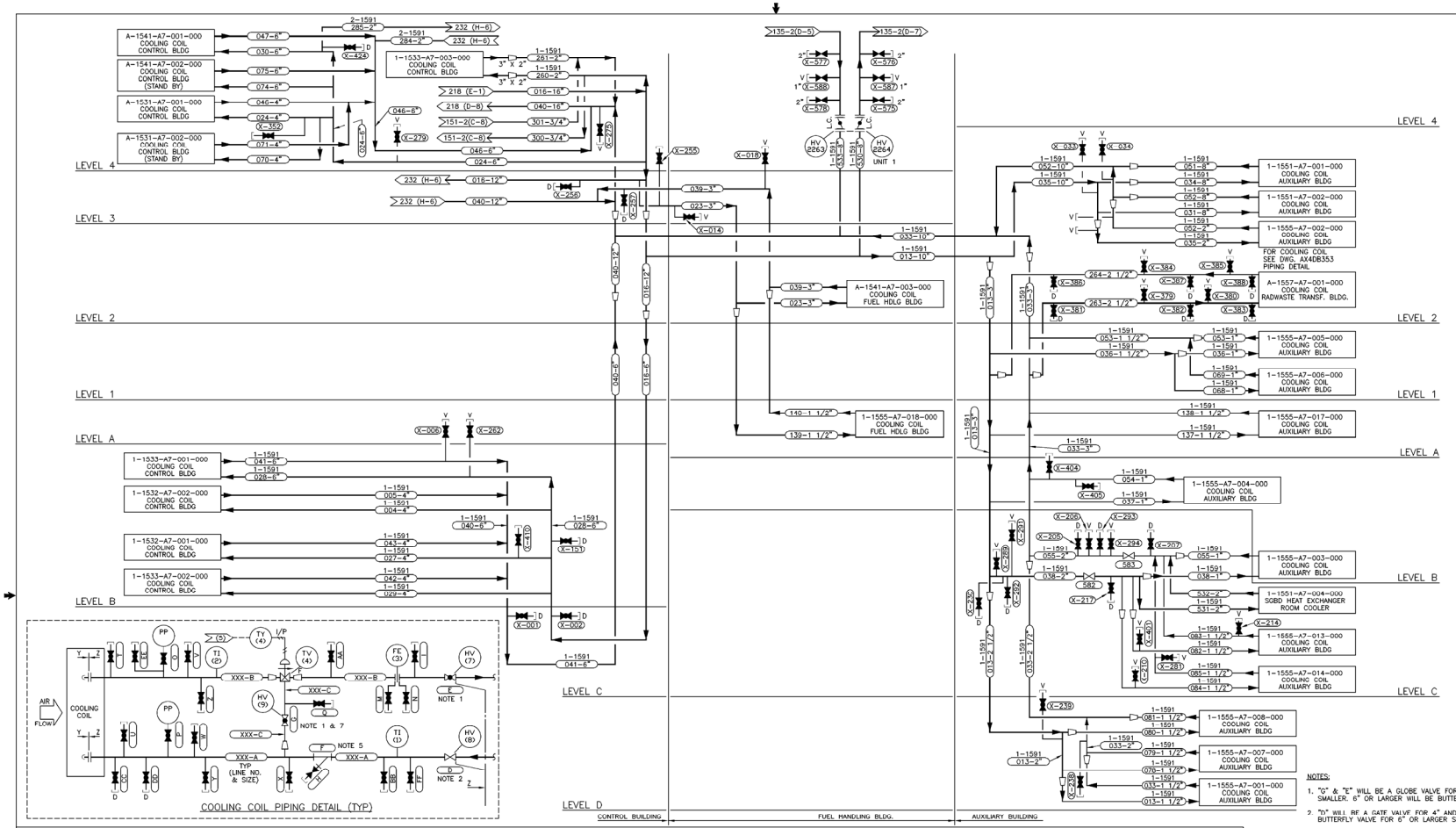
NO.		DATE	DR	CHK	APPV	DTL
16.0	REDRAWN PER ABN-V70059, VER. 1.0	6-6-06	CYN	LCF	WPE	X

SCALE: NONE
SIZE E 34X44

DRAWING NO. AX4DB225 VER. 16.0
JOB NO.10604

DRAWING CATEGORY: CRITICAL



CHILLED WATER COIL REFERENCE TABLE

✱SEE NOTE 3

EQUIPMENT NO.	INSTRUMENTATION NUMBER ✱ TI (1)	TI (2)	FE (3)	TY (4)	CONTROL VALVE ✱ TV (4)	P & ID (5)	HV (7)	HV (8)	PIPE & VALVE TAG SEQUENCE NUMBERS AND SIZE IN INCHES A NO.	A SIZE	B NO.	B SIZE	C NO.	C SIZE	D NO.	D SIZE	E NO.	E SIZE	F NO.	F SIZE	G NO.	G SIZE	H NO.	H SIZE	J NO.	K NO.	L NO.	ROOT VALVES (✱✱) M NO.	N NO.	O NO.	P NO.	EQUIP AREA NO	CLNG COIL CODE CLASS	PIPING PROJ MATERIAL CL	Y	Z	I	Q	T	U	V	W	X	Y	Z	AA	BB	CC	DD	EE	FF
																												ROOT VALVES: (✱✱) V=VENT & D=DRAIN																														
A-1531-A7-001-000	22000	22001	22002	12161A	12161A	206-2 (F-6)			024	4	046	4	024	3	309	4	307	4	025	4	100	3	034	—	—	—	035	036	096	075	22B4	626	424/LL0	V-347				V-341	V-339		D-343				D-345								
A-1531-A7-002-000	22003	22004	22005	12161B	12161B	206-2 (D-6)			070	4	071	4	070	3	310	4	308	4	026	4	101	3	112	—	—	—	037	038	097	076	22B4	626	424/LL0	V-348				V-342	V-340		D-344				D-346								
A-1541-A7-001-000	22006	22007	22008	2538A	2538A	204-1 (C-7)	22009	22010	030	6	047	6	030	3	—	6	—	6	003	6	054	3	036	—	—	—	039	040	098	077	12B4	626	424/LL0					V-353	V-355													
A-1541-A7-002-000	22012	22013	22014	2538B	2538B	204-1 (B-7)	22015	22016	074	6	075	6	074	3	—	6	004	6	108	3	114	—	—	—	—	—	041	042	099	078	22B4	626	424/LL0					V-354	V-356													
A-1541-A7-003-000	22066	22067	22068	—	—	—	—	—	023	3	039	3	—	—	023	3	075	3	—	—	—	—	—	—	—	—	043	044	100	079	13A1	626	424/LL0					D-376														
1-1532-A7-001-000	21997	21998	21999	12744	12744	207-1 (F-8)	—	—	027	4	043	4	027	2	212	4	213	4	013	4	064	2	046	—	—	—	045	046	101	080	12DB	313	424/LL0							V-198	V-194					D-199		◉ D-155			
1-1532-A7-002-000	22114	22115	22116	12745	12745	207-1 (B-7)	—	—	004	4	005	4	005	2	089	4	092	4	014	4	065	2	047	—	—	—	047	048	102	081	12DB	313	424/LL0							V-196	V-334					D-197		◉ D-157			
1-1533-A7-001-000	22117	22118	22119	2739	2739	210 (D-3)	22042	22043	028	6	041	6	028	2	—	6	—	6	015	6	062	2	044	—	—	—	049	050	103	082	12FB	626	424/LL0															D-411			
1-1533-A7-002-000	22105	22106	22107	12758	12758	211 (E-4)	—	—	029	4	042	4	029	2	090	4	093	4	016	4	063	2	045	—	—	—	051	052	104	083	12BB	626	424/LL0	D-203	D-201													D-412			
1-1551-A7-001-000	22048	22049	22050	12678	12678	208-1 (G-5)	22051	22052	153	8	154	8	153	4	—	6	—	6	033	8	214	4	037	—	—	—	028	029	030	032	13B2	626	424/LL0					D-318	V-316				◉ D-122								
1-1551-A7-002-000	22054	22055	22056	12682	12682	208-1 (G-3)	22057	22058	155	8	156	8	156	4	—	6	—	6	034	8	215	4	038	—	—	—	025	026	027	031	13B2	626	424/LL0					V-317		◉ D-123		◉ D-136								
1-1555-A7-001-000	22129	22130	22131	—	—	—	—	—	013	1 1/2	033	1 1/2	—	—	033	1 1/2	085	1 1/2	—	—	—	—	—	—	—	—	053	054	105	084	13GD	313	415/LL2					◉ V-166	◉ D-165				V-240	D-188					◉ D-050	
1-1555-A7-002-000	22045	22046	22047	—	—	—	—	—	035	2	052	2	—	—	216	2	217	2	—	—	—	—	—	—	—	—	055	056	106	085	13F2	313	424/LL0					◉ V-252	V-251												
1-1555-A7-003-000	22126	22127	22128	—	—	—	—	—	038	1	055	1	—	—	218	1	219	1	—	—	—	—	—	—	—	—	057	058	107	086	13GC	313	415/LL2					D-296	D-423	V-164			V-249								
1-1555-A7-004-000	22108	22109	22110	—	—	—	—	—	037	1	054	1	—	—	220	1	221	1	—	—	—	—	—	—	—	—	059	060	108	087	13HB	313	424/LL0																		
1-1555-A7-005-000	22078	22079	22080	—	—	—	—	—	036	1	053	1	—	—	222	1	223	1	—	—	—	—	—	—	—	—	061	062	109	088	13E1	313	424/LL0													D-366					
1-1555-A7-006-000	22075	22076	22077	—	—	—	—	—	068	1	069	1	—	—	224	1	225	1	—	—	—	—	—	—	—	—	063	064	110	089	13F1	313	424/LL0																		
1-1555-A7-007-000	21994	21995	21996	—	—	—	—	—	078	1 1/2	079	1 1/2	—	—	168	1 1/2	169	1 1/2	—	—	—	—	—	—	—	—	065	066	111	090	13ED	313	415/LL2										◉ D-171						◉ V-172	
1-1555-A7-008-000	21973	21974	21975	—	—	—	—	—	080	1 1/2	081	1 1/2	—	—	173	1 1/2	174	1 1/2	—	—	—	—	—	—	—	—	024	023	022	091	13ED	313	415/LL2																◉ V-177	
1-1555-A7-013-000	22202	22203	22204	—	—	—	—	—	082	1 1/2	083	1 1/2	—	—	178	1 1/2	179	1 1/2	—	—	—	—	—	—	—	—	067	068	112	092	13HC	313	415/LL2					◉ V-182	V-402	D-213			D-295				◉ D-176			
1-1555-A7-014-000	22208	22209	22210	—	—	—	—	—	084	1 1/2	085	1 1/2	—	—	183	1 1/2	184	1 1/2	—	—	—	—	—	—	—	—	070	069	113	093	13HC	313	415/LL2					V-280	V-209	◉ D-187				D-187			◉ D-181			
1-1555-A7-017-000	21991	21992	21993	—	—	—	—	—	137	1 1/2	138	1 1/2	—	—	024	1 1/2	076	1 1/2	—	—	—	—	—	—	—	—	071	072	114	094	13BA	313	414/LL0														◉ D-186			
1-1555-A7-018-000	22153	22154	22155	—	—	—	—	—	139	1 1/2	140	1 1/2	—	—	226	1 1/2	227	1 1/2	—	—	—	—	—	—	—	—	073	074	115	095	13DA	313	424/LL0														◉ D-032			
1-1533-A7-003-000	21946	21947	21948	12320	12320	241 (G-3)	—	—	260	2	261	2	262	1	302	2	303	2	—	—	304	1	—	—	—	—	335	336	337	338	—	626	424/LL0																		
A-1557-A7-001-000	—	—	—	—	—	—	—	—	263	2 1/2	264	2 1/2	265	2 1/2	SEE DRAWING AX4DB353 FOR COOLING COIL DETAILS												—	—	—	—	—	626	424/LL0																		
1-1551-A7-004-000	—	—	7394	—	—	—	—	—	531	2	532	2	—	—	580	2	581	2	—	—	—	—	—	—	—	—	582	583	581	579	13GC	626	424/LL0	V-584														D-580			

✱✱ ALL ROOT VALVES HAVE AN "X" PREFIX EXCEPT WHERE NOTED ◉

NOTES:

1. "G" & "E" WILL BE A GLOBE VALVE FOR SIZES 4" OR SMALLER. 6" OR LARGER WILL BE BUTTERFLY VALVES.
2. "D" WILL BE A GATE VALVE FOR 4" AND SMALLER AND BUTTERFLY VALVE FOR 6" OR LARGER SIZE.
3. FIRST CHARACTER OF INSTRUMENTATION TAG NO. (UNIT NO.) IS SAME AS ASSOCIATED EQUIPMENT'S UNIT NO.
4. STRAINERS 2" AND SMALLER SHALL BE SOCKET WELDED.
5. ALL STRAINERS SHALL BE WYE TYPE EXCEPT AS MARKED.
6. DELETED.
7. VALVE "G" SHALL BE THROTTLED AS REQUIRED TO SIMULATE THE PRESSURE DROP ACROSS THE COOLING COIL. THIS WILL ENABLE SYSTEM FLOW TO REMAIN CONSTANT IN THE EVENT THAT CHILLED WATER FLOW MUST BYPASS THE COOLING COIL TO MAINTAIN TEMPERATURE REQUIREMENTS.
8. THE ISOLATION VALVES FOR EQUIPMENT 1-1551-A7-001 & 002 TAG NOS. HV-22051, HV-22052, HV-22057 & HV-22058 AT COORDINATE (A-7) ARE 6" IN SIZE.
9. ALL PIPING IS PROJECT CLASS 424 LLO UNLESS OTHERWISE NOTED.
10. VALVE 050 INSTALLED AS DRAIN VALVE. VDL SHOWS 11591V4050 AS SURPLUS VALVE.
11. VENTS & DRAINS SHALL BE GATES OR GLOBES AS INDICATED IN VDL OR RVDL.

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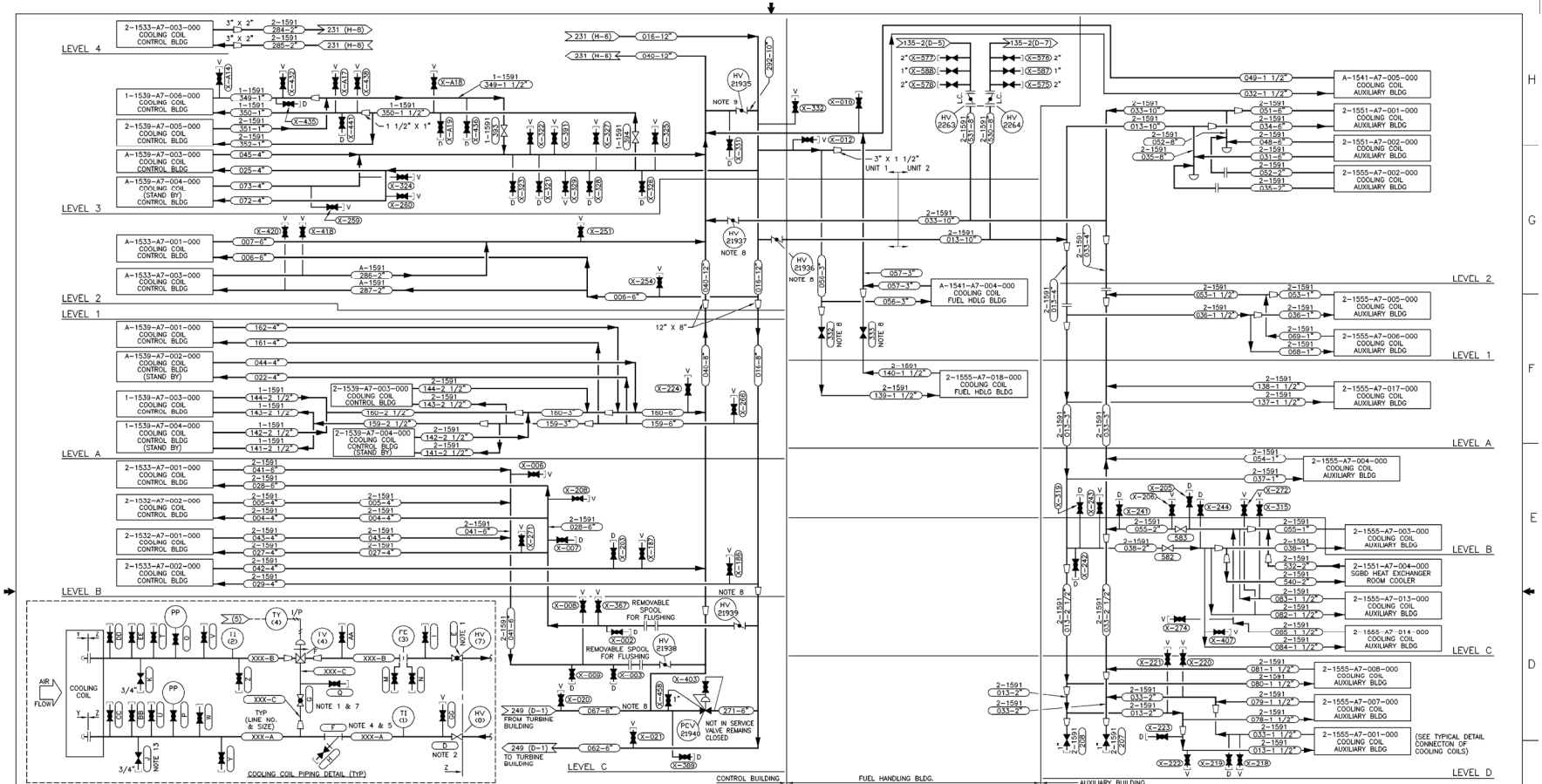
NO.	VERSIONS	DATE	DR	CHK	APPV	DTL
21.0	REDRAWN PER ABN- V70111, VER. 1.0	08/21/08	NJ	LCF	SHS	X

CAD AX4DB231
ACAD2004 NJ – 01

SOUTHERN COMPANY
GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT
P & I DIAGRAM
NORMAL CHILLED WATER
COOLING COILS
SYSTEM NO. 1591

SCALE: NONE
DRAWING NO. AX4DB231 VER. 21.0
JOB NO.10604
SIZE E 34044
DRAWING CATEGORY: CRITICAL



CHILLED WATER COIL REFERENCE TABLE

EQUIPMENT NO.	INSTRUMENTATION NUMBER TI (1)	TI (2)	FE (3)	TY (4)	CONTROL VALVE TV (4)	P & ID (5)	HV (7)	HV (8)	PIPE & VALVE TAG SEQUENCE NUMBERS AND SIZE IN INCHES A NO.	SIZE	NO.	B SIZE	NO.	C SIZE	NO.	D SIZE	NO.	E SIZE	NO.	F SIZE	NO.	G SIZE	NO.	H NO.	J NO.	K NO.	ROOT VALVES (NOTE 11) L NO.	M NO.	N NO.	Q NO.	P NO.	EQUIP AREA NO	CLNG COIL COIL CLASS	PIPING PROJ MATERIAL CL	Z	ROOT VALVES: (NOTE 11) VALVENT & DnDRAIN I	Q	T	U	V	W	Y	Z	AA	BB	CC	DD	EE	GG
A-1533-A7-001-000	22039	22040	22041	12764	12764	215 (E-2)	22218	22219	006	6	007	6	006	3	—	6	—	6	006	6	094	3	091	125	140	118	119	143	134	22B2	626	424/LL0	—	D-253	V-250	V-252	—	—	—	D-267	—	—	—	—	—	—	—	—	—
A-1539-A7-001-000	22084	22085	22086	12809A	12809A	225 (D-7)	—	—	161	4	162	4	161	2	240	4	241	4	007	4	061	2	043	167	150	120	121	144	135	22BA	626	424/LL0	D-232	—	—	—	—	—	V-233	—	V-236	—	—	—	—	—	—
A-1539-A7-002-000	22087	22088	22089	12809B	12809B	225 (B-7)	—	—	022	4	044	4	022	2	242	4	243	4	008	4	111	2	117	—	—	122	123	145	136	22BA	626	424/LL0	D-228	—	—	—	—	—	V-229	—	V-235	—	—	—	—	—	—
A-1539-A7-003-000	22019	22020	22021	12900A	12900A	216 (C-3)	—	—	025	4	045	4	025	2	244	4	245	4	009	4	053	2	035	—	—	124	125	146	137	22A3	626	424/LL0	D-230	—	—	—	—	—	V-234	—	V-330	—	—	—	—	—	—
A-1539-A7-004-000	22021	22022	22023	12900B	12900B	216 (C-2)	—	—	072	4	073	4	072	2	246	4	247	4	010	4	107	2	113	121	136	126	127	147	138	22A3	626	424/LL0	D-258	—	—	—	—	—	—	—	—	—	—	—	—	—	—
A-1541-A7-004-000	22063	22064	22065	—	—	—	—	—	056	3	057	3	—	—	051	3	104	3	—	—	—	—	—	—	—	128	129	148	139	23A1	626	424/LL0	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
A-1541-A7-005-000	22072	22073	22074	12520	12520	204-1 (B-4)	—	—	032	1 1/2	049	1 1/2	032	1	022	1 1/2	074	1 1/2	012	1 1/2	248	1	039	—	—	130	131	149	140	23B1	626	424/LL0	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
A-1532-A7-001-000	21927	21928	21999	12744	12744	207-1 (F-3)	—	—	027	4	043	4	027	2	212	4	213	4	013	4	064	2	346	155	—	045	046	101	080	22DB	313	424/LL0	—	—	—	—	—	—	V-198	—	V-194	—	V-199	—	—	—	—
A-1532-A7-002-000	22114	22115	22116	12745	12745	207-1 (B-2)	—	—	004	4	005	4	004	2	089	4	092	4	014	4	065	2	047	—	—	047	048	102	081	22DB	313	424/LL0	D-197	—	—	—	—	—	V-196	—	V-192	—	—	—	—	—	—
A-1533-A7-001-000	22117	22118	22119	2739	2739	210 (F-3)	22042	22043	028	6	041	6	028	2	—	6	—	6	015	6	062	2	044	153	154	049	050	103	082	22FB	626	424/LL0	D-193	D-376	—	—	—	—	V-204	—	V-195	—	—	—	—	—	—
A-1533-A7-002-000	22105	22106	22107	12758	12758	211 (E-4)	—	—	029	4	042	4	029	2	090	4	093	4	016	4	063	2	045	161	—	051	052	104	083	22BB	626	424/LL0	D-212	D-201	—	—	—	—	V-202	—	V-200	—	—	—	—	—	—
A-1551-A7-001-000	22048	22049	22050	12678	12678	208-1 (G-5)	22051	22052	034	6	051	6	034	4	—	6	—	6	017	6	214	4	037	—	—	028	029	030	032	23B2	626	424/LL0	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
A-1551-A7-002-000	22054	22055	22056	12682	12682	208-1 (G-3)	22057	22058	031	6	048	6	031	4	—	6	—	6	018	6	215	4	038	—	—	025	026	027	031	23B2	626	424/LL0	—	—	D-395	—	—	—	—	—	—	—	—	—	—	—	—	—
A-1555-A7-001-000	22129	22130	22131	—	—	—	—	—	013	1 1/2	033	1 1/2	—	—	033	1 1/2	085	1 1/2	—	—	—	—	165	166	053	054	105	084	23G0	313	424/LL0	—	—	—	—	D-188	D-189	D-190	—	—	—	—	—	—	—	—	—	—	—	—
A-1555-A7-002-000	22045	22046	22047	—	—	—	—	—	035	2	052	2	—	—	216	2	217	2	—	—	—	—	—	—	—	055	056	106	085	23F2	313	424/LL0	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
A-1555-A7-003-000	22126	22127	22128	—	—	—	—	—	038	1	055	1	—	—	218	1	219	1	—	—	—	—	—	—	—	057	058	107	086	23GC	313	424/LL0	—	—	D-430	D-261	V-460	V-459	—	—	—	—	—	—	—	—	—	—	—
A-1555-A7-004-000	22108	22109	22110	—	—	—	—	—	037	1	054	1	—	—	220	1	221	1	—	—	—	—	—	—	—	059	060	108	087	23HB	313	424/LL0	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
A-1555-A7-005-000	22078	22079	22080	—	—	—	—	—	036	1	053	1	—	—	222	1	223	1	—	—	—	—	—	—	—	062	061	109	088	23E1	313	424/LL0	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
A-1555-A7-006-000	22075	22076	22077	—	—	—	—	—	068	1	069	1	—	—	224	1	225	1	—	—	—	—	—	—	—	063	064	110	089	23F1	313	424/LL0	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
A-1555-A7-007-000	21994	21995	21996	—	—	—	—	—	078	1 1/2	079	1 1/2	—	—	168	1 1/2	169	1 1/2	—	—	—	—	171	172	065	066	111	090	23ED	313	424/LL0	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
A-1555-A7-008-000	21973	21974	21975	—	—	—	—	—	080	1 1/2	081	1 1/2	—	—	173	1 1/2	174	1 1/2	—	—	—	—	176	177	023	024	022	091	23ED	313	424/LL0	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
A-1555-A7-013-000	22203	22204	22205	—	—	—	—	—	082	1 1/2	083	1 1/2	—	—	178	1 1/2	179	1 1/2	—	—	—	—	181	182	067	068	112	092	23HC	313	424/LL0	—	—	—	—	D-405	D-213	—	—	V-273	—	—	—	—	—	—	—
A-1555-A7-014-000	22208	22209	22210	—	—	—	—	—	084	1 1/2	085	1 1/2	—	—	183	1 1/2	184	1 1/2	—	—	—	—	186	187	069	070	113	093	23HC	313	424/LL0	—	—	—	—	V-406	D-703	D-408	—	—	—	—	—	—	—	—	—	—	—	—
A-1555-A7-017-000	21991	21992	21993	—	—	—	—	—	137	1 1/2	136	1 1/2	—	—	024	1 1/2	076	1 1/2	—	—	—	—	—	—	—	071	072	114	094	23DA	313	424/LL0	—	—	—	—	V-A23	—	—	—	—	—	—	—	—	—	—	—	—	—
A-1555-A7-018-000	22153	22154	22155	—	—	—	—	—	139	1 1/2	140	1 1/2	—	—	226	1 1/2	227	1 1/2	—	—	—	—	—	—	—	074	073	115	095	23DA	313	424/LL0	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
1-1539-A7-003-000	21979	21980	21981	—	—	—	—	—	143	2 1/2	144	2 1/2	—	—	230	2 1/2	231	2 1/2	—	—	—	—	—	—	—	116	117	132	141	12BA	626	424/LL0	D-246	—	V-248	V-263	—	—	D-371	—	D-247	D-264	V-414	V-413	D-372	—	—	—
1-1539-A7-004-000	21976	21977	21978	—	—	—	—	—	141	2 1/2	142	2 1/2	—	—	232	2 1/2	233	2 1/2	—	—	—	—	—	—	—	152	153	133	142	12BA	626	424/LL0	D-225	—	V-227	V-265	—	—	D-368	—	D-226	D-282	V-416	V-415	D-370	—	—	—
2-1539-A7-003-000	21979	21980	21981	—	—	—	—	—	143	2 1/2	144	2 1/2	—	—	230	2 1/2	231	2 1/2	—	—	—	—	—	—	—	116	117	132	141	12BA	626	424/LL0	D-246	—	V-248	V-263	—	—	D-264	—	D-247	—	—	—	—	—	—	—
2-1539-A7-004-000	21976	21977	21978	—	—	—	—	—	141	2 1/2	142	2 1/2	—	—	232	2 1/2	233	2 1/2	—	—	—	—	—	—	—	152	153	133	142	12BA	626	424/LL0	D-225	—	V-227	V-265	—	—	D-368	—	D-226	—	—	—	—	—	—	—
2-1533-A7-003-000	21946	21947	21948	12320	12320	241 (H-8)	—	—	285	2	284	2	290	1	311	2	313	2	—	—	—	—	315	1	—	357	358	359	360	—	626	424/LL0	—	—	—	—	—	—	—	—	D-429	—	—	—	—	—	—	—
A-1533-A7-003-000	21941	21942	21943	12989	12989	215 (H-5)	—	—	287	2	286	2	291	1	319	2	320	2	—	—	—	—	321	1	—	361	362	363	364	—	626	424/LL0	—	—	D-419	—	V-417	V-421	D-422	—	—	—	—	—	—	—	—	—	—
1-1539-A7-006-000	13157	13158	13159	—	—	—	—	—	350	1	349	1	—	—	395	1	396 ✱	1	—	—	—	—	—	—	—	A01	A02	A03	A04	12A3	313	424/LL0	—	—	D-A09	D-A10	—	—	—	—	—	—	—	—	—	—	—	—	—
1-1539-A7-005-000	13154	13155	13156	—	—	—	—	—	352	1	351	1	—	—	397	1	398	1	—	—	—	—	—	—	—	A05	A06	A07	A08	12B3	313	424/LL0	—	—	—	—	—	—	V-443	—	D-446	—	D-444	—	V-445	—	—	—
2-1551-A7-004-000	—	—	7394	—	—	—	—	—	540	2	532	2	—	—	580	2	581	2	—	—	—	—	—	—	—	582	583	586	579	23CC	626	424/LL0	V-584	—	—	—	—	—	—	—	—	D-585	—	—	—	—	—	—

✱ SEE NOTE 12

NOTES:

1. "G" & "E" WILL BE A GLOBE VALVE FOR SIZES 4" OR SMALLER, 6" OR LARGER WILL BE BUTTERFLY.
2. "D" WILL BE A GATE VALVE FOR 4" AND SMALLER AND BUTTERFLY VALVE FOR 6" OR LARGER SIZE.
3. FIRST CHARACTER OF INSTRUMENTATION TAG NO. (UNIT NO.) IS SAME AS ASSOCIATED EQUIPMENT'S UNIT NO.
4. STRAINERS 2" AND SMALLER SHALL BE SOCKET WELDED.
5. ALL STRAINERS SHALL BE WYE TYPE EXCEPT AS NOTED.
6. DELETED.
7. VALVE "G" SHALL BE THROTTLED AS REQUIRED TO SIMULATE THE PRESSURE DROP ACROSS THE COOLING COIL. THIS WILL ENABLE SYSTEM FLOW TO REMAIN CONSTANT IN THE EVENT THAT CHILLED WATER FLOW MUST BYPASS THE COOLING COIL TO MAINTAIN TEMPERATURE REQUIREMENTS.
8. VALVES SHALL BE USED TO ISOLATE UNIT 2 SYSTEMS DURING CONSTRUCTION AND PERMIT OPERATION OF THE UNIT 1 AND COMMON SYSTEMS. VALVES SHALL BE LOCKED OPEN WHEN UNIT 2 SYSTEMS ARE IN OPERATION.
9. VALVE SHALL BE USED TO BALANCE UNIT 1 AND COMMON SYSTEMS DURING UNIT 2 CONSTRUCTION. VALVE SHALL BE LOCKED CLOSED WHEN UNIT 2 SYSTEMS ARE IN OPERATION.
10. ALL PIPING IS PROJECT CLASS 424/LLO UNLESS OTHERWISE NOTED.
11. ALL ROOT VALVE NOS. SHALL HAVE PREFIX-X.
12. VALVE 1-1591-U4-396 IS A GATE VALVE.
13. FOR EQUIPMENT # 2-1555-A7-013-000, VALVE J IS LOCATED DOWNSTREAM OF VALVE BB.

Q

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NO.	VERSIONS	DATE	DR	CHK	APPV	DTL
31.0	REDRAWN PER ABN-V70114, VER. 1.0	08/21/06	NJ	LCF	SHS	X

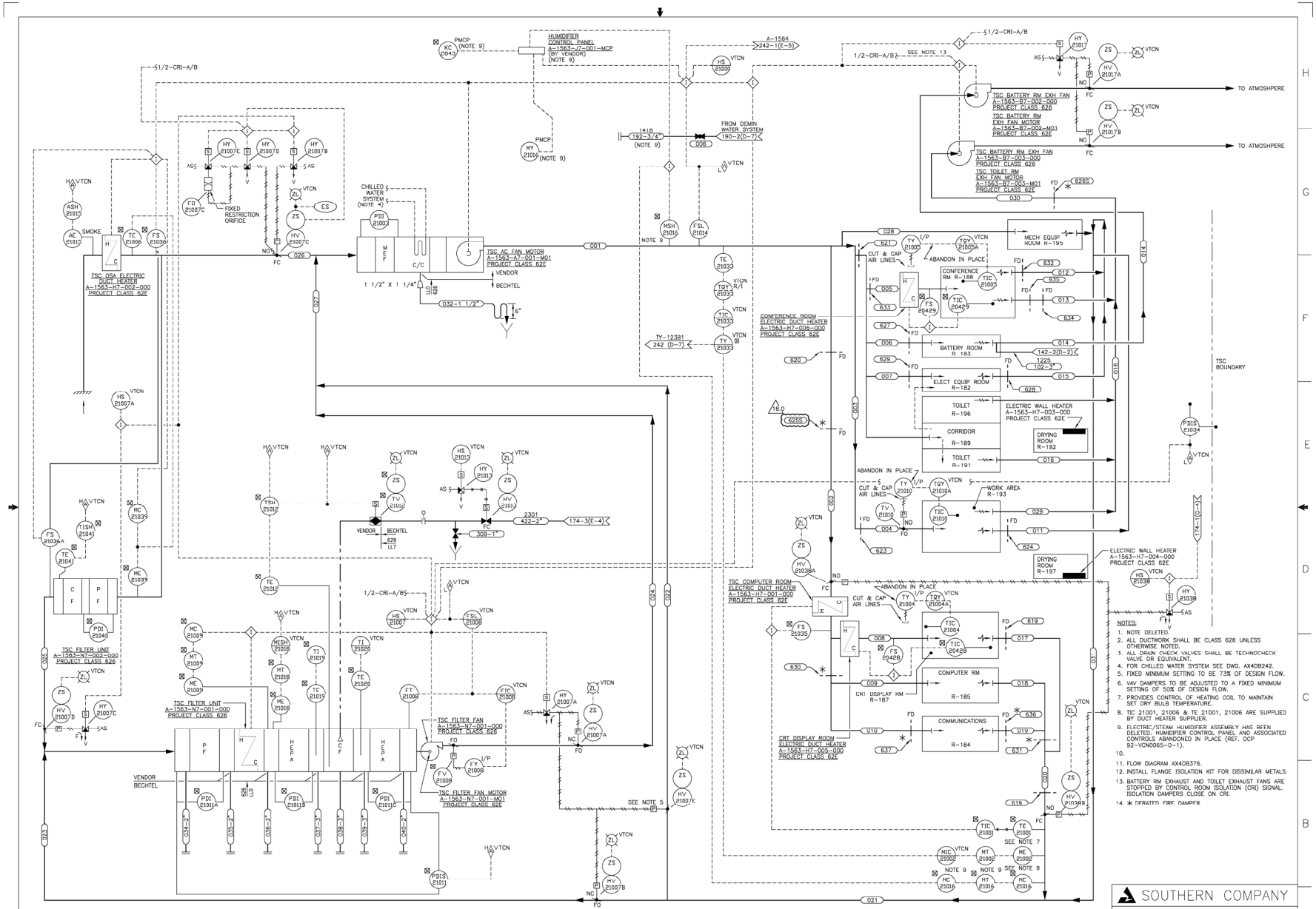
CAD AX4DB232
ACAD2004 NJ ~ 01

SOUTHERN COMPANY

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
NORMAL CHILLED WATER
COOLING COILS
SYSTEM NO. 1591

SCALE: NONE
DRAWING NO. AX4DB232
VER. 31.0
JOB NO.10604
SIZE E 34044
DRAWING CATEGORY: CRITICAL



NOTES:

1. NOTE DELETED.
2. ALL DUCTWORK SHALL BE CLASS 626 UNLESS OTHERWISE NOTED.
3. ALL DRAIN CHECK VALVES SHALL BE TECHNOCHECK VALVE OR EQUIVALENT.
4. FOR CHILLED WATER SYSTEM SEE DWG. AX4DB242.
5. FIXED MINIMUM SETTING TO BE 73% OF DESIGN FLOW.
6. VAV DAMPERS TO BE ADJUSTED TO A FIXED MINIMUM SETTING OF 50% OF DESIGN FLOW.
7. PROVIDES CONTROL OF HEATING COIL TO MAINTAIN SET DRY BULB TEMPERATURE.
8. TIC 21001, 21006 & TE 21001, 21006 ARE SUPPLIED BY DUCT HEATER SUPPLIER.
9. ELECTRIC/STEAM HUMIDIFIER ASSEMBLY HAS BEEN DELETED. HUMIDIFIER CONTROL PANEL AND ASSOCIATED CONTROLS ABANDONED IN PLACE (REF. DCP 92-VCN0065-0-1).
10.
11. FLOW DIAGRAM AX4DB376.
12. INSTALL FLANGE ISOLATION KIT FOR DISSIMILAR METALS.
13. BATTERY RM EXHAUST AND TOILET EXHAUST FANS ARE STOPPED BY CONTROL ROOM ISOLATION (CRI) SIGNAL. ISOLATION DAMPERS CLOSE ON CRI.
14. * DENOTES FIRE DAMPER

CAD AX4DB235
Acad2004 TJS - 02

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NO.	VERSIONS	DATE	DR	CHK	APPV	UTL
18.0	REVISED PER ABN-V00596, VER. 1.0.	3-15-07	TJS	MWD	WLN	X

SOUTHERN COMPANY

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
ONSITE TECHNICAL SUPPORT CENTER
HVAC SYSTEM
SYSTEM NO. 1563

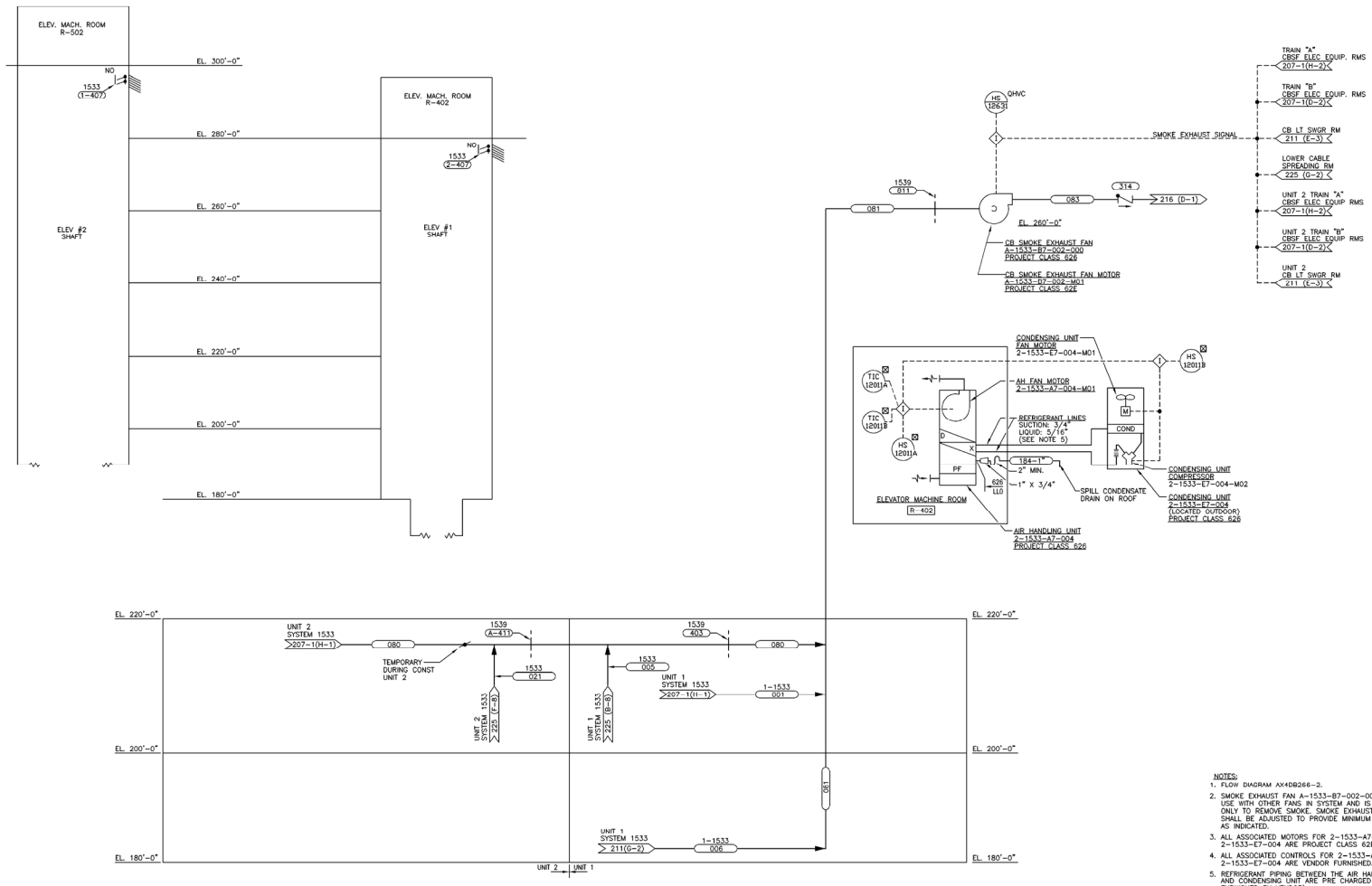
SCALE: NONE

DRAWING NO. AX4DB235 VER. 18.0

JOB NO.10604

SIZE E 34X44

DRAWING CATEGORY: CRITICAL



NOTES:
1. FLOW DIAGRAM AX4DB266-2.
2. SMOKE EXHAUST FAN A-1533-B7-002-000 IS NOT IN USE WITH OTHER FANS IN SYSTEM AND IS OPERATED ONLY TO REMOVE SMOKE. SMOKE EXHAUST QUANTITY SHALL BE ADJUSTED TO PROVIDE MINIMUM AIR QUALITIES AS INDICATED.
3. ALL ASSOCIATED MOTORS FOR 2-1533-A7-004 AND 2-1533-E7-004 ARE PROJECT CLASS 62E.
4. ALL ASSOCIATED CONTROLS FOR 2-1533-A7-004 AND 2-1533-E7-004 ARE VENDOR FURNISHED.
5. REFRIGERANT PIPING BETWEEN THE AIR HANDLING UNIT AND CONDENSING UNIT ARE PRE CHARGED (TO BE FURNISHED BY VENDOR).
6. FIRE ZONE INTERLOCK FROM LOCAL ZONE INDICATING PANEL NO. LZIP IS34 TO TORNADO DAMPER 1-1533-W7-407 HAS BEEN DISABLED PER DCP 87-V1E0219-1-1.

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NO.	VERSIONS	DATE	DR	CHK	APPV	DTL
13.0	REDRAWN PER ABN-V70117, VER. 1.0	08/22/06	CYN	LCF	EPD	X

CAD AX4DB237
ACAD2004 | CYN-01

SOUTHERN COMPANY

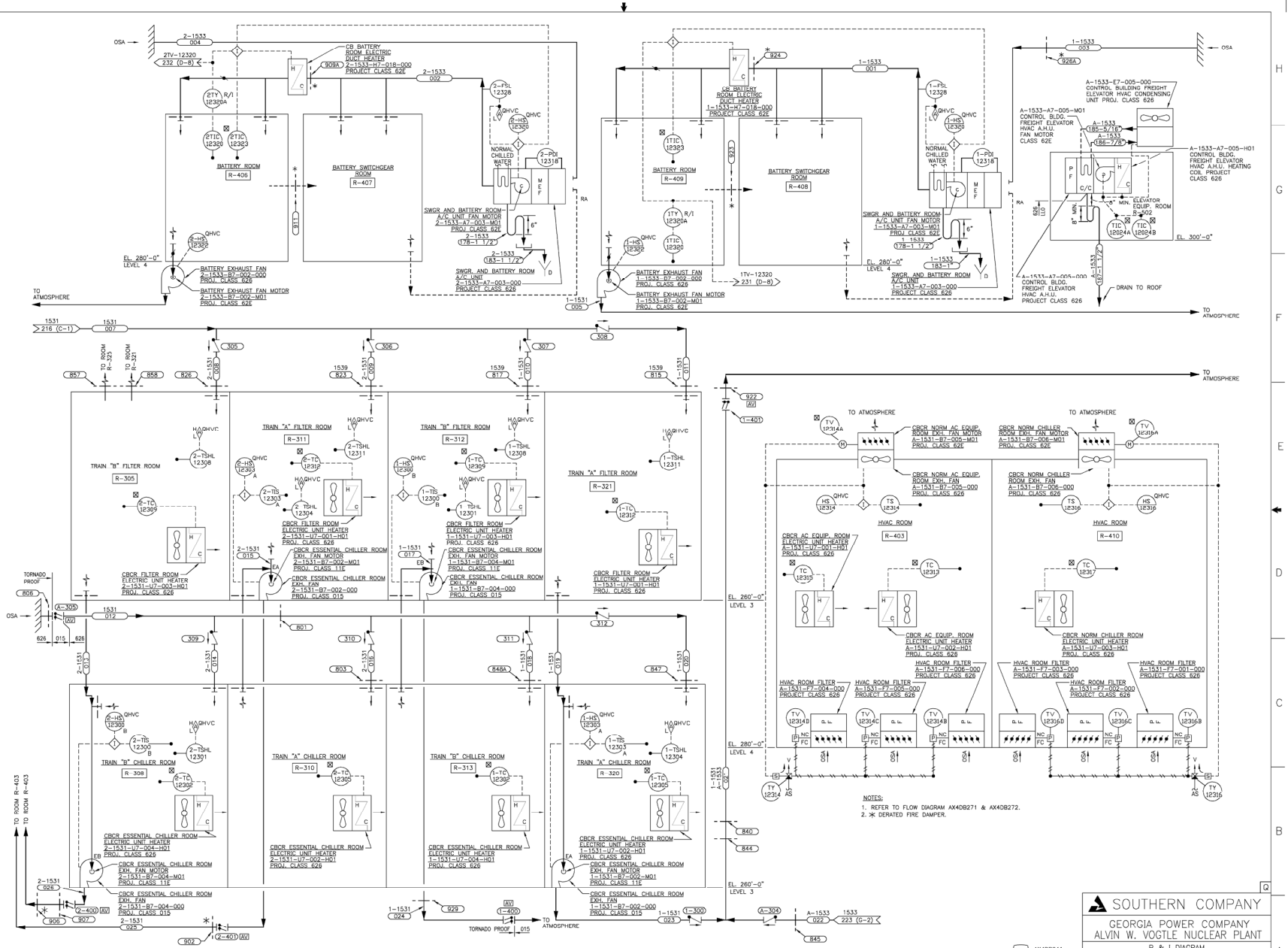
GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
CONTROL BUILDING
SMOKE EXHAUST SYSTEM
SYSTEM NO. 1533

SCALE: NONE
JOB NO.10604
SIZE E 34044

DRAWING NO. AX4DB237
VER. 13.0

DRAWING CATEGORY: CRITICAL



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NO.		DATE	DR	CHK	APPV	DTL
13.0	REDRAWN PER ABN-V70052, VER. 1.0	6-1-06	CYN	LCF	EPD	X

SIZE E 34X44

CAD AX4DB241
ACAD2002 CYN- 01

SOUTHERN COMPANY

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

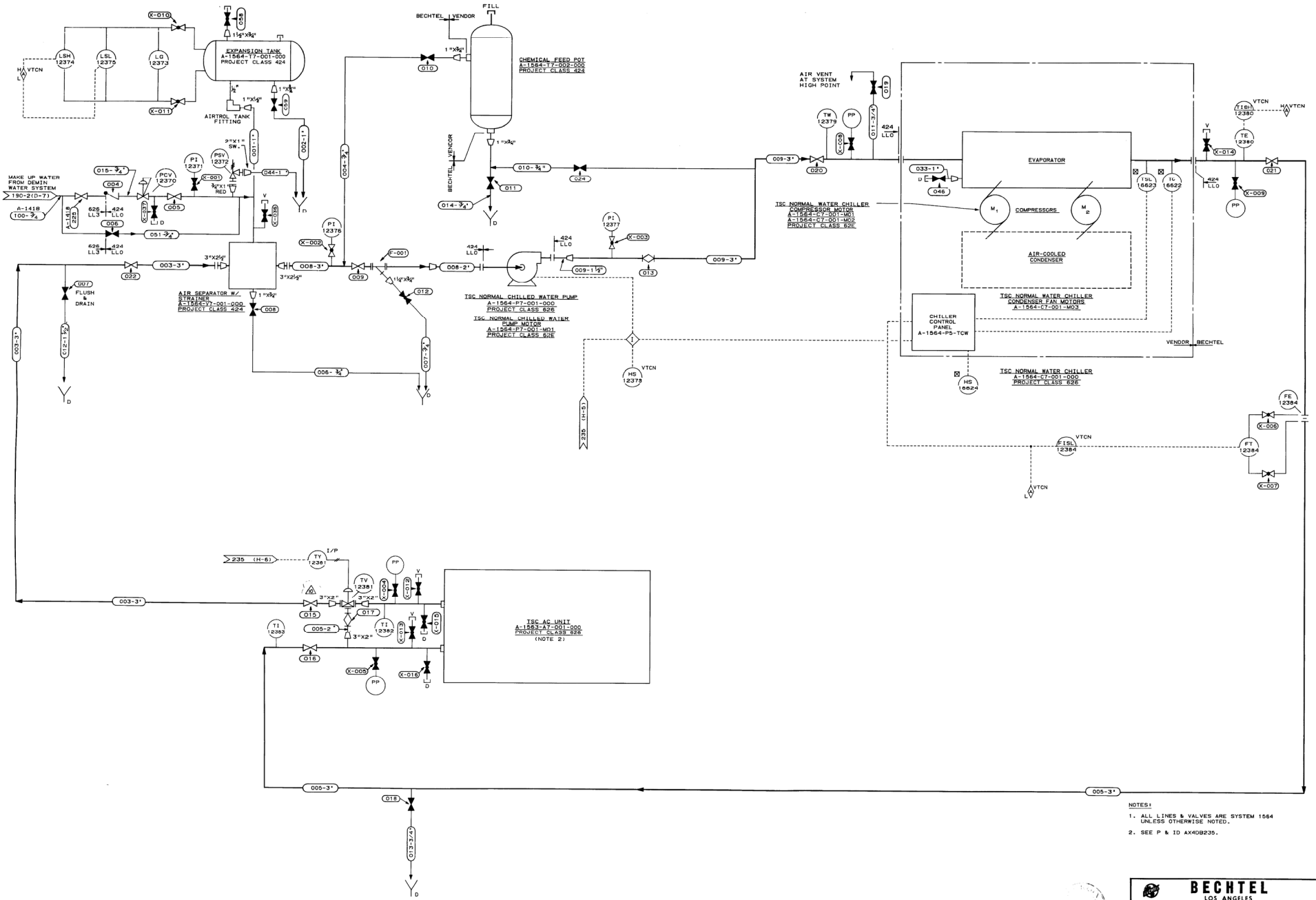
P & I DIAGRAM
CONTROL BUILDING HVAC AND ELECTRICAL EQUIPMENT ROOMS LEVEL 3 & 4
SYSTEM NO. 1531 & 1533

SCALE: NONE

DRAWING NO. AX4DB241 VER. 13.0

JOB NO.10604

1 DRAWING CATEGORY: CRITICAL



NOTES:

1. ALL LINES & VALVES ARE SYSTEM 1564 UNLESS OTHERWISE NOTED.
2. SEE P & ID AX4DB235.

NO.	REVISIONS	DATE	DR.	CHK.	SUPV.	LLG	EGS	CHF.E	P.E.	Q.A.E.
1	INCORPORATED M-FCRB-15423	12-8-86	RK	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
2	INCORPORATED M-FCRB-5868-P AND M-FCRB-14,235	2-24-94	RK	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
3	INCORP. DCN'S 17(M-FCRB-13126) AND 18(PFE)	10-26-94	MFK	SEE MICROFILM FOR SIGNATURES						
4	INCORPORATE DCN NO. 18: TRANSFER TO PFE.	9-30-86	SEE MICROFILM FOR SIGNATURES							
5	INCORPORATED DCN 14 & 15 (M-FCRB-10806)	11-7-85	SEE MICROFILM FOR SIGNATURES							
6	INCORP DCN 11 (M-FCRB-13551 & 13 DCN 12 INCORP IN REV 4	06-06-85	SEE MICROFILM FOR SIGNATURES							
7	INCORPORATED DCN 9 & 10	03-11-85	SEE MICROFILM FOR SIGNATURES							
8	INCORPORATED DCNS 4,5,6,7 & 8	12-28-84	SEE MICROFILM FOR SIGNATURES							
9	INCORP DCN 3	06-07-84	SEE MICROFILM FOR SIGNATURES							
10	INCORP DCN 1&2	12-28-83	SEE MICROFILM FOR SIGNATURES							
11	INCORPORATED DCN 0 & 1 ISSUED FOR CONSTRUCTION	01-13-83	SEE MICROFILM FOR SIGNATURES							

ENG. MGR. DATE

BECHTEL
LOS ANGELES

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
ONSITE TECHNICAL SUPPORT CENTER
CHILLED WATER SYSTEM
SYSTEM NO. 1564

SCALE: NONE | DRAWING NO. AX4DB242-1 | REV. 10
JOB NO. 9510

34X44 "E" SIZE

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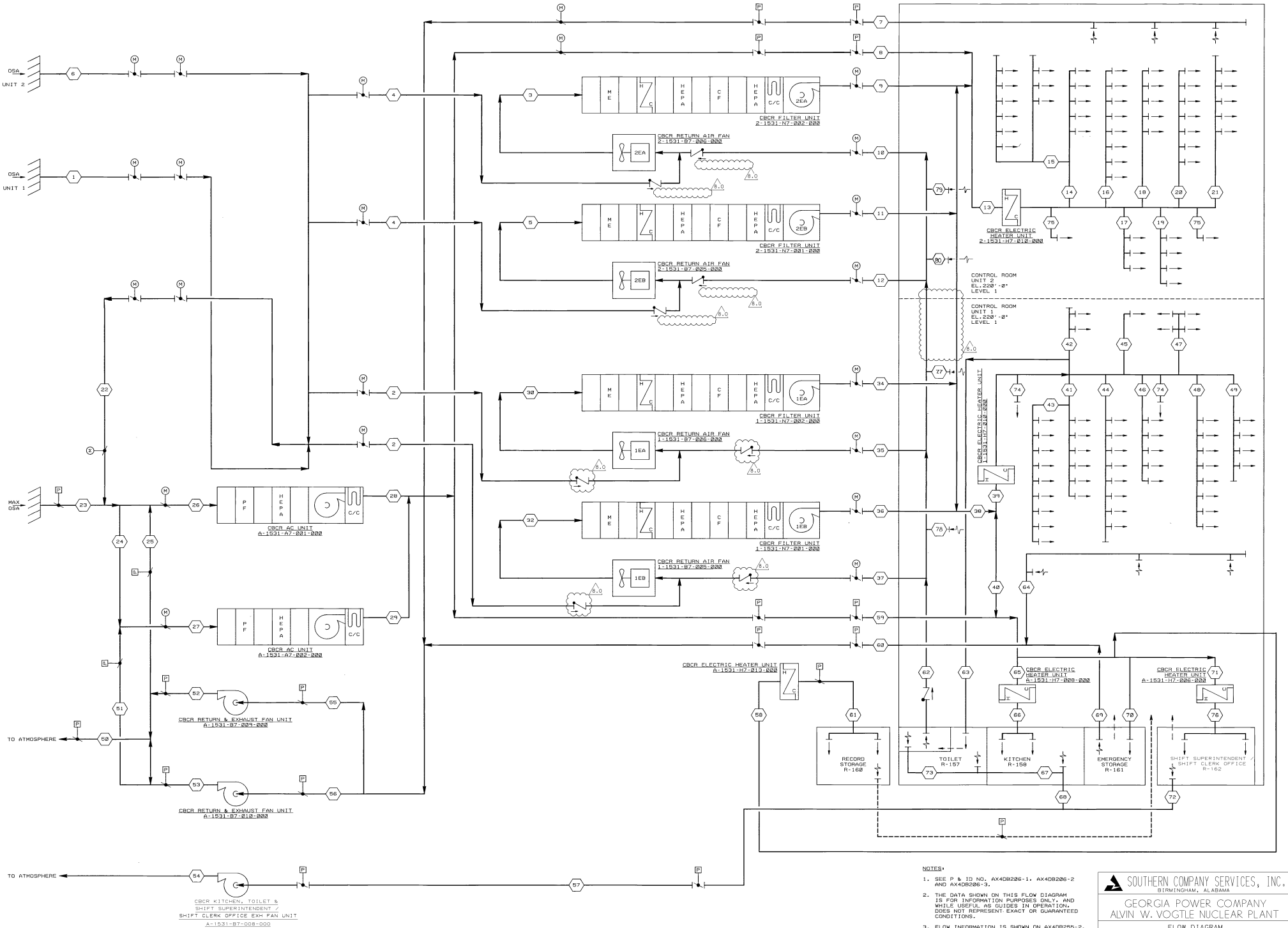

NOTES:

1. SEE P & ID NO. AX4DB206-1, AX4DB206-2 AND AX4DB206-3.
2. THE DATA SHOWN ON THIS FLOW DIAGRAM IS FOR INFORMATION PURPOSES ONLY, AND WHILE USEFUL AS GUIDES IN OPERATION, DOES NOT REPRESENT EXACT OR GUARANTEED CONDITIONS.
3. FLOW INFORMATION IS SHOWN ON AX4DB255-2.

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NO.	VERSIONS	DATE	DR	CHK	APPV
8.0	REVISED PER ABN-V00466, VER.1.0	1-19-07	BgT	MWD	WLN

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

FLOW DIAGRAM
CONTROL BUILDING
CONTROL ROOM HVAC SYTEM LEVEL 1
SYSTEM NO. 1531

SCALE: NONE	DRAWING NO.	VER.
JOB NO.10604	AX4DB255-1	8.0

SIZE E 34X44

DWG CATEGORY: CRITICAL

Vogtle Electrical Generating Plant
Unit No. 1 / 2

Document Number:

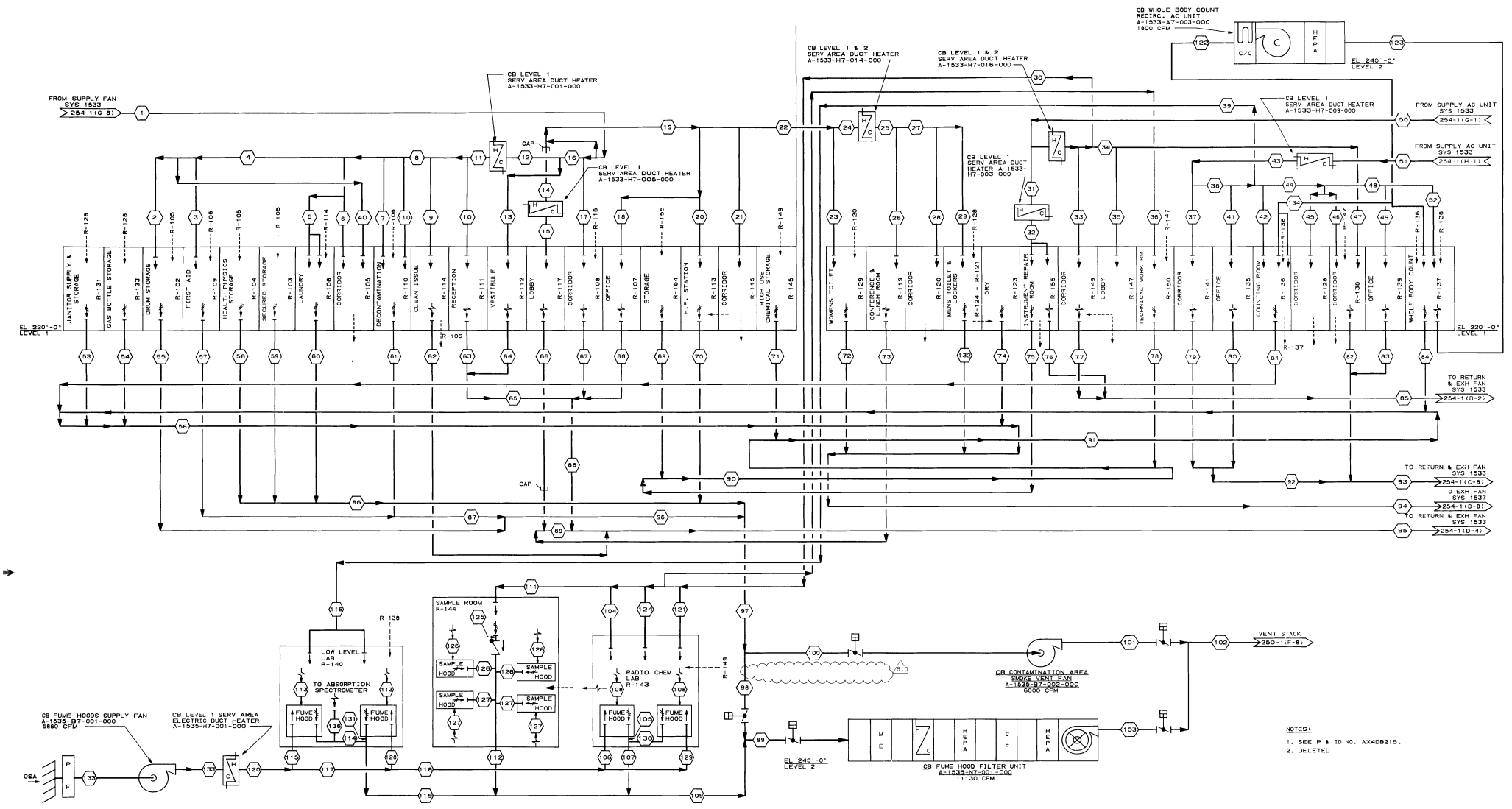
AX4DB255-3

Document Title:
AIR QUANTITY FOR AIR BALANCE CONTROL BLDG CONTROL RM HVAC SYS LVL 1 PIPING AND INSTRUMENTATION

VOIDED

Voided Per ABN-V00466, VER.1.0

Ver. 6.0



MODE	PARAMETER		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50
SUMMER	FLOW	CFM	10925	765	400	1690	1000	525	330	4175	845	510	5530	5530	165	580	580	5695	520	650	4130	600	520	2360	200	2160	2160	1250	910	610	300	1750	1050	1050	545	1570	770	400	200	4180	1020	525	420	840	4380	1900	625	700	800	575	200	4895
	TEMP	°F	57	59	59	59	59	59	59	58	59	59	58	58	59	59	59	58	59	59	57	59	59	57	59	58	58	59	59	59	59	58	59	59	59	59	59	59	59	58	58	59	59	59	58	58	59	59	59	59	59	58
	PRESSURE	INCH OF WATER																																																		
WINTER	FLOW	CFM	10925	765	400	1690	1000	525	330	4175	845	510	5530	5530	165	580	580	5695	520	650	4130	600	520	2360	200	2160	2160	1250	910	610	300	1750	1050	1050	545	1570	770	400	200	4180	1020	525	420	840	4380	1900	625	700	800	575	200	4895
	TEMP	°F	57	69	69	69	69	69	69	69	69	69	69	58	59	59	59	58	59	59	57	59	59	57	59	58	58	59	59	59	59	58	59	59	59	59	59	59	59	65	64	58	69	65	65	64	64	64	64	59	64	64	58
	PRESSURE	INCH OF WATER																																																	

MODE	PARAMETER		51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90	91	92	93	94	95	96	97	98	99	100	101	102	103	104
SUMMER	FLOW	CFM	4480	375	150	125	840	2175	440	860	210	1425	730	715	510	165	675	580	800	650	100	625	150	250	1250	365	95	790	460	215	645	420	925	1550	200	415	1750	2495	1170	2125	1830	195	540	1065	2815	3145	4670	2010	5130	5130	11130	0	0	11130	11130	310
	TEMP	°F	56	59	73	88	73	NA	75	85	86	75	78	73	76	72	NA	76	68	76	78	78	70	75	75	80	76	76	67	78	78	74	75	76	74	75	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	59
	PRESSURE	INCH OF WATER																																																			
WINTER	FLOW	CFM	4480	375	150	125	840	2175	440	860	210	1425	730	715	510	165	675	580	800	650	100	625	150	250	1250	365	95	790	460	215	645	420	925	1550	200	415	1750	2495	1170	2125	1830	195	540	1065	2815	3145	4670	2010	5130	5130	11130	0	0	11130	11130	310
	TEMP	°F	58	64	79	94	64	NA	70	95	96	85	72	83	70	70	NA	72	68	76	78	78	68	75	70	80	76	76	65	75	75	70	75	76	72	75	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	59
	PRESSURE	INCH OF WATER																																																		

MODE	PARAMETER		105	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130	131	132	133	134	135	136
HOOD DOOR OPEN (SUMMER TEMPS)	FLOW	CFM	1225	715	2450	510	6000	335	400	1000	535	1250	715	1020	2145	1430	2550	2860	310	1800	1800	310	0	250	250	715	715	1225	1250	355	2860	1400		50
	TEMP	°F	90	98	NA	79	NA	59	59	NA	74	90	98	59	98	98	NA	98	59	60	75	59	——	75	75	98	NA	NA	NA	77	98	62			
	PRESSURE	INCH OF WATER																																		
HOOD DOOR CLOSED (WINTER TEMPS)	FLOW	CFM	1225	715	2450	510	6000	335	400	1000	535	1250	715	1020	2145	1430	2550	2860	310	1800	1800	310	1000	0	0	715	715	1225	1250	355	2860	1400		50
	TEMP	°F	62	56	NA	72	NA	69	59	NA	72	62	56	65	56	56	NA	56	59	60	75	59	70	——	——	56	NA	NA	NA	77	17	62			
	PRESSURE	INCH OF WATER																																		

MODE	PARAMETER		98	99	100	101	102	103	122	123
SMOKE EXH.	FLOW	CFM	0	0	6000	6000	6000	0	0	0
	PRESSURE	INCH OF WATER		——				——	——	——
NORMAL	FLOW	CFM	5130	11130	0	0	11130	11130	1800	1800
	PRESSURE	INCH OF WATER			——	——				

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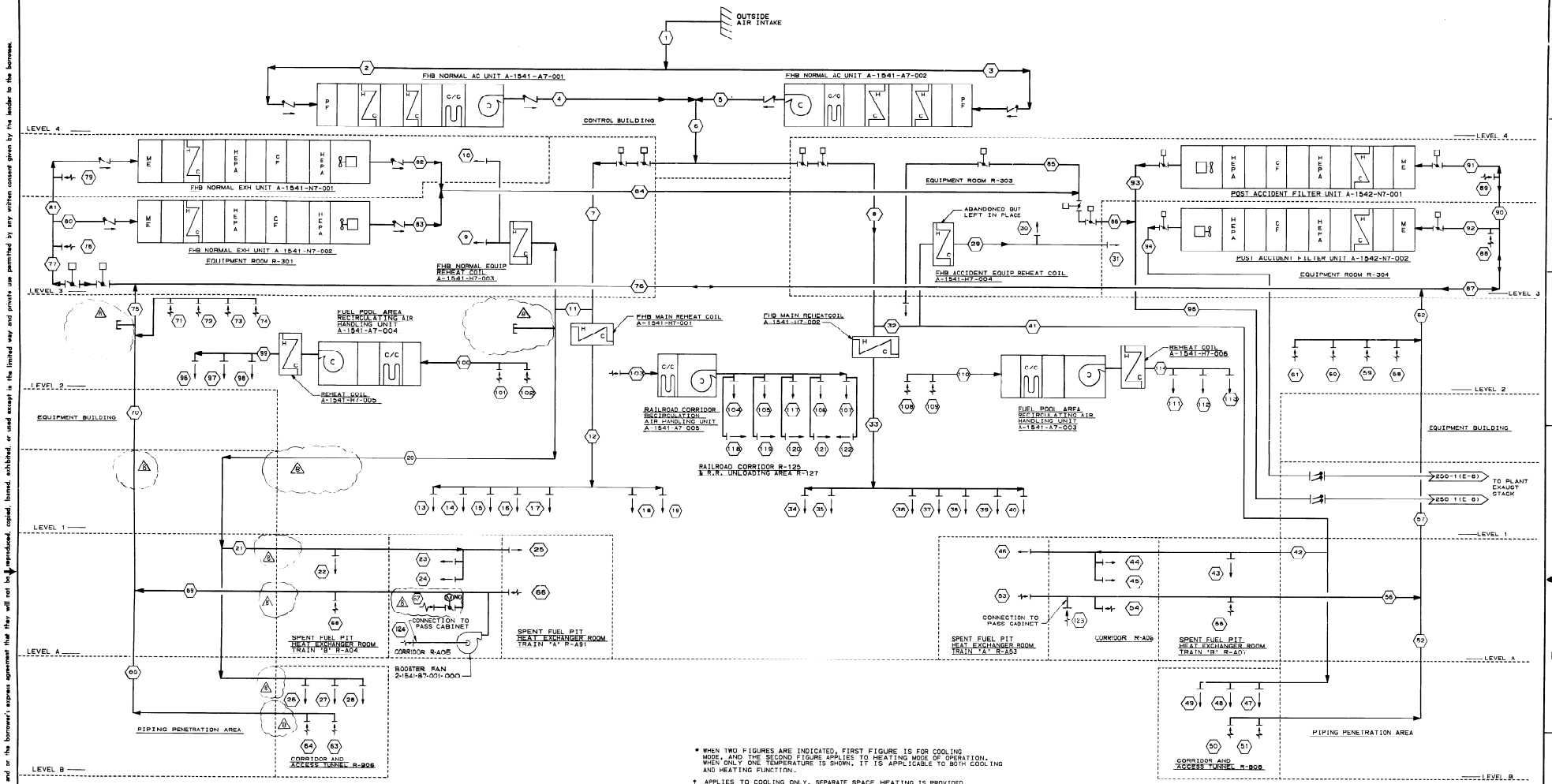
NO.	VERSIONS	DATE	DR	CHK	APPV
8.0	REVISED PER ABN-V00466, VER.1.0	1-19-07	BGT	MWD	WLN

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

FLOW DIAGRAM
CONTROL BLDG HVAC SYSTEM
LEVEL 1
SYSTEM No. 1533, 1535, 1537

SCALE: NONE
JOB NO. 10604
DRAWING NO. AX4DB256-1
VER. 8.0
SIZE E 34X44
DWG CATEGORY: CRITICAL



• WHEN TWO FIGURES ARE INDICATED, FIRST FIGURE IS FOR COOLING MODE, AND THE SECOND APPLIES TO HEATING MODE OF OPERATION. WHEN ONLY ONE TEMPERATURE IS SHOWN, IT IS APPLICABLE TO BOTH COOLING AND HEATING FUNCTION.

† APPLIES TO COOLING ONLY. SEPARATE SPACE HEATING IS PROVIDED.

MODE	PARAMETER		(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)	(17)	(18)	(19)	(20)	(21)	(22)	(23)	(24)	(25)	(26)	(27)	(28)	(29)	(30)	(31)	(32)	(33)	(34)	(35)	(36)	(37)	(38)	(39)	(40)	(41)	(42)
NORMAL	FLOW	CFM	25300	25300	25300	25300	25300	25300	12690	12690	750	750	2790	9900	1400	1400	1420	1420	1420	1420	1420	1290	780	220	170	170	220	170	170	170	1500	150	150	2790	9900	1420	1420	1420	1420	1420	1400	1400	1290	780
	TEMP *	°F	98/17	98/17	98/17	56	56	56	56	56	56/76	56/76	56	89	89	56	89	89	89	89	89	56	56	56	56	56	56	50	50	50	50/97	50/97	50/97	50	89	89	89	89	89	89	89	89	56	56
	PRESSURE	IN.OF WATER	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	◄	—	—	—	RANGE	—	—	—	—	►+0.5	—	—	—	—	—	—	—	—	—	—	—	◄RANGE►	
FUEL ACCIDENT	FLOW	CFM	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	TEMP	°F	◄						◄													SEE NOTE 2																							
	PRESSURE	IN.OF WATER	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
UNIT 1 & COMMON	FLOW	CFM	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
	TEMP	°F	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
	PRESSURE	IN.OF WATER	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—

MODE	PARAMETER		(43)	(44)	(45)	(46)	(47)	(48)	(49)	(50)	(51)	(52)	(53)	(54)	(55)	(56)	(57)	(58)	(59)	(60)	(61)	(62)	(63)	(64)	(65)	(66)	(67)	(68)	(69)	(70)	(71)	(72)	(73)	(74)	(75)	(76)	(77)	(78)	(79)	(80)	(81)	(82)	(83)	(84)
NORMAL	FLOW	CFM	220	170	170	220	170	170	170	325	325	650	248	430	280	1158	1808	2745	2700	2700	2745	12698	325	325	650	280	430	280	1190	1840	2745	2700	2700	2746	12730	14348	27078	825	825	28728	27903	28728	28728	28726
	TEMP *	°F	56	56	56	56	56	56	56	104/40	104/40	104/40	104/40	104/40	104/40	104/40	104/40	104/92	104/92	104/92	104/92	104/40	104/40	104/40	104/40	104/40	104/40	104/40	104/40	104/40	104/92	104/92	104/92	104/19	104/40	104/40	104/40	104/40	104/40	104/40	104/40	104/40	104/40	104/40
	PRESSURE	IN.OF WATER	-0.55	—	—	—	—	—	►+0.5	►-0.25	►-1.75	—	►-1.75	—	►-0.25	►-1.75	—	—	—	—	—	—	►-0.25	—	—	RANGE	—	►-1.75	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
FUEL ACCIDENT	FLOW	CFM	0	0	0	0	0	0	0	◄	SEE NOTE 4	0	0	SEE NOTE 4	0	0	0	0	0	0	0	0	SEE NOTE 4	SEE NOTE 4	0	(NOTE 5)	►	0	0	SEE NOTE 4	►	►	►	►	0	NOTE 4	0	0	0	►	►	►	►	►
	TEMP	°F	◄																						SEE NOTE 2																				
	PRESSURE	IN.OF WATER	—	—	—	—	NOTE 3	—	—	-0.25	►-1.75	—	►-0.25	—	—	—	—	—	—	—	—	►-1.75	►-0.25	—	—	RANGE	—	—	—	—	►-1.75	—	—	—	—	—	—	—	—	—	—	—	—	—	—
UNIT 1 & COMMON	FLOW	CFM	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
	TEMP	°F	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
	PRESSURE	IN.OF WATER	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—

MODE	PARAMETER		(85)	(86)	(87)	(88)	(89)	(90)	(91)	(92)	(93)	(94)	(95)	(96)	(97)	(98)	(99)	(100)	(101)	(102)	(103)	(104)	(105)	(106)	(107)	(108)	(109)	(110)	(111)	(112)	(113)	(114)	(115)	(116)	(117)	(118)	(119)	(120)	(121)	(122)	(123)	(124)	(125)	(126)
NORMAL	FLOW	CFM	28728	28728	1650	200	200	780	0	0	0	0	28728	850	825	825	2500	2500	1250	1250	3100	310	310	310	310	1250	1250	2900	650	625	625	2500	DELETED	DELETED	310	310	310	310	310	200	100	
	TEMP *	°F	104/40	104/40	104/40	104/40	104/40	◄	NOTE 2	►	104/40	104	104	104	104	104	104	104	104	104	104	56	56	56	56	104	104	104	104	104	104	104	104			56	56	56	56	56	56	104/40	104/40	
	PRESSURE	IN.OF WATER	—	—	—	—	—	-1.75	-1.75	-1.75	0	0	►+0.50	►+0.60	►+0.70	►+0.80	-0.70	-0.60	-0.60	-0.60	—	—	—	—	—	-0.50	-0.60	-0.70	-0.70	-0.60	-0.60	-0.60	—	—	—	—	—	—	—	—	—	NOTE 6	
FUEL ACCIDENT	FLOW	CFM	27000	0	◄	—	NOTE 4	►	5000	5000	5000	650	625	625	2500	2500	1250	1250	3200	600	600	600	600	600	1250	1250	2900	650	625	625	2500	—	—	—	—	—	—	—	—	—	NOTE 5	
	TEMP	°F	NOTE 2	◄			NOTE 2									NOTE 5			►	104†	56†	56†	56†	56†	◄				NOTE 5			►									NOTE 2	
	PRESSURE	IN.OF WATER	0	0	-2.0	-2.0	-2.0	-2.0	-3.0	+3.0	+3.0	+0.50	+0.60	-0.70	+0.80	-0.70	+0.60	0.50	0.50	-0.60	-0.60	-0.55	-0.50	0.00	-0.60	-0.70	-0.70	-0.60	-0.50	-0.60	—	—	—	—	—	—	—	—	—	—	NOTE 6	
UNIT 1 & COMMON	FLOW	CFM	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	
	TEMP	°F	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	
	PRESSURE	IN.OF WATER	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	

GENERAL NOTES:

1. DELETED
2. TEMPERATURE IS NOT CONTROLLED DURING ACCIDENT CONDITIONS.
3. DELETED
4. FHB MAXIMUM TOTAL ALLOWABLE INLEAKAGE FLOW IS 3000 CFM UNMITIGATED MODE. FHB NORMAL EXHAUST UNIT MAY BE ALLOWED TO OPERATE FOR RECIRCULATION UNDER ACCIDENT CONDITIONS.
5. REFER TO P&ID DWG NO. AX4DB204-1 & AX4DB204-2.
6. DELETED
7. RUN THE FOLLOWING UNITS TO PROVIDE THE FLOWS INDICATED:
   - A-1541-A7-001 25,380 CFM
   - A-1541-A7-002 28,380 CFM
   - A-1541-N7-001 28,728 CFM
   - A-1541-N7-002 28,728 CFM
8. SLIGHTLY NEGATIVE RELATIVE TO ITS SURROUNDING SPACE.
9. DURING PASS PANEL SAMPLING, PASS BLOWER 2-1541-B7-001 SHALL BE RUN WITH PASS BLOWER OPERATING. FLOW AT NODE (68) WILL BE 250 CFM AND 0 AT NODE (67).

NO.	REVISIONS	DATE	DR.	CHK.	DM.	EQ.	EGS	CHF.E	P.E.	Q.A.E
4	INCORPORATED DCN# 4	12-87	MR	MK	REP					
3	INCORPORATED DCN # 3	4-87		MK						
2	INCORPORATED DCN # 2	8-87		MK						
1	INCORPORATED M-FCRB-16,401 (MOD)	6-86		MK						
A	INCORP. ABN 88-VIE099A006 (MODIFIED)	11-86	MR	MK						

NO.	REVISIONS	DATE	DR.	CHK.	SUPV.	EQ.	EGS	CHF.E	P.E.	Q.A.E
4	INCORPORATED M-FCRB-15,718 AND DCN 1	4-87				SEE MICROFILM FOR SIGNATURES				
3	INCORP. M-FCRB-14,632 & M-FCRB-14,697 MOD (PFE) REVISED Q-LIST & ADDED NOTES 8,9 AND DATA FOR UNIT 1 & COMMON	8-87				SEE MICROFILM FOR SIGNATURES				
2	ISSUED FOR CONSTRUCTION	10-86				SEE MICROFILM FOR SIGNATURES				
1										

ENG. MGR. DATE

BECHTEL
LOS ANGELES

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

FLOW DIAGRAM
FUEL HANDLING BLDG. HVAC SYSTEM
SYSTEM 1541 & 1542

SCALE: NONE | DRAWING NO. AX4DB260-1 | REV. 8
JOB NO. 9510

30044 "E" SIZE

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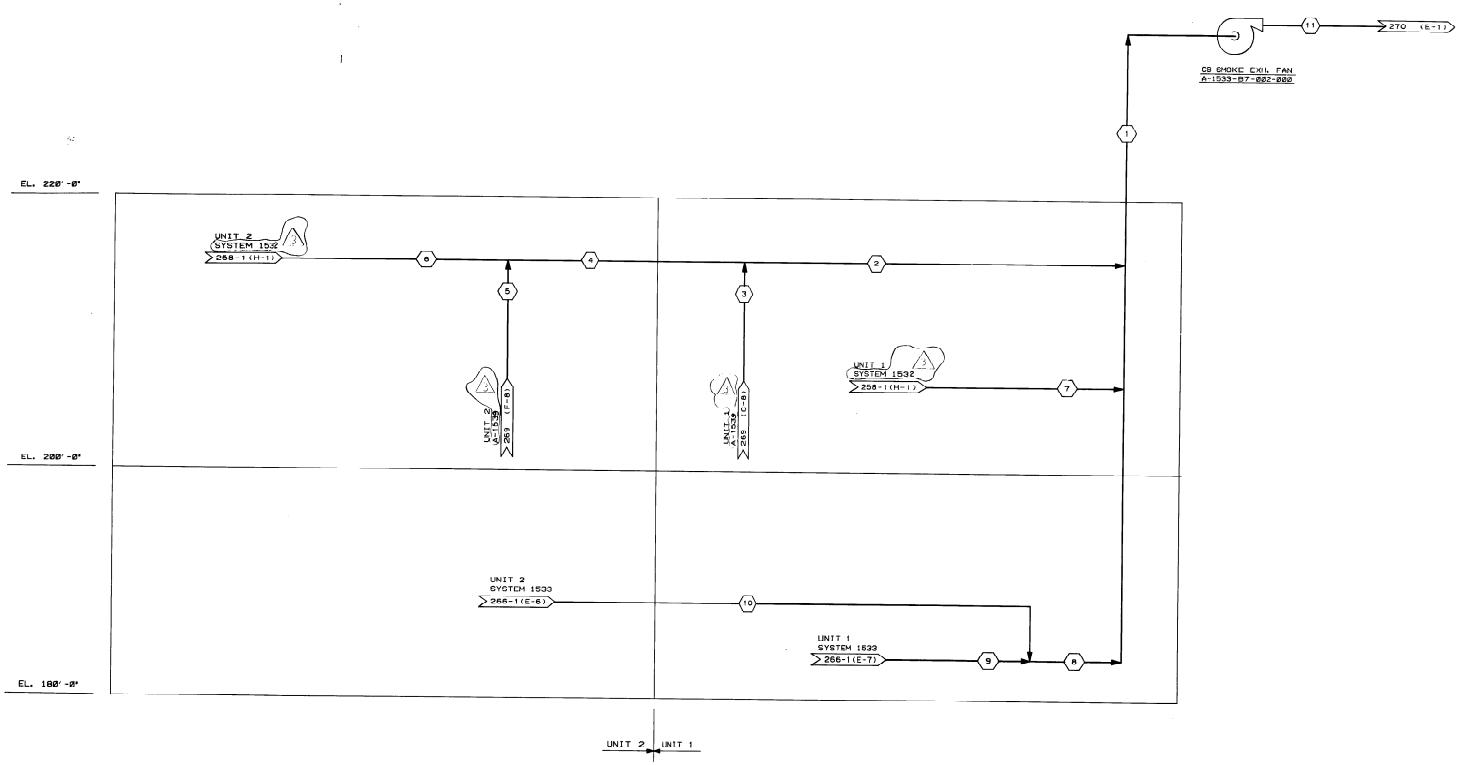


MODE	PARAMETER	1	2	3	4	5	6	7	8	9	10	11
SMOKE EXHAUST	FLOW CFM	28000	11000	5765 △2	11000	5460 △3	11000	9900 △3	20000	20000	18250 △3	20000
	TEMPERATURE °F	SEE NOTE 3	SEE NOTE 3	SEE NOTE 3	SEE NOTE 3	SEE NOTE 3	SEE NOTE 3	SEE NOTE 3	SEE NOTE 3	SEE NOTE 3	SEE NOTE 3	SEE NOTE 3
	PRESSURE INCH OF WTR	3.6	0.25	3.25	3.0	3.0	2.75	3.0	3.25	3.25	3.25	1.5

NOTES:

1. SEE P & ID AX4DB237, 1X4DB207-1, 2X4DB207-1 & AX4DB225.
2. DELETED
3. TEMPERATURE DURING SMOKE REMOVAL OPERATION IS NOT CONTROLLED.
4. FLOW VALUES SHOWN FOR THE SMOKE EXHAUST SYSTEM ARE MINIMUM QUANTITIES. NO MAXIMUM VALUES ARE REQUIRED.

NO.	REVISIONS	DATE	DR.	CHK.	SUPV.	EQ.	EGS	CHF. E	P.E.	Q.A.E.
△	INCORPORATED DCN-7	3-1888	BMS	KBN						
△	INCORPORATED G-PCRB-0934 & DCN # 3 (PFR)	12-5-87	PFR	GG	WA					
△	INCORPORATED DCN 1 & 2	3-9-88	CAG	SEE MICROFILM FOR SIGNATURES						
△	ISSUED FOR CONSTRUCTION	11-8-84	CAG	SEE MICROFILM FOR SIGNATURES						

ENG. MGR. DATE

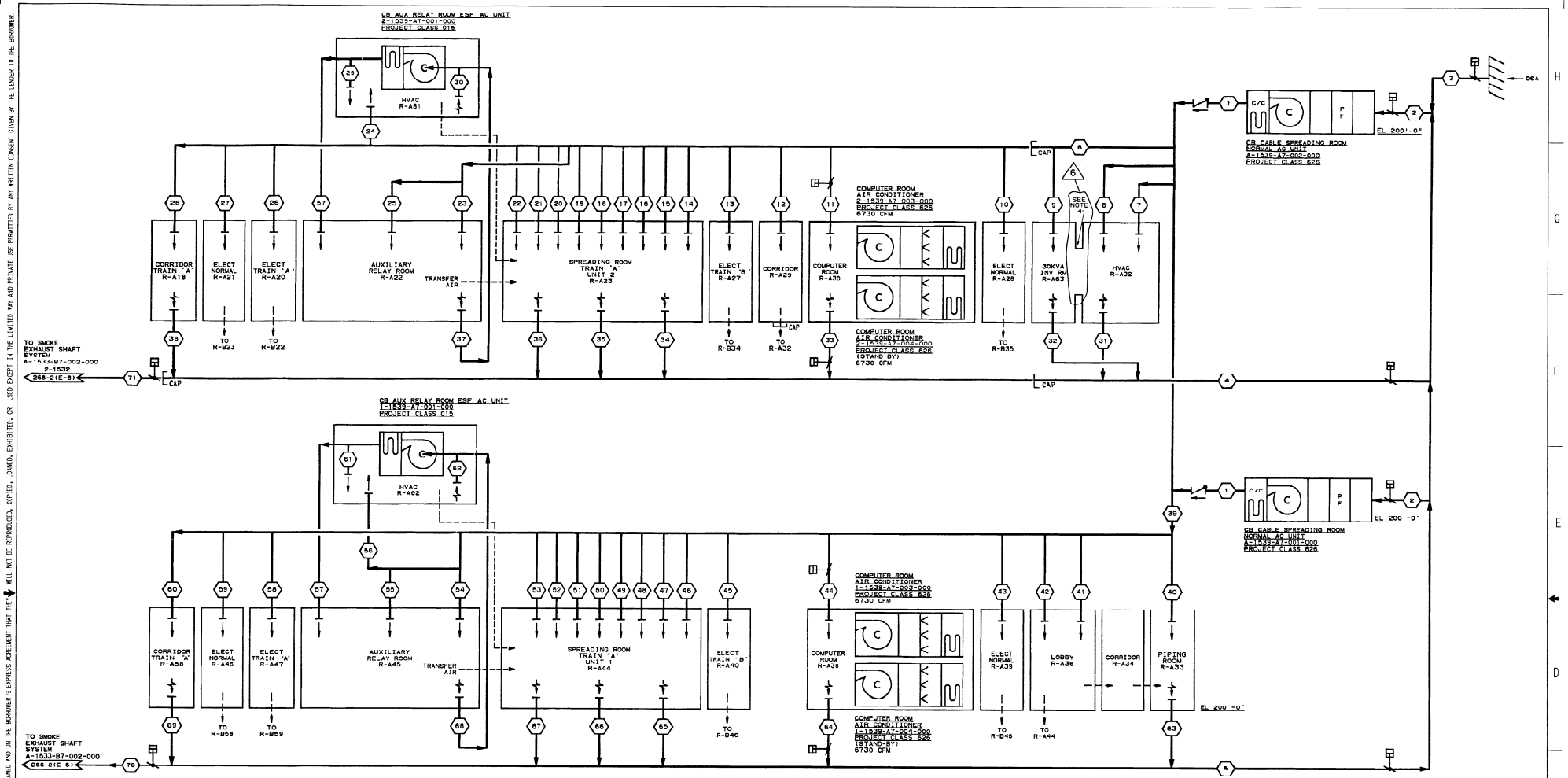
BECHTEL
LOS ANGELES

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

FLOW DIAGRAM
CONTROL BUILDING
SMOKE EXHAUST SYSTEM
SYSTEM NO. 1533

SCALE: NONE | DRAWING NO. AX4DB266-2 | REV. 3

JOB NO. 9510



NOTES

1. P & ID DWG AX4DB225.
2. THE DATA SHOWN ON THIS FLOW DIAGRAM IS FOR INFORMATION PURPOSES ONLY, AND WHILE USEFUL AS GUIDES IN OPERATION, DOES NOT REPRESENT EXACT OR GUARANTEED CONDITIONS.
3. THE AIR FLOWS SHOWN ARE FOR UNIT 1 AND UNIT 2 COMBINED OPERATION. AIR FLOWS FOR UNIT 1 OPERATION ONLY ARE INDICATED ON DRAWING NO. 1A4DB273.
4. DOOR V2211L1A65 REMOVED BETWEEN ROOMS A63 AND A32 TO EQUALIZE THE COOLING LOAD.

MODE	PARAMETER		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35
SUMMER	FLOW	CFM	12645	12645	1420	5460	5765	6320	445	445	300	100	200	470	100	350	200	400	350	200	400	350	200	400	215	75	210	100	100	600	200	200	1160	300	200	1120	1115
	TEMP	°F	57	76	98	73	73	58	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	62	65	71	96	75	64	64
	PRESSURE	INCH OF WTR																																			
WINTER	FLOW	CFM	12645	12645	1420	5460	5765	5320	445	445	300	100	200	470	100	350	200	400	350	200	400	350	200	400	215	75	210	100	100	600	200	200	1160	300	200	1120	1115
	TEMP	°F	57	68	17	73	73	58	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	62	65	71	96	75	64	64
	PRESSURE	INCH OF WTR																																			

MODE	PARAMETER		3	70	71
SMOKE EXHAUST	FLOW	CFM	12645	5765	5640
	PRESSURE	INCH OF WTR			
NORMAL	FLOW	CFM	1420	—	—
	PRESSURE	INCH OF WTR		0	0

MODE	PARAMETER		36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70	71
SUMMER	FLOW	CFM	1115	1200	450	6435	825	500	500	100	200	100	520	175	400	520	175	400	260	400	230	230	100	1200	100	100	600	200	200	1415	200	1250	1150	1150	1200	600	0	0
	TEMP	°F	64	75	83	58	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	62	59	59	59	59	64	106	75	64	64	64	75	83	—	—
	PRESSURE	INCH OF WTR																																				
WINTER	FLOW	CFM	1115	1200	450	6435	825	500	500	100	200	100	520	175	400	520	175	400	260	400	230	230	100	1200	100	100	600	200	200	1415	200	1250	1150	1150	1200	600	0	0
	TEMP	°F	64	75	83	58	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	59	62	59	59	59	59	64	106	75	64	64	64	75	83	—	—
	PRESSURE	INCH OF WTR																																				

NO.	REVISIONS	DATE	DR	CHK	RE	DM	DM	DM	PDM
3	INCORP. ARN 08743	1-14-0	TJ						
2	REVISIONS	DATE	DR	CHK	RE	DM	DM	DM	PDM
1	ISSUED FOR CONSTRUCTION	06/08/93	SEE MICROFILM FOR SIGNATURES						
NO.	REVISIONS	DATE	DR	CHK	SUPV	EGL	EGS	CHFE	PE	KPE

ENG. MGR. DATE

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

FLOW DIAGRAM
CONTROL BUILDING NORMAL
HVAC SYSTEM LEVEL A
SYSTEM NO. 1539

SCALE: NONE
JOB NO. 10604
DRAWING NO. AX4DB269
REV. 6
SIZE E 34X44

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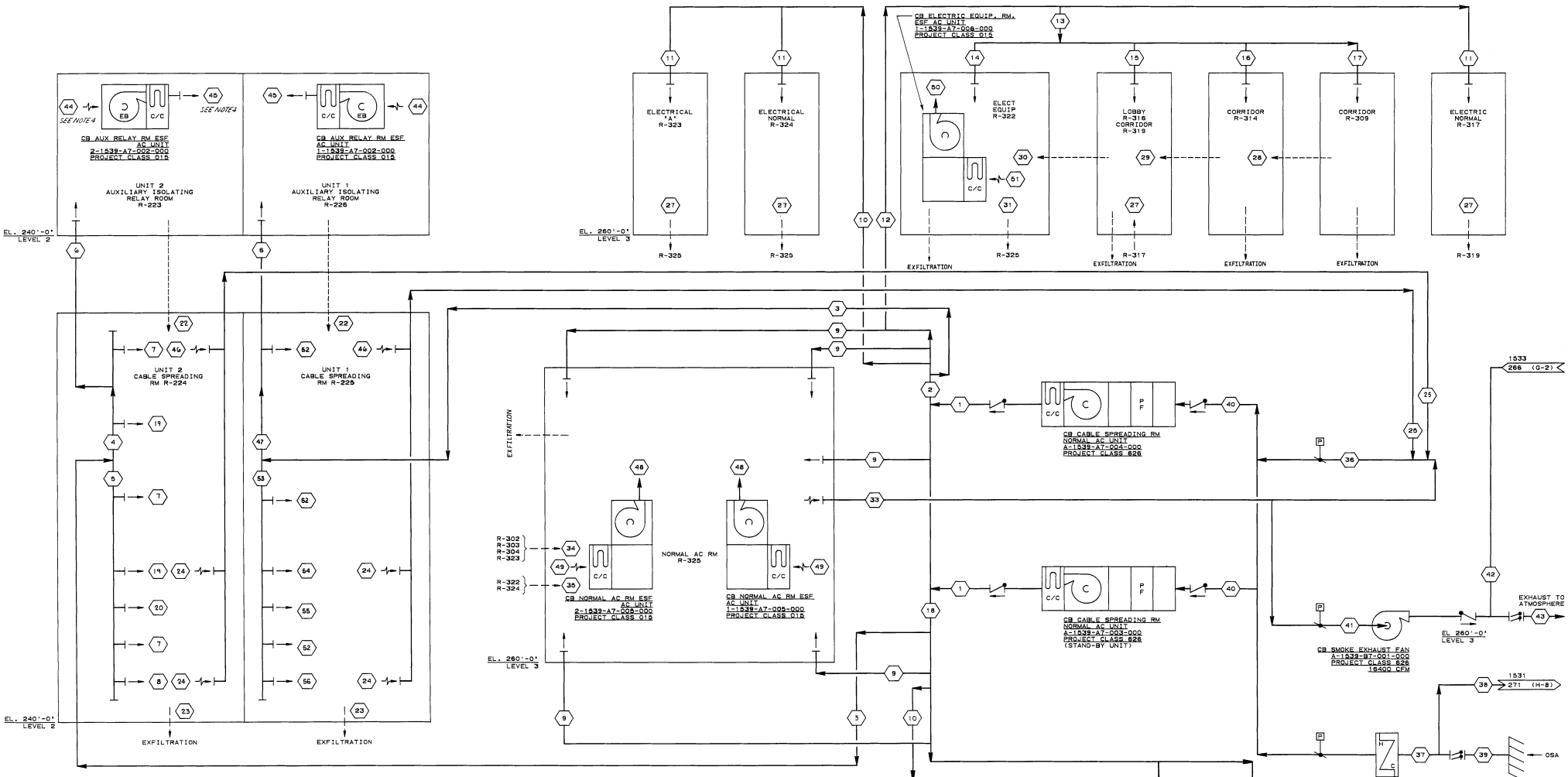

MODES	PARAMETER		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25
SUMMER	FLOW	CFM	15510	8185	3050	960	2090	200	470	590	1560	110	55	12005	11135	1300	250	350	450	6935	290	270	300	200	200	900	2850
	TEMP	°F	57	57	58	59	59	59	59	59	59	58	59	58	58	59	59	59	59	57	58	58	59	75	70	70	70
WINTER	FLOW	CFM	15510	8185	3050	960	2090	200	470	590	1560	110	55	12005	11135	1300	250	350	450	6935	290	270	300	200	200	900	2850
	TEMP	°F	57	57	58	59	59	59	59	59	59	58	59	58	58	59	59	59	59	57	58	58	59	75	70	69	69
SMOKE REOVAL	FLOW	CFM	15510	8185	3050	960	2090	200	470	590	1560	110	55	12005	11135	1300	250	350	450	6935	290	270	300	200	200	900	2850
	TEMP	°F	17/98	17/98	17/98	17/98	17/98	17/98	17/98	17/98	17/98	17/98	17/98	17/98	17/98	17/98	17/98	17/98	17/98	17/98	17/98	17/98	17/98	NOTE 3	NOTE 3	NOTE 3	NOTE 3

MODES	PARAMETER		26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56
SUMMER	FLOW	CFM	55	55	350	675	955	1005	250	8975	415	1060	15375	1300	16400	2000	1810	—	—	—	600	600	1050	720	4300	4300	2000	2000	520	2330	340	320	630
	TEMP	°F	100	81	76	78	75	90	89	90	100	90	86	98	98	98	88	—	—	—	75	59	59	58	55	100	59	100	58	58	58	58	58
WINTER	FLOW	CFM	55	55	350	675	955	1005	250	8975	415	1060	15375	1300	16400	2000	1810	—	—	—	600	600	1050	720	4300	4300	2000	2000	520	2330	340	320	630
	TEMP	°F	100	81	67	68	53	67	68	78	90	80	78	17	17	17	75	—	—	—	75	59	59	58	55	100	59	100	58	58	58	58	58
SMOKE REOVAL	FLOW	CFM	55	55	350	675	955	1005	250	8975	415	1060	15375	1300	16400	2000	1810	16380	20000	36380	—	—	1050	720	—	—	—	—	520	2330	340	320	630
	TEMP	°F	NOTE 3	NOTE 3	NOTE 3	NOTE 3	NOTE 3	NOTE 3	NOTE 3	NOTE 3	NOTE 3	NOTE 3	NOTE 3	17/98	17/98	17/98	17/98	NOTE 3	NOTE 3	NOTE 3	—	—	NOTE 3	17/98	—	—	—	—	17/98	17/98	17/98	17/98	17/98

NOTES:

1. SEE P & ID NOS. AX4DB216, AX4DB237 & AX4DB211.
2. DELETED
3. TEMPERATURE VARIES DURING SMOKE REMOVAL OPERATION.
4. FOR UNIT 2, AIR FLOW ASSOCIATED WITH NODES 44 & 45 IS MINIMUM 11000CFM (FAN TAG # 2-1539-A7-002).

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NO.	VERSIONS	DATE	DR	CHK	APPV
9.0	REVISED PER 1070199801M002, VER. 1.0	10-07-09	KB	RBH	JMR

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

FLOW DIAGRAM
CONTROL BLDG CABLE SPREADING ROOM
LEVEL 2 & 3 HVAC
SYSTEM NO. 1539

SCALE: NONE

DRAWING NO. AX4DB270

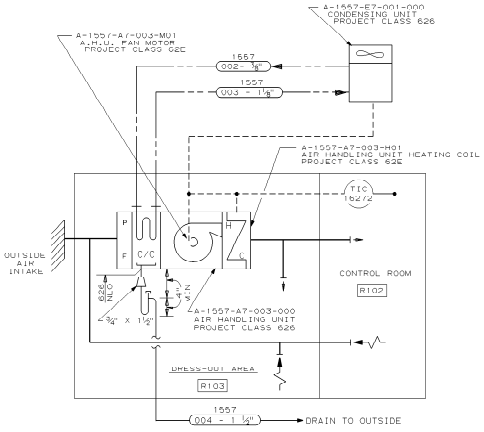
VER. 9.0

JOB NO.10604

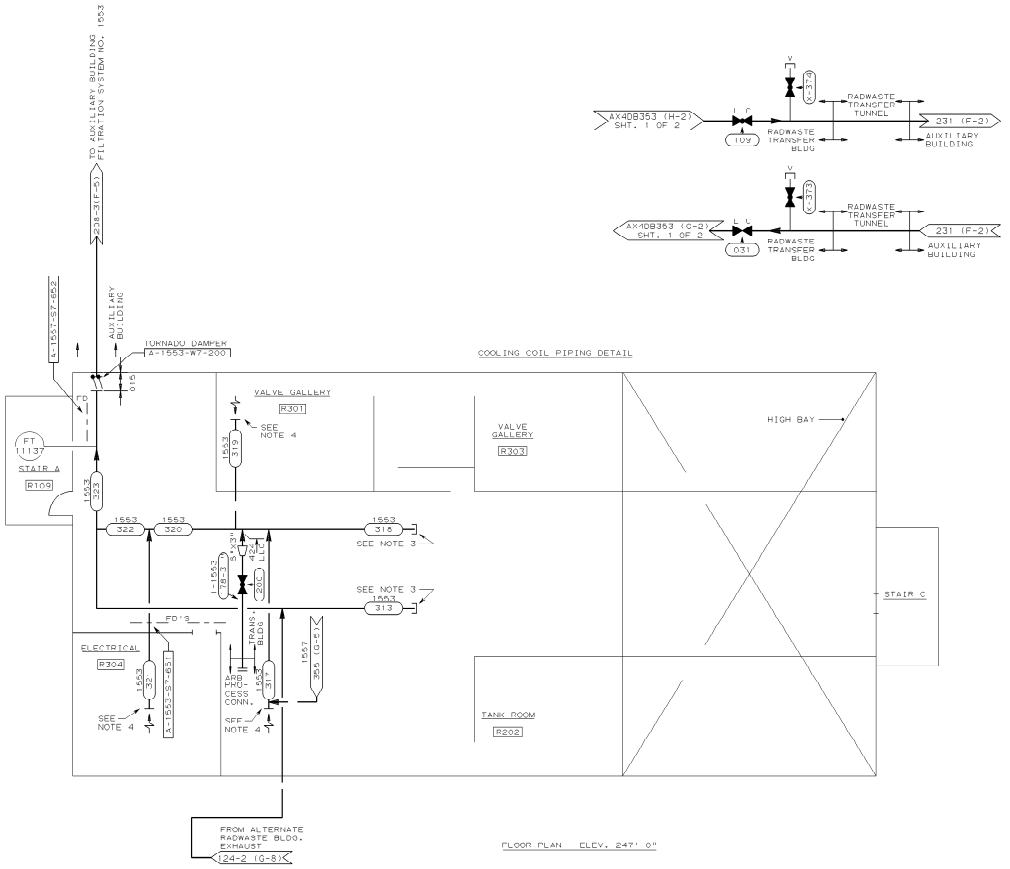
SIZE E 34X44

DRAWING CATEGORY: CRITICAL

\DCP\1070199801-1\AX4DB270.DGN 10/13/2009 7:38:30 AM

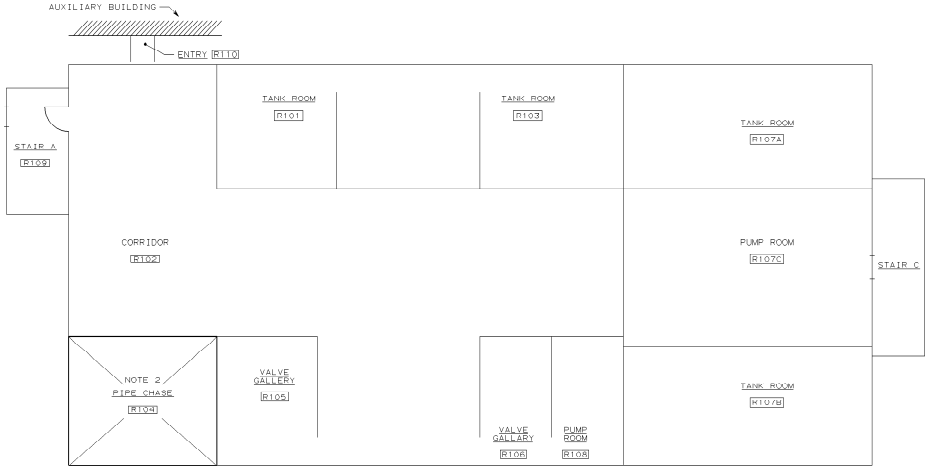


ARB CONTROL RM. AND DRESS-OUT AREA
FLOOR ELEVATION 220'-0"



FLOOR PLAN ELEV. 247' 0"

COOLING COIL PIPING DETAIL



FLOOR PLAN ELEV. 220' 0"

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT
P & I DIAGRAM
RADWASTE TRANSFER BUILDING
HVAC SYSTEM SYSTEM NO. 1557 & 1553
SHEET 2 OF 2

SCALE: NONE	DRAWING NO.	REV.
JOB NO. 10604	AX4DB353	17

SIZE E 34x44
CAD NAME AX4DB3532

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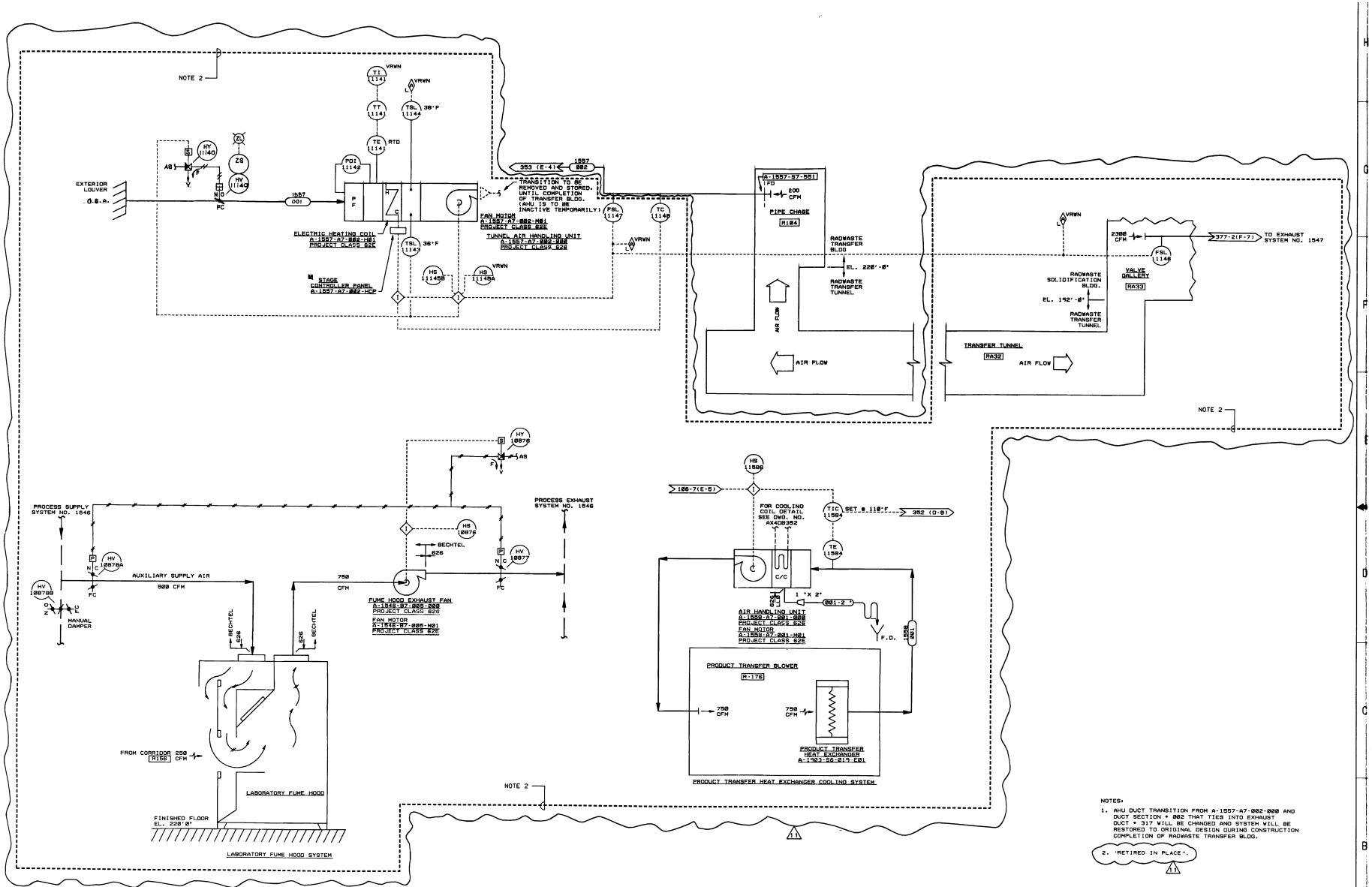


NOTES:

1. AHU DUCT TRANSITION FROM A-1557-A7-882-H00 AND DUCT SECTION # 002 THAT TIES INTO EXHAUST DUCT # 317 WILL BE CHANGED AND SYSTEM WILL BE RESTORED TO ORIGINAL DESIGN DURING CONSTRUCTION COMPLETION OF RADWASTE TRANSFER BLDG.
2. "RETIRED IN PLACE".

Daniel K. Hawkins 7/9/96

NO.	REVISIONS	DATE	DR	CHK	SUPV
1	REVISED PER REA 94-VAA043	5/11/94	JLH	RS	SD
0	ISSUED FOR CONSTRUCTION	9-25-92	SEE MICROFILM FOR SIGNATURES		

ENG. MGR. DTL EOS CHFE PE QAE DATE

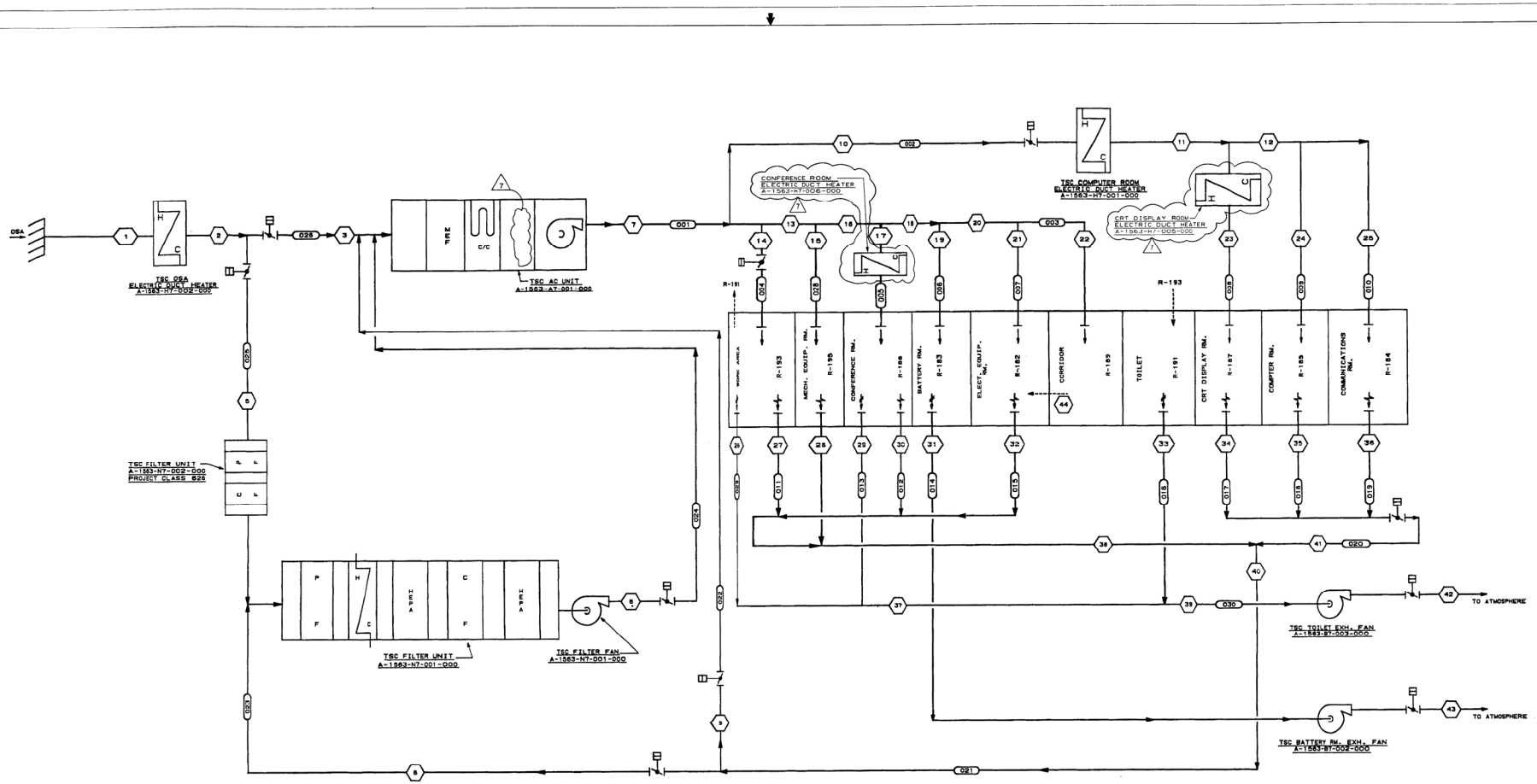
SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
RADWASTE TRANSFER TUNNEL H & V AND LABORATORY FUME HOOD SYSTEMS
SYSTEM NO. 1557 & 1546

SCALE: NONE
JOB NO.10604
DRAWING NO. AX4DB355
REV. 11

SIZE E 34x44



MODE (NORMAL)	PARAMETER	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43		
SUMMER	FLOW CFM	1250	1250	1250	—	—	—	12880	—	11630	8085	8085	5890	2575	2220	420	2155	870	1285	55	1220	1220	0	2195	4050	1640	50	1575	870	480	470	280	1200	340	2175	3700	1700	530	4055	870	11830	7575	870	280		
	TEMPERATURE °F	98	98	98				56		77	56	56	56	57	57	57	57	57	57	57	57	57	57	57	57	57	75	75	85	75	75	75	85	80	75	75	75	—	80	—	77	75	—	—		
	PRESSURE IN. OF W.G.																																												
WINTER	FLOW CFM	1250	1250	1250	—	—	—	12880	—	11630	8085	8085	5890	2575	2220	420	2155	870	1285	55	1220	1220	0	2195	4050	1640	50	1575	870	480	470	280	1200	340	2175	3700	1700	530	4055	870	11830	7575	870	280		
	TEMPERATURE °F	17	68	68				56		77	56	56	56	57	57	57	57	57	57	57	57	57	57	57	57	57	75	75	85	75	75	75	85	80	75	75	75	—	80	—	77	75	—	—		
	PRESSURE IN. OF W.G.																																												

MODE (EMERGENCY)	PARAMETER	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43		
SUMMER	FLOW CFM	500	500	—	—	500	2500	12880	3000	9880	8085	8085	5890	2575	2220	420	2155	870	1285	55	1220	1220	0	2195	4050	1840	—	1765	1070	0	870	0	1400	—	2175	3700	1700	—	4605	—	12380	7575	0	0		
	TEMPERATURE °F	98	98			98	77	56		77	56	56	56	57	57	57	57	57	57	57	57	57	57	57	57	57	—	75	85	—	75	—	85	—	75	75	75	—	80	—	77	75				
	PRESSURE IN. OF W.G.																																												
WINTER	FLOW CFM	500	500	—	—	500	2500	12880	3000	9880	8085	8085	5890	2575	2220	420	2155	870	1285	55	1220	1220	0	2195	4050	1840	—	1765	1070	0	870	0	1400	—	2175	3700	1700	—	4605	—	12380	7575	0	0		
	TEMPERATURE °F	17	68			68	77	56		77	56	56	56	57	57	57	57	57	57	57	57	57	57	57	57	57	—	75	85	—	75	—	85	—	75	75	75	—	80	—	77	75				
	PRESSURE IN. OF W.G.																																												

MODE	PARAMETER	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43		
ISOLATION	FLOW CFM	—	—	—	—	—	3000	12880	3000	9880	8085	8085	5890	2575	2220	420	2155	870	1285	55	1220	1220	0	2195	4050	1840	—	1765	1070	0	870	0	1400	—	2175	3700	1700	—	4605	—	12380	7575	0	0		
	TEMPERATURE °F	—	—	—	—	—	77	56		77	56	56	56	57	57	57	57	57	57	57	57	57	57	57	57	57	—	75	85	—	75	—	85	—	75	75	75	—	80	—	77	75				
	PRESSURE IN. OF W.G.																																												

NOTES:

1. SEE P & ID NUMBER AX4DB235
2. THE DATA SHOWN ON THIS FLOW DIAGRAM ARE FOR INFORMATION ONLY, AND WHILE USEFUL AS GUIDES IN OPERATION, DO NOT REPRESENT EXACT OR GUARANTEED OPERATING CONDITIONS.
3. THE COMMUNICATION ROOM, CRT DISPLAY ROOM, CONFERENCE ROOM AND WORK AREA SHALL BE BALANCED TO MINIMUM 1/8" POSITIVE PRESSURE ABOVE ATMOSPHERIC. THE REST OF THE AREAS WITHIN THE TSC BOUNDARY SHALL BE SLIGHTLY POSITIVE.

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

FLOW DIAGRAM
ONSITE TECHNICAL SUPPORT CENTER
HVAC SYSTEM 1563

NO.	REVISIONS	DATE	DR	CHK	RE	DM	DM	DM	PDM
3	INCORP. ABN-18366.	5-1-17	CD	[illegible]	[illegible]	[illegible]			
2	REVISIONS				SEE MICROFILM FOR SIGNATURES				
1	ISSUED FOR CONSTRUCTION								
NO.	REVISIONS	DATE	DR	CHK	SUPV	EGL	EGS	CHFE	PE	RPE

ENG. MGR. DATE

SCALE: NONE

JOB NO. 10604

DRAWING NO. AX4DB376

REV. 1

SIZE E 34X44

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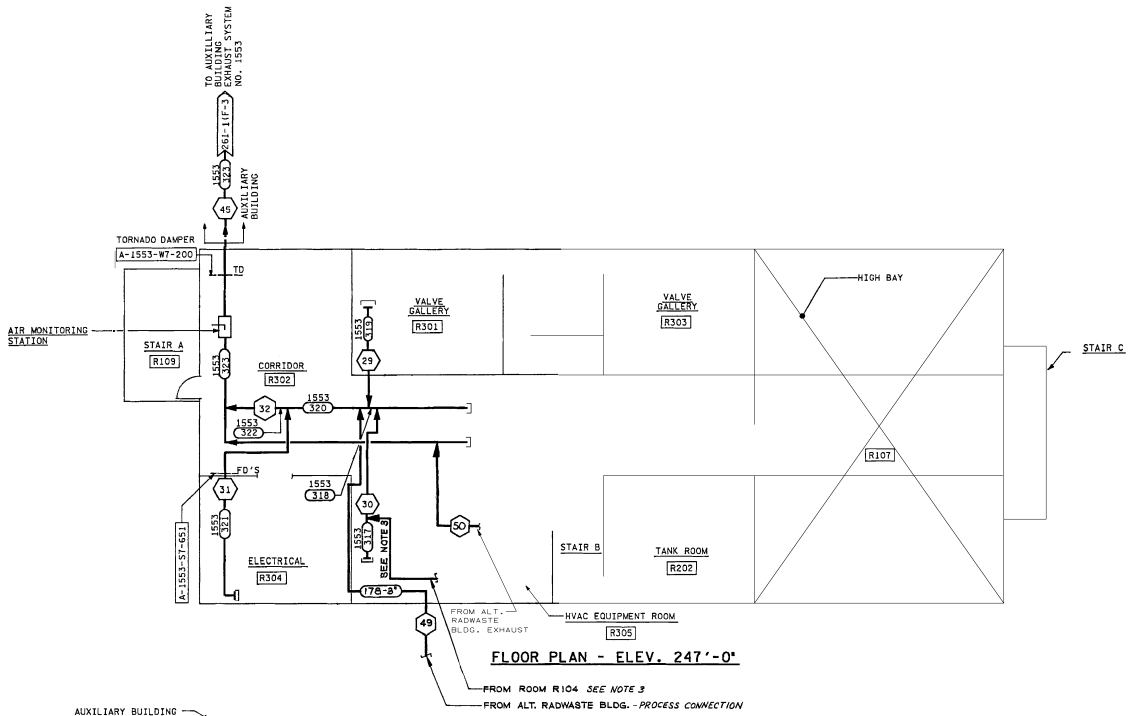
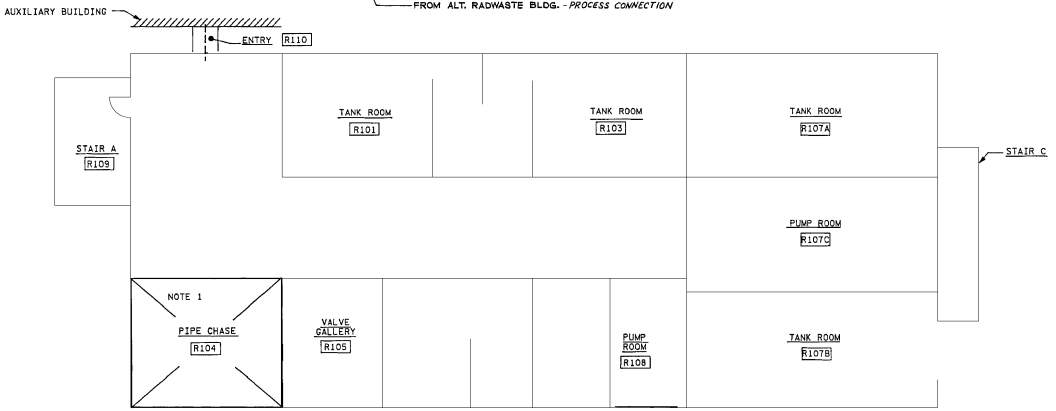
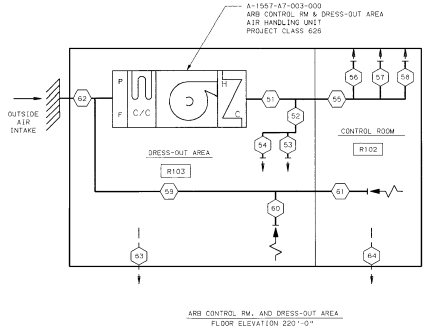


FLOOR PLAN - ELEV. 247'-0"



FLOOR PLAN - ELEV. 220'-0"



ARB CONTROL RM. AND DRESS-OUT AREA
FLOOR ELEVATION 220'-0"

MODES	PARAMETER		51	52	53	54	55	56	57	58	59	60	61	62	63	64
NORMAL	FLOW	CFM	1800	720	360	360	1080	360	360	360	1300	445	855	500	275	225
	TEMP	°F (HIGH)	—	—	—	—	—	—	—	—	75	75	75	98	75	75
		°F (LOW)	—	—	—	—	—	—	—	—	75	75	75	17	75	75

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

FLOW DIAGRAM
RADWASTE TRANSFER BLDG. HVAC SYSTEM
SYSTEM NO. 1557 & 1553
SHEET 2 OF 2

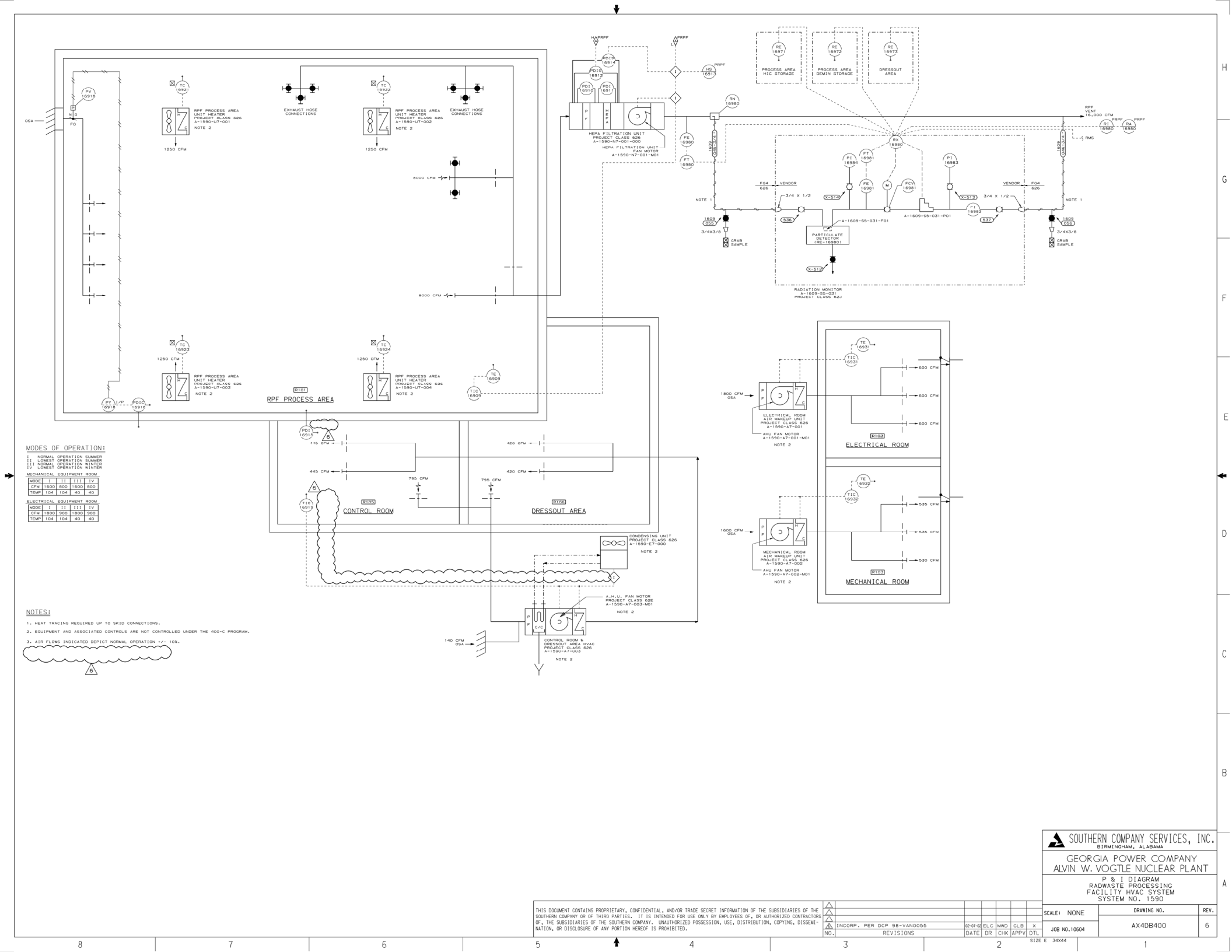
5

ENG. MGR.		
DATE	SCALE: NONE	DRAWING NO. AX4DB378 REV. 5
	JOB NO.10604	SIZE E 34X44

NO. NO. NO.

8 7 6 5 4 3 2 1

H G F E D C B A



MODES OF OPERATION:

I NORMAL OPERATION SUMMER
II LOWEST OPERATION SUMMER
III NORMAL OPERATION WINTER
IV LOWEST OPERATION WINTER

MECHANICAL EQUIPMENT ROOM

MODE	I	II	III	IV
CFM	1600	800	1600	800
TEMP	104	104	40	40

ELECTRICAL EQUIPMENT ROOM

MODE	I	II	III	IV
CFM	1800	900	1800	900
TEMP	104	104	40	40

NOTES:

1. HEAT TRACING REQUIRED UP TO SKID CONNECTIONS.
2. EQUIPMENT AND ASSOCIATED CONTROLS ARE NOT CONTROLLED UNDER THE 400-C PROGRAM.
3. AIR FLOWS INDICATED DEPICT NORMAL OPERATION +/- 10%.

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NO.	REVISIONS	DATE	DR	CHK	APPV	DTL
3	INCORP. PER DCP 98-VAN0055	02-07-02	ELC	MWD	GLB	X

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
RADWASTE PROCESSING
FACILITY HVAC SYSTEM
SYSTEM NO. 1590

SCALE: NONE
DRAWING NO. AX4DB400
REV. 6
JOB NO. 10604
SIZE E 34X44